

Enhancing conceptual modeling through multimodal data analysis and mixed reality

DISSERTATION

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I hereby declare that I have written this thesis independently, that I have completely specified the utilized sources and resources and that I have definitely marked all parts of the work – including tables, maps and figures – which belong to other works or to the internet, literally or extracted, by referencing the source as borrowed.

I further declare that I have used generative AI tools only as an aid, and that my own intellectual and creative efforts predominate in this work. In the appendix “Overview of Generative AI Tools Used” I have listed all generative AI tools that were used in the creation of this work, and indicated where in the work they were used. If whole passages of text were used without substantial changes, I have indicated the input (prompts) I formulated and the IT application used with its product name and version number/date.

Vienna, March 1, 2026

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Abstract

Conceptual modeling provides a structured means of formalizing domain knowledge. Traditionally, it relies on declarative knowledge elicited through interviews, workshops, and documentation. However, conceptual models typically capture what experts can articulate or reenact, rather than what they actually enact in practice. As a result, a substantial portion of domain knowledge remains tacit and is rarely reflected in formal representations.

This thesis addresses that limitation by introducing a method that integrates multimodal data analysis and Mixed Reality–based elicitation into conceptual modeling.

First, artificial intelligence for multimodal data analysis supports modeling what has actually occurred in practice—rather than relying on retrospective explanation—by examining video recordings, emails, chat transcripts, system logs, and related artifacts.

Second, Mixed Reality serves as an egocentric elicitation substrate that moves beyond asking experts what they think they would do or asking them to verbally reenact past behavior. Instead, it places them inside controlled counterfactual situations in which they must act. These enacted responses generate new multimodal traces, which are again subjected to multimodal data analysis.

Together, observation of factual behavior of domain experts and enactment of counterfactual scenarios create a closed analytical loop through which tacit domain knowledge becomes systematically observable and integrable into conceptual modeling practice.

The method is evaluated through controlled studies and structured expert assessments across multiple modeling languages and scenarios within the emergency room domain. Empirical results demonstrate measurable improvements in conceptual model precision and conciseness compared to modeling sessions relying solely on traditional elicitation approaches.

Kurzfassung

Die konzeptuelle Modellierung stellt ein strukturiertes Mittel zur Formalisierung von Domänenwissen dar. Traditionell basiert sie auf deklarativem Wissen, das durch Interviews, Workshops und Dokumentation erhoben wird. Konzeptionelle Modelle erfassen jedoch typischerweise das, was Expert:innen artikulieren oder nachstellen können – nicht jedoch das, was sie in der Praxis tatsächlich tun. Infolgedessen bleibt ein wesentlicher Teil des Domänenwissens implizit (tacit) und wird nur selten in formalen Repräsentationen abgebildet.

Diese Dissertation adressiert diese Einschränkung durch die Einführung einer Methode, die multimodale Datenanalyse und Mixed Reality–basierte Elicitierung in die konzeptuelle Modellierung integriert.

Erstens ermöglicht künstliche Intelligenz zur Analyse multimodaler Daten eine Modellierung dessen, was in der Praxis tatsächlich geschehen ist – anstatt sich auf retrospektive Erklärungen zu stützen – indem Videoaufzeichnungen, E-Mails, Chatprotokolle, Systemlogs und verwandte Artefakte untersucht werden.

Zweitens fungiert Mixed Reality als egozentrisches Elicitierungsmedium, das über die bloße Befragung von Expert:innen hinausgeht, was sie glauben, getan zu haben oder tun zu würden. Stattdessen werden sie in kontrollierte kontrafaktische Situationen versetzt, in denen sie handeln müssen. Diese ausgeführten Handlungen erzeugen neue multimodale Daten, die wiederum einer multimodalen Analyse unterzogen werden.

Die Kombination aus der Beobachtung faktischen Verhaltens von Domänenexpert:innen und der Enaktierung kontrafaktischer Szenarien schafft einen geschlossenen analytischen Kreislauf, durch den implizites Domänenwissen systematisch beobachtbar und in die konzeptuelle Modellierung integrierbar wird.

Die Methode wird durch kontrollierte Studien und strukturierte Expert:innenbewertungen über mehrere Modellierungssprachen und Szenarien hinweg im Anwendungsbereich der Notaufnahme evaluiert. Die empirischen Ergebnisse zeigen messbare Verbesserungen hinsichtlich der Präzision und Konsistenz konzeptueller Modelle im Vergleich zu Modellierungssitzungen, die ausschließlich auf traditionellen Elicitierungsansätzen beruhen.

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Introduction

Long before humans learn to speak, reason, or formalize their thoughts, they accumulate thousands of lived fragments: looking, reaching, failing, adjusting, being corrected, trying again. From this slow sedimentation of embodied experience, certain patterns become stable enough that we stop noticing them — they become the implicit rules by which we navigate the world without ever having to say them aloud.

As argued by Mylopoulos, Guizzardi, and Guarino [MGG25], human cognition emerges from our continuous exposure to physical reality. We engage with this reality through our bodies and limited perceptual apparatus. Across our lifetime—crucially during early development—reality discloses itself to us through temporally atomic sensorimotor episodes. From this stream of situated experiences, and under the formative influence of our social environment (family, education, culture), we gradually abstract regularities that become cognitively relevant objects and events, forming what is called our cognitive domain [BEF⁺56]. Each perceptual presentation corresponds to a particular world, understood as a concrete arrangement of those relevant objects and events. Through repeated exposure, we learn not only which objects and events matter, but also the properties and relations that structure them.

Early philosophical thought already highlights that **concepts** stand at the interface between mind and language [Kar24]. For instance, Clement of Alexandria stated that humans can attain secure knowledge of the external world, and that such knowledge is conceptual in nature (middle of the 2nd century). Karamanolis observes that in early thinkers concepts were treated both as mental representations and as meanings of words, and that our access to reality is mediated by concepts in these two inseparable senses [Kar24].

In a complementary line, Guarino characterizes concepts as intensional entities [Gua98]: they function as rules for individuating instances of properties or relations across possible worlds, hence as functions from possible worlds to extensional relations over the cognitive

domain. These intensional entities—conceptual properties and conceptual relations—are what we simply call concepts.

This view is consolidated in the account by Mylopoulos, Guizzardi, and Guarino [MGG25], who describe concepts as cognitively relevant regularities that track universals. Importantly, on this view concepts are mental phenomena; they do not reside in external reality, nor do they inherently live inside computers.

To appreciate the tension between our need for conceptual structure and the unruliness of reality, consider the words of a famous author:

Deep in the human unconscious is a pervasive need for a logical universe that makes sense. But the real universe is always one step beyond logic.

— Frank Herbert

Despite the fact that concepts do not inhabit physical reality nor computational artifacts, we are nevertheless compelled to construct faithful computational representations of (external) reality that are sufficiently aligned with the conceptual structures that guide human understanding and decision-making.

However, even when we succeed in capturing explicit and shareable conceptual structures, a significant part of what experts know never becomes explicit in the first place. As noted by Cairó Battistutti, and Bork [CBB17], much knowledge is tacit in nature: it resides within the minds of individuals, tied to their embodied experience and situated practice, and therefore resists transfer through written or verbal communication alone.

This motivates the central ambition of this thesis: to enhance conceptual modeling by enriching models with *tacit knowledge*¹ that human experts apply but rarely state explicitly.

We target two complementary sources to collect such knowledge. First, **multimodal data**, such as video, human–system interaction logs, and conversations, which we hypothesize to encode latent conceptual regularities that can be uncovered through algorithmic analysis as well as human introspection. Second, **Mixed Reality (MR)** as an *egocentric elicitation substrate*², which captures knowledge directly from the expert’s own first-person perspective and enables perturbing the environment through live what-if counterfactuals, thereby revealing implicit decision rules that are only manifest when the world is varied.

¹Tacit knowledge is personal, experiential, and hard to articulate [Wik25b]. Tacit and implicit both mean understood without being stated, but implicit knowledge is suggested, inferred, or built into a system/process (like an unwritten rule or a software’s underlying logic), and can often be explained or codified with effort, making tacit knowledge a deeper, harder-to-share subset of the broader concept of implicit knowing. For further study on the role and nature of tacit knowledge in organizations and knowledge creation, we refer the reader to [NTU96, LG03, Hoe06, Tso05].

²Elicitation is the synthesis of opinions of experts on a subject where there is uncertainty due to insufficient data [Wik25a]. It is used in various fields like intelligence gathering, psychology, and software development to uncover what is not readily available or to get a specific response from someone.

Multimodal data provides information that exists across multiple different formats or modalities simultaneously, including text, audio, image, video, and sensory or specialized data. From a technical perspective, multimodal data presents significant design challenges because each modality has its own structure, scale, and semantic properties [LNR⁺25]. For example, text data is typically processed as sequences of tokens, images as pixel matrices, audio as waveforms, and time series as sequential numerical values, each requiring different pre-processing techniques, and analytical approaches that can be successfully handled using Artificial Intelligence (AI) [FHJZ24].

The concept of Mixed Reality first appeared in 1994 thanks to Paul Milgram and Fumio Kishino [MK94]. Both defined MR as a technology that allowed at the same time: introducing a real object/person into the virtual world and introduce a virtual object/person into the real world.

The combination of automatic multimodal analysis and interactive MR-based elicitation opens a systematic way to discover and formalize conceptual regularities that would otherwise remain unspeakable. In this way, conceptual modeling is no longer limited to what experts say, but can also incorporate what they *actually do* — and what they *would do if the situation were different*. This may close a crucial epistemic gap between the concepts that live in human cognition and the representations we build inside computational systems.

Section 1.1 motivates the thesis and concludes with the research problem statement. Section 1.2 defines the research goals and questions. Section 1.3 summarizes related research contributions by the author. Section 1.4 presents the main outcomes of the thesis. Section 1.5 outlines the remainder of the dissertation.

1.1 Motivation

Tacit knowledge is essential for understanding how expertise and organizational capability emerge, as it encompasses experiential, embodied, and context-dependent knowing that cannot be fully articulated or codified [Hoe06]. Nonaka’s theory of organizational knowledge creation emphasizes that innovation arises through the dynamic interaction between implicit (tacit), and explicit knowledge, mediated by social processes such as shared practice and dialogue [NTU96]. Later work stresses that tacit knowledge is not merely a precursor to explicit knowledge but a fundamental and irreducible form of knowing that underpins skilled action [LG03, Hoe06]. From a critical perspective, attempts to fully extract or formalize tacit knowledge risk overlooking its inherently situated and practice-bound nature [Tso05].

To illustrate these challenges concretely, we anchor our motivation in a representative example domain for a system under investigation (SUI): the emergency room reception front desk. This setting is rich in multimodal interactions—visual cues, verbal exchanges, human-system logs, and rapid situational assessments—all of which influence process outcomes but are rarely captured in formal models explaining the process [LNR⁺25].

1.1.1 An illustrative Domain

The emergency room (ER) reception front desk represents a complex, high-stakes environment in which administrative tasks, clinical sensemaking, and multimodal situational awareness converge, a setting that has been explicitly studied since the mid-20th century in foundational work on emergency department decision-making [Tis66]. Receptionists operate at the intersection of patient communication, digital systems, and real-time coordination with medical staff. Their work requires continuous interpretation of visual cues from CCTV³, tone and urgency conveyed in speech, and fast-changing operational signals such as alarms, alerts, and intercom⁴ requests. For example, during patient check-in, ER receptionists verify identity, and create or retrieve electronic health record (EHR)⁵ entries. Certain situations require immediate escalation, such as *Code Silver*, an event that indicates a violent or threatening situation within the hospital [Wik25c], where the receptionist must respond to alarms or phone instructions, follow security protocols, and coordinate with clinical staff and security teams with calm procedural clarity. An ER receptionist faces situations where a patient, visitor, or accompanying individual may pose a hidden risk despite appearing calm on the surface. Such scenarios include cases where a person may be carrying a concealed weapon, exhibiting controlled but dangerous behavior, or coercing a patient in ways that resemble kidnapping or forced accompaniment. Those domain experts rely on subtle multimodal cues—unusual posture, restricted movement, guarded tone, avoidance of eye contact, or inconsistencies between verbal statements and observed behavior [LNR⁺25].

1.1.2 Importance of Conceptual Modeling

Many of the scenarios encountered at the reception front desk combine explicit procedures with tacit, multimodal decision-making. Formalizing these situations in conceptual models allows organizations to document practices that are otherwise learned only through experience [GBP25h]. Such models provide a foundation for developing clear guidelines that standardize responses across staff, ensuring consistency in fast-changing and high-pressure contexts. Beyond documentation, these models serve for training and simulation. They aim to enable new staff to rehearse rare but critical events, explore what-if scenarios, and build confidence in applying appropriate escalation paths [GBP24b]. By choosing well-known notations such as BPMN⁶ or UML⁷ Sequence Diagrams, we can capture both the temporal order of events and the responsibilities of different stakeholders (e.g.,

³CCTV stands for closed-circuit television and is commonly known as video surveillance.

⁴A security intercom is a communication system designed to enable two-way communication between a person at an entry point (such as a door, gate, or entrance) and someone inside a building or facility, often for the purpose of security and access control.

⁵An electronic health record is the systematized collection of electronically stored patient and population health information in a digital format.

⁶Business Process Model and Notation (BPMN), is a global standard for creating graphical diagrams of business processes [Obj08].

⁷The Unified Modeling Language (UML) is a general-purpose, object-oriented, visual modeling language that provides a way to visualize the architecture and design of a system [Obj07].

receptionist, security, clinical staff, and technical systems). For instance, modeling a Code Silver scenario as a sequence diagram aims to clarify how observations lead to an alert, how the receptionist escalates to security, how alarms propagate, and how clinical areas are secured. Figure 1.1 provides a simplified but formalized sequence diagram depiction of how a Code Silver situation unfolds from the perspective of the ER reception front desk. The diagram illustrates the flow of information between actors and systems.

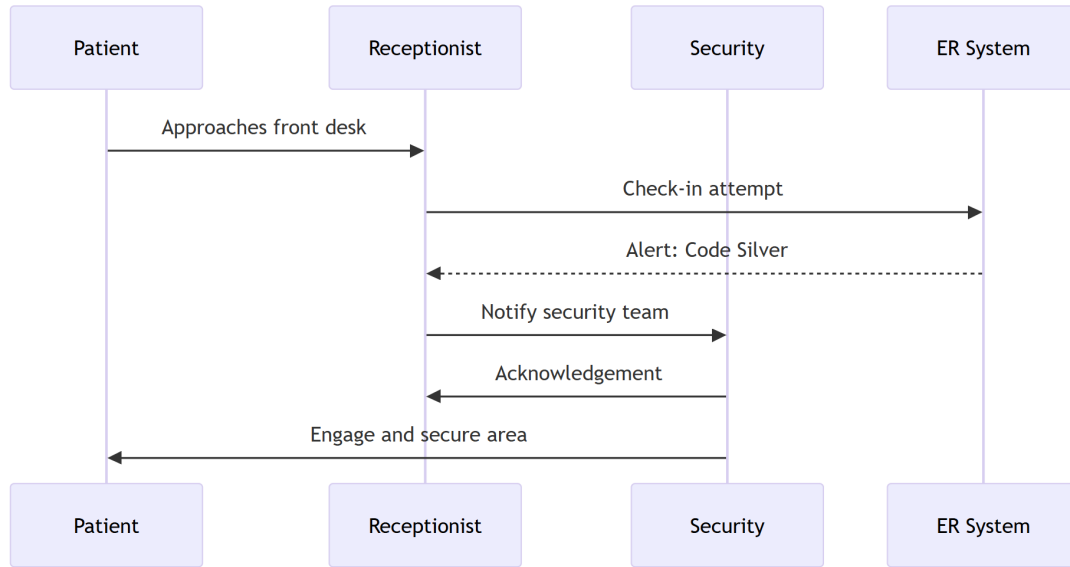


Figure 1.1: Illustrative sequence diagram of a Code Silver handling at the ER reception front desk. Disclaimer: This example is purely a visual illustration and does not reflect or analyze the operational use of this model.

1.1.3 The Problem of Knowledge Elicitation in Conceptual Modeling

A conceptual modeling session brings together two complementary forms of expertise: the *domain expert*, who possesses deep practical knowledge of emergency-room reception workflows, and the *domain modeler*, who is skilled in applying formal modeling languages such as BPMN. In the context of the ER reception front desk, the goal of such a session would be to translate experience-based operational knowledge into conceptual models. The session typically begins with the domain expert narrating how the process unfolds in everyday practice, often using concrete, recent situations to illustrate typical and exceptional events. These descriptions form the raw material from which the modeler begins to identify actors, activities, decision points, and communication channels. The modeler then proposes initial abstractions using a formal modeling notation, checking with the domain expert whether the structure faithfully reflects the real process. As the conversation progresses, the two collaborators converge on a shared representation of the workflow.

1. INTRODUCTION

A core difficulty in conceptual modeling is that experts routinely rely on knowledge they never think to verbalize [vdAADM⁺11] – humans filter the world through habits and assumptions shaped by repeated experience. To address this challenge, structured knowledge-engineering methodologies such as CommonKADS were developed [Sch00]. In everyday domains we instinctively rely on shared common sense [For89]. However, in highly complex and safety-critical domains like emergency-room reception, what feels like “common sense” is only common among those with long practical experience [PBLV12].



Figure 1.2: Illustrative SAP Scenes setup for the ER front-desk domain arranged to reenact a situation. Work created with ScenesTM by SAP AppHaus (<https://sapapphaus.wpenginepowered.com/scenes>).

Traditional elicitation techniques such as *SAP Scenes* [SAP20] use story-driven methods to help stakeholders externalize and communicate their work practices. With SAP Scenes, participants use printed cardboard figures, environment props, and simple comic-style panels to construct narrative scenes. By arranging characters, objects, and settings on a table or board, domain experts can playfully reenact typical situations, illustrate exceptions, and jointly explore alternative process paths. This hands-on, visual approach aims to lower the barrier to communication and encourages creativity. An illustrative scene in Figure 1.2 was created using SAP Scenes, which—under the Creative Commons Attribution–Non-Commercial–Share-Alike license—allows the use of ScenesTM elements.

1.1.4 Research Problem Statement

Foundational account of conceptualization, first articulated in [GG95] and refined in [Gua09], frames the cognitive mechanism underlying concept learning as a structured abstraction from experienced reality: A conceptualization is given as a triple $\langle D, W, R \rangle$, where D denotes a cognitive domain of discourse, W is the set of all possible arrangements (or states) of that domain, and R is a set of conceptual properties and relations defined over D . Conceptualizations, as defined, are mental constructs that reside in the cognitive apparatus of human agents.

The practice of conceptual modeling has exerted structural influence across disciplines such as Modeling and Simulation (M&S), Information Systems (IS), and Software Engineering (SE), which motivated the expert panel chaired by Robinson [RAB⁺15], which convened researchers from multiple communities specifically to examine whether a *unified cross-disciplinary definition* of a conceptual model could be articulated, and to debate its purpose and benefits beyond the boundaries of any single field.

For this thesis, we adopt the definition by Arbez and Birta (as in [RAB⁺15]), who describe a conceptual model as

“a concise and precise consolidation of all goal-relevant structural and behavioral features of the system under investigation presented in a predefined format.”

As shown in Fig. 1.3, we explicitly bridge the thesis ambition with the definition of conceptual modeling so that an enhancement criterion can be meaningfully stated.

The *thesis ambition* constrains the mechanisms of enhancement to two classes: (i) multimodal data analysis, which exposes latent conceptual regularities in situated conduct, and (ii) Mixed Reality, which serves as an egocentric elicitation substrate enabling direct counterfactual perturbation of the environment.

Finally, the *enhancement criterion* specifies the desired value for conceptual modeling outcome in terms of two properties: **precision** and **conciseness**, for a removal of ambiguity by grounding insights in behaviorally evidenced regularities.

This definition is adopted in this thesis for three reasons.

First, it interprets the term “system under investigation (SUI)” in the most general and domain-agnostic sense, without restricting it to engineered or physically realized entities. Under this view, the system under investigation may be a socio-technical process, an organizational routine, a regulated workflow, a proposed design, or even a purely hypothetical construct. This level of generality is desirable in conceptual modeling, which routinely targets not only extant systems but also intended, normative, or counterfactual ones.

Second, the definition does not presuppose a physical instantiation of the SUI. In design-oriented disciplines, conceptual models frequently precede implementation.

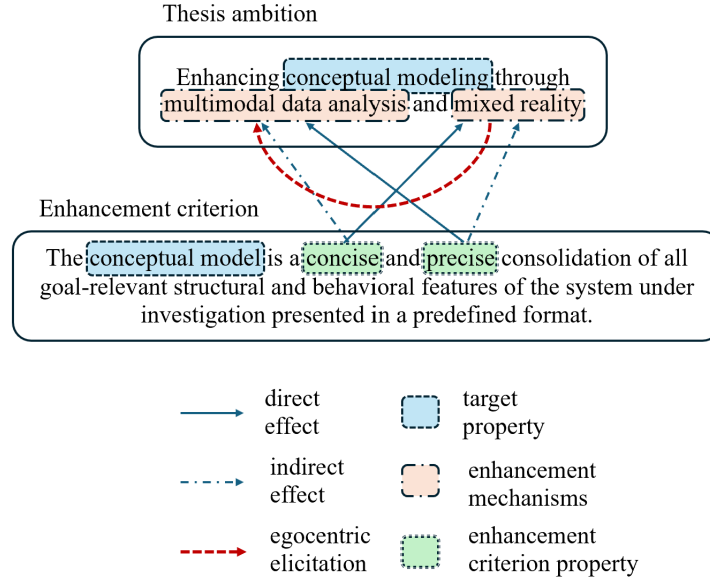


Figure 1.3: Connecting the thesis ambition to the formal definition of conceptual modeling [RAB⁺15] in order to articulate the enhancement criterion.

Third, the definition implicitly assumes that the SUI exhibits temporal or behavioral regularities, rather than being a static aggregation of structural features. By requiring the consolidation of both structural and behavioral aspects, it frames conceptual models as artifacts about systems that unfold over time, evolve under conditions, and support explanation or prediction. This is aligned with the epistemic role of conceptual modeling in disciplines where understanding dynamic behavior is a prerequisite for analysis, reasoning, and intervention.

The problem addressed in this thesis can now be stated:

We currently lack systematic methods to enhance the precision and conciseness of conceptual models by incorporating tacit knowledge that human experts rely on in practice, but rarely articulate in documentation or interviews.

1.2 Research Goals

The overarching goal of this thesis is to enhance conceptual modeling by developing a theoretical framework, method, and technical implementation that capture and incorporate tacit domain knowledge into conceptual models and by evaluating these improvements using established quality criteria, namely precision and conciseness, over a multi-scenario end-scale application domain.

1.2.1 Research Objectives and Research Questions

Research Objectives (ROs). The research objectives structure the research underpinning this thesis along the progression from theory to evaluation. The thesis addresses following research objectives:

RO1 is to develop a theoretical framework that conceptualizes tacit domain knowledge and establishes its role in conceptual modeling.

RO2 is to develop a method for the systematic capture, elicitation, and interpretation of tacit domain knowledge in conceptual modeling, grounded in multimodal data analysis and counterfactual dispositions.

RO3 is to design and implement a technical system that operationalizes the proposed method in an end-scale application domain, supporting multiple scenarios and conceptual modeling languages.

RO4 is to empirically evaluate the impact of the proposed approach on conceptual model quality using established criteria, with a focus on precision and conciseness across scenarios and conceptual modeling languages.

Together, these objectives reflect a design-science-oriented research trajectory [Wie14] in which conceptual foundations, methodological developments, and technical artifacts are tightly coupled. Rather than addressing isolated modeling tasks, the thesis targets end-scale application settings in which multiple scenarios, heterogeneous data sources, and different conceptual modeling languages coexist.

Research Questions (RQs). The research questions operationalize the objectives and guide the investigation. The thesis addresses following research questions:

RQ1: How can tacit domain knowledge be conceptualized and integrated into conceptual modeling frameworks?

RQ2: Which methods enable the reliable capture and elicitation of tacit domain knowledge into representations suitable for conceptual model construction?

RQ3: How can reliable capture and elicitation of tacit domain knowledge methods be realized in a technical implementation that supports multi-scenario, language-independent conceptual modeling in an end-scale application domain?

RQ4: To what extent does the integration of tacit domain knowledge improve the precision and conciseness of conceptual models across scenarios and modeling languages?

Alignment Between ROs and RQs. Each research question directly corresponds to one research objective, ensuring traceability from problem formulation to evaluation. RQ1 addresses the theoretical grounding required by RO1, RQ2 operationalizes RO2 at the method level, RQ3 realizes RO3 through technical implementation and application, and RQ4 validates RO4 through empirical evaluation. This alignment ensures that the

overarching goal of enhancing conceptual modeling through is addressed in a systematic, and evaluable manner.

1.3 Author's Related Research Contributions

This section outlines the author's related research contributions that provide the conceptual, methodological, technical, and evaluation foundations upon which the present thesis is built. The trajectory of the author's related contributions is summarized chronologically in Table 1.1, providing a compact overview of the conceptual and technical evolution underlying this thesis through author's related research contributions.

Table 1.1: Overview of author's related research contributions

Peer-reviewed research papers	Reference
Enhancing Process Understanding through Multimodal Data and Extended Reality (doctoral consortium paper)	[Gav23]
Conceptual Models in Machine Learning (systematic review)	[AGPB23]
Multimodal Process Mining (MMPM)	[GBP24c]
Enriching Event Logs with Multimodal Evidence (BICycle)	[GBP24a]
Beyond Conventional Event Logs (RelaxPM)	[GBP25b]
Stakeholder-Specific and Jargon-based Multimodal Conversations	[GBP24d]
How Does UML Look and Sound?	[GBP24b]
Turning Process Models into Videos	[GBP25j]
Petri Net of Thoughts	[GBP25e]
Petri Net Structure-Driven Video Generation (Short paper)	[GBP25f]
Toward Multimodal Conceptual Modeling (MMeCM)	[GBP25i]
Surgery AI for Process Mining and Conformance Checking	[GBP25h]
Research papers with undergoing peer reviewing or revision phase	
Process Adversarial Networks	[GBP25g]
Aligning AI Knowledge and Conceptual Model Symbols (ALeKS)	[GBP25a]
From Multimodal to Crossmodal Process Mining	[GBP25d]
Comics as a Process Model Notation (ViEnNa)	[GBP25c]
Multimodal Process Analysis	[GPB26]
Multimodal Business Process Simulation	[GBP26]

Enhancing process understanding through multimodal data analysis and extended reality. A first articulation of the problem investigated in this thesis was presented in the doctoral consortium paper by Gavric [Gav23]. That work introduced the agenda of enriching process understanding by exploiting *multimodal data* and *extended reality* as complementary carriers of latent conceptual content. While its scope was framed in terms of business processes (and implicitly process models rather than conceptual models in the general sense adopted in this thesis), it already identified two key gaps in the state of the art: (i) the lack of systematic mechanisms to extract conceptual regularities

from situated multimodal traces of expert behavior, and (ii) the absence of interactive elicitation substrates capable of revealing counterfactual decision rules. Crucially, the paper concluded that bridging multimodal analysis with conceptual modeling requires encoding conceptual models in a AI-usable form.

A systematic review of conceptual models encoding for machine learning.

That requirement motivated the systematic review [AGPB23], which surveyed and analyzed existing encodings of conceptual models for machine learning as an AI approach. The review cataloged representational forms currently used to turn models into numerical or symbolic structures suitable for computational analysis. The resulting taxonomy provides the methodological substrate upon which this thesis builds its multimodal analysis pipeline.

Multimodal process mining. The work on *Multimodal Process Mining* [GBP24c] marks the first implemented pipeline that operationalizes the agenda of extracting conceptual regulations from heterogeneous behavioral traces present in multimodal data. Classical process mining such as [vdAWM04] presupposes the existence of event logs that faithfully record the execution of process activities; yet many processes leave sparse digital traces and omit the manual, physical, and analog segments of work [KKR22, LNR⁺25]. In this contribution, multimodal data streams (video) were algorithmically fused to discover conceptual (process) models, thereby demonstrating that conceptual artifacts can be induced from *what experts do* rather than solely from what they state. The paper not only instantiated the technical feasibility of such multimodal inference, but also coined the term *multimodal process mining*, which rapidly gained traction and subsequently led to the establishment of the *International workshop on Multimodal Process Mining*⁸, that has been organized in conjunction with the International Conference on Advanced Information Systems Engineering (CAiSE).

Enriching event logs with multimodal evidence. A subsequent extension of the multimodal data analysis agenda was realized in the paper on *Enriching Business Process Event Logs with Multimodal Evidence* [GBP24a]. This contribution addressed structural blind spots in event logs by proposing the **B**usiness-knowledge **I**ntegration **C**ycles (BICycle) method together with the *mm-proc-miner* tool, which converts raw multimodal signals (video, audio, sensors) into a structured, unified event log with a human-in-the-loop. A key design decision is the use of *embedding spaces* as the computational substrate for integrating heterogeneous evidence. An embedding space is a high-dimensional vector space in which semantically related entities—be they words, visual patches, acoustic segments, or process activities—are mapped to nearby points. In contrast to symbolic alignment, which requires explicit schema matching, embedding-based alignment allows heterogeneous modalities to be brought into a common metric space where semantic proximity can be measured geometrically. This enables, for example, linking a spoken

⁸<https://conferences.big.tuwien.ac.at/caise2025/mmppm.php>

instruction to a visually observed action, or matching a gesture to a textual activity label by comparing their vector representations rather than their literal forms. In the context of this thesis, embedding spaces are the mathematical mechanism that turns multimodal traces into *comparable*, *computable*, and *searchable* carriers of latent conceptual regularities, forming the basis for both model discovery and model enrichment from non-linguistic evidence. The pipeline from [GBP24a] performs semantic alignment across modalities within a self-hosted joint embedding space, while explicitly engineering for (i) privacy retention, (ii) real-time performance and scalability, and (iii) prevention of AI hallucination. Beyond the methodology, the paper introduced a novel held-out evaluation dataset in the domain of DNA home-kit testing to ensure clean generalization assessment. The results provide empirical support that multimodal evidence can enrich event logs without displacing human oversight, further reinforcing the feasibility of using multimodal traces as carriers of latent conceptual regularities.

Beyond conventional event logs: using embeddings as first-class conceptual regularities evidence. In our further research [GBP25b], we *formalize* the role of embedding spaces not merely as auxiliary representations, but as *legitimate carriers of conceptual regularities (process) evidence* on par with event logs. It demonstrates that traditional process mining algorithms—whose semantics hinge on crisp activity labels and exact event correspondences—can be systematically *relaxed* when operating over continuous embedding spaces. This relaxation enables discovery of conceptual models even when no explicit or complete event log exists, using proximity in representation space as a surrogate for symbolic equality. The same principle is extended to define *relaxed conformance checking*, where compliance is not binary but graded by the geometric distance between observed behaviors and model-implied behaviors in the embedding space.

Linguistic representation of multimodal data that is stakeholder-specific and conceptual modeling jargon-based. Building on multimodal pipelines, our work turns to the problem of *human consumption* of multimodal conceptual evidence [GBP24d]. Instead of treating multimodal data solely as a machine input, the paper argues that experts reason and negotiate about processes through *linguistic surrogates*—domain-specific jargon, metaphors, and tacit shorthand. The contribution proposes stakeholder-specific, jargon-based representations of multimodal traces to align what the multimodal pipeline *detects* with how stakeholders *talk* about process reality. In doing so, it initiates the line of research in this thesis that treats conversations about multimodal evidence as a first-class element of conceptual modeling, i.e., as the interface where tacit, multimodal, and formal knowledge are reconciled.

From interpretation to synthesis of multimodal surrogates - How does a conceptual model look and sound? The work on *How Does UML Look and Sound?* [GBP24b] constitutes the first transition from *interpreting* multimodal data to *synthesizing* multimodal surrogates *for the purpose of explaining conceptual regularities*.

The approach uses AI to generate visual and auditory data that accompany conceptual models (UML diagrams), thereby externalizing the implicit semantics that the model presupposes but does not encode procedurally. Beyond explanation, the paper demonstrates the use of such synthesized multimodal surrogates to simulate counterfactuals—i.e., to render how a model would *sound* or *look* under hypothetical changes. In this way, the work introduces multimodal synthesis as a means not only to interpret models but to probe their latent decision logic via controlled “what-if” articulation.

From multimodal surrogates to full video representations of conceptual models. Where earlier work generated isolated visual or auditory surrogates of conceptual structures, the contribution *Turning Process Models into Videos* [GBP25j] performs the first end-to-end synthesis of complete video narratives directly from process models. The approach composes scenes, actors, temporal structure, and narration to produce a temporally coherent audiovisual rendering of the model’s implied execution. This moves from partial “look and sound” augmentations to full video realization of process semantics. The resulting videos serve three functions central to this thesis: (i) they externalize implicit conceptual regularities in a human-digestible multimodal form, (ii) they create a bridge between formal models and multimodal evidence by enabling direct visual comparison, and (iii) they establish the technical substrate for counterfactual video-based simulation of alternative model variants. In doing so, this work closes the loop from *interpreting models through multimodal data* to *expressing models as multimodal data*.

Petri net structure as a generative control signal. We demonstrate that the transition from models to videos need not be purely heuristic or template-based: the *structure* of a formal process model—in this case, a Petri net—can serve as the generative control signal that constrains and orchestrates video synthesis [GBP25f]. We show how places and transitions induce a causal-temporal scaffold that dictates scene ordering, actor movement, and resource availability in the generated video. In contrast to free-form video generation, the model functions as a *semantic prior* ensuring that the video is not merely plausible but *aligned* with the underlying conceptual specification. This establishes a principled mechanism by which conceptual models can act not only as targets of multimodal interpretation but as drivers of multimodal generation, enabling faithful, explainable, and counterfactually manipulable audiovisual realizations of conceptual regulatory structure.

Toward conceptual model-aware artificial intelligence through Petri Net structure-enhanced prompting. Having shown that Petri net structure can guide video synthesis, the work on *Petri Net of Thoughts* generalizes the same insight from a generative video task to *AI reasoning itself*. Instead of feeding large language models unstructured prompts, the paper proposes to condition AI systems on explicit process structure (places, transitions, incidence relations) as a cognitive scaffold for inference. The Petri net here plays the role of a *thinking geometry*: it constrains the admissible flow

of plausible answers in the same way it constrains the flow of tokens in execution. In doing so, the work positions conceptual models not merely as descriptive artifacts but as computational priors that can shape the behavior of general-purpose AI systems.

The broader implication of this approach is that AI can benefit from *structured conceptual constraints* to avoid hallucination, enforce domain-compatibility, and reason over counterfactual variants in a controlled manner. We develop a Petri-net-of-thoughts framework [GBP25e], that shows that conceptual structure can function as a *bias injector* into foundation models without retraining: by embedding model structure into the prompt, AI systems are nudged to produce outputs that respect process semantics. This provides a scalable pathway from task-specific multimodal generation to a general principle: conceptual models are not only objects to be discovered or visualized, but also *interfaces that make AI process-aware*, thereby aligning the internal reasoning of AI with the latent regularities that conceptual modeling seeks to externalize.

Toward enrichment of the conceptual models. To close the arc by abstracting the insights from processes to conceptual modeling at large, we articulate the explicit formulation of the *multimodal conceptual modeling* [GBP25i]: that conceptual models can and should be enriched by invariants extracted from multimodal traces of expert behavior and counterfactual elicitation, rather than relying solely on declarative elicitation and structural refinement. It synthesizes the core claims demonstrated across the preceding works: that (i) multimodal evidence is a carrier of latent conceptual regularities, (ii) embedding spaces provide a unifying analytic substrate for aligning such evidence with formal languages, (iii) interactive environments such as MR reveal counterfactual dispositions that static observation omits, and (iv) these enrichments are not merely representational but can drive generation, validation, and AI reasoning.

Domain instantiation in surgery as final application. Where earlier works established the theoretical, methodological, and representational foundations, the *Surgery AI* contribution serves as the concrete domain instantiation discussion in which these ideas are exercised end-to-end. We analyze how the full stack developed throughout this line of work — multimodal inference, embedding-based alignment, model-driven synthesis, and relaxed conformance concepts — can be deployed in a single, high-stakes application domain. The *Surgery AI* work [GBP25h] demonstrates how multimodal process mining and Mixed Reality can jointly support real-time conformance checking and operative guidance. Multimodal traces from sensors, video, speech, tool telemetry, and physiological signals are fused to detect latent conceptual invariants of expert surgical conduct, while Mixed Reality headsets provide an egocentric interface for just-in-time guidance. This illustrates in a single, consequential use case the core thesis of this dissertation: that conceptual modeling can be enriched, validated, and operationalized through multimodal data and Mixed Reality, enabling not only better models but model-driven, real-time interventions in complex human activity.

Although the surgical application demonstrates multimodal inference, embedding-based

alignment, and real-time conformance checking, it leaves out one ingredient that is central to the thesis as a whole: *counterfactual elicitation via Mixed Reality*. In the surgical context, ethical and safety constraints preclude probing “what-if” behaviors in vivo during actual procedures. Thus, the demonstrated system instantiates the stack up to real-time multimodal conformance, but not the MR-based perturbation of the environment to reveal counterfactual dispositions.

Ongoing extensions of the research. Taken together, the subsequent work further extend and consolidate the trajectory established above along multiple reinforcing axes. *Process Adversarial Networks* [GBP25g] proposes the first adversarial training framework in which a generator synthetically proposes behavioral traces and a discriminator tests their conceptual admissibility against a reference model, thereby operationalizing model conformance as a learnable objective rather than a post-hoc check. *Multimodal Process Analysis* [GPB26] generalizes the discovery pipeline beyond mining to full-cycle analytical reasoning over multimodal traces, reframing multimodal inference as a rich space of diagnostic, comparative, and predictive tasks rather than a single discovery step. The work *Aligning AI Model’s Knowledge and Conceptual Model’s Symbols* [GBP25a] addresses the semantic interoperability problem by defining principled alignment layers between learned latent knowledge and formal modeling vocabularies, enabling AI outputs to be mapped onto conceptual artifacts in a symbolically grounded manner. *From Multimodal to Crossmodal Process Mining* [GBP25d] advances from simple fusion of modalities to crossmodal consistency constraints, where evidence in one modality is used to predict, correct, or refute evidence in another, elevating multimodality from mere redundancy to epistemic coupling. Across these works, the throughline is preserved: conceptual modeling is progressively redefined as a discipline that (i) learns from multimodal evidence, (ii) elicits counterfactual dispositions interactively, (iii) generates multimodal explanations and simulations from models, and (iv) integrates AI’s internal representations as both input and output of conceptual reasoning. The work on *Comics as Process Model Notation* [GBP25c] explores the communicative boundary between formal modeling and human understanding. Instead of using abstract diagrammatic syntax, it renders processes as sequences of realistic narrative panels in which actors, objects, and context are visually situated. Each panel is grounded in object-centric event logs and multimodal traces, so that what is shown is not hypothetical but evidence-based. The result is a stakeholder-aligned surrogate representation that preserves the semantics of the process model while expressing it in a form that non-modelers can interpret without learning a notation. Finally, *Multimodal Business Process Simulation* [GBP26] constitutes an earlier and more constrained instantiation of the ideas further developed in this thesis. It focuses on grounding business process simulation in multimodal execution data without employing Mixed Reality as an elicitation or interaction medium. Its primary emphasis is on the analysis and replay of event-based execution traces, with the resulting simulation artifact designed for inspection and analytical validation. In contrast, the present thesis broadens the scope toward immersive, human-centered exploration of processes, placing greater emphasis on interaction, interpretation, and refinement of conceptual models.

1.3.1 Impact and Scholarly Adoption

A growing body of research has begun to adopt and extend the foundational ideas introduced in our earlier contributions on multimodal process mining and Mixed Reality for conceptual modeling. These studies demonstrate that the methodological bridge between multimodal data analysis, MR, and conceptual model-oriented abstraction, first proposed in our prior work, has been cited in peer-reviewed publications across domains such as robotics, human-computer interaction, and cognitive systems.

Extraction of Conceptual Models from Multimodal Data

Among the notable examples is the work of Corradini *et al.* [CPR⁺25b, CPR⁺25a] that addresses the challenge of bridging fine-grained multimodal sensor data and high-level activity representations within robotic environments. Using the dataset which provides synchronized video and audio recordings of a robot manipulating objects, the authors demonstrate that combining modalities yields event logs that more faithfully represent the robot’s behavior and mitigate imprecision inherent in unimodal pipelines. This direction exemplifies how multimodal reasoning can serve as an interface between embodied sensor data and conceptual representations—a core ambition of the present thesis.

Berezovsky *et al.* [BSV⁺25] extend the vision of model-enhanced reasoning by combining large language models with conceptual modeling techniques to automate organizational structure redesign. By using generative AI for extracting and optimizing regulatory process descriptions, the study advances model clarity and transparency—objectives closely aligned with our thesis. The authors’ emphasis on fusing linguistic and structural knowledge mirrors our own integration of multimodal data and modeling formalisms to improve conceptual precision and actionable insight.

Reinhard *et al.* [RLW⁺25] propose a reference architecture for constructing event logs from multimodal data in IT service management. Their design-science approach aligns closely with our thesis vision, as it operationalizes the integration of heterogeneous evidence sources—videos, documents, and system traces—into unified process representations. This work further validates the growing shift toward multimodal foundations for process mining and conceptual modeling that our prior research helped initiate.

Sanchez-Ferriz *et al.* [SFELB25] introduce a two-stage natural language processing (NLP) and embedding-based approach to derive event logs from unstructured construction reports. By transforming heterogeneous textual inputs into semantically clustered activity representations, their method enhances event log completeness and conceptual coherence. This direction resonates with our thesis in extending multimodal evidence—here textual and semantic embeddings—toward more precise and structurally aligned process representations.

Jitvirat *et al.* [JCM⁺25] present an AI-driven framework for voice-activated modeling of innovations in future businesses, integrating human-in-the-loop research with multimodal data collection and analysis. Their approach—combining qualitative observation with

large language models and NLP—illustrates how interactive, voice-based modalities can inform conceptual representations of organizational behavior. This aligns with our thesis emphasis on using multimodal and interactive evidence to refine and extend conceptual modeling practices.

Accessibility in Conceptual Modeling

Sarioğlu *et al.* [SMB25] conduct a systematic review and propose a keyboard-only UML modeling module to advance accessibility in conceptual modeling environments. Their research highlights how inclusivity and (multimodal) interaction design can broaden participation in modeling activities. This aligns with our thesis in emphasizing embodied and multimodal interfaces—such as Mixed Reality—as vehicles for capturing a wider range of human cognitive and perceptual contributions to conceptual modeling.

Similarly, Bork *et al.* [BKM⁺25] present a vision for inclusive model-driven engineering (MDE), emphasizing accessibility as a fundamental design principle across modeling tools, methods, and processes. By advocating inclusivity not only in generated software but within the modeling act itself, their work resonates with our thesis aim to expand conceptual modeling through embodied, multimodal, and interactive means—ensuring that modeling environments become both cognitively and physically accessible to all stakeholders.

Teaching Conceptual Modeling

Snoeck and Pastor [SP25] argue that the rise of large language models requires a pedagogical shift in conceptual modeling education—from emphasizing model creation to cultivating skills of model evaluation, refinement, and critical interpretation. Their view complements our thesis by highlighting how human expertise remains essential in judging and contextualizing models generated or enhanced through AI and multimodal systems. This aligns with our focus on integrating human interpretive judgment within multimodal and Mixed Reality modeling workflows.

Kunz *et al.* [KS25] review advances in intelligent tutoring systems (ITS) for teaching graphical modeling, emphasizing adaptive feedback, learning analytics, and cognitive assistance in model construction. Their synthesis highlights the role of AI-driven, multimodal feedback in supporting learners’ modeling competence. This aligns with our thesis perspective that intelligent, interactive systems—particularly those grounded in multimodal and Mixed Reality interfaces—can enhance human understanding and precision in conceptual modeling tasks.

Encoding Semantics for AI

López *et al.* [LDC24] experiment with domain-specific word embeddings tailored for conceptual modeling tasks. By analyzing linguistic regularities in modeling discourse, they show how specialized embeddings can capture semantic nuances of modeling constructs.

This resonates with our use of multimodal representations to align symbolic and behavioral dimensions of conceptual models, enhancing their precision and interpretability.

Glaser *et al.* [Gla25] explore methods for encoding semantic information within conceptual models to make them interoperable with machine learning workflows. By structuring model semantics for algorithmic consumption, they bridge human-understandable models and computational reasoning. This aligns with our thesis ambition to operationalize conceptual knowledge extraction from multimodal data streams and enable machine reasoning over conceptual abstractions.

Müller *et al.* [Mül24] present *CM2ML*, a portable and extensible framework for encoding conceptual model structures for downstream AI applications. The framework supports consistent mapping of model elements into machine-readable form, fostering hybrid reasoning between human modeling intent and machine inference. This direction complements our thesis by advancing the interoperability of conceptual models with multimodal analytical pipelines and intelligent systems.

Llorente *et al.* [LVCFA25] propose a non-parametric identifier for systems with uncertain models, employing a Lyapunov-based continuous form of backpropagation to achieve stable learning and adaptive control. Their contribution demonstrates how continuous learning mechanisms can approximate complex system behaviors without explicit parametric assumptions—an approach conceptually related to our thesis aim of inferring latent conceptual regularities from multimodal evidence without prespecified structural models.

End-Case Applications in AI and Mixed Reality

Wiranata *et al.* [WA24] develop a deep learning-based prediction and control system for co-firing combined cycle power plants, improving operational performance through adaptive inference on multimodal sensor data. Their work exemplifies how continuous multimodal monitoring can reveal process regularities and optimize system behavior, paralleling our thesis motivation to mine structured conceptual insights from dynamic, real-world multimodal processes.

Agarwal *et al.* [ASS⁺24] introduce *SafeRoutes*, an AI-driven framework that integrates clustering algorithms with GPS data to enhance women’s safety through spatial risk modeling. Their approach combines multimodal data sources—geolocation, contextual patterns, and behavioral analytics—to construct dynamic safety models. This resonates with our thesis in its goal of transforming heterogeneous, real-world evidence into structured, interpretable conceptual representations.

Vasić [Vas25] explores AI-based methods for enhancing digital heritage through virtual reality, focusing on perceptual optimization, user attention tracking, and semantic enrichment via ontologies and large language models. By integrating behavioral data, spatial modeling, and semantic reasoning, this work exemplifies how multimodal and Mixed Reality environments can bridge human cognition with formal representations—an ambition that directly resonates with our thesis on enhancing conceptual modeling through multimodal analysis and interactive MR-based elicitation.

1.3.2 Consolidation of insights from prior work

The sequence of works summarized above collectively establishes four converging insights that form the epistemic foundation of this thesis. First, multimodal traces of expert behavior are not merely auxiliary “context data” but constitute a *first-class carrier of conceptual regularities*. They reveal what experts do when not asked, and therefore expose invariants that are routinely omitted from declarative elicitation. Second, embedding spaces provide a unifying analytic substrate in which heterogeneous modalities and formal conceptual structures can co-exist and be compared. This enables conceptual alignment, discovery, enrichment, and conformance checking even in the absence of explicit event logs or symbolic annotation. Third, reality simulators and linguistic conversations offer an elicitation and explanation interface that is epistemically stronger than passive observation: it enables egocentric capture of tacit operations and controlled perturbation of the environment to expose counterfactual dispositions that cannot be observed in static recordings. Fourth, conceptual models can serve not only as representational endpoints but as *structuring priors* that constrain multimodal generation, guide AI reasoning, and enable enhanced validation of the conceptual model utilization.

1.4 Outcomes

This thesis delivers a coherent set of original and consolidated contributions that directly address the research problem formulated in Section 1.1.4, and formalized in Section 1.2. The central integration, formalization, and large-scale empirical validation presented here constitute novel results that emerge only in the unified approach developed throughout this dissertation, while several foundations build upon prior research work, that we summarize in the proceedings.

1.4.1 Results Overview

At the theoretical level (Chapter 4), the thesis introduces a formal account of tacit domain knowledge as a dual structure consisting of (i) observed factual regularities derived from multimodal traces and (ii) counterfactual dispositional rules elicited through controlled perturbations. Our earlier works laid important groundwork in multimodal conceptual modeling [GBP25i], embedding–symbol alignment [GBP25a], multimodal process mining [GBP24c, GBP24a, GBP25d, GBP25b], and narrative-based modeling with ViEnNa [GBP25c], the unified epistemic model that integrates latent representations, symbolic constructs, and counterfactual reasoning is a novel synthesis developed in this thesis. In particular, the explicit alignment between multimodal embeddings and conceptual modeling primitives, and their joint interpretation within a single formal enhancement framework, extends beyond the scope of the individual prior publications.

At the level of the proposed method (Chapter 5), the thesis introduces an original human–AI co-modeling cycle that integrates multimodal data analysis and Mixed Reality simulation into a structured conceptual modeling enhancement approach. Earlier

contributions on simulation and video-based reasoning [GBP25j, GBP25f, GBP26] and stakeholder elicitation structures [GBP24d] provided partial building blocks, for which extended discussions are provided in Chapter 6 (multimodal data analysis) and Chapter 7 (Mixed Reality simulation).

Technologically (Chapter 8), the thesis delivers a real-time, browser-based, multi-agent architecture that operationalizes the theoretical constructs and the proposed method in an integrated runtime environment.

Empirically (Chapter 9), the thesis provides the first comprehensive evaluation of this integrated approach in a heterogeneous, high-stakes domain. Using precision and conciseness as normative evaluation criteria [RAB⁺15] and applying a controlled A/B design grounded in causal inference principles [Rub74], the study systematically compares the proposed approach (Condition A) with a baseline elicitation method, *SAP Scenes* [SAP20] (Condition B). While prior publications independently validated foundational components (as summarized in Sections 6.7 and 7.7), the integrated evaluation of the proposed approach as a whole with novel end-scale domain application across multiple conceptual modeling languages and contexts, together with the explicit linkage to research questions (as stated in Section 1.2.1), constitutes a novel empirical contribution.

1.4.2 Deliverables

To support this new mode of AI-assisted and multimodal-driven elicitation, we developed a novel conceptual modeling approach specifically designed for narrative, situational, and embodied forms of modeling. This approach represents processes as comic-like panels that depict actors, context, interactions, and tacit cues in a visually intuitive form. This visual narrative structure is highly aligned with AI-based generation, multimodal data extraction, and Mixed Reality simulation: AI can create or refine panels based on observed behavior, while human experts can interpret, critique, and modify the scenes. The result is a modeling paradigm that is inherently suited for capturing complex domains where tacit knowledge and situational awareness are just as important as formal process logic.

Figure 1.4 illustrates the conceptual modeling approach developed as part of this thesis, representing a conceptual model as a *Virtually Enhanced Narrative (ViEnNa)*, and the immersive introspection through interactive SUI. Our approach aims to enable both the modeler and the AI to produce rich narrative scenes, jump directly into specific situations, and generate new what-if scenarios grounded in realistic context. The integration of interactive digital-system elements and AI-generated panorama environments aims to provide an immersive substrate for elicitation, allowing experts to explore complex situations while the AI continuously learns from their reactions and refines the emerging conceptual model.

This thesis delivers a method that transforms heterogeneous multimodal evidence into conceptual models and interactive representations of SUI. Figure 1.5 provides a high-level

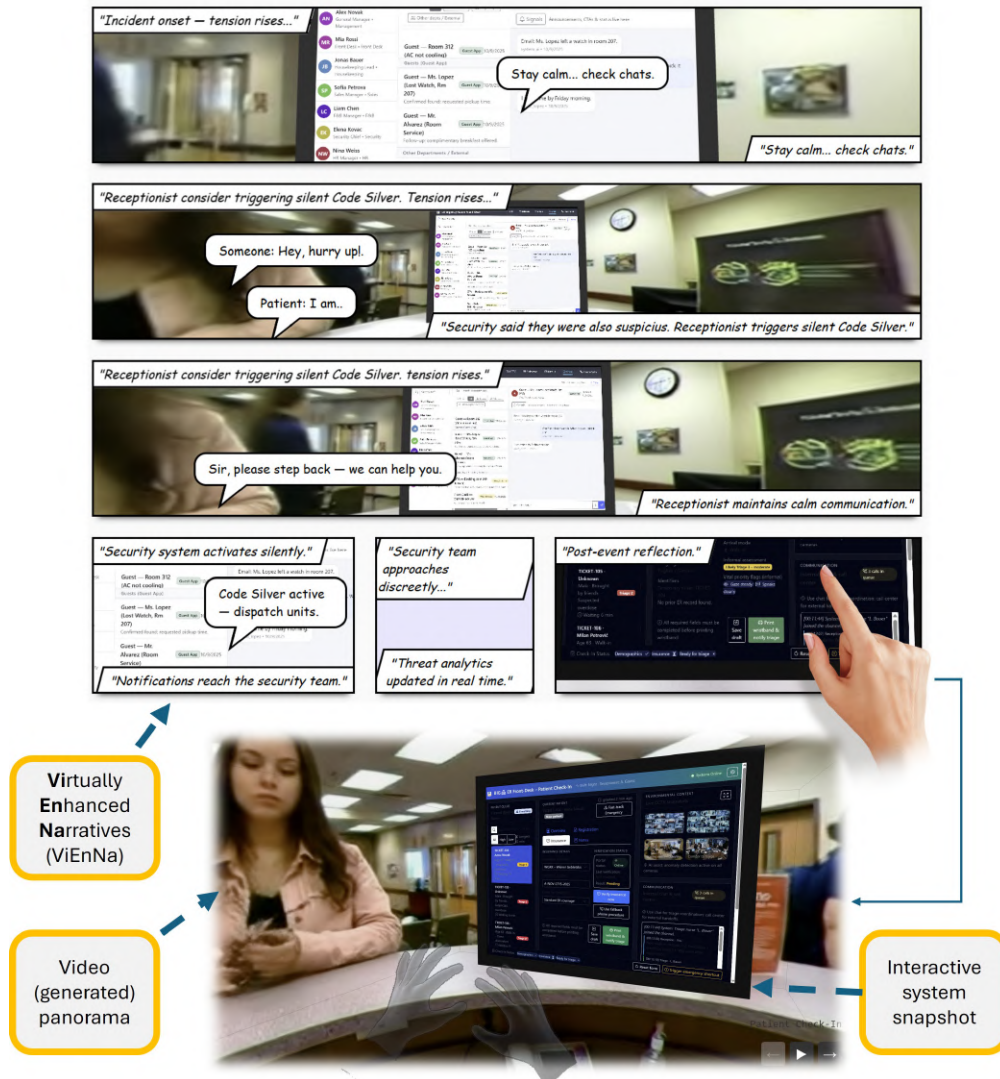


Figure 1.4: Overview of our novel conceptual modeling approach, designed for AI-assisted enhancement of conceptual modeling using multimodal data analysis and Mixed Reality.

overview of this workflow, structured around three main components: the *Input Space*, the *SUI Contextualization* stage, and the resulting *Output Space*.

Input Space. The methodology begins with three complementary types of multimodal data inputs: (i) *event logs* describing observed control-flow behavior, (ii) *process evidence data* such as videos, images, audio, documents, and sensor traces that reveal tacit and contextual aspects of real-world work, and (iii) *reference data* including floor plans, spatial layouts, or domain schemata that provide structural grounding. Together, these inputs form a multimodal foundation that goes beyond purely symbolic or textual sources

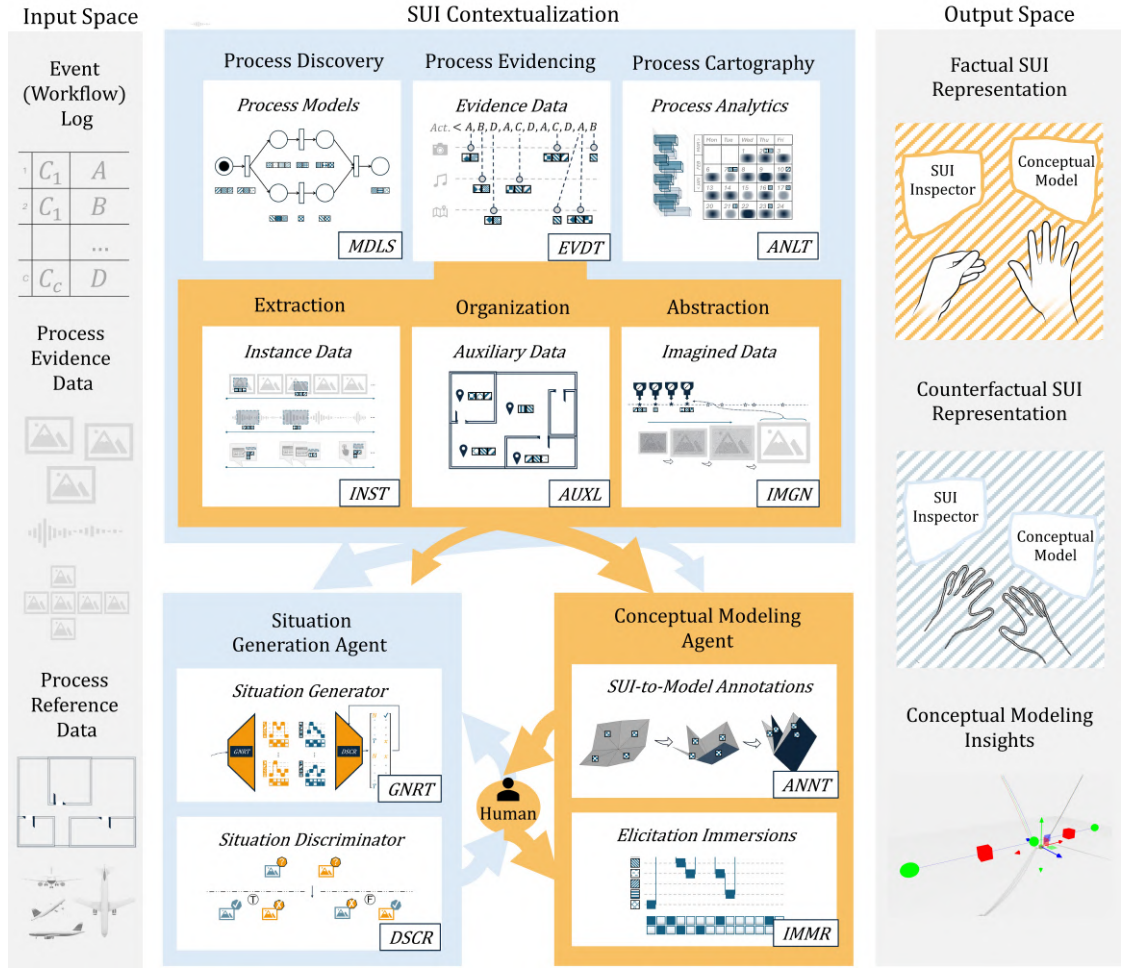


Figure 1.5: High-level methodological overview. Reflecting Figure 5.4.

typically used in conceptual modeling.

Figure 1.5 thus captures the high-level overview of the developed method: a multi-modal, agent-supported approach that systematically elevates raw observational data into conceptual models and MR situational representations.

SUI Contextualization. At the core of the approach is a contextualization pipeline that incrementally organizes, enriches, and abstracts multimodal data into SUI representations. This pipeline is structured along three methodological layers.

First, the *Process Discovery* layer (MDLS) extracts behavioral structures from logs and evidence, producing process models and analytics that highlight sequences, frequencies, and deviations. Second, the *Process Evidencing* layer (EVDI) reconciles heterogeneous cues—temporal, spatial, behavioral, and communicative—into interpretable evidence

structures. Third, the *Process Cartography* layer (ANLT) integrates these insights into analytical maps that guide model interpretation.

These process-oriented activities are complemented by an orthogonal triad of data-handling operations. *Extraction* (INST) identifies instance-level data such as activity traces, observed interactions, and local perceptual cues. *Organization* (AUXL) aligns auxiliary context—including spatial layouts, resource configurations, or communication pathways—to enrich the situational grounding of the process. *Abstraction* (IMGN) produces imagined or counterfactual data that fill observational gaps and support generative reasoning during elicitation.

Agents for Generation and Modeling. A couple of interacting agents operationalize this pipeline. The *Situation Generation Agent*, composed of a generator (GNRT) and a discriminator (DSCR), synthesizes or discriminates candidate situations, ensuring that both factual and counterfactual variants remain coherent with domain evidence. In parallel, the *Conceptual Modeling Agent* connects SUI content to conceptual modeling artifacts through SUI-to-model annotations (ANNT) and immersive elicitation stimulation (IMMR). Human modelers remain central throughout, steering both agents, validating interpretations, and injecting tacit domain expertise.

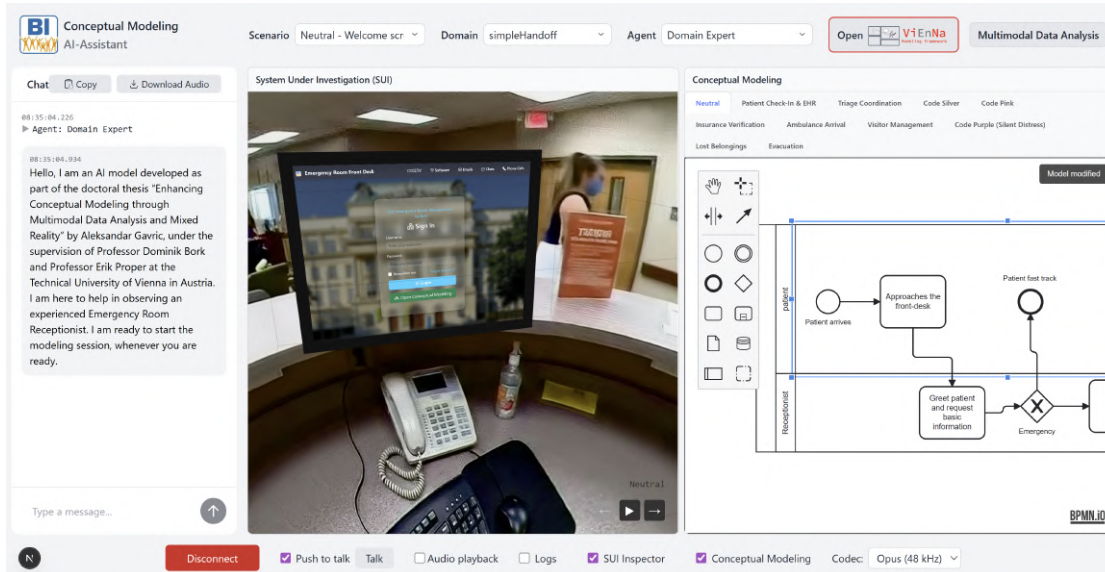


Figure 1.6: Prototype of the conceptual modeling environment developed as part of this thesis.

Output Space. The outcome of the methodology is a set of *factual* and *counterfactual* SUI representations aligned with conceptual models. These representations enable inspectors and modelers to explore real and hypothetical process situations, supporting deeper insight into domain behavior, variability, and tacit knowledge. Ultimately, the

methodology yields *conceptual modeling insights* that integrate multimodal evidence, human expertise, and AI-assisted situational reasoning.

Technical implementation. As illustrated in Figure 1.6, this thesis introduces a novel technical implementation of a tool that tightly integrates multimodal observation, conceptual modeling, and AI-assisted elicitation. This tool enables the modeler to analyze both real-world and what-if situations, inspect tacit cues, interact with domain experts or AI agents, and construct or refine conceptual models in real time. The system operationalizes the proposed method by making multimodal data, MR simulation, and conceptual modeling coexist within a single coherent workspace.

1.5 Thesis Outline

This dissertation is organized into eleven core chapters and four appendices. Each chapter builds upon the previous one to progressively develop, instantiate, and evaluate the proposed approach for enhancing conceptual modeling through multimodal data analysis and Mixed Reality. The structure follows a logical flow from theoretical grounding and problem analysis, through methodological and technical developments, to evaluation and domain-scale application.

Figure 1.7 provides an overview of the thesis deliverables and their connection to the thesis outline.

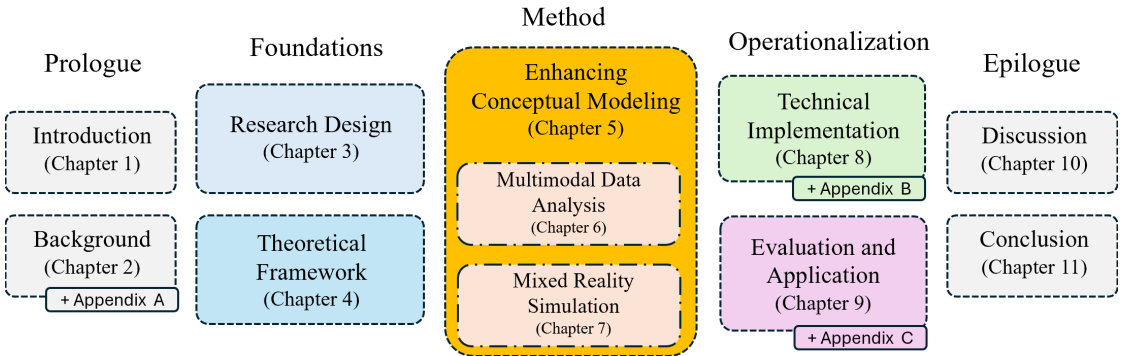


Figure 1.7: Overview of the thesis outline.

Chapter 1 — Introduction. Introduces the overall context of the thesis, motivating the research, defining the core problem, outlining key objectives and questions, summarizing prior foundational work, delimiting the scope, presenting the main contributions, and providing a roadmap for the structure of the dissertation.

Chapter 2 — Background. This chapter establishes the theoretical and technological foundations upon which the thesis is built. It introduces the three pillars of the thesis:

conceptual modeling, multimodal data analysis, and Mixed Reality.

Chapter 3 — Research Design. Describes our methodology, and analyzes the key challenges that motivate this research, including evidence scarcity versus evidence overload, grounding difficulties between models and real-world execution, and the absence of effective stakeholder feedback loops. It formally derives requirements for multimodal- and MR-enhanced conceptual modeling. This chapter also builds upon insights from our prior publications [Gav23, AGPB23], which provided the initial empirical and analytical framing of the problem.

Chapter 4 — Theoretical Framework. Presents the epistemic grounding of tacit knowledge and formalizes key constructs used throughout the thesis, including the modeling process, conceptual modeling languages, and conceptual models. It also introduces the notion of *counterfactual dispositions* and their elicitation as a mechanism for surfacing unspoken but cognitively stable decision regularities. The chapter consolidates and extends contributions from our prior works [GBP25i, GBP25a, GBP25e, GBP24c, GBP24a, GBP25d, GBP25b, GBP25c].

Chapter 5 — Enhancing Conceptual Modeling. Outlines the proposed method for enhancing conceptual modeling, data acquisition protocols, and ethical and privacy considerations. It describes the multimodal synchronization procedures, the MR-based elicitation protocol, and the embedding and representation layers that integrate heterogeneous data sources.

Chapter 6 — Multimodal Data Analysis. Describes the perception and discovery components that convert multimodal data into structured process/event representations and latent conceptual regularities. The chapter consolidates and extends contributions from our prior works [GBP25i, GBP25a, GBP25e, GBP24c, GBP24a, GBP25d, GBP26, GBP25b].

Chapter 7 — Mixed Reality Simulation. Defines the MR-based elicitation mechanisms that enable interactive “what-if” scenarios for capturing counterfactual behavior. It details how elicited responses are measured and how decision rules are inferred and integrated with observational traces. The chapter draws on and extends our previous contributions [GBP24b, GBP25j, GBP25f, GBP25c, GBP24d, GBP26].

Chapter 8 — Technical Implementation. Details the architecture of the system developed to support this thesis and describes its prototype components, including the integration with existing modeling environments and explanation facilities. The focus is on practical realization, interoperability, and tool-assisted exploration of multimodal and MR-enhanced models.

Chapter 9 — Evaluation and Application. Discuss the end-scale application of the approach in the medical hospital domain, covering system deployment, real-time conformance checking, and MR-based interaction. This chapter extends the contributions from our domain-specific publication [GBP25h]. For the evaluation, it specifies the evaluation metrics, datasets, and protocols used to assess the proposed approach. It presents experimental studies and user evaluations demonstrating gains in precision and conciseness compared to declarative-only modeling baselines.

Chapter 10 — Discussion. Synthesizes the key findings and theoretical implications of the thesis. It discusses practical relevance, limitations, and outlines future directions and directions for broader application and impact.

Chapter 11 — Conclusion. Summarizes the answers to the research questions, and revisits the core contributions.

Appendices. The appendices are covering the extended theoretical background in Appendix A, the supplementary implementation details in Appendix B, and the supplementary evaluation details in Appendix C—provide additional depth that complements the core contributions of this thesis. Appendix D—provides a transparent overview of the generative artificial intelligence tools that were used during the preparation of this thesis.

The full implementation, all evaluation materials, and a promotional demo video are available in the public repository: <https://github.com/aleksandargavric/enhancing-conceptual-modeling>.

Background

This chapter establishes the related theoretical and technological foundations upon which this thesis is built. It introduces the three central pillars of the work: *conceptual modeling* and its enhancement, *multimodal data analysis*, and *Mixed Reality*. Each of these pillars contributes a complementary dimension to the overall aim of *enhancing conceptual modeling through multimodal data analysis and Mixed Reality*.

Section 2.1 introduces the *fundamentals of conceptual modeling*. It discusses domain modeling and conceptualization, emphasizing how domain abstractions are socially constructed to represent relevant aspects of reality for a particular purpose. Section 2.1.1 introduces domain modeling and conceptualization while Section 2.1.2 discusses the formal foundations of conceptual modeling languages, outlining their syntax, semantics, and expressive capabilities. Next we discuss roles and uses of conceptual models (Section 2.1.3).

Section 2.2 is about approaches for enhancing conceptual modeling practices by establishing how process models can be connected to conceptual models to enable precise, dynamic, and behaviorally grounded representations of organizational reality. It discusses the contribution of *process mining* as a bridge between data and (process) model structure through the process discovery, conformance checking and process simulation and replay (Section 2.2.1) and the practices of validation and interaction with the model (Section 2.2.2). Section 2.2.3 then presents the role of *artificial intelligence* in enabling multimodal data analysis with the role of *AI-supported conceptual modeling*.

Next, Section 2.3 focuses on how *multimodal data analysis* contributes to conceptual modeling, including how conceptual models can be constructed from multimodal evidence (Section 2.3.1), how stakeholders can interact with conceptual models through multiple modalities (Section 2.3.2).

Section 2.4 discusses how *Mixed Reality* technologies enhance conceptual modeling contexts by enabling immersion and interaction. It covers the deployment of AR, VR, and MR in modeling environments (Section 2.4.1), and examines the underlying *cognitive*

and usability foundations that make MR an effective tool for elicitation and validation (Section 2.4.2).

Subsequently, Section 2.5 surveys *related work* that bridges the identified domains, including approaches to eliciting tacit knowledge (Section 2.5.1), multimodality in process mining (Section 2.5.2), encoding conceptual models for AI (Section 2.5.3), and Mixed Reality applications for simulation in regard to the conceptual modeling practices (Section 2.5.4).

Finally, Section 2.6 synthesizes the main insights, and outlines the positioning of the approach proposed in this thesis, thereby motivating the methodological and theoretical developments that follow in subsequent chapters.

Appendix A consolidates the extended theoretical and technical background underlying the thesis. Appendix A.1 outlines the foundations of conceptual modeling, including major formal languages and modeling tools that provide the structural basis for the proposed enhancements. Appendix A.2 presents the formal background of process mining, covering discovery, conformance checking, and simulation as analytical foundations for process-oriented modeling. Appendix A.3 addresses privacy-aware data analysis as an essential prerequisite for trustworthy multimodal and Mixed Reality-based modeling. Finally, Appendix A.4 introduces the artificial intelligence foundations—including hybrid intelligence, generative AI, embedding spaces, and video generation—that support the multimodal and model-driven contributions of this thesis.

2.1 Conceptual Modeling

Human cognition, as argued by Mylopoulos, Guizzardi, and Guarino [MGG25], emerges from our continuous interaction with the physical world. We perceive and interpret our surroundings through embodied sensorimotor experiences, which, over time, sediment into stable patterns that guide how we think and act. These patterns are not merely episodic memories but abstractions that capture recurring structures in experience. Through socialization—shaped by family, culture, and education—we progressively distill regularities from this lived experience, forming what is called our *cognitive domain*. Each momentary perception constitutes a distinct “world,” a concrete configuration of relevant objects and events. Over time, repeated exposure to such configurations allows us to internalize both *what* elements of reality are significant and *how* they relate to one another.

This connection between cognition and the external world has been recognized since antiquity. As noted by Karamanolis [Kar24], early philosophical thought—tracing back to figures in the second century—treated concepts as the essential mediators between mind and language. For these thinkers, human beings could attain reliable knowledge of reality only through conceptual understanding. Concepts thus occupied a dual role: as mental representations in the mind and as meanings of words in language. This duality

remains central to the epistemological foundations of conceptual modeling: access to reality is always *conceptually mediated*.

From a modern philosophical and logical standpoint, Guarino refines this view by interpreting concepts as *intensional entities* in Carnap’s sense [Gua98]. A concept, on this account, is not just an extensional set of instances in the world, but a rule or function that determines which instances in each possible world count as belonging to it. In other words, concepts are functions from possible worlds to extensional relations defined over the cognitive domain. These functions encapsulate the invariant meanings that allow human beings to recognize the “same” entity or property across different circumstances. Conceptual properties and relations—understood as such intensional entities—constitute the building blocks of conceptual structures.

This intensional interpretation of concepts is consolidated in the account of Mylopoulos, Guizzardi, and Guarino [MGG25], who emphasize that concepts are mental regularities that capture universals discerned through experience. They exist in the cognitive apparatus of agents rather than in the external world or in computers. Importantly, conceptual modeling does not attempt to *create* concepts but to externalize and structure them in a communicable form.

Guarino’s foundational formalization of conceptualization, first proposed in [GG95] and refined in [Gua09], offers a precise cognitive account of this process. A conceptualization is represented as a triple $\langle D, W, R \rangle$, where D is a cognitive domain of discourse containing the relevant entities, W denotes the set of possible configurations or “worlds” of that domain, and R is the set of conceptual relations and properties defined over D . This formal structure highlights that conceptualizations are internal to human cognition—they are the organized mental scaffolds that make sense of experience. Conceptual models, in contrast, are *external artifacts* that approximate or communicate aspects of these conceptualizations for specific purposes.

The distinction between mental conceptualizations and external conceptual models was articulated early by Mylopoulos [Myl98] and reaffirmed in later syntheses [MGG25]. Conceptual models are *information artifacts*—intentional representations crafted to serve communicative, analytical, and design-oriented functions. They enable communities of stakeholders to share, reason about, and refine their understanding of complex domains. As such, conceptual models play a pivotal role in disciplines like Information Systems (IS), Modeling and Simulation (M&S), and Software Engineering (SE). Their cross-disciplinary relevance motivated the expert panel convened by Robinson [RAB⁺15], which sought to unify diverse perspectives into a shared understanding of what a conceptual model is and how it should be evaluated. The panel’s deliberations underscored that, regardless of the specific language or domain, the essence of a conceptual model lies in its ability to convey conceptualizations in a precise yet cognitively accessible manner.

Yet, despite this progress, a profound epistemic challenge remains: much of what experts know never becomes explicit in the first place. As Cairó Battistutti, and Bork [CBB17] emphasize, a significant portion of human knowledge is *tacit*. It is embedded in experience,

embodied in skillful performance, and situated within practice. Tacit knowledge often resists codification because it is enacted rather than described—it manifests in what experts *do* rather than what they *say*. Consequently, the most decisive determinants of expert performance—contextual awareness, subtle judgments, situational cues—are precisely those that elude conventional modeling techniques.

This view aligns with established knowledge-creation and management theories, which argue that tacit knowledge is not merely an incomplete or immature form of explicit knowledge, but a foundational mode of knowing that underpins learning and innovation. Nonaka and Takeuchi highlight that organizational knowledge emerges through continuous interaction between tacit and explicit forms, driven by socialization and shared practice rather than formal instruction alone [NTU96]. Subsequent analyses reinforce that tacit knowledge is irreducible, deeply contextual, and inseparable from the situations in which it is exercised, making its full articulation inherently limited [LG03, Hoe06, Tso05].

This recognition motivates the central ambition of this thesis: to expand the epistemic basis of conceptual modeling beyond verbalized or symbolic evidence, by leveraging multimodal and embodied forms of data and interaction. To achieve this, we must integrate insights from cognitive science, philosophy, and AI into a new methodological foundation capable of capturing not only explicit conceptual structures but also the implicit regularities underlying expert cognition and behavior.

2.1.1 Domain Modeling and Conceptualization

Conceptual modeling is ultimately grounded in our ability to form and externalize structured conceptions of a domain. The activity of *domain modeling* seeks to capture the essential abstractions of some portion of reality—existing, envisioned, or imagined—so that they can be analyzed, communicated, and refined. As Guizzardi and Proper [GP21] emphasize, a *domain model* is not merely a technical construct but a *social artifact*: it gains its meaning only through collective recognition by a community of agents who acknowledge it as representing a certain abstraction of a domain for a given purpose. This acknowledgment positions modeling squarely within a social and epistemic context—one where meaning arises not from syntax alone but through shared interpretation and acceptance.

In this sense, domain models extend beyond the familiar “boxes-and-lines” diagrams typical of information systems and enterprise engineering. Depending on the modeling purpose, they may take the form of textual descriptions, mathematical formalisms, simulations, games, or even physical objects. Regardless of representation, what makes an artifact a model is its recognized correspondence to a domain abstraction under an intended purpose. The model is thus both a representation and an act of communication: a negotiated symbolization of a shared conceptualization.

The notion of a *domain* itself is broad, referring to “anything one can explicitly speak or reflect about” [GP21]. In enterprise and information systems contexts, this encompasses business processes, value exchanges, information structures, organizational transactions,

and other systemic entities. Crucially, the domain need not already exist—the model may describe a desired future system or an imagined scenario. Models, therefore, acquire value by serving specific purposes, and these purposes anchor their epistemic and practical utility. The benefits of modeling—clarity, shared understanding, or computational use—are thus directly tied to the purposes for which a model is created and accepted.

An important theoretical grounding for this interpretation is found in the *semiotic triangle* of Ogden and Richards [OR27], which articulates the triadic relationship between *symbol*, *thought*, and *referent*. In this view, symbols (such as elements in a model) do not directly correspond to real-world entities but are mediated through mental conceptions. The relations among these three poles—symbol to thought, thought to referent, and symbol to referent—determine whether communication and understanding are achieved. Ogden and Richards argue that symbols are powerful precisely because they operate as instruments of thought, allowing humans to “weave together contexts into higher contexts.” Domain models inherit this semiotic character: they are symbolic representations that mediate between thought (the conceptualization) and the referent (the domain), thereby enabling both reflection and coordinated action.

Building on this semiotic foundation, Proper and Guizzardi [PG21] elaborate the cognitive process underlying domain conceptualization. They distinguish between two forms of composed thought: *perceptions*, which are sensory experiences of referents, and *conceptions*, which are structured abstractions derived from those perceptions. When modelers create conceptions, they apply successive filters to manage the complexity of the world—selecting and generalizing aspects they deem essential while omitting others. This act of filtering is not a defect but a necessity: every model, by its very nature, simplifies reality to make it intelligible and useful. Conceptualization thus represents the disciplined abstraction process through which perceptions are transformed into coherent conceptual structures that can be expressed symbolically.

Proper and Guizzardi [PG21] also position *domain conceptions*—and the underlying process of domain conceptualization—as the critical bridge between semiotic theory and practical domain modeling. Domain conceptions capture the internalized understanding of a domain that precedes and informs the construction of models. They form the cognitive substrate from which domain models are generated, ensuring that modeling remains anchored in meaning rather than degenerating into mere diagrammatic formalism. This insight also connects domain modeling to broader theories in enterprise engineering, such as the Factual Information (FI) [Die15] theory and Model Universe (MU) [DM20] theory, both of which investigate how information and models mediate between reality and interpretation.

A key distinction emerging from this work is that between *conceptual models* and *utilization design models*. Conceptual models, according to Proper and Guizzardi [PG21], are those whose primary ambition is to remain as faithful as possible to the original domain conception. Their goal is conceptual purity—to preserve the integrity of meaning. Utilization design models, by contrast, intentionally introduce *compromises* to serve practical computational or experiential purposes, such as simulation, implementation,

or visualization. These two kinds of models occupy a continuum rather than a strict dichotomy: a single conceptual model may yield multiple utilization design models, each adapted to a specific purpose. Nevertheless, awareness of the conceptual compromises introduced during this transformation is essential to maintaining the traceability between domain understanding and its various operational instantiations.

2.1.2 Conceptual Modeling Languages and Frameworks

A wide range of established conceptual modeling languages and frameworks underpin the foundation of this thesis. These include general-purpose formalisms such as UML [Obj07], BPMN [Obj08, vRWCdM15], and ArchiMate [The09, LPJ09]; ontologically grounded approaches such as OntoUML [Gui05, GFB⁺18]; mathematically rigorous behavioral formalisms like Petri Nets [PR08]; and domain-specific languages (DSMLs) that encapsulate specialized domain semantics [Fra13, JRD⁺17, KKP⁺14]. Complementing these formalisms are frameworks that describe how conceptual models function in practice, including their roles in communication, reasoning, implementation, and as dynamic substrates for digital twins [MVS23, OZH⁺24]. Contemporary modeling tools [BDC23, MB23, GB21, MF24b] and the broader Model-Driven Engineering paradigm [Ken02, dS15, GDD09, ZP23] further contextualize how these languages operate within modern modeling ecosystems. Appendix A.1 includes detailed descriptions of major conceptual modeling languages (UML, BPMN, ArchiMate, OntoUML, Petri Nets), an exposition of the roles and uses of conceptual models, an extended discussion on digital twins as dynamic conceptual models, an overview of graphical modeling tools, and a synthesis of the Model-Driven Engineering paradigm. These sections together offer the full theoretical background that supports the methodological and technical contributions of this thesis.

2.1.3 Roles and Uses of Conceptual Models

Conceptual models serve as the primary means by which human understanding of a domain is externalized, shared, and reasoned about [Obj07]. They translate cognitive conceptualizations—mental abstractions of reality—into explicit, structured artifacts that enable communication, analysis, and design. Their value lies not only in the final product (the model itself) but in the *process* of modeling, which prompts reflection, negotiation, and alignment among stakeholders.

Within information systems, software engineering, and business process management, conceptual models fulfill several distinct yet interrelated roles [RAB⁺15]. First, they act as *representational surrogates*: they abstract away from the full complexity of the real world to highlight only those entities, relations, and constraints relevant to a particular purpose. Second, they serve as *communication interfaces*, bridging cognitive and disciplinary boundaries between domain experts, analysts, and technical implementers. Third, they provide a *reasoning substrate* upon which automated or semi-automated analysis can be performed—for instance, checking for consistency, completeness, or compliance with

constraints. Finally, conceptual models are often used as *blueprints for implementation*, linking high-level conceptual understanding to executable system designs.

Digital Twins (DTs)

As systems increasingly intertwine with their operational environments, conceptual models have evolved from static representations into dynamic counterparts that continuously synchronize with real-world data—a transition embodied in the rise of *digital twins*. A digital twin is a virtual representation of real-world entities and processes, synchronized at a specified frequency and fidelity [MVS23]. It extends the traditional conceptual model by coupling it to live sensor data, simulation environments, or behavioral logs, thereby maintaining a bidirectional link between the physical and the conceptual realms. This continuous synchronization transforms the model from a descriptive artifact into an operational one, enabling predictive analysis, anomaly detection, and decision support grounded in up-to-date contextual information. The conceptual foundations of modeling thus become directly actionable, supporting both *what-is* and *what-if* reasoning.

A comprehensive exploration by Oakes et al. [OZH⁺24] underscores the multifaceted nature of digital twins through four case studies analyzed using a structured DT description framework.

Milošević and Van Schalkwyk [MVS23] emphasize that as digital twins evolve toward greater autonomy, they must also embody principles of responsibility and governance. Beyond technical precision, responsible DT design requires explicit modeling of socio-economic and ethical dimensions, including data privacy, regulatory compliance, and organizational value systems. Their work shows how policies and ethical norms can be systematically translated into governance rules that constrain DT behavior and guide decision-making in real time.

Model-Driven Engineering (MDE)

The paradigm of **Model-Driven Engineering** emerged as a response to the growing complexity of software systems and the need to elevate the level of abstraction in system development. Rather than focusing on source code as the primary artifact, MDE emphasizes models as first-class citizens that capture domain-specific knowledge and system behavior in a structured and analyzable form [Ken02]. As argued by Kent, the fundamental premise of MDE lies in the automation of transformations between models and code, thereby aligning the software development process more closely with human reasoning and design intent. This shift enables engineers to abstract away from technological details and instead concentrate on the semantics of the problem domain.

Over time, the MDE paradigm has matured into a comprehensive methodological and technological framework. Da Silva [dS15] presents a unified conceptual model of MDE, offering a systematic overview of its constituent concepts—models, metamodels, transformations, and tooling ecosystems. According to this view, MDE facilitates the separation

of concerns between platform-independent and platform-specific representations, promoting reuse, traceability, and consistency across lifecycle artifacts. The survey also identifies the proliferation of domain-specific modeling languages as a central factor driving the adaptation of MDE across diverse engineering and information system contexts.

The integration of MDE with ontological engineering has further expanded its semantic depth and interoperability potential. As explored by Gašević, et al. [GDD09], aligning MDE with ontology development provides a formal semantic foundation that bridges the gap between model-based representations and domain ontologies. This alignment ensures that conceptual structures in models are not merely syntactic abstractions but semantically grounded artifacts that can support reasoning, consistency checking, and automated knowledge integration. Consequently, ontology-driven MDE fosters interoperability among heterogeneous systems and enables richer forms of model validation.

Recently, the scope of MDE has extended into areas such as trustworthy simulation and intelligent systems design. Zschaler and Polack [ZP23] argue for the use of domain-specific modeling languages to enhance transparency, reproducibility, and accountability in agent-based simulations. Their work demonstrates how MDE principles can structure and formalize simulation design processes, ensuring that domain assumptions, behavioral rules, and data provenance are explicitly encoded within models. Such approaches illustrate the growing recognition of MDE not only as a technical methodology but also as a paradigm for ensuring trustworthiness and explainability in complex socio-technical systems.

2.2 Enhancing Conceptual Modeling

The quality of conceptual models has long been a central concern in information systems research. In terms of Business Process Optimization (BPO), it concerns making optimal operational decisions during process redesign or process execution, including decisions on which resource to assign to which task, how many resources to schedule at which point in time, and when to plan the processing of which case [Dij26]. These decisions must balance objectives like cost, time, and customer satisfaction while respecting constraints like availability, deadlines, and budgets often under uncertainty and across cases that are running at the same time [Dij26]. The SEQUAL framework and its subsequent refinements provide a comprehensive theoretical basis for understanding and assessing model quality across syntactic, semantic, and pragmatic dimensions [Kro15]. In particular, the specialization of SEQUAL to business process models clarifies how process representations function as knowledge artifacts that guide action and communication within organizations [Kro16]. Specialized version of the framework emphasizes that process models represent *knowledge for action*, thereby linking representational fidelity to practical usability and organizational performance [KSJ06]. The next section narrows this discussion to process modeling, where the interplay between action, representation, and multimodal evidence becomes particularly salient.

Conceptual models aim to capture, structure, and communicate the essential regularities

of a domain, enabling shared understanding, analysis, and reasoning across stakeholders. Process models, on the other hand, provide a formalized description of dynamic behavior—how entities interact and evolve over time through sequences of activities, decisions, and events. The close alignment between these two modeling paradigms allows the dynamic traces of process execution to serve as evidence for refining and validating conceptual structures. In this sense, process models can be viewed as the operational semantics of conceptual models: they make explicit the temporal and causal dependencies that underlie abstract domain concepts.

The ontological foundation of the connection between process models and conceptual models can be deepened through the theory of *variable embodiments* proposed by Kit Fine [Fin82] and revisited by Guarino and Guizzardi [GG24]. In Fine’s framework, an *embodiment* is an entity whose identity depends on its parts, and whose persistence through time can be understood either as *rigid*—when its parts are fixed—or *variable*—when its parts may change while the entity itself maintains identity over time. For example, the process of erosion of a cliff is a variable embodiment: its manifestations at different moments in time represent distinct temporal parts of one enduring process.

As process models not only describe how activities unfold over time but also instantiate the operational semantics of the conceptual structures that govern them, we unlock the direction of conceptual model simulation and replay through process models. The bridge between process models and conceptual models thus lies in how dynamic behavior—captured as event data—can be played out, evaluated, and generalized to refine the underlying conceptual representation of a domain. As Van der Aalst observes, event data are ubiquitous across modern information systems, recording the activities executed by resources at particular times and for particular cases [vdA18]. Through process mining, these data can be transformed into models that reveal the real operational logic of a system, providing a behavioral counterpart to the static conceptual schema.

2.2.1 Process Mining

Process mining, as defined in [vdAADM⁺11], offers a methodological bridge between empirical observation and formal conceptualization. Its core premise is that event data—captured from information systems or multimodal sensors—encode implicit behavioral regularities that can be systematically analyzed to discover, monitor, and improve real processes. Activities in a business process are completed in a non-uniform manner and in settings where fault occurrences are inevitable [CWD22].

By mapping observed actions, decisions, and contextual cues onto structured process representations, such as Petri Nets, BPMN, or declarative models, we obtain a storyline that exposes both the order and logic of human and system behavior. This integration allows conceptual models to be not merely static abstractions, but empirically grounded narratives that evolve from lived experience. Hence, **enhancing conceptual modeling** through multimodal data and MR elicitation can be approached through process models that guide the storyline towards **precision**—by capturing what actually happens—and

conciseness—by abstracting only the invariant causal structure. In this view, process models act as mediating artifacts between raw multimodal evidence and formal conceptualization.

Process discovery techniques such as the Inductive Miner [LFvdA13], implemented in toolkits like PM4Py [BvS23], derive process models from event logs and evaluate them along dimensions such as soundness, fitness, precision, and generalization. Some proposed approaches such as [VGL⁺23], are iterative and interactive, thus taking experts' knowledge into account during the discovery process. Conformance checking methods [vdA96, BvdA21] then compare these models against execution data to quantify deviations and iteratively refine the conceptualization of the underlying processes. Finally, process simulation and replay [vdA18] connect empirically grounded models with forward-looking *what-if* analyses, enabling the exploration of alternative process behaviors and generalization properties. To keep the main narrative focused, the detailed formal definitions (e.g., of event logs, process trees, language functions, and quality metrics) and algorithmic descriptions (including token-based replay and backward state-space exploration) are provided in Appendix A.2.

Sense-making of Process Mining Results

While process mining techniques excel at uncovering patterns and deviations within event data, their results are not inherently self-explanatory. They describe *what* has happened in a process but not necessarily *why* it occurred. Grisold *et al.* [GvdAF⁺24] emphasize that meaningful interpretation of process mining outcomes requires contextual grounding and managerial sense-making. To address this gap, they introduce a *context framework for sense-making of process mining results*, which systematically categorizes the contextual factors influencing process dynamics across three levels: process-immediate, organization-internal, and organization-external.

At the **process-immediate context** level, analysts examine aspects directly tied to process execution, including activity changes, resource availability, and IT system behavior. The **organization-internal context** captures broader structural and procedural dynamics, as well as cultural and identity-related factors shaping organizational practices. Finally, the **organization-external context** encompasses environmental disruptions, inter-organizational relations, and regulatory influences that affect process performance.

Building on this foundation, Franzoi *et al.* [FHG⁺25] advance the discussion by proposing a comprehensive *Process Mining Context Taxonomy* designed to explain the underlying dynamics and changes observed in business processes. The authors outline two practical usage paths for applying the taxonomy: one emphasizing its diagnostic role in explaining process anomalies, and another focusing on its interpretive value for identifying managerial and organizational implications.

One key challenge is to discover and correct deviations among different process representations [SWD25]. Sai *et. al* (2025) [SWD25] propose a three-phase approach. First, a Large Language Model (LLM) converts the process description into an intermediate

representation. Second, process structure trees are compared to this representation to detect deviations. Using process trees allows minimal alterations by identifying the smallest sub-tree containing the deviation. Third, the parts of the description referring to this sub-tree are replaced, for which we rely on a combination of sentence templates and an LLM for refining and correcting the sentences. The effectiveness of the proposed method is demonstrated based on experimental results evaluated at each phase.

Process Mining and Visual Analytics (VA) can mutually benefit from each other by combining their strengths. VA perspectives complement these developments, with Miksch [Mik21], Di Ciccio et al. [CMS⁺24], and Van der Aa et al. [vdADCD⁺24] highlighting the importance of human-in-the-loop interpretability. From the perspective of visual analytics and simulation, works such as Alman et al. [AAB⁺24] and Kirchdorfer et al. [KBK⁺24] extend this vision toward interactive, spatial, and simulation-driven process analytics. An approach by Van der Linden et al. (2025) [vdLFP⁺25] emphasizes interactive visualization and analysis for a more flexible and iterative conformance checking process by, for example, allowing to dynamically refine and define additional constraints based on insights from multiple perspectives and making all deviations explainable and understandable.

2.2.2 Enhancing Validation and Interaction with Conceptual Models

Validation plays a central role in conceptual modeling, ensuring that the constructed models not only capture domain semantics accurately but also remain usable and interpretable by stakeholders [Kro15]. In this thesis, validation is approached as a bidirectional interaction between *model* and *modeler*, in which the model is treated as both a representation and a conversational artifact. Traditional validation techniques—such as expert walkthroughs, simulations, and conformance checking—primarily assess whether the model correctly reflects observed or intended behavior. However, in the context of multimodal and Mixed Reality data, validation extends beyond correctness to include how effectively a model can be *experienced*, *queried*, and *refined* through embodied interaction.

Model validation

Model validation also requires examining how abstraction mechanisms preserve the intended conceptual meaning while reducing representational complexity. Romanenko *et al.* [RCG24] address this issue by evaluating the quality of ontology-driven conceptual model abstractions produced by automated algorithms. From a cognitive standpoint, the authors first elaborate on the notion of conceptual model quality and the mental processes involved in abstraction. By distinguishing between syntactic compression and semantic preservation, they clarify that a high-quality abstraction must summarize the model while maintaining the “gist” of the conceptualization it represents. To test these principles empirically, the proposed algorithm was implemented and evaluated using the FAIR catalog of OntoUML conceptual models—an openly accessible repository enabling reproducibility and large-scale benchmarking [BSF⁺22]. Testing across this catalog demonstrated the algorithm’s correctness (faithful implementation of transformation

rules) and effectiveness (achieving substantial summarization rates without semantic distortion).

Model Interaction and Automation

Model interaction lies at the heart of validation and understanding in conceptual modeling. It refers to the dynamic engagement between stakeholders and the model, where the model becomes not merely an artifact to be inspected but an interactive medium for exploration, negotiation, and reasoning.

Interactive modeling also benefits from increasing levels of automation. Vu, Leopold and van der Aa [VLvDA25] provide a systematic analysis of the current landscape of *business process automation*, highlighting that automation extends far beyond robotic process automation (RPA). It includes workflow orchestration, integration platforms, decision automation, and intelligent agents that together ensure the continuous synchronization of models with operational data. Their market analysis identifies three key capabilities that shape the lifecycle of automated artifacts: **monitoring**, **governance**, and **content marketplaces**. Monitoring ensures visibility into process performance and compliance during run-time, governance enforces quality and security standards across automated workflows, and content marketplaces promote reuse and rapid deployment through curated automation templates.

A central insight from [VLvDA25] is that automation has become increasingly *human-centered*. With the rise of low-code and no-code environments, business users and domain experts—those closest to process reality—can now design, execute, and maintain automation artifacts themselves. This democratization parallels the MR-based model interaction envisioned in this thesis: both paradigms aim to close the gap between technical formalism and human intuition. Whereas automation platforms allow experts to *instantiate and observe* models in real operational contexts, MR-based elicitation enables them to *simulate and experience* these models within controlled virtual environments.

Traditionally, the optimal resource allocation in a business process is determined by manually exploring a number of options, but whether this leads to the optimal solution is questionable; it is possible that an unexplored option produces even better results [PDG21]. Therefore, an automated approach is needed that explores all possible options in a structured manner to find the best one. The experiments show that the proposed search strategies such as [PDG21] scale very well and obtain the optimal value in only a fraction of the number of steps needed by an exhaustive exploration of the solution space.

2.2.3 Artificial Intelligence

The evolution of artificial intelligence can be traced through the historical development of *Decision Support Systems* (DSS), which aimed to augment human reasoning with computational assistance. Alter’s seminal taxonomy [Alt77] provides one of the earliest and clearest articulations of how such systems evolved along two complementary trajectories: *model-driven* and *data-driven* decision support. In his view, these trajectories differ by

the “degree of action implication of system outputs,” forming a spectrum ranging from highly analytical systems grounded in formal models to empirically oriented systems that rely on the abundance and structure of available data.

Model-driven DSS place a theoretical or analytical model at their core. Their strength lies in explicit reasoning mechanisms that enable the simulation or optimization of well-understood systems. A sports analytics model optimizing a hockey team’s decision on when to pull the goaltender, or a football play-calling optimization model, are canonical examples of model-driven reasoning. In contrast, data-driven DSS leverage large volumes of organized data, extracting patterns and insights through retrieval, summarization, and analysis—early precursors to what we now call data mining or machine learning.

Artificial Intelligence today can be seen as the convergence of these two paradigms. On the one hand, AI inherits the formal rigor of model-driven systems, embedding explicit knowledge representations and reasoning rules; on the other, it capitalizes on the scale and adaptability of data-driven systems, particularly through deep learning architectures that autonomously derive patterns from multimodal data sources. This duality forms the conceptual bridge between *model-based understanding* and *data-based adaptation*, a synergy that underlies the methodological foundations of this thesis. Within this broader trajectory, AI serves as both an analytical engine and an empirical learner, mediating between conceptual models, multimodal data, and Mixed Reality environments.

Today’s advances in artificial intelligence are predominantly implemented through deep learning frameworks that enable efficient training, experimentation, and deployment of large-scale models. Among these, PyTorch has emerged as one of the most widely adopted libraries, providing an imperative style of programming that combines flexibility with high performance [PGM⁺19]. Its dynamic computation graphs and extensive ecosystem make it the foundation for both academic research and industrial applications across domains such as computer vision, natural language processing, and multimodal learning.

Symbolic, Subsymbolic, and Generative Paradigms

The theoretical basis of this thesis draws on several complementary strands of contemporary artificial intelligence. At a high level, AI can be decomposed into symbolic, subsymbolic, and human-centered layers, each contributing structure, adaptivity, and interpretive grounding to hybrid intelligence systems. Symbolic representations—including knowledge graphs and conceptual models—provide explicit, ontologically grounded structures for reasoning, whereas subsymbolic approaches such as large language models (LLMs) learn implicit semantic patterns from multimodal data [VFQP25]. Generative AI [FHJZ24] extends this landscape with transformer-, diffusion-, and GAN-based architectures capable of synthesizing text, images, and video [VSP⁺17, GYS⁺24]. Embedding spaces [GENL⁺23] unify these modalities into continuous semantic manifolds that support cross-modal alignment, reasoning, and grounding. Recent advances in text-to-video diffusion further demonstrate how abstract descriptions can be animated as temporally coherent visual narratives [RPG⁺21]. To maintain focus in the main exposition, the

detailed discussion of these topics—including intuitions behind generative and embedding models—is provided in Appendix A.4.

The intersection of AI and Conceptual Modeling

The intersection of AI and conceptual modeling has recently emerged as one of the most active and transformative research frontiers in the field of information systems engineering. Early works have begun to systematically map how AI techniques, ranging from symbolic reasoning to machine learning, can support or even automate activities across the conceptual modeling lifecycle, like proposed by Bork et al. [BAR23]. The general ambition is to leverage AI to augment human modeling capabilities—enhancing efficiency, consistency, and accessibility—while maintaining the semantic rigor and expressiveness that are central to conceptual models.

Process management systems support key decisions about the way work is allocated in organizations, including decisions on which task to perform next, when to execute the task, and who to assign the task to [BDvJ26]. Suitable software tools are developed to support these decisions in a way that is optimal for the organization. An example is GymPN [BDvJ26], that supports optimal decision-making in business processes using Deep Reinforcement Learning [MSH16].

Efforts towards *AI-assisted domain modeling* have explored how intelligent agents can support the elicitation and refinement of domain knowledge. For instance, Feltus et al. [FMPK21] proposed an approach that integrates AI reasoning mechanisms to assist modelers in structuring domain ontologies and conceptual schemas. Such systems can reduce the cognitive burden on modelers by suggesting relevant entities, relationships, and constraints, derived from either structured repositories or unstructured data sources.

The rapid rise of LLMs has intensified research on their potential role in conceptual modeling. Storey, Pastor, Guizzardi, Liddle, Maaß, Parsons, Ralyté, and Santos [SPG⁺25] provided an extensive assessment of how LLMs can be integrated into modeling environments, showing their promise for tasks such as generating textual model descriptions, suggesting conceptual structures, and facilitating natural language-based model querying. Complementary experiments by Fill et al. [FFK23] demonstrated how models like ChatGPT can interpret, explain, and extend conceptual models through conversational interaction, highlighting both opportunities and current limitations.

However, these advances are not without challenges. Muff et al. [MF24a] showed that while LLMs exhibit impressive linguistic competence, their understanding of metamodeling constructs and formal semantics remains limited. To address these gaps, Fill et al. [FHV⁺24] introduced the *CMAG framework* (Conceptual Model–Augmented Generative AI), which enriches generative models with domain ontologies and metamodel constraints to improve coherence and validity in AI-generated models. Similar ideas have emerged in process modeling research, where LLMs are employed to interpret event logs, generate process descriptions, or construct process models from textual inputs, by Kourani, Berti, Schuster, and Van der Aalst [KBSvdA24], and Kampik et al. [KWR⁺24].

From a broader perspective, the integration of AI into modeling is extending beyond static models toward dynamic and hybrid paradigms. For example, AI-driven conceptual modeling has been linked to digital twins [KPZ24], where models continuously learn and adapt from multimodal data streams to maintain synchronization with their physical counterparts. Fettke et al. [FFK25] further argued that the evolution from *LLM* (Large Language Model) to *LAM* (Large Action Model) and ultimately to *LxM Agents* represents a shift from passive text generation to active model co-construction—where intelligent agents can autonomously reason, design, and refine conceptual representations based on contextual understanding.

2.3 Multimodal Data Analysis

Multimodal data analysis has become increasingly relevant for conceptual modeling, as modern information systems are immersed in rich streams of heterogeneous data, including audio, video, and sensor modalities. Within the conceptual modeling community, this line of inquiry investigates how meaning can be systematically extracted and abstracted from multimodal observations to inform or refine conceptual artifacts.

The earliest research on conceptual modeling from multimodal data can be traced to the works of Day et al. [DDI⁺95] and Escobar-Molano et al. [EMG95] in 1990s, who pioneered object-oriented and structured approaches to representing video data. These works aimed to provide semantic abstractions over complex multimedia content, marking an important step toward understanding how conceptual models could serve as knowledge structures for interpreting visual narratives. Similarly, Smith and Benitez [SB00] proposed in 2000s conceptual representations for audio-visual content, introducing schemas that describe objects, scenes, and temporal relationships within multimedia streams.

2.3.1 Conceptual Modeling from Multimodal Evidence

The increasing digitalization and datafication of everyday activities have resulted in vast amounts of multimodal data—ranging from text and audio to video and sensor streams—that continuously capture human and organizational behavior. As Grisold et al. [GSHB24] emphasize, this pervasive data collection, often described as digital surveillance, arises from the growing tendency to transform human actions into quantifiable data objects, making them visible, analyzable, and actionable through digital technologies. Within organizations, such data streams are not only used for monitoring efficiency but also offer rich empirical foundations for understanding, modeling, and improving work processes. In this sense, multimodal data analysis becomes a bridge between raw behavioral traces and conceptual abstractions: it enables the identification of patterns, events, and relationships that can be systematically translated into conceptual models. This section explores how multimodal evidence informs conceptual modeling, how human-model interaction evolves through multimodal interfaces, and what advantages and limitations arise when integrating such heterogeneous data sources into the modeling process.

Over time, multimodal modeling shifted from descriptive representations toward analytical interpretations. Xie, Sundaram, and Campbell [XSC08] introduced event mining techniques to automatically extract semantically meaningful occurrences from video and audio data, bridging signal-level processing with high-level conceptual abstraction. In the context of process analysis, Pasquadibisceglie et al. [PACM20] integrated computer vision into predictive process mining, showing how visual observations could enhance event log data by providing additional contextual features such as task duration, human motion, or environmental cues.

2.3.2 Multimodal Interaction with Conceptual Models

Recent works emphasize the role of multimodality not only in data extraction but also in human-model interaction. Lepsien, Bosselmann, and Koschmider [LBMK22] demonstrated that process mining can be performed directly on video data by identifying activity boundaries and mapping them to event logs, while Kratsch, König, and Röglinger [KKR22] proposed a reference architecture that enables enterprises to systematically leverage video data in process mining pipelines. Building upon these foundations, Lepsien, Koschmider, and Kratsch [LKK23] presented an analytics pipeline for integrating multimodal streams into model-based reasoning, effectively allowing process models to become interactive visual and analytical objects.

Koschmider et al. [KAR⁺23] further extended this perspective by identifying challenges and research directions in applying process mining to unstructured data such as text, images, and videos. Their study highlights the need for multimodal pipelines that can bridge perception-level data with semantic-level modeling constructs. This resonates with the notion that conceptual models themselves could become interactive artifacts—dynamic representations that evolve through human–AI collaboration across modalities such as speech, gesture, and gaze.

The integration of multimodal data into conceptual modeling offers numerous advantages. As shown by Restrepo et al. [RWVV⁺24], multimodal fusion frameworks enable the unification of heterogeneous signals into coherent process representations, supporting richer contextualization and data-driven abstraction. Wornow et al. [WN⁺24] recently proposed the *Wonderbread* benchmark, which systematically evaluates multimodal foundation models on business process management tasks, demonstrating that multimodal embeddings can outperform unimodal baselines in recognizing process semantics.

2.4 Mixed Reality

Mixed Reality (MR) represents the convergence of the physical and virtual environments into a single, interactive continuum, where users can seamlessly engage with both real and digital artifacts [SHN19]. Within the broader field of Human–Computer Interaction, MR has been conceptualized as part of an extended reality (XR) spectrum encompassing Augmented Reality (AR) and Virtual Reality (VR), each differing in their level of

immersion and mediation [RSNC20]. As immersive technologies evolve, their potential for conceptual modeling becomes increasingly evident, especially as they enable spatial cognition, embodied interaction, and multimodal communication of models.

2.4.1 AR/VR/MR in Conceptual Modeling Environments

The integration of AR, VR, and MR technologies into conceptual modeling environments has grown significantly over the past two decades, motivated by their ability to make abstract representations more tangible and spatially grounded. Early work demonstrated that immersive visualization could enhance comprehension of mathematical and geometric relations through embodied interaction [KS02]. Similarly, in the context of conceptual modeling, Zhang et al. [ZC05] showed how virtual reality could augment UML-based modeling by offering 3D representations that promote intuitive understanding of structural and behavioral relations.

In recent years, the conceptual modeling community has increasingly explored MR for enterprise and domain modeling. Muff et al. [MF22] conducted a morphological analysis identifying multiple use cases for AR in enterprise modeling, while [MF23b] and [FM24] synthesized past achievements and envisioned future opportunities for combining VR/AR with modeling activities. The development of spatially anchored modeling languages, such as the AR Workflow Modeling Language (ARWFMM) and M2AR [MF24b], as well as the WebXR-based visual modeling languages proposed by Muff et al. [MF23a], exemplifies the technical convergence between metamodeling and XR technologies.

Fill et al. [Fil24] introduced the notion of *spatial conceptual modeling*, which explicitly anchors conceptual entities in the real world, bridging abstract representations with their physical referents. This line of work continues with the metamodeling perspective outlined by [Muf25], who formalized XR-oriented modeling paradigms as extensions of traditional metamodeling frameworks. Furthermore, recent explorations of AI-assisted modeling in XR environments, such as the CMAG framework [MF25], highlight a new generation of intelligent, language-model-supported modeling tools.

Complementary research in immersive visualization and simulation involving multi-modal data, broadens this context. The work of Lautenbach et al. [LPBK25] and Prohaska et al. [PPSK25] explores geospatial data visualization and dense indoor scene capture for immersive telepresence—both of which contribute to the technical backbone required for MR-based modeling. Similarly, studies in VR-enabled conversational agents [BKK24, Bos25], immersive analytics [MKS25], and multimodal simulation of human behavior [ZRH⁺25] showcase how immersive systems can support collaborative, data-driven reasoning. Collectively, these studies establish MR as a powerful medium for conceptual modeling, not merely as a visualization tool but as an interactive epistemic workspace that merges symbolic, visual, and embodied forms of knowledge [Lee25].

2.4.2 Cognitive and Usability Foundations

The effectiveness of MR for conceptual modeling depends fundamentally on cognitive and usability principles that govern human interaction with immersive systems. Research shows that Mixed Reality environments can enhance cognitive engagement by leveraging spatial memory, embodied cognition, and multimodal perception [RSNC20]. Speicher et al. [SHN19] emphasize that MR’s defining strength lies in its ability to blur perceptual boundaries, enabling users to reason through spatial alignment and object manipulation in ways that transcend traditional screen-based modeling interfaces.

Cognitive studies such as those by Mugisha et al. [MA25] demonstrate that MR-based procedural learning tasks improve performance while maintaining manageable cognitive load, suggesting strong applicability to complex reasoning tasks like model construction or validation. In conceptual modeling contexts, Oberhauser et al. [Obe21] showed that immersive modeling with VR-UML leads to higher engagement and improved comprehension due to embodied interaction and immediate feedback. Similarly, Fill et al. [FM24] argue that AR-mediated modeling activities bridge the mental and physical domains, allowing modelers to “see” abstract constructs within their real-world context—a crucial step toward anchoring conceptual representations in the environment where they are meant to operate.

From a usability standpoint, XR-based modeling environments such as M2AR [MF24b] and WebXR-driven domain-specific languages [MF23a] emphasize accessibility and collaboration. These developments converge with human-centered design principles to ensure that MR tools not only visualize but also cognitively scaffold conceptual modeling activities. As discussed by Lee et al. [Lee25], immersive experiences depend on the balance between technological fidelity and human perceptual coherence—an equilibrium that future MR-based modeling environments must deliberately design for.

2.5 Related Work

This section situates the thesis within existing research on tacit knowledge elicitation, multimodal process mining, model encoding for AI, and Mixed Reality for simulation. The reviewed works collectively motivate the proposed integration of multimodal data and MR-based elicitation as complementary evidence sources for enriching conceptual models.

2.5.1 Tacit Knowledge and Elicitation

Tacit knowledge represents experiential, embodied, and often unarticulated understanding that experts acquire through practice and interaction. Traditional elicitation techniques [CBB17], such as interviews or document analysis, tend to capture only explicit fragments of this knowledge, leaving underlying decision rules and perceptual cues unmodeled. Recent developments in spatial and visual conceptual modeling extend the

representational scope of models by embedding knowledge in physical and perceptual contexts.

In parallel, visualization and visual instruction methods provide new means for elicitation and refinement. Fill and Muff [FM23] show how large language models can generate structural visualizations from natural prompts, pointing toward the integration of generative AI in model communication and elicitation. Moreover, visual instruction tuning [LLWL23] and image-language alignment models such as ImageBind [GENL⁺23] suggest a pathway for aligning multimodal cues (text, image, sound, and motion) into unified embedding spaces that can encode tacit conceptual regularities. These advancements inform this thesis’s approach to capturing tacit knowledge through multimodal and egocentric MR elicitation.

2.5.2 Multimodal Process Mining

The fusion of process mining and multimodal data analysis has gained significant momentum, especially for domains where sensor data, video, and robotic telemetry complement event logs. Foundational works such as Lepsien et al. [LBMK22] and Kratsch et al. [KKR22] establish reference architectures for leveraging video data in process mining, while Lepsien et al. [LKK23] propose a full analytics pipeline for unstructured visual data. Building on these efforts, Corradini et al. [CPR⁺25b, CPR⁺25a] extend process mining to multimodal robotic systems, demonstrating zero-shot activity recognition and multimodal log construction. Similarly, Reinhard et al. [RLW⁺25] propose an architecture for event log construction from multimodal IT service data, while Sanchez-Ferriz et al. [SFELB25] extract events using text embeddings and natural language processing.

Further, Koschmider et al. [KAR⁺23] outline a comprehensive research agenda for process mining on unstructured data, emphasizing challenges in integrating multimodal evidence streams.

Zschaler et al. [ZMH⁺26] examine the challenges and opportunities of simulation reuse in healthcare applications, emphasizing modularity, interoperability, and the systematic integration of simulation artifacts across clinical contexts to improve decision support and operational efficiency.

Pasquadibisceglie et al. [PDA⁺26] propose a multimodal predictive process monitoring approach applied to explainable clinical pathways, demonstrating how heterogeneous data sources can be combined to enhance predictive accuracy while maintaining interpretability in healthcare settings.

Collectively, these efforts motivate our integration of multimodal evidence capture and MR-based elicitation as a path toward more complete, context-rich conceptual models.

2.5.3 Conceptual Model Encoding for AI

Conceptual modeling is increasingly intersecting with artificial intelligence, both as a beneficiary and as a contributor to intelligent reasoning. The systematic mapping

by Storey et al. [SPG⁺25] identifies growing synergies between conceptual modeling and large language models, emphasizing their potential to support model generation, validation, and semantic enrichment. Feltus et al. [FMPK21] propose AI-assisted domain modeling, leveraging reasoning and pattern recognition capabilities to guide modelers in identifying relevant domain constructs. At the same time, Glaser [Gla25] explores how semantic information from conceptual models can be encoded into vector spaces to support machine learning applications, paving the way for a new class of AI-readable conceptual artifacts.

Parallel efforts in video and image generation also inform model encoding strategies. Diffusion-based models for photorealistic video generation [GYS⁺24], text-to-image models [RPG⁺21], and segmentation foundations such as Segment Anything [KMR⁺23] reveal how structural and visual semantics can be embedded in large-scale generative models. In creative domains, comic generation systems [KBS24, KBN25, SSTS25] and multimodal story generation frameworks [YGL⁺25] show how visual narratives can represent structured conceptual information concisely. This thesis extends these ideas by using comic-like diagrammatic representations as compact, precise visual surrogates of conceptual models that bridge human interpretability and AI-encoded semantics.

Comic-like diagrams are adopted in this thesis as an expressive medium to bridge formal conceptual representations and intuitive human understanding. Unlike textual specifications or static process diagrams, comics naturally convey temporal flow, context, and emotion—making them particularly effective for depicting situated expert behavior and tacit decision-making. Prior studies demonstrate that visual narratives can significantly improve comprehension and engagement. For instance, Sharma and Guleria [SG25] show that comics enhance readers’ ability to grasp complex procedural information compared to plain text, while Kumar et al. [KBS24] and Singh et al. [SSTS25] highlight how AI-driven comic generation systems can translate structured narratives into visually coherent scenes. Moreover, Krishnaraj et al. [KBN25] present automatic video-to-comic conversion as a method for summarizing temporal information in compact, accessible form. Inspired by these insights, this thesis employs comic-like diagrams not for entertainment but as concise, semantically grounded surrogates of conceptual models—preserving precision while enhancing interpretability and recall.

2.5.4 Mixed Reality Simulation and Conceptual Modeling

Mixed Reality provides a dynamic substrate for simulation, model exploration, and experiential learning. Early inspirations come from the process mining community, where De Leoni et al. [DLShvdA16] introduced the idea of “process movies,” enabling animation of historical event logs for intuitive understanding. Contemporary MR research extends this paradigm to immersive, bidirectional simulation. Muff [Muf25] details the use of MR environments for metamodeling and structural alignment, while Fill [Fil24] discusses spatial anchoring of models in physical environments.

Recent advances in multimodal and embodied simulation technologies align well with this

vision. Gupta et al. [GYS⁺24] and Yang et al. [YGL⁺25] demonstrate how generative AI can produce interactive, context-sensitive simulations that respond to user behavior. By integrating MR with multimodal process evidence, this thesis proposes a novel mechanism for eliciting counterfactual decision behavior—enabling experts not just to observe models, but to inhabit, manipulate, and refine them through embodied interaction.

MM-AR [MBF26] introduces a web-based open-source metamodeling platform for spatial conceptual modeling, demonstrating how augmented reality and metamodel-driven engineering can be combined to support interactive, spatially grounded model creation and manipulation. Kajtazi [Kaj25] investigates AR-supported knowledge transfer through motion tracking and process visualization, focusing on how augmented reality environments can capture and externalize tacit process knowledge for improved understanding and training. Both [MBF26] and [Kaj25] complement this thesis by exploring AR-based extensions of conceptual modeling and tacit knowledge elicitation, yet they do not provide a systematic method for enhancing the precision and conciseness of conceptual models across multiple modeling languages and scenarios as pursued in this research.

Video generation is employed in this thesis as a foundation for creating immersive 360° Mixed Reality elicitation environments, enabling experts to re-experience and reflect on real or simulated scenarios from an egocentric perspective. Unlike static 3D models or textual descriptions, video-based MR scenes preserve motion, spatial cues, and environmental dynamics essential for uncovering tacit decision rules. Recent advances in generative video modeling have made this approach increasingly feasible: Gupta et al. [GYS⁺24] demonstrate photorealistic video synthesis through diffusion models, allowing controllable scene generation, while Yang et al. [YGL⁺25] introduce multimodal long-story generation that maintains temporal coherence across frames. In process analysis, De Leoni et al. [DLSHvdA16] show how animating event data into visual sequences aids interpretability and process understanding. Building upon these developments, our use of video generation supports 360° MR elicitation by transforming process traces into spatially immersive narratives—where experts can “step inside” their own processes, relive decision moments, and articulate reasoning that would otherwise remain implicit.

2.6 Synthesis and Positioning

The synthesis presented in this section provides a consolidated view of the research landscape and highlights several open challenges at the intersection of conceptual modeling, multimodal data analysis, and Mixed Reality elicitation, while the following Chapter 3 builds directly upon this summary. It systematically analyzes the limitations of current methods in capturing tacit knowledge, aligning multimodal evidence with conceptual abstractions, and operationalizing MR for elicitation. By doing so, it establishes the formal problem space and conceptual gap that this thesis seeks to close.

Across the surveyed literature, several gaps emerge. First, while multimodal process mining advances data capture and interpretation, its integration with conceptual modeling

remains limited—process models are extracted, but not semantically linked to domain concepts. Second, current model encoding efforts focus on static representations, leaving dynamic, embodied, and counterfactual aspects of expert cognition underexplored. Third, MR technologies are used mainly for visualization or training, not as active elicitation tools capable of revealing tacit decision structures. Lastly, the use of visual surrogates—such as comics or immersive diagrams—has yet to be systematically adopted as a means of enhancing model precision and communicability.

These identified gaps directly inform this thesis’s contribution. We propose to enhance conceptual modeling by integrating multimodal process evidence and MR-based elicitation into a unified framework. Multimodal data (e.g., video) serve as traces of tacit expertise, while MR environments enable controlled manipulation and observation of expert behavior, exposing counterfactual decision rules. Through this synthesis, conceptual models are refined for greater *precision* and *conciseness*. Moreover, the thesis introduces comic-like visualizations as evaluative artifacts that render complex model dynamics accessible, verifiable, and transferable between human and artificial intelligences.

Research Design

This chapter presents the overall research design of the thesis and explicates how the previously formulated research problem (Section 1.1.4) is systematically addressed through a structured methodological and technical approach. As stated earlier, the central challenge of this work lies in the lack of systematic methods for enhancing the *precision* and *conciseness* of conceptual models by incorporating tacit domain knowledge that experts rely on in practice but rarely articulate explicitly.

The chapter is structured as follows. Section 3.1 provides an investigation of the problem context. Section 3.2 introduces the methodological framework guiding the research, including the design science research paradigm, iterative artifact development, and evaluation strategy. Section 3.3 derives and formalizes the requirements of the proposed system. Finally, Section 3.4 summarizes the research design and transitions to the subsequent chapters that detail the developed theoretical framework (Chapter 4), method (Chapter 5, with elaborations in Chapter 6 and 7), technical realization (Chapter 8) and empirical application (Chapter 9).

3.1 Problem Investigation

Building upon the synthesized gaps identified in Section 2.6, this chapter deepens the analysis by examining the structural and epistemological tensions that hinder the enhancement of conceptual modeling through multimodal data and Mixed Reality. While prior research has advanced the technical capabilities for data collection, visualization, and automation, there remains a fundamental dissonance between what can be captured empirically and what can be meaningfully conceptualized. The purpose of this chapter is, therefore, to articulate the *research gap*—the disconnect between raw multimodal evidence and the formal, abstract representations required for precise and communicable conceptual models. To this end, we dissect the dual problem of evidence management: the

simultaneous scarcity and overload of evidence in modeling practice, and the challenges of mapping human sensemaking to computational formalism.

To optimize a business process, as in [Dij26]: first, problems are constrained by the constructs that are commonly used to model business processes; Second, a strong emphasis need to be put on using real-world data to gain insights into how processes actually work; While this data-driven approach enhances real-world applicability, it also introduces high uncertainty, which results in complex and irregular probability distributions that are problematic for traditional optimization techniques; Third, business process optimization problems are typically dynamic in the sense that the business process evolves over time, and new decisions may need to be made before the outcomes of previous decisions are fully realized.

As process behavior evolves, the process model and constraints might need to be adjusted over time, however from a Visual Analytics perspective, despite the existence of visualization techniques for this data type, these do not focus on conformance checking problems [vdLFP⁺25].

It is noted that in many organizations, documentation of process knowledge is scattered around various process information sources which introduces considerable problems [Gav23], but we are introducing a new concept of information fragmentation through various modalities.

Typically, information from heterogeneous sources is extracted during a data pre-processing or engineering phase. However, this step is often performed in an ad-hoc manner and is time-consuming and labor-intensive [vdPDA⁺25]. A reference model such as [vdPDA⁺25] can be used to provide a standard way for storing and extracting production event data, which opens new opportunities in interpretation of the data fragmented in various sources.

To achieve this interpretation, we can capture the digital footprint of activities and transactions, allowing businesses to streamline their processes, identify bottlenecks, and enhance efficiency. However, it is important to recognize that not all aspects of real-world processes are adequately represented in any representation of the business process. To underscore the limitations of current Process Mining methods, let's consider a familiar scenario – the repair process for a mobile phone. Figure 3.1 represents a simplified business process for phone repair, specifically focusing on repairing a phone with a broken camera. This figure employs a flowchart format to visually map out the sequence of activities involved in the repair process, with the manual tasks highlighted in grey. This visualization aids in comprehending the steps and stages essential for conducting phone repair within a business setting, facilitating both a high-level overview and a detailed understanding of the repairment procedure. Such visual representations are valuable tools for enhancing process efficiency, quality control, and workforce training within businesses that offer phone repair services.

Building valuable representations of a process can be based on event logs like presented in Table 3.1. In this example, the event log serves as a detailed record of activities,

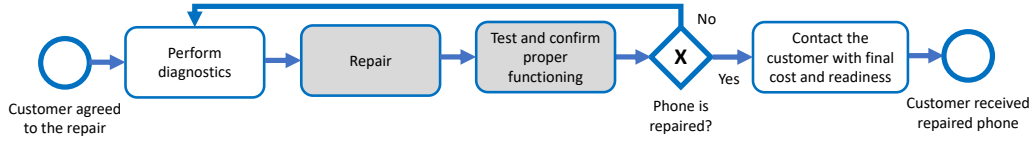


Figure 3.1: Simplified repair process of a phone with a broken camera. The grey activities are manual activities. Adopted from Gavric [Gav23].

their timestamps, sources of information, actors involved, objects in use, duration, and the respective environments in which these activities took place. We refer to it as a *pre-enhanced* event log as we will introduce the *enhanced* model in Section 3.1 and an enhanced event log in Table 3.2. The symbol "?" or red-colored text represent an unknown piece of information.

Table 3.1: Example of a pre-enhanced event log for phone repair business process. Adopted from Gavric [Gav23].

Timestamps	[00:18-00:28]	[00:28-00:33]	[00:33-?]	[?-00:59]	[00:59-01:04]
Event	Start Re-pair	Camera Diagnosis	Replacement Part Order	Camera Replacement	Quality Check
Source of Info.	Manual	Machine log	Manual	Manual	Manual
Actor	Technician A	Technician A	Admin	Technician A	Quality Inspector
Objects in Use	Phone, Screw-driver	Phone, Diagnostic Tool	None	Phone, Camera Module, Screw-driver	Phone
Duration	10 minutes	5 minutes	? minutes	? minutes	5 minutes
Environment	Workshop	Workshop	Office	Workshop	Workshop

One aspect highlighted within these event logs is the source of information for each event. It is noted that most information sources are manual, indicating that a human actor manually entered data into the system to record these events. Another noteworthy element in this table is the presence of unknown timestamps, denoted by question marks, which occur when there is no interaction with a digital system to record a specific event. Consequently, this lack of digital tracking makes it challenging to determine the exact duration of certain activities, resulting in estimates provided as a range of minimal to maximal values. Additionally, the table reveals that tracking the objects in use during

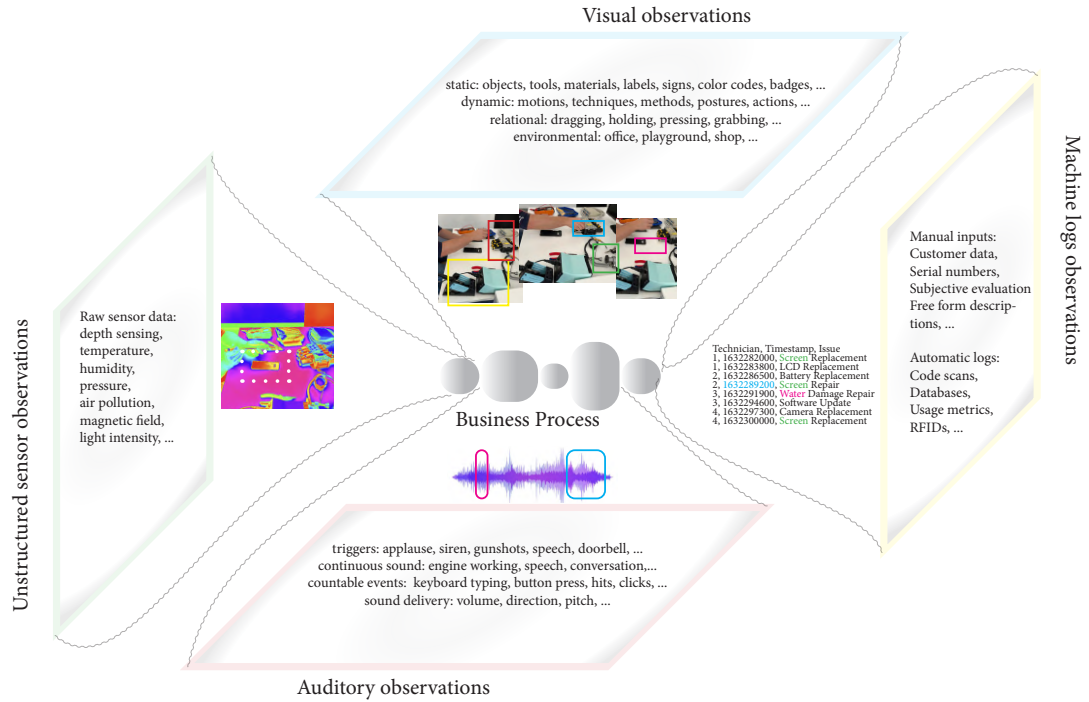


Figure 3.2: Multimodal enhancement of event logs for a business process. Adopted from Gavric [Gav23].

each event is a somewhat vague aspect of this process. The list of objects is not exhaustive, and it is unclear whether some objects can be used in parallel. This ambiguity can complicate resource allocation and potentially affect the efficiency of the repair process.

Integrating multimodal data sources into event logs can offer innovative solutions to address the challenges of enriching event logs. By supplementing manual data entry with diverse data streams such as video recordings from multiple perspectives, sensor data like depth sensing maps, and audio recordings capturing specific sound events, businesses may significantly enhance the accuracy, completeness, and richness of their event logs, as illustrated in Figure 3.2.

Multiple camera angles can capture different aspects of the repair procedure, allowing for precise and real-time documentation of technician actions and the condition of the device. Video data also offers the advantage of providing a visual timeline, eliminating the need for uncertain timestamps. With this visual evidence, the duration of each repair step can be accurately determined, enabling more precise process analysis. Moreover, video data can be used to verify the usage of tools and objects, enabling the tracking of the resources in parallel and optimizing the resource allocation. Sensor data, particularly unstructured ones, offers a wealth of information that can complement traditional event logs. These data streams can provide insights into the intricate details of a repair, such

as the depth and dimensions of components, the accuracy of alignments, and the quality of connections. Audio data recording of specific sound events can add another layer of context to event logs. For instance, the detection of particular sounds, such as clicks, whirrs, or engine sounds during the repair process, can serve as additional markers for events and their timing. These audio cues can be correlated with the corresponding visual and sensor data, offering a holistic understanding of the repair workflow. Furthermore, audio data can assist in identifying potential issues or anomalies during the repair, facilitating real-time intervention and quality assurance. Table 3.2 shows sample event logs by adding multimodal data sources of information to the system. The measurable exact duration of specific activities is a crucial factor in evaluating manual processes, as it often indicates efficiency, identifies areas for improvement, and guides resource allocation. Shorter durations suggest streamlined and optimized processes, while longer durations may signal inefficiencies. Analyzing activity durations may help benchmarking, compliance, quality control, and ensuring a positive customer experience.

Table 3.2: Example of an enhanced event log for phone repair business process. Adopted from Gavric [Gav23].

Timestamps	[00:11-00:23]	[00:23-00:32]	[00:33-00:40]	[00:40-00:52]	[00:52-1:11]
Event	Removing phone screen	Heating up the cleaner	Camera Cleaning	Module Replacement	Quality Check
Source of Info.	Video	Audio, Thermal camera	Video	Video	Video, Manual
Actor	Technician A	Technician B	Technician A	Technician A	Quality Inspector
Objects in Use	Phone, Screw-driver	Phone, Diagnostic Tool	None	Phone, Camera Module, Screw-driver	Phone
Duration	12.8 minutes	8.7 minutes	7.1 minutes	12.3 minutes	19.2 minutes
Environment	Work-desk1	Cleaner's corner	Work-desk1	Work-desk2	Workshop

To fully harness the potential of incorporating business logic on top of enriched event logs, there is a pressing need for a specialized approach that can deal with unique demands of working with video and volumetric data in conjunction with temporal and spatial information. We illustrate it in Table 3.2 with **Video** marks. Such an approach may involve applications of Mixed Reality for immersive data manipulation and precise domain-specific modeling. Incorporating MR into the analysis of enriched event logs may

not only enhances the understanding of complex processes may also offer a more intuitive and interactive approach to data analysis.

The enrichment of conceptual models for conceptual modeling applications aims to create representations that are empirically grounded, semantically precise, and cognitively accessible. By integrating multimodal data and MR-based elicitation, the thesis proposes a methodological pathway to close the research gap, advancing BPM from descriptive documentation toward dynamic, evidence-based, and human-aligned modeling that starts with event log creation from videos.

The underlying principle of event log creation in videos involves the extraction and analysis of meaningful semantic information from low-level event logs. While low-level logs capture detailed data such as object detection, sound recognition, and speech detection, the higher-level analysis focuses on incorporating the underlying semantics into these logs. This process involves identifying and categorizing events based on their semantic context, such as recognizing specific events composed of actions, identifying objects or people in different contexts and understanding their interactions. By incorporating semantics into the low-level event logs, the resulting higher-level event logs provide a more comprehensive and abstract representation of the video content, enabling advanced applications like activity recognition, video summarization, and semantic search.

Scene Logging involves creating a knowledge graph of entities present in videos and their spatio-temporal changes within the video content. By constructing a knowledge graph, entities can be represented as nodes and their relationships and interactions can be modeled as edges connecting these nodes with additional data which may include the date, time, location, duration, and participants of a given interaction. By analyzing the changes in spatial proximity, relative positions, and trajectories of objects, meaningful relationships can be inferred.

Ontology Modeling can organize objects, people, actions and events into a hierarchical structure. For example, in a scene, different types of vehicles can be categorized into subclasses such as cars, trucks, or motorcycles. Maintaining this hierarchical organization is crucial for efficient object identification, as it allows for a more detailed understanding of the scene and facilitates higher-level reasoning. Hierarchical object recognition can be achieved through hierarchical classifiers or by employing deep learning architectures that incorporate hierarchical features. These approaches enable the system to identify objects at different levels of abstraction, from general object categories to fine-grained subclasses.

Layered Approach. The expected architecture to fill-in the research gap is illustrated in Figure 3.3. It offers a comprehensive, layered approach to integrating and analyzing multimodal data sources in the context of process mining. This design embodies a holistic strategy, aiming to enable the conversion of raw data inputs into enriched conceptual models. The goal of this architecture is its applicability in a multitude of fields. The solution should show transformative impact across domains of Process Mining, where the model aids in deciphering complex processes, or Process Monitoring/Tracking, where real-time insights are extracted, or even Process Guidance, where the model provides direction

and recommendations that can be valuable for both training purposes or real-time on-field guidance.

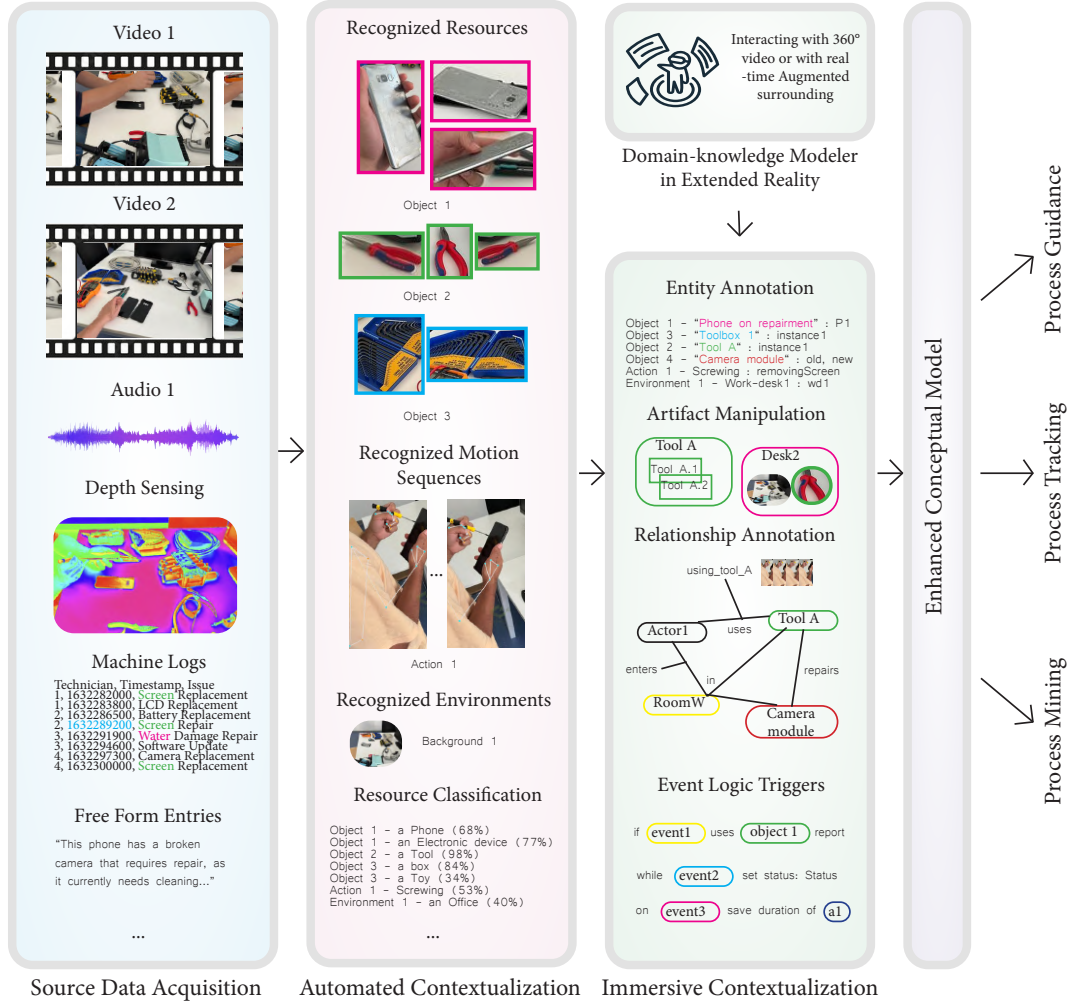


Figure 3.3: Problem analysis and solution design architecture. Adopted from Gavric [Gav23].

Source Data Acquisition. The foundation of our architecture begins with the Source Data Acquisition phase. This section accumulates a wide array of input sources, ranging from videos captured from diverse vantage points to audio recordings, in-depth sensor data maps, and detailed machine logs that track all interactions and database entries with the associated software. Given the heterogeneity of these data sources, it is imperative to have a unified architecture to ensure that all these disparate data streams are captured, processed, and made for subsequent layers of analysis.

Automated Contextualization. Following the data collection phase, the architecture delves into the Automated Contextualization. This segment is predominantly AI-driven and boasts of capabilities like object recognition (initially unnamed), identification of people (who are not yet recognized as system actors), motion tracking that detects sequences of postures of individuals and objects, classifiers for predicting the categories of various resources or activities, and background isolation to discern between diverse environments and settings. The power of artificial intelligence in this phase ensures that the raw data is transformed into recognizable entities and contexts, paving the way for deeper analysis and integration.

Immersive Contextualization. The subsequent tier, Immersive Contextualization, is manual and calls upon domain-knowledge experts to embed business logic into the evolving model. This immersive experience is facilitated through Mixed Reality headsets, enabling experts to navigate through 360-degree videos. At this juncture, the Entity Naming process comes to the fore, serving as a verification mechanism for the automated classifications performed earlier. This is complemented by Artifact Manipulations, which provide the flexibility to either group entities into more abstract categories or further specify them. For instance, an environment can be identified as a derivative of another setting, contingent on the availability of specific tools. Relationship Annotation further enriches the model, charting out a graph of pertinent links between artifacts and activities. The culmination of this phase is the Event Triggers module, where intricate business logic rules and advanced connections between recognized entities and activities are established.

Enhanced Conceptual Model. After traversing the intricate layers of the architecture, what emerges is the Enhanced Conceptual Model. This model is not only aware of the enriched multimodal event log, but is also primed for diverse applications. It encapsulates the depth and breadth of the information from the varied data sources and the insights gained through automated and manual contextualization.

Domain Specialization. Generally, there is a trade-off between providing standardized solutions that apply across many domains and creating domain-specific ones. Providing the necessary flexibility with solutions enables users to incorporate domain knowledge and iteratively refine their analyzes [vdLFP⁺25]. Pufahl et al. (2022) [PZWW22] advocate that healthcare process modeling tasks require systematic guidance in terms of modeling tools and methods, especially to support shared understanding by different stakeholders. Indeed, process modelers often lack advanced healthcare knowledge, while domain experts may perceive the richness of the conceptual modeling language as a drawback and may benefit from a demonstration by concrete examples [PZWW22].

3.2 Methodology

This research follows the *Design Science* paradigm, as articulated by Wieringa in his Design Science Methodology (DSM) for information systems and software engineer-

ing [Wie14]. Design Science is concerned with the systematic design, construction, and evaluation of artifacts intended to solve identified practical problems, while simultaneously generating scientifically grounded knowledge about those solutions. In alignment with this view, the present thesis develops and evaluates an artifact—a method and its supporting system—for enhancing conceptual modeling through multimodal data analysis and Mixed Reality–based elicitation.

Problem Investigation. In accordance with [Wie14], the research begins with a structured *problem investigation*. The central problem, introduced in Chapter 1 and elaborated in Section 3.1, concerns the limited ability of traditional conceptual modeling practices to systematically capture tacit domain knowledge, thereby constraining model precision and conciseness. The investigation involved (i) analysis of existing conceptual modeling approaches, (ii) examination of elicitation practices, and (iii) identification of technological opportunities offered by multimodal data and Mixed Reality. This stage resulted in a formalized research problem statement and a set of research questions that guide the remainder of the study.

Treatment Design. Following DSM’s *treatment design* phase [Wie14], a theoretical framework (Chapter 4) was constructed to conceptualize how multimodal evidence, alignment mechanisms, and immersive elicitation can contribute to conceptual modeling enhancement. This framework was operationalized into a concrete method (Chapter 5), specifying the enhancement cycle, the roles of AI agents, alignment processes, and simulation-based elicitation mechanisms. The method integrates prior validated components (cf. Chapters 6 and 7) into a coherent treatment for the identified problem.

Artifact Implementation. Consistent with DSM’s engineering orientation, the proposed method is instantiated as an implemented system (Chapter 8). The implementation adheres to the requirements derived in Section 3.3, ensuring traceability between the problem analysis, theoretical design decisions, and technical realization. The system operationalizes multimodal logging, crossmodal alignment, Mixed Reality simulation, and conceptual model construction within a unified architecture.

Evaluation Design. The evaluation follows DSM’s *empirical evaluation* principle, where artifacts are assessed with respect to their effectiveness in addressing the original problem [Wie14]. The central evaluation question is whether the proposed approach improves the *precision* and *conciseness* of conceptual models compared to a baseline elicitation technique.

The theoretical understanding of model quality in this thesis is inspired by prior work on the SEQUAL quality framework for process models by Krogstie et al. [KSJ06], that provides a foundation by conceptualizing process models as knowledge representations for action and by structuring quality along semiotic dimensions.

To this end, the study employs a controlled *A/B testing* setup [Rub74]. Condition A corresponds to our proposed approach for enhancing conceptual modeling through multimodal data analysis and Mixed Reality, while Condition B employs a baseline elicitation method. Domain experts and domain modelers collaboratively produce conceptual models under both conditions across multiple scenarios and modeling languages. The produced models are then evaluated using structured, scenario-specific metrics operationalizing the quality dimensions of precision and conciseness. The subjective assessment items underlying these metrics are measured using Likert-scale instruments grounded in the attitude measurement technique introduced by Likert [Lik32], enabling systematic and statistically analyzable expert judgments.

The evaluation design includes:

- structured pre-sessions for scenario and language alignment,
- controlled session rotations to mitigate bias,
- post-session expert-based quality assessment,
- comparative analysis of aggregated metric scores across conditions.

This setup enables a systematic comparison between the proposed treatment and the baseline, ensuring that observed differences in model quality can be attributed to the introduced method rather than uncontrolled contextual factors.

Knowledge Contribution. In line with [Wie14], the thesis contributes both a *technical artifact* (the method and system) and *design knowledge*. The latter concerns principles for integrating multimodal data analysis, alignment mechanisms, and Mixed Reality elicitation into conceptual modeling workflows. The evaluation thus serves not only to test the artifact’s effectiveness but also to refine and justify the underlying design decisions.

3.3 Requirements Specification

This section specifies the requirements that guide the *technical implementation* of the proposed approach. The subject of these requirements is the concrete system realization through the technical implementation described in Chapter 8. That implementation operationalizes the method introduced in Chapter 5, which in turn instantiates the theoretical framework introduced in Chapter 4. In this sense, the requirements serve as the formal bridge between theory, method, and system realization.

The section is structured as follows. Section 3.3.1 derives requirements for multimodal data analysis and semantic alignment. Section 3.3.2 specifies requirements for Mixed Reality-based elicitation as an epistemic instrument for surfacing tacit knowledge.

3.3.1 Requirements for Multimodal Data Analysis

To overcome the limitations identified in Section 3.1—ranging from evidence imbalance and ambiguity to missing stakeholder feedback loops—a new generation of conceptual modeling approaches must integrate multimodal data and interaction. The envisioned paradigm, referred to in this thesis as *Enhanced Conceptual Modeling*, establishes methodological and technological requirements that bridge empirical observation with conceptual abstraction. These requirements span three interrelated dimensions as follows.

MM-1. Data integration and synchronization. Conceptual modeling must evolve to incorporate multimodal data streams—video, and contextual metadata—captured during authentic task performance. These heterogeneous sources need to be temporally synchronized and semantically aligned, forming a unified evidence base from which model elements can be inferred. This requirement calls for standardized data schemas, alignment algorithms, and pipelines capable of translating raw multimodal signals into interpretable behavioral traces that preserve both action and intent.

MM-2. Semantic grounding and model alignment. To ensure conceptual integrity, multimodal evidence must be grounded in domain ontologies and formal modeling constructs. The system should support bidirectional mappings between observed behavior and model semantics, enabling models to both interpret and constrain data. Such grounding mitigates the risk of data-driven overfitting while preserving the explanatory and normative role of conceptual models. It also enables counterfactual reasoning—understanding not only what happened, but what *could* or *should* have happened.

MM-3. Continuous validation and experiential evaluation. Multimodal-enhanced modeling must include mechanisms for ongoing validation. Techniques such as video-based storytelling, or comic-like diagrams can serve as evaluative artifacts—bridging cognitive, visual, and experiential feedback channels. These mechanisms ensure that models remain living representations, adaptable to changing practices and reflective of human understanding.

3.3.2 Requirements for Mixed Reality Elicitation

While multimodal data integration enhances the empirical grounding of conceptual models, *elicitation* remains the critical mechanism through which tacit domain knowledge is externalized. To function as an effective elicitation environment within the framework, MR systems must fulfill a set of three interdependent requirements as follows.

MR-1. Spatial and contextual fidelity. An MR elicitation environment must reproduce or augment the expert’s operational context, including spatial arrangements, tools, and dynamic interactions. Fidelity is not limited to visual realism—it extends to environmental cues, and task affordances that trigger authentic behavior. High-fidelity reconstruction ensures that experts act naturally, allowing implicit procedural and perceptual knowledge to emerge without verbal mediation. Such fidelity can be

achieved through 360° video backdrops, or context-aware overlays anchored to physical space.

MR-2. Cognitive and behavioral capture. To reveal tacit cognition, MR elicitation must enable the capture of multimodal signals during the user’s interaction with virtual elements. These data streams provide complementary insights into attention, intention, and reasoning patterns. The MR setup thus becomes an active sensorium for conceptual modeling—transforming immersive interaction into a structured observation of thought-in-action.

MR-3. Bidirectional interaction and model feedback. Elicitation should not be a one-way act of demonstration but an interactive dialogue between expert and AI representing the model. MR environments must provide real-time feedback—visualizing inferred process structures, highlighting inconsistencies, or proposing alternative sequences. Such bidirectionality allows experts to validate, correct, or refine conceptual interpretations while remaining immersed in their task context. This continuous feedback loop transforms modeling from a post-hoc interpretation into a co-creative process of discovery and validation.

3.4 Conclusion

This chapter established the systematic research design that structures and legitimizes the thesis. Starting from the articulated research problem in Section 1.1.4, the chapter translated the identified epistemic and structural gaps into a coherent design-science-oriented investigation. Section 3.1 analyzed the core tensions underlying conceptual modeling enhancement, including evidence scarcity versus overload, grounding challenges, ambiguity and bias, missing stakeholder feedback loops, and representational misalignment. These analyses clarified why existing approaches are insufficient for systematically integrating tacit domain knowledge into precise and concise conceptual models.

Section 3.2 positioned the thesis within the Design Science paradigm [Wie14], framing the contribution as the construction and empirical evaluation of a method and supporting system. The subsequent requirements specification (Section 3.3) operationalized the problem analysis into concrete multimodal data analysis and MR-specific requirements, ensuring traceability between identified gaps and architectural design decisions. The envisioned architecture and enhancement pipeline demonstrated how multimodal logging, semantic grounding, immersive contextualization, and model enrichment jointly address the research gap.

The specified requirements directly operationalize and structure the investigation of the four research questions. RQ1, which asks how tacit domain knowledge can be conceptualized and integrated into conceptual modeling frameworks, is informed primarily by the semantic grounding, ontology modeling, scene logging, and Enhanced Conceptual Model requirements, as these define how empirical multimodal evidence is transformed into structured, conceptually aligned representations. RQ2, concerned with reliable capture

and elicitation, is shaped by the requirements for multimodal data integration and synchronization (MM-1), semantic grounding and model alignment (MM-2), continuous validation (MM-3), as well as the Mixed Reality requirements for spatial fidelity (MR-1), cognitive and behavioral capture (MR-2), and bidirectional model feedback (MR-3). RQ3 is directly influenced by the architectural requirements articulated in the translation of research gaps into system requirements, including source data acquisition, automated contextualization, immersive contextualization, and the layered solution architecture, which collectively define how the method is realized in a scalable, multi-scenario, language-independent technical implementation. Finally, RQ4 is anchored in the validation-oriented requirements—continuous experiential evaluation, ontology-driven abstraction, enriched event log construction, and the deployment of the Enhanced Conceptual Model in mining, monitoring, and guidance settings—since these establish the measurable conditions under which improvements in precision and conciseness can be empirically assessed across modeling languages and scenarios. In this way, the requirements not only guide system design but also concretely delimit the empirical and analytical scope of each research question.

The following chapters build upon this design blueprint: Chapter 4 formalizes the theoretical foundations of enhanced conceptual modeling; Chapter 5 specifies the enhancement method and its operational cycle (with detailed elaborations in Chapters 6 and 7); Chapter 8 presents the technical realization; and Chapter 9 reports the empirical evaluation in the end-scale application domain. In this way, the research design chapter serves as the structural bridge between the identified problem and its realized theoretical, methodological, and empirical contributions.

Theoretical Framework

This chapter establishes the theoretical framework that underpin the conceptual modeling enhancement method presented in Chapter 5, as well as its elaborations through the underlying approaches from the author’s related publications, grouped according to the foundational technologies of multimodal data analysis and Mixed Reality, presented in Chapters 6 and 7, respectively.

The theoretical framework serves three purposes. First, it clarifies the epistemic status of tacit knowledge and latent conceptual regularities, positioning them as legitimate modeling targets. Second, it formalizes the representational structures required to connect multimodal evidence and counterfactual elicitation to conceptual artifacts. Third, it defines the normative criteria—most notably precision and conciseness—through which enhancement effects are evaluated.

Section 4.1 introduces the overarching notion of enhanced conceptual modeling and reframes modeling as an evidence-informed abstraction process that integrates tacit and counterfactual knowledge. Building on this foundation, Section 4.1.4 introduces *Multimodal Enrichment of Conceptual Models* and formalizes the attachment function that links symbolic model elements with multimodal evidence, then establishes the embedding-based alignment framework, as adapted from Gavric et al. [GBP25i]. It leads to defining joint embedding spaces and semantic shift operations that connect natural language, symbolic constructs, and perceptual data, as adapted from Gavric et al. [GBP25a]. Section 4.2 formalizes *Multimodal Process Mining* and *Multimodal Conformance/Compliance Analysis*, extending classical process mining into heterogeneous sensory domains. This is adapted from Gavric et al. [GBP24c]. This is followed by a structured progression from monomodal to panmodal process mining (Section 4.2.1, adapted from Gavric et al. [GPB26]) and the formal introduction of relaxed process mining in embedding spaces (Section 4.2.2, adopted from Gavric et al. [GBP25b]). Section 4.2.3 subsequently introduces the *Petri Net of Thoughts* as a process-guided reasoning formalism for structuring multimodal and language-based inference. This is adopted from Gavric

et al. [GBP25e]. Section 4.3 introduces *Virtually Enhanced Narratives* as adapted from Gavric et al. [GBP25c]. The conclusion is provided in Section 4.4.

4.1 Enhanced Conceptual Modeling

As elaborated in Section 1.1, conceptual modeling has traditionally been grounded in declarative elicitation: interviews, documentation, and stakeholder articulation are translated into structured artifacts within a formal modeling language.

Frameworks such as *SEQUAL* [KSJ06, Kro15] conceptualizes model quality as the alignment between several interrelated sets: the modeling language (L), the model (M), the domain (D), the knowledge of participants (K), the interpretations of actors and tools (I, T), and the goals of modeling (G), all situated within a broader socio-technical context of actors (A). Central to this view is the notion of *model activation*, i.e., the process by which a model affects reality through automated, manual, or interactive interpretation. In its specialization for business process models [Kro16], *SEQUAL* emphasizes how process models represent knowledge for action and how their quality depends on physical, syntactic, semantic, pragmatic, and social dimensions. In particular, physical quality concerns the externalization and internalizability of models—whether knowledge can be effectively expressed in a modeling language, persistently stored, distributed, and interpreted by relevant audiences and tools. This perspective provides a foundation for understanding how conceptual models mediate between domain knowledge and coordinated action.

Building upon this foundation, we position multimodal data analysis and Mixed Reality not merely as technological add-ons, but as theoretical extensions that expand the interrelated sets. Multimodal evidence enriches the relationship between D , K , and M by grounding models in enacted domain behavior rather than solely articulated knowledge, thereby strengthening semantic and pragmatic quality. Mixed Reality environments further transform model activation by enabling interactive and immersive interpretation loops in which actors simultaneously perceive, enact, and refine models *in situ*.

This thesis presents an approach that challenges an assumption that relevant domain knowledge is either declaratively accessible or can be reconstructed through linguistic interaction alone, by treating tacit knowledge and counterfactual dispositions as primary modeling resources. From a theoretical perspective, this requires re-framing conceptual modeling as an epistemic integration problem: how can observable carriers of tacit knowledge (Definitions 8–13) be systematically transformed into structured model commitments? This reframing necessitates precise definitions of:

- The forms of conceptual knowledge (declarative vs. tacit),
- The observable carriers through which tacit knowledge manifests,
- The representational artifacts of conceptual modeling,

- And the normative quality criteria governing model evaluation.

The following subsections introduce these foundations in a formally grounded manner, beginning with conceptual knowledge and its forms.

4.1.1 Conceptual knowledge and its forms

Conceptual knowledge comprises the set of domain-relevant distinctions, relations, rules, and patterns that experts rely on to perceive, structure, and act within a domain. It enables both interpretation of situations and the formation of coherent conceptual models. Following established perspectives in knowledge representation and conceptual modeling, we distinguish two complementary forms of such knowledge: declarative and tacit.

Definition 1 (Declarative conceptual knowledge) *Declarative conceptual knowledge is the subset of domain knowledge that is explicitly stated in linguistic, symbolic, or diagrammatic artifacts. Formally, let E be a set of experts and $\mathcal{K}$ their total conceptual knowledge; the declarative subset is $K_{\text{decl}} \subseteq \mathcal{K}$ such that each $k \in K_{\text{decl}}$ has a canonical representation in some artifact language L . K_{decl} is the traditional input to conceptual modeling.*

Definition 2 (Tacit conceptual knowledge) *Tacit (implicit) conceptual knowledge is the complement $K_{\text{tacit}} := \mathcal{K} \setminus K_{\text{decl}}$ comprising cognitively available but non-explicitly stated knowledge items. Tacit items guide behavior and judgments but lack an authoritative declarative encoding at the time of observation; they may be partially articulable ex post through elicitation.*

Definition 3 (Conceptual regularities) *Conceptual regularities are stable patterns over a domain's states and processes that support explanation or prediction relative to modeling goals.*

Definition 4 (Latent conceptual regularities) *Latent conceptual regularities are regularities that are present in expert cognition and/or behavior but absent from K_{decl} . Formally, let $\mathcal{R}$ be all goal-relevant regularities and $\mathcal{R}_{\text{decl}} \subseteq \mathcal{R}$ those represented in K_{decl} ; then $\mathcal{R}_{\text{lat}} := \mathcal{R} \setminus \mathcal{R}_{\text{decl}}$.*

In this work, we treat $\mathcal{R}_{\text{lat}}$ as the primary analytical target, since tacit knowledge itself is not directly observable, whereas its induced regularities can be detected from multimodal traces and counterfactual elicitation.

The latent conceptual regularities are the primary target of interest in this thesis.

4.1.2 Evidence and carriers of tacit knowledge

Tacit conceptual knowledge, by its nature, leaves no direct symbolic trace; it must be inferred from the *evidence it produces* and the *carriers through which it manifests*. This subsection formalizes the structures that enable such inference. We distinguish between declaratively elicited knowledge, multimodal observations of expert conduct, and counterfactual responses elicited through controlled perturbations in Mixed Reality. These artifacts constitute the observable carriers whose statistical and structural properties should provide access to latent conceptual regularities that experts employ in practice but rarely articulate. By grounding each concept in precise definitions, we establish a rigorous foundation for identifying, extracting, and operationalizing tacit knowledge within the broader enhancement pipeline developed in this thesis.

Definition 5 (Declarative elicitation) *Declarative elicitation is a process El_{decl} that maps interactions with experts to declarative items, with the aim of populating K_{decl} . It is language- and protocol-dependent and typically does not observe behavior under perturbation.*

Definition 6 (Multimodal data) *Multimodal data are time-indexed observations drawn from $m \geq 2$ sensing/interaction channels. Let $\mathcal{T}$ be a time domain; a multimodal record is a tuple $D = (X^{(1)}, \dots, X^{(m)})$ with $X^{(i)} : \mathcal{T} \rightarrow \mathcal{X}^{(i)}$. Optionally, a synchronization map ϕ aligns modalities to a common temporal grid.*

Definition 7 (Multimodal data analysis) *Multimodal data analysis is a mapping $A : \mathcal{D}_{\text{mm}} \rightarrow \mathcal{H}$ from multimodal datasets $\mathcal{D}_{\text{mm}}$ to structured hypotheses or features $\mathcal{H}$ that can support inference of conceptual regularities. Typical A composes detection, classification, segmentation, alignment, representation learning, and pattern mining.*

This thesis is primarily concerned with the role of multimodal data analysis in **process mining**, i.e., the extraction, alignment, and reconstruction of process-relevant regularities from heterogeneous process traces.

Definition 8 (Multimodal traces of expert conduct) *Multimodal traces of expert conduct are multimodal data (Def. 6) specifically capturing expert actions in-situ, yielding behavioral event streams $E = \langle e_1, \dots, e_n \rangle$ with $e_i = (a_i, t_i, c_i)$ for action label a_i , time t_i , and context c_i derived from multimodal datasets, $\mathcal{D}_{\text{mm}}$. Such traces operationalize the “what experts do” part of tacit knowledge.*

Definition 9 (Mixed Reality) *Mixed Reality is an interactive system that renders virtual content projected to the physical environment and supports bidirectional sensing and actuation. Formally, an MR session is a tuple $\text{MR} = (\mathcal{E}, \mathcal{V}, \mathcal{I}, \text{Sync})$ of environment states $\mathcal{E}$, virtual states $\mathcal{V}$, interaction channels $\mathcal{I}$, and a synchronization mechanism Sync*

that maintains spatial-temporal coherence. MR enables controlled perturbations to elicit counterfactual responses.

Definition 10 (Counterfactual disposition) A counterfactual disposition describes how an expert would act under hypothetical (i.e., non-actual) conditions. Formally, let $\mathcal{C}$ be a set of intervention contexts (e.g., removal of a resource, change of risk level, added constraint), and let $\mathcal{A}$ be a set of admissible actions. A counterfactual disposition is a function $\delta : \mathcal{C} \rightarrow \Delta(\mathcal{A})$, where $\Delta(\mathcal{A})$ is the set of probability distributions over $\mathcal{A}$. Thus, for any imagined intervention $c \in \mathcal{C}$, $\delta(c)$ returns the (possibly stochastic) action the expert intends or would select if that intervention were realized. It captures “what an expert would do if conditions differed,” even though those conditions are not observed in the actual world.

Example. Consider a senior surgical nurse deciding whether to temporarily take over an instrument-passing task when a junior assistant slows down. The actual surgery video never shows such a slowdown event (it did not occur), yet the nurse can still state how she *would* act if it did. Let $c_{\text{slow}} \in \mathcal{C}$ denote the counterfactual context where the assistant becomes delayed; This mapping exists even though the hypothetical context is unobserved in the real trace.

Definition 11 (Elicited counterfactual response) An elicited counterfactual response is an observed realization of a counterfactual disposition (Def. 10) under an explicit perturbation. In an MR session, present $c \in \mathcal{C}$ and record $r = (a, \tau, \mathbf{z})$, where $a \in \mathcal{A}$ is the chosen action, τ is reaction time, and $\mathbf{z}$ contains auxiliary signals (gaze, speech, biometrics). The dataset $\mathcal{R} = \{(c_j, r_j)\}_j$ is used to infer latent regularities.

Definition 12 (Carriers of tacit conceptual knowledge) Carriers of tacit conceptual knowledge are observable or inferable structures whose statistics are informative about K_{tacit} (Def. 2). Let Ξ denote the carrier space; typical elements include conduct multimodal data traces (Def. 8) and elicited counterfactual responses (Def. 11). A carrier family Ξ is valid if there exists an identification mapping $\text{ld} : \Xi \rightarrow \mathcal{P}(\mathcal{R}_{\text{lat}})$ (Def. 4).

Definition 13 (Observable carriers of tacit conceptual knowledge) Observable carriers are carrier instances $\Xi_{\text{obs}} \subseteq \Xi$ (Def. 12) that can be measured with specified fidelity constraints and provenance metadata.

Observable carriers of tacit conceptual knowledge are the concrete inputs to enhancement pipelines and serve as evidence for inferring $\mathcal{R}_{\text{lat}}$ (Def. 4).

4.1.3 Conceptual Modeling and Its Artifacts

Conceptual modeling rests on cognitive foundations. As argued by Mylopoulos, Guizzardi, and Guarino [MGG25], human understanding arises from embodied interaction with reality, through which we abstract cognitively relevant regularities that form a cognitive domain [BEF⁺56]. Concepts mediate between mind and language [Kar24] and, in Guarino’s ontological account [Gua98, GG95, Gua09], function as intensional entities structuring possible worlds. Conceptual modeling can therefore be understood as the disciplined externalization of such conceptualizations into intersubjectively interpretable artifacts.

In line with the cross-disciplinary discussion summarized by Robinson [RAB⁺15], and adopting the definition by Arbez and Birta [RAB⁺15], we treat a conceptual model as “a concise and precise consolidation of all goal-relevant structural and behavioral features of the system under investigation presented in a predefined format.” This emphasizes goal-dependence, structural and behavioral scope, and representation within a modeling language. Precision and conciseness therefore become normative quality criteria of the resulting artifact.

We now formalize the modeling process, the modeling language, and the conceptual model.

Definition 14 (Conceptual modeling) *Conceptual modeling is the goal-driven externalization process*

$$\text{CM} : (\mathcal{C}, K_{\text{decl}}, \Xi_{\text{obs}}, G, L) \mapsto M \in \mathcal{M}(L),$$

where $\mathcal{C} = \langle D, W, R \rangle$ denotes a conceptualization in the sense of [GG95], K_{decl} represents declarative knowledge, Ξ_{obs} observational or enacted evidence, G encodes modeling goals, and L is a modeling language. The process transforms cognitively grounded structures and empirical inputs into a formal artifact through elicitation, abstraction, formalization, and validation.

Definition 15 (Formal conceptual modeling language) *A formal conceptual modeling language is a triple $L = (\Sigma, \text{Syntax}, \text{Semantics})$ in which Σ is a vocabulary of modeling primitives, Syntax specifies the well-formed constructions over Σ , and Semantics : Syntax $\rightarrow \mathcal{S}$ assigns each well-formed construction an interpretation in a semantic domain $\mathcal{S}$.*

Typical instances include UML [Obj07], BPMN [Obj08], ArchiMate [The09], OntoUML [Gui05], and Petri Nets [PR08], each of which constrains admissible constructions and determines how resulting models are interpreted independently of their diagrammatic layout.

Definition 16 (Conceptual model) *Let $L = (\Sigma, \text{Syntax}, \text{Semantics})$ be a formal conceptual modeling language with semantic domain $\mathcal{S}$. A conceptual model in L is a pair*

(α, M) such that $M \in \text{Syntax}$ is a well-formed construction over Σ , $\alpha : \Sigma \rightarrow D$ assigns domain referents from the cognitive domain D , and the semantic interpretation $\text{Semantics}(M_\alpha) \in \mathcal{S}$ is defined. We write $\mathcal{M}(L)$ for the set of all such models.

Two conceptual models are goal-equivalent under modeling goals G if they induce identical answers for the goal-relevant query set Q_G when evaluated in $\mathcal{S}$. In this sense, conceptual models are language-constrained externalizations of conceptualizations whose quality is judged relative to goals; enhancements grounded in multimodal evidence aim to improve the precision and conciseness of this transformation without altering the ontological status of the underlying concepts. Suppose L is BPMN. The syntactic component **Syn** contains diagrams formed from tasks, events, and gateways. An analyst builds a model M of a hospital intake process and fixes an interpretation α that maps symbols (e.g., a “Triage” task node) to concrete referents in the hospital domain (the actual triage station and activity). The BPMN execution semantics then yields M_α as a set of executable paths or a transition system. If another modeled variant N with interpretation β yields exactly the same answers to all compliance-related goal queries Q_G (e.g., “Is triage always performed before medication?” “Is revisiting of registration possible?”), then the two models are goal-equivalent despite possibly differing structure.

Definition 17 (Precision of conceptual models) *Let (α, M) be a conceptual model in the sense of Def. 16, interpreted into the semantic domain $\mathcal{S}$ as M_α . Fix a set of goal queries Q_G and a family Φ of admissible semantic states (e.g., all runs allowed by regulation). We say that (α, M) is precise (w.r.t. G) if for every query $q \in Q_G$ the answer induced by M_α is single-valued over Φ , i.e., M_α leaves no goal-relevant ambiguity: for any two admissible semantic states $s, t \in \Phi$, the induced answers coincide.*

This notion of precision aligns with the fact-oriented tradition of conceptual modeling as articulated in Halpin’s Object-Role Modeling framework [Hal01] and in the earlier NIAM approach [NH89]. In these works, conceptual schemas are required to satisfy the *100% principle*: all relevant facts about the universe of discourse must be captured explicitly in the schema, and nothing essential should be left implicit. Precision, in this sense, corresponds to eliminating semantic underspecification that would allow multiple admissible interpretations of goal-relevant facts. A model that violates precision permits different admissible states to yield different answers to the same goal query, indicating that some fact types, constraints, or role commitments remain insufficiently articulated.

Definition 18 (Conciseness of conceptual models) *Let (α, M) be a conceptual model as in Def. 16. We say that (α, M) is concise (w.r.t. G) if its syntactic form and semantic commitments make the goal-relevant content concise to a competent reader — that is, the constructs that determine the answers to Q_G are made salient rather than obscured by dispersion, indirection, or articulation choices that impair immediate understanding.*

Conciseness, in this sense, is not the minimization of size, but the minimization of cognitive opacity: among all goal-equivalent models, a concise model is one whose structure renders the goal-relevant conceptual commitments easiest to grasp.

From the perspective of fact-oriented modeling [Hal01, NH89], conciseness is achieved not by compressing representation, but by structuring it so that elementary facts, constraints, and role relationships are expressed in a cognitively transparent manner. Halpin emphasizes verbalization, role clarity, and constraint explicitness as mechanisms that reduce ambiguity while preserving understandability. Thus, conciseness complements precision: whereas precision eliminates semantic multiplicity across admissible states, conciseness reduces cognitive effort in apprehending the commitments that determine those states.

Taken together, these notions echo the dual requirement in fact-oriented approaches: (i) completeness and constraint-explicitness (precision), and (ii) intelligible, fact-based articulation of domain semantics (conciseness). In this thesis, both are evaluated relative to goal queries Q_G , thereby operationalizing classical conceptual modeling quality principles within a goal-oriented semantic framework.

Definition 19 (Conceptual modeling enhancement) *Let CM be the conceptual modeling process of Def. 14, producing models in $\mathcal{M}(L)$. An enhancement of conceptual modeling is any transformation of CM or of its inputs $(K_{\text{decl}}, \Xi_{\text{obs}}, G, L)$ that yields models (α, M) with strictly improved quality under the chosen criteria—such criteria include but are not limited to improved precision (Def. 17), improved conciseness (Def. 18), increased completeness of goal-relevant content, or improved faithfulness to expert conduct extracted from Ξ_{obs} . Formally, a procedure **Enh** is a conceptual modeling enhancement if for the output $M^+ = \text{Enh}(\text{CM}(\cdot))$ we have that M^+ strictly dominates M along at least one quality dimension and is not worse in the others.*

4.1.4 Multimodal Enrichment of Conceptual Models

We introduce multimodal enrichment at the level of conceptual models as defined in Definition 16. Rather than extending the metamodel structure directly, enrichment is formulated as an augmentation of the semantic grounding of a model $(\alpha, M) \in \mathcal{M}(L)$ through explicit linkage to multimodal evidence.

Let $L = (\Sigma, \text{Syntax}, \text{Semantics})$ be a formal conceptual modeling language with semantic domain $\mathcal{S}$. Let $\Gamma = \{\gamma_1, \gamma_2, \dots, \gamma_m\}$ be a finite set of multimodal data types (e.g., image, audio, video), and let $\mathcal{E}_\Gamma$ denote the universe of multimodal evidence instances typed by Γ .

Given a conceptual model $(\alpha, M) \in \mathcal{M}(L)$, we denote by $\text{Elem}(M)$ the set of syntactic elements occurring in the well-formed construction M .

Definition 20 (Multimodal Attachment Function) *Let $(\alpha, M) \in \mathcal{M}(L)$ be a conceptual model. A multimodal attachment function is a mapping*

$$\text{Att} : \text{Elem}(M) \rightarrow 2^{\mathcal{E}_\Gamma}$$

such that for each syntactic element $e \in \text{Elem}(M)$, $\text{Att}(e)$ is a (possibly empty) set of multimodal evidence instances grounding or contextualizing e .

The attachment function does not alter the syntactic validity of M nor the definition of its semantic value $M_\alpha \in \mathcal{S}$. Instead, it introduces an explicit linkage between model elements and empirical evidence, thereby strengthening their semantic and pragmatic grounding.

Definition 21 (Multimodal-Enriched Conceptual Model (MMeCM)) *Let $(\alpha, M) \in \mathcal{M}(L)$ be a conceptual model. A multimodal-enriched conceptual model is a triple*

$$(\alpha, M, \text{Att})$$

where Att is a multimodal attachment function as defined above.

We write $\mathcal{M}^+(L)$ for the set of all multimodal-enriched conceptual models over L .

Under this formulation, multimodal enrichment preserves the formal properties of conceptual models in L while extending their epistemic basis. Whereas (α, M) establishes a mapping from syntactic constructs to domain referents and induces a semantic value $M_\alpha \in \mathcal{S}$, the enriched structure (α, M, Att) additionally anchors these constructs in multimodal observations. In this sense, multimodal enrichment augments the interpretative layer of the model without modifying its formal semantics, enabling tighter alignment between symbolic representation and empirically observable domain phenomena.

4.1.5 Aligned Embeddings of Knowledge and Symbols

We define the principal constructs underlying our alignment methodology between conceptual modeling artifacts and AI-driven semantic spaces. The following definitions establish the foundational terminology used to formalize the Aligned Embeddings of Knowledge and Symbols (ALeKS) framework, further described in Section 6.2 .

Definition 22 (Embedding Space) *Let $\mathbb{R}^d$ denote a d -dimensional real-valued vector space, where $d \in \mathbb{N}^+$. An **embedding space** $\mathcal{E} \subseteq \mathbb{R}^d$ is a continuous vector space in which elements—such as symbols, textual labels, or perceptual inputs—are represented as dense vectors.*

Definition 23 (Joint Embedding Space) *A **joint embedding space** $\mathcal{E}$ is an embedding space in which heterogeneous modalities (e.g., text, vision, audio, structured data) are aligned into a shared semantic space. Formally:*

$$\mathcal{E} = \mathcal{E}_{\text{TEXT}} \cup \mathcal{E}_{\text{VISION}} \cup \mathcal{E}_{\text{AUDIO}} \cup \mathcal{E}_{\text{SYMBOLIC}} \subseteq \mathbb{R}^d.$$

Definition 24 (Natural Language Term (NLT)) A *natural language term* is a textual element $a \in C_{\text{NLT}}$ extracted from a conceptual model, usually from attributes such as names or labels. The set $C_{\text{NLT}} = \{a_1, a_2, \dots, a_k\}$ captures the linguistic surface forms used in the modeling domain.

Definition 25 (Conceptual Model Term (CMT)) A *conceptual model term* $a' \in C_{\text{CMT}}$ is an enriched version of a natural language term $a \in C_{\text{NLT}}$, extended with semantics from the modeling language (e.g., stereotypes, roles, taxonomies). C_{CMT} includes domain-specific and structural knowledge encoded in conceptual artifacts.

Definition 26 (Semantic Shift) Given $x, x' \in \mathcal{E}$, the *semantic shift* from x to x' is the vector:

$$\vec{s}(x \rightarrow x') = E(x') - E(x),$$

where $E(\cdot)$ denotes the embedding function. The shift encodes how meaning evolves through semantic enrichment or modality transformation.

Definition 27 (Shift Function) The *shift function* $s : \mathcal{E} \rightarrow \mathcal{E}$ maps between embeddings to quantify semantic transformation. Formally:

$$s(x \rightarrow x') = E(x') - E(x),$$

where x is a natural language or perceptual term, and x' its conceptualized or target form.

Definition 28 (Set of Semantic Shifts (S_e)) The total set of semantic shifts is defined as:

$$S_e = \{s(x \rightarrow x')\} \cup \{s(m(x) \rightarrow x)\} \cup \{s(m(x) \rightarrow m')\},$$

where $x \in C_{\text{NLT}}$, $x' \in C_{\text{CMT}}$, and $m(x), m' \in C_{\text{MM}}$. This set captures how terms evolve through symbolic and perceptual transformation within the joint embedding space.

4.2 Multimodal Process Mining and Analysis

This section introduces the formal foundations of *Multimodal Process Mining* (MMPM) and its extension to *Multimodal Conformance/Compliance Analysis* (MMCCA). Together, these definitions describe how heterogeneous sensory and data modalities can be integrated into process mining pipelines and subsequently evaluated for behavioral alignment and compliance.

Definition 29 (Multimodal Process Mining (MMPM)) Let $M = \{m_1, m_2, \dots, m_k\}$ represent the set of modalities. The comprehensive event log E is constructed as $E = \bigcup_{i=1}^k E_i$, where $E_i = \{(e_{i1}, t_{i1}, a_{i1}), (e_{i2}, t_{i2}, a_{i2}), \dots, (e_{in}, t_{in}, a_{in})\}$ captures the

events, timestamps, and attributes for each modality m_i . The event log E integrates these individual logs as:

$$E = \bigcup_{i=1}^k \{(e, t, a) \mid (e, t, a) \in E_i\}.$$

The discovery function, defined as $\mathcal{P} = \text{Discover}(E)$, applies process mining algorithms to analyze E and generate a process model $\mathcal{P}$. Thus, MMPM captures multimodal contributions to knowledge gain in process mining and is formally defined as:

$$\text{MMPM} : (M, E) \mapsto \mathcal{P} = \text{Discover}(E),$$

where M denotes the set of modalities, E is an enriched event log obtained by integrating modality-specific evidence $E_1, \dots, E_k$, and $\mathcal{P}$ is the set of discovered process models.

The enriched event log E , as defined in Definition 29, serves as the foundational input for **Multimodal Conformance/Compliance Analysis (MMCCA)**. By utilizing the detailed event logs generated through MMPM, MMCCA applies conformance functions to evaluate how well the observed behavior aligns with predefined process models and complies with organizational or regulatory rules. The resulting conformance score C provides a quantitative measure of behavioral alignment and compliance. Hence, while MMPM facilitates the collection and integration of multimodal data, MMCCA ensures that these observations are consistent with both structural models and normative requirements, ultimately enhancing the accuracy and trustworthiness of process analyses.

Definition 30 (Multimodal Conformance/Compliance Analysis (MMCCA))

Let $M = \{m_1, m_2, \dots, m_k\}$ be the set of modalities. For each modality m_i , let $E_i = \{(e_{i1}, t_{i1}, a_{i1}), (e_{i2}, t_{i2}, a_{i2}), \dots, (e_{in}, t_{in}, a_{in})\}$ represent the event log. The comprehensive event log E is then constructed as $E = \bigcup_{i=1}^k E_i$. Given a process model P and a set of conformance rules R , the conformance function $\text{Conform}(E, P, R)$ evaluates the degree to which the observed behavior E adheres to the process model P and complies with the rules R . Define C as the conformance score:

$$C = \text{Conform}(E, P, R).$$

Thus, Multimodal Conformance/Compliance Analysis is defined as:

$$\text{MMCCA} : (M, E, P, R) \mapsto C = \text{Conform}(E, P, R),$$

where M denotes the set of modalities, E is an enriched event log obtained by integrating modality-specific evidence $E_1, \dots, E_k$, P is the process model, R is the set of conformance rules, and C is the resulting conformance score capturing the degree of alignment between observed behavior and the prescribed process and constraints.

4.2.1 From Monomodal to Panmodal Process Mining

To position our contribution within the evolving landscape of process mining, we formalize a systematic progression from classical monomodal process mining to its panmodal extension—encompassing multimodal, crossmodal, and transmodal variants. This progression reflects the gradual enrichment of process mining methodologies to handle, integrate, and reason over heterogeneous data modalities, ultimately fostering modality-invariant process understanding.

Definition 31 (Monomodal Process Mining, based on [vdA12]) *In the conventional process mining paradigm, event data is assumed to be represented in a single, uniform modality. Let $\mathcal{L} = \{(c_i, \langle e_{i1}, e_{i2}, \dots, e_{in} \rangle)\}$ be an event log where each event e_{ij} is described using a single modality (e.g., textual labels). **Monomodal Process Mining** refers to the analysis of such event logs using a single, uniform data modality for process mining tasks, including discovery, conformance, and enhancement. This represents the classical setting in which each event is symbolically represented and modality-specific context is not explicitly modeled.*

Multimodal Process Mining as defined in Definition 29, extends traditional process mining by integrating these heterogeneous modalities to enhance model accuracy, contextual grounding, and interpretability.

Definition 32 (Panmodal Process Mining) ***Panmodal Process Mining** refers to the overarching paradigm in which process mining tasks are performed across and between multiple modalities. It subsumes both crossmodal and transmodal strategies, thereby enabling higher degrees of representational abstraction, knowledge fusion, and modality-invariant process understanding.*

Definition 33 (Crossmodal Process Mining) *Let $\mathcal{M} = \{m_1, \dots, m_k\}$ be a set of data modalities and let $S \subseteq \mathcal{M}$ with $|S| \geq 2$. **Crossmodal Process Mining** creates a joint latent space Z_S from the modalities in S , enabling events from those modalities to be aligned, translated, and mutually reinforced—while modalities outside S remain decoupled. This facilitates knowledge transfer and inference across selected modalities without requiring full modality integration.*

Definition 34 (Transmodal Process Mining) *Let $\mathcal{M} = \{m_1, \dots, m_k\}$ be the full set of available modalities. **Transmodal Process Mining** embeds every $m \in \mathcal{M}$ into a single shared space Z , ensuring full interchangeability and seamless joint reasoning over all modalities. This highest level of integration allows for complete semantic alignment and inferential generalization across modalities.*

4.2.2 Relaxed Process Mining

To handle the variability and uncertainty inherent in multimodal data, we introduce the concept of *Relaxed Process Mining*. This framework generalizes classical process mining to continuous embedding spaces, where events and activities are represented by dense vectors rather than discrete labels. It accounts for semantic variability, multimodal representations, and noise tolerance by defining equivalence and alignment in terms of spatial proximity in the embedding space.

Definition 35 (Relaxed Process Mining) *Let $\mathcal{E} = \{e_1, e_2, \dots, e_m\}$ be an event log, where each event e_i is represented by an embedding in $\mathbb{R}^n$. In relaxed process mining, events are treated as similar (and thus potentially interchangeable) if their embeddings lie within a specified distance $\delta > 0$, i.e., if $|e_i - e_j| \leq \delta$. This relaxed notion of equivalence supports:*

1. **Semantic variability.** *Words or labels that share similar embeddings (e.g., synonyms) are considered equivalent.*
2. **Multimodal data.** *Events derived from video, images, or audio are compared in a continuous space; frames or segments sufficiently close in this space are grouped as one activity.*
3. **Noise tolerance.** *Small perturbations in event embeddings do not alter the process structure, mitigating issues arising from measurement or labeling noise.*

Hence, relaxed process mining generalizes classical process mining tasks (e.g., discovery and conformance checking) to operate over neighborhoods in the embedding space rather than strict, discrete labels.

Definition 36 (Relaxed Process Discovery) *Relaxed process discovery specializes relaxed process mining to the task of constructing a process model (e.g., a Petri Net, BPMN diagram, or process tree) from an event log $\mathcal{E} \subseteq \mathbb{R}^n$. Formally, given:*

- *A threshold parameter $\delta > 0$ defining when two embeddings are considered equivalent,*
- *A noise tolerance parameter $\alpha \in [0, 1]$ allowing a fraction of events to be disregarded,*

a relaxed process discovery algorithm partitions and clusters events in $\mathcal{E}$ by grouping those whose embeddings lie within distance δ of a representative reference. These clusters form the basis for high-level subprocesses and control-flow relations (e.g., sequence, parallelism, choice) in the discovered model. Events or clusters failing to meet the relaxed equivalence criteria may be treated as noise (up to $\alpha \cdot 100\%$ of the log) or penalized. The resulting process model tolerates lexical variation and multimodal representation shifts while preserving the behavioral semantics of the process.

Definition 37 (Relaxed Conformance Checking) Let $T = \langle e_1, e_2, \dots, e_m \rangle$ be an observed trace whose events are represented by embeddings in $\mathbb{R}^n$, and let $\mathcal{M} = \langle r_1, r_2, \dots, r_L \rangle$ be a modeled process whose activities are also represented by embeddings in $\mathbb{R}^n$. A relaxation parameter $\delta > 0$ specifies when an event is considered close enough to match a modeled activity; that is, $|e_i - r_k| \leq \delta$ implies equivalence between e_i and r_k .

A relaxed alignment is a mapping:

$$\mathcal{A} : \{e_1, e_2, \dots, e_m\} \rightarrow \{r_1, r_2, \dots, r_L\} \cup \{\epsilon\},$$

where ϵ denotes a skip symbol representing a deviation. Let $\gamma > 0$ be a penalty for skipping an event. The cost of aligning e_i with r_k is defined as:

$$c(e_i, r_k) = \begin{cases} |e_i - r_k|, & \text{if } |e_i - r_k| \leq \delta, \\ +\infty, & \text{otherwise,} \end{cases} \quad c(e_i, \epsilon) = \gamma.$$

The total alignment cost is:

$$C(\mathcal{A}) = \sum_{i=1}^m c(e_i, \mathcal{A}(e_i)),$$

subject to the constraint that $\mathcal{A}$ preserves event order (i.e., $i < j \Rightarrow k \leq k'$ for aligned pairs). The optimal relaxed alignment $\mathcal{A}^*$ minimizes $C(\mathcal{A})$ over all valid alignments.

Given a normalization constant $C_{\max}$ (e.g., $C_{\max} = m \cdot \max\{\delta, \gamma\}$), the relaxed fitness of trace T with respect to model $\mathcal{M}$ is:

$$F(T) = 1 - \frac{C(\mathcal{A}^*)}{C_{\max}}.$$

If $F(T) = 1$, the observed trace perfectly fits the model within distance δ . A noise tolerance $\alpha \in [0, 1]$ may also be applied, allowing partial mismatches:

$$\frac{|\{e_i \in T : \forall k, |e_i - r_k| > \delta\}|}{|T|} \leq \alpha.$$

If this condition holds, T is still accepted as conformant. This formulation naturally accommodates linguistic synonyms and multimodal phenomena (e.g., multiple frames of the same scene) by treating embeddings representing equivalent concepts as proximate in $\mathbb{R}^n$.

4.2.3 Foundational Definitions for the Petri Net of Thoughts (PNoT)

To formally ground the Petri Net of Thoughts (PNoT) paradigm, we introduce a set of definitions that characterize its representational structures, execution semantics, and the mapping between Petri Net constructs and LLM-driven reasoning. These definitions mirror the rigor of traditional process modeling while extending it to cover the orchestration of structured prompting chains.

Definition 38 (Petri Net, based on [Rei85]) A Petri Net is a tuple $\mathcal{PN} = (P, T, F, M_0)$ where P is a finite set of places, T is a finite set of transitions with $P \cap T = \emptyset$, $F \subseteq (P \times T) \cup (T \times P)$ is the flow relation, and $M_0 : P \rightarrow \mathbb{N}$ is the initial marking. A transition $t \in T$ is enabled under marking M if every input place p of t satisfies $M(p) \geq 1$. Firing t yields a new marking M' according to the standard token-flow semantics.

Definition 39 (Thought state) A thought state is the cognitive analogue of a Petri Net place. Formally, a thought state is a pair $s = (p, \sigma)$ where $p \in P$ is a place of the Petri Net and σ is a structured summary of all reasoning-relevant information accumulated up to that point, derived from the event log and prior LLM outputs. The set of all thought states is denoted $\mathcal{S}_{\text{PNoT}}$.

Definition 40 (Thought transition) A thought transition is an annotated Petri Net transition $\tau = (t, \rho)$ where $t \in T$ and ρ is a directive specification describing the reasoning step the LLM must perform when t fires. The annotation ρ encodes: (i) the decision point, (ii) constraints from the process logic, and (iii) the goal of the transition, providing semantic grounding for the reasoning action. The set of all annotated transitions is $\mathcal{T}_{\text{PNoT}}$.

Definition 41 (PNoT marking) A PNoT marking is a marking $M : P \rightarrow \mathbb{N}$ together with a reasoning trace $\Gamma = \langle r_1, r_2, \dots, r_k \rangle$, where each r_i is an LLM output generated at a place or transition. PNoT evolves both the token distribution and the accumulated reasoning trace.

Definition 42 (Place-level prompt) Given a thought state $s = (p, \sigma)$, a place-level prompt is a function $\pi_{\text{place}} : \mathcal{S}_{\text{PNoT}} \rightarrow \mathcal{X}$ that maps s to a system prompt summarizing (i) the events leading to p , (ii) insights extracted from the event log, and (iii) the contextual rationale for subsequent decisions. This prompt anchors the LLM in the reasoning accumulated up to the current state.

Definition 43 (Transition-level prompt) Given a thought transition $\tau = (t, \rho)$, a transition-level prompt is a function $\pi_{\text{trans}} : \mathcal{T}_{\text{PNoT}} \rightarrow \mathcal{X}$ that maps τ to a directive prompt instructing the LLM to perform the reasoning operation encoded in ρ , thereby determining how the thought process proceeds from the current state to the next.

Definition 44 (Prompt composition operator) The prompt composition operator is a function

$$\text{ComposePrompt} : (\mathcal{S}_{\text{PNoT}}, \mathcal{T}_{\text{PNoT}}) \rightarrow \mathcal{X}$$

that, for an enabled transition t under current marking M , returns a prompt produced by composing the place-level summary π_{place} and the transition-level directive π_{trans} . Formally:

$$\text{ComposePrompt}(s, \tau) = \pi_{\text{place}}(s) \oplus \pi_{\text{trans}}(\tau)$$

where $\oplus$ is a concatenation with role-preserving formatting (e.g., system-then-instruction).

Definition 45 (Petri Net of Thoughts (PNoT)) *The Petri Net of Thoughts corresponding to a Petri Net $\mathcal{PN}$ and a prompting architecture Π is the tuple*

$$\text{PNoT} = (\mathcal{PN}, \mathcal{S}_{\text{PNoT}}, \mathcal{T}_{\text{PNoT}}, \text{ComposePrompt}, \Gamma),$$

whose controlled execution yields a structured reasoning trace that is transparent, reproducible, and constrained by the formal semantics of the underlying process model.

4.3 Virtually Enhanced Narratives

Virtually Enhanced Narratives (ViEnNa) constitute the central conceptual modeling representational layer of this thesis: a visually grounded, comic-like conceptual modeling approach designed to bridge symbolic process structures with multimodal evidences obtained from real or simulated environments. Building on the methodological foundation of object-centric event data representation [BKA⁺24], ViEnNa leverages structured event traces as the backbone for lifting raw perceptual cues—actions, spatial configurations, object interactions—into coherent process elements.

Within this section, we formalize the representational principles that enable ViEnNa to function as both a modeling language and an elicitation substrate. To this end, we introduce two complementary definitions. First, the notion of a *Process Comic*, which provides a formal account of comic-based process depictions as sequential, multimodal narrative frames aligned with process semantics. Second, the broader *Multimodal Conceptual Modeling (MMCM)* framework, which generalizes how conceptual models can be systematically connected to multimodal evidences through mappings, constraints, and embedding-space transformations. These definitions jointly establish the theoretical foundation for ViEnNa as a hybrid modeling paradigm that integrates symbolic, subsymbolic, and perceptual layers of intelligence.

Definition 46 (Process Comic) *A Process Comic is a visual process representation defined as a tuple*

$$\mathcal{C} = \langle F, O, A, L \rangle,$$

where F is a sequence of comic frames illustrating process steps, O is a set of depicted process-relevant objects or actors, A is a set of annotated actions or events, and L is a layout function that arranges F in a coherent temporal or logical order. Each frame in F may integrate multimodal elements (e.g., images, text, or symbols) to express contextual and behavioral information.

Definition 47 (Multimodal Conceptual Modeling (MMCM) Framework) *Let C be a set of process models, where each $c \in C$ is defined as a tuple $c = (E, R)$, with E denoting a set of concept elements and R a set of relationships between elements in E .*

Let M be a set of multimodal evidences, where each $m \in M$ is a tuple $m = (T, V)$, with T being the type of modality (e.g., image, video, text) and V the actual content. We define a cross-modal retrieval function $f : E \rightarrow \mathcal{P}(M)$, linking each concept element to a set of multimodal evidences. Human-to-model communication is represented by the function $h : M \rightarrow E$, mapping evidence to a corresponding concept element, while model-to-human communication is represented by the function $g : E \rightarrow \mathcal{P}(M)$, mapping concept elements to sets of multimodal evidences. The complete framework $\mathcal{F}$ for MMCM is thus

$$\mathcal{F} = (C, M, f, h, g),$$

enabling bidirectional multimodal communication that allows modelers to describe concepts using multimodal evidences, and models to respond by providing reference examples for concept elements.

4.3.1 Counterfactual Disposition and Elicitation

A central claim of this thesis is that expert knowledge relevant for conceptual modeling extends beyond observable conduct. Experts possess not only routinized behavioral patterns but also latent *counterfactual dispositions*: systematic tendencies to act or decide differently under perturbed or hypothetical conditions. These dispositions encode domain logic that rarely appears in documentation but nonetheless constrains correct or expected behavior. Mixed-reality elicitation and multimodal scenario perturbation should provide access to these latent structures. This section formalizes counterfactual dispositions, characterizes elicited counterfactual responses, and discusses validity and reliability considerations for integrating such knowledge into conceptual models.

4.3.2 Formalization of Dispositions

Counterfactual dispositions generalize the notion of behavior from the space of observed situations to the space of hypothetical situations. Let Σ denote the set of possible situational states relevant to a domain, and let A denote the set of admissible expert actions. A disposition can then be conceptualized as a mapping from situations to actions.

Definition 48 (Counterfactual Disposition) A counterfactual disposition is a mapping

$$d_{\text{cf}} : \Sigma \rightarrow A$$

from hypothetical or perturbed situational states Σ to admissible actions A , such that $d_{\text{cf}}(\sigma)$ represents the action an expert would take if σ were realized. Counterfactual dispositions encode latent domain rules and decision tendencies not directly inferable from observed conduct.

Observable behavior provides only a partial view of this mapping, limited to situations an expert has actually encountered.

Definition 49 (Dispositional Completion) *Let d_{cond} denote a conduct-based disposition defined on observed situations, and let d_{cf} denote a counterfactual disposition defined on hypothetical situations. The dispositional completion of an expert is the total mapping*

$$d_{\text{total}} = \text{complete}(d_{\text{cond}}, d_{\text{cf}}),$$

which integrates observed behavior with elicited counterfactual tendencies.

Mixed reality allows construction of controlled situational perturbations that reveal how experts would adapt their actions, thereby directly informing d_{cf} .

Definition 50 (Multimodal Counterfactual Scenario) *A multimodal counterfactual scenario is a structured situational variant*

$$\sigma' = (\mathbf{v}, \mathbf{a}, \mathbf{s}, \mathbf{c})$$

composed of video features $\mathbf{v}$, audio cues $\mathbf{a}$, system-state variables $\mathbf{s}$, and contextual metadata $\mathbf{c}$, generated by perturbing a real-world scenario while preserving ecological coherence.

These scenarios are central stimuli for elicitation within this thesis.

4.3.3 Elicited Counterfactual Responses

Because counterfactual dispositions do not appear directly in empirical traces, they must be elicited. Traditional methods rely on interview-based what-if questioning, whereas this thesis employs mixed reality to present experts with ecologically valid counterfactual variants of observed situations.

Let $\mathcal{E}$ denote the elicitation function that maps an expert and a hypothetical scenario to a response.

Definition 51 (Counterfactual Elicitation) *Counterfactual elicitation is a process*

$$\mathcal{E} : E \times \Sigma \rightarrow R$$

that maps an expert $e \in E$ and a hypothetical situational state $\sigma \in \Sigma$ to an elicited response $r \in R$. The response provides explicit evidence about the expert's counterfactual disposition.

Elicited responses can take multiple modalities—linguistic, diagrammatic, gestural, or embodied interaction via mixed reality.

Definition 52 (Elicited Counterfactual Response) *An elicited counterfactual response is an explicit expert judgment*

$$r = \mathcal{E}(e, \sigma')$$

that encodes the expert's intended action or rationale under the hypothetical scenario σ' . A response r is valid evidence for the disposition if and only if it satisfies the entailment condition

$$r \models d_{\text{cf}}(\sigma').$$

Mixed-reality-based elicitation enhances this process in two ways:

1. It enables egocentric immersion, narrowing the gap between real and hypothetical scenarios.
2. It supports systematic perturbation, revealing how decision rules change under controlled variations.

These elicited responses thus form a bridge between tacit cognitive structures and their integration into formal conceptual models.

4.3.4 Validity and Reliability Considerations

Elicited counterfactual knowledge requires methodological safeguards to ensure its epistemic soundness.

Construct validity. To ensure elicitation captures genuine dispositions rather than artifacts, counterfactual scenarios derive directly from multimodal recordings of real-world interactions. Perturbations are limited to targeted variables to preserve ecological realism.

Definition 53 (Construct Validity of Counterfactual Elicitation) *Construct validity refers to the extent to which an elicited response r reflects the underlying counterfactual disposition $d_{\text{cf}}(\sigma)$ rather than elicitation-induced bias. Elicitation is construct-valid when*

$$r \approx d_{\text{cf}}(\sigma)$$

under controlled perturbations and cross-modal consistency checks.

Internal reliability. Repeated elicitation of equivalent or closely related scenarios should yield stable responses.

Definition 54 (Reliability of Elicited Knowledge) *The reliability of elicited counterfactual knowledge is the degree to which elicited responses remain stable across repeated*

presentations of equivalent scenarios. Formally, responses $r_1, \dots, r_n$ for the same σ are reliable if

$$\text{Var}(r_1, \dots, r_n) < \theta$$

for a method-dependent threshold θ .

4.4 Conclusion and Implications for Method Operationalization

This chapter has established a theoretical foundation for enhancing conceptual modeling through the systematic integration of multimodal evidence and counterfactual elicitation. The framework repositions conceptual modeling from a predominantly declarative abstraction activity to an epistemically enriched transformation process that incorporates observable carriers of tacit knowledge and elicited counterfactual dispositions.

The central operational implication is that conceptual modeling can no longer be treated as a function of declarative elicitation alone (cf. Def. 5). Instead, the modeling process **CM** (Def. 14) must be instantiated as a pipeline that systematically incorporates observable carriers Ξ_{obs} (Def. 13) derived from two complementary sources: (i) multimodal traces of expert conduct (Def. 8), and (ii) elicited counterfactual responses (Def. 11). Operationally, this entails:

1. A **data acquisition layer** that captures synchronized multimodal records and constructs enriched event logs suitable for multimodal and relaxed process mining (Defs. 29–35);
2. An **analysis layer** that extracts latent conceptual regularities $\mathcal{R}_{\text{lat}}$ (Def. 4) through embedding-based alignment, semantic shift analysis, and panmodal reasoning;
3. A **elicitation layer** in which Mixed Reality scenarios (Def. 50) operationalize controlled perturbations to reveal counterfactual dispositions (Def. 48);
4. A **model integration layer** that materializes these regularities as structured commitments within multimodal-enriched conceptual models $(\alpha, M, \text{Att}) \in \mathcal{M}^+(L)$ (Def. 21);
5. And a **reasoning and validation layer** in which process-guided inference (Def. 45) and conformance analysis (Def. 30) ensure semantic consistency and goal alignment.

From a methodological standpoint, the framework implies that enhancement (Def. 19) must be operationalized as an iterative refinement cycle in which empirical evidence and symbolic commitments mutually constrain each other. Multimodal enrichment strengthens semantic and pragmatic grounding without altering the formal semantics of the modeling language (Def. 15), while relaxed and panmodal mining techniques allow variability and multimodal heterogeneity to be systematically reconciled within embedding

spaces. Counterfactual elicitation complements this by extending the evidential base beyond historically observed states to hypothetical yet domain-relevant situations.

Crucially, the operationalization of this framework remains anchored in the normative criteria of precision (Def. 17) and conciseness (Def. 18). Each methodological component—data integration, alignment, elicitation, enrichment, and reasoning—serves the overarching objective of producing models that (i) eliminate goal-relevant semantic ambiguity and (ii) render their commitments cognitively transparent to competent readers. In this sense, the theoretical constructs introduced in this chapter collectively define not merely a set of definitions, but a structured method blueprint: a formally grounded, empirically informed, and technologically instantiated approach to conceptual modeling enhancement.

The subsequent chapter translates this framework into a concrete realization of a method, detailing the artifact architecture, tool support, and evaluation design through which the theoretical framework is enacted in practice.

Enhancing Conceptual Modeling

“The greatest value of a picture is when it forces us to notice what we never expected to see.”

– Dr. John Wilder Tukey

Conceptual models aim to represent the essential structure and behavior of a domain [PG21]. Yet, traditional modeling practices rely predominantly on interviews, documentation, and retrospective explanation—sources that capture what experts can articulate, but often miss what they enact in practice [vdAWM04, GBP24c, LNR⁺25]. As stated in Section 1.1.4, we currently lack systematic means to enhance the precision and conciseness of conceptual models by incorporating tacit knowledge that experts rely on but rarely verbalize.

This chapter introduces a structured method for enhancing the conceptual modeling designed to address that gap, and builds on the theoretical framework introduced in Chapter 4. The method consolidates and scopes prior author’s research contributions, for which extended discussions are provided in Chapter 6 (multimodal data analysis) and Chapter 7 (Mixed Reality simulation). The realized method is subsequently implemented according to the requirements defined in Section 3.3, with implementation details reported in Chapter 8, and empirically applied and evaluated as reported in Chapter 9.

The central assumption guiding the proposed method is that domain knowledge manifests along two complementary dimensions: what experts *do* in situated environments and what they *would do* under altered or counterfactual conditions. The first dimension becomes observable through multimodal traces of human–system interaction, such as video, audio, spatial movement, and event logs. The second dimension surfaces when experts are placed into modified situations that challenge habitual patterns and reveal implicit decision rules. Capturing both dimensions requires an integration of empirical observation and controlled elicitation.

To operationalize this integration, the method combines multimodal data analysis with mixed reality simulation in a closed-loop architecture centered around the System Under Investigation (SUI). Multimodal observations provide factual grounding by exposing invariant behavioral regularities embedded in situated action. Mixed reality elicitation extends this grounding by enabling systematic perturbation of environmental and procedural conditions, thereby surfacing counterfactual reasoning that remains inaccessible through passive observation alone.

The method does not treat conceptual modeling as a purely representational task but as an iterative process of alignment between perception, interpretation, and formal abstraction. Through semantic junctions, embedding-based crossmodal alignment, adversarial situation formation, and immersive navigation operators, the proposed framework links sensory-level evidence with higher-level conceptual constructs. In doing so, it transforms conceptual modeling into an evidence-informed and dialogical process in which human expertise and artificial reasoning co-evolve.

This chapter details the architecture, components, and operational mechanisms of the proposed method. It formalizes how multimodal evidence is structured, how semantic alignment is achieved, how counterfactual situations are generated and evaluated, and how iterative human–AI co-modeling refines conceptual artifacts. Together, these elements establish the technical and conceptual backbone for enhancing precision and conciseness in conceptual modeling.

This chapter is structured as follows. Section 5.1 introduces the overall methodological architecture and positions the SUI as the empirical anchor of the framework. Section 5.2 details the enhancement cycle for conceptual modeling, formalizing the iterative human–AI co-modeling loop. Section 5.3 presents the Crossmodal Alignment Agent and its junction-based integration of multimodal data, analytics, and model constructs. The subsequent sections elaborate the core representational components of the SUI interpretation model, including process models (MDLS), evidence data (EVDT), analytics (ANLT), and semantic annotations (ANNT), as well as immersive elicitation mechanisms (IMMR). Section 5.4 introduces Mixed Reality Simulation and the Adversarial Situation Formation (ASF) mechanism, detailing the interplay between the Situation Generator (GNRT) and Situation Discriminator (DSCR). Together, these components form a proposed method that systematically integrates factual and counterfactual knowledge into conceptual models. Section 5.5 presents the validation strategy of the proposed method, outlining how its effectiveness in enhancing conceptual modeling is empirically assessed. Section 5.6 concludes the chapter by synthesizing the core components of the method.

5.1 Method Overview

An overview of the proposed methodical framework is supported by the Figure 5.1. At the core of the framework lies the *System Under Investigation* (SUI), which the process participant—an individual possessing tacit domain expertise—interacts with. The SUI

represents the operational environment in which conceptual structures are enacted rather than merely described.

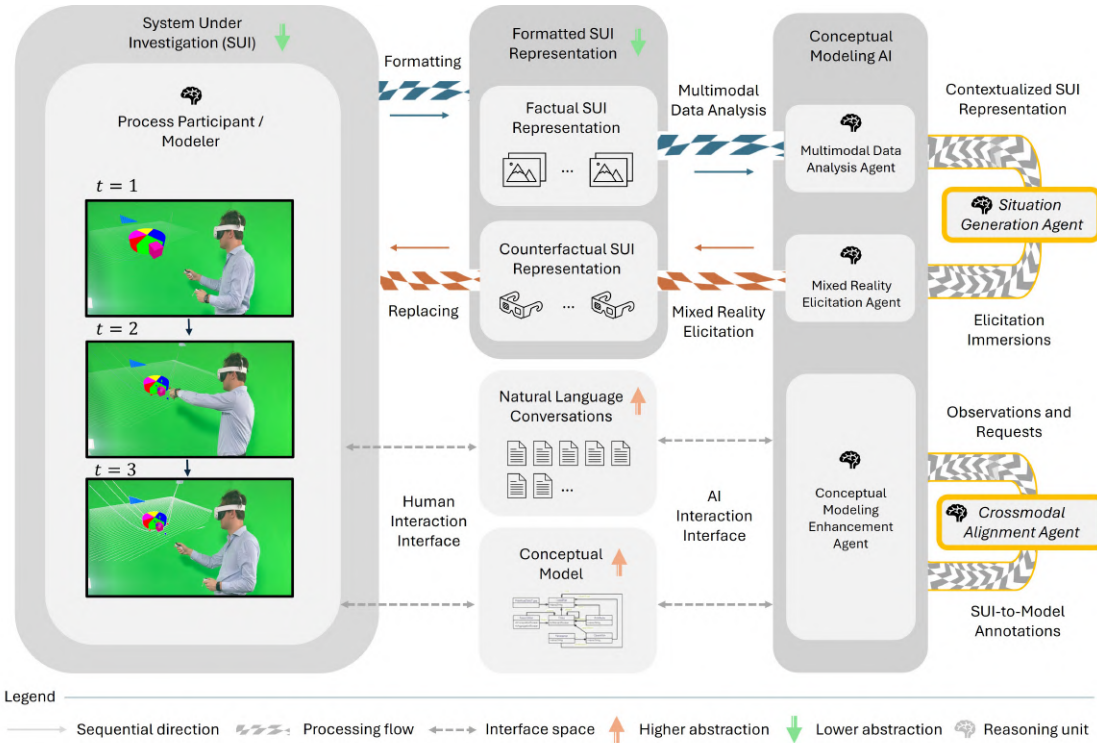


Figure 5.1: Overview of the proposed methodology integrating multimodal data analysis and mixed reality elicitation to enhance conceptual modeling.

The framework begins with the collection of multimodal observations during real interactions with the SUI. These data streams—such as video, audio, depth, and motion—are synchronized and preprocessed through temporal and spatial alignment, noise filtering, and anonymization. This ensures that multimodal recordings are both analytically consistent and ethically compliant.

After preprocessing, the data are transformed into a *formatted representation* of the SUI, suitable for computational analysis. This representation may originate from: (1) **Factual interactions**, recorded from real-world multimodal observations; or (2) **Synthetic interactions**, gathered when the modeler explores counterfactual variants of the same SUI through a mixed reality headset. These formatted traces constitute the empirical basis for detecting the tacit regularities underlying domain conduct.

The next stage involves the *Multimodal Data Analysis Agent*, which performs the core analytical operations. It fuses multimodal signals to derive contextualized, process-oriented representations of the SUI through discovering structured patterns of action from multimodal event data; linking observed behaviors to model constructs; and mapping how multimodal events and conceptual elements co-occur in context.

The outcome is a contextualized process representation of the SUI—rich in both behavioral and semantic dimensions—that captures not only what experts do, but how those actions relate to conceptual elements of their domain.

Beyond factual observation, the methodology extends to eliciting counterfactual cognition through the *Mixed Reality Elicitation Agent*. Here, a *Situation Generation Agent* constructs immersive, altered representations of the SUI—effectively substituting parts of factual reality with synthetic, disposed variants.

Process participants are then re-immersed into these generated situations via mixed reality headsets, where they can respond to modified environmental or procedural conditions. This interactive perturbation surfaces decision logics that would otherwise remain implicit, revealing the counterfactual layer of expert reasoning.

Once both factual and counterfactual evidences have been gathered, they become inputs to the *Conceptual Modeling Enhancement Agent*. Through natural language dialogues, the process participant and the modeler collaboratively refine the conceptual model, by directly manipulating model elements and structures.

During this interaction, the agent mediates between linguistic expressions, observed behaviors, and model semantics, ensuring alignment between what is described, what is done, and what is conceptualized. The agent delegates crossmodal reasoning to an agent who integrates multimodal observations, textual expressions, and model entities into a unified embedding space. Within this joint semantic space: (1) **Lower-level abstractions** correspond to the SUI and the observed reality; (2) **Intermediate abstractions** correspond to multimodal evidence and process structures; and (3) **Higher-level abstractions** correspond to natural language dialogue and conceptual models.

This layered representation supports bi-directional reasoning—linking grounded sensory data to abstract conceptual models—thus closing the gap between practical enactment and formal conceptualization.

The methodology integrates five reasoning units, each functioning as an autonomous yet interdependent module in the overall architecture:

1. **Conceptual Modeling Enhancement Agent** — Facilitates the integration of multimodal evidence and linguistic feedback into the conceptual model (Section 5.2).
2. **Crossmodal Alignment Agent** — Aligns SUI observations, natural language, and model constructs within a shared multimodal embedding space (Section 5.3).
3. **Situation Generation Agent** — Produces synthetic, counterfactual scenarios for elicitation and immersive testing (Section 5.4).
4. **Multimodal Data Analysis Agent** — Executes multimodal process mining, evidencing, and cartography to derive structured behavioral representations (Chapter 6).

5. Mixed Reality Elicitation Agent — Manages immersive elicitation processes for capturing counterfactual cognition (Chapter 7).

Together, these reasoning units establish a closed loop between observation, elicitation, interpretation, and model enhancement. Each unit contributes a distinct layer of reasoning—ranging from perception and simulation to abstraction and alignment—thereby operationalizing the thesis vision of embedding tacit and counterfactual knowledge within conceptual models.

5.2 Enhancing Conceptual Modeling: The Method

To construct a coherent storyline for mixed reality elicitation, the process begins with translating the SUI into a structured process representation. This transformation is essential because mixed reality elicitation does not occur in a narrative vacuum: it relies on a structured understanding of how the SUI operates—its activities, dependencies, and observable dynamics. By first formatting the SUI into a process-oriented form, we establish a concrete foundation upon which the mixed reality storyline can unfold and evolve.

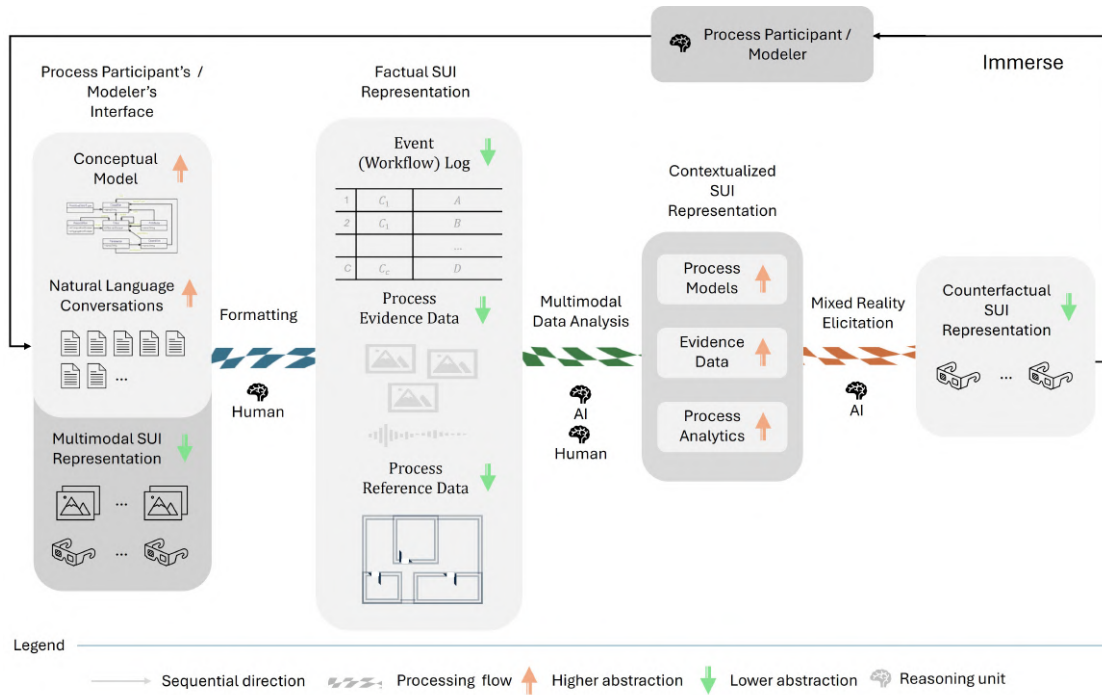


Figure 5.2: Progressive transformation from multimodal SUI observations toward contextualized and mixed-reality-enhanced representations.

As illustrated in Figure 5.2, conceptual models, natural language conversations, and multimodal SUI representations exist at different levels of abstraction. Conceptual models

and natural language interactions operate at higher abstraction levels—they express intentions, structures, and semantics of the domain. In contrast, multimodal SUI data occupy the lower abstraction layer, providing direct, sensory-level evidence of how those conceptual structures are enacted in reality.

These heterogeneous representations are formatted by a human-in-the-loop process into what is termed the *Factual SUI Representation*. Within this representation, the domain is expressed through three complementary sources: *event (workflow) logs*, which encode temporal sequences of activity; *process evidence*, which links observed behavior to conceptual constructs; and *process references*, which connect specific data segments to the contextual meaning they embody. Together, these constitute the lower abstraction level where situated knowledge is most directly observable.

The factual SUI representation is then jointly processed by both human and artificial intelligence components to produce a *Contextualized SUI Representation*. This stage involves the triangulation of behavioral, linguistic, and conceptual perspectives through a combination of *process mining*, *process evidencing*, and *process cartography*. The details of these operations and their semantic alignment mechanisms are discussed in Section 5.3. The outcome of this step is a structured and explainable synthesis that connects multimodal experience with domain semantics.

Once the contextualized SUI representation has been established, the next methodological step involves the generation of a mixed reality disposition of the SUI (as described in Chapter 7). In this stage, a modeler or process participant is immersed in an augmented version of the same environment—one that introduces deliberate perturbations, counterfactual variants, or hypothetical constraints. These elicited situations serve not only to test the stability of conceptual assumptions but also to expose latent decision patterns that would not emerge under passive observation.

The process thus forms an iterative loop: multimodal observations of the factual SUI are formatted and analyzed, re-synthesized into contextualized representations, re-enacted through mixed reality elicitation, and fed back into the conceptual modeling cycle. Each iteration enriches the model with new evidence and refines its explanatory precision, ultimately aligning conceptual abstractions with the lived, embodied realities they are meant to represent.

Iterative Human–AI Co-Modeling Loop

At the core of the proposed methodology lies an iterative *human–AI co-modeling loop* that continuously refines the conceptual model through alternating cycles of human interpretation and machine reasoning. Each iteration builds upon previous representations of the SUI, integrating multimodal observations, natural language interactions, and formal model structures into a unified reasoning process.

As shown in Figure 5.3, the co-modeling loop is structured around two interacting perspectives: the *modeler’s interface*, which includes the conceptual model, natural

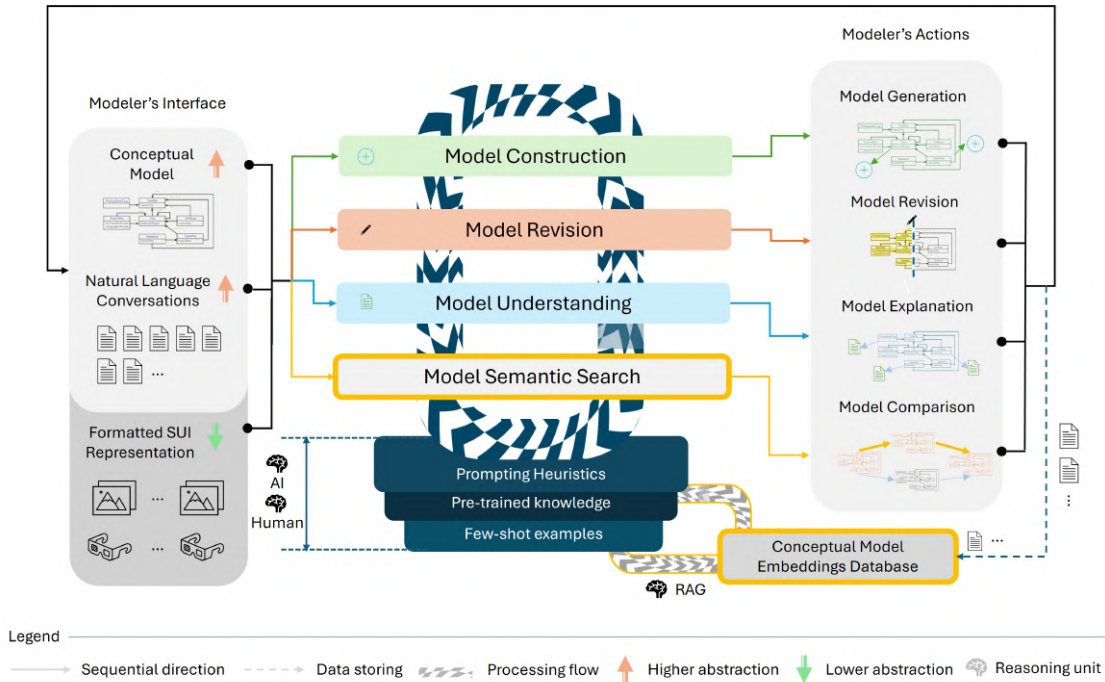


Figure 5.3: Iterative human–AI co-modeling loop connecting multimodal data, natural language, and conceptual models through reasoning agents.

language conversations, and formatted SUI representations; and the *modeler's actions*, which involve model generation, revision, explanation, and comparison. Between these two layers operates a set of AI reasoning components that facilitate understanding, construction, and alignment of models.

At the upper level of abstraction, the **Conceptual Model** and **Natural Language Conversations** represent the semantic and linguistic dimensions of the modeling process. The conceptual model encapsulates formalized structures—entities, relationships, and behavioral rules—while the natural language layer captures human discourse, intentions, and clarifications that guide modeling activities. Together, they form the communicative substrate through which model meaning evolves.

At the lower abstraction level lies the **Formatted SUI Representation**, which encodes multimodal evidence (e.g., event logs, visual and spatial data) that links real-world phenomena to conceptual elements. This layer provides the factual grounding that ensures the model remains aligned with observed and elicited realities.

Bridging these layers are four principal reasoning operations:

Model Construction supports the creation of new conceptual elements based on either natural language input or multimodal evidence. It translates human utterances or detected patterns into formal model constructs.

Model Revision allows both human and AI to modify existing models through iterative refinement, ensuring that the evolving conceptualization remains coherent with observed or counterfactual behavior.

Model Understanding serves as the interpretive mechanism, enabling the AI to parse model structures, detect inconsistencies, and contextualize them in relation to the SUI and prior modeling sessions.

Model Semantic Search provides a retrieval capability across model repositories and prior interactions, leveraging embeddings to find semantically related concepts or model fragments.

These reasoning operations are informed by deeper AI layers that supply cognitive context: *prompting heuristics* (rules guiding reasoning strategies), *pre-trained knowledge* (conceptual priors from large language models), and *few-shot examples* (domain-specific adaptation patterns). Together, they form the adaptive intelligence substrate that supports interactive modeling sessions.

The resulting reasoning outcomes are stored in a **Conceptual Model Embeddings Database**, a multimodal memory that encodes conceptual models, linguistic utterances, and multimodal evidence in a shared embedding space. This database enables semantic search, analogy reasoning, and retrieval-augmented generation (RAG) during later modeling cycles, ensuring that knowledge from past sessions can inform future refinements.

On the right-hand side of Figure 5.3, the *Modeler's Actions* depict the explicit interventions through which human participants interact with the AI.

Together, these human and AI contributions form a continuous reasoning cycle: multimodal observations inform the SUI representation; linguistic dialogue guides interpretation; AI-driven reasoning units translate insights into model updates; and the resulting models feed back into subsequent elicitation and analysis rounds. This loop transforms conceptual modeling from a static design task into a living, dialogical process of evidence-based refinement.

In this configuration, each iteration enhances not only the model's descriptive accuracy but also its epistemic depth—gradually revealing the tacit, embodied, and counterfactual dimensions of expert knowledge that traditional modeling approaches often overlook.

5.3 Multimodal Data Analysis

The *Crossmodal Alignment Agent* constitutes the semantic and analytical core of the proposed methodology. It is responsible for connecting multimodal observations, natural language dialogues, and conceptual models into a unified interpretive space. Through this agent, empirical traces of behavior, speech, and perception are translated into semantically grounded model elements that reflect both factual and counterfactual aspects of the SUI.

The agent thereby ensures that conceptual modeling evolves as an evidence-informed and contextually grounded activity.

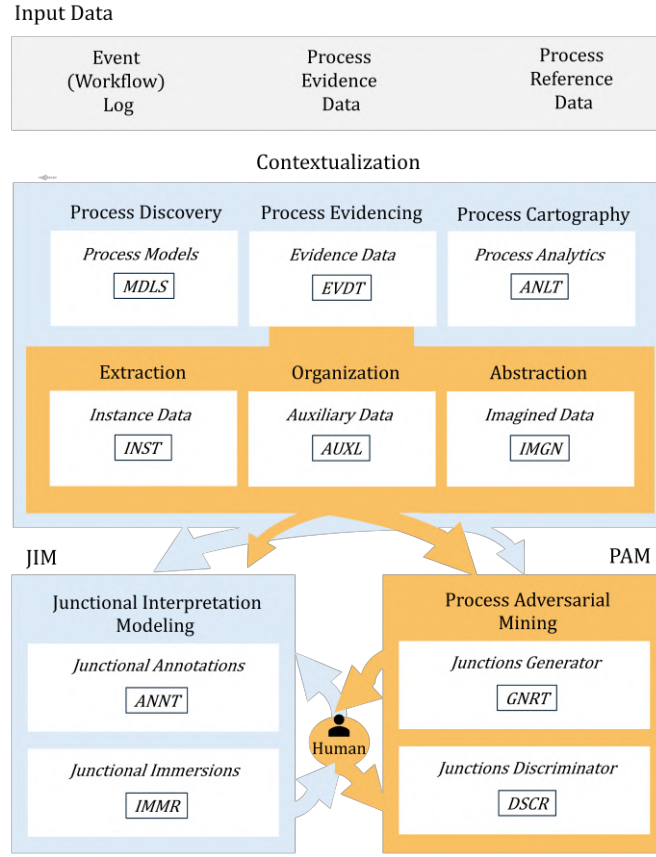


Figure 5.4: Overview of the Crossmodal Alignment Agent connecting multimodal process data, evidencing, and conceptual model elicitation across factual and counterfactual representations.

The Crossmodal Alignment Agent is an AI-enabled modeling mechanism, designed to create junctions¹ between process models, evidence data, and analytic visualizations. Within this framework, junctions serve as the foundational anchors in the multimodal embedding space that connects three modalities: (1) multimodal observations of the SUI, (2) linguistic descriptions and modeling dialogues, and (3) conceptual representations within the model. Through these junctions, the agent enables bidirectional reasoning—translating concrete, sensor-level data into symbolic structures and contextualizing abstract model elements back into perceptual experience.

In complement, the Situation Generation Agent operates as an adversarial system²

¹Derived from the Latin *jungere*, meaning “to join”. Refers to a semantic binding between perception and abstraction.

²Derived from the Latin *adversus* (“turned against”).

composed of a generator and a discriminator. The generator synthesizes counterfactual situations and model variants from multimodal evidence, while the discriminator evaluates their plausibility against human-authored or empirically grounded junctions. This iterative adversarial process produces a gradual convergence between factual and generated representations: initially, the discriminator is more accurate at detecting artificial constructs, but as training progresses, the generator improves, eventually producing scenarios indistinguishable from those informed by real human elicitation.

As depicted in Figure 5.4, the Crossmodal Alignment Agent mediates between the input space (event and workflow logs, process evidence, and reference data) and the output space (factual and counterfactual SUI representations and conceptual modeling insights).

5.3.1 Junctional Integration of Multimodal Data

For making interpretations of SUI, we focus on the integration of process models, evidence data, analytics, annotations, and immersions. Our solution is designed as a knowledge-assisted method, which uses domain expertise to refine the interconnection of multiple visual regions of interest in event logs and additional process-relevant data.

The solution, as illustrated in Figure 5.4, encompasses several stages, from input data acquisition to the generation of insightful output data.

Input Data

The solution’s flow begins with the collection of various types of input data:

- **Event (Workflow) Log:** This includes detailed logs of events or activities captured during a process. These logs are foundations for discovering process models.
- **Process Evidence Data:** This includes data that provides evidence of the execution of processes, such as images, sensor data, and video and audio recordings.
- **Process Reference Data:** This encompasses data that serves as a reference for understanding the context and environment in which the processes occur, such as blueprints, maps, and diagrams.

Contextualization

The contextualization phase involves three main activities: *Process Discovery*, *Process Evidencing*, and *Process Cartography*.

Process Discovery. In this step, process models (MDLS) are discovered from the event logs. These models visually represent the sequence and flow of activities in the process, highlighting key steps, decision points, and parallel paths.

Process Evidencing. Here, evidence data (EVDT) is integrated to substantiate the discovered process models. This data aims at providing a factual basis for the models.

Extraction, Organization, and Abstraction are phases of evidencing for structuring and abstracting the data with an objective of interpretation and analysis.

Process Cartography. Process cartography (ANLT) involves creating analytical visualizations of the processes. This step uses analytics to provide aggregated insights into the process dynamics, performance metrics, etc. while representing it using cartography.

Extraction. Instance data (INST) is extracted from the input data, capturing specific instances or occurrences of the process activities, resources, and objects. This step ensures that relevant execution data points that are collected in multimodal EVDT data are ready for analysis with AI.

Organization. The process reference data from the input is organized into auxiliary data (AUXL), which includes additional context and metadata for process analysis. This organized data provides a structured foundation for subsequent abstraction and modeling. Examples of AUXL data are 3D models or maps that can be marked at some of their spatial points and referred to later.

Abstraction. In the abstraction step, imagined data (IMGN) is generated to simulate potential scenarios and outcomes through content generation that is made possible by generative AI capabilities. This step uses image generation techniques to create hypothetical EVDT data points that help in understanding the process behavior under different conditions.

Integration of the Components

The method involves two main components:

SUI-to-Model Annotations. SUI-to-Model Annotations (ANNT) are the essence of the proposed semantic junctions for providing contextual information. With annotations, we link marked concepts between a broad set of available markable candidates among concepts in MDLS, EVDT, and ANLT.

Elicitation Immersions. Immersions (IMMR) involve creating immersive visual experiences that allow users to interact with the process interpretation models. This component aims at enhancing the user's understanding by providing a way to explore the process data with engagement and intuition.

The Situation Generation Agent contributes the generation and interpretation of process interpretation models through two main components:

Situation Generator. The generator (GNRT) produces crossmodal junctions from event logs by performing content generation, retrieval, and referencing. This component is trained to automatically create detailed and accurate process interpretation model for the given input data.

Situation Discriminator. The discriminator (DSCR) is trained to differentiate between authentic junctions and those altered with noise and human-made versus machine-made ones. This component is used as a teaching companion for the generator component.

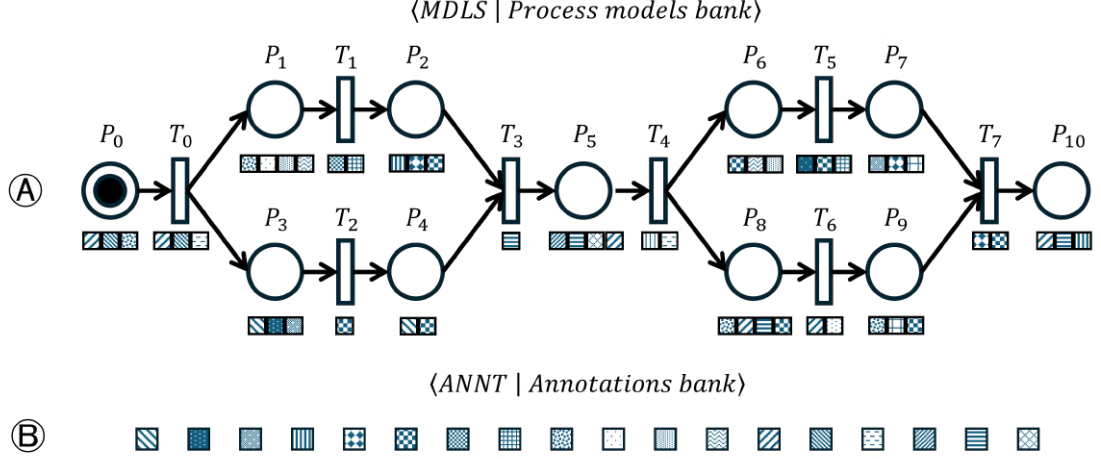


Figure 5.5: Process model (MDLS) elements, such as places and transitions, associated with assigned SUI-to-Model annotation(s) (ANNT).

Human Involvement. Human experts (HUMN) play an important role in our solution by providing feedback and annotations during the training of the discriminator in the Situation Generation Agent and contributing to the refinement of the interpretation models by the Crossmodal Alignment Agent. This human-in-the-loop approach aims to ensure that the models are not only technically accurate but also aligned with expert knowledge and practical insights.

Next, we will cover steps of contextualization for creating SUI interpretation model, particularly - process models (Section 5.3.2), evidence data (Section 5.3.3), analytic data (Section 5.3.4), annotations (Section 5.3.5), and immersions (Section 5.3.6).

5.3.2 Process Models (MDLS)

The base elements of the SUI interpretation model are the process models, denoted as MDLS. In our framework, MDLS are implemented as Petri Nets, although the methodology is designed to be extensible to other process modeling languages such as BPMN.

Process Models Bank

Figure 5.5 illustrates the management of process models and their corresponding annotations in a process interpretation model. Section A (upper) of Figure 5.5, labeled <MDLS>, shows an example process model from a *process models bank* that aggregates collected/discovered process models. These models consist of places (P), transitions (T), and annotations that are used for marking the various stages and pathways within a process. Each model begins with an initial place (P_0) and proceeds through a series of transitions and places, ultimately reaching a terminal place (i.e. P_{10} in the example model).

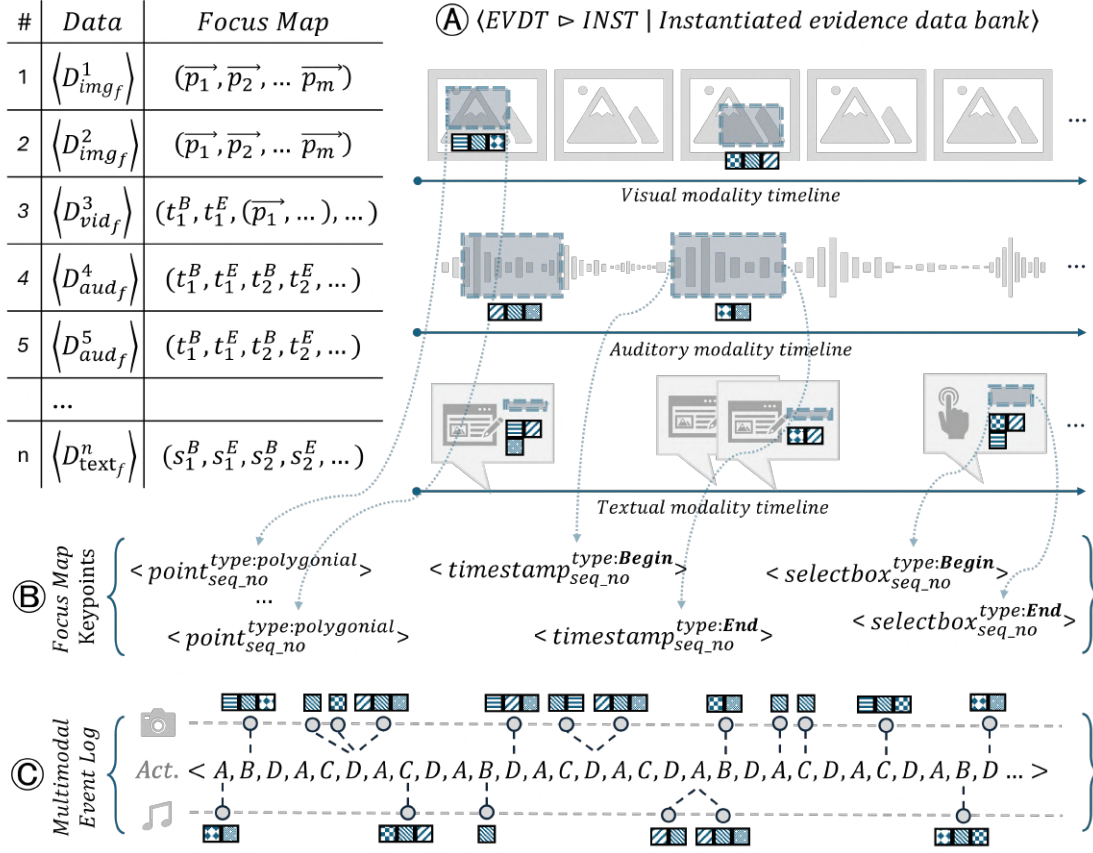


Figure 5.6: Instance data (INST) derived from the execution instances of processes.

We enhance the traditional Petri Net model by allowing each element (places and transitions) to be associated with an SUI-to-Model annotation (ANNT). The ANNT-assigned transitions (marked T) represent the activities or steps within the process linked with a certain ANNT concept, while the ANNT-assigned places (marked P) denote the states or conditions between these activities linked with a certain ANNT concept. Annotations serve as semantic links that connect MDLS elements to other semantic nodes within the framework. These annotations can be different to provide additional context, explanations, and insights into the process stages. Section B (lower) of Figure 5.5 organizes the various annotations (patterns) systematically. The concept of annotations is further described in Section 5.3.5.

5.3.3 Evidence Data (EVDT)

Evidence data comprises various forms of multimedia such as images, audio, and 3D models. These data types serve as empirical or supportive materials that enhance the understanding, analysis, and visualization of processes. In our method, EVDT can originate from three primary sources: *instance* data, *auxiliary* data, and *imagined* data.

Instance Data (INST)

Instance data (INST) are derived from the execution instances of processes. These data represent the real-time and historical records of process executions, capturing specific occurrences and their outcomes. Examples of instance data include:

- **Images or video** captured during the execution of a process, providing visual documentation of the process states.
- **Audio recordings** of events or interactions within the process, offering auditory evidence of the process flow.
- **Sensor data** attributed with a temporal and spatial value, generated from the process environment, illustrating the physical layout within the process.

Section A of Figure 5.6 illustrates the structure and organization of the *Instantiated Evidence Data Bank* ($\langle EVDT \triangleright INST \rangle$). This data bank captures and organizes multimodal data to support process analysis and interpretation. The upper-left part of Figure 5.6 lists different data types and their corresponding *focus maps*. Each data type (D) is annotated with a subscript indicating its modality, such as images (img), videos (vid), audio (aud), and text ($text$). The focus map captures keypoints and segments relevant to the data, structured to isolate regions of interest. The *visual modality timeline* displays sequences of visual data frames (D_{img}^1, D_{img}^2 , etc.). Each frame is linked to specific points or regions of interest defined by a focus map. This timeline helps in visualizing the flow and transitions within visual data, important for understanding processes involving visual inspection or monitoring. *Points* in the focus map (point type polygonal) are important for identifying specific *regions of interest* within the data. These points are mapped to corresponding data segments in the timelines, ensuring that significant features are highlighted and available for detailed referral in the later analysis. The *auditory modality timeline* represents sequences of audio data (D_{aud}^4, D_{aud}^5 , etc.). Key timestamps (t_1^B, t_1^E) mark the beginning and end of significant audio events. This timeline aids in analyzing auditory signals, which can be used in processes where sound plays a critical role, such as monitoring machinery or detecting anomalies through audio cues. *Timestamps* (timestamp type begin>, timestamp type end) mark the start and end of relevant events in the data. They provide temporal context, helping to align and synchronize data across different modalities. This temporal mapping is important for understanding the sequence and timing of events within the process. The *textual modality timeline* captures sequences of text data (D_{text}^n). This timeline is designed for inspecting processes involving documentation, communication, or textual/numerical analysis. Labels of conventional event log data, including sensor data, fit here.

The *multimodal event log* (section C of Figure 5.6) aggregates activities from different data modalities, creating a unified log of events. Each event is annotated with symbols representing the data source, such as cameras or audio recordings. This timeline is

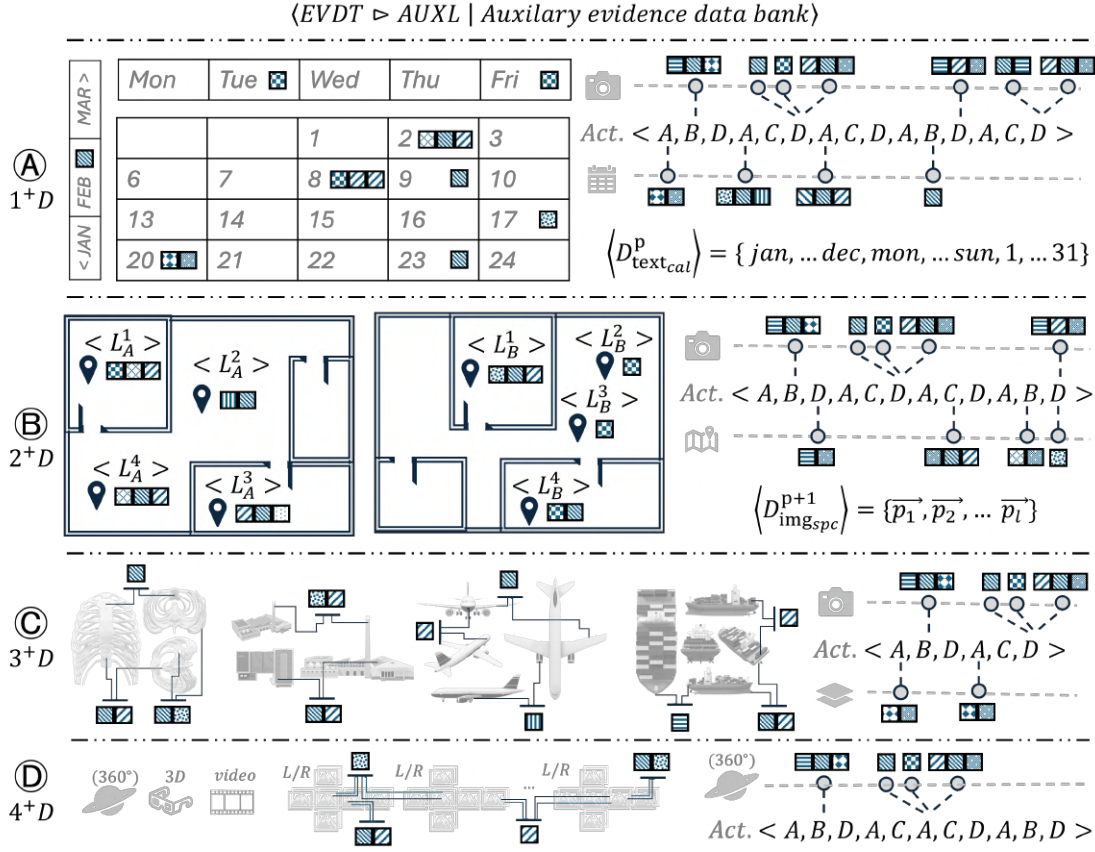


Figure 5.7: Auxiliary data (AUXL) representing organizational data that provide additional context and support for the process models. A-D: at least one dimensional data to at least four dimensional data, respectively.

important for providing a view of the process, integrating multiple data streams into a cohesive narrative. By synchronizing visual, auditory, and textual data, analysts can uncover patterns and correlations that may not be apparent when examining each modality in isolation. The instantiated evidence data bank supports evidence-based analysis by providing a structured repository of multimodal data. Analysts can refer to this data bank to validate their interpretations and ensure that their conclusions are grounded in factual evidence. This approach aims at enhancing the reliability and credibility of process insights. The visual representation of data in timelines and focus maps facilitates interaction and exploration. Analysts can navigate through the data, zooming in on specific segments or examining the temporal relationships between events, enabling analysts to track the evolution of events and their impact on the process over time.

Auxiliary Data (AUXL)

Auxiliary data (AUXL) consists of organizational data that provide additional context and support for the process interpretation models. These data are not directly generated from process executions but are important for an understanding of the processes. Examples of auxiliary data include location maps that illustrate the geographical distribution of process activities and resources, and 3D models of organizational facilities or equipment, providing spatial context and aiding in the planning and optimization of processes. By incorporating auxiliary data in our framework, we aim to bridge the gap between process models and their real-world environments, enabling more accurate and context-aware analysis. Figure 5.7 illustrates the structure and organization of the *Auxiliary Evidence Data Bank* ($\langle EVD \triangleright AUXL \rangle$). By integrating multi-dimensional data such as calendar, spatial, domain-specific, and advanced visualization data, the auxiliary evidence data bank supports the identification of trends, optimization of workflows, and improvement of overall process efficiency. In section, A of Figure 5.7, the first data example (1^+D) is at least one-dimensional data (like calendar categorical data), but represented by a calendar grid (introducing more dimensions) covering weeks and months. The textual calendar data ($D_{textcal}^p$) is represented by labels such as months (*jan, feb, ..., dec*), day names (*mon, tue, ..., sun*) and numerical day values (1, ..., 31). These labels are linked to specific dates and events in the calendar, providing a textual reference that complements the visual representation. This calendar includes annotations ANNT with different patterns that correspond to marks of specific events or activities. The activities (*Act.*) are mapped onto the *calendar data timeline*, indicating the sequence of actions (i.e. *A, B, D, A, C, D, A, C, D, ...*) linked with concepts on specific days. This mapping provides a visual representation of activity patterns over time, aiding in the identification of trends, busy periods, and potential bottlenecks. In Section B of Figure 5.7, the second data example (2^+D) includes spatial layouts ($\langle L_A^1, \langle L_A^2, \langle L_A^3, \langle L_A^4 \rangle \rangle \rangle$). These layouts depict different floor plans with marked locations using ANNT markers. Each layout is annotated with patterns that indicate significant areas or points of interest within the spaces. The activities (*Act.*) are mapped onto the spatial layouts, showing the sequence of actions performed at different locations within the layouts. This mapping can help in understanding the spatial context of activities, identifying frequently used areas, and analyzing movement patterns within the space. The image space data (D_{imgspc}^{p+1}) comprises sets of points ($\{p_1, p_2, \dots, p_l\}$) that represent specific locations within the images. These points are linked to the spatial layouts and provide a visual reference for understanding the spatial distribution of activities and events. In section C of Figure 5.7, the third data example (3^+D) represents domain-specific data, exemplified by medical imagery, factory plan, and lastly, airplane and a ship 3D model. The image shows a skeleton and other medical visuals annotated with ANNT markers that highlight critical areas. In the medical domain i.e., data like this can benefit PM for processes involving specific spatially referred medical diagnostics, treatment planning, and patient monitoring. In the transportation domain, AUXL data that includes visuals of airplanes, ships, and logistics facilities, annotated with ANNT marks can highlight significant areas relevant to transportation processes. This AUXL data aims at benefiting PM for understanding logistics, supply chain

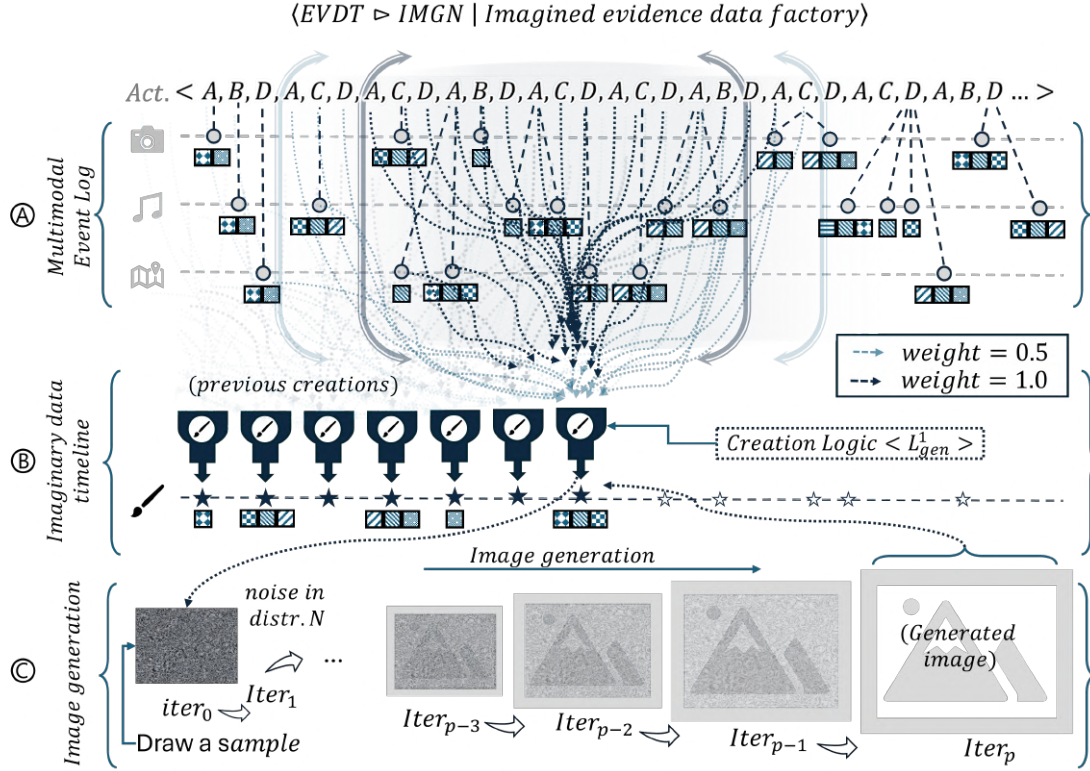


Figure 5.8: Imagined data (IMGN) generated data that can be synthesized based on scenarios, simulations, or envisioning of processes. (A) Multimodal event log timelines, getting morphed into (B) imaginary data timelines through (C) an image generation process.

management, and transportation planning. In section D of Figure 5.7, the fourth example (4^+D) illustrates the visualization of four-or-more dimensional data, including 3D 360° (panorama) video data. Such data modality, offers left/right (L/R) stereo views and 360° panoramic views, providing different perspectives and detailed visual information about the processes. The activities (*Act.*) are mapped onto the *360 visualizations timeline*, indicating the sequence of actions in complex processes.

Imagined Data (IMGN)

Imagined data (IMGN) are generated data that can be synthesized based on scenarios, simulations, or envisioning processes. These data are particularly useful for (1) creating hypothetical scenarios to test the robustness and flexibility of process models, (2) generating visual or auditory representations for processes that have not yet been executed or for which real data are unavailable, and (3) enhancing training and education by providing immersive and illustrative examples of process operations. Imagined data expand the analytical and exploratory capabilities of our framework by allowing users to

experiment with and visualize potential process scenarios. As mentioned, the integration of EVDT with MDLS and ANLT in our framework is facilitated through annotations (ANNT). Imagined data (IMGN) turns those links into an image (using AI for image generation) and therefore provide even more evidence data, EVDT.

Figure 5.8 illustrates the structure and workflow of the *Imagined Evidence Data Factory* ($\langle EVDT \triangleright IMGN \rangle$). This component generates synthetic data to support process analysis and simulation. Section A of Figure 5.8 represents the multimodal event log. This log integrates data from various sources such as visual, auditory, and textual data, capturing a sequence of activities ($A, B, D, A, C, D, A, C, D$). The integration of multiple modalities provides a view of the events and activities occurring within the process. The activities (*Act.*) are mapped across different modalities, with each activity linked to specific visual, auditory, and textual data. This mapping provides a detailed representation of the process, highlighting how different data sources contribute to the understanding of each activity. The illustration in section A of Figure 5.8 depicts weights ($weight = 0.5$, $weight = 1.0$) associated with different data paths. These weights indicate the significance or relevance of specific data sources in the process of image generation. Higher weights suggest more critical data contributions, which we use as an instruction (natural language-based) for prompting an image creation. Those weights are assigned in accordance to the *content window* that can be defined as a parameter and represents the number of neighboring concepts that will be considered for the image generation. Section B of Figure 5.8 illustrates the *imaginary data timeline*, which includes previous synthetic data creations. This timeline captures the evolution of synthetic data generated over time, providing a historical context for the current data generation process. The synthetic data generation process involves creating new data points based on previous creations. This iterative process ensures that the generated data is grounded in historical patterns while introducing novel variations to *simulate different scenarios*. The *creation logic* (L_{gen}^1) governs the synthetic data generation process. This logic defines the rules and parameters for creating new data points, ensuring that the generated data is both realistic and relevant to the process being analyzed.

The *image generation process* (section C of Figure 5.8) initiates image generation that begins with drawing (or pulling) a sample from a noise distribution (i.e. N). This noise represents random variations that serve as the starting point for generating new synthetic data. The introduction of noise ensures diversity and prevents overfitting to historical (training) patterns. The synthetic data generation process is iterative, with each iteration ($Iter_0, Iter_1, \dots, Iter_p$) refining the initial noise sample. Each iteration applies the creation logic to progressively transform the noise into a coherent and realistic data point. The final output of the synthetic data generation process is the generated image ($Iter_p$). This image represents a fully refined *synthetic data point*, ready for use in process analysis and simulation. The generated image aims to capture the important characteristics and patterns identified by the creation logic.

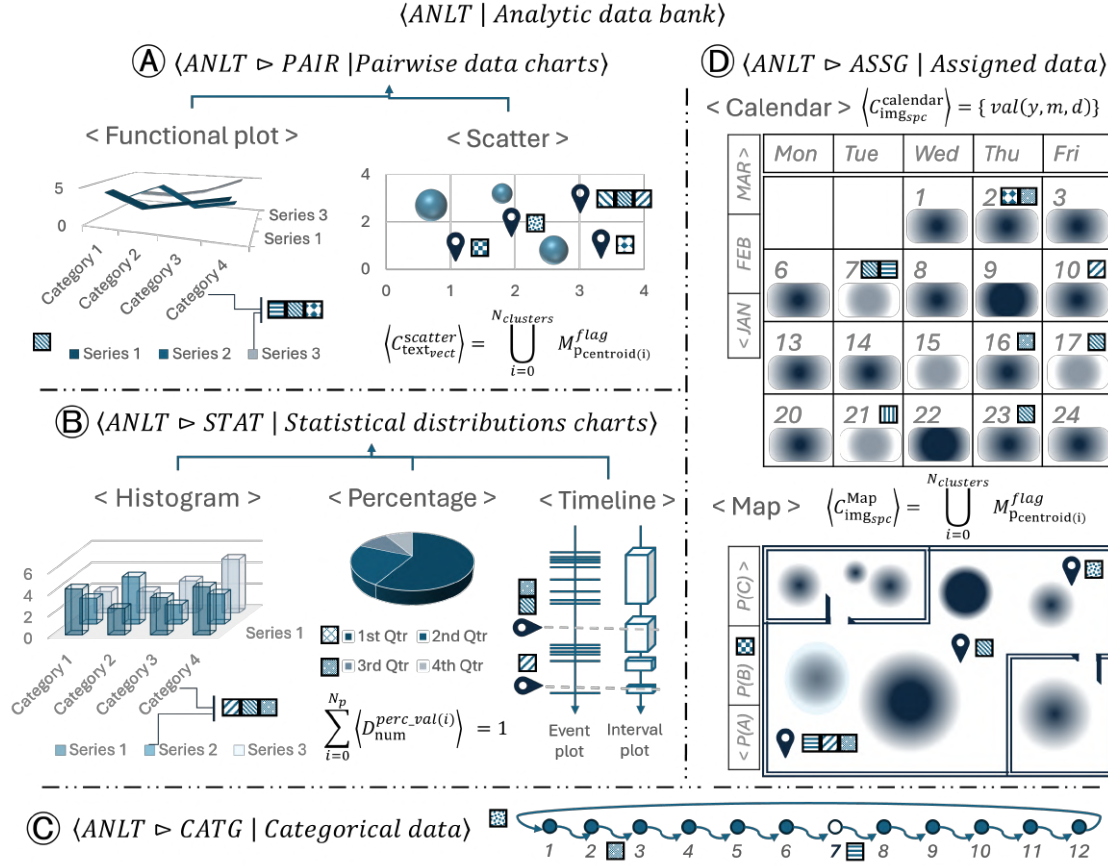


Figure 5.9: Analytics data (ANLT) encompassing various forms of data visualization and analysis techniques for interpretation of process models (MDLS), evidence data (EVDT), and process performance. A and B: Pairwise, and statistical distribution charts, categorical and assigned data. C and D: Categorical (CATG) and assigned data (ASSG).

5.3.4 Analytics Data (ANLT)

Analytics data encompass various forms of data visualization and analysis techniques that help in interpreting and understanding the process models (MDLS), evidence data (EVDT), and process performance. ANLT is divided into several *plot categories*, each focusing on a specific type of analytic data representation: *pairwise data*, and *statistical distributions charts*, as well as, categorical and *assigned data*.

Pairwise Data Charts (PAIR)

Pairwise data charts (PAIR) are graphical representations that display the relationship between two variables. In our method, pairwise data charts are categorized into two main types: *functional plots* and *scatter plots*.

Functional plots illustrate how one variable changes as a function of another variable. These plots are useful for identifying trends, patterns, and relationships between variables. For instance, (1) line plots showing the variation of a process metric over time, and (2) bar charts comparing different categories of data.

Scatter plots display individual data points on a two-dimensional plane, representing the values of two variables. These plots are particularly effective for detecting correlations, clusters, and outliers. Examples include (1) scatter plots showing the relationship between process duration and cost, and (2) bubble charts where the size of the bubbles represents an additional variable, such as the number of occurrences.

Statistical Distribution Charts (STAT)

Statistical distribution charts (STAT) are used to represent the frequency and distribution of data points. These charts are important for understanding the variability, central tendency, and spread of process metrics. Our method includes several types of statistical distribution charts:

Histograms display the distribution of a single variable by dividing the data into bins and plotting the frequency of data points in each bin. For instance, they are useful for (1) identifying the shape of the data distribution (e.g., normal, skewed), and (2) detecting outliers and unusual data patterns. Histograms provide a clear overview of the data's distribution, helping to inform decisions based on the spread and central tendencies of the data. Percentage pie charts illustrate the proportion of different categories within a dataset. Each slice of the pie represents a category, with the size of the slice proportional to the percentage of the category. These charts are useful for (1) visualizing the composition of categorical data, and (2) comparing the relative sizes of different categories.

Percentage pie charts offer an intuitive way to understand the relative distribution of categorical data. Timeline charts, including *event plots* and *interval plots*, represent events or intervals over a period. These charts are important for analyzing temporal aspects of processes. Examples include (1) event plots showing the occurrence of specific events over time, and (2) interval plots depicting the duration of activities within a process. Timeline charts help in understanding the temporal dynamics of processes, identifying patterns, and detecting anomalies in event sequences.

Other Analytical Data Categories

Assigned data (ASSG) charts are used to represent data that are associated with specific time periods or locations. Our framework includes two main types of assigned data charts: *calendar charts* and *map charts*. Calendar charts display data over a calendar period, such as days, weeks, or months. For instance, these charts are useful for (1) visualizing trends and patterns over specific time periods, and (2) identifying peak periods and seasonality effects. Map charts represent data spatially, showing how data points are distributed across different geographic locations. These charts are useful for (1) visualizing the geographic distribution of process activities, and (2) identifying spatial

patterns and trends. Categorical data charts (CATG) are used to represent and analyze data that are divided into distinct categories. The integration of ANLT with process models in our framework is achieved through the use of annotations (ANNT).

5.3.5 SUI-to-Model Annotations (ANNT)

SUI-to-Model Annotations (ANNT) enable the semantic connection of contextualized elements through links. Each annotation bridges elements within the process models (MDLS), evidence data (EVDLT), and analytics (ANLT). This linking is achieved through the following methods:

1. *Contextual Connections*: Annotations create semantic links between contextual elements, providing additional information and context that enhance the understanding of each element. For example, a place in a Petri Net might be annotated with operational details, performance metrics, geographical location, or relevant historical data.
2. *Cross-Element Linking*: Each annotation can be assigned to multiple elements, allowing for the creation of semantic networks. This approach ensures that related elements are interconnected, facilitating a comprehensive view of concepts across different parts of the process. For example, an annotation containing a performance metric might link multiple places and transitions that are part of the same operational workflow.

Different types of annotations can be utilized to serve various purposes within the process models. Examples include (1) *descriptive annotation* (providing textual descriptions or notes that explain the function or significance of an element), (2) *analytical annotation* (linking elements to analytical data, such as performance metrics or statistical analyses, to provide insights into process efficiency and effectiveness), and (3) *multimodal data annotation* (associating images, audio files, or 3D models with elements to offer a richer, multi-sensory understanding of the process).

Junction Operations

The process of semantic linking within a junctional interpretation model involves the concepts of *Process Semantics Junction* and *Junction Alignment*, which together describe how multimodal observations are anchored to conceptual model constructs. Figure 5.10 illustrates this mechanism, providing a visual overview of how the Crossmodal Alignment Agent aligns perceptual traces with symbolic representations to ensure semantic coherence across modalities. Section A in Figure 5.10 (top) focuses on the *Process Semantics Junction*. This part of the semantic linking process integrates multiple data concepts (from multiple sources) and their representations, ensuring that the semantic meaning of each process element, that is marked with a particular ANNT mark, is connected. The example process begins at an initial place (P_0) and transitions (T_0) to the next state (P_1),

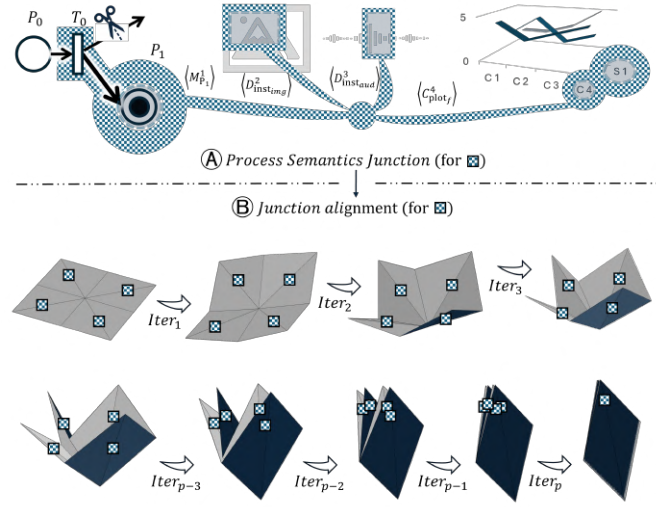


Figure 5.10: The process of semantic linking by aligning combined embedding representation of all marked elements in MDLS, EVDT, and ANLT. (A) integration of multiple data concepts, ensuring the connection of semantic meanings of each annotated process element. (B) An iterative approach ensures that the semantic links are continuously improved and aligned with the evolving understanding of the process.

in the MDLS scope. This transition is marked by an annotation that (in this case) signifies the semantic shift from one state to another, laying the groundwork for further semantic linking. At P_1 , various data sources are integrated, including INST visual data ($D_{instimg}^2$), auditory data ($D_{instaud}^3$), and lastly ANLT's functional plot (C_{plotf}^4). This integration exemplifies that the semantic junction encompasses a view of the process, incorporating multiple modalities (and sources). Visual data ($D_{instimg}^2$) is represented as images and diagrams that capture the visual aspects of the process. This data is important for understanding the visual context and ensuring that the semantic links accurately reflect the visual elements of the process (as it comes from evidence data, EVDT). Similarly, auditory data ($D_{instaud}^3$) includes sound recordings and audio signals associated with the process. On the other hand, analytics data brought functional plots (C_{plotf}^4) to represent the computed pairwise relationships between different process elements.

Semantic contextualization involves aligning the integrated data sources with the semantic meaning of the process elements. This step ensures that the semantic links are not just superficial connections but reflect the true meaning and context of the process. As our underlying AI model, which fuels the understanding of semantics, is organized as a mapping from raw data to a latent embedding space where each data entry is represented as a vector in a multi-dimensional space, we can consider our process of semantic alignment as a transformation of this space. This transformation can be compared to paper folding, where the embeddings are adjusted to achieve alignment. By marking concepts from MDLS, EVDT, and ANLT with ANNT, we performed categorization of the process elements

based on their semantic attributes. Section B in Figure 5.10 (bottom) illustrates the iterative refinement of semantic links through the *Junction Alignment* process. This iterative approach ensures that the semantic links are continuously improved and aligned with the evolving understanding of the process. The first iteration (*Iter1*) begins with an initial alignment of semantic links. This step involves establishing preliminary connections based on the initial understanding of the process elements and their relationships. In the second iteration (*Iter2*), the semantic links are further refined by incorporating additional data and insights. This step ensures that the links become more accurate and reflective of the true semantic relationships within the process. The third iteration (*Iter3*) involves a more detailed alignment of semantic links. This step focuses on resolving any inconsistencies and ensuring that the links are robust and well-defined. Progressive refinement continues with the *Iter_{p-3}* stage, where the semantic links are further adjusted based on the evolving understanding of the process. This step ensures that the links remain relevant and accurate. Subsequent iterations (*Iter_{p-2}*, *Iter_{p-1}*) involve fine-tuning the semantic links. These steps focus on resolving any remaining issues and ensuring that the links are well-aligned with the overall process model. The final iteration (*Iter_p*) completes the alignment process. At this stage, the semantic links are (preferably) refined and aligned, providing a robust and accurate representation of the semantic relationships within the process.

5.3.6 Elicitation Immersions (IMMR)

Elicitation Immersions (IMMR) are the most advanced feature of our SUI interpretation model, designed to guide user interaction with process models through navigation within *Regions of Interest* (ROIs). Immersions aim to create interactive navigation through process models by creating immersive environments that highlight key areas of interest. This is achieved by pulling all annotated links into a single scene, creating a cohesive and engaging experience for the user. Regions of Interest are specific areas within the process models that are of particular importance or relevance. These regions are identified based on the annotations linked to them, and they guide the user's exploration and interaction. We differentiate the term 'scene' from 'regions of interest' because we define a scene as what we see at the moment, while a region of interest can encompass more than just the current scene. Examples of using ROIs include (1) critical process steps (key steps within the process that have a significant impact on the overall performance), (2) bottlenecks and constraints (areas where the process experiences delays or faces limitations), and (3) high-impact elements (elements that are linked to important analytical or contextual data).

Two types of immersive visualizations are introduced to support the interpretation of multimodal process data: the *Immersive Log Replay Diagram* ($\langle\text{LGRP}\rangle$) and the *Immersive Navigate Diagram* ($\langle\text{NAV}\rangle$). These visualizations provide interactive and immersive representations that enhance the understanding of process behavior and contextual relationships. Figure 5.11 illustrates both visualization types and their complementary roles in the analysis workflow.

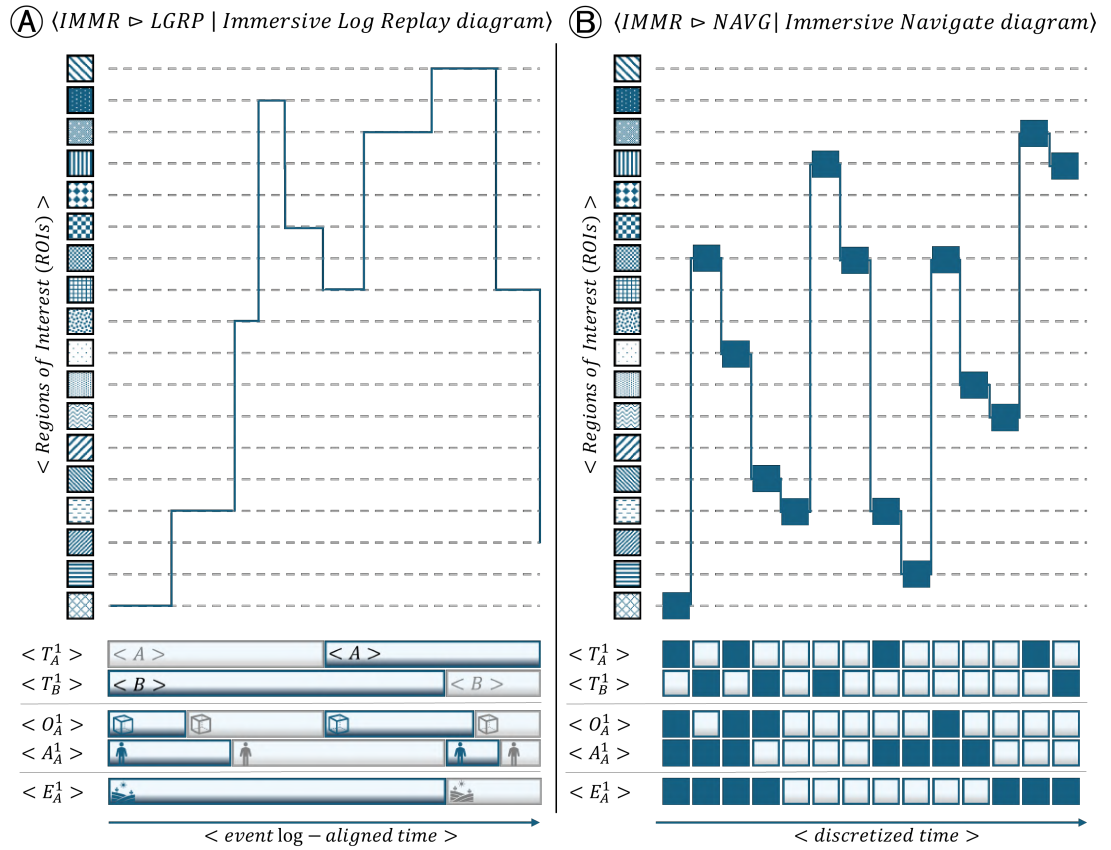


Figure 5.11: (A) A replay of the event log, aligned over time, and corresponding regions of interest. (B) A discretized view of the event log, focusing on the navigation through regions of interest.

Immersive Log Replay Diagram (LGRP)

The *Immersive Log Replay Diagram* ($\langle \text{LGRP} \rangle$) represents a visualization technique designed to replay event logs in an immersive, time-aligned environment. It highlights regions of interest to reveal salient moments within the process execution. Section A in Figure 5.11 (left) illustrates this concept, showing how temporal alignment and spatial emphasis support a deeper understanding of process dynamics. ROIs are represented with a preview of the elements that they connect (marked with different patterns in figure examples) and are aligned along the vertical axis. These regions represent significant areas within the process interpretation model and guide attention in the human-model interaction. The horizontal axis presents the event log aligned over time, providing a chronological view of the process activities and the extracted evidence data EVDT (naturally, instance data, INST). This alignment helps in understanding the sequence and duration of events, facilitating the analysis of temporal relationships and dependencies within the process. The components of the LGRP include various types of data, such as

tasks (T_A^A, T_B^A), objects (O_A^A), actors (A_A^A), and environments (E_A^A). Each component is aligned along the timeline, offering a detailed view of how different elements interact over time.

Immersive Navigate Diagram (NAVIG)

The *Immersive Navigate Diagram* ($\langle \text{NAVIG} \rangle$) offers a discretized representation of the event log, emphasizing the navigation between distinct regions of interest (ROIs) rather than the continuous temporal replay. It allows users to explore process behavior through spatial transitions and contextual groupings of activities. Section B in Figure 5.11 (right) illustrates this concept, with ROIs vertically aligned to maintain semantic correspondence with the $\langle \text{LGRP} \rangle$. The horizontal axis in NAVIG represents the event log in a discretized way, breaking down the continuous timeline into distinct intervals. This discretization helps in analyzing the frequency and distribution of events, providing a granular view of the process dynamics - through simple information of present/not present for each semantic concept that is contextualized. Time in such an interpretation model is irrelevant in terms of duration, as only the order of ROIs matters. The components of the NAVIG also include tasks (T_A^B, T_B^B), objects (O_A^B), actors (A_A^B), and environments (E_A^B), however, each component is discretized and aligned along the timeline, offering a view of the process at different intervals.

While LGRP and NAVIG focus on visualizing process data, they differ in their approach. LGRP provides a continuous, aligned view of the event log, emphasizing the temporal sequence of events. In contrast, NAVIG offers a discretized view, focusing on the distribution and frequency of events. Together, these diagrams provide complementary insights into the process dynamics.

5.4 Mixed Reality Simulation

In immersive analytical environments, the exploration of complex process models transitions from a purely visual activity to an embodied, spatial experience. Within such environments, users no longer inspect static diagrams but navigate through dynamically rendered process landscapes. Each path represents a distinct analytical or experiential trajectory—whether followed by a trainee to understand procedural logic, a manager to assess efficiency, or a stakeholder to evaluate decision outcomes.

However, as models grow in complexity, the act of navigating them requires structured mechanisms to guide attention, manage cognitive load, and preserve interpretative coherence. Similar to how navigational aids structure the traversal of a physical environment, immersive operators formalize the ways in which users can move through, segment, and recombine parts of a process within mixed reality. These operators establish a grammar of interaction—defining how readers transition between different regions of interest and how multiple process narratives can be selectively followed or integrated into a cohesive understanding.

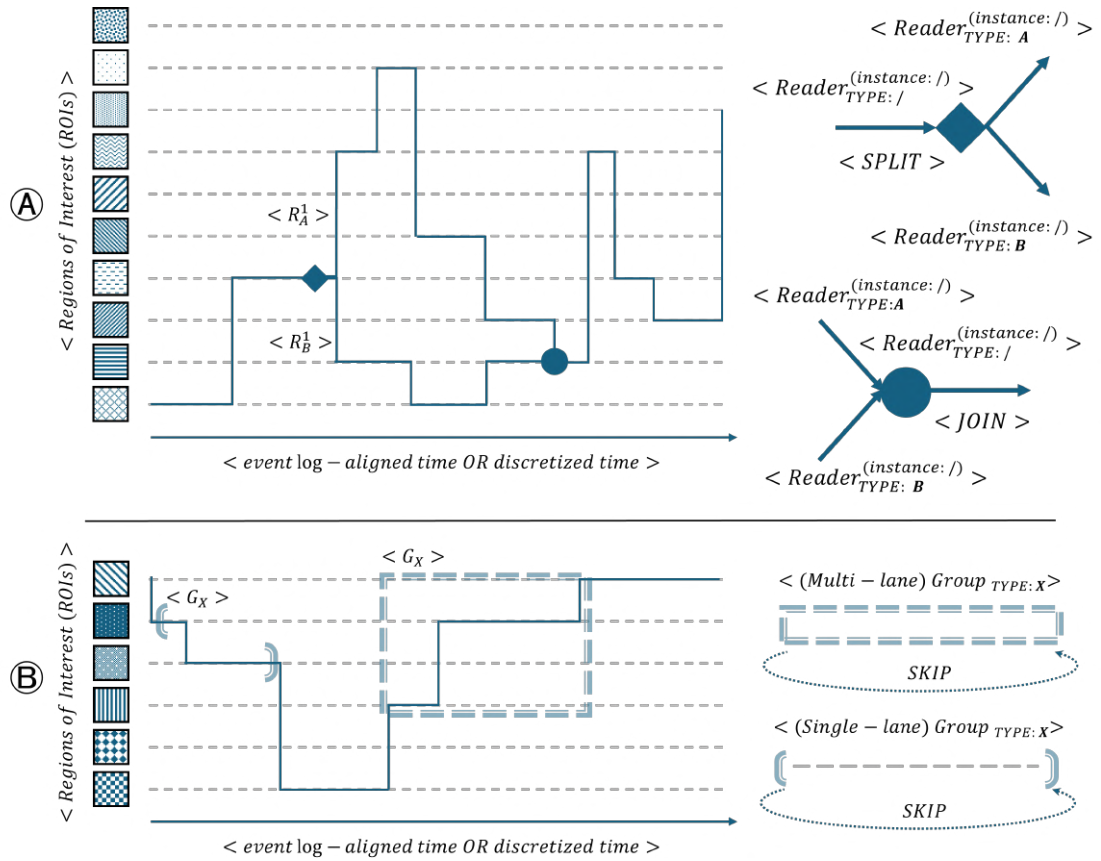


Figure 5.12: Navigation operators: (A) Reading Split and Reading Join to facilitate focused analysis, and (B) Groups and Skip to enhance the efficiency of navigating through large NAVG or LGRP diagrams.

5.4.1 Immersive operators

Our method includes a set of navigation features designed to enhance the exploration and interaction with Mixed Reality Simulations. This section discusses the key immersive operators—*Reading Split* and *Reading Join*—and additional operators—*Groups* and *Skip*—that facilitate an efficient and intuitive navigation experience.

Immersive operators define how users can traverse the models. These operators aim to help in dissecting complex models into manageable parts, making it easier to focus on specific elements or flows within the process, based on the specific characteristics of the model readers. The *Reading Split* operator is used to divide a process model into multiple paths, allowing users to explore different branches or scenarios separately. This operator is particularly useful in scenarios where a process reading can follow multiple parallel or alternative paths. For instance, splitting a process at a decision point to explore the outcomes of different choices based on the profile of the reader (such as someone who needs to be trained on the process, someone who needs to make an executive

decision about the process, someone who is a client, or someone who is a server in the process). The Reading Split operator helps users focus on specific parts of the process, enabling detailed analysis of each branch independently. The *Reading Join* operator is the counterpart to the Reading Split operator. It is used to merge multiple paths or branches of a process model back into a single flow. This operator is important for understanding how different branches converge and how their combined outcomes affect the overall process. The Reading Join operator ensures that users can comprehensively understand the integrated process after exploring its individual branches. Beyond the split-join operators, our framework also includes immersive operators such as *groups* and *skip*, which further enhance the navigation and analysis capabilities of the process interpretation model. The *groups* operator allows users to cluster related elements within a process model into cohesive groups. This operator is useful for managing complexity by organizing elements based on specific criteria, such as (1) functional similarity (where elements performing similar functions are grouped together), (2) process stages (where elements belonging to the same stage of the process are clustered), and (3) performance characteristics (where elements with similar performance metrics are grouped). Grouping elements aim at simplifying navigation by allowing users to focus on specific clusters, making it easier to analyze and interpret the process model. On the other side, the *skip* operator enables users to bypass certain elements or sections of a process model. This operator is useful when specific parts of the process are not relevant to the current analysis or when users want to quickly navigate to a particular section. Examples include (1) skipping over routine or non-critical activities to focus on key decision points, and (2) bypassing sections of the process that have already been faced in the human-model interaction, to avoid redundancy. Figure 5.12 illustrates the Immersive Navigate Diagram (NAVIG), which visualizes process navigation through different regions of interest. Section A of Figure 5.12 (upper part) highlights different ROIs, and an immersive diagram with the event log in either aligned time or discretized time (LGRP or NAVIG). Figure 5.12 illustrates a split operation, where the process diverges into multiple paths. This is depicted by the diamond symbol, indicating a point where the process can take different routes depending on certain conditions - for instance, the type of the reader. Conversely, join operations are depicted by the circle symbol, indicating points where multiple paths converge into a single route. The example in Figure 5.12 includes various reader types: ($\langle \text{Reader}_{\text{TYPE: A}}^{(\text{instance: /})} \rangle$, $\langle \text{Reader}_{\text{TYPE: B}}^{(\text{instance: /})} \rangle$). These readers represent different instances and types of data, providing a detailed view of how different data streams are processed and analyzed within the NAVIG, or LGRP. Section B in Figure 5.12 (bottom) illustrates grouping operations, where multiple human-model interaction checkpoints are grouped together. This is represented by the dashed boxes, indicating groups of human-model interaction checkpoints that are presented collectively. Grouping operations are useful for summarizing and understanding complex processes. Multi-lane groups ($\langle \text{Multi-lane Group}_{\text{TYPE: X}} \rangle$) represent parallel streams of events (therefore applicable to reading paths that are split, for instance), while single-lane groups ($\langle \text{Single-lane Group}_{\text{TYPE: X}} \rangle$) represent a sequential stream of human-model interaction checkpoints. Skip operations are depicted by the graphic symbol of parentheses (opened and closed pair) placed on the human-model

interaction line, and indicate points where certain human-model interactions or groups are skipped. This is useful for focusing on specific parts of the process and ignoring less relevant events, simplifying the analysis and visualization of the process.

5.4.2 Adversarial Situation Formation

In the context of our methodology, we developed the *Adversarial Situation Formation* (ASF) mechanism as a core component to systematically enhance the generation and evaluation of complex, multimodal process situations. ASF is designed to enhance the robustness and interpretative depth of visual process understanding. It is composed of two primary components—the *generator* and the *discriminator*—that engage in an iterative contest to refine situation representations. The generator is responsible for producing synthetic or augmented scenarios derived from multimodal evidence, while the discriminator evaluates these outputs against authentic samples to determine their plausibility and consistency within the modeled domain. Through this adversarial interplay, both components co-evolve: the generator learns to produce increasingly realistic and contextually coherent representations, and the discriminator becomes more adept at detecting subtle inconsistencies. This dynamic fosters the development of more accurate and generalizable visual process interpretation models, capable of capturing nuanced situational patterns that might otherwise remain latent in the data.

The generation action in ASF is responsible for producing SUI models from event logs and evidence data, which starts from a process model that can be either provided or discovered using integrated support. After the process model is obtained, the generation action operates through three primary functions: (1) **Nodes Generation:** The generator synthesizes junction nodes that represent the process interpretations; (2) **Nodes Retrieval:** It retrieves relevant data from event logs, integrating this data into the process interpretation models to provide a rich, evidence-based depiction of processes; and (3) **Referencing:** It collects references between different components of the model, linking related elements and ensuring the coherence of the overall representation into immersive regions of interest. These functions enable the generator to produce detailed SUI models, which are then subjected to evaluation by the discriminator.

The discrimination action in ASF differentiates between authentic junctions (created by a human) and those altered with noise or inaccuracies. This way, the training process of the discriminator is augmented by human-in-the-loop involvement. The training of the discriminator starts with training data preparation: creating a combination of authentic and intentionally altered junctions, enabling the discriminator to learn the distinguishing features of accurate process interpretation models. It is important to note that the discriminator doesn't know which samples are real or not before finalizing its prediction. Human experts interact with the discriminator, providing annotations that help connect concepts from event logs (process models) and evidence data. This interaction is facilitated through natural language human-model dialogues, allowing for intuitive and effective communication, powered by a large language model. The training process then moves towards the discriminator's refinement. Based on the feedback from

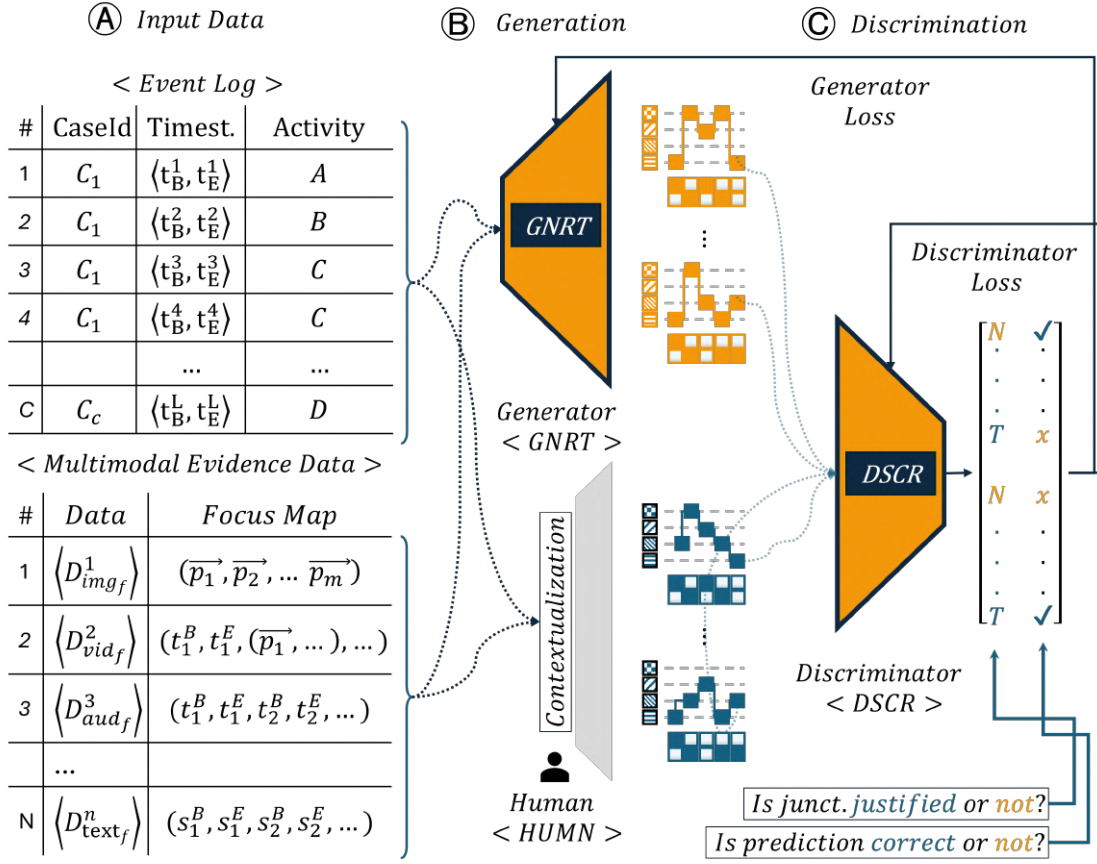


Figure 5.13: Adversarial Situation Formation (ASF) mechanism overview. (A) Input data that provide the foundation for SUI interpretation. (B) Human-made SUI models generation process (bottom part); and Artificial SUI models generation process (upper part). (C) Distinguishing between human-made and artificially-made SUI models.

human experts, the discriminator continually improves its ability to detect inaccuracies and noise in the generated models.

The adversarial process in ASF is illustrated in Figure 5.13, and operates as follows:

1. The generator produces a SUI model from the given event logs, incorporating content generation, retrieval, and referencing.
2. The discriminator evaluates the generated model, distinguishing between authentic junctions and those with inaccuracies or noise (given a human-approved reference model).
3. Feedback from the discriminator, augmented by human-in-the-loop interactions, is used to refine the generator's approach, improving the quality and accuracy of the generated models.

4. This cycle repeats, with the generator continually attempting to outsmart the discriminator by creating increasingly sophisticated and accurate visual process interpretations. The discriminator has a natural advantage, and therefore, it has more success over the generator initially.

5.4.3 Situation Generator (GNRT)

The Situation Generator (GNRT) is performing the generation task and is therefore responsible for producing SUI models from the event log and multimodal evidence data. *Junction nodes generation* involves synthesizing process interpretation models. At the heart of this process is the embedding of multimodal data into a unified space where semantic distances between embeddings are calculated and interpreted.

To achieve this, we utilize a multimodal embedding space, and sample from its learned distribution. In such a space, different types of data (e.g., text, images, audio, video) are mapped into a high-dimensional vector space where semantically similar data points are close to each other. The embedding function $f : X \rightarrow \mathbb{R}^d$ maps an input $x \in X$ from any modality into a d -dimensional embedding space.

For two data points, x_i and x_j , from possibly different modalities, their semantic distance in the embedding space are calculated using a distance metric such as cosine similarity. The cosine similarity $\text{sim}(x_i, x_j)$ is defined as:

$$\text{sim}(x_i, x_j) = \frac{f(x_i) \cdot f(x_j)}{\|f(x_i)\| \|f(x_j)\|}$$

where $f(x_i)$ and $f(x_j)$ are the embeddings of x_i and x_j , respectively, and $\cdot$ denotes the dot product.

The generator leverages these distances to determine the relevance and contextual alignment of various multimodal data points. For example, given an event log $\mathcal{E}$ and a set of multimodal evidence data $\mathcal{D}$, the generator first embeds each event $e_i \in \mathcal{E}$ and each evidence datum $d_j \in \mathcal{D}$ into the multimodal embedding space:

$$\mathcal{E} \mapsto \{f(e_i) \mid e_i \in \mathcal{E}\}, \quad \mathcal{D} \mapsto \{f(d_j) \mid d_j \in \mathcal{D}\}$$

The semantic relevance of evidence data d_j to an event e_i is then determined by calculating the semantic distance $d(f(e_i), f(d_j))$. A lower distance indicates higher relevance. The generator uses these distances to select and integrate the most relevant evidence data into the process interpretation models, ensuring that each model is contextually enriched and semantically coherent.

Furthermore, the generator incorporates temporal and spatial annotations by leveraging time-series and spatial embeddings. For an event e_i occurring at time t_i and location l_i , the generator combines these embeddings into a unified (concatenated) representation:

$$f(e_i) = f(e_i^{\text{content}}) + f(t_i) + f(l_i)$$

where $f(e_i^{\text{content}})$ is the content embedding, and $f(t_i)$ and $f(l_i)$ are the temporal and spatial embeddings, respectively. This combined embedding aims at ensuring that the generated models accurately reflect the temporal and spatial contexts of the events.

Junction Node Retrieval is how we interact with a multimodal embedding space to identify and retrieve relevant content based on a given textual input, known as a domain point. By performing nearest neighbors search, we identify the closest embedded modalities (EVDT, ANLT, or MDLS) to form contextually relevant junction nodes. Let $\mathcal{E} = \{e_1, e_2, \dots, e_n\}$ be the set of event logs and $\mathcal{D} = \{d_1, d_2, \dots, d_m\}$ be the set of evidence data. The LLM generates a set of domain points $\mathcal{P} = \{p_1, p_2, \dots, p_k\}$ from these inputs. Each domain point p_i is a textual representation derived from the input data, $\mathcal{P} = \text{LLM}(\mathcal{E}, \mathcal{D})$.

Next, we map each domain point $p_i \in \mathcal{P}$ to the multimodal embedding space using the embedding function f : $\mathbf{p}_i = f(p_i)$ for all $p_i \in \mathcal{P}$. Simultaneously, we have the embeddings of the event logs $\mathcal{E}$ and evidence data $\mathcal{D}$ already mapped into the embedding space: $\mathcal{E} \mapsto \{\mathbf{e}_i = f(e_i) \mid e_i \in \mathcal{E}\}$, and $\mathcal{D} \mapsto \{\mathbf{d}_j = f(d_j) \mid d_j \in \mathcal{D}\}$;

To retrieve the most relevant junction nodes, we perform a nearest neighbors search in the embedding space. For each domain point embedding $\mathbf{p}_i$, we find the nearest neighbors among the embeddings of EVDT, ANLT, and MDLS modalities. The nearest neighbors search is performed using the cosine similarity- formulated as:

$$\text{NN}(\mathbf{p}_i) = \arg \max_{\mathbf{e}_j, \mathbf{d}_k} \frac{\mathbf{p}_i \cdot \mathbf{e}_j}{\|\mathbf{p}_i\| \|\mathbf{e}_j\|} + \frac{\mathbf{p}_i \cdot \mathbf{d}_k}{\|\mathbf{p}_i\| \|\mathbf{d}_k\|}$$

The result of this search provides the set of nearest neighbor embeddings that are closest to the domain point $\mathbf{p}_i$:

$$\text{NN}(\mathbf{p}_i) = \{\mathbf{e}_{j^*}, \mathbf{d}_{k^*}\}$$

where $\mathbf{e}_{j^*}$ and $\mathbf{d}_{k^*}$ are the embeddings of the nearest event log and evidence data instances, respectively. These nearest neighbors are then used to form the junction nodes, which integrate the relevant EVDT, ANLT, and MDLS modalities.

Referencing to a specific *Focus Map* within evidence data is designed to enhance the precision and relevance of the generated process interpretation models. Focus Maps are calculated using pixel-level image segmentation techniques (such as Meta’s Segment Anything model [KMR⁺23]). This segmentation process allows us to reference distinct segments of our evidence data, providing a detailed and contextually accurate visualization. Although the segmentation process is primarily automated, human intervention can refine the results by adding positive and negative points to guide the segmentation with a sense of intention.

Pixel-level image segmentation involves partitioning an image I into multiple segments $\{S_1, S_2, \dots, S_k\}$, where each segment S_i corresponds to a focus map within the image. The segmentation function $\mathcal{S}$ maps each pixel p in the image to a segment label, $\mathcal{S} : \mathbb{R}^2 \rightarrow \{1, 2, \dots, k\}$. This referencing mechanism allows us to associate specific segments of the evidence data with corresponding junction nodes and ROIs in our model, enhancing the contextual accuracy and relevance of the visual representations.

5.4.4 Situation Discriminator (DSCR)

The *Situation Discriminator* (DSCR) constitutes the evaluative counterpart within the *Adversarial Situation Formation* framework. It is implemented as a multimodal large language network, initially pre-trained on a comprehensive corpus of textual, visual, and structural data to ensure broad interpretative capability. In our methodology, the DSCR serves to distinguish between human-authored (*justified*) and AI-generated (*constructed*) interpretation models, thereby ensuring that synthesized situations remain semantically coherent and epistemically grounded. As illustrated in Section C of Figure 5.13 (right), the discriminator performs a comparative evaluation of model junctions, identifying subtle divergences between authentic and generated structures. The discriminator network D is tasked with differentiating between two interpretation models: those created by humans (justified junctions) and those generated by the Generator ($\langle\text{GNRT}\rangle$). The discriminator operates by comparing two inputs, junctions A and junctions B , and produces a natural language output detailing the differences. Given two sets of junctions, A and B , the discriminator evaluates them using a function $D(A, B)$ which outputs a textual description $T = D(A, B)$. This textual description T is then reviewed by a human, who identifies which junctions are generated (G) and which are justified (J). The human reviewer can also revise the textual description to improve its accuracy and clarity, creating a feedback loop that we term as Discriminator Loss. It encompasses several key factors to ensure the effectiveness and accuracy of the generated models. These criteria include the justification of junctions and the correctness of the predictions made by the discriminator. The evaluation process involves distinguishing between junctions A and B , assessing the quality of the natural language outputs, and translating these assessments into concrete, discretized values.

The evaluation relies on structured prompts that elicit interpretable discriminations along different analytical dimensions. A *semantic coherence prompt* asks, “Compare the semantic relevance and contextual coherence between junction A and junction B . Describe key differences in intent, structure, and interpretation.” A *justification and evidence prompt* requests, “Determine which junction (A or B) appears more justified given the underlying evidence data. Support your conclusion with explicit references to relevant data points.” Finally, a *visual and structural clarity prompt* asks, “Assess the clarity of the visual representation in junction A versus junction B . Indicate which junction provides a more accurate depiction of the process flow and explain why.” These prompts encourage the Discriminator to produce interpretable reasoning traces that facilitate human evaluation and iterative correction. To enable quantitative assessment, qualitative judgments are mapped to discrete scales.

5.4.5 Altering data to Initialize ASF

To ensure the initial predictive advantage of the Discriminator over the Generator, we alter (justified) positive human-annotated samples to create negative samples.

Figure 5.14 illustrates the role and functioning of the *altering data* to question the

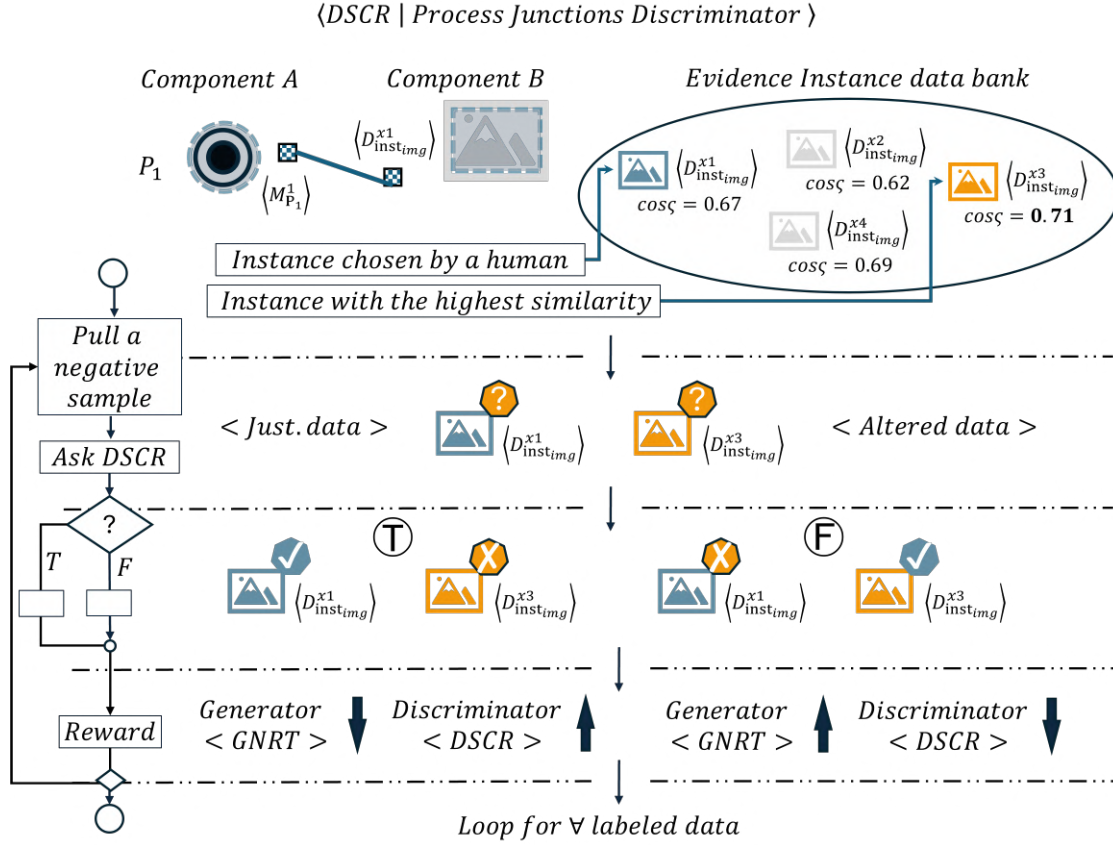


Figure 5.14: The role and functioning of the data alternation for situation discriminator.

discriminator within the ASF.

This process begins with Components A and B, which represent different (exemplary) concepts of the process model (MDLS), evidence data (EVDT) or analytics (ANLT). In the example from Figure 5.14, Component A is associated with a specific place (P_1) of the process model ($M_{P_1}^1$), while Component B includes an evidence data instance ($\langle D_{instimg}^{x1} \rangle$), INST. These components provide the contextual data needed for discrimination.

The *Evidence Instance Data Bank* contains multiple instances of evidence data, each associated with a *cosine similarity score* ($\cos$). This score indicates the degree of similarity between different evidence instances. The discriminator uses these scores to identify the instance with the highest similarity, which serves as the basis for further training-evaluation. The process involves selecting an instance based on two criteria: the instance chosen by a human expert and the instance with the highest similarity score (in this example, we assume those are different instances). This dual selection approach ensures that the training is grounded in both human expertise and data-driven metrics, enhancing the accuracy and reliability of the discrimination process. To gather more training examples, we can use other candidates that have a similarity score close to the

instance chosen by a human. We call these candidates *negative samples*. The negative sample represents data that is intentionally altered or incorrect, providing a contrast for the discriminator’s assessment. As we are technically taking the ‘good’ example and slightly modifying it to create a negative example, we call this process *noise adding* (a.k.a *data altering*). The framework pulls a negative sample so a discriminator can compare it against the chosen instance(s). The discrimination process involves asking the Discriminator to evaluate the justifiability and correctness of the data. The Discriminator assesses whether the chosen instance is justified and whether the prediction is correct. This evaluation helps to validate the accuracy and reliability of the generated models.

5.5 Validation

The investigation conducted in Chapter 3 led to the formulation of the theoretical framework presented in Chapter 4. This framework is operationalized through the method introduced in Chapter 5, which constitutes the proposed approach for enhancing conceptual modeling through multimodal data analysis and Mixed Reality-based elicitation. The method consolidates and scopes the author’s prior research contributions, whose extended technical discussions are provided in Chapter 6 (multimodal data analysis and alignment) and Chapter 7 (Mixed Reality elicitation and simulation).

Individual components of the overall approach have been evaluated independently in the author’s related publications. The accuracy and robustness of multimodal process mining from video and crossmodal evidence integration have been validated in [GBP24a, GBP24c, GBP25d]. The ViEnNa representation and its suitability for narrative-based conceptual modeling have been evaluated in [GBP25c]. The dialogue structures underlying elicitation Condition A have been investigated in [GBP24d]. Conceptual modeling-relevant multimodal simulation and process-to-video generation have been assessed in [GBP25j, GBP25f, GBP26]. Finally, the reliability of multimodal alignment between learned representations and symbolic conceptual models has been validated in [GBP25a, GBP25i, GBP25b].

These prior evaluations are summarized in Sections 6.7 and 7.7, where each foundational technology employed by the method is validated individually. A consolidated overview of experimental settings, datasets, metrics, and results is provided in Appendix C. Together, these studies establish the technical soundness of the constituent components that underpin the proposed method.

The realized method is subsequently implemented in accordance with the requirements specified in Section 3.3, with architectural and technical details reported in Chapter 8. Building upon this foundation, the present validation focuses on the evaluation of the integrated system that operationalizes the complete approach for conceptual modeling enhancement within an end-scale application domain.

Following the research design outlined in Section 3.2, the evaluation adopts an *A/B testing* setup [Rub74]. Condition A (**ours**) corresponds to the proposed multimodal

and Mixed Reality–assisted conceptual modeling approach, whereas Condition B represents a baseline elicitation technique that does not incorporate multimodal analytics, alignment mechanisms, or immersive simulation. The resulting conceptual models are evaluated comparatively with respect to the normative quality dimensions of *precision* and *conciseness*, across multiple scenarios and modeling languages.

In this way, validation proceeds on two complementary levels: (i) prior component-level validation establishing the correctness and reliability of the underlying technologies, and (ii) integrated system-level evaluation assessing whether their combination leads to measurable improvements in conceptual model quality.

5.6 Conclusion

This chapter introduced a systematic method for enhancing conceptual modeling through the integration of multimodal data analysis and mixed reality elicitation. The central premise guiding this approach is that domain expertise manifests not only in what experts explicitly articulate, but also in how they act within situated environments and how they would act under counterfactual conditions. Capturing these complementary dimensions of knowledge requires a structured method capable of bridging sensory-level evidence, linguistic interpretation, and formal conceptual abstraction.

To address this need, we formalized the System Under Investigation as the empirical anchor of modeling activities and structured its interpretation through layered representations: factual SUI representations grounded in multimodal traces; contextualized SUI representations derived through process discovery, evidencing, and cartography; and mixed-reality dispositions that surface counterfactual decision logics. The Cross-modal Alignment Agent operationalizes semantic junctions between process models, evidence data, and analytics, enabling bidirectional reasoning between perceptual data and symbolic constructs.

The introduction of SUI-to-Model annotations (ANNT) and junction alignment mechanisms ensures that conceptual elements remain semantically anchored in observable or elicited phenomena. Through iterative embedding-based alignment, the method transforms heterogeneous multimodal signals into structured, explainable process representations. This integration establishes a shared embedding space in which lower-level perceptual traces and higher-level conceptual abstractions can be jointly reasoned about.

Beyond factual observation, the Mixed Reality Simulation component extends the method into the counterfactual domain. Immersive operators structure navigation within complex process landscapes, while the Adversarial Situation Formation (ASF) mechanism introduces a principled generator–discriminator interplay. This adversarial dynamic systematically refines synthetic situation generation, ensuring that generated junctions converge toward human-justified interpretations. The inclusion of human-in-the-loop feedback preserves epistemic grounding and prevents drift toward purely statistical plausibility.

Collectively, the proposed method transforms conceptual modeling from a static diagramming exercise into a dynamic, evidence-informed, and dialogical process. By embedding multimodal traces and counterfactual elicitation into the modeling cycle, the framework enhances not only descriptive fidelity but also epistemic depth. Conceptual models evolve as living artifacts—continuously refined through observation, simulation, semantic alignment, and adversarial validation.

Most importantly, the method provides a structured pathway toward the thesis objective: improving the *precision* and *conciseness* of conceptual models. Precision is strengthened through explicit grounding of model elements in multimodal evidence and adversarial validation, reducing ambiguity and unsupported abstractions. Conciseness is supported through junction alignment and semantic consolidation, eliminating redundant or weakly justified constructs.

In this way, the chapter establishes the backbone of the thesis: a closed-loop architecture that integrates perception, simulation, interpretation, and formalization. The subsequent chapters build upon this foundation by detailing technical implementations, empirical evaluations, and domain applications that demonstrate the effectiveness of this approach in real-world modeling scenarios.

Multimodal Data Analysis

In [GBP25i], we proposed an architecture for linking multimodal data analysis outcomes with conceptual modeling environments; [GBP25a] extends this notion by formalizing alignment between AI latent spaces and modeling constructs, enabling semantic synchronization between learned and symbolic representations. *Petri Net of Thoughts* [GBP25e] introduced a structure-enhanced prompting approach for process-aware AI, demonstrating how formal process structures can guide multimodal reasoning and generation. In [GBP24c], we introduced the first end-to-end approach for integrating multimodal evidence into process mining pipelines; [GBP24a] demonstrated how video, audio, and textual streams can be linked with classical event logs to enhance contextual interpretation. Building upon these, [GBP25d] generalized the concept by defining crossmodal relationships and shared embedding spaces for aligning semantically related signals; [GPB26] further extended this work by formalizing methods for discovering latent conceptual structures from multimodal data and integrating them with process models. Finally, [GBP25b] explored how neural activations and AI embeddings can serve as process evidence, moving the field beyond traditional event logs toward representation-based reasoning.

Multimodal Data Analysis (MMDA) forms the analytical foundation of this thesis, bridging raw perceptual data and higher-level conceptual representations. In contemporary modeling environments, human activities, decisions, and interactions increasingly occur across multiple channels—visual, auditory, textual, and spatial. Each of these modalities captures a distinct aspect of reality, and when analyzed jointly, they reveal a richer, more nuanced understanding of complex processes than any single source alone.

This chapter consolidates and extends the method presented in Chapter 5 with our prior work on multimodal conceptual modeling and process analysis.

We begin with *Multimodal-Enriched Conceptual Modeling* (Section 6.1), grounded in [GBP25i], and extend semantic alignment between symbolic and latent representations through

Aligned Latent Embeddings of Knowledge and Symbols (Section 6.2), based on [GBP25a]. Structured reasoning over these representations is introduced via the *Petri Net of Thoughts* paradigm (Section 6.3), following [GBP25e].

We then operationalize multimodal evidence through *Multimodal Process Mining* (Section 6.4), building on [GBP24c, GBP24a], and its crossmodal extension (Section 6.4.5) from [GBP25d]. Human-in-the-loop refinement is formalized through Business-Knowledge Integration Cycles (Section 6.5), derived from [GBP24a]. Finally, we relax classical process mining into embedding space (Section 6.6);, extending [GBP25b], and conclude with validation (Section 6.7) and synthesis of the unified multimodal pipeline (Section 6.8).

6.1 Multimodal-Enriched Conceptual Modeling (MMeCM)

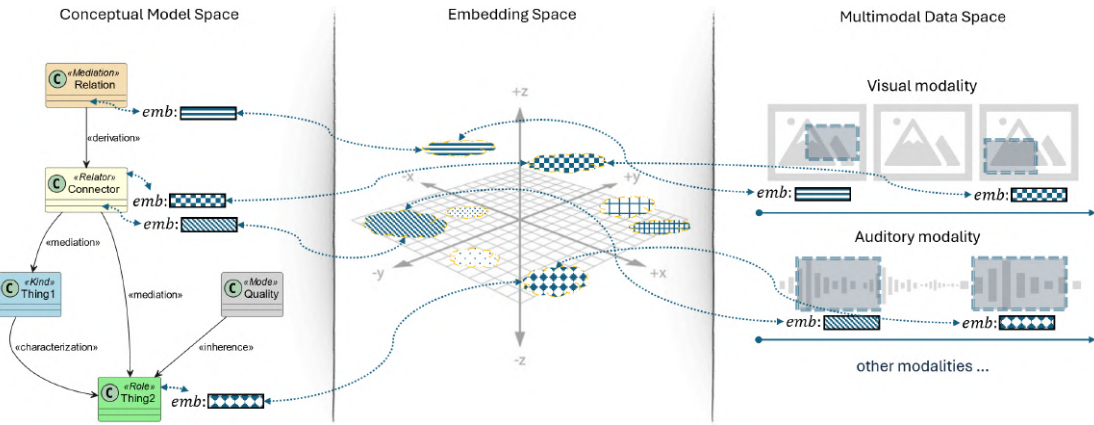


Figure 6.1: Overview of the methodology for enriching conceptual models with multimodal data. Patterned regions illustrate embeddings. Adopted from Gavric et al. [GBP25i].

The aim of this section is to introduce the principles and techniques of MMDA that support the proposed research methodology for *Multimodal-enriched Conceptual Modeling (MMeCM)*. In this context, MMDA provides both the evidence and structure necessary for enriching conceptual models with grounded, contextual information extracted from multimodal sources such as video recordings, speech transcripts, sensor logs, and digital traces. Through alignment, synchronization, and fusion of these heterogeneous data streams, MMDA enables the identification of patterns, behaviors, and dependencies that are otherwise difficult to formalize through traditional elicitation methods.

MMeCM connects directly to the thesis' overarching goal: improving the fidelity and expressiveness of conceptual models. MMDA is positioned as a crucial intermediary step between data capture and model representation—serving as the analytical layer that interprets real-world evidence and translates it into model-ready knowledge. The results of multimodal analysis feed into subsequent stages of the research, including

situation generation, model enrichment, and mixed reality elicitation, thereby establishing a continuous feedback loop between observation and abstraction.

By combining algorithmic insights with human interpretability, MMDA not only strengthens the empirical grounding of conceptual models but also lays the technical foundation for the next methodological components of MMeCM. The chapter therefore sets the stage for how data-driven analysis can become an integral, reproducible part of conceptual modeling practice—shifting the focus from purely symbolic design toward evidence-based model evolution.

6.1.1 MMeCM: The General Perspective

In this section, we detail our research methodology for enriching conceptual models with multimodal data. The methodology is structured into several components that build on one another: we first formalize the structure of conceptual modeling languages, then provide a general method for multimodal enrichment of conceptual models, and finally discuss its specific instantiation for a modeling language as a running case.

Figure 6.1 presents an overview of the methodology for integrating multimodal data into conceptual models. The process spans three interconnected spaces: the *Conceptual Model Space*, the *Embedding Space*, and the *Multimodal Data Space*. The conceptual model consists of structured elements, such as kinds, roles, qualities, and relations, which are linked through mediation, characterization, and inherence relationships. These elements are projected into an embedding space, where they are transformed into distributed vector representations. Simultaneously, multimodal data—such as images and audio—are encoded into corresponding embeddings. The embedding space serves as an intermediary, aligning the conceptual structures with multimodal data representations, thereby enabling enriched semantic connections.

Preliminaries. Let L be a set of conceptual modeling languages. For any language $\ell \in L$, its metamodel is defined as $\mathcal{M}_\ell = (E_\ell, R_\ell, \alpha_\ell)$, where E_ℓ denotes the set of modeling elements (e.g., classes, entities, activities), R_ℓ denotes the set of relationships (e.g., associations, generalizations), and α_ℓ is the set of syntactic and semantic constraints. A concrete model M_ℓ conforming to $\mathcal{M}_\ell$ is represented as $M_\ell = (E_\ell^*, R_\ell^*, \alpha_\ell^*)$, with $E_\ell^* \subseteq E_\ell$, $R_\ell^* \subseteq R_\ell$, and $\alpha_\ell^* \subseteq \alpha_\ell$.

The concept of the *Multimodal-Enriched Conceptual Model* extends the conventional structure of conceptual models by introducing the notion of multimodal enrichment through the *attachment function*. As stated in Definition 20, the attachment function Att establishes a mapping between model elements and the corresponding multimodal data types (e.g., image, audio, or video) that provide contextual evidence for these elements. Building upon this, Definition 21 integrates the attachment function into the model structure, resulting in the formal definition of the MMeCM as $M_\ell^+ = (E_\ell^*, R_\ell^*, \alpha_\ell^*, \text{Att})$. In essence, the MMeCM unifies symbolic model representations with their multimodal

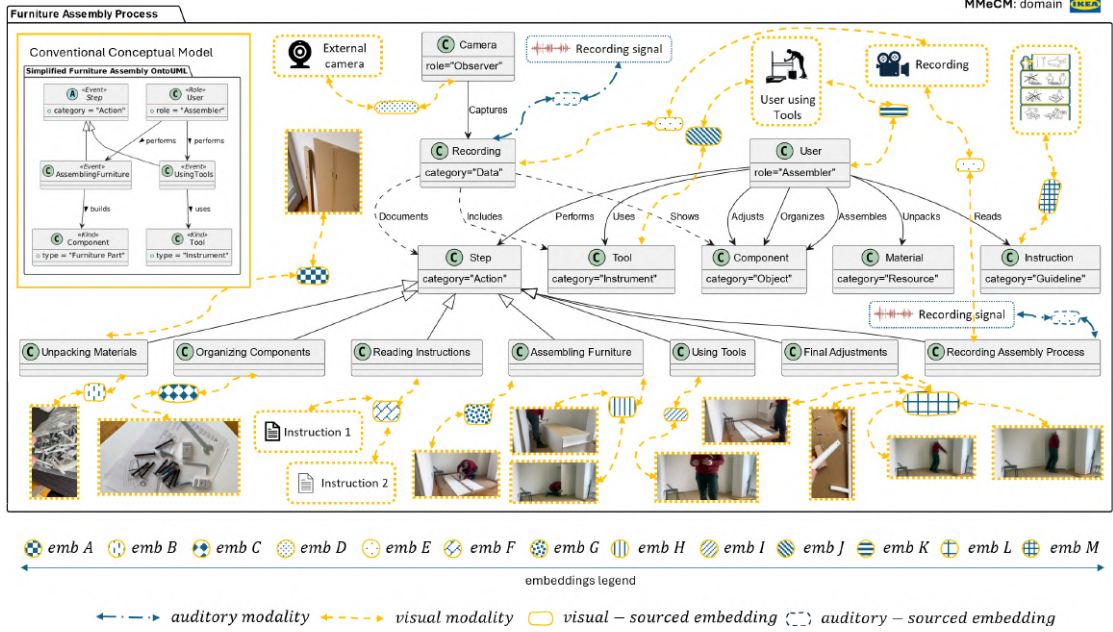


Figure 6.2: Illustrative scenario: Multimodal enrichment of the furniture assembly process. Adopted from Gavric et al. [GBP25i]. (If labels are not readable in print, focus on the layout and structure—refer to the supplementary material for a high-resolution version.)

manifestations, enabling a richer semantic grounding of conceptual models in real-world observations.

To instantiate this methodology, we outline the following steps: (1) Extend existing modeling tools to allow each element $e \in E_\ell^*$ to carry an attribute $\text{Att}(e)$ that references external multimedia resources. (2) Define a metadata function $\mu : \Gamma \rightarrow \mathcal{M}$, where for any $\gamma \in \Gamma$, $\mu(\gamma)$ provides essential metadata such as the URI, file format, and checksum. (3) Ensure that the enriched model M_ℓ^+ adheres to the original constraints α_ℓ by validating $\text{Att}(e)$ for all $e \in E_\ell^*$.

6.1.2 An Illustrative Scenario for MMeCM

To demonstrate the practical application of MMeCM, we present an illustrative scenario focused on the furniture assembly process. This scenario provides an in-depth view of how conceptual models can be enriched with multimodal data, including visual and auditory modalities, enabling enhanced semantic representation and reasoning.

Fig. 6.2 showcases a structured conceptual model of a furniture assembly process, integrating multiple modalities into an enriched representation. The model consists of key entities such as User, Step, Tool, Component, and Material, each categorized under specific conceptual roles (e.g., Action, Instrument, Object, Resource). These elements are interconnected through relations, such as performs, uses, and adjusts,

forming the structural foundation of the assembly process.

The conceptual model elements are projected into an embedding space, where each entity e_i is mapped to a corresponding vector representation $\mathbf{v}_i$. In particular, let $\mathcal{E}$ denote the set of entities, and let $f : \mathcal{E} \rightarrow \mathbb{R}^d$ be the embedding function mapping each entity e_i to a d -dimensional space: $\mathbf{v}_i = f(e_i)$, $\forall e_i \in \mathcal{E}$. This embedding allows semantic comparisons and proximity-based retrieval of related concepts.

Multimodal data sources, including visual and auditory signals, are aligned with the conceptual model through their respective embeddings. The embeddings from images (visual modality) and audio recordings (auditory modality) are extracted via pretrained model: $\mathbf{v}_j^{(vis)} = g_{\text{vision}}(I_j)$, $\mathbf{v}_k^{(aud)} = g_{\text{audio}}(A_k)$, where I_j and A_k denote image and audio samples, and $g_{\text{vision}}, g_{\text{audio}}$ are modality-specific embedding functions.

Each step in the assembly process—such as `Unpacking Materials`—is documented through multimodal evidence. The model captures visual cues (e.g., images of screws, planks, and assembly instructions) and auditory recordings (e.g., the sound of a screwdriver or user commentary). The embeddings are dynamically updated based on new observations: $\mathbf{v}_t^{(step)} = \alpha \mathbf{v}_t^{(step)} + (1 - \alpha) \mathbf{v}_t^{(new)}$, where $\mathbf{v}_t^{(step)}$ represents the embedding of an assembly step at time t , and $\mathbf{v}_t^{(new)}$ is the new multimodal update. The weighting factor α determines the balance between prior knowledge and new data.

By integrating multimodal embeddings, the model enables richer semantic reasoning. For example, the action `Using Tools` is associated with a set of tools (e.g., screwdriver, hammer), each linked to corresponding embeddings from prior tasks. Given a new tool T_x , similarity-based reasoning can determine its function by computing its distance to known tool embeddings. If sim exceeds a predefined threshold, the model can infer the tool’s function and suggest appropriate usage instructions.

6.1.3 A Framework for Integrating Multimodal Data into Conceptual Modeling

In this section, we present our proposed framework for integrating multimodal data into conceptual models, producing *Multimodal-Enriched Conceptual Models* (MMeCM). Our framework consists of three primary stages: (1) extraction of all natural language elements from conceptual models while preserving their contextual links, (2) computation of multimodal embeddings for each extracted element, and (3) performing similarity matching between these embeddings and multimodal data.

In our approach, the similarity matching is facilitated by a joint multimodal embedding space, such as [GENL+23]. It is a method that learns a joint embedding space across multiple modalities (in particular, images, text, audio, depth, thermal, and inertial measurement units such as accelerometer and gyroscope, when it comes to [GENL+23]). We design our similarity matching to act as the enrichment step by *linking any of these multiple modalities to the corresponding natural language elements in conceptual models*.

Framework Walkthrough. In MMeCM, we (1) extract each natural language term from the model, (2) clean and normalize the extracted term, (3) convert each term into a vector via the multimodal encoder, and (4) compute similarity scores between vectors to attach matching multimodal data to conceptual regularities.

Let a conceptual model be denoted as M , which contains a set of elements E and their associated natural language descriptions. The goal is to extract each natural language description while keeping track of its original position in M . Let $\mathcal{N}$ be the set of all extracted natural language values, and let $\mu : \mathcal{N} \rightarrow E$ be the mapping that associates each natural language description with its corresponding model element. After extraction, each $n \in \mathcal{N}$ is transformed into an embedding $v \in \mathbb{R}^d$ using a multimodal model. The similarity function $\sigma : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ then computes the affinity between any two embeddings, thereby enabling the enrichment with modalities from the set Γ .

Extraction of Natural Language Elements. To initiate the enrichment process, we extract all natural language elements from the conceptual model while preserving their contextual links. This step lays the foundation for linking the extracted textual descriptions with the corresponding multimodal data. Alg. 1 outlines the extraction of these elements from the model.

Algorithm 1 Extract Natural Language Elements from Conceptual Models

Input: Conceptual model M with element set E

Output: Set of natural language values $\mathcal{N}$ and mapping $\mu : \mathcal{N} \rightarrow E$

```

1 Initialize  $\mathcal{N} \leftarrow \emptyset$  and mapping  $\mu \leftarrow \emptyset$ 
2 foreach element  $e \in E$  do
3   Identify natural language properties  $P(e)$  associated with  $e$ 
4   foreach property  $p \in P(e)$  do
5     Extract natural language value  $n$  from  $p$   $\mathcal{N} \leftarrow \mathcal{N} \cup \{n\}$   $\mu(n) \leftarrow e$  // Maintain
     provenance
6   end
7 end
8 return  $(\mathcal{N}, \mu)$ 

```

Following the extraction, the set of natural language values $\mathcal{N}$ and the mapping μ serve as the basis for the *Embedding and Similarity Matching* stage in our pipeline.

Embedding and Similarity Matching for Enrichment. The framework proceeds to compute multimodal embeddings for each natural language element. This embedding step leverages a multimodal model to transform natural language into a joint embedding space, where similarity matching can be performed across different data modalities. Alg. 2 illustrates how the textual data is preprocessed, embedded, and subsequently aligned with multimodal data via similarity scoring.

Algorithm 2 *Multimodal-Enriched Conceptual Models (MMeCM)*

Input: Set of natural language values $\mathcal{N}$ (from Alg. 1); multimodal model F ; modality set Γ ; batch size B ; conceptual model M

Output: Embedding mapping $\eta : \mathcal{N} \rightarrow \mathbb{R}^d$ and similarity matrices for multimodal enrichment

```

9 Partition  $\mathcal{N}$  into batches  $\{N_1, N_2, \dots, N_k\}$  of size  $B$ 
10 foreach batch  $N_i$  do
11   Prepare list  $T \leftarrow N_i$  Transform  $T$  using context-aware preprocessing:  $T' \leftarrow \mathcal{T}(T \mid M)$ 
      Compute embeddings  $E \leftarrow F(T')$ , with  $E \subset \mathbb{R}^d$ 
12   foreach embedding  $e_j \in E$  associated with text  $t_j \in T$  do
13      $\eta(t_j) \leftarrow e_j$  // Store linguistic embedding
14   end
15   Compute similarity matrix  $S$  such that for all  $(t_j, t_k) \in T \times T$ :
      
$$S_{jk} = \sigma(e_j, e_k) = \text{softmax}(e_j^\top e_k)$$

16   Store  $S$  and  $\eta$  for batch  $N_i$  for alignment with modalities in  $\Gamma$ 
17 end
18 return  $\eta$  and all computed similarity matrices

```

Computation of embeddings using a multimodal model, followed by the construction of similarity matrices, facilitate matching between the natural language embeddings and the various modalities in Γ , thus enabling the multimodal enrichment of conceptual models through a flexible approach that can be adapted to diverse conceptual modeling languages and extended to integrate additional modalities as required.

6.2 Aligned Latent Embeddings of Knowledge and Symbols (ALeKS)

Artificial intelligence and conceptual modeling each offer distinct yet complementary strengths. AI models, such as generative pre-trained transformers, are capable of learning from vast amounts of unstructured data, capturing statistical patterns and contextual knowledge. In contrast, conceptual models, such as BPMN, UML, ArchiMate, and OntoUML provide precise, human-understandable representations of domains through symbolic elements, structured relationships, and formal constraints [BAR23]. As AI models grow more capable of interpreting real-world data in multiple formats—text, images, video, even audio—there’s a disconnect between how AI understands a concept, and how that same concept is represented in modeling languages. Large AI models trained on massive datasets across modalities—often referred to as *foundation models*—learn to represent different types of inputs (like a paragraph or a photo) as high-dimensional

vectors in a shared embedding space. In that space, items that have *similar meaning* are placed close together, regardless of whether they were originally text, images, or something else. These embeddings effectively serve as the AI model’s internal representation of knowledge. On the other hand, conceptual models encode structured knowledge through items like *classes*, *flows*, *actors*, and *stereotypes*. While symbolic, these elements can also be embedded using specialized encoders [AGPB23], or prompting techniques [RSGvDA24], that take into account their textual labels, graphical features, and meta-model constraints.

In this section, we propose a method to align the embedding space of AI models with that of conceptual models, in order to support joint reasoning on tasks relevant to decision-makers and stakeholders. Aligning these representations enables new forms of collaboration between AI and CM. For example, it leads towards the possibility to recognize conceptual structures—such as processes or architectures—directly within video recordings or other multimodal observations, or comparison of formal norms (e.g., GDPR [Eur16] rules) with observed behaviors or textual policies, facilitating conformance checking at scale. Ultimately, this may result in conceptual models more grounded in practice, and AI models more transparent in meaning.

We propose a methodology to systematically align symbols from conceptual modeling artifacts with latent knowledge representations within AI models, named Aligned Embeddings of Knowledge and Symbols (ALeKS).

Preliminaries. The *Aligned Embeddings of Knowledge and Symbols* framework, defines a series of constructs that together describe how conceptual modeling elements interact with multimodal and semantic AI representations. We refer to theoretical concepts formally introduced in Section 4.1.5: Definitions 26–28 introduce the notion of *semantic shifts*, which quantify transformations between multimodal and conceptual entities within the joint embedding space. Together, these definitions provide the theoretical basis for aligning conceptual model semantics with data-driven embeddings in ALeKS.

6.2.1 ALeKS: The General Perspective

In this section, we explain the general perspective on ALeKS. As illustrated in Fig. 6.3, CM-STEPS process primarily conceptual models, while MM-STEPS multimodal data.

Collection of Conceptual Models. Our approach begins with a curated *Collection of Conceptual Models* (C_{CM}), which includes heterogeneous modeling diagrams from real-world domains, such as [BSF⁺22]. Each model serves as both a symbolic knowledge structure and a grounding point for semantic alignment. For illustration, we use a UML sequence diagram from the healthcare domain, as represented in Fig. 6.3, which models the procedural steps of a DNA collection homekit. This diagram specifies the user’s interactions with a *test kit*, *tube*, and *biohazard bag*, and forms a narrative structure embedded with task semantics. These conceptual models, which can range from class diagrams to sequence diagrams, provide the foundational symbols (e.g., stereotypes, classes, attributes, messages) that represent domain knowledge.

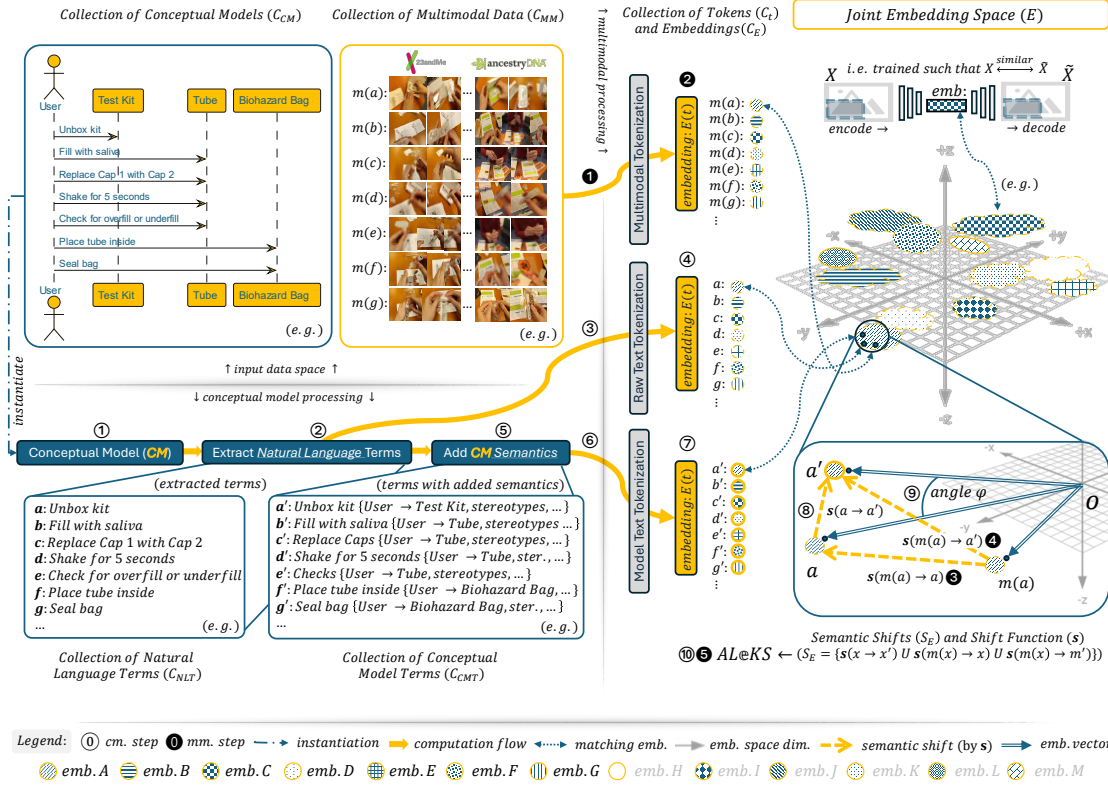


Figure 6.3: Overview of the methodology that aligns conceptual modeling terms with multimodal embeddings in a joint semantic space. Patterned regions (A–M) denote embedding areas. The illustrated pipeline operates in parallel along two streams: one driven by conceptual modeling semantics (cm. steps) and the other by perceptual grounding via multimodal data (mm. steps). Adopted from Gavric et al. [GBP25a].

cm-Step 1: Instantiating a Conceptual Model To process each model within the collection C_{CM} , we instantiate a data structure that explicitly represents its symbolic elements—this serves as the operationalization of a `foreach` function over the model collection. Each instantiation captures the syntactic and semantic features of the modeling language in use, enabling structured traversal and downstream processing.

cm-Step 2: Extracting Natural Language Terms From each instantiated model, we extract a *Collection of Natural Language Terms* (C_{NLT}) by identifying attributes that carry semantically meaningful text. This typically involves selecting the "name" attribute from modeling elements, though the specific attribute may vary across modeling languages.

cm-Step 3-4: Tokenizing and Embedding Natural Language Elements Once the natural language terms (C_{NLT}) are collected, we proceed with tokenization, wherein

each textual term is decomposed into a sequence of tokens. This step prepares the text for numerical representation and aligns with standard NLP preprocessing workflows. Following tokenization, we transform each tokenized sequence into a high-dimensional vector space using embedding models. These models are pre-trained for semantic embedding tasks and convert text into arrays of numbers, or embeddings, that encode the contextual meaning of the terms.

cm-Step 5-7: Collection of Conceptual Model Terms With the *Collection of Natural Language Terms* (C_{NLT}) prepared and semantically embedded, we enrich this collection by incorporating structural semantics derived from the modeling language, resulting in the *Collection of Conceptual Model Terms* (C_{CMT}). For each term in C_{NLT} , we add contextual metadata such as stereotypes and types from [BSF⁺22], classifying whether an element represents, for instance, a Kind, Role, or Relator. Furthermore, model-specific annotations from diverse modeling repositories (e.g., [LCIC22]) are included, such as *eClass* identifiers and fully qualified names, which position elements within broader domain taxonomies. From enterprise architecture datasets (e.g., [GSB23]), we incorporate higher-level organizational features like *ArchiMate* concept of *layer* and *type*. The resulting C_{CMT} encapsulates both the natural language expressions and their conceptual modeling semantics. Once the C_{CMT} is assembled, each enriched term—now combining natural language content with modeling semantics—undergoes the same tokenization and embedding process as described for C_{NLT} . This ensures consistency in how linguistic and structural elements are numerically represented. By treating both C_{NLT} and C_{CMT} with the same embedding pipeline, we maintain a shared representational space, which is used for reliable comparison, clustering, and semantic alignment in downstream tasks.

mm-Step 1-2: Collection of Multimodal Data To complement the symbolic and linguistic representations, we compile a *Collection of Multimodal Data* (C_{MM}), comprising images, audio, and video segments that are semantically linked to the conceptual model terms in C_{NLT} . Each modality is preprocessed and embedded a foundation model that creates a unified embedding space for multiple data modalities. This enables a direct comparison between, for instance, an image of a test kit and the term "test kit" extracted from a UML sequence diagram. Tokenization in this context involves encoding non-textual data into fixed-length representations via modality-specific encoders, after which all embeddings are projected into a shared latent space using modality-agnostic transformers. Through this unified embedding approach, C_{MM} is mapped into the same vector space as C_{NLT} and C_{CMT} , allowing semantic comparison across modalities. For each natural language term $a \in C_{NLT}$, we identify its multimodal counterpart $m(a) \in C_{MM}$, such as a corresponding image or video clip. This mapping enables the cross-modal embedding $E(m(a)) \in \mathbb{R}^d$, which can now be directly compared to the text-based embeddings $E(a)$ and $E(a')$ through vector operations in $\mathbb{R}^d$.

Why Multimodal Data Matters. Incorporating multimodal data allows to generalize beyond the inherently biased and limited textual information extracted from conceptual models. While C_{NLT} captures terms grounded in modeling practices, it reflects a narrow linguistic perspective shaped by domain-specific conventions. By retrieving the closest multimodal match $m(a)$ for each term $a \in C_{NLT}$ —even if no explicit image or video exists in the original model—we tap into real-world representations that enrich and diversify semantic understanding. Although not every term will have a perfect multimodal counterpart, the use of a shared embedding space (e.g., ImageBind) ensures that the retrieved $m(a)$ is the most semantically relevant instance available. To maintain control and interpretability, we weight each multimodal contribution proportionally to its embedding similarity $\cos(E(a), E(m(a)))$, effectively modulating its influence based on relevance. This strategy enhances generalization by leveraging perceptual grounding while preserving alignment with the conceptual model’s symbolic core.

cm-Step 8: Computing the Semantic Shift. To evaluate the impact of introducing conceptual modeling semantics, we define the semantic shift vector $\vec{s}(a \rightarrow a') = E(a') - E(a)$, where $E(a)$ and $E(a')$ are the respective embeddings of the natural language term a and its enriched conceptual model form a' . This shift vector quantifies the transformation imposed by conceptual modeling, capturing how embedding semantics evolve when a term is recontextualized within formal modeling constructs (e.g., stereotypes, qualified names, abstraction layers). The vector $\vec{s}(a \rightarrow a')$ is used for tracing how symbolic enrichment repositions a term in the embedding space. As shown in Fig. 6.3, we interpret this semantic shift as a directed edge in $\mathbb{R}^d$ from a to a' . If the modeling process adds significant semantic grounding, we expect the vector norm $\|\vec{s}(a \rightarrow a')\|$ to be non-trivial, and its direction to align with the conceptually meaningful transformation path. Importantly, a and a' remain semantically linked, and the shift encodes this transformation’s nature—be it abstraction, disambiguation, or specialization.

cm-Step 9: Cosine Similarity as Alignment Indicator To measure how much semantic content is retained or realigned through modeling, we compute the cosine similarity between the embeddings of a and a' . This angle φ serves as a metric of conceptual proximity. A smaller angle implies semantic preservation, whereas a larger angle implies a substantial shift due to modeling. Hence, φ becomes an empirical indicator of how much the modeling semantics "reframe" the original natural term.

By calculating $\cos(\varphi)$ across all (a, a') pairs, we can generate a statistical profile of modeling impact. Terms with high similarity are already semantically close to their conceptual representation, while those with low similarity indicate deeper semantic reinterpretation introduced by modeling languages or ontologies.

mm-Step 3–4: Multimodal Matching from $m(a) \rightarrow a'$ In parallel to the conceptual path $a \rightarrow a'$, we compute the multimodal shift from a natural language term’s multimodal representation $m(a)$ to its modeled form a' . This is denoted as $\vec{s}(m(a) \rightarrow a') = E(a') - E(m(a))$, to capture how much conceptual modeling repositions a term relative to its

embodied (e.g., visual or auditory) representation. The embedding $E(m(a))$ arises from the shared embedding space, ensuring compatibility with $E(a')$.

For illustrative purposes, Fig. 6.3 visualizes the vector $\vec{s}(m(a) \rightarrow a) = E(a) - E(m(a))$, which represents the shift between a multimodal instance and its textual form. For example, an image might evoke associations or affordances not present in the original textual label.

cm-Step 10 + mm-Step 5: Defining Semantic Shifts S_E and Shift Function s
To integrate both the conceptual and multimodal perspectives, we define the full set of semantic shifts as:

$$S_E = \{s(x \rightarrow x')\} \cup \{s(m(x) \rightarrow x)\} \cup \{s(m(x) \rightarrow m')\}$$

where x is a natural language term, x' is its modeled version, $m(x)$ is its multimodal instantiation, and m' is the modeled multimodal target. The shift function $s: \mathbb{R}^d \rightarrow \mathbb{R}^d$ maps embedding vectors to their target transformations, capturing both the symbolic (linguistic $\rightarrow$ conceptual) and perceptual (multimodal $\rightarrow$ linguistic or conceptual) evolution of meaning.

In practice, the three types of shifts— $s(x \rightarrow x')$, $s(m(x) \rightarrow x)$, and $s(m(x) \rightarrow m')$ —form the semantic backbone for model evaluation. The magnitude and direction of each shift vector, as well as their aggregated statistics, offer insights into where alignment succeeds, where ambiguity persists, and where new semantics are introduced by modeling processes.

6.2.2 An Illustrative Scenario for ALeKS

Fig. 6.4 showcases two key conceptual events—*Shaking* and *SealingBag*—drawn from a UML activity model for a DNA collection test kit¹. Multimodal data and conceptual modeling semantics coalesce to generate semantically aligned embeddings, with alignment forces visualized via attractive (green) and repulsive (red) vectors.

Conceptual Model Serialization. To formalize the enrichment from natural language to conceptual modeling terms, we instantiate each conceptual model term using a controlled sentence template. We employ a templating mechanism to serialize conceptual model terms by enriching extracted natural language terms with conceptual modeling semantics. The following template is used for illustration in Fig. 6.4 to generate structured sentences:

*The event «**EventName**» is triggered by an instance of «**KindName**», which plays the role «**RoleName**» within the context of «**RelatorName**». During this event, it affects the «**AffectedSubkind**», which plays the role*

¹Note: The conceptual model in the upper left is scaled down to emphasize the overall layout. For a detailed view, please refer to the supplementary material.

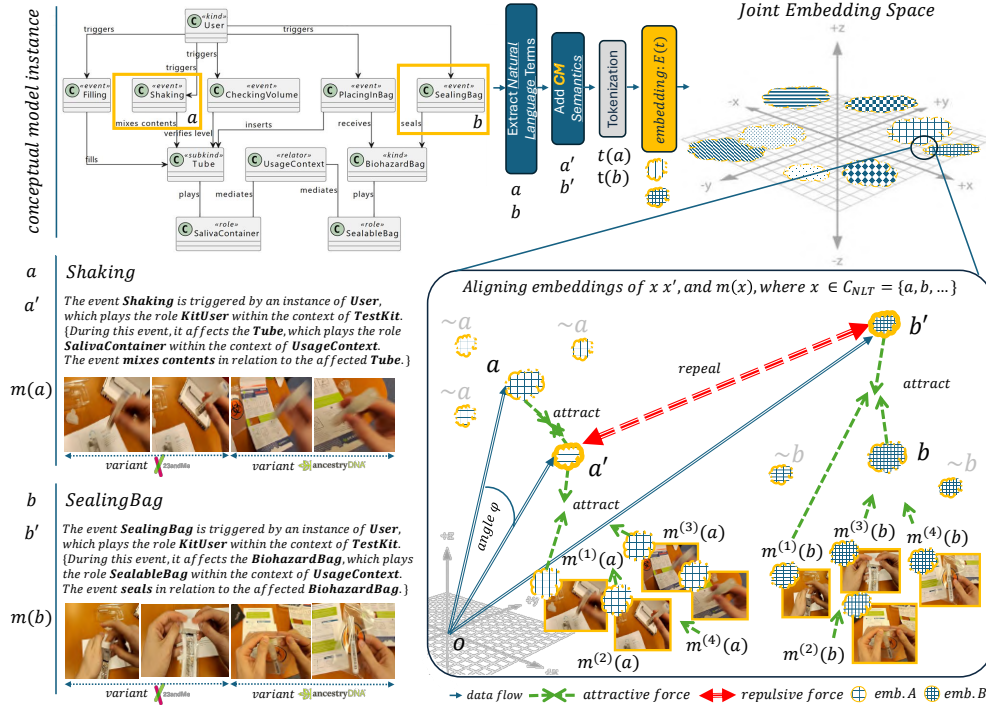


Figure 6.4: An Illustrative Scenario for ALeKS, showing how natural language terms (a , b), their semantically enriched versions (a' , b'), and corresponding multimodal instances ($m(a)$, $m(b)$) are aligned in a joint embedding space. Two exemplar events—Shaking and SealingBag—from a conceptual model of a DNA test kit usage scenario are shown. Adopted from Gavric et al. [GBP25a].

«**RoleName**» in the context of «**findContext()**». The event «**Function**» in relation to the affected «**AffectedSubkind**».

The template serves as a domain-agnostic, syntactically consistent structure to convert symbolic conceptual model instances into textual descriptions. The placeholders (e.g., EventName, KindName, etc.) are dynamically replaced with values derived from the instantiated model. The section enclosed in braces ... denotes a `foreach` loop iterating over all AffectedSubkind elements that participate in the same event instance. For each such element, a corresponding sentence is generated to describe its involvement in the event. This approach ensures that the generated text reflects the multiplicity and contextual semantics present in the conceptual model.

As a result, each enriched term $a' \in C_{CMT}$ becomes a semantically grounded textual artifact that not only embeds linguistic content but also captures the structural and behavioral roles defined in the model. These serialized texts are then fed into our embedding pipeline, for the purpose of alignment with their corresponding natural language term a and multimodal instance $m(a)$ within a shared vector space. This

example of serialization grounds the term a' with modeling semantics, explicitly enriching its embedding relative to the plain term a , and similarly, producing b' as the semantically contextualized version of b .

Aligning Embeddings in the Shared Space. Natural language terms (a, b) , their conceptually enriched forms (a', b') , and multimodal instances $(m(a), m(b))$ are jointly embedded as illustrated in Fig. 6.4. Conceptual enrichment pulls each a' and b' closer to their corresponding natural term (a, b) via attractive forces, while simultaneously pushing them away from unrelated terms (e.g., $a' \rightarrow \text{repel}(b')$).

Semantic Shifts. Each event (a, b) is paired with a collection of multimodal data entries $m(a)$ and $m(b)$, composed of image sequences sourced from real-world DNA test kit usage. These images span multiple domains (e.g., 23andMe, AncestryDNA), highlighting visual variance across kits while maintaining semantic coherence. The multimodal embedding model projects each $m^{(i)}(a)$ and $m^{(j)}(b)$ into the same space as their symbolic counterparts, and the figure shows a clustering behavior around a' and b' . Semantic transitions are geometrically encoded via vector shifts. These shifts— $s(a \rightarrow a')$, $s(m(a) \rightarrow a')$ —are drawn as arrows within the joint embedding space. By computing and visualizing angles φ between a and a' , or distances between $m(a)$ and a' , we gain semantic similarity insight.

6.2.3 A Framework for Aligning Embeddings of Knowledge and Symbols

In this section, we present the ALeKS framework. ALeKS is designed to bridge symbolic conceptual models and machine-readable embeddings, enabling downstream tasks such as similarity search, model comparison, and intelligent querying over conceptual structures. Our implementation specifically targets the OntoUML modeling language, known for its ontologically grounded constructs and formal semantics.

We begin by demonstrating the serialization procedure, which transforms OntoUML models—comprised of modeling terms and properties—into structured natural language sentences. This step is critical to ensuring the symbolic data can be meaningfully embedded. Following serialization, we describe the alignment mechanism, which trains a joint vector space where symbolic and statistical representations are semantically aligned.

The serialization phase handles two major model constructs: *terms* (e.g., classes or entities) and *properties* (e.g., attributes or associations). Each element is decomposed into a set of human-readable, formally structured sentences that preserve the semantics encoded in the modeling language. This transformation enables the use of language models and embedding techniques to learn aligned representations.

Conceptual Model Serialization: Terms and Properties

The serialization procedure converts raw modeling elements into readable and semantically meaningful sentences. This structured textual representation acts as an intermediate form between symbolic models and their embedding. In Table 6.1, we detail each step of the serialization process for OntoUML *terms* and *properties*, illustrating them with sentence patterns.

Table 6.1: Serialization Rules for Properties and Terms. Adopted from Gavric et al. [GBP25a].

Aspect	Serialization Pattern
Parent Term Linkage	PROPERTY n (id: ID_N) BELONGS TO m (id: ID_M).
Type Declaration	PROPERTY n (id: ID_N) IS OF TYPE T .
Boolean Flags	PROPERTY n IS DERIVED; IS READ-ONLY; IS ORDERED.
Cardinality	PROPERTY n (id: ID_N) HAS CARDINALITY [MIN..MAX].
Nested Property Type	PROPERTY n (id: ID_N) HAS PROPERTY TYPE T .
Aggregation Kind	PROPERTY n (id: ID_N) HAS AGGREGATION KIND K .
Property Assignments	PROPERTY n (id: ID_N) HAS PROPERTY ASSIGNMENTS { $k:v$, ... }.
Stereotype	PROPERTY n (id: ID_N) HAS STEREOTYPE S .
Subsetted Properties	PROPERTY n (id: ID_N) HAS SUBSETTED PROPERTIES { p_1 , p_2 , ... }.
Redefined Properties	PROPERTY n (id: ID_N) HAS REDEFINED PROPERTIES { r_1 , r_2 , ... }.
Serialization of Terms	
Identification	TERM n (id: ID_N).
Parent-Term Relationship	TERM n (id: ID_N) BELONGS TO m (id: ID_M).
Type Declaration	TERM n (id: ID_N) IS OF TYPE T .
Boolean Flags	TERM n (id: ID_N) IS ABSTRACT; IS DERIVED.
Stereotype and Assignments	TERM n (id: ID_N) HAS STEREOTYPE S ; HAS PROPERTY ASSIGNMENTS { $k:v$, ... }.

6.2.4 Embedding Alignment and Dimensionality Reduction

Once conceptual modeling elements are serialized into CMT and paired with their corresponding NLT, we align their representations in a shared vector space. The alignment process begins with reducing the dimensionality of high-dimensional embeddings to enable qualitative visual inspection and statistical analysis.

Dimensionality Reduction. To visualize how NLT and CMT embeddings relate in latent space, we apply t-distributed stochastic neighbor embedding. First, we take two sets of aligned vectors: NLT_EMBEDDING and CMT_EMBEDDING, and vertically stack them into a single matrix for joint processing.

Pretrained Embeddings Models. To populate the aligned space with semantic vectors, we employ an embedding model via the `ollama` interface. Specifically, we use the `ALL-MINILM` model to embed natural language descriptions of OntoUML models.

6.3 Petri Net of Thoughts (PNoT)

As established in the preceding sections, embedding spaces provide a powerful substrate for aligning heterogeneous modalities—text, vision, audio, and structured data—into a shared representational continuum. Within these spaces, semantic shifts emerge as dynamic transformations that capture how meanings evolve across modalities, contexts, or interactional turns. For instance, a gesture observed in video, a spoken utterance, and a textual annotation may all project into neighboring regions of the joint embedding space, yet their local trajectories reveal how the underlying concept unfolds or changes during interpretation.

However, while embedding spaces allow for continuous semantic reasoning, they remain inherently static in representing the temporal and causal dynamics of thought. In real-world analytical scenarios—such as interpreting a clinical conversation, a design workshop, or a human–AI interaction session—reasoning unfolds not as a trajectory through a vector space, but as a structured process involving dependencies, feedback loops, and decision gateways. These dynamic aspects of cognition and communication cannot be fully captured through embeddings alone.

To address this gap, we introduce *Petri Net of Thoughts (PNoT)*, a formal mechanism that bridges semantic embeddings with process-oriented reasoning. PNoT provides a discrete, analyzable structure in which each token movement corresponds to a shift in semantic activation, allowing us to model the interplay between multimodal evidence and cognitive flow. By combining the fluidity of embedding-based representation with the rigor of process modeling, PNoT transforms abstract reasoning traces into executable and interpretable models.

Conversations—both human–human and human–machine—play a particularly important role in this transition. They exemplify multimodal data streams where verbal, visual, and contextual signals intertwine over time. In such settings, each utterance or gesture can be viewed as a transition firing within a cognitive process, redistributing attention and knowledge across conceptual nodes. Mapping these conversational shifts onto Petri Net structures enables a richer form of multimodal data analysis—one that captures not only what is being discussed, but how thought itself propagates and transforms across modalities.

Conceptual Modeling Perspective. From a conceptual modeling perspective, process representations are indispensable because they provide the structural backbone through which meaning, intention, and causality can be systematically encoded. Conceptual models are not mere collections of static entities; they describe how phenomena evolve and interact within a domain. By adopting process formalisms such as Petri

Nets, we can translate the dynamics of reasoning and communication into analyzable constructs—places, transitions, and tokens—that correspond to conceptual states, operations, and knowledge flows. This alignment ensures that the resulting models are not only semantically grounded in multimodal data but also structurally consistent with established modeling practices, bridging the gap between cognitive reasoning and formal conceptual representation.

6.3.1 PNoT: The General Perspective

Prompt engineering has emerged as a critical technique for enhancing the reasoning capabilities of Large Language Models / AI. Techniques such as *Chain-of-Thought (CoT)* prompting guide LLMs in breaking down problems into sequential intermediate steps [WWS⁺23]. More recent paradigms, such as *Tree-of-Thoughts (ToT)* [YYZ⁺23] and *Graph-of-Thoughts (GoT)* [BBK⁺24], expand CoT by allowing branching and richer dependency structures in the reasoning process. In parallel, the field of *process mining* has developed formal methods for modeling and analyzing processes using *Petri Nets* [Rei85, vdA12]. Petri Nets consist of *places* and *transitions* connected by arcs, providing a mathematically rigorous framework for modeling concurrency, causality, and synchronization [Mur89]. Process discovery techniques, such as the Alpha Miner algorithm, analyze event logs to infer a Petri Net model that represents the underlying process [vdAWM04]. Despite the advancements in LLM prompting strategies within the context of business processes [FHV⁺24, CBS23], there remains a gap in methodologies that explicitly integrate formal process modeling techniques to enhance LLM reasoning—enabling traceable, logically consistent, and domain-aware decision-making.

In this section, we propose **Petri Net of Thoughts (PNoT)**—a prompting approach that marries process mining with LLM prompting. PNoT treats the reasoning process as an event log, uses process discovery to derive a Petri Net, and then employs this net to guide the LLM during inference. This approach aims to capture the business logic of a process in a formal sense which replicates how decisions are made by participants in real-world scenarios. Unlike CoT, ToT, and GoT, where the reasoning schedule might be abstract or loosely structured, PNoT uses well-defined (discovered) places and transitions to simplify and focus decision-making. This concrete structure aims to help isolate moments of sequential and concurrent reasoning, allowing for a more organized and contextual aware reasoning. PNoT aims to ensure that an LLM’s reasoning follows a process structure modeled on evidence (via process discovery) or confidence (via domain experts) levels, thereby aimed at enhancing the transparency and verifiability of its intermediate steps. By mapping the internal states of the LLM onto Petri Net components, PNoT aims to enable a systematic exploration of possible reasoning paths. This formalism facilitates error analysis and debugging by allowing practitioners to trace back through the net’s transitions, thereby illuminating where and why an inference may have gone astray. Moreover, the integration of process mining techniques into LLM prompting opens the door for adaptive reasoning, where the system can dynamically adjust its strategy based on feedback extracted from the event log, ultimately aiming at

enhancing both robustness and performance on complex, multi-step tasks.

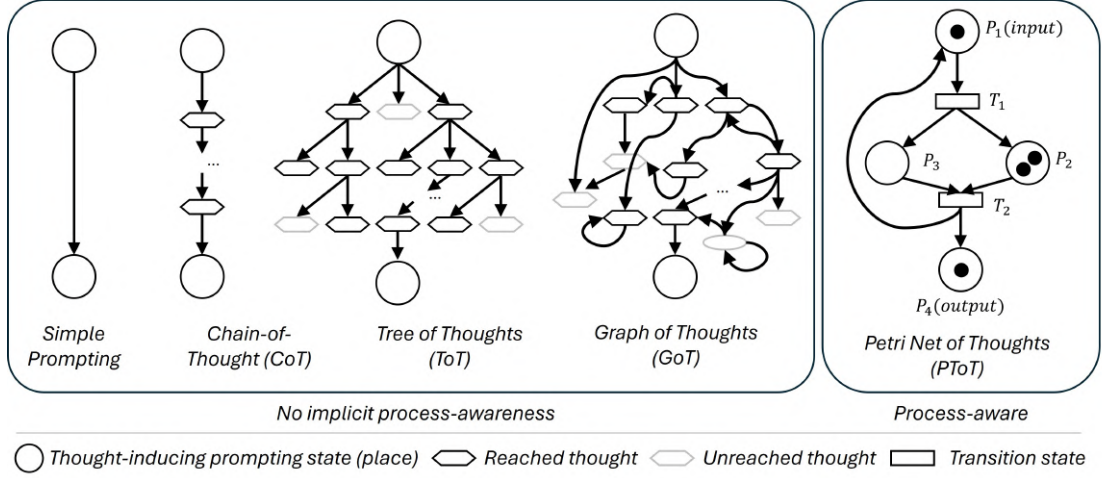


Figure 6.5: Petri Net of Thoughts Conceptual Overview: Orchestrating LLM Prompting with a Petri Net compared to other structure-enhanced prompting approaches. Adopted from Gavric et al. [GBP25e].

Fig. 6.5 conceptually illustrates how the Petri Net guides the prompting process.

6.3.2 An Illustrative Scenario for PNoT

Over the past few years, a variety of prompting paradigms have been introduced to enhance the reasoning capabilities of LLMs. Early work such as *Chain-of-Thought (CoT)* prompting [WWS⁺23] laid the groundwork by guiding LLMs to generate intermediate reasoning steps. Building on this idea, the *Tree-of-Thought (ToT)* framework was proposed in multiple forms. For instance, one variant [YYZ⁺23] focuses on a systematic tree structure that encapsulates reasoning hierarchies. Subsequent methods have further diversified this line of inquiry. The *Tree of Uncertain Thought* [MX23] and *Tree-of-Mixed-Thought* [HQL⁺23] approaches tackle ambiguity and the integration of heterogeneous reasoning pathways, respectively. Similarly, *Skeleton-of-Thought* [NLZ⁺24] strips down the reasoning process to its essential components, aiming for a minimalist yet effective structure. More recently, researchers have introduced the *Graph of Thought (GoT)* method [BBK⁺24], which represents reasoning as interconnected nodes and edges to better capture complex dependencies, while the *Everything of Thoughts (XoT)* approach [DZW⁺24] proposes an even more comprehensive framework that leverages pre-trained reinforcement learning and Monte Carlo Tree Search. The *Hypergraph-of-Thought* model [YTL⁺23] generalizes these ideas by accounting for higher-order relationships between reasoning elements. Parallel to these innovations, techniques emphasizing *Thought Propagation* [YHY24] and *Socratic Questioning* [QXS⁺23] have been developed to foster iterative refinement of reasoning. These methods encourage the LLM to continuously

reexamine its assumptions and validate each inference step, thereby improving both transparency and robustness.

In contrast to the aforementioned approaches that primarily focus on the structure of the reasoning process itself, our work introduces the Petri Net of Thoughts. PNoT leverages process mining techniques to embed business logic within a formal model. By combining the strengths of formal process modeling with advanced prompting techniques, we aim to more effectively capture and replicate real-world decision-making in a controlled, analyzable way.

Consider a clinical decision-support system designed to assist surgeons during emergency operations. Traditional reasoning frameworks such as Chain-of-Thought or Tree-of-Thought would guide the model through a series of verbalized reasoning steps—e.g., identifying symptoms, assessing vital parameters, and deciding on an intervention. However, these approaches lack an explicit formalism for representing concurrency, synchronization, and feedback loops inherent in real surgical workflows.

In contrast, the PNoT models each reasoning element as a *place* and each inference or decision as a *transition* within a Petri Net. Tokens represent evolving states of knowledge—such as available patient data, visual cues from the operating room, or model-generated hypotheses. When a new observation arrives (e.g., a sudden drop in oxygen saturation), it triggers a transition that redistributes tokens across the network, activating relevant subnets corresponding to potential diagnoses or interventions.

For example, one subnet may represent the process of evaluating airway obstruction, while another models the decision to administer anesthesia or call for an additional surgical team. These subnets can execute concurrently, reflecting the parallel reasoning typical of an expert surgeon’s cognitive process. The resulting trace of token flows can be analyzed post-hoc to reveal why certain decisions were made, how evidence influenced the flow of reasoning, and where critical bottlenecks or misjudgments occurred.

By explicitly structuring the reasoning process as a dynamic system rather than a linear chain, PNoT provides both interpretability and control. It not only records the *sequence* of thoughts but also their *causal dependencies* and potential for synchronization—making it possible to replay, simulate, or even refine reasoning pathways using process mining techniques.

6.3.3 A Framework for The Petri Net of Thoughts Paradigm

Our novel paradigm integrates formal process modeling with LLM prompting to create the Petri Net of Thoughts. This approach is composed of two major components:

1. **Process Model Construction:** We capture the business logic either by directly incorporating an expert-designed process model or by discovering one from event logs. The resulting Petri Net consists of (A) **places (states)** that represent the distinct states or conditions in the business process, and (B) **transitions** that denote the events or decision points that lead to state changes.

2. **Orchestration of the Prompting Chain:** We orchestrate the prompting chain of the LLM by executing the Petri Net. Each token replay step corresponds to the progression through a place and the activation of a transition, where the LLM is guided by system prompts that encapsulate the semantics of the corresponding state or event.

The definitions introduced in Section 4.2.3 formalize the Petri Net of Thoughts (PNoT) as a structured reasoning paradigm grounded in the execution semantics of Petri Nets. A Petri Net (Definition 38) provides the underlying control-flow structure, whose places and transitions are reinterpreted as cognitive constructs through thought states (Definition 39) and thought transitions (Definition 40). Reasoning proceeds over enriched markings (Definition 41) that combine token distributions with accumulated LLM outputs. At each step, the system generates either a place-level summary prompt (Definition 42) or a transition-level directive prompt (Definition 43), composed via a principled operator (Definition 44) that integrates contextual memory with decision logic. A PNoT execution then unfolds as the iterative application of these operators while firing enabled transitions, ultimately yielding a transparent and reproducible reasoning trace. Collectively, these components define the PNoT framework (Definition 45) as a formally grounded mechanism for orchestrating LLM reasoning in alignment with domain-specific process models.

Algorithm 3 Petri Net of Thoughts (PNoT)

Input: Initial marking M_0 ; Petri Net $\mathcal{N}$; event log $\mathcal{L}$

Output: Execution trace recorded in $\mathcal{L}$

```

19 Initialize current marking  $M \leftarrow M_0$ 
20 while not IsFinalMarking( $M, \mathcal{N}$ ) do
21   Determine enabled transitions  $T_{\text{enabled}} \leftarrow \text{GetEnabledTransitions}(M, \mathcal{N})$ 
22   if  $T_{\text{enabled}} = \emptyset$  then
23     // No enabled transitions available
24     break
25   end
26   foreach transition  $t \in T_{\text{enabled}}$  do
27     Compose prompt  $p \leftarrow \text{ComposePrompt}(M, t)$ 
28     Query language model  $r \leftarrow \text{LLM\_Prompt}(p)$ 
29     Log execution event  $\text{LogEvent}(\mathcal{L}, M, t, r)$ 
30     Fire transition  $M \leftarrow \text{FireTransition}(M, t, \mathcal{N})$ 
31   end
32 end
33 return  $\mathcal{L}$ 

```

The *token replay algorithm* [vdA12] simulates the execution of the Petri Net by sequentially activating transitions based on the current marking (i.e., distribution of tokens over

places). At each transition, the LLM receives a system prompt that reflects both the state (place) and the decision (transition) to be executed. Below is the pseudocode for prompt orchestration based on the token replay:

The function `ComposePrompt` in the pseudocode constructs a prompt that integrates both the information from the current place (state) and the specifics of the transition (decision), ensuring that the LLM receives a clear, context-rich directive at every step.

At the **place level**, the `ComposePrompt` function aggregates information that captures a summary of the events that have occurred up to the current state. This involves extracting key insights from the event log, synthesizing the rationale behind past decisions, and presenting a concise overview of the context. By summarizing what has happened, the prompt helps the LLM recall and maintain coherence over the reasoning process, ensuring that the model is well-informed about the established business logic and the current state of the process. This summarization acts as an anchor, reinforcing the sequence of events and the contextual background necessary for making informed decisions.

For the **transition level**, the `ComposePrompt` function is designed to articulate the reasoning required to move from the current state to the next business state. This step involves formulating a clear, directive prompt that specifies the decision point at hand, the potential options, and the criteria or constraints drawn from the business process model. Here, the prompt explicitly challenges the LLM to evaluate how to bridge the current state to a desired future state, engaging in a form of guided reasoning that mimics Socratic questioning and thought propagation. The resulting prompt not only instructs the LLM on the necessary analytical steps but also ensures that the transition is executed in alignment with the underlying process logic. Unlike the Graph of Thoughts approach, which allows for flexible and expressive reasoning paths, PNoT uses the formal semantics of Petri Nets to encode decision logic, concurrency, and causality explicitly—offering more structured and transparent reasoning aligned with real-world business processes.

Translation to the reasoning process. In the prompt orchestration, a system prompt is responsible for interpreting the initial context and adding a system prompt that corresponds to a place or a transition. This aim to ensure that the LLM appropriately structures its response based on the nature of the state within the process: (1) if it is a place, summarize the accumulated knowledge up to this state; (2) if it is a transition, generate a directive to guide the next step in the reasoning process. Unlike Graph of Thoughts, which allows arbitrary branching and often lacks grounded semantics, PNoT constrains the reasoning trajectory through formally defined transitions and decision logic—aimed at supporting more transparent, reproducible, and domain-aligned reasoning paths.

Example: From MMPM Model to PNoT Prompts. To illustrate the integration of Multimodal Process Mining (MMPM) with the Petri Net of Thoughts (PNoT), consider a simple assembly scenario discovered from synchronized video, audio, and depth data. MMPM identifies recurrent actions such as *component preparation*, *assembly*, and

inspection, which are translated into a Petri Net where places represent stable operational states and transitions capture observable actions or decision points.

This discovered Petri Net is directly reused as a PNoT. When a token resides in a place, the LLM receives a state-level prompt summarizing the accumulated multimodal evidence and prior reasoning. When a transition becomes enabled, the LLM is issued a directive prompt that guides the next reasoning step in accordance with the encoded process logic. Table 6.2 summarizes the mapping between PNoT elements and their corresponding prompt roles.

Element	Name	Prompt Role
Place	Components Prepared	Summarize observed preparation actions and restate the current process context.
Transition	Assemble Part	Generate a directive to perform the assembly step under discovered constraints.
Place	Assembly Completed	Condense multimodal evidence of the executed assembly.
Transition	Inspect Quality	Evaluate the outcome and decide whether to proceed or rework.

Table 6.2: Mapping from an MMPM-discovered Petri Net to PNoT place- and transition-level prompts.

6.4 Multimodal Process Mining (MMPM)

To operationalize the Petri Net of Thoughts and similar reasoning frameworks, it is necessary to first derive explicit process structures from multimodal data. In other words, before thoughts or decisions can be represented as token flows in a Petri Net, the underlying activities, transitions, and dependencies must be extracted from the raw evidence of human behavior and other observations of SUI. This discovery phase is the domain of **Multimodal Process Mining (MMPM)**—a methodology that extends traditional event-log-based process mining to incorporate heterogeneous data sources such as video, audio, and depth streams.

Unlike conventional logs, multimodal traces are not prestructured; they must be interpreted and synchronized across temporal and semantic dimensions. For instance, a single conceptual action (e.g., “assembling a component”) might simultaneously manifest as an auditory signal (“click” sound), a visual pattern (object alignment), and a depth change (hand motion). Capturing such relationships enables the reconstruction of process models that more faithfully represent real-world activity sequences, rather than symbolic simplifications. Figure 6.6 illustrates how actions are recognized and fused across modalities to form process-level evidence. In this example, auditory cues are used to identify discrete actions such as a “click,” while depth and visual data capture continuous gestures like “dragging.” Each recognized multimodal event is timestamped and aligned within a unified timeline, which is then processed by the multimodal process miner to

produce structured event representations. These representations serve as the empirical foundation for constructing formal models—ultimately feeding into PNoT for reasoning, analysis, and simulation.

By discovering process structures directly from multimodal streams, we bridge the gap between human perception and conceptual modeling. The extracted models not only reveal how actions unfold in time but also provide a precise substrate upon which higher-level reasoning mechanisms, such as PNoT, can operate. This integration allows conceptual modeling to move beyond static diagramming toward data-driven, dynamically grounded representations of behavior and cognition.

6.4.1 MMPM: The General Perspective

Given a process P , let there be a set of N modalities $M = \{m_1, m_2, \dots, m_N\}$, where each modality m_i provides a unique perspective of the process. Let, for $1 \leq i \leq N$, the set D_i represent the dataset of raw data (typically comprising unstructured data such as depth sensing maps, audio, and raw pixels of video frames) associated with modality m_i , reporting on a sequence of events $E_i = \{e_{i,1}, e_{i,2}, \dots, e_{i,O_i}\}$ observed through modality m_i . Each event $e_{i,j}$ in E_i is characterized by a tuple $(a_{i,j}, t_{i,j}, v_{i,j})$, where $a_{i,j}$ is the activity, $t_{i,j}$ is the timestamp, and $v_{i,j}$ is a vector of modality-specific attributes observed for the event.

Our objective is to construct a comprehensive event log EL that accurately represents the sequence and characteristics of activities in the process as observed across all modalities in M . Formally, we seek to optimize the following objective function:

$$\max_{EL} \sum_{i=1}^N \lambda_i \cdot \Phi(PM, D_i)$$

where $\Phi(EL, D_i)$ is a function that measures the fidelity of the event log EL with respect to the dataset D_i of modality m_i , and λ_i is a weighting factor that denotes the importance or reliability of modality m_i in the overall process understanding.

The challenge lies in accurately integrating the disparate and possibly conflicting information from different modalities to construct a process model that is both comprehensive and faithful to the observed data. This involves not only aligning events across modalities but also reconciling differences in the granularity, scale, and interpretation of the data.

Definitions 29 and 30 establish a theoretical foundation for the integration of multimodal evidence into process analysis pipelines. While Definition 29) focuses on the discovery and synthesis of process models from diverse multimodal sources, Definition 30) extends this by systematically assessing conformance and compliance.

6.4.2 An Illustrative Scenario for Multimodal Process Mining

To concretize how multimodal process mining contributes to conceptual modeling, consider an illustrative scenario involving the assembly of a medical instrument in a labo-

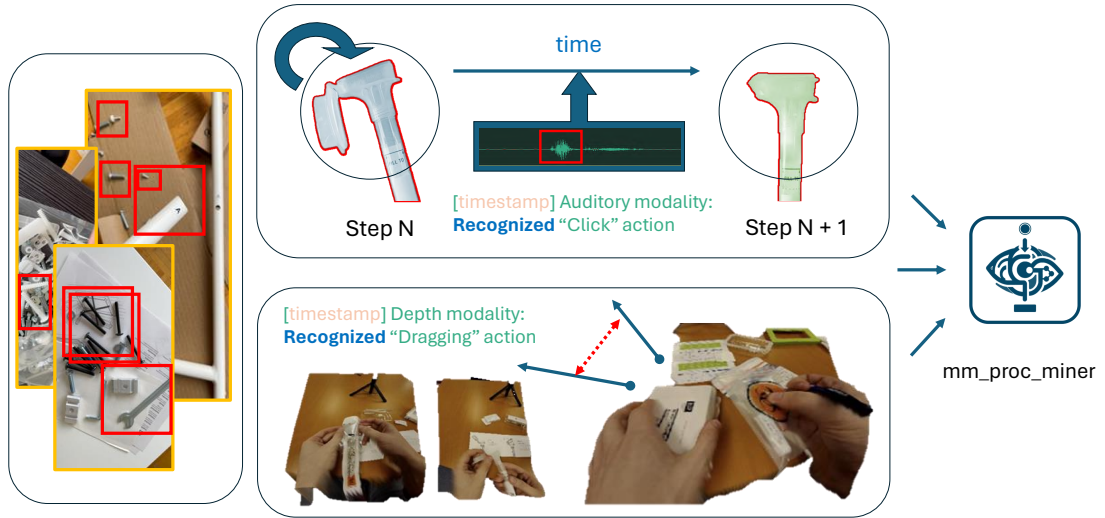


Figure 6.6: Recognition and temporal alignment of multimodal conceptual regularities (auditory, visual, and depth) for process model discovery using the multimodal process miner. Adopted from Gavric et al. [GBP24a].

ratory setting. The task requires a technician to follow a predefined yet flexible procedure—unboxing components, aligning parts, applying adhesive, and verifying connections. Each of these activities unfolds as a sequence of observable multimodal events: hand movements recorded by depth sensors, object manipulations detected in video streams, and characteristic sounds (such as a “click” or “snap”) captured through microphones.

During execution, the multimodal process mining pipeline continuously fuses these heterogeneous data streams into temporally aligned observations. As illustrated in Figure 6.6, the system identifies discrete and continuous actions across modalities—for example, a short auditory spike classified as a *click*, or a prolonged 3D trajectory detected as a *dragging* motion. Each action is assigned a timestamp and modality tag, forming a unified multimodal event trace.

This trace is then processed by the multimodal process miner, which performs two key operations. First, it segments the continuous multimodal input into semantically meaningful units—transforming raw sensor readings into symbolic activity instances. Second, it applies discovery algorithms that detect causal relations, concurrency, and loops among these activities. The output is a process model that represents the actual execution flow observed in the real world, bridging perceptual data and formal modeling constructs.

Such a scenario highlights how multimodal process mining extends beyond classical event logs. Instead of relying on manually recorded data, it autonomously constructs event-level representations from sensor and perceptual input, effectively linking low-level observations to high-level conceptual entities. The resulting process models can be

visualized, analyzed, or further transformed into reasoning structures such as the Petri Net of Thoughts, enabling both behavioral analysis and cognitive interpretation.

Conceptual Modeling Perspective. From a conceptual modeling perspective, multimodal process mining plays a crucial role in grounding abstract model elements in empirical evidence. Traditional conceptual models are often based on interviews, documentation, or subjective elicitation, which can omit subtle procedural nuances and contextual dependencies. In contrast, multimodal traces capture the embodied and temporal aspects of domain activities, providing a data-driven foundation for constructing or validating models. This ensures that conceptual models not only describe intended system behavior but also reflect how processes actually unfold in practice—thereby bridging the gap between design-time abstractions and real-world enactments. By integrating these discovered structures into conceptual modeling frameworks, we enable a new generation of models that are both semantically rich and empirically verifiable.

6.4.3 A Framework for Multimodal Process Mining

To effectively bridge multimodal data streams and conceptual process representations, we developed a unified framework for MMPM. This framework formalizes how perceptual signals—such as video frames, audio cues, or sensor readings—can be systematically transformed into structured event logs suitable for process discovery and analysis. Its design responds to a fundamental challenge in process modeling: while traditional process mining assumes well-structured logs, real-world activities are often captured through unstructured, continuous, and overlapping multimodal observations.

The proposed framework addresses this gap by establishing a modular pipeline that extracts, synchronizes, and interprets multimodal evidence in a way that aligns with process mining principles. Each modality contributes a distinct but complementary perspective on the observed behavior: video captures object interactions, audio reflects temporal markers and verbal cues, while depth and motion data convey embodied actions. By combining these perspectives, the framework enables the reconstruction of process models that more accurately mirror the temporal and cognitive structure of real activities.

Figure 6.7 provides an overview of the architecture. At its core lies the *Vid2log* tool, designed and implemented to incorporate multimodal data in the way processes are documented and analyzed. Large Language Models (LLMs) use tokenization to break down the input data into smaller, manageable pieces, which allows us to leverage the pre-trained capabilities of multimodal LLMs to process and analyze multimodal data effectively.

Tokenization is the process of converting a sequence of characters into a sequence of tokens (e.g., words, subwords, or characters). Multimodal LLMs take the concept of tokenization and apply it to different types (modalities) of data. For simplicity, assume we have a vocabulary V . For an example, given an input text $\mathbf{T} = \text{"Process is finished!"}$, the tokenization process maps this text into a sequence of tokens (assumable): $\mathbf{T} =$

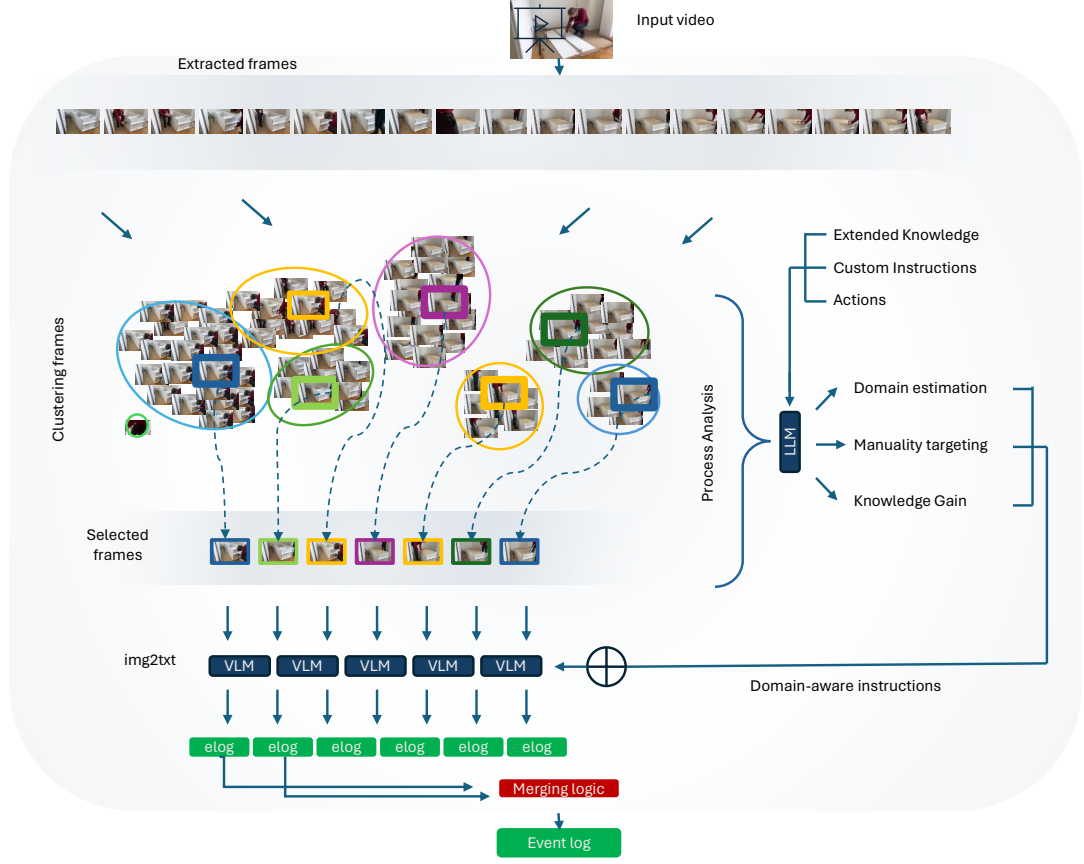


Figure 6.7: Overview of the *Vid2log* pipeline for multimodal process mining. The framework integrates multiple sensing modalities—such as video, audio, and depth data—into a unified processing flow. Adopted from Gavric et al. [GBP24c].

["Process", "is", "finished", "!"]. Each token is then mapped to an integer ID using the vocabulary V : $\mathbf{x} = [x_1, x_2, x_3, x_4]$, where $x_i \in \mathbb{N}$ and corresponds to the index of the token in the vocabulary.

These token IDs are converted into dense vectors using an embedding matrix $\mathbf{E} \in \mathbb{R}^{|V| \times d}$, where d is the embedding dimension. The embedding for each token is $\mathbf{e}_i = \mathbf{E}[x_i]$. The input sequence is then represented as $\mathbf{X} = [\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4]$.

The machine learning architecture that is used for building the predictive model is a *transformer architecture* which primarily consists of self-attention mechanisms and feed-forward neural networks. Self-attention computes a representation of each token in the context of all other tokens in the sequence. Given the input matrix $\mathbf{X}$, self-attention works as follows:

- (1) *Linear Transformations*: Compute the query $\mathbf{Q}$, key $\mathbf{K}$, and value $\mathbf{V}$ matrices: $\mathbf{Q} = \mathbf{X}\mathbf{W}^Q$, $\mathbf{K} = \mathbf{X}\mathbf{W}^K$, and $\mathbf{V} = \mathbf{X}\mathbf{W}^V$, where $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V \in \mathbb{R}^{d \times d_k}$.

(2) *Scaled Dot-Product Attention*: Compute the attention scores:

$$\mathbf{A} = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} \right).$$

Lastly, it applies the attention scores to the values: $\mathbf{Z} = \mathbf{A}\mathbf{V}$.

(3) *Concatenation and Linear Transformation*: If using multi-head attention, concatenate the outputs from multiple attention heads and apply a final linear transformation: $\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_h) \mathbf{W}^O$, where $\mathbf{W}^O \in \mathbb{R}^{hd_k \times d}$ and h is the number of attention heads. After the attention mechanism, the output is passed through a feed-forward neural network, applied independently to each position: $\text{FFN}(\mathbf{z}_i) = \max(0, \mathbf{z}_i \mathbf{W}_1 + \mathbf{b}_1) \mathbf{W}_2 + \mathbf{b}_2$.

A single transformer layer consists of the following components: *Self-Attention Mechanism*, *Add & Norm* (incorporating residual connection and layer normalization), *Feed-Forward Network (FFN)*, and *Add & Norm* (incorporating residual connection and layer normalization). The complete transformer model is formed by stacking multiple transformer layers. The output from the final transformer layer is used for downstream tasks, such as classification or sequence generation. For language modeling, the final output is often a probability distribution over the vocabulary, which is computed using a softmax function applied to the linear transformation of the final hidden states: $\mathbf{y}_i = \text{softmax}(\mathbf{h}_i \mathbf{W}^P + \mathbf{b}^P)$, where $\mathbf{W}^P \in \mathbb{R}^{d \times |V|}$ and $\mathbf{b}^P \in \mathbb{R}^{|V|}$.

We focus on using an LLM to tokenize and analyze text and images. We start with a clustering algorithm to remove similar frames, reducing redundancy and making the analysis more efficient. After filtering redundant frames, we select the most informative frames from video data using a video frame selection logic that focuses on identifying blind spots, especially in activities that are very manual or physical, as described in the following. This ensures that we choose frames rich in relevant information for the business process we are studying. Then our solution employs a multimodal LLM to infer the domain of the process. Leveraging this inferred domain knowledge, the system then utilizes instructions aimed at maximizing *Knowledge Gain* that we will define as follows. Our solution enriches both, videos and conventional event logs, with responses to event-related questions like *who*, *when*, *where*, *how*, and *why*. Subsequently, these partial event logs, derived from various segments of the video and conventional logs, are combined into a final, unified event log. This amalgamation is achieved through a process of *Temporal semantic matching*, where the system aligns and merges logs based on their occurrences in time and contextual relations.

Video Frame Selection

The first step in focusing on business-relevant parts of video is done through a reduction of the number of frames to be considered from the video, for which we designed a clustering-based selection.

Let N be the total number of frames available, and let K represent the number of clusters to be formed. We define C_i as the i -th cluster, where i ranges from 1 to K , and let F_k denote the selected frame from cluster C_k , where k ranges from 1 to K .

The clustering-based selection process consists of the following steps: (1) First, clustering using K-Means, where the N frames are clustered into K clusters using the K-Means algorithm: $C_1, C_2, \dots, C_K$. (2) Then, selecting centroids as selected frames, where the centroid of each cluster is chosen as a selected frame: $F = \{\text{Centroid}(C_1), \dots, \text{Centroid}(C_K)\}$.

Our objective function for clustering and centroid selection is thus:

$$G(F) = \sum_{i=1}^K \sum_{j \in C_i} \text{similarity}(\text{Centroid}(C_i), \text{Frame}_j).$$

This objective function calculates sum of similarities between each selected centroid and all frames within the cluster it represents.

Search for Business-Relevant Activities

We use the concept of *Domain Estimation* and *Manuality* as a way of communication with an LLM to narrow the scope of all activities and resources present in raw data to a set of business-relevant process evidence (see Fig. 6.7).

Firstly we prompt an LLM to estimate a business domain of our data. Then we retrieve a set of business-relevant activities and resources for the estimated domain, again, by prompting an LLM. Our search for business-relevant activities continues through prompts about activities with high Manuality. Manuality is a metric value that measures the level of difficulty in digitally tracking a process without employing multimodal analysis techniques. Manuality seeks to quantify the extent to which a process necessitates the integration of multiple data streams or modes to gain a comprehensive understanding.

The concept of Manuality stems from the recognition that many real-world processes involve complex interactions that cannot be fully captured by single-dimensional data sources. For instance, tracking the quality of a manufacturing process might require the simultaneous analysis of sensor data, visual inspection, and audio feedback. In such cases, the level of Manuality would be high, indicating a significant reliance on multimodal analysis to monitor and control the process effectively. Processes with high Manuality values require more sophisticated data collection methods and technology investments to ensure accurate monitoring and control. Conversely, processes with low Manuality values are more amenable to straightforward, unimodal tracking. Understanding the Manuality of different processes within an organization can guide technology adoption strategies and help prioritize data analytics initiatives.

Image-to-Text Conversion

Once the clustering step is finished and the business-relevance filters are applied, we use extracted evidence representatives as a prompt to an LLM to perform *Image-to-text*

conversion (img2text) in accordance to the evidence-relevant information (*who*, *when*, *where*, *how*, and *why*) for each step of a process.

Firstly, the 'who' aspect of prompting focuses on identifying actors involved in each step of the process. By querying the LLM about the individuals or entities participating in a step, we gain insights into roles, responsibilities, and interactions. This aspect of information is used for understanding the human or systemic agents driving the process. Secondly, addressing the 'when' component involves determining the timing and sequence of each process step. Prompting the LLM to extract temporal data ensures a chronological understanding of the process flow. This is used for analyzing the timeline of the process. By identifying when each step occurs, the LLM helps in constructing a temporal map of the process, which is essential for optimization and monitoring. We structure this temporal information not to be defined by a timestamp (as a timestamp is already available from the frame position in the video) but to be relational, giving an answer to more specific question - *after what* and *before what* this particular activity happens. The 'where' prompt searches into the spatial context of each process step. Understanding the location or environment where each step occurs is used for assessing resource allocation, environmental constraints, and logistical considerations. The 'how' aspect is concerned with the methods, tools, and procedures employed in each step. By prompting the LLM to analyze the modes of operation, techniques, and resources utilized, we obtain a detailed understanding of the execution of each step. Lastly, the 'why' prompt aims to uncover the rationale or purpose behind each step. Understanding the objectives and reasons for each action provides a strategic perspective of the process. By prompting the LLM to analyze the underlying motivations and goals, we gain insights into the process's relevance.

6.4.4 Multimodal Knowledge Gain

Another step of structuring evidence data into a more business-relevant event log representation involves filtering activities and resources based on the knowledge they can contribute if they are included in the log. We perform this using a *Knowledge Gain* filter. Knowledge Gain is the potential for acquiring additional information, insights, or data about a particular process or activity by incorporating various digital technologies and data sources. High Knowledge Gain potential are associated with processes that have a high level of Manuality. Knowledge Gain considers the increase in information or data obtained by incorporating additional digital tools or data sources.

Let K_{initial} represent the initial level of knowledge or information available about a process through traditional IT system interactions or event logs. Let K_{enhanced} represent the increased level of knowledge or information gained by incorporating additional digital technologies, such as cameras, sensors, and other data sources. Then, Knowledge Gain (KG) are represented as: $KG = K_{\text{enhanced}} - K_{\text{initial}}$.

The difference between K_{enhanced} and K_{initial} quantifies the increase in knowledge due to the incorporation of digital technologies and additional data sources. We can represent

Knowledge Gain using equations based on entropy, which is often used to measure uncertainty or information content. Let P_{initial} be the probability distribution of events or states before incorporating additional data sources or digital tools. Let P_{enhanced} be the probability distribution of events or states after incorporating additional data sources or digital tools. The entropy H is then defined as:

$$H(P) = - \sum [p(x) \log_2(p(x))]$$

where: $H(P)$ is the entropy of the probability distribution P , and $p(x)$ represents the probability of a specific event or state x in the distribution.

Then, Knowledge Gain (KG) is represented as the difference in entropy: $KG = H(P_{\text{initial}}) - H(P_{\text{enhanced}})$. We can prompt an LLM to estimate P_{initial} , while P_{enhanced} we can calculate from the video.

Merging Logic for Unifying Multimodal Event Logs

The merging logic for unifying multimodal event logs involves methods to integrate information from various event-log sources while maintaining semantic coherence and temporal accuracy. Temporal stamps (available directly from videos) play a critical role in this process, as they allow for the chronological ordering of events. By aligning events based on their timestamps, the merging logic ensures that the sequence of occurrences is accurately preserved. This temporal alignment is essential for understanding the flow of activities and their interdependencies over time, which is crucial for subsequent analysis and decision-making on case identifiers (IDs).

Beyond merely comparing timestamps, the merging logic also examines semantic relationships between concepts. This involves analyzing the content and context of events to identify meaningful connections and associations. For instance, events that describe similar or related activities, even if they occur at different times, are grouped together to provide a more comprehensive understanding of a process. This semantic analysis helps in clustering related events, thereby enhancing the interpretability of the event logs. It also aids in detecting patterns and trends that might not be evident through temporal analysis alone.

After ordering events and identifying semantic relationships, the merging logic estimates case IDs based on resource usage. Resources, such as personnel or equipment involved in specific events, provide clues for linking related events. By tracking the involvement of resources across different events, the merging logic infers which events belong to the same case or process instance. This estimation of case IDs ensures that events are correctly attributed to their respective cases, facilitating accurate process analysis and reporting.

Scaling from Video to Other Modalities

While the initial implementation of the Vid2log framework focused on video as the primary modality, the architecture was designed from the outset to accommodate additional

data sources such as audio, text, and physiological or motion sensors. Each modality contributes a complementary perspective on human activity, and their integration allows for a richer and more robust reconstruction of process behavior.

In video-based analysis, temporal sequences of frames are segmented into atomic motion units that represent distinct actions or state changes—such as “grasping,” “placing,” or “rotating.” These visual primitives serve as the foundation of the multimodal event log. When extending to audio, temporal synchronization becomes essential: short bursts of sound (e.g., clicks, knocks) are interpreted as discrete events, while continuous patterns (e.g., speech or ambient noise) are represented through sliding-window features that preserve prosodic and semantic variation over time. Textual inputs—such as transcripts, operator annotations, or system messages—are directly tokenized into linguistic units aligned with the corresponding temporal intervals.

Tokenization serves as the unifying mechanism across all modalities. For textual data, conventional subword or byte-pair encodings (BPE) are used to decompose sentences into semantically meaningful tokens that maintain linguistic granularity. For audio, the process involves converting waveforms into spectrogram representations and subsequently applying vector quantization or learned audio tokenizers (e.g., WHISPER or AUDIOLM encoders) to derive tokens that correspond to short acoustic segments or phonetic features.

Once tokenized, these heterogeneous representations are projected into a shared embedding space, enabling crossmodal reasoning through multimodal LLMs (MLLMs). This unified token-level abstraction allows the framework to leverage pre-trained capabilities of MLLMs—such as pattern recognition, semantic inference, and event summarization—across modalities without the need for modality-specific retraining.

In conceptual modeling terms, this scaling enables models to capture both what happens (through visual and sensor evidence) and why it happens (through linguistic and contextual cues). The multimodal tokenization thus forms the basis of an integrated representational layer that links perceptual data to conceptual abstractions—paving the way for process discovery and reasoning methods such as the PETRI NET OF THOUGHTS to operate over truly multimodal evidence.

Back in Chapter 3, we provided a Figure 3.2 that illustrates how heterogeneous data sources converge within the proposed multimodal process mining framework. At the center lies the *business process*, which acts as the semantic anchor linking all streams of observation. The figure distinguishes five primary modalities:

- **Visual observations** capture both static and dynamic information, including tools, materials, movements, and interactions between humans and objects. These data streams form the spatial and relational basis for detecting process-relevant activities.
- **Auditory observations** provide temporal markers and communicative context, such as speech, clicks, alarms, or engine sounds, which can indicate the initiation, progression, or completion of actions.

- **Unstructured sensor observations** (e.g., temperature, depth, light, pressure, motion) enrich the interpretation by offering environmental or physiological context—crucial for understanding conditions under which processes occur.
- **Machine log observations** combine automatically recorded system data (e.g., database entries, RFID scans, software logs) with manually entered textual notes, representing the digital trace of operational systems.
- **Textual and linguistic data**—often embedded within logs or transcripts—act as semantic carriers that align human intent with observed behavior, connecting empirical activity data with conceptual abstractions.

In conceptual modeling terms, the framework provides the necessary infrastructure to link *what is observed* (empirical reality) with *what is modeled* (conceptual abstraction), thus enabling data-driven enrichment, validation, and evolution of conceptual models.

6.4.5 Crossmodal Process Mining

Despite the advances in multimodal process mining, most existing techniques rely on *multimodal fusion*, processing each data stream in parallel and combining outputs only at a late stage. As surveys on unstructured process data highlight [KAR⁺23, KEK⁺25], this late fusion approach can struggle when modalities are noisy, incomplete, or missing. Drawing inspiration from crossmodal and transmodal analysis—where one modality can augment or replace another via joint embedding spaces [WLZ⁺25, SWR25]—we advocate a paradigm shift toward *crossmodal process mining*: an approach that enables the alignment, substitution, and enrichment of process-related information across heterogeneous modalities. In this setting, missing video frames could be inferred from textual logs, or ambiguous sensor readings clarified through scene understanding, yielding robustness akin to human perception.

In this section, we develop a crossmodal process mining framework that leverages joint embedding spaces to reason across heterogeneous modalities, such as video and sensor logs. Definitions 31–34 delineate a clear hierarchy of modality integration within process mining. As illustrated in Figure 6.8, the transition from *Monomodal* (Definition 31) to *Multimodal* (Definition 29) extends classical event logs with heterogeneous sensory and contextual data. *Panmodal Process Mining* (Definition 32) generalizes this view by encompassing both *Crossmodal* (Definition 33) and *Transmodal* (Definition 34) reasoning. While crossmodal strategies promote partial alignment across selected modalities, transmodal approaches achieve full unification, allowing seamless interpretation and inference across all available data sources. Together, these stages outline the conceptual evolution of process mining toward panmodal intelligence—bridging symbolic, perceptual, and behavioral representations in unified analytical frameworks.

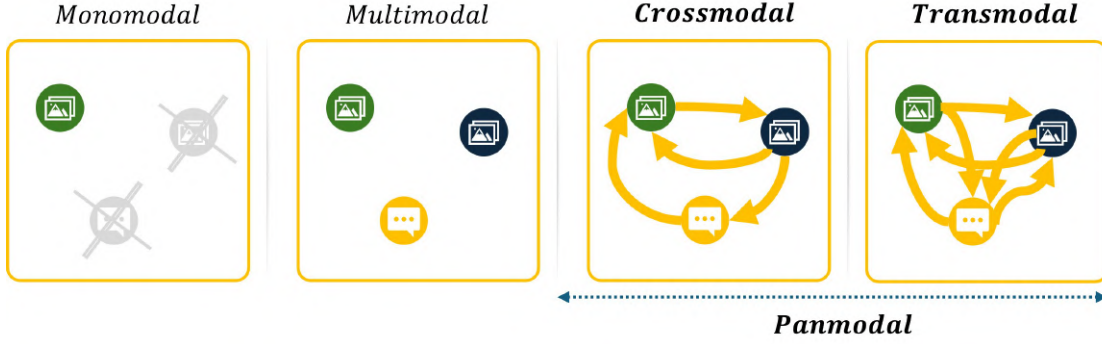


Figure 6.8: Modalities and task support schema adapted for Process Mining: from Monomodal to Transmodal Process Mining. Adopted from Gavric et al. [GBP25d].

6.4.6 Perceptual Tasks and Multimodal Signals

In real-world process scenarios, multiple data modalities capture complementary aspects of activity execution. For instance, a video stream may reveal body posture and object interaction, while sensor logs record machine states, and textual logs document operator decisions.

These perceptual primitives are often processed independently in multimodal pipelines, where each modality is analyzed separately, and fusion occurs at the decision level. For example, a process discovery algorithm may use textual logs for event sequencing, video frames for action classification, and sensor data for status monitoring—each interpreted in isolation and integrated post hoc.

Limitations of Late Fusion and the Case for Crossmodal Reasoning

While multimodal fusion can increase robustness and resolution, it is often brittle when one modality is noisy or missing. Moreover, it treats each data stream as semantically self-contained, neglecting the rich interdependencies between modalities. For instance, a failure to detect an object in video may not be recoverable unless correlated textual evidence is leveraged.

Crossmodal reasoning aims to address this limitation by learning *joint embedding spaces* in which semantically equivalent information from different modalities is mapped to nearby vectors. Formally, let $x_v \in \mathcal{X}_v$ be a video-based representation and $x_t \in \mathcal{X}_t$ a textual representation. A crossmodal encoder maps these into a shared latent space $\mathcal{Z}$ via $f_v : \mathcal{X}_v \rightarrow \mathcal{Z}$ and $f_t : \mathcal{X}_t \rightarrow \mathcal{Z}$ such that $d(f_v(x_v), f_t(x_t))$ is minimized for aligned instances, where d is a suitable distance metric (e.g., cosine or Euclidean distance).

This embedding-based approach allows modalities to inform and substitute for each other. For example, if the video segment x_v is missing or ambiguous, a matching textual trace x_t can be used to retrieve semantically similar video embeddings or generate a surrogate visual representation. Importantly, this does not require the modalities to be aligned

temporally but rather semantically and contextually—a key distinction from traditional fusion methods.

Implications for Process Mining

In the context of process mining, crossmodal reasoning allows to move beyond mere co-occurrence of multimodal events and instead infer latent process states from partial or heterogeneous evidence. This is particularly relevant in object-centric process scenarios, where the same artifact (e.g., a product or document) may be referenced across modalities in fragmented and asynchronous ways.

By using crossmodal embeddings, we aim to align these references, reconstruct missing segments of a process trace, and generate a more coherent and resilient process model. For example, when a process model lacks visual confirmation of an activity due to occlusion or camera dropout, textual descriptions or sensor activations may serve as substitutes—retrieved or interpreted via the embedding space. Thus, crossmodal process mining offers a pathway toward robust, semantically grounded, and context-aware process intelligence.

6.5 Business-Knowledge Integration Cycles (BICycle)

The integration of multimodal data into conceptual modeling pipelines introduces new opportunities, but also new challenges—particularly regarding interpretability, contextual relevance, and the inclusion of tacit business knowledge. Automated pipelines, while effective in structuring raw observations, often lack the capacity to discern subtle, domain-specific nuances or to validate whether extracted process patterns correspond to meaningful business realities. To bridge this gap, we propose the incorporation of a *human-in-the-loop* approach, whereby domain experts and process analysts actively guide, validate, and refine the data-to-model transformation.

This human participation does not merely serve as a corrective mechanism but as a co-evolutionary force that ensures alignment between algorithmic inferences and organizational knowledge. By engaging humans in iterative feedback loops—during event identification, activity labeling, and semantic alignment—the system should become progressively more accurate, interpretable, and value-aligned. Such integration should foster a deeper understanding of the interplay between formal models and lived practices, aiming to enable organizations to capture and operationalize tacit expertise that would otherwise remain hidden from digital traces.

Within this paradigm, we introduce the **Business-Knowledge Integration Cycles (BICycle)**, a framework designed to address the challenges of digital traceability in manually intensive or low-digital-footprint domains. The BICycle framework operationalizes these human-machine feedback loops to generate process-mining-ready event logs from video, audio, and unstructured data. Its human-centered design not only enhances accuracy

and interpretability but also anchors the discovered models in real business semantics and practices.

To meet these challenges, we have designed the BICycle method and developed a BICycle-enabled tool for the creation of event logs, which employs a joint embedding space across modalities together with LLMs, and demonstrates the capability of identifying moments in video, audio, and sensor data that correspond to process-relevant activities.

6.5.1 BICycle: A General Perspective

To transform modalities into an event log, we compute the matching scores between the querying modality and keys of collected modalities. Both keys are queries embeddings over which we perform similarity matching.

To querying modality, we choose text to keep it compatible with conventional event logs and their textual labels (numeric or characters). Framework obtain the embedding vector for the querying text using a pre-trained embedding space, denoted as E_q , and the embedding vector for the key modality, denoted as E_k . The matching score is calculated by performing a dot product between these embeddings, given by $S = E_q \cdot E_k$. This dot product serves as a similarity measure between the two embeddings. Given multiple key embeddings $\{E_k^1, E_k^2, \dots, E_k^n\}$, we compute the dot product for each pair, resulting in similarity scores $\{S_1, S_2, \dots, S_n\}$, where $S_i = E_q \cdot E_k^i$. The softmax function is then applied to these similarity scores to obtain the normalized probabilities P_i for each key modality. This is computed as:

$$P_i = \frac{\exp(S_i)}{\sum_{j=1}^n \exp(S_j)}$$

where $\exp(S_i)$ is the exponential of the similarity score S_i , and the denominator is the sum of the exponentials of all similarity scores. This process ensures that the resulting probabilities $\{P_1, P_2, \dots, P_n\}$ sum to one, providing a robust measure of the likelihood that each key modality matches the querying text. Higher values of P_i indicate a better match between the querying text and the corresponding key modality. Thus, by using the dot product of embeddings and the softmax function, we compute and normalize the matching scores across different modalities, and select the best match.

6.5.2 An Illustrative Scenario for BICycle

Consider the example of a home kit for sampling DNA, a process steeped in the need for accuracy and reliability. As users engage with the kit, they follow a sequence of steps, one of which involves closing a funnel lid to secure a sample. The visual action of closing the lid may seem straightforward, yet the auditory *click* sound is what conclusively indicates the lid has been securely fastened as illustrated in Figure 6.6. This sound not only assures the user of the successful completion of this step but also serves as an auditory event log, marking a crucial point in the process.

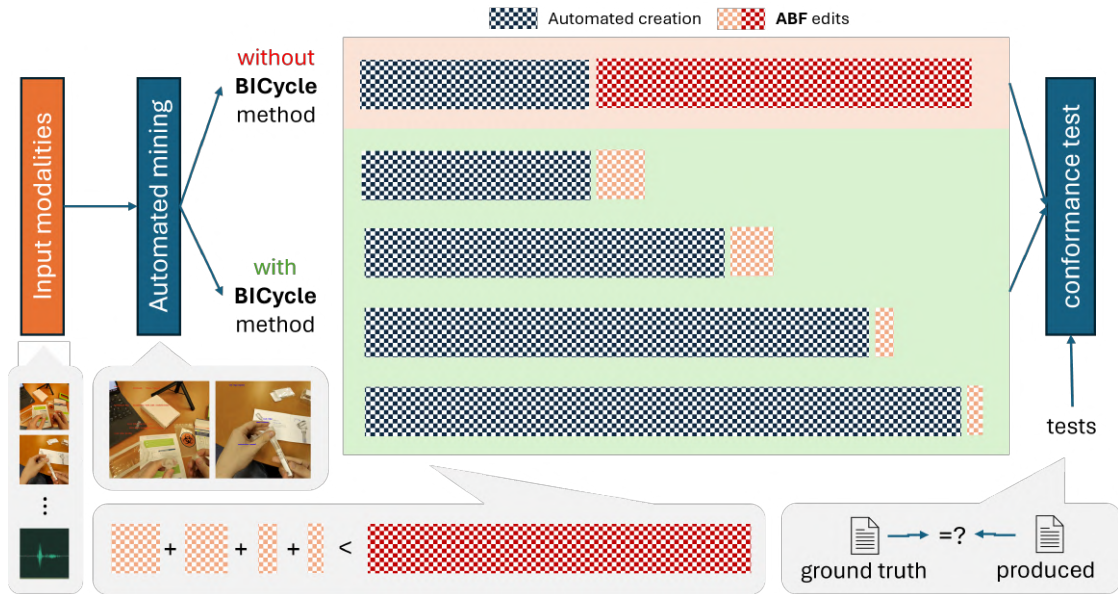


Figure 6.9: The BICycle method. ABF - Atomic Business Feedback operations. Adopted from Gavric et al. [GBP24a].

Capturing auditory cues extends the capability of process mining tools by enabling them to involve humans more directly in the interpretation and supervision of process execution. Sounds such as verbal confirmations, tool interactions, or characteristic acoustic signals act as natural feedback channels through which humans signal progress, uncertainty, or successful task completion. When these cues are integrated alongside visual recordings, multimodal event logs no longer represent only observable actions, but also the communicative and interactional aspects of human behavior. This allows analysts and domain experts to remain actively involved in validating, correcting, and contextualizing extracted process events, thereby strengthening the interpretability and trustworthiness of the resulting models.

Building upon the integration of visual, depth, and auditory modalities, the incorporation of sensor data introduces an additional layer of human-centered control into multimodal event logs. Sensors measuring humidity, temperature, and pressure are particularly relevant in domains where humans must operate within strict environmental constraints, such as DNA sampling. By capturing these measurements and importing them through textual modality, the system enables both automated checks and human oversight of environmental compliance. This supports human-in-the-loop supervision, where experts can inspect deviations, explain anomalies, and intervene when necessary, ensuring that observed human actions are interpreted within their correct physical context.

To evaluate the business-oriented applicability of multimodal conformance checking under human-in-the-loop conditions, we recorded new video data and made the complete dataset publicly available (refer to [GBP24a]). The recordings consist of out-of-internet wild samples, which we guarantee were not used in training any of the underlying LLMs.

Evidence data was collected across two domains, with a primary focus on DNA sampling in home environments. We recorded two instances of DNA sample collection processes using Ancestry DNA test kits and 23andMe DNA test kits. These recordings illustrate how real-time multimodal monitoring can support guided and supervised home-based medical procedures, allowing humans—both end users and experts—to remain actively involved in process execution, verification, and correction. An overview of the collected dataset is provided in Figure 6.10.

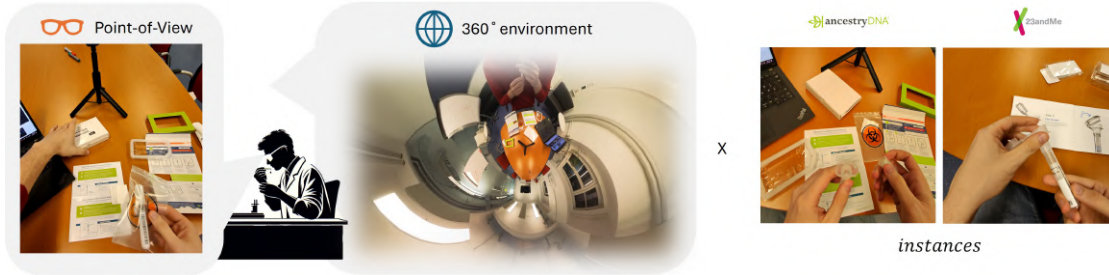


Figure 6.10: An overview of the dataset in the domain of DNA home kit collecting that we recorded, processed, and used for BICycle. Adopted from Gavric et al. [GBP24a].

6.5.3 A Framework for Business-Knowledge Integration Cycles

To facilitate the importance of incorporating domain-specific knowledge into our system, we developed a human-in-the-loop framework which is designed to systematically integrate human (usually a domain expert) feedback into our process mining framework (as illustrated in Figure 6.9). This approach ensures that the generated event logs are not only rich in data from various modalities but are also related to relevant concepts of the business domain of the process.

Atomic Business-Knowledge Feedback (ABF)

The integration of human expertise within multimodal process mining requires mechanisms that allow fine-grained, interpretable interaction between users and AI systems. In this regard, conversational feedback becomes a powerful form of collaborative intelligence—enabling domain experts to express corrections, contextual nuances, and domain semantics in a natural, iterative manner. To formalize this interaction, the *Business-Knowledge Integration Cycle* (BICycle) introduces **Atomic Business-knowledge Feedback (ABF)** operations as its core human-in-the-loop mechanism. These atomic interventions represent the minimal units of expert knowledge that can modify or validate system-generated event logs. Rather than relying on post-hoc manual corrections or opaque retraining, ABF operations embed expert insight directly into the modeling loop—allowing human guidance to dynamically steer the learning and interpretation process. In conversational terms, ABFs act as dialogue acts between human and machine: each correction, renaming, or removal constitutes a semantic utterance that refines the shared understanding of the observed business process.

ABF operations are designed as a granular set of edits that domain experts can perform to refine and enhance the accuracy of the event logs. These operations are crucial for tailoring the automated log generation to the specific contexts and requirements of various fields, making the data more relevant and actionable for users. ABF operations include three actions:

- *Instanting* ABF operation involves specifying a general tool or action mentioned in the event log to a particular named instance. For example, if the automated system identifies a “cutting” action, a domain expert may instantiate this action to a more specific “cutting with a surgical scalpel” in a medical context. This operation increases the specificity and contextual relevance of the event logs.
- *Renaming* ABF operation is design for a fix when terms or labels used by the automated system do not align perfectly with the domain-specific terminology. The renaming operation allows experts to replace these with the correct terminology, enhancing the clarity and usability of the event logs. For example, if the automated system labels an action as “data input,” a domain expert might rename this to “patient information entry” in a healthcare setting.
- *Removing* When certain actions or events captured might be irrelevant or noisy in the context of the specific process being analyzed, system can use the *Removing* ABF operation. The removing operation enables experts to delete these entries, ensuring that the event logs remain focused and pertinent. For example, if the event log contains entries for “background noise detection” in an audio processing task, these can be removed to focus solely on relevant audio events.

The BICycle method incorporates these ABF operations into iterative cycles of feedback and refinement. After an initial set of event logs is generated by the system, domain experts review the logs and apply ABF operations where necessary. The system then integrates this feedback, refining its algorithms and improving the accuracy and relevance of the generated logs. Each ABF operation is encoded as a structured constraint over the LLM’s event abstraction and labeling outputs (e.g., abstraction granularity, label equivalence, or event suppression). These constraints are injected into subsequent LLM inference cycles via updated system prompts, controlled decoding rules, and post-generation validation checks that align generated events with expert-approved semantics. Across iterations, frequently applied ABF constraints are promoted to persistent prompt templates and validation schemas, progressively biasing the LLM toward domain-consistent event log generation. This process is not a one-time effort but rather a continuous cycle of improvement. As more feedback is integrated, the system’s understanding of domain-specific processes deepens, leading to progressively more accurate and actionable event logs.

Incorporating ABF within conversational human–AI workflows transforms the act of model correction into a structured dialogue between the model and its human counterpart. Each

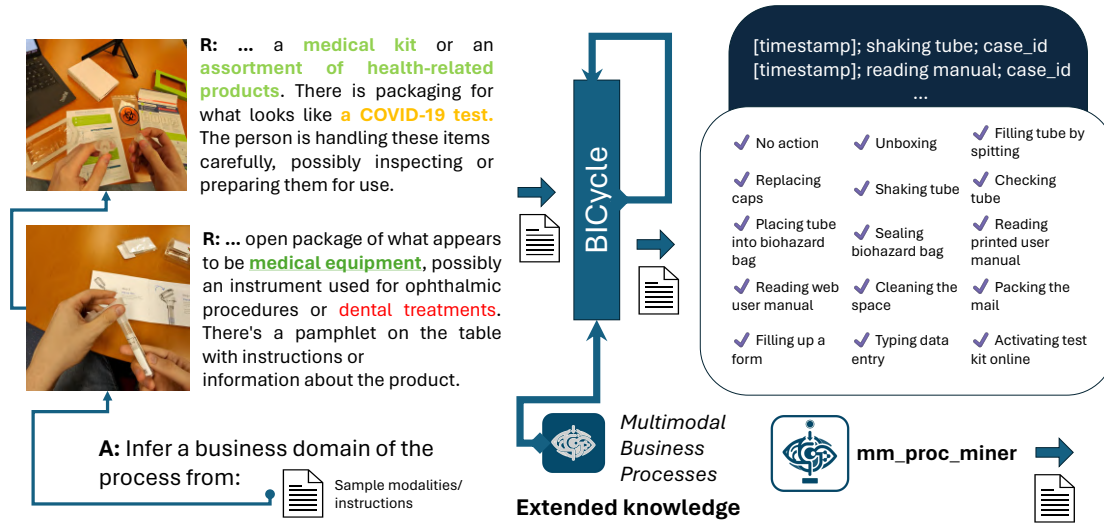


Figure 6.11: Atomic Business-Knowledge Feedback (ABF) operations in BICycle as a dialogical approach that aligns the technical process of event log refinement with human cognitive processes. Adopted from Gavric et al. [GBP24a].

ABF operation—whether instancing, renaming, or removing—acts as a conversational turn that grounds the system’s reasoning in real business semantics. This dialogical approach aligns the technical process of event log refinement with human cognitive processes of negotiation, clarification, and contextualization. Consequently, the ABF mechanism not only enhances model fidelity but also reinforces the broader goal of the BICycle framework: to ensure that multimodal process mining and conceptual modeling evolve through continuous, human-centered integration cycles that capture both the explicit and tacit dimensions of business knowledge.

In the illustrated test scenario in Figure 6.11, the LLM initially abstracts observed actions into generic events such as *Handle equipment* and *Dispose item*. Through ABF instancing and renaming, a domain expert corrects these abstractions by refining *Handle equipment* into *Shaking tube* and *Filling tube by splitting*, and by renaming *Dispose item* to *Placing tube in a biohazard bag*. These ABF operations are encoded as structured prompt constraints and decoding rules, biasing subsequent LLM inference toward healthcare-consistent event labels and yielding more precise and domain-aligned event logs in the next BICycle iteration.

6.6 Relaxing Process Mining into the Embedding Space (RelaxPM)

Traditional process mining operates over discrete, event-based representations, where activities, timestamps, and resources are explicitly recorded in structured logs. While this paradigm has proven effective for digitally mature domains, it assumes that the

underlying events are already well-defined and discretized—a condition that does not hold in many multimodal or cognitively rich contexts. Real-world processes, especially those inferred from videos, audio, or sensor data, often lack clear boundaries between actions or explicit symbolic annotations. Their dynamics unfold continuously across multiple modalities, each capturing complementary fragments of the same underlying behavior.

To address these challenges, we propose to *relax* process mining into the embedding space—performing discovery, alignment, and conformance checking directly on continuous vector representations derived from multimodal evidence. Instead of relying on symbolic event logs, each observation is represented as an embedding vector within a shared latent space, where semantic and temporal similarities between instances can be quantitatively assessed. In this relaxed formulation, process relations emerge not through explicit control-flow constructs but through geometric patterns such as clustering, trajectory continuity, and manifold topology.

This paradigm shift enables process mining to operate closer to the raw perceptual data, aiming to bridge the gap between data-driven representation learning and conceptual abstraction. By embedding both human and machine observations into the same semantic manifold, it becomes possible to identify recurring behavioral motifs, deviations, and causal dependencies even when explicit event boundaries are unavailable. Moreover, relaxing the process model into the embedding space opens new opportunities for integrating reasoning techniques such as the Petri Net of Thoughts, where token movements can be conceptualized as flows through embedding trajectories rather than discrete transitions.

Ultimately, this section argues that the embedding space itself can serve as the new substrate for process discovery—a continuous process model in which multimodal data, linguistic reasoning, and human feedback coalesce into a unified, geometry-aware representation of processes.

6.6.1 Relaxed Process Mining: A General Perspective

Recent developments indicate that problem-solving can be carried out directly in an embedding space [LSR⁺19]. Simultaneously, approximate process discovery [vDSL23] and approximate conformance checking [GKM⁺25] suggest that process mining tasks can be made more efficient by partially offloading computation to vector representations. While current techniques encode inputs to vectors, they do not leverage the rich semantics present in embedding spaces learned by large pre-trained models.

An important open question emerges here: if mathematical and logical reasoning can be performed in embedding spaces [LSR⁺19], is it possible to execute key process mining operations—namely process discovery and conformance checking—directly in these spaces as well? Our work addresses this question by proposing a *framework that relaxes process mining tasks to function on AI embeddings*. We hypothesize that this method can *i)* exceed the accuracy of symbolic-only approaches, *ii)* handle real-world unstructured

multimodal data as event logs, and *iii*) can utilize knowledge enraptured in AI models for process mining tasks.

As organizations increasingly seek to mine processes from raw data-text, images, and beyond—our approach has the potential to bridge the gap between the *representational power of AI models* and the *practical needs of process mining tasks*.

Building on the premise that process-relevant information can be encoded in the latent spaces of modern AI models, we present a general perspective that extends classical process mining and conformance checking methods to operate within an embedding space. Traditional process mining algorithms, such as the inductive miner, and conformance checking techniques are typically defined over discrete events with crisp boundaries. However, real-world data is often noisy, heterogeneous, and semantically ambiguous—conditions under which symbolic techniques struggle. Our perspective accommodates semantic equivalence across synonyms by considering sets of embeddings for different words and by ensuring that events with embeddings within a defined distance are treated as equivalent. Moreover, in the context of multimodal event logs derived from video data, multiple frames or scenes that are similar enough are aggregated into single activities using the same relaxed notion of similarity.

The first part of our perspective develops a relaxed process discovery algorithm that recursively partitions the event log into subprocesses based on relaxed cuts, where each cut is validated by checking that events in one partition have corresponding events in the other within a prescribed distance. The second part extends this methodology to conformance checking by aligning observed traces with a modeled process; here, each observed event is matched against a process activity by minimizing a cost function based on the Euclidean distance in the embedding space, with an allowance for skipped events at a fixed penalty.

Definitions 35–37 formalize a continuum-based approach to process mining in embedding spaces. *Relaxed Process Mining* (Definition 35) establishes the foundational tolerance for semantic and multimodal variability. *Relaxed Process Discovery* (Definition 36) leverages this principle to generate process models from continuous event representations, while *Relaxed Conformance Checking* (Definition 37) provides a mathematically grounded means of aligning observed and modeled behavior with graded similarity.

Relaxed Process Discovery

We describe a relaxed process discovery using an inductive miner algorithm adapted to an embedding space where each event is represented by an embedding $e \in \mathbb{R}^n$, and the event log is given by $E = e_1, e_2, \dots, e_m$. For each embedding e , we define its neighborhood as the ball $B(e, \delta) = \{x \in \mathbb{R}^n : |x - e| \leq \delta\}$, where $\delta > 0$ is a relaxation parameter that accommodates noise and minor variations. We also define a similarity function between two embeddings e_i and e_j by $\text{sim}(e_i, e_j) = \exp(-|e_i - e_j|^2 / \sigma^2)$, where σ is a scaling parameter; two embeddings are considered to represent the same activity if $\text{sim}(e_i, e_j) \geq \tau$, with $\tau \in (0, 1]$ as the similarity threshold.

In the inductive miner, rather than enforcing strict cuts, we relax the cut condition by partitioning the log E into segments E_1 and E_2 such that for every $e_i \in E_1$, there exists a reference embedding r_2 for E_2 satisfying $e_i \in B(r_2, \delta)$, i.e., $|e_i - r_2| \leq \delta$, and similarly for events in E_2 . When considering synonyms, assume each word w is represented by a set of embeddings $S_w = e_1^{(w)}, e_2^{(w)}, \dots, e_{k_w}^{(w)}$, and require that for any two synonyms w and w' , there exists a pair $e^{(w)}i \in S_w$ and $e^{(w')}j \in S_{w'}$ such that $|e^{(w)}i - e^{(w')}j| \leq \delta$, so that activities associated with different synonyms are considered equivalent. In the context of video data as a multimodal event log, each activity may be represented by multiple frames; let the set of frame embeddings for a video segment be $F = f_1, f_2, \dots, f_p$ and define the similarity between two video segments F_1 and F_2 as $d(F_1, F_2) = \min_{f \in F_1, g \in F_2} |f - g|$, with the segments considered the same if $d(F_1, F_2) \leq \delta$. The algorithm proceeds recursively by seeking a relaxed cut that partitions the event log into sublogs E_A and E_B representing different subprocesses, while allowing a fraction $\alpha \in [0, 1]$ of events to be treated as noise. The algorithm recurses on the sublogs E_A and E_B until a base case is reached (when a sublog corresponds to a single activity or no further meaningful partition exists). At each recursive step, the process model is constructed as a tree whose internal nodes correspond to relaxed cuts (interpretable as control-flow operators such as sequence, parallelism, or choice) and whose leaf nodes represent atomic activities via their embeddings or groups of similar embeddings.

Finally, robustness of the relaxed miner is evaluated by applying it to various sets of synonyms and video segments; if for all pairs of embeddings $e_i^{(w)}$ and $e_j^{(w')}$ corresponding to the same activity we have $|e_i^{(w)} - e_j^{(w')}| \leq \delta$, then the miner consistently produces the same process model structure. This framework thus generalizes the traditional inductive miner to accommodate embedding space relaxations, handling noise and multimodal event logs through the careful tuning of the parameters δ , τ , σ , and α .

Relaxed Conformance Checking

We now describe a relaxed conformance checking approach in an embedding space, where each event in an observed trace is represented by an embedding $e \in \mathbb{R}^n$ and the process model is represented as a sequence or structure of activities with representative embeddings $r_1, r_2, \dots, r_L$, each associated with a neighborhood $B(r_k, \delta) = \{x \in \mathbb{R}^n : |x - r_k| \leq \delta\}$, where $\delta > 0$ is a relaxation parameter that accounts for noise. Given an observed trace $T = e_1, e_2, \dots, e_m$ from a multimodal event log (where a video segment or multiple frames are condensed into one event when similar enough), we first define a similarity measure between an observed event e_i and a modeled activity r_k by $\text{sim}(e_i, r_k) = \exp(-|e_i - r_k|^2 / \sigma^2)$ with σ a scaling parameter, and we consider e_i conformant with r_k if $\text{sim}(e_i, r_k) \geq \tau$, where $\tau \in (0, 1]$ is a similarity threshold. To allow for the possibility that events corresponding to synonyms (each represented by a set of embeddings $S_w = e^{(w)}1, e^{(w)}2, \dots, e^{(w)}k_w$ for a word w) are treated as equivalent, we require that for any two synonyms w and w' , there exists $e^{(w)}i \in S_w$ and $e^{(w')}j \in S_{w'}$ satisfying $|e^{(w)}i - e^{(w')}j| \leq \delta$, thus allowing the process model to be matched against traces containing varied synonyms.

Conformance checking is performed by aligning the observed trace T with the modeled sequence $r_1, r_2, \dots, r_L$ via a relaxed alignment function; let $\mathcal{A} : T \rightarrow r_1, r_2, \dots, r_L \cup \epsilon$ be an alignment that assigns to each observed event either a corresponding modeled activity or a skip symbol ϵ representing a deviation. The cost of aligning an event e_i with an activity r_k is defined as $c(e_i, r_k) = |e_i - r_k|$ if $|e_i - r_k| \leq \delta$, and $c(e_i, r_k) = +\infty$ otherwise; the cost of skipping an event is set to a fixed penalty $c(e_i, \epsilon) = \gamma$, where $\gamma > 0$ is chosen to balance the trade-off between matching and skipping. The overall alignment cost for a given alignment $\mathcal{A}$ is then $C(\mathcal{A}) = \sum_{e_i \in T} c(e_i, \mathcal{A}(e_i))$, and the optimal alignment is determined by $\mathcal{A}^* = \arg \min \mathcal{AC}(\mathcal{A})$, subject to the constraint that the order of activities is preserved in the alignment, i.e., if e_i is aligned to r_k and e_j to $r_{k'}$ with $i < j$, then $k \leq k'$.

The fitness of the observed trace with respect to the process model is then quantified by a normalized measure $F(T) = 1 - \frac{C(\mathcal{A}^*)}{C_{\max}}$, where $C_{\max}$ is a normalization constant representing the worst-case cost, for example $C_{\max} = |T| \cdot \max \delta, \gamma$; a fitness value of 1 indicates perfect conformance. In the case of video data, where an event may be represented by multiple frame embeddings $F = f_1, f_2, \dots, f_p$, we define the distance between the video event and the model activity r_k as $d(F, r_k) = \min_{f \in F} |f - r_k|$, and the conformance condition becomes $d(F, r_k) \leq \delta$. The relaxed conformance checking framework is further robustified by allowing a noise tolerance $\alpha \in [0, 1]$, such that if the fraction of events in T for which no acceptable alignment exists (i.e., where $c(e_i, r_k) = +\infty$ for all r_k) is below α , the trace is still considered overall conformant. If $\frac{|\{e_i \in T : \min_{r_k} |e_i - r_k| > \delta\}|}{|T|} \leq \alpha$, then the trace is accepted as conformant with the process model.

6.6.2 A Framework for Relaxed Process Mining

We describe a framework to support our approach in which the embedding space itself can serve as the new substrate for process discovery—a continuous process model in which multimodal data, linguistic reasoning, and human feedback coalesce into a unified, geometry-aware representation of processes. Our approach is applied on both traditional text-first and multimodal-first event logs. The former maintains compatibility with established benchmarks, while the latter (comprising videos with images and audio) showcases the method’s capabilities in an emerging field where standardized benchmarks are still under development.

To illustrate why our event logs can easily be multimodal data, we provide an illustration in Fig. 6.12 that presents our overall pipeline. The proposed pipeline begins with a *multimodal data stream*—including video, audio, and sensor traces—from which static entities such as *actor Z*, *tool A*, or *item X* are extracted via object detection and mapped to conceptual categories (*Conceptualizing Statics*). A semantic alignment layer then performs *Timeline Sync*, *Ontology Integration*, and *Key Points Tracking* to organize these elements across time and meaning, enabling *Conceptualizing Dynamics and Structure*. The resulting representations are embedded into a joint *concept space*, where observed entities are aligned with curated *business knowledge points*, allowing reasoning over

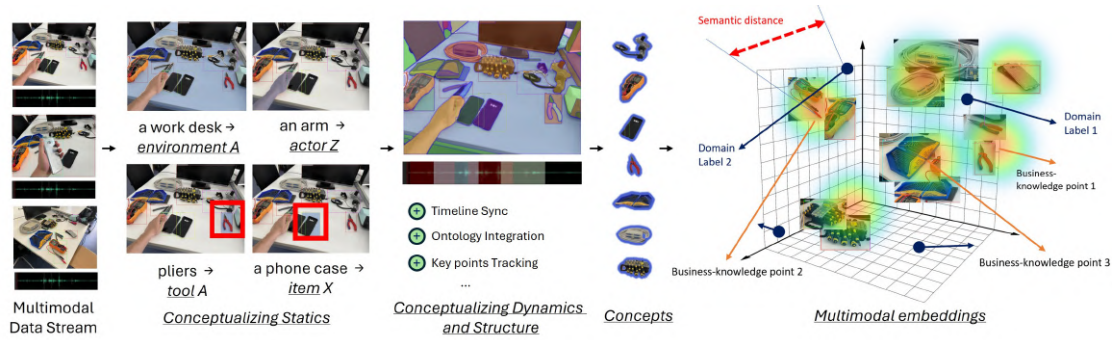


Figure 6.12: Overview of the multimodal observations and static concept mapping to temporal-semantic alignment and embedding-based reasoning with respect to business knowledge anchors. Adopted from Gavric et al. [GBP25b].

semantic distances and classification into *domain labels*. As in Fig. 6.12, various input sources—such as video frames, audio signals, and text annotations—are first detected or segmented (e.g., identifying a phone, a pair of pliers, and a hand in the scene). These individual elements are then converted into continuous embeddings using a unified embedding model, thereby mapping potentially diverse data into the same semantic space. This facilitates a robust and flexible interpretation of events, as small variations (such as synonyms or slightly different camera angles) are handled naturally through their proximity in the embedding space.

Embedding Space Operations

To capture the semantic meaning of each activity, we convert the event descriptions (in particular, activity and resource attributes) into embeddings. Before embedding, we ensure that every event description meets the context length requirement. We compute the embeddings by iterating over each event description and invoking the embedding function. Algorithm 4 details the embedding computation processes.

Event Embeddings Similarity Check via Cosine Similarity

To validate the usefulness of the embeddings, we compare them using cosine similarity.

Clustering and Visualization

The embeddings enable us to cluster similar events in the n -dimensional space. We perform hierarchical clustering. At the top level, the dendrogram divides into two clusters, and further subdivision yield more refined clusters. Although hierarchical clustering appears promising for subsequent process discovery (by preparing data for the Inductive Miner who naturally searches for cuts in the event logs), a detailed evaluation of clustering methods is out of the scope for this thesis and left for future work.

Visualization is achieved by reducing the embedding dimensions to two using dimensionality reduction techniques. We apply T-distributed Stochastic Neighbor Embedding (t-SNE) [CCR⁺20]. Additionally, we utilize an LLM to **generate cluster labels** based on its foundational knowledge from extensive training data.

Algorithm 4 Compute Embeddings for Event Descriptions with Uniform Modality

Input: Set of event descriptions $\mathcal{E} = \{e_1, e_2, \dots, e_n\}$; modality flag `is_multimodal`;
text embedding model $\mathcal{M}_{\text{text}}$; multimodal embedding model $\mathcal{M}_{\text{multi}}$

Output: Set of embedding vectors $\{emb(e) \mid e \in \mathcal{E}\}$

```

33 if is_multimodal then
34   | Select embedding model  $\mathcal{M} \leftarrow \mathcal{M}_{\text{multi}}$ 
35 else
36   | Select embedding model  $\mathcal{M} \leftarrow \mathcal{M}_{\text{text}}$ 
37 end
38 foreach event description  $e \in \mathcal{E}$  do
39   |  $emb(e) \leftarrow \mathcal{M}(\text{prompt} = e)$            // Encode event description
40 end
41 return  $\{emb(e) \mid e \in \mathcal{E}\}$ 

```

Process Mining Tasks on Embeddings

Once embeddings have been generated and clustered, we apply process mining tasks. Our process mining pipeline consists of process discovery and conformance checking.

Framework then employs Inductive process discovery algorithm from [BvS23] that take the event logs, now enriched with clustering and embedding information, to generate process models that capture the underlying workflow dynamics. After process models are discovered, framework performs conformance checking to compare the discovered models with the actual event logs. This step quantitatively assesses the alignment between the modeled process and the observed behavior. Algorithm 5 outlines the complete process mining pipeline that has been employed here.

Figure 6.13 illustrates a pipeline that begins by projecting raw events into a unified embedding space, where proximity denotes semantic similarity. These embeddings are then clustered based on distance thresholds or density criteria, and domain knowledge is applied to label each cluster with a descriptive name. The events in the log are subsequently replaced with these cluster labels, producing a discrete event log where items that share a latent concept are unified. This transformed log is finally processed by a process mining tool, allowing for conformance checking, and discovery, in a conventional event-log format despite originating from diverse, multimodal sources.

Algorithm 5 Process Mining Pipeline on Embeddings with Cluster Label Augmentation

Input: Event log E ; embedding generation procedure (Alg. 4); clustering method; language model LLM; set of discovery algorithms $\mathcal{A}$

Output: Set of discovered process models and conformance scores $\{(M_A, score_A) \mid A \in \mathcal{A}\}$

```

42 Extract activities and resource attributes from  $E$  to obtain  $\mathcal{E}$  Generate embeddings
    $embs \leftarrow \{emb(e) \mid e \in \mathcal{E}\}$  using Alg. 4 Cluster embeddings to obtain  $Clusters \leftarrow$ 
   Clustering( $embs$ )

   // Cluster label augmentation
43 foreach  $cluster\ C \in Clusters$  do
44   | Collect original event labels  $labels_C$  from events in  $C$   $name_C \leftarrow \text{LLM}(labels_C)$ 
   | // Generate semantic cluster name
45   | Replace the cluster label of each event in  $C$  with  $name_C$ 
46 end

47 Augment the original event log to obtain  $E'$  Transform  $E'$  into a process-mining-
   compatible log  $L$ 

   // Process discovery and evaluation
48 foreach  $algorithm\ A \in \mathcal{A}$  do
49   |  $M_A \leftarrow A(L)$  // Discover process model
50   |  $score_A \leftarrow \text{ConformanceCheck}(M_A, L)$  // Evaluate model quality
51 end
52 return  $\{(M_A, score_A) \mid A \in \mathcal{A}\}$ 

```

6.7 Validation

The validation of the multimodal data analysis (MMDA) pipeline is grounded in the empirical evaluations conducted across the individual research contributions that constitute its components. Since this chapter consolidates these contributions into a unified architecture, we reflect on each prior evaluation and justify its relevance for the present integrated framework.

Multimodal-Enriched Conceptual Modeling [GBP25i]. In [GBP25i], we evaluated the feasibility of enriching conceptual model elements with multimodal evidence via shared embedding spaces. The study demonstrated that natural language labels extracted from conceptual models can be aligned with visual and auditory embeddings in a semantically meaningful manner. Quantitative similarity measures and qualitative inspection confirmed that enriched model elements exhibit coherent semantic grounding. Although the focus was on enrichment rather than full process discovery, this evaluation validates the representation layer of the MMDA pipeline: symbolic modeling constructs can be reliably connected to multimodal perceptual evidence. Details are provided in Appendix C.3.

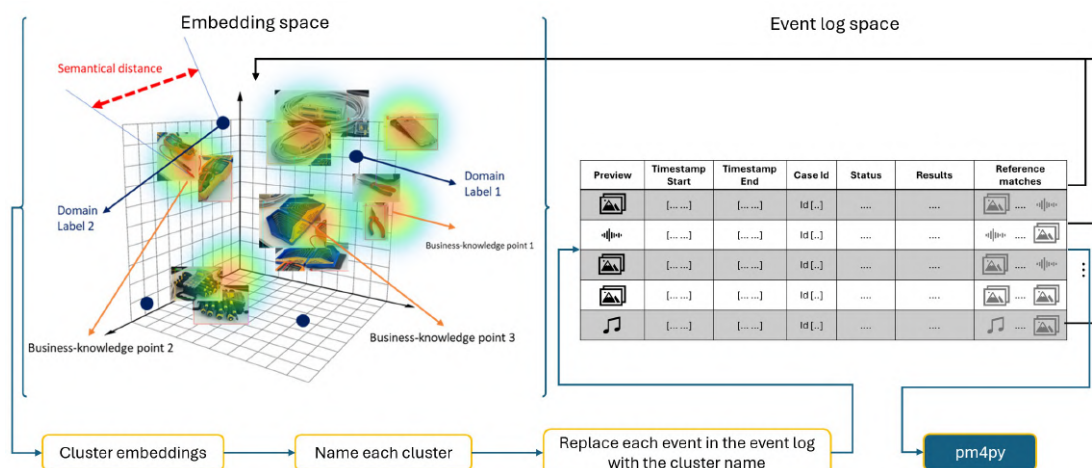


Figure 6.13: Overview of transforming embeddings into discrete event labels for process mining. On the left, various concepts reside in a shared embedding space, where proximity indicates semantic similarity. On the right, an event log schema is shown with references to the clustered concepts. Adopted from Gavric et al. [GBP25b].

Aligned Latent Embeddings of Knowledge and Symbols (ALeKS) [GBP25a].

The ALeKS framework was empirically assessed in [GBP25a] through the analysis of semantic shifts between natural language terms, enriched conceptual model terms, and multimodal instances. Using cosine similarity, clustering, and dimensionality reduction techniques, the study demonstrated that conceptual enrichment induces measurable and interpretable geometric transformations in embedding space. This evaluation substantiates the semantic alignment layer of the present chapter, particularly the mechanisms for computing semantic shifts and crossmodal alignment between symbolic and perceptual representations. Details are provided in Appendix C.4.

Petri Net of Thoughts (PNoT) [GBP25e]. In [GBP25e], we evaluated the Petri Net of Thoughts paradigm by comparing structured, process-based prompting against alternative reasoning approaches such as Chain-of-Thought and Graph-of-Thought. The results showed improved traceability, logical consistency, and interpretability of reasoning traces when guided by formally defined Petri Net structures. While that study operated primarily in textual reasoning settings, it validates the abstraction and reasoning layer of the MMDA pipeline: once multimodal evidence is transformed into process structures, formal execution semantics provide reproducible and analyzable reasoning trajectories.

Multimodal Process Mining and BICycle [GBP24c, GBP24a]. The perception-to-log transformation stage was evaluated in [GBP24c, GBP24a]. These works introduced the Vid2log pipeline and the Business-Knowledge Integration Cycles (BICycle), demonstrating that multimodal video, audio, and sensor data can be converted into structured event logs suitable for classical process mining. Empirical evaluations on publicly re-

leased datasets confirmed that multimodal enrichment increases contextual completeness and improves interpretability of discovered models. These results directly validate the event-log construction and human-in-the-loop refinement mechanisms integrated into the MMDA pipeline. Details are provided in Appendix C.5.

Crossmodal Process Mining and Drift Analysis [GBP25d]. In [GBP25d], we evaluated crossmodal process mining by demonstrating that joint embedding spaces enable alignment and substitution across heterogeneous modalities. Experimental analyses showed increased robustness of process reconstruction under noisy or incomplete modalities when compared to monomodal baselines.

Relaxed Process Mining in Embedding Space [GBP25b]. The embedding-based relaxation of process discovery and conformance checking was evaluated in [GBP25b]. The study demonstrated that similarity-based alignment in embedding space can approximate symbolic process mining results while handling semantic variability, synonyms, and multimodal ambiguity. Experiments conducted on both traditional text-first and multimodal-first logs confirmed the viability of embedding-based discovery and conformance checking. This evaluation substantiates the continuum-based abstraction layer of the MMDA pipeline, where discrete process reasoning emerges from continuous latent representations.

Synthesis of Validation Across Components. Taken together, the prior evaluations provide multi-layered validation of the unified MMDA architecture presented in this chapter:

- [GBP25i] validates multimodal grounding of conceptual model elements.
- [GBP25a] validates semantic alignment and measurable embedding shifts between symbolic and perceptual representations.
- [GBP25e] validates process-structured reasoning and traceability via Petri Net-guided prompting.
- [GBP24c, GBP24a] validate multimodal event-log construction and human-in-the-loop refinement.
- [GBP25d] validates crossmodal robustness and drift-aware alignment mechanisms.
- [GBP25b] validates embedding-based relaxation of discovery and conformance checking.

Although each study evaluated a specific layer under distinct experimental settings, their collective results establish the empirical feasibility, architectural coherence, and robustness of the end-to-end MMDA pipeline. The present thesis integrates these

validated components into a unified framework and evaluates their combined impact on conceptual model precision and conciseness in Chapter 9.

6.8 Conclusion

The multimodal data analysis pipeline provides an end-to-end architecture that transforms raw, heterogeneous evidences into structured conceptual representations and latent regularities that can be incorporated into high-quality process models. At its core, the pipeline operationalizes the methodological principle that *tacit knowledge becomes observable through systematic multimodal perception and formalizable through process-oriented abstraction*.

The pipeline begins with *multimodal perception*, where raw textual, auditory, visual, and spatiotemporal streams are preprocessed into aligned symbolic and subsymbolic feature spaces. Building on our work on multimodal conceptual modeling foundations [GBP25i], this stage harmonizes heterogeneous evidences and provides unified crossmodal embeddings suitable for downstream reasoning. The second stage, *semantic alignment*, maps multimodal cues to domain concepts, drawing on alignment mechanisms developed in [GBP25a]. Here, entities, activities, and contextual factors are identified and linked to conceptual model elements.

Subsequently, the *discovery layer* extracts process-relevant structures. Through the multimodal process mining techniques introduced in [GBP24c, GBP24a], the pipeline reconstructs event sequences and behavioral constraints from continuous multimodal streams rather than symbolic logs alone. Crossmodal drift detection and behavioral change analysis [GBP25d] are applied to uncover latent regularities, deviations, and dispositions that remain hidden in unimodal settings.

Finally, the *abstraction and formalization layer* transforms extracted patterns into structured models. Our method for Petri Net of Thoughts construction [GBP25e] formalizes token-level reasoning paths, while multimodal predictive alignment frameworks [GPB26] and internal-representation-based process discovery [GBP25b] elevate subsymbolic invariants into conceptual modeling artifacts. The result is a pipeline that systematically captures, aligns, and formalizes both explicit and tacit knowledge, thereby enabling more precise and concise conceptual models across diverse domains.

Mixed Reality Simulation

“A story is not an accumulation of information strung into a narrative, but a design of events to carry us to a meaningful climax.”

— *Robert McKee, Storylogue*

This chapter builds upon and extends several of our previous works focused on generating multimodal data for conceptual modeling, immersive process representation, and visual elicitation. In [GBP24b], we described how multimodal evidence (e.g., narration, and image cues) can enhance the interpretation of conceptual models such as UML; [GBP25j] introduced a generative framework that translates process models into explanatory videos, creating a direct link between process semantics and visual storytelling - we demonstrated how structural properties of Petri Nets can guide video generation and facilitate visual process understanding [GBP25f]. Importantly, [GBP25c] explore the use of comic-style narratives and visual scenes for domain elicitation and crossmodal explanation, illustrating the expressive potential of multimodal storytelling in modeling contexts. As a consolidation of those contributions, [GBP26] address a non-mixed reality, event log-centric variant of the proposed simulation approach, whereas this thesis focuses on immersive, human-centered conceptual model refinement. Lastly, [GBP24d] contributed to stakeholder-centered model enrichment by embedding domain-specific terminology and perspectives into the modeling process.

Building upon these works, we develop the *Mixed Reality Simulation* method, which enables interactive “what-if” scenarios to capture tacit and counterfactual decision-making behaviors. Through immersive simulations, experts interact with realistic process reconstructions, revealing implicit rules and situational reasoning that are otherwise inaccessible through data observation alone. The elicited behavioral traces are analyzed, quantified, and integrated with prior observational data to construct richer, cognitively grounded process models.

The chapter begins with interactive *what-if* scenarios (Section 7.1), which motivate the need for dynamic exploration of alternative process configurations and expert decision-making. The chapter then introduces *modSense* (Section 7.2) based on [GBP24b], which describes how conceptual models are operationalized into multimodal data representations that enable observation, interaction, and analysis. Building on this foundation, Section 7.3 outlines the role of conversational AI as an interface for model inspection, explanation, and refinement within mixed reality environments, based on [GBP24d, GBP24b]. Next, Section 7.4 presents PN-VidGen [GBP25j, GBP25f], which transforms formal process models into video-based simulations to support embodied and temporal understanding of process behavior. These simulations are contextualized and structured using Virtually Enhanced Narratives (ViEnNa) [GBP25c] in Section 7.5, which provide coherent, scenario-driven representations across modalities. The chapter then introduces Multimodal Business Process Simulations (MM-BPS) [GBP26] in Section 7.6, which unify formal models, multimodal enactments, and human interaction into an executable simulation paradigm. Validation mechanisms and expert-based assessment strategies are discussed in Section 7.7. Finally, Section 7.8 concludes the chapter by synthesizing the presented components and highlighting their collective contribution to data-grounded and immersive conceptual modeling.

7.1 Interactive What-If Scenarios

Interactive *what-if* scenarios serve as the core elicitation mechanism, enabling participants to explore, simulate, and respond to hypothetical process variations in an immersive environment. These scenarios are designed to reveal tacit knowledge and counterfactual reasoning—aspects of human decision-making that often remain implicit in routine operations or traditional data collection methods. By prompting experts to engage with dynamically changing process states, the system captures situated judgments, exception-handling strategies, and anticipatory reasoning that extend beyond the boundaries of observed evidence.

7.1.1 Scenario Construction

The design of *what-if* scenarios is grounded in the idea that process understanding emerges not only from past execution traces but also from projected alternatives—the situations that could have occurred but did not. In practice, these scenarios allow domain experts to interact with multimodal reconstructions of processes, modifying parameters or conditions and observing the effects in real time. This setup provides a means to uncover the causal and dispositional relations underlying human decision-making, as experts externalize their reasoning while reacting to hypothetical stimuli. The approach thus complements the purely observational analysis from multimodal data pipelines by transforming users into active participants in the modeling process.

Each *what-if* scenario is constructed around a baseline model instance, typically derived from event logs or multimodal traces generated in the analysis phase. From this baseline,

variations are introduced through the modification of contextual variables or through alternative event sequences. These perturbations are either automatically generated by the AI or manually scripted by analysts to target specific reasoning phenomena.

The MR interface presents the scenario through synchronized visual, auditory, and spatial cues. As the participant acts within this environment, their verbal explanations, and system interactions are recorded as multimodal evidence for further interpretation. The resulting data capture both the *action layer* (observable responses) and the *cognitive layer* (verbalized rationale and justification), forming a comprehensive basis for model enrichment.

The interaction model follows three complementary dimensions: *parameter exploration*, *decision intervention*, and *narrative branching*. In parameter exploration, the user manipulates variables to observe emergent effects on the simulated process. Decision intervention focuses on the user’s ability to alter the flow—for instance, deciding whether to escalate a case, reassign a task, or postpone an activity. Narrative branching extends this idea into structured alternatives, allowing users to switch between parallel process outcomes and verbalize the reasoning behind each choice. Together, these interactions constitute a structured elicitation loop where the MR system not only visualizes consequences but also invites reflection and explanation.

7.1.2 Elicited Data and Counterfactual Capture

A key goal of these scenarios is the elicitation of *counterfactual knowledge*¹—insights about how experts think processes should evolve under altered circumstances. Each interaction produces a set of annotated multimodal streams linked to corresponding process elements. Through temporal alignment and semantic tagging, these are converted into a structured representation that captures “dispositions”—mappings between conceptual situations and possible actions. The resulting counterfactual models extend the traditional event log with hypothetical variants, thus serving as a new form of process evidence.

While counterfactual capture yields rich, disposition-oriented knowledge about how experts reason under alternative conditions, its analytical value depends on how systematically these insights can be represented, aligned, and reused across modeling activities. To move beyond isolated scenario enactments, the elicited multimodal streams and their counterfactual annotations must be transformed into a coherent data structure that preserves their temporal, semantic, and conceptual relationships. This requires a mechanism that operationalizes conceptual models as sensing and alignment artifacts for multimodal evidence. The following section introduces *modSense*, which addresses this challenge by turning conceptual models into structured multimodal data representations that enable inspection, querying, and downstream simulation.

¹A counterfactual is a hypothetical situation or statement that considers what could have happened if things had been different, contrary to the actual facts.

7.2 Turning Conceptual Models into Multimodal Data (modSense)

This section explores the inverse direction of multimodal enrichment—transforming conceptual models themselves into multimodal data representations. By doing so, we aim at enabling visual, auditory, and interactive interpretations of modeling elements, making conceptual knowledge more accessible to both humans and machines.

As an early contribution to this vision, we refer to our work “How Does UML Look and Sound” [GBP24b], where the Unified Modeling Language (UML) serves as an illustrative case for conceptual models in general. In this study, UML diagrams are reinterpreted through visual and auditory modalities to examine how structural and behavioral model features can be experienced beyond static notation.

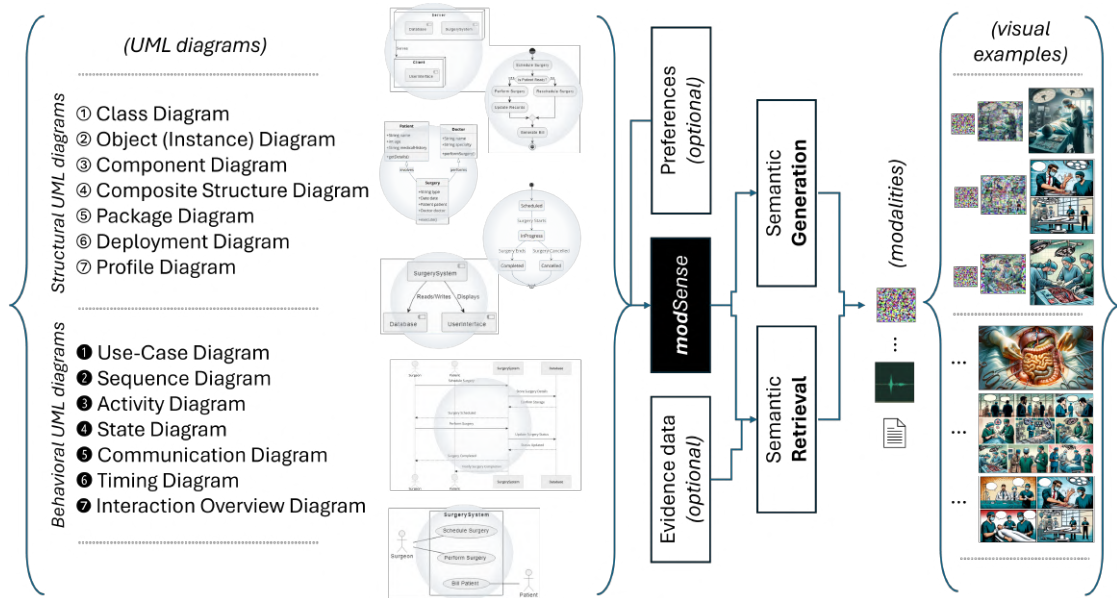


Figure 7.1: Overview of the *modSense* framework for transforming UML conceptual models into multimodal data representations. Adopted from Gavric et al. [GBP24b].

7.2.1 The *modSense* Framework

The *modSense* framework bridges the gap between traditional symbolic modeling languages and sensory modalities such as vision and sound, thus enabling the perceptual exploration of modeling structures and semantics. At its core, *modSense* operates on input conceptual models, encompassing both *structural* and *behavioral* diagram types. UML structural diagrams—such as Class, Component, and Deployment Diagrams—define static relationships between entities, while UML behavioral diagrams—such as Sequence, Activity, and State Diagrams—capture dynamic system behaviors. In [GBP24b], these models serve as the symbolic foundation for generating multimodal data.

Figure 7.1 illustrates the overall architecture of the *modSense* framework, developed to translate conceptual models into multimodal representations.

The process begins by parsing and embedding the semantic content of the UML elements. This information is then optionally enriched with user preferences or evidence data (e.g., visual references, textual annotations, or domain recordings). The embedded representations are processed by the *modSense* core, which performs two key operations:

1. **Semantic Retrieval** — searches across multimodal databases to retrieve existing examples (images, videos, audio clips) that correspond to the semantic meaning of model elements.
2. **Semantic Generation** — synthesizes new multimodal content through generative models, allowing the creation of visual and auditory artifacts aligned with the conceptual semantics of the model.

The resulting outputs form a multimodal corpus that reflects the modeled concepts through diverse sensory forms. As depicted on the right side of Figure 7.1, these outputs include realistic visual renderings, abstract textures, comic-style panels, and even corresponding sounds or narrated elements.

By transforming conceptual knowledge into perceivable media, *modSense* demonstrates how modeling languages like UML can become bridges between symbolic reasoning and human sensory understanding.

7.2.2 Understanding Conceptual Models

In [GBP24b], conceptual models are a set of UML models in their graphical representations that help in visualizing, specifying, constructing, and documenting the artifacts of a SUI. UML models are broadly categorized into two types: structural diagrams and behavioral diagrams. Structural diagrams focus on the static aspects of the SUI, portraying the physical and conceptual elements and their relationships. Common structural diagrams include class diagrams, object diagrams, component diagrams, composite structure diagrams, deployment diagrams, and package diagrams. These diagrams help in detailing the system’s architecture, such as classes and their relationships, system components, and the arrangement of software deployments.

Behavioral diagrams, in contrast, represent the dynamic aspects, detailing how the system behaves over time and in response to events. This category includes use case diagrams, sequence diagrams, activity diagrams, state machine diagrams, communication diagrams, interaction overview diagrams, and timing diagrams. These diagrams map out the processes, workflows, and operational sequences within a SUI, providing insights into the flow of data and control among different system components.

Both types of models are constructed from a variety of elements that provide the building blocks for modeling software systems. We have identified 59 such elements across all 14

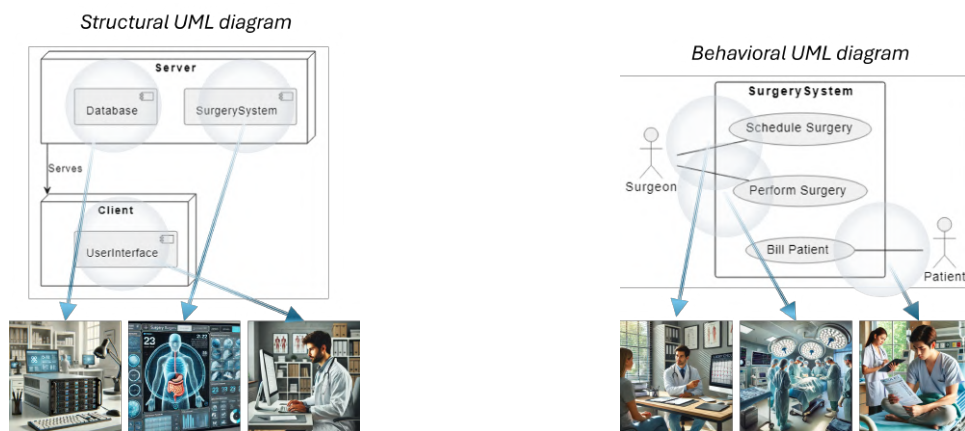
Table 7.1: Conceptual Models and Key Elements for Generating *modSense* interpretations in [GBP24b] where conceptual models are a set of UML models.

No.	Conceptual Model Type	Key Elements for Generating <i>modSense</i> interpretations
1	Class Diagram	Classes, Interfaces, Associations, Generalizations, Dependencies
2	Object Diagram	Objects, Links, Fields
3	Use Case Diagram	Actors, Use Cases, Relationships, System Boundary
4	Sequence Diagram	Objects, Lifelines, Messages, Activation Bars, Time Constraints
5	Collaboration Diagram	Objects, Links, Messages, Sequence Numbers
6	Activity Diagram	Activities, Transitions, Swimlanes, Forks, Joins, Start/End Points
7	State Machine Diagram	States, Transitions, Events, Actions, Initial and Final States
8	Component Diagram	Components, Interfaces, Ports, Connections
9	Deployment Diagram	Nodes, Artifacts, Links
10	Timing Diagram	Objects, Time Rulers, State or Value Lifelines, Events
11	Interaction Overview Diagram	Interaction Frames, Activity Nodes, Control Flows, Decisions
12	Communication Diagram	Objects, Messages, Links, Sequence Numbers
13	Composite Structure Diagram	Parts, Connectors, Ports, Collaborations
14	Package Diagram	Packages, Dependencies, Mergings, Access

diagram types of UML. The elements include, but are not limited to, classes, interfaces, components, use cases, actors, and various types of interaction units such as messages and transitions. Table 7.1 categorizes these elements under their respective models:

Understanding UML structural diagrams is designed for comprehending the static aspects of a system, as illustrated in Figure 7.2a. Structural diagrams are primarily concerned with illustrating the system’s architecture. For instance, class diagrams provide a visual representation of the classes within a system and their relationships, such as inheritance and associations. This visualization aids in grasping the underlying structure and relationships among system components, making it easier for both technical and non-technical stakeholders to understand how system elements interact and are organized. By visualizing these concepts, learners can more readily appreciate the distribution of responsibilities within the system, how objects are interconnected, and how changes in one area might affect another.

Behavioral diagrams, on the other hand, focus on the dynamic aspects of systems, illustrating how software behaves in operation. Each diagram type aims to help in understanding different facets of system behavior. Our solution can interpret UML behavioral diagrams as illustrated in Figure 7.2b. For example, sequence diagrams demonstrate the order of operations and the interaction between system components over time, providing a step-by-step visualization of the processes. Activity diagrams help in visualizing the flow from one activity to another within a system, highlighting parallel processes and decision points. These visual tools are instrumental in modeling the flow of control and data, offering insights into the temporal and conditional dependencies within the system. Through behavioral diagrams, developers and analysts can ensure that the system’s functionality aligns with user requirements and expected behaviors, ultimately aiding in the design of more robust systems.



(a) Structural conceptual model mapped to a generated content.

(b) Behavioral conceptual model mapped to a generated content.

Figure 7.2: Overview of the *modSense* framework linking UML structures to generative multimodal interpretations. Adopted from Gavric et al. [GBP24b].

Classes represent the fundamental building blocks of object-oriented systems, encapsulating data and behavior into a single entity. They are represented as rectangles with three compartments: the class name, attributes, and methods. Objects represent instances of classes at runtime, each possessing its own state and behavior. They are represented as rectangles with underlined names within sequence diagrams, indicating their lifelines. Attributes represent the properties or characteristics of a class, describing the state of objects belonging to that class. They are typically listed within the second compartment of class rectangles in class diagrams. Methods, also known as operations, represent the behaviors or functionalities provided by a class. They define the operations that objects of the class can perform and are listed within the third compartment of class rectangles in class diagrams. Associations represent the relationships between classes, indicating how objects of one class are connected to objects of another class. They are represented as lines connecting the participating classes, often annotated with multiplicity indicators to

specify the cardinality of the association. Messages represent the communication between objects in sequence diagrams, indicating the exchange of information or invocation of methods during a particular scenario. They are represented as arrows between lifelines, with labels specifying the content of the message and the timing of its delivery. Activities represent individual tasks or actions within an activity diagram, representing the flow of control from one activity to another. They are represented as rounded rectangles, with arrows indicating the transitions between activities based on specified conditions or triggers. States represent the different conditions or modes that an object or system component can be in, within a state machine diagram. They are represented as rounded rectangles, with arrows indicating the transitions between states triggered by events or actions.

7.2.3 Visual Generative Capabilities

By taking the advanced capabilities of discrete Variational Autoencoders (dVAE), autoregressive transformers, and diffusion models ([RPG⁺21]), we employ in [GBP24b] a system capable of generating dynamic visual and auditory content that contextualizes conceptual models.

Conditioned Sample Generation

The conditioned sample generation commences with the encoding of a textual description y , which specifies the desired image attributes. This text is converted into a sequence of Byte Pair Encoding (BPE) tokens [SKF⁺99] and then embedded into the model. The autoregressive transformer models the generation process where each image token x_i is sequentially predicted based on the previously generated tokens $x_{<i}$ and the input text y .

The expression for the conditional probability of generating an image given a text description is given by:

$$p(x \mid y) = \prod_{i=1}^N p(x_i \mid x_{<i}, y),$$

where N is the total number of image tokens to be generated. This formulation illustrates the sequential nature of generating image tokens, where the probability of each token x_i depends on its preceding tokens $x_{<i}$ and the text y .

Each token's probability is determined using the softmax function on the logits produced by the transformer for each possible token:

$$p(x_i = k \mid x_{<i}, y) = \frac{\exp(g_\theta(x_{<i}, y)_k)}{\sum_j \exp(g_\theta(x_{<i}, y)_j)},$$

where g_θ denotes the function of the transformer that generates logits for each token k at position i , and j indexes all possible image tokens.

Application to Conceptual Models Business Logic

For businesses, this generative model is useful for creating visually appealing and tailored content. In e-commerce, for instance, product images can be dynamically generated to reflect specific customer descriptions, enhancing product presentations and potentially increasing engagement and sales. In advertising, the model allows for the creation of targeted visuals that resonate with diverse customer demographics, adapting imagery to suit regional tastes and cultural contexts.

7.2.4 Auditory Generative Capabilities

For generating audio, we incorporate the principles from a solution by [RAC⁺24] into our platform's pipeline.

Exact Autoregressive Decomposition: In the exact autoregressive modeling, the sequence of tokens U from the quantized representation Q is modeled such that each token U_t at time t is predicted based on all previous tokens:

$$p_t(U_t|U_{t-1}, \dots, U_0) = P(U_t|U_{t-1}, \dots, U_0)$$

Here, U_0 is often a special token indicating the start of the sequence, and p_t is the autoregressive probability model at time t .

Inexact Autoregressive Decomposition: This approach allows for simultaneous prediction of multiple codebooks to reduce complexity and increase efficiency. If the sequence V represents tokens with K codebooks at time t , then the probability of observing $V_{t,k}$, the token from the k -th codebook at time t , given all previous tokens is:

$$p_{t,k}(V_{t,k}|V_{t-1}, \dots, V_0) = P(V_{t,k}|V_{t-1}, \dots, V_0)$$

where $V_{t,k}$ is the token at time t from codebook k , and $p_{t,k}$ is the autoregressive model specific to codebook k .

To manage the complexity due to multiple codebooks, it employs different interleaving patterns, as described. Here are two specific patterns:

Parallel Pattern: All codebooks at a given time step are predicted simultaneously:

$$\text{Parallel Pattern: } Ps = \{(s, k) : k \in \{1, \dots, K\}\}$$

Delay Pattern: Introduces a delay between codebooks, allowing the prediction of each codebook at a different step based on a calculated delay:

$$Ps = \{(s - k + 1, k) : k \in \{1, \dots, K\}, s - k \geq 0\}$$

Text Conditioning: When the model is conditioned on text, the following tensor C is used to modify the probabilities in the Transformer model:

$$C \in \mathbb{R}^{T_C \times D}$$

Here, T_C is the number of time steps in the conditioning text sequence, and D is the dimensionality of the text embeddings.

Auditory Conditioning: Similar to text conditioning, melody features can also modify the sequence processing in the transformer model. These features are derived from a pre-processed version of another audio input or directly embedded within the sequence.

Output Logits Transformation: Finally, the output from the Transformer model is transformed into logits that predict the next token in the sequence, which is essential for generating the audio output during inference:

$$\text{Logits} = W \cdot \text{Transformer_Output} + b$$

where W and b are the weights and biases of the final output layer, transforming the Transformer output to logits corresponding to the next token probabilities.

7.2.5 Focus Masks

To customize generation of the content, we employ a concept of *Focus Masks* (illustrated in Figure 7.3).

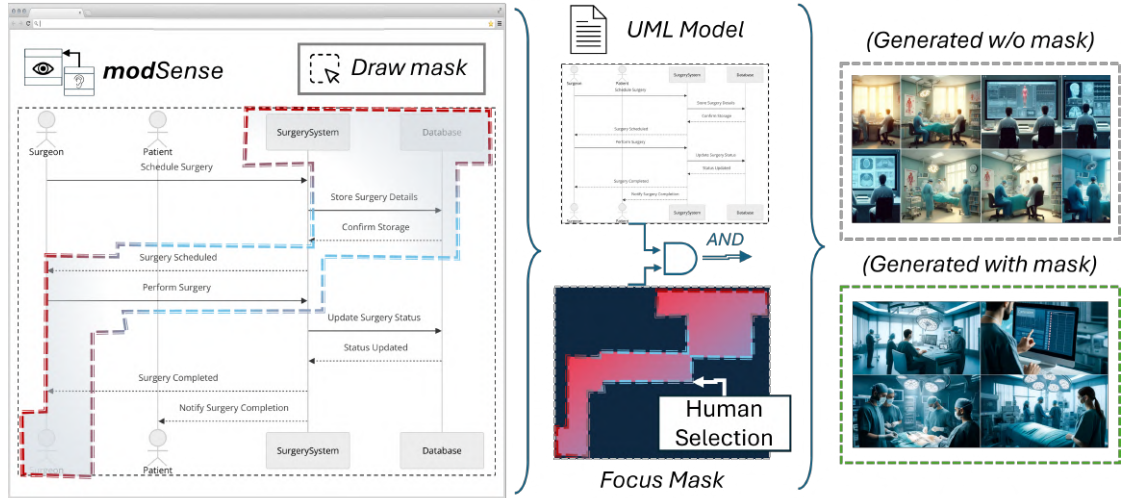


Figure 7.3: Focus mask concept for human-guided selective model interpretation in *modSense* framework. Adopted from Gavric et al. [GBP24b].

Implementing a focus mask in the context of multimodal representation of a conceptual model offers a highly personalized and interactive approach to learning and understanding complex systems. A focus mask is essentially a user-defined selection within a conceptual model that highlights specific parts or elements of the diagram for deeper exploration. This functionality aims at enhancing the educational experience by allowing users to concentrate on particular components of a system, with a goal to simplify the learning process by isolating complex interactions or structures. Users can apply a focus mask to any part of a UML diagram—be it a cluster of classes in a class diagram, specific interactions in a sequence diagram, or particular states in a state machine diagram.

Upon defining a focus mask, the system then generates multimodal data based on a specific part of the conceptual model highlighted by the mask. This generation process dynamically creates or adjusts the visual representation of the selected conceptual model portion based on user inputs and predefined modeling rules. For example, if a user selects a series of interactions in a sequence diagram, the AI generates an enhanced view of these interactions, possibly expanding on details like message sequencing, timing, and participant lifelines.

Moreover, the focus mask approach integrates with conversational AI, which guides users through the newly generated or highlighted parts of the diagram. This AI part explains the roles and relationships of the focused elements, suggests possible modifications, and even answers queries related to the diagram's specifics. For instance, in a class diagram, the AI could elucidate the inheritance hierarchy or the implications of various associations between the classes within the focus mask.

7.3 Conversational AI Capabilities for Conceptual Modeling in Mixed Reality

Conversational Artificial Intelligence extends the way conceptual models are created, explored, and refined by introducing natural, dialog-driven interactions between humans and modeling systems. In the context of mixed reality, these capabilities become even more immersive—allowing modelers to engage with conceptual structures through speech, gestures, and spatial references rather than static editing interfaces. Within this section, we illustrate how conversational AI can facilitate the understanding and generation of different conceptual models by supporting multimodal reasoning, adaptive explanations, and guided co-construction of models. The presented examples, integrated within the *modSense* framework, demonstrate how mixed-reality environments can transform modeling sessions into collaborative and contextually aware experiences where both human and AI agents participate in shaping the conceptual representation of a system.

In the context of reading, learning, and creating conceptual models, conversational agents, or chatbots, can play a significant role in simplifying complex concepts, providing instant feedback, and guiding users through step-by-step processes. Incorporating conversational AI into the learning and analysis of conceptual models offers a dynamic and adaptive tool

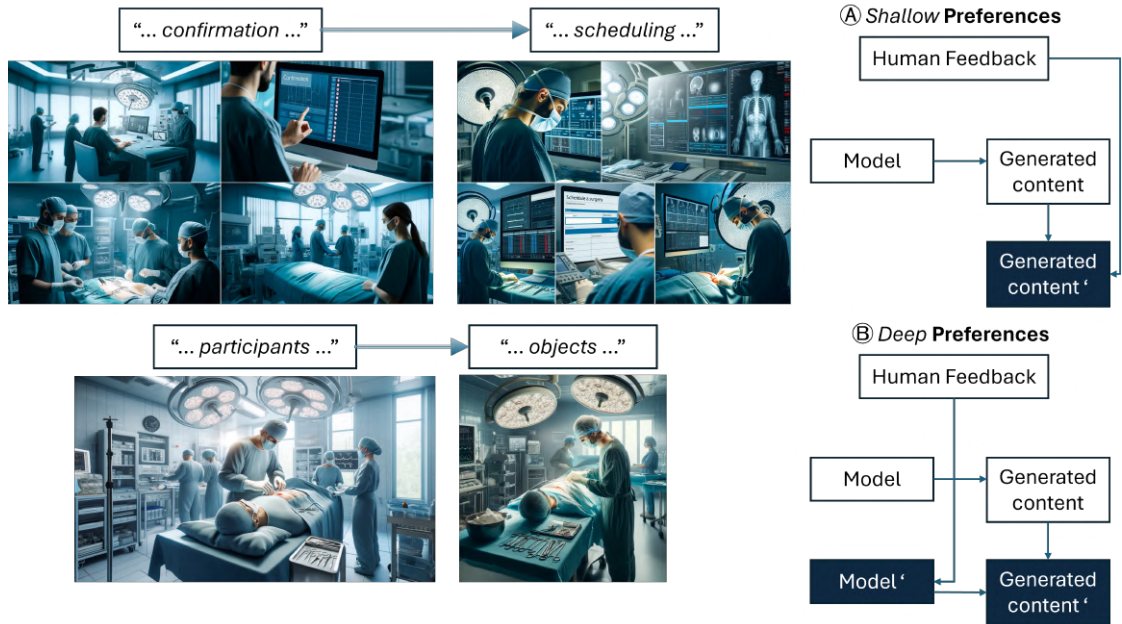


Figure 7.4: Illustration of human feedback through (A) *Shallow* and (B) *Deep* approach in *modSense* framework. The before–after comparison of the generated images highlights a shift in the focus of the generation, reflecting how human feedback steers the model toward conceptually relevant regions and relations. Adopted from Gavric et al. [GBP25e].

that can tailor its interactions based on user feedback. This feedback can be categorized into two types: *shallow* and *deep*, each serving distinct purposes and enhancing the learning experience in unique ways, as illustrated in Figure 7.4.

7.3.1 Shallow Feedback Incorporation

Shallow feedback involves immediate, surface-level reactions from users about their interactions with the conversational AI. This type of feedback is typically operational and straightforward, focusing on the usability and accessibility of the system. For example, users might provide input on the clarity of the explanations provided, the speed of interaction, or the relevance of the content generated in response to their queries. Shallow feedback helps to fine-tune the conversational interface, ensuring that responses are not only accurate but also presented in a user-friendly manner. It facilitates quick adjustments to the AI’s delivery method, such as simplifying language or altering the pace of information delivery. This level of feedback is designed for maintaining engagement and ensuring that the instructional content is accessible to users with varying levels of expertise and background knowledge.

7.3.2 Deep Feedback Incorporation

Deep feedback involves a search into the substantive content of the learning material and the user's conceptual understanding of the UML diagrams. This type of feedback is more analytical and reflective, involving users' feedback on how well the AI explanations align with their learning objectives or how effectively their conceptual questions are being answered. Deep feedback might include suggestions on additional topics for exploration, requests for more detailed examples, or critiques of the AI's interpretations of complex diagram elements. This feedback is used to adapt the conversational AI's responses to better meet the educational needs of the user, tailoring the complexity and depth of the content to match the user's comprehension level and learning progress. For instance, if a user struggles with understanding the interactions in a sequence diagram, the AI can offer more detailed step-by-step breakdowns or analogies to improve comprehension.

Both shallow and deep feedback are integral to refining the capabilities of conversational AI in teaching UML diagrams. By incorporating both types of feedback, the AI can evolve from a static informational tool into a dynamic, responsive educational assistant that adjusts its instructional approach based on real-time user interactions and preferences.

7.3.3 Enhancing Comprehension of Conceptual Models through Conversations

Conversational agents aid users in understanding the various elements and structures of conceptual models. The dialogues summarized in Tables 7.2–7.5 illustrate the spectrum of human–agent interactions supported within the *modSense* framework. Each example demonstrates a distinct conversational scenario aligned with a specific conceptual model: class diagrams, use case diagrams, and sequence diagrams. Through these interactions, *modSense* showcases its ability to guide users in both understanding and constructing conceptual models by engaging in multimodal, context-aware dialogue.

Guiding Users Through UML Diagram Creation

Creating UML diagrams can be daunting, especially for beginners. Conversational agents can provide step-by-step guidance, aiming at helping users to construct diagrams correctly and efficiently.

Providing Real-Time Feedback and Examples

Conversational agents can also offer real-time feedback on user-created diagrams, suggesting improvements or pointing out errors.

Tailoring Learning Experiences to User Preferences

By incorporating user preferences and feedback, conversational agents can tailor the learning experience, ensuring that it meets the specific needs and comprehension levels of each user.

Table 7.2: Example Interaction between User and Agent

User	Can you explain what a class diagram is?
Agent	A class diagram is a type of UML diagram that represents the static structure of a system. It shows the system's classes, their attributes, operations, and the relationships among objects. Would you like to see an example?
User	Yes, please.
Agent	Here is a simple class diagram for a library system. It includes classes like <i>Book</i> , <i>Member</i> , and <i>Library Staff</i> , with attributes such as <i>Title</i> for the <i>Book</i> class and <i>Name</i> for the <i>Member</i> class. Would you like to dive deeper into any specific class?
User	Tell me more about the <i>Book</i> class.
Agent	The <i>Book</i> class has attributes like <i>Title</i> , <i>Author</i> , <i>ISBN</i> , and <i>Published-Date</i> . It also has operations such as <i>CheckOut</i> and <i>Return</i> , representing actions that can be performed on a book in the system.

Table 7.3: Example Interaction for Creating a Use Case Diagram

User	I want to create a use case diagram for an online shopping system. Can you help?
Agent	Of course! Let's start by identifying the primary actors. Who are the main users or systems interacting with your online shopping system?
User	Customers and administrators.
Agent	Great. Next, let's outline the main use cases. What are the primary functions that customers and administrators will perform?
User	Customers will browse products, add products to the cart, and make payments. Administrators will manage inventory and process orders.
Agent	Perfect. Now, let's sketch the diagram. We'll place <i>Customer</i> and <i>Administrator</i> as actors. <i>Browse Products</i> , <i>Add to Cart</i> , and <i>Make Payment</i> will be use cases connected to the <i>Customer</i> actor. <i>Manage Inventory</i> and <i>Process Orders</i> will be linked to the <i>Administrator</i> actor. Would you like to see a visual representation?
User	Yes, that would be helpful.
Agent	Here is a basic use case diagram. Do you need any further details or modifications?

Table 7.4: Example Interaction for Reviewing a Sequence Diagram

User	I've created a sequence diagram for a login process. Can you review it?
Agent	Sure, please upload your diagram.
User	[Uploads diagram]
Agent	Let's see. Your sequence diagram looks good overall, but the <i>Authenticate User</i> message is not clearly linked to the correct actor. It should be directed from the <i>User</i> to the <i>Authentication System</i> . Would you like to correct this now?
User	Yes, I'll fix it.
Agent	Once you've made the correction, you can upload the updated diagram for another review.

Table 7.5: Example Interaction for Learning Class Diagrams through Examples

User	I find it difficult to understand class diagrams. Can we focus more on examples and less on definitions?
Agent	Absolutely. Let's work through an example. Imagine we are designing a class diagram for a university system. We have classes like <i>Student</i> , <i>Professor</i> , and <i>Course</i> . Would you like to see how these classes interact with each other?
User	Yes, show me an example with interactions.
Agent	Here's a diagram showing the relationships between <i>Student</i> , <i>Professor</i> , and <i>Course</i> . <i>Student</i> is enrolled in <i>Course</i> , and <i>Professor</i> teaches <i>Course</i> . Each <i>Course</i> can have multiple students, but only one professor. Does this example help clarify class diagrams for you?
User	Yes, this is much clearer. Can we do another example with a different system?
Agent	Certainly! Let's create a class diagram for a hospital management system next. We'll identify classes like <i>Patient</i> , <i>Doctor</i> , and <i>Appointment</i> .

7.3.4 An Illustrative Scenario in Emergency Room Example

Consider a scenario where a surgeon employs a novel technique requiring a secondary incision. The system detects a deviation (X_t^{img} and X_t^{inst} differ from the expected process). Two major phases of human-in-the-loop involvement in our framework are:

1. Shallow Feedback (A):

- The system continuously captures data from multiple sources (e.g., gaze tracking, instrument usage, sensor logs).
- It compares the *as-is* process to the *desired* BPMN model.
- When a potential discrepancy or question arises, the system prompts the user (surgeon, nurse, etc.) for feedback.

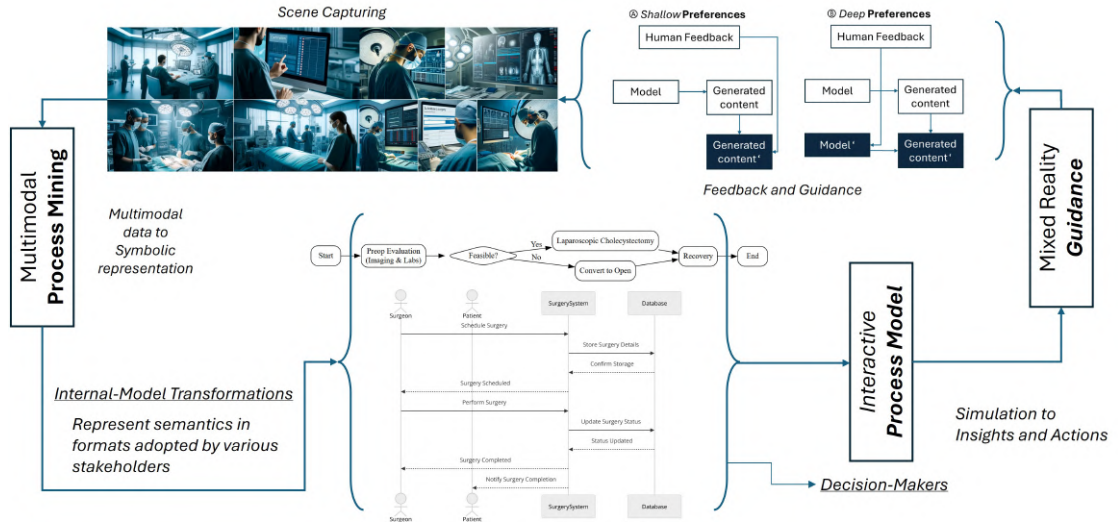


Figure 7.5: (Top) Illustration of human feedback via (A) Shallow and (B) Deep approaches, and a multimodal scene tracking. (Bottom) The solution's pipeline and illustrated (simplified) conceptual model representation adjusted for conformance checking and guidance. Adopted from Gavric et al. [GBP24b].

- The user provides confirmation, correction, or small clarifications. This feedback is used to adjust or annotate the *current* run-time process.
- The feedback is applied at the *instance level*, refining or steering the generated content (e.g., annotations, highlights, or visual focus) for the current execution without altering the underlying model.

2. Deep Feedback (B):

- In-depth reflections by human experts are used to refine the *model* itself or the methods that interpret the captured data.
- For instance, if the current process model does not account for a new device or step introduced by the surgical team, deep feedback cycles can lead to an updated BPMN model or a reconfiguration of the data capture pipeline.
- Over time, repeated deep feedback loops result in an evolving knowledge base that is more robust and better tailored to each specific surgery or environment.
- This feedback operates across instances, influencing how future content is generated by modifying the conceptual model, generation rules, or interpretation mechanisms rather than a single execution.

Figure 7.5 illustrates the integration of conversational AI into the mixed-reality pipeline for multimodal process mining and interactive guidance. The process begins with scene capturing, where multimodal inputs—such as visual streams, audio cues, and sensor

data—are transformed into symbolic representations that feed into the Multimodal Process Mining module. This enables the system to interpret human activities and decisions as structured events aligned with a process model. Through internal model transformations, these representations are converted into formats understandable by diverse stakeholders, such as surgeons, nurses, and system operators.

The **Interactive Process Model** acts as a conversational mediator, dynamically updated through **human feedback loops**. Shallow preferences refine generated content through immediate corrections, and deep preferences, reshape the model’s underlying reasoning and structure. These interactions exemplify how large language models can engage in reflective dialogue with domain experts, transforming user utterances and contextual signals into model revisions.

Finally, the updated process model informs the MR guidance environment, where simulations and AI-driven recommendations are projected back into the workspace. This establishes a continuous feedback cycle between perception, conversation, and action—enabling mixed-reality systems to act as intelligent collaborators that not only visualize but also discuss, adapt, and co-create conceptual models with human users in real time. A MR component of our setup serves multiple purposes:

- **Precision of Multimodal Recordings:** By using MR headsets, the system can track the surgeon’s gaze in relation to specific instruments or areas of the patient’s body. Likewise, position and orientation of surgical staff can be recorded.
- **Spatial Conceptual Modeling for Feedback:** It builds on *Spatial Conceptual Modeling* [Fil24], which allows to overlay real-time process conformance data directly into the OR environment. For example, a soft highlight (visible in the MR headset) might appear over the next instrument to be used, or an alert icon might appear above a piece of equipment that must be sanitized.

Chain-of-Thought LLM Prompting

The proposed methodology employs a *conversation engine* powered by LLMs that can: (1) **Parse sensor events** and interpret them in the context of the conceptual model, (2) **Generate feedback prompts** when conformance might be violated, (3) **Solicit clarifications and deeper insights** from the surgeon or nurse (for refining the model), and (4) **Provide intermediate “food-for-thought”** to guide the surgical team or system designers on why certain steps are suggested or flagged.

An important component of the conversational capabilities of AI for conceptual modeling is the construction of *well-curated prompts* that merge domain knowledge with real-time sensor data (e.g., the last tool recognized by a vision sensor was a cauterizing instrument). Table 7.6 presents two illustrative examples of how conversational prompts operate in mixed-reality settings. The first example demonstrates how the system interprets multimodal contextual cues—such as gaze fixation and temporal sequences of surgical

actions—to reason about the current process state. The second example shows the agent’s capability to generate structured modeling feedback, adapting the conceptual model representation based on user responses and detected procedural deviations. In this way, simulations give rise to *insights and actions* that are continuously fed back into the mixed-reality guidance component, establishing a closed-loop between multimodal perception, conceptual modeling, and situated intervention.

Table 7.6: Example Prompts for Understanding and Feedback Generation in Mixed Reality Contexts

Example Prompt for Understanding Mixed Reality Inputs

“The surgeon’s gaze has been fixated on the laparoscope for 5 seconds, and the nurse passed the laparoscope 10 seconds ago. The BPMN model indicates we are in the ‘Insert Laparoscope’ task. Confirm if this step is complete. If uncertain, ask for feedback from the user.”

Example Prompt for Generating Feedback

“The users indicated that they are testing a new technique requiring a secondary incision. The current BPMN model does not include this step. Rather than adding an optional sub-process, this should be modeled as an alternative path. Insert an exclusive Gateway with the existing and new technique subprocesses as subsequent elements. Record new recommended tasks and provide a revised BPMN snippet.”

7.3.5 Stakeholder-specific and jargon-based representation of multimodal data

Beerepoot et al. [BDR⁺23] have highlighted that the struggle between human involvement and task automation in managing work processes points to the significant impact that resolving these challenges will have on knowledge-intensive work. One of the significant challenges in process management is effectively communicating the insights gained from these analyses to stakeholders, who often come from diverse backgrounds with varying levels of familiarity with the technical and domain-specific language. The precision required in process models leads to the use of jargon, which, while transparent to domain experts, can be confusing or opaque to others. This communication barrier can hold back the adoption of process mining insights, limiting their impact on decision-making and process improvement.

For instance, let’s imagine a project manager, Sarah, who oversees the implementation of a new digital healthcare system in a large hospital. Sarah has extensive experience in project management, but her familiarity with healthcare-specific jargon is limited. She works closely with a team of doctors, nurses, and IT specialists, each fluent in their domain language. During meetings, the medical professionals often discuss processes in terms that are second nature to them—terms like “EHR integration,” “clinical workflows,” and “patient pathway optimization.” To them, these phrases precisely capture the complexity of the processes involved in patient care. However, to Sarah, who lacks a clinical background, these discussions often feel like a foreign language, making it

difficult for her to grasp the important details of the system she’s managing. Now, let’s imagine another scenario where Sarah is working on a similar project, but this time, the communication has been tailored to her level of understanding. Instead of using healthcare-specific jargon, the process models are described in more general project management terms. For instance, rather than discussing “EHR integration,” the conversation revolves around “aligning the digital system with existing hospital processes.” Instead of “clinical workflows,” they talk about “task sequences in patient care.” In this scenario, Sarah could feel more confident and engaged in the project because the information is presented in a way that resonates with her background and expertise. This contrast between jargon-heavy communication and language tailored to the listener’s experience is not just a hypothetical situation—it’s an identified challenge in many industries [FA10, Pet94, PBLV12], especially those that rely on complex processes and specialized knowledge, such as healthcare, finance, manufacturing, or education. The problem of jargon-laden communication is intensified in process mining, where the interpretation and representation of process models are essential for understanding and improving organizational workflows. Process mining involves extracting knowledge from event logs to visualize and analyze processes. However, when these process models are presented in technical or domain-specific language that stakeholders may not understand, the benefits of process mining can be significantly diminished. Existing methods have made progress in addressing similar challenges by processing language [BRSL23] or using visual aids [AAB⁺24, BPB21] to make process models more accessible.

In this section, we propose an approach that leverages transformer-based architectures to create personalized representations of process models. Our goal is to bridge the communication gap by developing two types of paraphrasing: one that aligns with the *domain-specific jargon* used by experts and another that adapts to the *individual communication styles* of different stakeholders. But language is only part of the equation. In complex fields like healthcare, education, and industrial operations, process models can benefit from integrating multimodal evidence. By understanding how different stakeholders interact with these multimodal representations, we can gain insights into their learning processes, which can improve how we teach and implement process mining techniques.

Domain-Specific Language Models

The use of domain-specific language models has been explored in several contexts, particularly in medical [LYK⁺20] and legal [CFM⁺20] domains, where the accuracy of language processing is critical due to the specialized terminology involved. These studies highlight the importance of tailoring language models to specific domains to improve performance. For instance, Lee et al. [LYK⁺20] developed BioBERT, a variant of BERT pre-trained on biomedical text, which significantly outperformed general-purpose models on tasks like named entity recognition and relation extraction in the medical domain. Our work draws inspiration from these works by training custom transformers on process model representations specific to different domains.

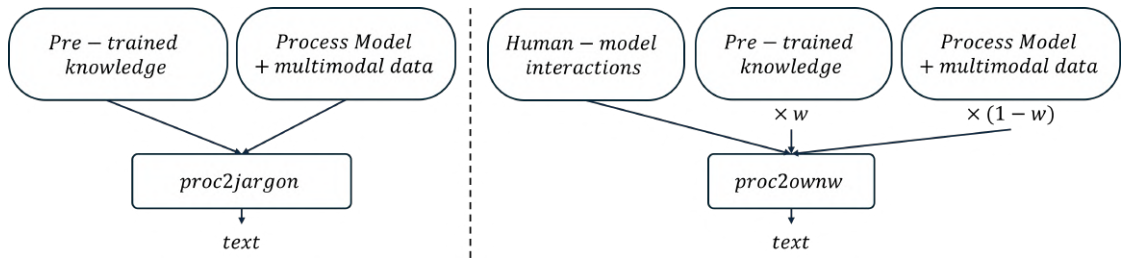


Figure 7.6: Illustration of (left) a jargon-specific paraphraser (`proc2jargon`) for domain-specific language and (right) a personalized paraphraser (`proc2ownw`) tailored to individual stakeholder preferences. Adopted from Gavric et al. [GBP24d].

In [GBP24d], we propose a transformer-based architecture to address the challenges of effectively communicating process models to stakeholders with varying levels of domain expertise, as detailed in Figure 7.6. This architecture consists of two core components: a jargon-specific paraphraser (`proc2jargon`) and a personalized paraphraser (`proc2ownw`); both are designed to translate process models and multimodal data into text that is accessible and meaningful to different audiences.

The component `proc2jargon` generates text that aligns with the specialized domain jargon. This model leverages pre-trained knowledge combined with the specifics of the process model and any associated multimodal data to produce outputs that maintain the technical rigor and precision expected by domain experts.

The component `proc2ownw` produces text personalized to individual stakeholders' communication styles and language preferences. This model balances integrating human-model interaction data with pre-trained knowledge and process model inputs to generate outputs that resonate with non-expert stakeholders.

7.4 Turning Conceptual Models into Video Simulation (PN-VidGen)

Business process simulation has long served as a tool for understanding, analyzing, and optimizing organizational workflows [MRLRD13], but developing simulation models manually is a labor-intensive and error-prone process, filled with numerous challenges. To overcome these limitations, researchers have proposed a range of automated techniques that extract process simulation models from historical event log data. These approaches include deep learning-based methods [CDGR22], quality evaluation frameworks [CCBB⁺23], runtime integration strategies [MDFG23], and agent-based discovery frameworks [KBK⁺24]. Traditional simulation techniques primarily focus on replaying event logs [vdA14]. However, the emergence of video generation models has enabled the creation of *realistic video simulations* that not only replicate the sequence of events but also provide lifelike scenarios that allow organizations to *enhance decision-making and training effectiveness* [SSLD18, Gag88]. Despite these advances, current video generation

approaches often fall short in capturing the *inherent dynamics* and *structural complexity* of **business processes**. The gap lies in their inability to reliably simulate the sequential nature of business workflows, which is important for producing consistent and actionable videos. This research narrows the video generation task’s focus to business process simulations, addressing the specific challenges: the lack of *business process-aware guidance* in video generation, and the need for *integrating actual process operational data* to drive the generation of consistent and realistic video simulations.

We propose a **Petri Net structure-driven video generation (PN-VidGen)** approach that builds on the formalism of discovered process models. In essence, the process model, comprised of places and transitions, *is played out* to construct a *storyboard*. This storyboard is then used as a structured instruction set to guide the video generation. The study employs a mixed-methods approach, integrating both qualitative and quantitative evaluations, to assess the simulation accuracy and applicability of the proposed approach. This line of work builds on the conceptual foundation laid in our previous work [GBP25e], where we introduced the Petri Net of Thoughts as a structure-enhanced prompting paradigm.

In the following, we provide an intuitive analysis of how video generation works, and how one can integrate process models. Thereafter, we detail our approaches for video business process simulation generation.

Video Generation

For video generation, we employ a video generation engine, described in [LZL⁺24]. Its interface allows (A) providing an input prompt that blends video *instance-specific information* with generic *video style* instructions, while the input prompt can be (A.1) textual or (A.2) textual with attached image; or by (B) *synthesizing a storyboard* which serves as the blueprint for generating a continuous video output.

Let $x \in \mathbb{R}^{H \times W \times 3}$ denote a single video frame and let a video be represented as $x = \{x_1, x_2, \dots, x_F\}$, with F frames. A video generation engine first encodes the video into a latent space using an encoder E : $z = E(x)$, $z \in \mathbb{R}^{T \times H' \times W' \times C}$, where T may be equal to F or a compressed temporal dimension, and H', W' are spatial dimensions in latent space. Then, it performs the diffusion process in this lower-dimensional latent space. The forward diffusion process gradually adds Gaussian noise, $q(z_t | z_{t-1}) = \mathcal{N}(z_t; \sqrt{1 - \beta_t} z_{t-1}, \beta_t \mathbf{I})$, $t = 1, \dots, T$, where $\beta_t \in (0, 1)$ is the noise schedule. After T steps, the latent becomes nearly pure noise: $z_T \sim \mathcal{N}(0, \mathbf{I})$. The noisy latent at step t can be written in closed form as: $z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, with $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$. The model learns to reverse this process using a Transformer-based denoiser. Given conditioning information c (e.g., a text prompt), the model approximates the reverse conditional distribution: $p_\theta(z_{t-1} | z_t, c) = \mathcal{N}(z_{t-1}; \mu_\theta(z_t, t, c), \Sigma_\theta(z_t, t, c))$. The training objective is to minimize loss: $L_{(\text{simplified})} = \mathbb{E}_{z_0, \epsilon, t} [\|\epsilon - \epsilon_\theta(z_t, t, c)\|^2]$, where $\epsilon_\theta(z_t, t, c)$ is the network’s prediction of the added noise at time step t . For video data, the latent

representation z is segmented into spatio-temporal patches: $\{p_i\}_{i=1}^N$, $p_i \in \mathbb{R}^d$, which act as tokens. A standard self-attention mechanism can be used to model the interactions between these tokens: $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$, mapping a query and a set of key-value pairs to an output, where the query (Q), keys (K), values (V), and output are all vectors. This serves the model to capture both spatial and temporal dependencies, aiming to ensuring consistency and coherence across frames. After the reverse diffusion yields the denoised latent z_0 , a decoder D transforms it back to the pixel space: $\hat{x} = D(z_0)$, resulting in the generated video frames.

Despite its capabilities, a video generation engine such as SORA currently operates on a primarily generic framework. Its design, while powerful for a broad range of applications [LZL⁺24, MLW24], does not inherently account for the structure and dynamics of business processes. In particular, several potentials are identified: (1) **Process Semantics Overlook**: Without explicit integration of process-specific information, the generated storyboard may fail to capture process activities, resources, and dependencies; (2) **Storyboard Inconsistencies**: The narrative flow, although coherent, might not align with the temporal/logical sequence inherent in business processes; and (3) **Limited Process Fidelity**: The absence of a process-aware evaluation (e.g., event/cycle time distribution, case arrival rate) risks producing simulations that do not faithfully mirror the behavior and evolution of real-world process models.

Integrating Discovered Process Models into Video Generation Pipeline

Given the stated limitations, we propose enhancing the video generation engine with process-aware guidance by incorporating discovered process models—such as those obtained via process mining. This integration aims at offering several advantages: (1) **Explicit Encoding of Process Dynamics**: Embedding process models (in particular, formal Petri Nets) into the storyboard aims at ensuring that every scene and transition is grounded in the actual operational logic of the business process; (2) **Temporal and Logical Consistency**: Aligning the storyboard with discovered process states and transitions aims at guaranteeing that the video accurately reflects the sequential and causal relationships of the process; and (3) **Enhanced Interpretability**: A process-aware storyboard provides insights into process behavior to the domain analysts, aiding in analysis and decision-making [SSLD18, Gag88].

Let a Petri Net be defined as $PN = (P, T, F, M_0)$, where P is a finite set of **places** (scenes or states), T is a finite set of **transitions** (actions or events), $F \subseteq (P \times T) \cup (T \times P)$ is the set of **arcs** (dependencies), M_0 is the **initial marking** (starting configuration). The evolution of the marking M over time (as tokens traverse the model places through transitions) represents the progression of the video narrative.

7.4.1 Three Strategies for Business Process-Guided Video Simulation

This section details the proposed video generation approaches, as illustrated in Fig. 7.7. In alignment with the naming conventions shown in the figure, we define three primary

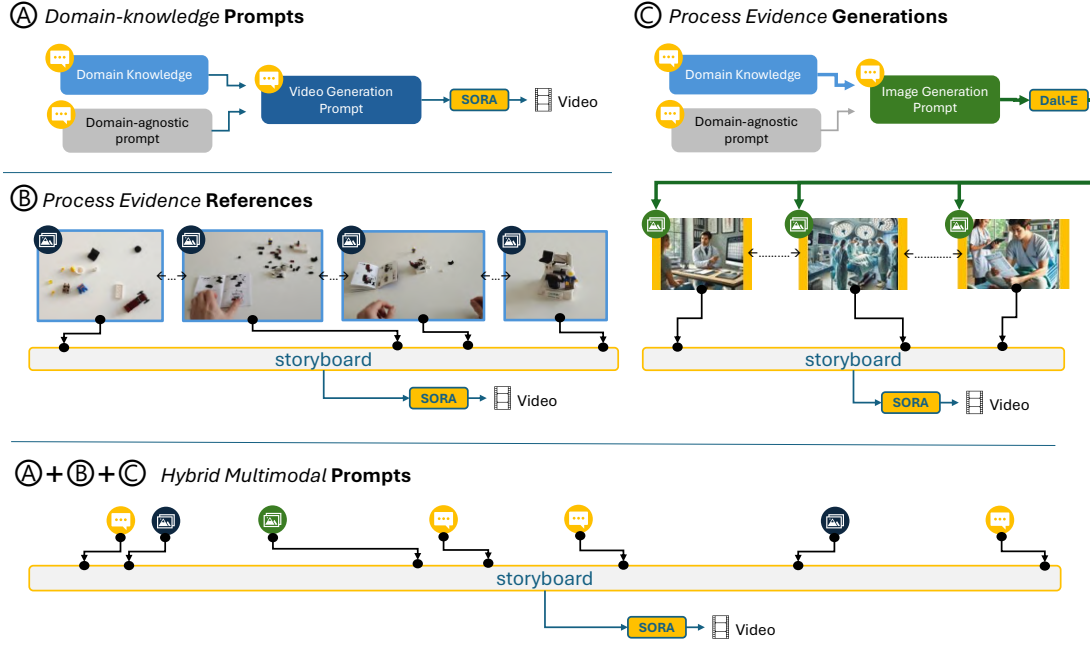


Figure 7.7: Overview of the proposed approaches for video generation. (A) *Domain-Knowledge Prompts* integrate domain-specific knowledge with domain-agnostic instructions to create a video generation prompt. (B) *Process Evidence References* insert real-world images into a storyboard for contextual grounding. (C) *Process Evidence Generations* rely on generative models (e.g., DALL-E) to produce synthetic images. Adopted from Gavric et al. [GBP25j].

strategies—approach A, B, and C—followed by a hybrid approach that unifies the best of all three.

Approach A: Domain-Knowledge Prompts.

Approach A (Fig. 7.7, top row) uses a *domain-knowledge-rich prompt*, enriched with domain-agnostic video generation instructions, to guide the video generation process. Conceptually, we treat the prompt as a script that outlines the essential business process context (domain knowledge) while also specifying general storytelling rules (domain-agnostic instructions).

Approach A Definition. Let I_A denote the composite prompt: $I_A = \mathcal{D} \oplus \mathcal{I}_{agnostic}$, where $\mathcal{D}$ is the set of domain-specific instructions and $\mathcal{I}_{agnostic}$ represents generic directives. The operator $\oplus$ concatenates these two sets of instructions into a single prompt. The video generation engine interprets I_A to generate a preliminary storyboard S_A , which is subsequently converted into a video. This approach is straightforward, yet could be

effective when domain experts can provide a sufficiently rich textual description of the process. We use this approach as our baseline method.

Approach B: Process Evidence References

Approach B (Fig. 7.7, middle row) emphasizes the integration of *process evidence references*, i.e., real-world images or snapshots from the actual business process execution. These references serve to the video generation engine as keyframes to anchor the storyboard, ensuring visual fidelity to the underlying process.

Approach B Definition. Let $\mathcal{I}_{proc}$ be a collection of process images. We treat these images as keyframes S_{key} within the storyboard: $S_{key} = \{img_1, \dots, img_n\} \subseteq \mathcal{I}_{proc}$. To achieve smooth transitions between keyframes, video generation engine realizes frame interpolation mechanism $\mathcal{F}_{interp}$: $S_B = \mathcal{F}_{interp}(S_{key})$. By using real operational images, Approach B grounds the simulation in authentic process visuals thereby enhancing the contextual relevance of the generated video.

Approach C: Process Evidence Generations

Approach C (Fig. 7.7, bottom row) introduces *process evidence generation* to handle scenarios where real operational images are unavailable or insufficient. Instead of relying on existing photos, it employs an image-generation model to synthesize visual references. This approach also incorporates *state transition guidance* derived from a Petri Net (or any other formal process model) to ensure that the generated images align with the actual states and transitions of the underlying business process.

Approach C Definition. Let us denote the set of states by $S = \{s_0, s_1, \dots, s_n\}$ and transitions by $T = \{t_1, t_2, \dots, t_n\}$, as discovered from a process model (e.g., a Petri Net). For each transition t_i , it generates a corresponding image by invoking a generative model $\mathcal{G}$ with an *image generation prompt* $\mathcal{P}_i$: $img_i = \mathcal{G}(\mathcal{P}_i)$. The prompts $\{\mathcal{P}_i\}$ incorporate domain-specific and domain-agnostic elements, ensuring contextually relevant visuals. It then assembles these generated images into a storyboard S_C : $S_C = \{img_1, img_2, \dots, img_n\}$, with each image corresponding to a state or transition in the process model. The interplay of formal process states (or transitions) with generative image synthesis aims to ensure that the resulting video accurately captures the logical flow of the business process, even when no real-world images are available.

Hybrid Approach (H): A + B + C

In the *Hybrid Approach*, we combine the strengths of Approaches A, B, and C to produce a robust, multimodal prompting pipeline (bottom line of Fig. 7.7). Specifically, we:

1. Use *domain-knowledge-rich prompts without multimodal reference augmentation* (core of Approach A) to encode high-level process logic.

2. Integrate *process evidence references* (core of Approach B) for tasks or segments where real images are available.
3. Employ *process evidence generations* (core of Approach C) via a generative model for tasks or transitions lacking real images.

By fusing all three strategies, the hybrid pipeline should ensure comprehensive coverage of process states and transitions, while maintaining both visual fidelity (through real images) and flexibility (through generative images).

Hybrid Approach Definition. Let $\mathcal{D}$, $\mathcal{I}_{agnostic}$, $\mathcal{I}_{proc}$, and $\mathcal{G}$ be the components from Approaches A, B, and C respectively. The *hybrid storyboard* S_H is constructed as: $S_H = (S_A \cup S_B \cup S_C)$, where S_A is derived from domain-knowledge prompts, S_B from real images, and S_C from generative images. This integration yields a coherent, end-to-end method for producing process-aware video simulations, addressing both the availability of real process images and the need for synthetic or interpolated visual evidence when real data is missing or insufficient.

Petri Net Structure-Driven Video Generation

A central pillar of the proposed approach is the use of Petri Nets to structure and guide the video generation process, aiming to ensure that the resulting simulation remains faithful to the underlying business process. This method is used in approaches B, C, and H. As depicted in Fig. 7.8, this method can be preceded by an *optional process discovery* phase that mines a Petri Net from event logs.

Event logs or Process Model as an input. If needed, a process discovery technique (e.g., inductive miner) can be used to extract a Petri Net from event logs. This step transforms real-world process data into a formal model $\mathcal{M}$ comprising $\mathcal{M} = (P, T, F, M_0)$, where P is the set of places, T is the set of transitions, F is the flow relation, and M_0 is the initial marking. In the illustrative example (Fig. 7.8), four places $\{P_1, \dots, P_4\}$ are illustrated, where P_1 and P_4 respectively denote the start and end of the process. Transitions $\{T_1, T_2, \dots\}$ connect these places according to the discovered behavior. Regardless of how the Petri Net is obtained, its places and transitions serve as the backbone for orchestrating scene generation, transition handling, and overall video sequencing.

Scene Definition. Each place P_i in the Petri Net is mapped to one or more *scenes* in the final video. A *scene* is a self-contained visual representation corresponding to the state of the process at P_i . For instance, in Fig. 7.8, P_1 and P_3 each link to specific scenes that depict the real or synthesized operational environment at those stages of the process.

Transition Handling via the Video Transition Agent Transitions T_j between places govern the movement of tokens in the Petri Net. In the video generation context, these transitions determine how the narrative flows from one scene to the next. As shown

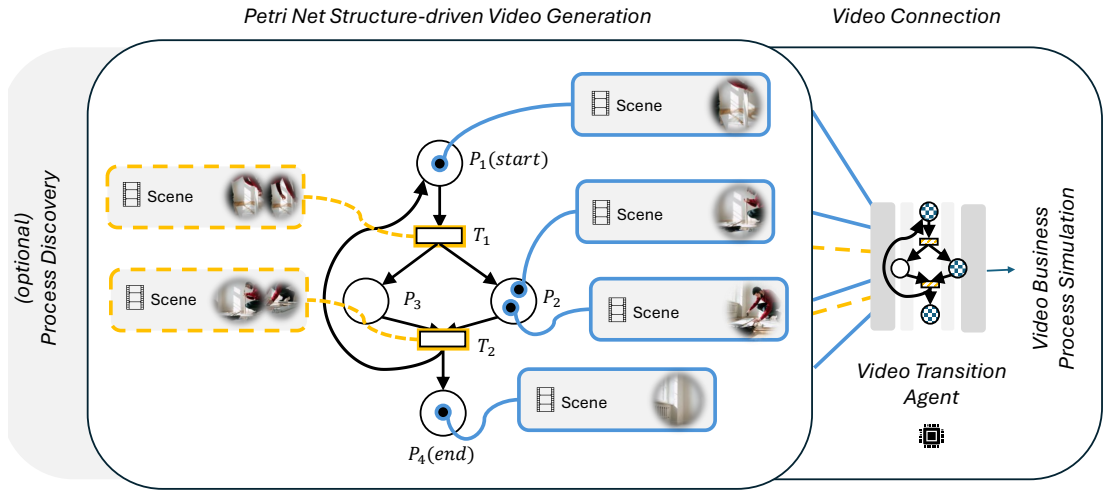


Figure 7.8: **Petri Net Structure-Driven Video Generation Architecture.** An optionally discovered Petri Net with places ($P_1, \dots, P_4$) and transitions ($T_1, T_2, \dots$). Each place corresponds to one or more *Scene(s)* in the final video. The *Video Transition Agent* then navigates through the Petri Net, invoking a video generation engine for scene generation and stitching these scenes together into a coherent *Video Business Process Simulation*. Adopted from Gavric et al. [GBP25].

in Fig. 7.8, a dedicated *Video Transition Agent* orchestrates these transitions in the video domain.

Video BPS End-to-End Flow The final output is a *Video Business Process Simulation* that reflects the structure of the Petri Net. The simulation begins at P_1 (start place), proceeds through transitions $\{T_1, T_2, \dots\}$, and concludes at P_4 (end place). Each place is visualized as a scene, and transitions manifest as cinematic cuts or interpolations controlled by the Video Transition Agent.

Conceptual Model Structure-Driven Video Generation

Figure 7.9 illustrates the workflow of a generalized approach for *Conceptual Model Structure-Driven Video Generation*, scaling beyond a Petri Net. This approach leverages the formal structure of a conceptual model—such as a UML use case or a BPMN process—to guide the generation of video sequences for domains where direct operational data or footage are unavailable.

The process begins with a conceptual model that outlines the major actors, interactions, and system functions. In the example shown, a UML model for the domain of *SurgerySystem* defines three top-level tasks: *Schedule Surgery*, *Perform Surgery*, and *Bill Patient*. Each task represents a node in the model’s state-transition graph and serves as a semantic anchor for generating visual content. From this structure, two types of textual prompts are composed: (i) *domain knowledge prompts*, which capture

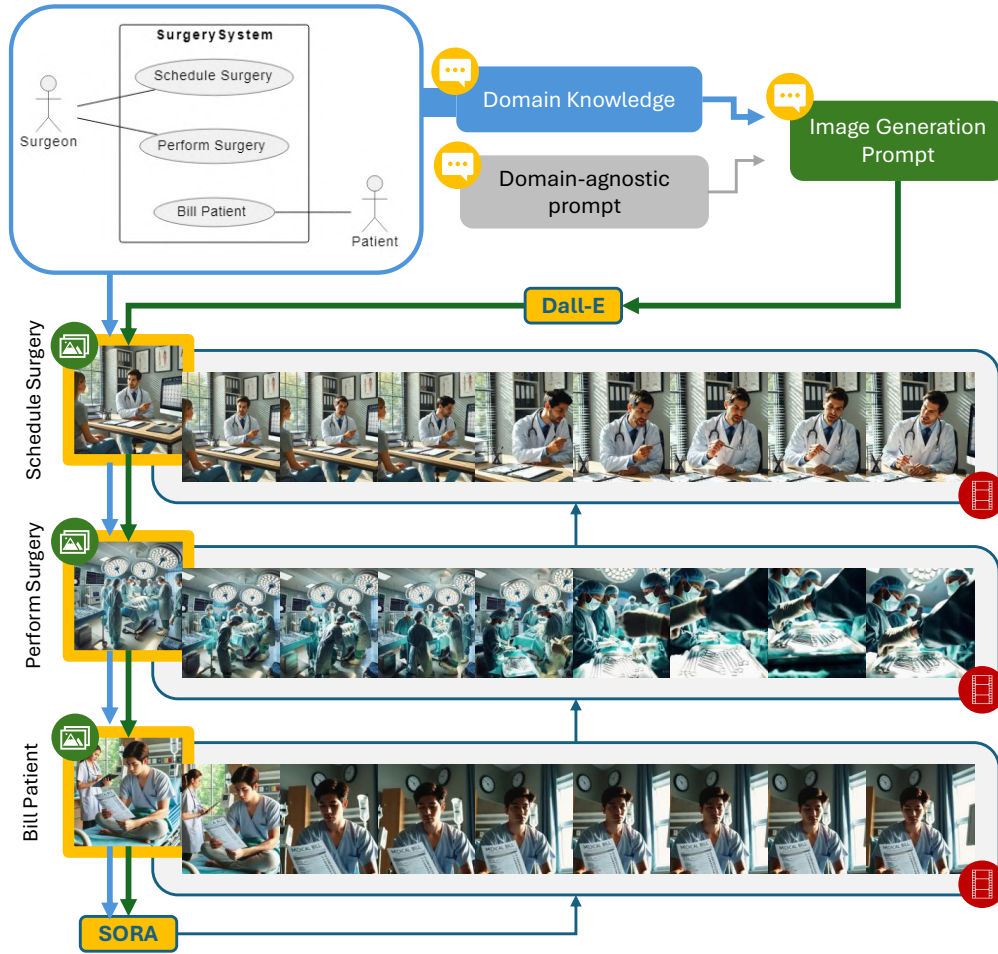


Figure 7.9: Conceptual Model Structure-Driven Video Generation in an example medical domain. The approach leverages the task decomposition from the conceptual model to guide prompt formation for image and video generation models. Adopted from Gavric et al. [GBP25j].

task-specific context (e.g., “a surgeon discussing a surgical plan with a colleague”), and (ii) *domain-agnostic prompts*, which describe general aesthetic or rendering properties (e.g., “cinematic lighting,” “first-person perspective,” or “hospital environment”).

The combined prompts are passed to a generative image model (e.g., DALL · E [RPG⁺21]) to produce a series of still images that visually represent the corresponding process state. These images are organized according to the task sequence defined in the conceptual model, thereby maintaining logical and temporal coherence. For a use case diagram in Figure 7.9, images are sequenced along the user–system interaction flow encoded by the use cases and their dependency relations, ensuring that actor intentions and system responses remain temporally and logically consistent. A video generation engine (e.g.,

Open Sora [LGC⁺24]) then interpolates between consecutive image sets to create smooth transitions, forming a continuous simulation of the modeled process.

This structure-driven approach ensures that each generated frame is both semantically aligned with the conceptual model and visually consistent across transitions. By grounding the generation in explicit model logic rather than purely textual narratives, the resulting videos can be systematically verified against the underlying model semantics. Furthermore, because the video synthesis process is modular, it can incorporate model updates, alternative pathways, or branching decisions, enabling flexible re-simulation of “what-if” scenarios without requiring new data collection [LZL⁺24].

7.5 Virtually Enhanced Narratives (ViEnNa)

Traditional conceptual models, often depicted as flowcharts or formal diagrams, aim to capture the sequence and logic of activities within a process. However, these representations frequently fall short of reflecting the richness and complexity of real-world processes. They typically lack contextual information, overlook the diversity of data sources involved, and fail to convey the dynamic, interactive, and multimodal nature of human work [BGP⁺24]. To overcome these limitations, research in process modeling has increasingly shifted toward representations that emphasize the relationships between data, activities, and contexts. For instance, the development of *object-centric event logs (OCEL)* and their more expressive successor *OCEL 2.0* [BKA⁺24] enables the fine-grained capture of process executions, tracking state changes and object interactions over time. Conceptual models can be derived from those event logs using techniques such as process discovery [vdA12]. This evolution demonstrates a movement toward more inclusive and reality-grounded representations of processes, particularly when combined with multimodal data sources such as video, audio, or sensor streams [LBS⁺23].

Building upon these advances, we propose a new conceptual modeling language—***Virtually Enhanced Narratives (ViEnNa)***—that blends the semantic rigor of object-centric models with the expressive richness of visual storytelling. Our goal is to design a modeling framework that not only integrates multimodal evidence into process models but also enhances the accessibility and appeal of process representations for diverse audiences, including both domain experts and non-experts.

In contrast to traditional conceptual modeling notations that rely solely on abstract symbols, ViEnNa adopts a *comic-like visual paradigm* to communicate process logic and context through narrative and imagery. The intent is to preserve the analytical depth of formal modeling while introducing a visual language that conveys human intention, object interaction, and situational nuance more intuitively.

Comics have proven to be powerful mediums for communication and cognition across disciplines. They have been successfully applied in scientific explanation [BB18], legal communication through *comic contracts* [KdRH24], and even software engineering, where code behavior is visualized through comic sequences [HST24]. Computational

approaches to comic generation [LD20, NRB19] and ontology-based interpretation of visual panels [GRBR17] further illustrate the potential for structured, machine-assisted understanding of comic narratives. Recent progress in generative AI has also made it feasible to automatically produce coherent comic panels from text or structured data [KBS24], enabling scalable creation of narrative-driven representations.

The motivation behind ViEnNa is to improve human–model interaction [FMPK21] by embedding process models within visually enriched narrative contexts that link model elements to multimodal evidence. Rather than merely visualizing multimodal data retrieved from memory or logs, ViEnNa seeks to *understand and synthesize* such data into coherent, evidence-aware storylines that communicate the semantics of process execution. This includes leveraging object- and character-driven visual metaphors to represent process participants, resources, and decisions in ways that are grounded in domain semantics but intuitively readable to humans and interpretable by AI agents.

ViEnNa thus introduces a new category of conceptual modeling artifacts: *comic-like process models*, where tasks, actors, and data objects are situated in sequential visual panels that together form a narrative aligned with process semantics. Each panel represents an activity or event instance, enriched with contextual cues such as environment, emotion, or spatial configuration. The resulting representation is both analytically structured and narratively expressive, bridging formal process logic with human-centered storytelling.

The **ViEnNa framework** encompasses three tightly integrated components: (1) a **modeling language** that specifies how process entities, objects, and multimodal cues are visually and semantically encoded in comic-like representations; (2) a **method** for transforming process and evidence data into coherent visual narratives, and (3) a **tool** that supports interactive creation and interpretation of ViEnNa models by both humans and AI agents.

Through this triadic structure, ViEnNa aims to position comic-like process modeling as a new paradigm in conceptual modeling—one that merges analytical rigor, multimodal evidence, and narrative visualization into a unified framework for enhanced understanding, engagement, and collaboration.

ViEnNa Conceptual Modeling Framework

The ViEnNa modeling framework is designed with built-in AI capabilities that enable multimodal data inputs to represent (real-world) data within the process model.

Our methodology blends structured event data with multimodal evidence to synthesize process models in the form of comic panels. This is achieved by integrating three core components: pixel-level image generation using diffusion-based models, multimodal data processing via AI-supported retrieval mechanisms (Section 7.5.2), and, a novel domain-specific modeling language tailored for narrative-first process representation, whose derivation is provided in Section 7.5.3.

Preliminaries. Definitions 46 and 47 together establish the theoretical foundation for ViEnNa as a multimodal conceptual modeling language. The *Process Comic* (Definition 46) specifies the visual and narrative structure of process representations, framing each activity as a multimodal scene composed of objects, actions, and spatial arrangements. In contrast, the *MMCM Framework* (Definition 47) provides a general mechanism for linking these visual-narrative elements to underlying multimodal evidence, ensuring bidirectional communication between human interpreters and computational models. In ViEnNa, the MMCM serves as the semantic layer that anchors each comic frame to the evidential data from which it was derived—closing the loop between visual storytelling and data-driven process understanding. Together, these formalisms provide a rigorous bridge between conceptual modeling, multimodal data analysis, and human-centered visual communication.

We developed ViEnNa as a concrete instantiation of a **process comic**, where the sequence F is generated using AI-based image synthesis conditioned on object-centric event data, the objects O and actions A are extracted from parsed OCEL logs, and the layout function L arranges frames according to temporal and logical process constraints. Furthermore, ViEnNa satisfies the properties of a **Multimodal Conceptual Modeling** framework by integrating multimodal evidences (e.g., text, visuals, and audio) into each panel. To establish ViEnNa as an MMCM framework, we developed a system that systematically integrates a suite of AI models and methods.

7.5.1 Ways of Using the ViEnNa Framework

Figure 7.10 illustrates three primary ways of using the ViEnNa framework for creating and interpreting process models. While these approaches represent distinct modes of engagement—ranging from free-form to data-driven modeling—they collectively form a unified methodological spectrum that connects creative visualization, conceptual modeling, and multimodal evidence integration.

Pixel-Level Generation. In the first approach (Fig. 7.10a), a modeler creates *ViEnNa process models* by employing pixel-level generation of comic panels through generative AI. This mode focuses on visual storytelling, allowing modelers to abstract business processes and complex concepts—including emotions, sound effects, or environmental cues—into expressive and engaging narratives. Such representations may be used to depict business situations, contextual conditions, or operational steps that would otherwise be difficult to convey in symbolic notation, thus enhancing intuitive understanding and audience engagement.

Conceptual Model Navigation. The second approach (Fig. 7.10b) focuses on the narrative interpretation of existing process models, such as BPMN or UML sequence diagrams. Here, ViEnNa serves as a *narrative wrapper* that visualizes the storyline behind the reading or exploration of diagrammatic models. This approach orchestrates the reader’s attention through the logical flow of the model, highlighting relevant concept

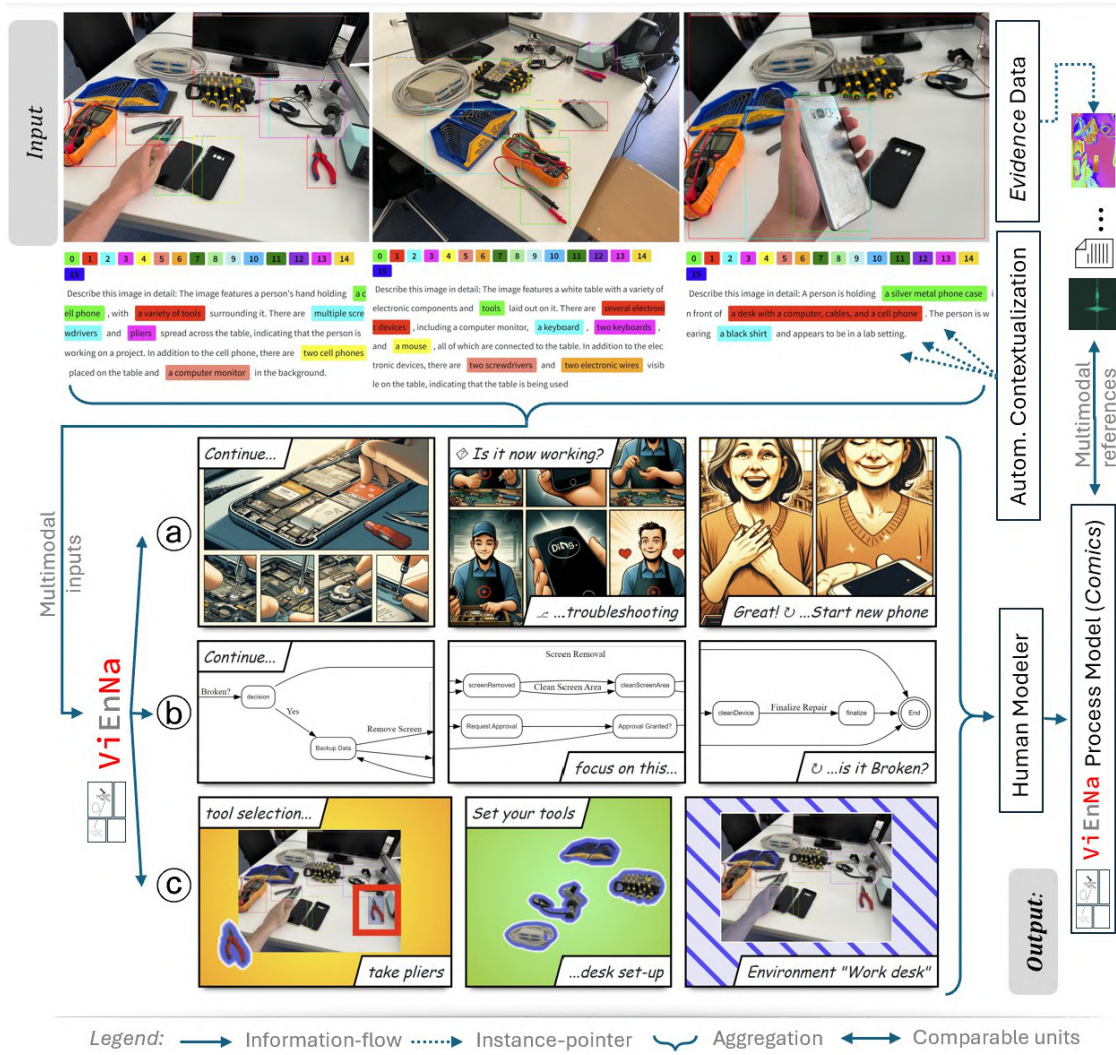


Figure 7.10: ViEnNa framework illustration. (a) Pixel-level generated panels using generative AI, (b) Conceptual model navigation, and (c) Panels modeled with semantic retrieval. Adopted from Gavric et al. [GBP25c]. (Example-domain labels are intentionally left unreadable.)

elements and their interconnections in a sequential, comic-like narrative. It thus adds an additional layer of abstraction that transforms static diagrams into guided experiences.

Semantic Retrieval Modeling. The third approach (Fig. 7.10c) introduces semantic retrieval to link visual representations directly with multimodal evidence data. In this mode, the framework interprets high-level conceptual descriptions and retrieves visual elements—such as frames, scenes, or actions—from an evidence pool containing videos, images, audio recordings, or textual narratives. By connecting modeled elements to

their real-world instances, this approach aims to enable the automatic generation of contextually grounded process panels.

Unified Practice. Although these three use cases are conceptually distinct, ViEnNa is designed to integrate them into a *unified modeling practice*. This allows traditional process modeling elements (e.g., BPMN tasks or data objects) to be seamlessly connected to multimodal evidence, while enabling narration-driven scenario modeling. The result is a dynamic modeling environment where evidence, visualization, and formal structure coexist and inform one another.

Automation. ViEnNa supports *semi-automated modeling*, where AI assists in interpreting multimodal inputs and generating visual narratives, but the modeler retains control over both the narrative flow and the structural semantics of the process. This aims to ensure that automation enhances, rather than replaces, human creativity and interpretive judgment in the modeling process.

Domains and Users. The framework is designed to be *domain-agnostic*, yet it is particularly effective in areas rich in multimodal evidence, such as healthcare, manufacturing, and education. It aims to accommodate a wide range of users—from process modeling experts to non-experts and AI agents—supporting both model creation and interpretation. This inclusivity reinforces ViEnNa’s objective to democratize process modeling and improve communication between human stakeholders and AI systems.

Usage by AI Agents. Beyond human use, ViEnNa enables AI-to-AI communication through the exchange of structured, interpretable visual narratives. AI agents can exchange large quantities of ViEnNa comics per second as transparent, process-oriented representations of reasoning and decision-making. Unlike conventional black-box information exchanges [BAR23], this approach maintains semantic interpretability, ensuring that both humans and AI systems operate within a shared, comprehensible modeling language.

7.5.2 ViEnNa as a Method

While the ViEnNa framework defines a novel way of representing processes through virtually enhanced narratives, its methodological foundation lies in the technical mechanisms that enable the creation, synthesis, and alignment of these visual representations with real-world data. *ViEnNa as a Method* describes how the framework operationalizes the principles of multimodal conceptual modeling to generate, retrieve, and integrate visual content into coherent comic-like process models. This methodological layer ensures that the ViEnNa language is not only expressive but also computationally grounded—bridging human creativity and data-driven precision.

Figure 7.11 illustrates how the ViEnNa method integrates *virtual enhancement* with *narrative structuring* to produce semantically grounded and intuitively comprehensible

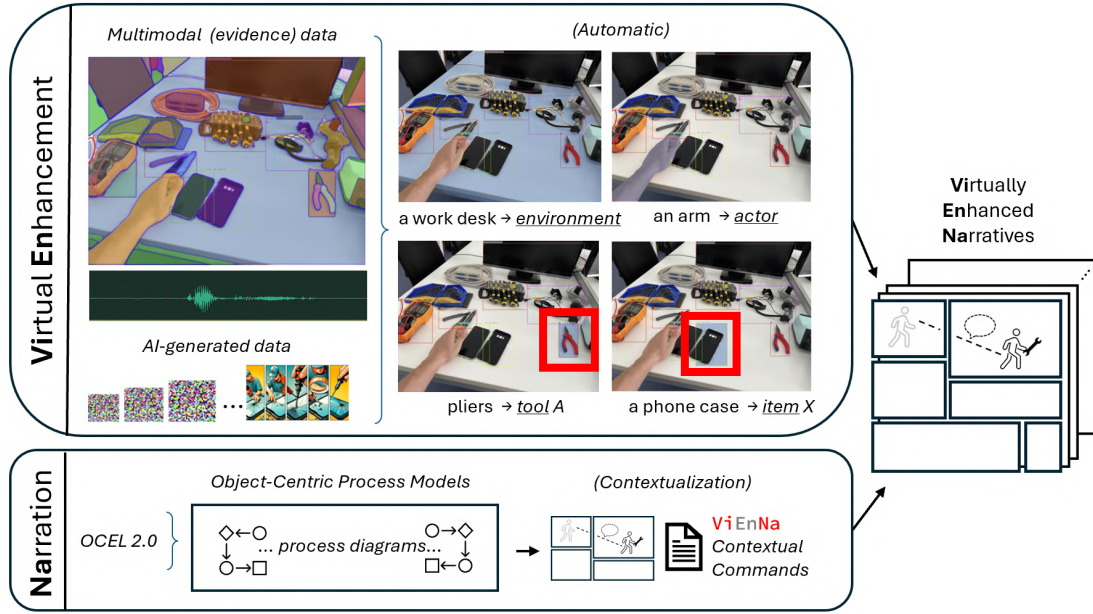


Figure 7.11: Overview of the ViEnNa method combining virtual enhancement and narrative structuring. Adopted from Gavric et al. [GBP25c].

process representations. In the *upper part* of the figure, AI components analyze multimodal data sources—such as images, videos, text, and audio—to extract relevant contextual and behavioral information. This analysis enables the automatic identification of key events, actors, and interactions, which are then translated into structured semantic elements suitable for modeling. In the *lower part*, the human modeler acts as the narrative designer, arranging these extracted elements into coherent comic-like panels that convey process logic and temporal flow. This human–AI collaboration mirrors the workflow of comic strip creation: while the AI contributes content understanding and visual generation, the human ensures story coherence, focus, and interpretability. Together, these layers exemplify ViEnNa’s central principle—combining data-driven evidence with human-centered narration to transform complex process knowledge into visually enriched, narrative-oriented conceptual models.

The method combines two key components. First, it employs **image synthesis** techniques based on denoising diffusion probabilistic models (Section 7.5.2), which generate comic panels directly from textual or structural model descriptions. These panels serve as the visual foundation of ViEnNa models, translating conceptual constructs into perceptually meaningful scenes. Second, it includes a **matching procedure** (Section 7.5.2) that aligns generated or retrieved images with *object-centric event logs* and other multimodal evidence sources. This ensures that each visual narrative remains traceable to factual data, thereby maintaining semantic consistency between modeled processes and their empirical manifestations.

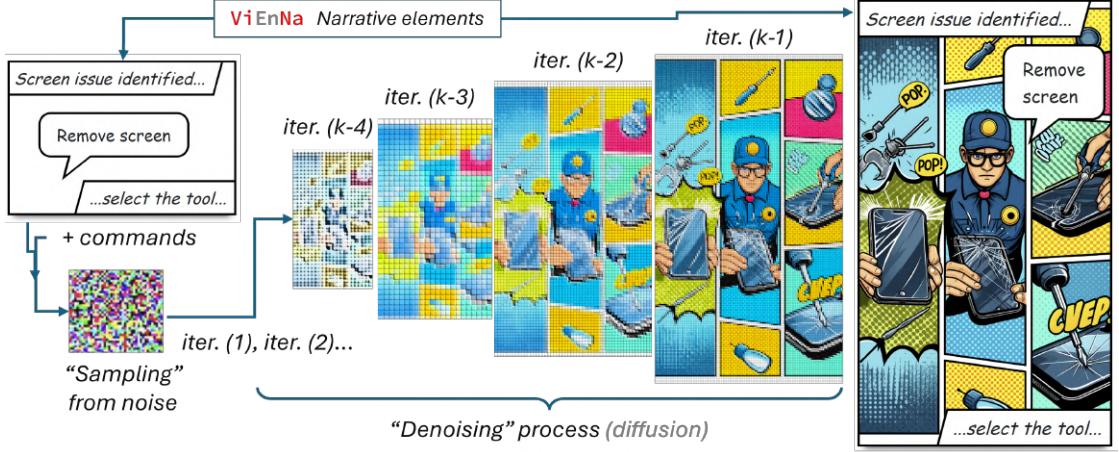


Figure 7.12: Panel generation (pixel-level) using generative diffusion model. Adopted from Gavric et al. [GBP25c].

Synthesizing Comic Image

For image synthesizing, we use the principles of denoising diffusion probabilistic models [RPG⁺21] for image generation. Starting from Gaussian noise, the process involves learning a series of denoising autoencoders to iteratively refine an image (as illustrated in Fig. 7.12). Let x_0 be the original image and x_T be the Gaussian noise. The forward diffusion process adds noise to the image in T steps, defined as $q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)\mathbf{I})$, where α_t is a noise schedule. The reverse process, parameterized by a neural network ϵ_θ , predicts the noise component at each step, given by $p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$, where $\mu_\theta(x_t, t)$ and $\Sigma_\theta(x_t, t)$ are functions learned during training. The loss function for training the network is the variational lower bound (VLB) of the data likelihood, expressed as

$$L = \mathbb{E}_q \left[\sum_{t=1}^T D_{KL}(q(x_{t-1}|x_t, x_0) || p_\theta(x_{t-1}|x_t)) - \log p_\theta(x_0|x_1) \right].$$

This training minimizes the difference between the predicted noise and the actual noise added during the forward process, allowing the model to generate high-quality images by effectively reversing the diffusion process.

Matching Real-World Processes and Event Data

One of the major challenges our framework faces is integrating multimodal evidence data with high-level event logs, particularly object-centric event logs, into a unified process model representation. ViEnNa aim to enable interpretation of high-level event logs based on setting a specific objective through natural language in semantic retrieval and image synthesizing.

Extracting from OCEL 2.0. At first, we focus on extraction of concepts from OCEL 2.0, in particular, *universes of strings* U_Σ , from which we derive various pairwise disjoint universes, as defined by [BKA⁺24]. These include $U_{ev} \subseteq U_\Sigma$ for events, $U_{etype} \subseteq U_\Sigma$ for event types (activities), $U_{obj} \subseteq U_\Sigma$ for objects, $U_{otype} \subseteq U_\Sigma$ for object types, $U_{attr} \subseteq U_\Sigma$ for attribute names, $U_{val} \subseteq U_\Sigma$ for attribute values, U_{time} for timestamps with $0 \in U_{time}$ as the smallest element and $\infty \in U_{time}$ as the largest, and $U_{qual} \subseteq U_\Sigma$ for qualifiers. Secondly, we use the structures provided by the OCEL 2.0 specification to instantiate a tuple $L = (E, O, EA, OA, evtype, time, objtype, eatype, oatype, eaval, oaval, E2O, O2O)$, where $E \subseteq U_{ev}$ is the set of events, $O \subseteq U_{obj}$ is the set of objects, $evtype : E \rightarrow U_{etype}$ assigns types to events, $time : E \rightarrow U_{time}$ assigns timestamps to events, $EA \subseteq U_{attr}$ is the set of event attributes, $eatype : EA \rightarrow U_{etype}$ assigns event types to event attributes, $eaval : (E \times EA) \rightarrow U_{val}$ assigns values to event attributes, $objtype : O \rightarrow U_{otype}$ assigns types to objects, $OA \subseteq U_{attr}$ is the set of object attributes, $oatype : OA \rightarrow U_{otype}$ assigns object types to object attributes, $oaval : (O \times OA \times U_{time}) \rightarrow U_{val}$ assigns values to object attributes, $E2O \subseteq E \times U_{qual} \times O$ are the qualified event-to-object relations, and $O2O \subseteq O \times U_{qual} \times O$ are the qualified object-to-object relations.

For instance, consider a manufacturing enterprise where different sensors, logs, and user inputs provide multimodal data. We map these inputs into our defined structure as follows: For any event $e \in E$ and event attribute $ea \in U_{attr}$, we map $eaval_{ea}(e) = eaval(e, ea)$ if $(e, ea) \in \text{dom}(eaval)$; otherwise, it is $\perp$. For example, an event could be the completion of a production task, with attributes like machine ID, operator ID, and timestamp. Similarly, for any object $o \in O$, object attribute $oa \in U_{attr}$, and time $t \in U_{time}$, we map $oaval_{oa}^t(o) = oaval(o, oa, t')$ if there exists $t' \in U_{time}$ such that $t' \leq t$ and no earlier t'' exists that meets the same criteria. If no such t' exists, then $oaval_{oa}^t(o) = \perp$. For example, an object could be a specific machine part with attributes like wear level, maintenance status, and temperature readings over time.

Blending OCEL 2.0 and Multimodal Embeddings. The blending of OCEL 2.0 and multimodal embeddings is achieved by setting a *task-centric objective*. This objective organizes the relationships and dynamics between multimodal and event data through human intervention via natural language. We prompt the LLM component of our framework with business-context questions, such as *Who*, *What*, *Why*, *When*, *Where*, *How*, and *How Much*, to analyze both multimodal data and OCEL. This is achieved through two primary methods: (1) retrieving references directly from the multimodal streams, and (2) generating abstractions of these streams through visual or textual synthesis. This way we provide detailed and context-rich answers. To combat hallucinations and false information, we employ principles from retrieval-augmented generation [LPP⁺21] to improve the accuracy and reliability of the generated outputs. This approach combines the strengths of retrieval-based systems and generative models, using the former to ground responses in factual, retrievable data. This approach aims to reduce the risk of generating plausible-sounding but incorrect information, which is a common issue in purely generative models [HYM⁺23].

By re-setting the task-centric objective, ViEnNa adapts and re-contextualizes inputs based on novel objectives. This dynamic re-creation process adjusts ViEnNa to capture different aspects and details extracted from the raw data sources. This approach aims to enable the creation of multi-perspective process models, tailored to various analytical and training needs. It is implemented by employing multimodal embeddings for process model annotations, instead of relying solely on textual labels. For instance, let multimodal embeddings E be created vectors in a high-dimensional space $\mathbb{R}^d$, where d is the dimensionality of the embedding space. Then, each event $e \in E$ and object $o \in O$ are represented as $\mathbf{e} \in \mathbb{R}^d$ and $\mathbf{o} \in \mathbb{R}^d$, respectively. The embedding approach is defined as follows: given an input data point x , the embedding function f maps x to a vector in $\mathbb{R}^d$, i.e., $f : x \mapsto \mathbf{v} \in \mathbb{R}^d$. This process should capture the essence of the input data in a lower-dimensional representation while preserving the critical information. For example, $f(e) = \mathbf{e}$ and $f(o) = \mathbf{o}$ for events and objects. By using embeddings, we do not commit to a single decoding of our labels in the universe of strings U_Σ . Instead, we introduce a new concept—the universe of embeddings $U_E \subseteq \mathbb{R}^d$, where elements can be iteratively decoded based on the set objectives. This iterative decoding process enables embeddings to dynamically adapt to different analytical needs without being constrained by predefined labels.

7.5.3 ViEnNa as a Modeling Language

The ViEnNa process modeling language is designed to be used either as a standalone visual process modeling language or as a meta-language to encapsulate other visual process modeling languages into a narrative-driven process model. It is specifically tailored as a domain-specific modeling language that is particularly focused on processes rich in narrative-intensive activities and blind spots introduced by manual work. It is ideally suited for modeling business operations that involve extensive physical and manual tasks with limited human-computer interactions, especially in fields like surgery, construction, and heavy industries. Several methodologies for developing domain-specific languages have been proposed, [Fra13, JRD⁺17, KKP⁺14]. The macro-process from [Fra13] is adopted, which outlines seven detailed phases (clarification of scope and purpose, analysis of generic requirements, analysis of specific requirements, language specification, design of graphical notation, development of modeling tool, and lastly, evaluation and refinement). Next, we provide a clarification of the scope and purpose, as well as an analysis of both generic and specific requirements.

Derivation of the ViEnNa Modeling Language

Clarification of Scope and Purpose. ViEnNa aims to make process modeling more accessible and useful, especially in narrative-driven scenarios where traditional models fall short. By emphasizing a narrative-driven approach and focusing on visual cues and the sequential flow of events, it helps all stakeholders—regardless of their technical background—understand complex processes more intuitively. This not only

improves communication and training but also enhances documentation and compliance for processes that are typically difficult to track.

Analysis of Generic Requirements. For the ViEnNa process modeling language, the development framework is structured around three essential General Requirements, corresponding to our three fundamental pillars – narration, deduction, and formation.

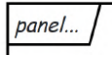
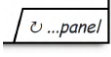
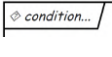
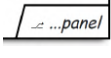
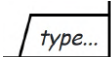
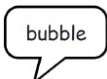
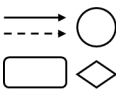
GR-1: Narration: *The language must effectively support narrative-driven processes, enhancing storytelling in environments that involve significant physical and manual work.*

GR-2: Deduction *The language needs to include analytical, predictive, and generative tools to aid in decision-making and effectively utilize multimodal evidence, thereby enabling users to extract actionable insights from the modeled scenarios.*

GR-3: Formation: *It should integrate seamlessly with existing major modeling languages and provide accurate representations of diverse system architectures, ensuring it is adaptable and precise for modeling complex systems.*

Analysis of Specific Requirements. To translate the broad goals set by the general requirements into actionable steps, we have outlined specific requirements (**SR-1** to **SR-13**). These detailed criteria help implement the principles of *Narration*, *Deduction*, and *Formation* in the modeling language, ensuring it meets the unique needs of different applications effectively. *GR-1* aims to enhance the narrative capability of the language, making it intuitive and engaging. *The modeling language must be designed to offer visual clarity and intuitiveness, facilitating immediate understanding and ease of use for all users (SR-1).* *It should incorporate features that promote interactivity and sustain user engagement throughout the modeling process (SR-2).* Additionally, *the language needs to support narrative flows centered around characters or key elements, ensuring coherence and engaging storytelling (SR-3).* *The modeling language must be easy to learn, allowing new users to quickly become proficient (SR-4).* *It must effectively represent time, allowing for an accurate depiction of scenarios as they evolve (SR-5).* *The language should accurately represent the dynamics within scenarios, highlighting changes and interactions (SR-6).* *GR-2* focuses on enabling users to analyze data and predict outcomes through the modeling process. *Predictive analytics capabilities must be integrated into the language, providing implicit predictions about enriching narrative scenarios based on provided data (SR-7).* *The language should be capable of incorporating multimodal evidence (text, images, video) to enhance the richness of the narratives and support modeling tasks (SR-8).* *It needs to possess generative capabilities to autonomously produce data and scenarios, enhancing its predictive and analytical utility (SR-9).* *GR-3* ensures the structural adaptability of the language, aligning it with existing frameworks and accurately depicting both, physical and digital, systems architectures. *The new modeling language must integrate with existing models and frameworks (SR-10).* *It must operate effectively across a full spectrum of abstraction, from high-level process overviews to detailed, specific artifacts (SR-11).* *The language must represent both, physical and*

Table 7.7: Narrative graphical notations in ViEnNa. Adopted from Gavric et al. [GBP25c].

Notation	Concept	Description
	<i>Declaration</i>	Used to instantiate a new panel within the narrative, introducing a specific scene. This element renders in the top left corner of a panel and has a structure: <code><panel_name><three_dots></code> .
	<i>Unconditional Jump</i>	Denotes a direct jump or transition in the narrative to another panel, occurring without any conditions. This element renders in the bottom right corner of a panel and has a structure: <code><jump_symbol><three_dots><destination_panel_name></code> .
	<i>Condition</i>	Represents the instantiation of a panel that is conditional; the narrative within this panel will proceed only if the condition is true. This element renders in the top left corner of a panel and has a structure: <code><condition_symbol><panel_name><three_dots></code> .
	<i>Conditional Jump</i>	Indicates a jump to another panel that is contingent on a specific condition being met; if the condition is false, the narrative follows the provided alternative path. This element renders in the bottom right corner of a panel and has a structure: <code><conditional_jump_symbol><three_dots><destination_panel_name></code> .
	<i>Typing</i>	Specifies the type or nature of elements within the narrative, categorizing or clarifying the roles and characteristics of narrative components. This element renders in the bottom right corner of a panel and has a structure: <code><type_name><three_dots></code> .
	<i>Bubble</i>	Used to display dialogue or thoughts of characters, typically in comics to show what a character is saying or thinking in a visually direct manner. This element renders anywhere inside a panel and has a structure: <code><bubble_text></code> .
	<i>Shapes, Arrows, Labels and Other</i>	Utilized for representing various process models to be incorporated into ViEnNa, aiding in the visualization and connection of different components and processes within the narrative. These elements can render anywhere inside a panel.

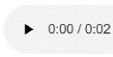
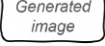
digital architecture of the systems it models (SR-12). It should include a precise digital versioning system to efficiently manage and track changes in models (SR-13).

Language Specification and Notation Design

ViEnNa process models are composed of panels arranged in a sequence that is read from left to right, starting from the top row and progressing to the bottom. Within each panel, a process model can be displayed with various parts zoomed in and focused to highlight specific details. The language specification for the ViEnNa process modeling language includes two pivotal components: *Narrative Graphical Notations* and *Virtual Enhancements*, each tailored to enhance the storytelling and visual representation of multimodal reference data within a modeling framework, respectively.

The Narrative Graphical Notations (as provided in Table 7.7) are designed to clarify and engage through visual storytelling conventions akin to those found in comics. The narrative begins with the *declaration of a panel*, used to instantiate new scenes within the model. A panel can be referred to by its unique name. To enable modeling concurrency and eliminate the limitation of modeling each state individually, we provide conditions and jumps. *Unconditional jumps* allow for direct transitions between panels without any condition, facilitating straightforward narrative progression or flashbacks.

Table 7.8: Notations for virtual enhancements with graphical examples in ViEnNa. Adopted from Gavric et al. [GBP25c].

Example	Concept	Description
	<i>Semantic Image Retrieval</i>	Involves retrieving images based on natural-language descriptions or other modality inputs, allowing users to find relevant images by describing them in words or using other media as prompts. Syntax: <code><visualize_image:><concept_descriptor></code> .
	<i>Semantic Video Retrieval</i>	Enables the retrieval of video content using natural-language queries or other types of media, making it possible to locate specific video segments by describing the content or providing related media as input. Syntax: <code><visualize_video:><concept_descriptor></code> .
	<i>Semantic Audio Retrieval</i>	Allows users to search for evidence audio clips by using natural-language descriptions or other modalities, enabling efficient access to relevant audio content based on descriptive queries or related media. Syntax: <code><visualize_audio:><concept_descriptor></code> .
	<i>Semantic Textual Retrieval</i>	Involves retrieving textual information such as documents, tables, or HTML content using natural-language queries or other modality inputs, facilitating the search for specific text-based information through descriptive prompts or related media. Syntax: <code><visualize_file:><concept_descriptor></code> .
	<i>Semantic Pattern Retrieval</i>	Allows for the retrieval of patterns, such as textures or designs, using natural-language descriptions or other modality inputs, enabling users to find relevant patterns by describing them or using related media. This is cashed from a key-value store. Syntax: <code><pattern:><concept_descriptor></code> .
	<i>Panel Generation</i>	Retrieves automatic creation (generation) of a visual panel based on natural-language descriptions or other modalities set-up on the scene, allowing users to generate illustrative content by providing descriptive prompts or related media inputs. Syntax: <code><design:>[<panel_id> <text_prompt>]</code> .

Conditional elements are introduced through labeled panels, where the scene within a panel describes a situation. Regular reading flow proceeds only if the condition is true, adding decision-making layers to the flow. A *conditional panel* is followed by a *conditional jump* element that prevents continuation to the next panel depending on the specific condition described within the panel. If the condition is met, an alternative path to a named panel is provided. Different flow structures are illustrated in Fig. 7.13. *Typing* within the narrative, categorizes and clarifies roles and characteristics of narrative components, and can be used to label or group panels. Typing can be performed either through a text-like graphical element from Fig. 7.7 or through pattern fill-ups as illustrated in Fig. 7.13. A *pattern* can be used to symbolize a specific environment where the scene is happening (*e.g., in the office*) or a specific situation (*e.g., a customer enters the shop*). A *legend of used patterns* can be provided. Dialogue or thoughts of characters are prominently displayed through a bubble notation, enhancing narrative interaction. Additionally, various *shapes, arrows, and labels* are utilized to guide the viewer's attention, illustrate connections, and provide additional information, but also for drawing other process models as our platform is providing multi-language modeling support.

Virtual Enhancements (as provided in Table 7.8) focus on visually representing multimodal evidence references to enrich the narrative within a panel. This involves

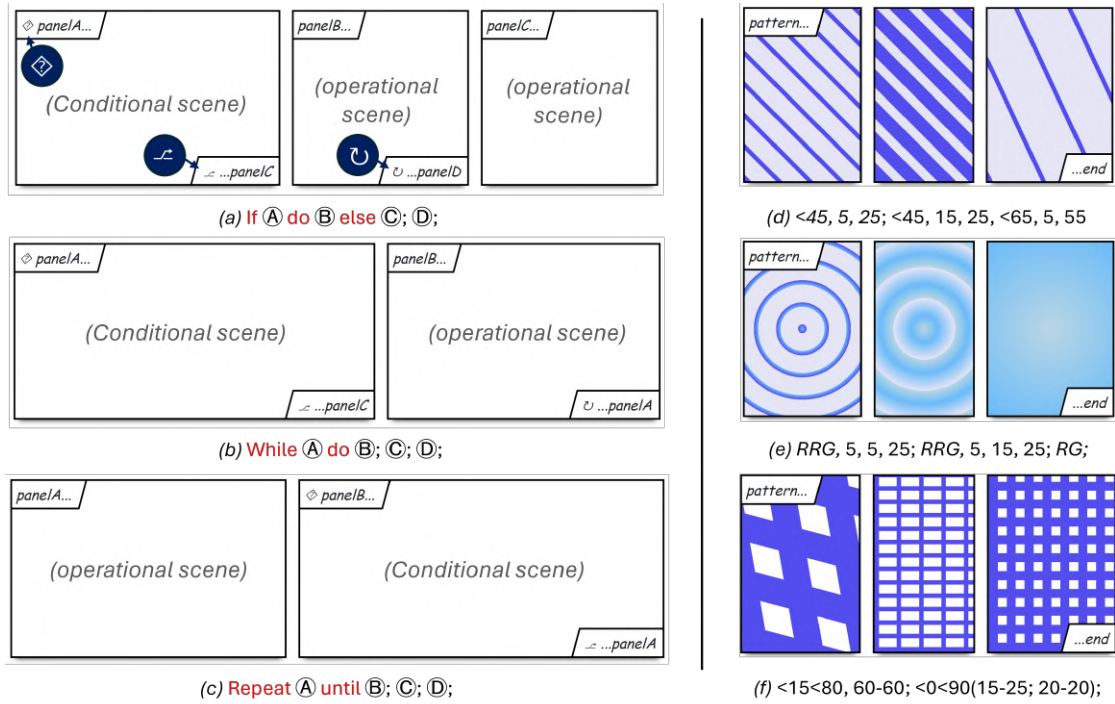


Figure 7.13: ViEnNa as a Conceptual Modeling Language. *Subfigures (a) to (c)* - branching, while and repeat-until control flows. Panels are ordered A, B, C, D (implicit if not visible); *Subfigures (d) to (f)* - pattern coloring in format: (d) and (f) repeating-linear-gradient - angles, positions (e) repeating-radial-gradient (RRG), radial-gradient (RG). Adopted from Gavric et al. [GBP25c].

embedding multimedia elements like images, videos, and audio clips directly into the narrative panels, setting scenes, or providing auditory context. *Semantic Retrievals* facilitate cross-referencing between different modalities, ensuring that textual descriptions can be linked to corresponding visual, auditory, or text (file) data. It also includes *Panel Generation* that uses generative AI (as described in Section 7.5.2) to create visual panels from natural-language descriptions with the support of other media inputs. When utilizing panel generation, various customizable options are available to suit specific needs. To name few, *perspectives* such as trainee, client, or decision-maker can be selected to align with the target audience and narrative. *Style choices* range from kid-friendly with vibrant colors and simple illustrations to professional with sleek, polished graphics, and thematic styles tailored to specific business contexts like technology, finance, or healthcare. Additionally, the *level of detail* can be adjusted from minimalistic and straightforward to highly detailed and intricate. These customizable options aims to enable the creation of engaging and relevant content that effectively communicates the intended message.

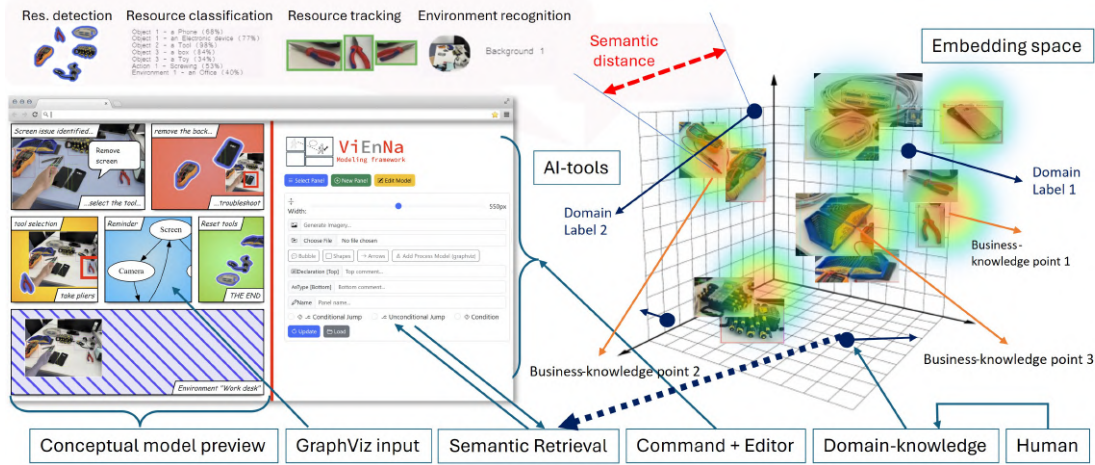


Figure 7.14: ViEnNa as a Conceptual Modeling Tool. (*left part*) Implemented modeling tool with a semantic prompt. (*right part*) *Embedding space* (a.k.a. *Latent space*). Adopted from Gavric et al. [GBP25c]. (Example-domain labels intentionally left unreadable)

7.5.4 ViEnNa as a Conceptual Modeling Tool

The implementation of the **ViEnNa** framework as a conceptual modeling tool focuses on making virtually enhanced narratives accessible, interactive, and executable across platforms. The visualization of ViEnNa models is developed using *JavaScript* for frontend operations, ensuring compatibility and responsiveness through standard web technologies. Each ViEnNa model is rendered in *Hypertext Markup Language (HTML)* and *Cascading Style Sheets (CSS)*, providing a lightweight, browser-based interface that can be easily integrated with existing process modeling environments or deployed independently. A custom parser for the ViEnNa meta-language was designed to interpret the syntax and commands specific to the ViEnNa notation, enabling efficient translation of model semantics into structured visual compositions.

By extending comics into immersive audiovisual experiences, tightening semantic control, and validating their impact in authentic contexts, ViEnNa as a conceptual modeling tool aspires to transform process knowledge into *living, instructive environments*—co-created and co-interpreted by both humans and AI agents. This evolution positions ViEnNa not only as a visualization technique but as an interactive medium for multimodal reasoning and communication in next-generation conceptual modeling.

Multimodal AI Capabilities of ViEnNa as a Tool

Essentially, systems are required to be capable of learning how the world operates [HL24] not solely from text but also from videos and other sensory inputs. Furthermore, these systems must have persistent memory about concepts and processes, a feature absent in current AI technologies [XML24]. They must be able to plan actions to fulfill objectives and be controllable and safe, possibly through the specification of guardrail objectives

which we can bring through process models - once the model is created, we adhere to it. This encapsulates the concept of objective-driven AI architectures [Sin23]. Central to the implementation of our solution is the concept of self-supervised learning. Self-supervised learning [EGLH22] involves taking a piece of data, such as text, and transforming or corrupting it in some manner. For instance, in the context of textual data, certain words may be replaced by blank markers. Subsequently, a large neural network is trained to predict the missing words, thereby reconstructing the original input. This methodology underpins the training of large language models, who, once trained, can take a sequence of words and predict the subsequent word. This iterative process continues, with each new prediction becoming part of the input, a method known as auto-regressive prediction [Mal23]. These models perform exceptionally well despite their process simplicity, and they are typically trained on vast amounts of data – often around 20 trillion tokens [DLY⁺23], where a token is a sub-word unit averaging three-quarters of a word.

To extend the power of self-supervised learning to other data modalities such as images, audio, and more, we use a method to tokenize these modalities similarly to how we tokenize text. At the core of our multimodal process retrieval is the concept of tokenization and embedding. Tokenization involves breaking down each modality into a set of discrete tokens, which are then embedded into a continuous vector space. After obtaining embeddings for each modality, the next step is to align them into a common vector space. This is achieved through a shared encoder network that maps modality-specific embeddings to a unified space. Let $\mathbf{E}_T$, $\mathbf{E}_I$, $\mathbf{E}_A$, etc., be the embeddings for text, image, audio, and other modalities, respectively. The tokenization function $\mathcal{T}$ converts raw data D into tokens $\mathbf{T}$ as $\mathbf{T} = \mathcal{T}(D)$. The embedding function $\mathcal{E}$ maps tokens to embedding vectors $\mathbf{E}$ as $\mathbf{E} = \mathcal{E}(\mathbf{T})$. Finally, the shared encoder function f aligns embeddings into a common vector space as $\mathbf{U} = f(\mathbf{E})$.

We can apply this method in cross-modal retrieval, where a query in one modality (e.g., text) retrieves relevant data in another modality (e.g., images). When a domain modeler queries a concept, we tokenize and embed the query Q (text) and the target data D (images) as $\mathbf{E}_Q = \mathcal{E}_T(\mathcal{T}_T(Q))$ and $\mathbf{E}_D = \mathcal{E}_I(\mathcal{T}_I(D))$ (as illustrated in Figure 7.14). Then, we map embeddings to the common space as $\mathbf{U}_Q = f(\mathbf{E}_Q)$ and $\mathbf{U}_D = f(\mathbf{E}_D)$. We envision a process modeling approach built upon the foundations of cross-modal retrieval, linking elements of process models to actual multimodal evidence of these concepts in practice. In such modeling, a modeler can show a picture or a video and use it to model concepts on an abstract level. This represents the human-to-model way of multimodal communication. Conversely, the model-to-human way of multimodal communication means that a model is capable of showing a reference example modality for a given concept element.

7.5.5 Comparing ViEnNa with Conventional and AI-Generated Representations

While recent advances in generative AI enable the automatic creation of visually appealing comic-style imagery [KBS24, NRB19, CJ24, KdRH24], such approaches typically lack semantic grounding, process logic, and traceability to real data. ViEnNa, by contrast, is not merely a tool for visual generation—it is a *conceptual modeling framework* that integrates multimodal evidence, process semantics, and human narrative design into a unified representation. This allows ViEnNa to produce visual narratives that are both aesthetically engaging and formally consistent with underlying process models [GBP25c]. In other words, ViEnNa bridges the gap between symbolic rigor and creative storytelling, offering structured visual reasoning capabilities that purely generative AI systems cannot achieve.

To illustrate this distinction, we compare multiple representations of the same process derived from identical multimodal data and object-centric event logs, as shown in Figure 7.15. Figure 7.15a presents the ViEnNa model, which integrates structured storytelling and semantic traceability; Figure 7.15b shows the same process in a conventional oc-BPMN diagram; and Figure 7.15c depicts a free-form comic-like visualization produced by students doing a PhD in AI related topics, referred as AI experts, without modeling support, as reported in [GBP25c]. This comparison highlights how ViEnNa’s structured meta-language and evidence-driven approach yield process narratives that are more interpretable, consistent, and data-grounded than unstructured generative outputs. In the Appendix C.7 we empirically demonstrate ViEnNa’s advantages over conventional process notations.

7.6 Multimodal Business Process Simulations (MM-BPS)

As the next step in extending ViEnNa beyond static or video-based process representations, we explore the integration of mixed reality simulations to immerse users directly within the modeled processes.

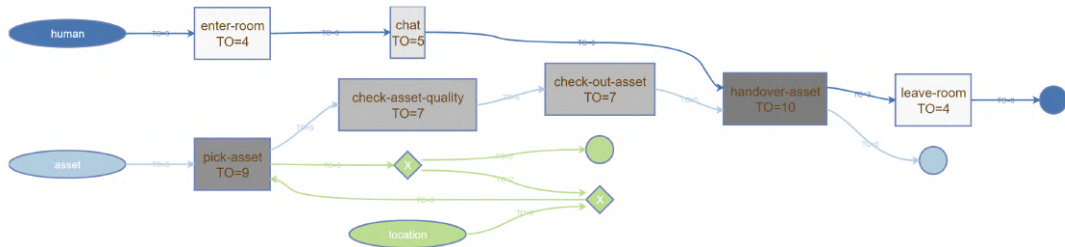
Business Process Simulation (BPS) is a well-established technique for analyzing and improving organizational processes by enabling the systematic exploration of alternative designs before their real-world deployment [vdA10]. Scenario generation allows organizations to create and evaluate *what-if* scenarios by adjusting an *as-is* model and observing its simulated behavior. By creating a virtual environment that mimics real-life business processes, BPS supports informed decision-making while avoiding the cost, risk, and disruption associated with direct experimentation in operational settings.

A recent systematic literature review by Khraiwesh and Pufahl provides a comprehensive overview of advances in the design of BPS models [KP25b], showing that little effort has been devoted to supporting scenario generation. Importantly, the review indicates that inaccuracies in BPS model construction significantly affect simulation outcomes, underscoring the need for richer and more faithful modeling approaches. A framework

7. MIXED REALITY SIMULATION



(a) ViEnNa model created over test multimodal data and OCEL.



(b) Conventional process model (oc-BPMN) created over the same test multimodal data and OCEL.



(c) Non-conventional (comic-like) models examples created by AI experts without using the ViEnNa framework.

from [KP25a], identifies key stages in BPS studies where Human-in-the-loop can be applied and the factors influencing the degree of human involvement and concludes that human involvement remains critical across all stages of data-driven simulation.

Building on the idea of turning conceptual models into video simulations (Section 7.4), we scale this concept to full *panoramic and 360-degree video environments*, where the camera is positioned at the center of the simulated space. This allows viewers to freely explore the process scene in all directions, creating an embodied experience of the modeled activities. A previous iteration of this work is reported in [GBP26] and can be seen as a non-mixed reality instantiation of the proposed approach. While it shares the same overarching goal of grounding business process simulation in multimodal execution evidence, it differs in several fundamental aspects: the treatment of the final simulation artifact is oriented toward inspectable, replayable executions rather than immersive, egocentric experience. The primary focus of that work lies in event log replay and multimodal alignment for simulation validation and analysis. By contrast, this thesis extends beyond log-centric replay toward mixed reality-based elicitation, counterfactual exploration, and human-in-the-loop refinement of conceptual models.

Depending on the level of visual generation and real-world context, ViEnNa supports three operational modes of simulation: *Actual Reality (ARL)*: full panoramic 360° footage from real environments; *Augmented Reality (AR)*: partial scene generation or overlay of specific elements (e.g., objects, agents, tools) to enrich recorded videos; *Virtual Reality (VR)*: fully synthetic 360° environments enabling complete virtual walkthroughs of modeled processes.

This tripartite design ensures a continuum between real, augmented, and virtual representations, all grounded in process semantics. As Muff’s work on *Metamodeling for Extended Reality* [Muf25] argues, metamodeling provides the necessary abstraction to connect conceptual models with their corresponding 3D and XR manifestations. Our approach applies these insights by linking ViEnNa’s narrative and object-centric representations to MR environments, effectively embedding process logic into spatially navigable simulations. Such an integration extends metamodeling “in and for” XR—supporting both comprehension and decision-making, and aligns with Muff’s XR-based model realization.

7.6.1 Converting 2D Videos into 360-Degree Simulations

In many real-world settings, process recordings are available only as traditional 2D perspective videos captured from a fixed viewpoint. To enable immersive simulation and training scenarios even under such data limitations, ViEnNa incorporates a procedure for transforming conventional 2D process recordings into 360-degree panoramic simulations. This process—illustrated in Figure 7.16—begins by identifying a *perspective anchor*, which represents the spatial reference point of the original camera. From this anchor, a full panoramic scene is synthesized to reconstruct the surrounding environment and extend the field of view beyond the captured frame.

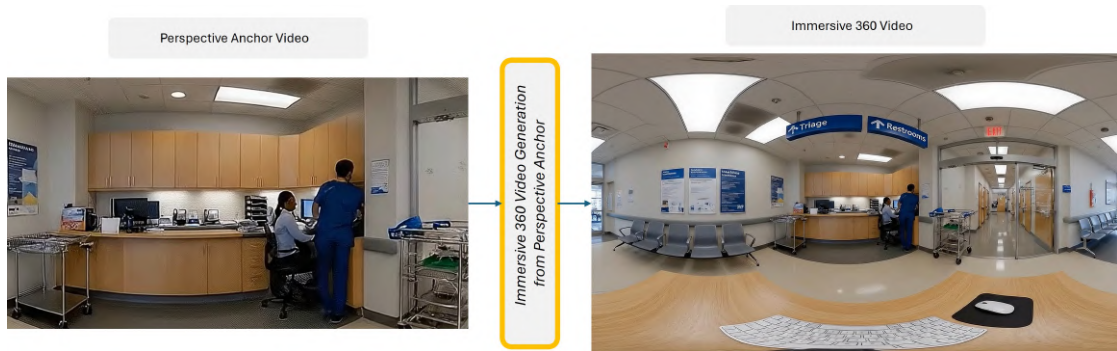


Figure 7.16: Conversion from a perspective anchor video to a 360-degree immersive simulation. The left side shows the original perspective view, while the right side illustrates the synthesized 360° environment generated around the central camera anchor.

The transformation pipeline relies on spatial inference techniques inspired by *Imagine360* [TYW⁺24], which demonstrated how large-scale generative models can hallucinate unseen regions and infer scene geometry from limited perspective inputs. In ViEnNa, we adapt this principle to the process simulation domain: using a combination of depth estimation, structure-from-motion (SfM), and diffusion-based scene completion, the system reconstructs the environment around the perspective anchor. First, depth and optical flow are extracted from the input video to recover approximate spatial relations. Next, the inferred geometry is projected onto a spherical panorama, where missing regions are filled by a conditional generative model trained for wide-angle context synthesis. Finally, the panoramic frames are temporally stabilized and rendered into an immersive 360° video format compatible with virtual reality viewers.

The right side of Figure 7.16 shows the resulting 360° reconstruction of a hospital triage area derived from a single 2D anchor video (left). This generated scene allows users to experience the environment as if they were standing in the center of the room, observing the process from all directions. Such conversions significantly expand the scope of ViEnNa’s mixed reality simulations, making it possible to reuse existing 2D process evidence for immersive exploration, behavioral analysis, and scenario training without requiring specialized 360° recording equipment.

7.6.2 Adding Interactive Mixed Reality Elements

Beyond passive visualization, ViEnNa extends mixed reality simulations with **interactive digital elements** that allow users to directly engage with the digital components of the process being modeled. Figure 7.17 illustrates how an immersive 360° environment (top left) is combined with interactive digital components (top right) through a *Mixed Reality Integration Layer*, producing an *Interactive Mixed Reality Simulation* (bottom). This setup creates a hybrid workspace where users can explore physical or simulated process contexts while simultaneously interacting with embedded digital systems.

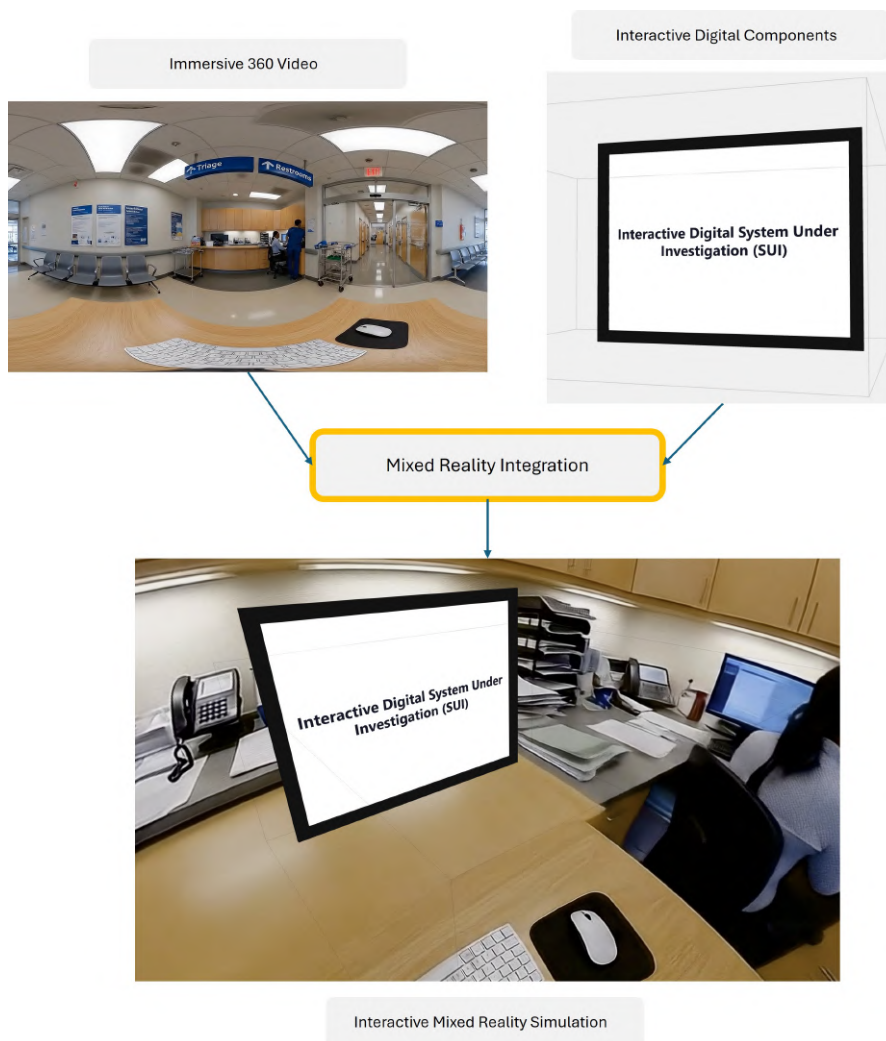


Figure 7.17: Integration of interactive digital components into mixed reality simulations. The immersive 360° video (top left) is combined with digital interfaces (top right) through a mixed reality integration layer, resulting in an interactive MR simulation (bottom) where users can engage with process-related digital elements directly within the immersive environment. Adapted from Gavric et al. [GBP26].

A key component of this setup is the *Interactive Digital System Under Investigation (SUI)*, a virtual interface to a digital SUI anchored within the MR environment. The SUI functions as a dynamic control and visualization hub—for example, as a projected PC screen or interactive dashboard—allowing users to monitor data, execute tasks, or annotate process models without leaving the immersive context. In the example depicted in Figure 7.17, the simulation displays an Interactive Digital System Under Investigation, enabling participants to manipulate or inspect software elements while immersed in the corresponding physical workspace.

This integration enables seamless interaction between the conceptual and operational dimensions of process modeling. Users can switch focus between observing the surrounding process and interacting with digital subsystems in real time. Such dual engagement supports both experiential understanding and analytic inspection—trainees, analysts, and modelers can not only visualize but also *act within* the modeled environment.

By embedding digital control interfaces inside panoramic scenes, ViEnNa extends beyond static process visualization toward an embodied form of conceptual modeling that fuses perception, action, and cognition. This approach aligns with recent advances in *metamodeling for extended reality* [Muf25], which emphasize the integration of modeling constructs into 3D and XR environments for improved comprehension and decision-making. In this way, ViEnNa’s MR integration bridges the gap between symbolic process logic and interactive operational experience, turning models into live, manipulable spaces for both human and AI collaboration.

7.6.3 Mixed Reality Interface



Figure 7.18: Interactive Mixed Reality view within the ViEnNa simulation environment. The user interacts with a digital system interface (SUI) embedded in a 360° video scene. The setup enables bidirectional manipulation between interactive elements, video-generated situations, and natural language conversations. Adapted from Gavric et al. [GBP26].

Figure 7.18 illustrates the *Interactive Mixed Reality Interface* designed to connect human

interaction, video-based simulation, and conversational AI components within a unified extended reality environment. In this view, the user is situated within an immersive 360° process scene—such as the hospital workspace shown in the figure—where a digital interface is embedded directly into the visual field. The SUI can represent any conceptual or operational digital system (for example, a patient information dashboard or process control panel) and can be manipulated as part of the simulation.

The lower part of Figure 7.18 depicts the triadic interaction model underlying ViEnNa’s MR environment. Users engage simultaneously with three interconnected dimensions: (1) *Interactive Elements* — tangible or virtual interfaces (e.g. software, or data overlays) that users can operate within the MR environment; (2) *Video-Generated Situations* — dynamic process scenes generated or captured as 360° videos that contextualize user actions within real or simulated workflows; (3) *Natural Language Conversations* — dialogic interactions with AI agents that interpret user commands, explain process behavior, or adapt simulation parameters in real time.

These components are connected through a bidirectional manipulation loop, where user actions on interactive elements influence both the ongoing video simulation and the conversational context, while the AI can, in turn, modify the displayed elements or generate situational responses. This configuration transforms the MR environment into a living model of process interaction, merging visual evidence, conceptual abstraction, and dialog-based reasoning.

The Interactive MR View exemplifies the next stage of ViEnNa’s methodological evolution—where process models become operational, participatory environments. By blending spatial visualization, interaction, and conversational intelligence, this setup enables users to not only observe or analyze processes but also to engage with them dynamically, fostering comprehension, exploration, and collaboration between humans and AI in shared, semantically rich MR spaces.

7.7 Validation

The Mixed Reality simulation approach presented in this chapter consolidates and extends a series of prior contributions. While each underlying component was evaluated in a context-specific manner in its respective publication, their validation results collectively justify the soundness of the integrated MR elicitation pipeline proposed in this thesis. In the following, we briefly revisit these evaluations and clarify their relevance for the present work.

Validation of Multimodal Model Operationalization (modSense). The *modSense* framework [GBP24b] was evaluated through qualitative user assessments and structured feedback sessions focusing on interpretability and multimodal consistency. Participants assessed whether generated visual and auditory artifacts meaningfully reflected UML semantics and whether multimodal renderings supported understanding beyond static diagrams. The results demonstrated that structural and behavioral elements of

conceptual models could be reliably transformed into perceptual representations without losing semantic alignment. For the present MR simulation framework, this validation is crucial: it establishes that symbolic model constructs can be operationalized into synchronized multimodal artifacts that preserve conceptual integrity—an essential prerequisite for immersive scenario reconstruction.

Validation of Process-Aware Video Generation (PN-VidGen). The Petri Net structure-driven video generation pipeline [GBP25j, GBP25f] was evaluated using both quantitative simulation metrics (e.g., NGram Distance, Absolute Event Distribution, Cycle Time Distribution) and controlled user studies assessing process comprehension and realism perception. These evaluations demonstrated that embedding formal process models into the storyboard significantly improves temporal and logical fidelity compared to domain-knowledge-only prompting. Within the MR elicitation context, these findings ensure that the immersive baseline scenarios presented to experts are not arbitrary cinematic narratives but semantically grounded enactments of formally defined process states. The quantitative conformance results provide confidence that the temporal backbone of MR simulations faithfully mirrors underlying process logic. Details are provided in Appendix C.6.

Validation of Narrative-Based Conceptual Modeling (ViEnNa). ViEnNa [GBP25c] was evaluated by comparing (i) structured comic-based models, (ii) conventional oc-BPMN models, and (iii) unconstrained AI-generated comic representations over identical multimodal data and OCEL inputs. The evaluation demonstrated that ViEnNa improves interpretability and stakeholder comprehension while maintaining traceability to formal process semantics. Although this evaluation was not conducted in a mixed-reality setting, it directly validates the representational layer used in MR elicitation. ViEnNa’s ability to preserve semantic grounding while enhancing cognitive accessibility justifies its use as the narrative interface within immersive simulations. Details are provided in Appendix C.7.

Validation of Stakeholder-Centered Conversational Integration. The transformer-based paraphrasing and stakeholder-specific representation mechanisms proposed in [GBP24d] were evaluated through domain-adapted language modeling experiments and stakeholder feedback studies. These evaluations demonstrated improved clarity and alignment of process explanations with user-specific jargon and comprehension levels. In the MR elicitation pipeline, conversational scaffolding captures experts’ verbalized reasoning during perturbations. The prior validation confirms that such conversational mediation can reliably adapt to stakeholder profiles while preserving semantic precision—ensuring that elicited explanations remain interpretable and structurally consistent with the underlying model.

Validation of Multimodal Business Process Simulation (MM-BPS). The non-mixed reality instantiation of Multimodal Business Process Simulation presented in [GBP26] focused on event log replay, multimodal alignment, and simulation accuracy.

It was evaluated using classical process simulation metrics and scenario-based validation experiments demonstrating alignment between simulated traces and real operational data. Although immersive embodiment was not the primary focus in that work, its evaluation establishes the correctness of the multimodal alignment pipeline and the integration of digital SUI interactions with simulated process states. This validation directly underpins the MR extension proposed here, where the SUI is embedded spatially rather than inspected externally.

Synthesis of Validation Across Components. While each prior evaluation targeted a distinct artifact—multimodal operationalization, process-guided video generation, narrative modeling, conversational adaptation, or log-centric simulation—their combined results provide layered evidence for the integrated MR simulation framework. Specifically:

- [GBP24b] validates semantic preservation across modality transformations;
- [GBP25j, GBP25f] validate temporal and logical fidelity of video-based enactments;
- [GBP25c] validates narrative-grounded conceptual modeling and multimodal traceability;
- [GBP24d] validates stakeholder-adaptive conversational mediation;
- [GBP26] validates multimodal event-log alignment and simulation consistency.

The present MR framework integrates these previously validated components into a unified elicitation environment. Although the immersive composition itself constitutes a methodological extension, it does not rely on untested building blocks. Rather, it synthesizes independently validated mechanisms into a coherent architecture aimed at capturing counterfactual expert behavior.

7.8 Conclusion

This chapter introduced the mixed reality simulation mechanisms that enable interactive “what-if” scenario exploration for capturing counterfactual behavior. It explains how expert responses to controlled perturbations are measured, how latent decision rules are inferred, and how these rules are subsequently integrated with observational traces and the interactive digital system under investigation.

The pipeline begins with the transformation of symbolic conceptual models into multimodal evidential material, following the methodology in [GBP24b]. This work introduced a structured way of converting process models into synchronized visual, textual, and auditory representations. In the context of elicitation, these multimodal assets serve as the foundation for reconstructing baseline scenarios inside MR: the model is not merely shown but *embodied*, enabling experts to perceive process logic as situated, spatial, and multimodal.

Building on this, the method from [GBP25j, GBP25f] extends the transformation pipeline to full-length *generated videos* of process executions. These synthetic videos instantiate the model’s behavior in a vivid, narrative, temporally coherent form. For the MR elicitation pipeline, these videos form the temporal backbone of the scenario, defining the default flow of events that experts will later confront, manipulate, or react to. This contributes with a lightweight mechanism to bind video segments to Petri Net executions, ensuring that every moment presented in MR aligns with a precise process-state semantics. Thus, the pipeline obtains a synchronized symbolic–subsymbolic correspondence: every frame of the reconstructed situation is mapped to a well-defined point in the underlying model.

The ViEnNa modeling framework [GBP25c] adds the next key capability: the ability to represent process knowledge in virtually enhanced, comic-like panels that can be entered, manipulated, or stepped through in MR. In the elicitation pipeline, ViEnNa provides the interface layer through which experts both, explore scenarios and respond to “what–if” variations. A perturbation applied in MR—such as changing a patient’s condition, removing a resource, or shifting event timing—is reflected both in the narrative panels and in the surrounding immersive environment. The expert remains inside the unfolding situation, and their embodied responses become elicited counterfactual traces.

To support this interactive flow, conversational scaffolding from [GBP24d] enables the expert to articulate their reasoning, explain decisions, and interact with the system through dialogue. This component embeds a stakeholder-centered conversational layer into the MR elicitation pipeline, allowing experts to clarify why they acted in a certain way, correct model assumptions, or request alternative perturbations. It closes the cognitive loop: MR provides the embodied “doing”, while the conversational interface captures the accompanying “thinking”.

Finally, the crucial component of the MR elicitation pipeline is the seamless integration of the interactive digital SUI, based on Multimodal Business Process Simulation [GBP26]. Within the MR setup, the SUI is not simulated abstractly but embedded as a fully functional, live window inside the immersive scene. This allows experts to interact with the actual digital system while simultaneously perceiving the surrounding physical and situational context reconstructed in MR. As perturbations unfold (e.g., a delayed arrival, a sudden resource shortage, a conflicting request), the embedded SUI reacts just as it would in reality, enabling the elicitation of counterfactual responses grounded directly in the expert’s authentic tool use. In this way, the SUI becomes a situated, multimodal anchor that links conceptual models, generated videos, ViEnNa comic panels, and conversational elicitation into a unified, evidence-rich environment where expert behavior can be observed, perturbed, and interpreted under highly realistic conditions.

Technical Implementation

This chapter presents the concrete software implementations, and multi-agent systems developed as part of this thesis. While the theoretical foundations established earlier formalise how multimodal conceptual knowledge can be elicited, refined, and represented, the implementations described here demonstrate how these ideas have been operationalized in practice.

At the core of the system is a set of real-time, browser-integrated AI agents, designed to collaborate across logging, process mining, elicitation, simulation, and conceptual model construction tasks. These agents are equipped with specialized tools that perform process mining, update the interactive user interface, construct diagrams, switch simulation contexts, and support dynamic mixed-reality elicitation.

This chapter is structured as follows. Section 8.1 introduces the overall real-time, agent-based system architecture and its decomposition of conceptual modeling responsibilities. Section 8.2 presents the user interface and interaction design that mediates between human experts and the underlying analytical components. The technical tooling and logging infrastructure is detailed in Section 8.3 and Section 8.4, explaining how multimodal interactions are captured across heterogeneous systems. Section 8.5 describes the multimodal data analysis pipeline that transforms these interactions into aligned symbolic and embedding-based representations. Mixed-reality simulation and counterfactual generation mechanisms are described in Section 8.6. Section 8.7 discusses the security architecture required to safely deploy the system in interactive and data-intensive settings. Finally, we conclude in Section 8.8. The application and evaluation of this implementation are provided in Chapter 9.

Appendix B provides an engineering-level complement to this chapter by detailing the software mechanisms that operationalize the proposed framework. Appendix B.1 describes the browser-based realtime client, including agent orchestration, audio streaming, and pane synchronization. Appendix B.2 presents the three.js-based Mixed Reality simulation

environment and its scenario parameterisation. Appendix B.3 outlines the ViEnNa editor and renderer, covering the panel-based syntax, live comic rendering, and linkage to Mixed Reality captures. Finally, Appendix B.4 specifies the crossmodal event logging subsystem and its integrations for process mining.

8.1 Conceptual Modeling Assistant

At the core of the implemented system is a cross-platform web browser-based architecture powered by real-time AI agents. This design choice is intentional: the browser provides a universal, installation-free execution environment capable of rendering mixed-reality scenes, hosting multimodal interaction panes, synchronizing conceptual model editors, and executing real-time agent workflows without requiring specialized hardware or local software deployments. By leveraging the modern browser as the execution substrate, the system ensures portability, ease of experimentation, and seamless integration with web-native simulation windows, embedded cameras, and MR overlays.

The system adopts an agent-based architecture because conceptual modeling in multimodal and MR settings requires continuous interpretation, context tracking, and parallel reasoning across heterogeneous tasks. A monolithic model would be inefficient in maintaining responsiveness while simultaneously performing logging, process mining, elicitation, diagram construction, UI orchestration, and simulation control. Instead, the implemented system decomposes functionality into a set of coordinated, browser-resident agents—each specialized for a distinct dimension of the implemented system. These agents communicate through shared embeddings and event streams, enabling low-latency reactions to crossmodal salience, user interactions, and changes in the simulated environment.

Together, the browser-based runtime and agent-based decomposition yield a system that can observe, analyze, and intervene inside mixed-reality situations in real time. Using `three.js`¹ for rendering, WebRTC² for multimodal streams, and interactive editors (e.g., BPMN, markdown-based renderers, and the ViEnNa modeling framework), the agents shape conceptual models directly from the evolving MR context.

8.2 Interaction & User Interface

The user interface (UI) of the Conceptual Modeling AI Platform is designed to support high-fidelity multimodal observation, real-time model co-creation, and seamless interaction between human experts and AI-driven assistants. As shown in Figure 8.1, the layout follows a three-pane architecture tailored to the workflow of conceptual modeling in data-rich environments:

- **Chat Pane (left):** This pane hosts the interactive dialogue between the user and the AI agents (e.g., Domain Expert Agent, AI Modeling Assistant). It supports

¹<https://threejs.org>

²<https://webrtc.org/>

text-based communication, audio playback, and real-time transcription, enabling fine-grained elicitation while maintaining a full communication trace for later analysis.

- **System Under Investigation (SUI) Inspector (center):** The central pane provides a panoramic, immersive rendering of the real-world environment under investigation. It supports 360° navigation, synchronized frame control, and the ability to inspect details of situated behavior, such as receptionist interactions, patient movement, or system use on shared screens. This enables precise observation of tacit and embodied knowledge that would otherwise remain unarticulated.
- **Panes for ViEnNa, Multimodal Data Analysis, and Conceptual Modeling (right):** The UI has additional pane and a toolbar to provide access to key functionalities, including the ViEnNa editor, multimodal data analytics panels, logs, conceptual modeling overlays, and SUI inspection tools. These modules can be activated at any point during observation, allowing the user to transition fluidly between inspection, elicitation, and model construction.

Altogether, the UI supports a tightly integrated workflow in which video-based evidence, expert reasoning, and structured model artifacts are co-evolving within a single environment. This design is particularly suited for eliciting counterfactual dispositions, identifying recurrent behavioral patterns, and validating conceptual models directly against real-world operational context.

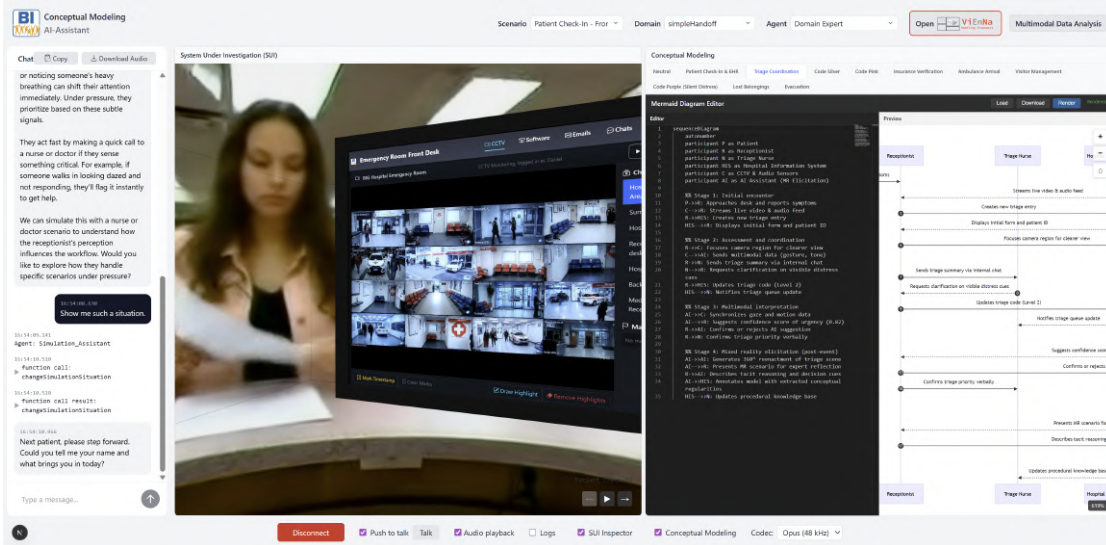


Figure 8.1: User interface of the implemented prototype, showing the AI chat, immersive System Under Investigation viewer, and modeling toolbars.

8.3 Architectural Overview

This section provides the details of our implementation of the multi-agent conceptual modeling system used throughout the thesis. The system is implemented using the `openai agents realtime`³ framework, extended with custom tools for switching simulation contexts, constructing conceptual models, etc.

The design follows a modular agent architecture in which each agent embodies a distinct perspective in the conceptual modeling AI workflow. Each agent is configured with specialized instructions, handoff behavior, and tool access, enabling collaborative modeling across modalities and representational formats.

The subsections below describe the implementation of the tools and agents that operationalize the thesis contributions.

8.3.1 Tooling Layer

To support a smooth elicitation and modeling workflow, three core tools were implemented: a tool to switch simulation situations, a tool to store and update markdown-based conceptual models, and a tool that captures salient interaction events and applies process-mining analyzes to them. These tools expose a structured interface through which AI agents can update the UI, persist artefacts, and align simulations with the modeling process.

Event Capture („Log Now“). The *Log Now* event capture generates a structured event entry in the multimodal event log, capturing the state of the crossmodal context at the moment an agent deems it salient. Rather than relying on user clicks alone, this tool can be invoked automatically by AI agents when drift-aware crossmodal analysis detects meaningful changes in the simulation window (e.g., a sudden shift in gaze patterns, proximity, or audio cues). Each captured event stores synchronized embeddings across modalities—video, audio, gaze, UI interactions—and is appended to the current log for downstream mining.

Multimodal Process Mining. The multimodal process mining tool performs process (re-)discovery and (re-)contextualization over the multimodal event log. Using the embedding-space representations of events, the tool clusters, aligns, and abstracts heterogeneous evidences into symbolic process structures. It supports classical discovery (e.g., inductive mining), enhancement (e.g., bottleneck detection), as well as multimodal contextualization—linking symbolic steps to visual behaviors, spatial patterns, or conversational cues. This tool is invoked by analytic agents to update the live process model whenever new events or refined embeddings become available.

³<https://www.npmjs.com/package/@openai/agents-realtime>

Counterfactual Disposition Simulation. The `counterfactual_disposition` tool enables AI agents to specify a hypothetical situation and automatically configure the simulation stack to instantiate it. Given a high-level situation description from a simulation agent (e.g., “calm but critically ill patient with delayed triage response”), the tool composes the required elements: a mixed-reality panorama video segment, synthetic or retrieved audio cues, and the appropriate digital system configuration. It then updates the simulation environment so that experts can respond to the scenario, allowing elicitation of counterfactual decision rules. These counterfactual responses are logged and later aligned with observational behavior to refine the underlying conceptual models.

Switching Simulations. The `change_simulation_situation` tool updates the mixed-reality simulation view by selecting the appropriate scenario based on the description or the identifier. This enables rapid context-switching during domain elicitation, aligning the visualization with the ongoing dialog.

Updating Conceptual Models. The `conceptual_model_update` tool extracts markdown code blocks (such as Mermaid⁴, more details in Section A.1.1) generated by the Modeling Assistant and writes them to the modeling workspace. The UI is then updated reactively, ensuring that each modeling step is visible and traceable. Similarly, the `ViEnNa_model_update` tool extracts vienna code blocks and stores them as `.vn` files. These files open in the dedicated ViEnNa viewer and allow agents (and humans) to collaboratively explore narrative-based conceptual models.

8.3.2 Agent Implementation

The implemented system relies on a coordinated set of three AI agents, each responsible for a distinct layer of the implemented system. Their design follows the methodological foundations established in Chapter 7 and Chapter 6, and extends the formal roles defined in Sections 5.2, 5.3, and 5.4. Together, these agents enable an integrated workflow that spans scenario control, multimodal analysis, conceptual model construction, and counterfactual elicitation.

Conceptual Modeling Assistant. The *Conceptual Modeling Assistant* implements and operationalizes the modeling methodology described in Section 5.2. It is responsible for transforming elicited domain knowledge—both observational and counterfactual—into formally structured conceptual models. The agent generates markdown-based representations such as Mermaid, or ViEnNa, and uses the `conceptual_model_update` tool to write these models into the workspace. The assistant also inspects previous model versions to ensure consistency, traceability, and alignment with newly captured events and salience markers. When new salient events are logged, the agent can refine the

⁴<https://mermaid.js.org/>

model incrementally, maintaining tight coupling between the running simulation and the symbolic representation being constructed.

Domain Knowledge Assistant. The *Domain Knowledge Assistant* operationalizes the crossmodal alignment and salience-analysis mechanisms introduced in Section 5.3. It continuously evaluates multimodal streams (video, audio, gaze, UI interactions) to determine when a change in the environment warrants an event capture. Through the `log now` tool, it records structured multimodal event entries enriched with embeddings and contextual metadata. These logs are then processed using the `multimodal process` mining tool, enabling real-time discovery, conformance checking, and contextualization of emerging processes. The assistant thus acts as the analytical backbone of the system: monitoring drift, detecting salient transitions, and delivering interpreted behavioral evidence back to the Conceptual Modeling Assistant for conceptual integration.

Although the `log now` tool can be invoked by any agent, the *Domain Knowledge Assistant* plays a broader epistemic role that extends beyond crossmodal salience detection. It serves as an always-available source of domain knowledge, enabling the modeler to query procedures, constraints, terminology, or best practices directly from the AI without interrupting the elicitation flow. This allows the modeler to triangulate expert statements, fill in gaps when the human expert is uncertain, or generate candidate questions that can guide the subsequent elicitation steps. In this sense, the assistant functions not only as an analytical observer but also as a domain-support agent that enriches the dialog, helps interpret ambiguous behaviors in the simulation, and supplies the modeler with AI-derived insights that complement human expertise.

Simulation Assistant. The *Simulation Assistant* implements the counterfactual situation-generation framework described in Section 5.4. Its primary responsibility is to manage the mixed-reality simulation environment. Based on expert input or agent-level hypotheses, it configures the simulation to express either a new observational context or a counterfactual condition. It uses the `change simulation situation` tool to load the correct scenario and the `counterfactual disposition` tool to assemble composite situations that blend panorama videos, layout configurations, system interfaces, and synthetic or retrieved cues. This agent ensures tight coupling between the elicitation process and the expert’s sensory field, enabling deliberate perturbations of the environment that reveal latent decision rules.

Agent Handoff and Tool-Oriented Collaboration. The coordinated workflow across agents typically begins with the *Domain Knowledge Assistant*, which either answers the modeler’s domain-related questions or identifies a need for further evidence—such as clarifying an ambiguous activity, resolving conflicting interpretations, or probing a potential counterfactual disposition. When additional situational evidence is required, the assistant delegates control to the *Simulation Assistant*, which configures or perturbs the mixed-reality scenario using the `change simulation situation` and `counterfactual disposition` tools. As the simulation unfolds, the Domain

Knowledge Assistant monitors multimodal signals and invokes `log now` on salient transitions, subsequently triggering multimodal process mining. The structured insights produced from these logs are then passed to the *Conceptual Modeling Assistant*, which integrates the newly surfaced knowledge into the evolving conceptual models and updates the workspace via the modeling tools. In this cycle, the tools serve as explicit coordination primitives that ensure each agent performs a well-defined operation, enabling a seamless flow from domain reasoning, to simulated evidence generation, to conceptual abstraction.

8.3.3 Pre-Real-Time Preparation and Agent Initialization

Before the system can operate in real time, a dedicated preprocessing phase constructs the analytical substrate on which all agents rely. This phase transforms raw multimodal event logs into structured representations, aligns them with the SUI, and configures the initial situation space from which real-time elicitation and simulation can unfold. The outcome of this phase is a fully instantiated set of agents—Conceptual Modeling Assistant, Domain Knowledge Assistant, and Simulation Assistant—each equipped with the contextual knowledge required to operate coherently during live interaction.

Processing Raw Multimodal Logs. The pipeline begins with the ingestion of raw multimodal event logs collected during baseline observations of the SUI. These logs include video recordings, audio streams, gaze trajectories, UI interactions, spatial metadata, and operational traces from the digital systems involved. In this phase, all signals are preprocessed into synchronized, aligned representations: embeddings are computed for each modality, temporal segmentation is applied, and preliminary salience markers are derived. This produces a canonical multimodal evidence base from which downstream agents can reason.

Contextualizing the SUI. Next, the system performs a first-pass contextualization of the SUI. Using the processed logs, it identifies stable patterns in user behavior, tool usage, interface flows, and environmental dynamics. These insights allow the system to construct an initial operational ontology of the SUI: the available system screens, typical activity sequences, common user roles, and the set of situational primitives that recur across observational traces. This contextualization forms the backbone against which later real-time reasoning and simulation are anchored.

Generating the Initial Situation Space. Based on the extracted multimodal regularities, the pipeline generates a library of canonical situations. These include prototypical scenes, baseline workflows, and contextually meaningful variations that can later be perturbed during counterfactual elicitation. Panorama video segments, interface states, environmental cues, and audio patterns are indexed and packaged into a structured situation taxonomy. This initial situation space establishes the range of scenes that the Simulation

Assistant can later manipulate through the change simulation situation and counterfactual disposition tools.

Agent Construction. Finally, the system instantiates the three core real-time agents. Each agent is parameterized using the processed multimodal representations, the contextualized SUI ontology, and the initialized situation space. The Domain Knowledge Assistant inherits the contextual semantics and crossmodal patterns needed for drift-aware salience detection and domain reasoning. The Simulation Assistant inherits the indexed situation taxonomy and environment configuration primitives. The Conceptual Modeling Assistant is initialized with the structural abstractions extracted from the logs and the organizational context of the SUI. Together, these agents form a coherent, prepared ecosystem ready for real-time elicitation, simulation, logging, and conceptual model construction.

8.3.4 Engineering Details

The methodological and empirical contributions of this thesis are supported by a fully implemented, browser-based software ecosystem that operationalizes real-time multi-agent elicitation, mixed reality simulation, multimodal event capture, and interactive conceptual modeling. While the present chapter focuses on the architectural principles and workflows that structure the system, the concrete engineering details are documented separately in Appendix B. That appendix provides a comprehensive, component-level description of all major modules: the real-time web client (Appendix B.1), the mixed reality simulation environment (Appendix B.2), the ViEnNa editor and renderer (Appendix B.3), and the crossmodal event capture mechanism implemented via the „Log Now“ function (Appendix B.4).

Together, these sections detail how the theoretical constructs developed throughout the thesis—such as agent handoff, situation switching, multimodal contextualization, and narrative-grounded modeling—are realized in practice within a cohesive implementation stack built on React, Next.js, three.js, OpenAI Realtime API, and custom process-mining pipelines.

8.4 Data Acquisition

To complement the mixed-reality elicitation and multimodal capture pipeline, the platform exposes a *logging-enabled interface* of the interactive digital system under investigation (SUI). This interface mirrors the actual operational tools used in the scenario (e.g., hospital information systems, email clients, CCTV dashboards), and attaches a lightweight logging layer to every relevant interaction. Each click, form submission, filter change, or communication event is time-stamped, linked to a case identifier, and written into the multimodal event log. The following figures illustrate the five main categories of supported systems: business software, CCTV dashboards, email clients, chat interfaces, and telephony.

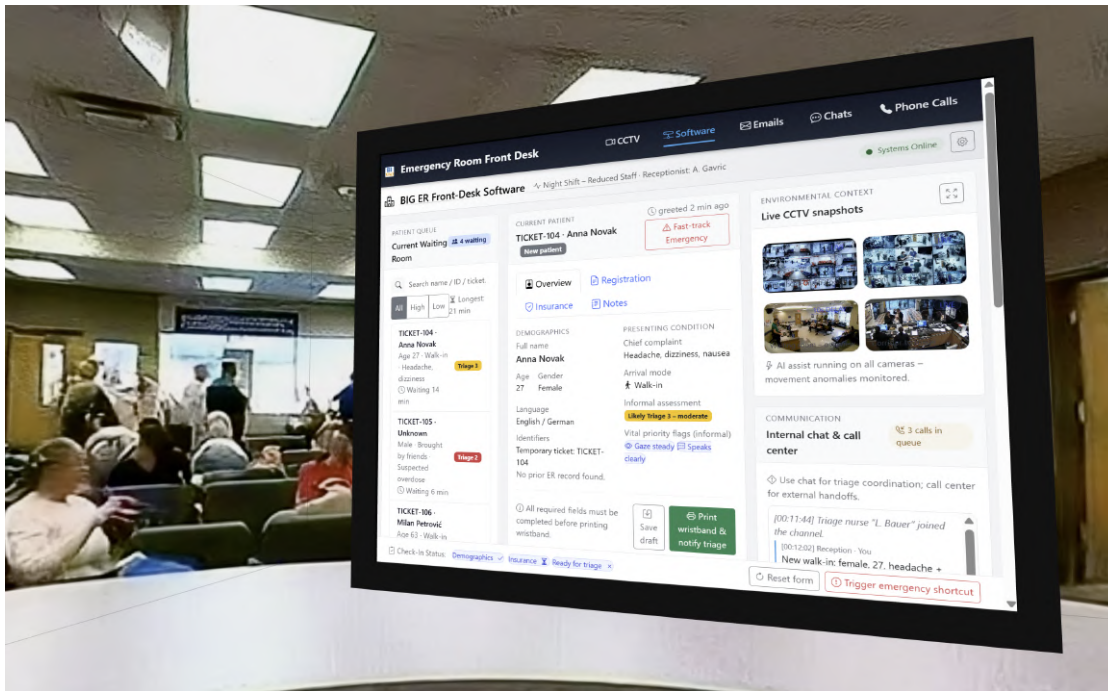


Figure 8.2: Logging-enabled business software user interface of the SUI.

Business Software Interfaces. Figure 8.2 shows a representative example of a business software interface as rendered inside the SUI pane. This interface includes full operational logic such as patient lookup, record creation, insurance verification, triage updates, workflow shortcuts, and cross-pane triggers (e.g., opening the chat pane when a medical query arises). The logging layer therefore records not only surface interactions—such as clicking buttons or switching tabs—but also the underlying interaction logic: which validation checks fire, which form sections become enabled or disabled, which automated workflows are triggered (e.g., fast-track routing), and how contextual panels (like environmental CCTV snapshots) react to the user’s actions.

Beyond these operational workflows, the system also exposes real-time state transitions that reflect broader hospital conditions. For example, if the user opens a patient record during a surge period, the software may automatically show waiting-room load indicators, triage capacity warnings, or reduced-staff notifications. The logging layer captures these system-driven context adaptations as explicit events, allowing the pipeline to reconstruct how software intelligence, staff decision-making, and environmental conditions jointly shape the interaction flow.

CCTV and Video Dashboards. The CCTV dashboard in Figure 8.3 embeds live or recorded camera streams with interaction logic supporting camera switching, synchronized time navigation, multi-view arrangement, and highlight annotations. Beyond simple UI clicks, the logging layer captures logical events such as when a stream enters “spotlight



Figure 8.3: CCTV dashboard view of the SUI with logging of operator interactions.

mode”, when the operator draws attention marks, when playback is scrubbed to align with a concurrent event (e.g., ambulance arrival), or when security-relevant channels are auto-suggested based on recent activity.

Additionally, the CCTV module features automated reasoning layers that respond to user choices. When an operator selects a specific camera cluster during a critical moment—such as a sudden patient collapse—the system may surface AI-generated prompts, anomaly detections, or recommended camera groups. The logging layer records these system prompts, user acceptances or dismissals, and the resulting view reconfigurations. This provides a rich account of not just *what* was viewed but *why* certain visual paths emerged, enabling alignment with symbolic decision points and situation transitions in the broader multimodal pipeline.

Email Interfaces. Figure 8.4 displays the email client pane embedded in the SUI, which supports interaction logic such as thread expansion, preview-pane synchronization, auto-tag suggestions based on message content, quick-reply templates, priority flagging, and escalation routing rules. The logging layer therefore records not only which messages are opened or sent but also the system-driven logic that accompanies these operations: when the system automatically classifies an email as triage-relevant, when it highlights missing patient identifiers, when a follow-up reminder is generated, or when an operator overrides a system-suggested label.

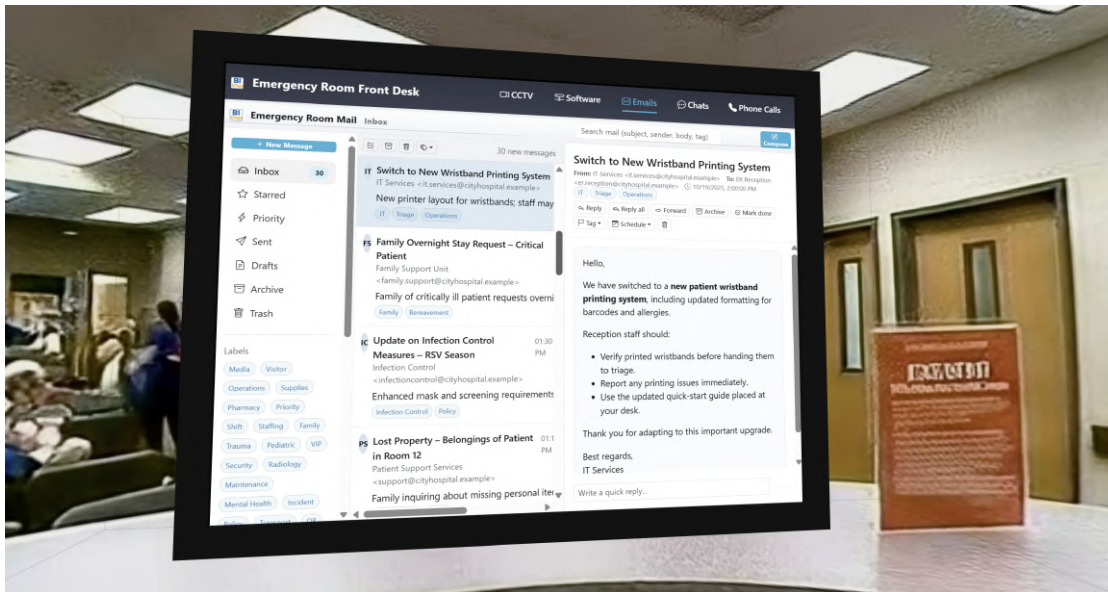


Figure 8.4: Email client pane with interaction logging for message handling in the SUI.

Emails also trigger cross-pane behaviors that the logging layer captures as explicit logical transitions. For example, opening a message about infection control may generate a contextual banner in the software pane, while an urgent-family-request email may cause the chat pane to auto-suggest contacting security or triage nursing. These coupled logic events reflect how information spreads across the operational environment and are critical for reconstructing decision provenance and operator attention flows.

Chat and Messaging Interfaces. The chat interface in Figure 8.5 supports multi-threaded communications between staff roles, with internal logic for thread pinning, tagging, role-based visibility filtering, live system signals, and cross-department escalation. The logging layer captures not only the sending and receiving of messages but also the underlying behavior logic: when a conversation becomes marked as “critical”, when an operator is mentioned and automatically navigated into a thread, when a message is drafted but not sent, when read receipts propagate across staff hierarchies, or when the system inserts prompts (e.g., “medical update available”).

The chat system also integrates event-driven logic that coordinates with other SUI components. For instance, when a security incident escalates, the software and CCTV panes may automatically shift into “incident mode”, displaying relevant patient or environmental data. These coordinated UI adjustments, along with their triggers and conditions, are logged as explicit state changes. This enables detailed reconstruction of emergent teamwork patterns and how communication events shape downstream actions and system configurations.

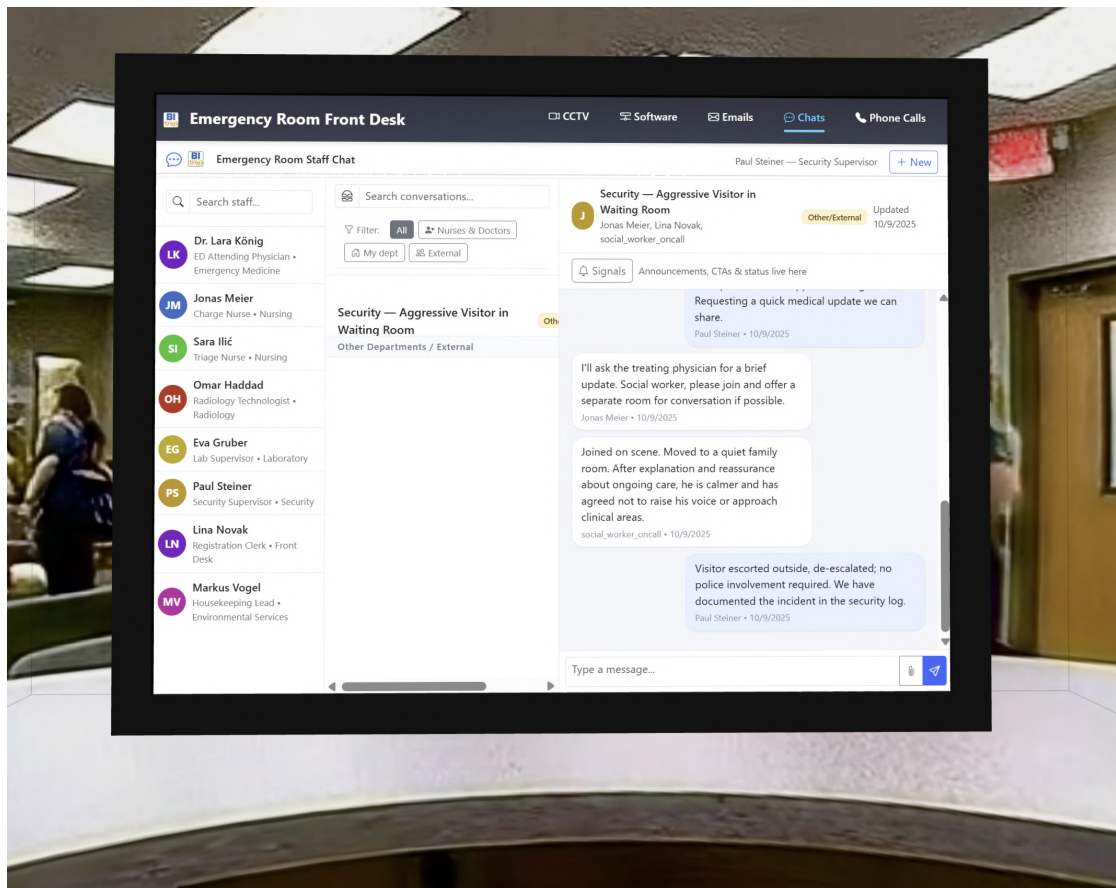


Figure 8.5: Chat and messaging interface of the SUI with structured interaction logging.

Telephony and Call Management Interfaces. The telephony pane in Figure 8.6 implements interaction logic for active call states, caller identification, triage-related quick actions, transfer and conference flows, and automatic transcript saving. The logging layer tracks changes in call logic such as when the call enters “priority mode”, when predefined triage phrases autofill during note-taking, when a call is placed on hold, when the system verifies caller identity based on internal directories, or when a shortcut triggers simultaneous notifications to security or triage nursing staff.

In addition to call handling, the system features context-linked call intelligence. For example, when an incoming call reports an unstable patient, the software pane may preload registration forms, the CCTV pane may automatically surface the entrance cameras, and the chat pane may prompt the operator to notify relevant staff. These orchestrated multi-pane responses are logged as synchronized logic events, providing a unified view of how telephony-driven incidents propagate through the entire SUI.

Together, these logging-enabled UI components provide a coherent, interaction-centric view of the SUI. For every supported system, the interface not only exposes realistic

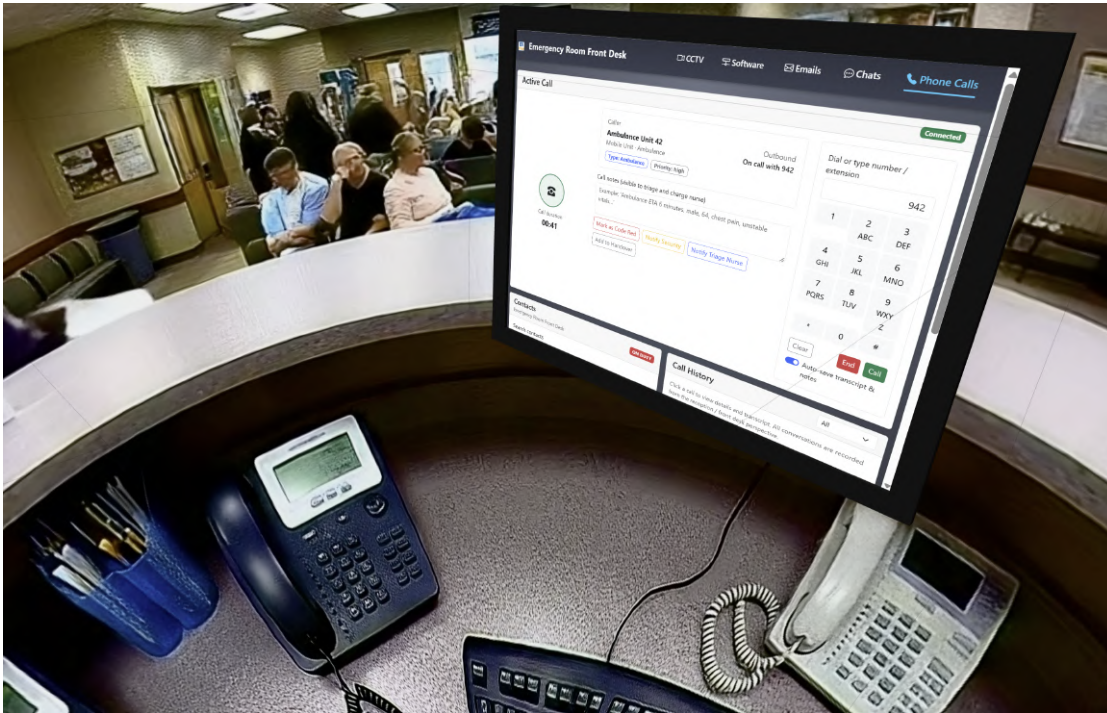


Figure 8.6: Telephony and call-management user interface of the SUI with logging hooks.

operational screens to participants but also transforms their interactions—including hidden logic transitions, system reactions, and cross-pane triggers—into structured, time-aligned events that feed into the multimodal process mining and conceptual modeling pipeline.

8.5 Multimodal Data Analysis

The multimodal data analysis component of the platform extends traditional event-log analysis by interpreting process behavior directly within the embedding space constructed from video, audio, UI interactions, and agent dialogues. Instead of relying solely on symbolic activity labels, the system treats each logged „event“—captured via the „Log Now“ mechanism, further described in Appendix B.4—as a high-dimensional point that represents the state of the system under investigation at a specific moment in time. These embeddings encode rich crossmodal cues such as gaze direction, screen interactions, environmental context, and agent role dynamics.

To analyze these multimodal logs, the system maps each event into a unified embedding space, often produced by a combination of computer vision encoders (for simulation frames), speech encoders (for audio), large language models (for transcript segments), and structural encoders (for active conceptual models). Each modality contributes a complementary view of the evolving emergency reception workflow, and the resulting joint

embedding serves as a dense, continuous representation of the process state. Once event embeddings are collected, we compute the *change vector* between consecutive events. Formally, given two consecutive embedded states e_t and e_{t+1} , the instantaneous process change is defined as the vector difference

$$\Delta_t = e_{t+1} - e_t.$$

These change vectors, when analyzed across full traces, reveal the shape of the trajectory that the real-world process follows through the multimodal state space. Large changes indicate transitions where tacit knowledge, attention shifts, or decision pivots occur. To make these change-based analyzes compatible with traditional process mining, the system extracts *semantic clusters* and *change-based activity labels*. A clustering algorithm, HDBSCAN [MHA⁺17], segments the embedding trajectory into coherent regions that correspond to recurring behavioural states such as „registration in progress“, „triage prioritisation“, „verification delay“, or „uncertainty resolution“. Each state is then associated with the most representative panel, frame, or conversational snippet. Transitions between clusters form the basis of a multimodal, data-driven event log, which is exported in a format compatible with pm4py [BvS23]. In pm4py, this log can be analyzed using standard discovery and conformance techniques. However, the underlying process model is no longer derived solely from declarative activity names—it is informed by the geometry of the embedding space. The result is a *behaviourally grounded* process model that captures tacit transitions, micro-interactions, and multimodal cues that would normally remain invisible in classical process mining. This method also enables the identification of *implicit routines* (e.g., second-by-second assessment behaviours), *soft decision boundaries*, and *variability hotspots* where human perception and judgement influence the process flow.

Multimodal Evidence Collection Interface. Figure 8.7 illustrates the *Multimodal Evidence Collection* component of the platform, which enables modelers to gather, inspect, and validate heterogeneous inputs before they enter the multimodal data analysis pipeline. This interface provides a unified, stage-based overview of all evidential sources—including video streams, software interaction traces, emails, chat threads, and audio recordings—allowing users to preview each modality, replace files when necessary, and assess evidential completeness. By centralizing these interactions, the interface ensures that the downstream perception, alignment, and conceptual extraction steps operate on high-quality, well-contextualized multimodal data.

8.6 Mixed Reality Simulation

A distinctive feature of the platform is its ability to dynamically generate or select the next mixed-reality situation video based on the unfolding elicitation conversation. Rather than navigating simulation scenes through fixed scripted paths, an AI agent continuously

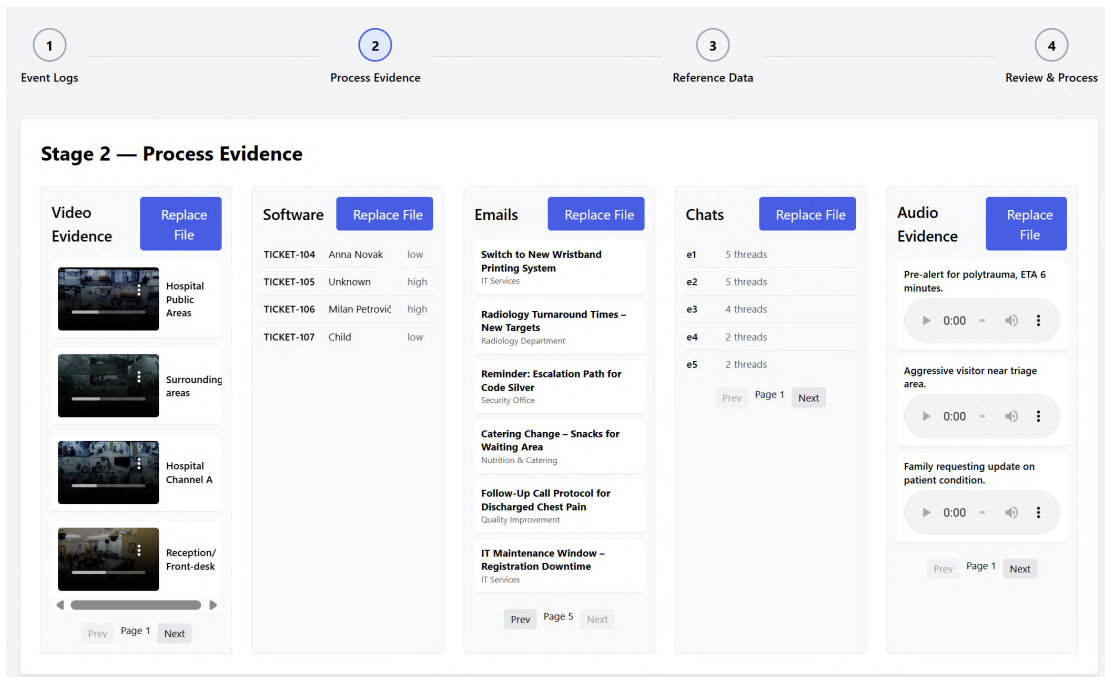


Figure 8.7: Interface for collecting and previewing multimodal inputs used in the multimodal data analysis pipeline.

interprets the dialogue between the *Domain Modeler* and the *Domain Expert*⁵, infers the underlying information need, and determines which situation should be explored next. This mechanism enables a conversationally guided traversal of simulated environments, ensuring that mixed-reality stimuli always align with the current elicitation focus.

The system uses a dedicated Simulation Assistant agent capable of monitoring the transcript in real time. As the participants discuss events, the agent identifies conversational cues that warrant a contextual shift in the simulation. For example, when the Domain Expert mentions: *This is where we often make early judgement errors if the patient suddenly becomes silent*, the agent determines that a situation highlighting silent distress cues is the most relevant next step. It then calls the `change simulation situation` tool with the appropriate identifier, causing the simulation pane to update automatically.

Situation transitions may involve one of two pathways:

1. **Selecting pre-recorded situation.** The framework includes a library of 360-degree videos and synchronized digital SUI captures, covering events such as patient check-ins, triage communication handoffs, insurance escalations, conflict incidents, and evacuation triggers. Based on linguistic and semantic context extracted from the dialogue, the agent selects the situation that best matches the described conditions.

⁵A domain expert is an individual with deep knowledge in a specific field, while a domain modeler is a developer who translates this knowledge into a conceptual model.

2. **Generating a situation.** When a conversational request points to a scenario variant not covered by the existing library—e.g., „Show me the same scene but with the patient accompanied by a paramedic“—the agent may trigger a video generation pipeline. This pipeline uses the stored camera parameters (yaw, pitch, zoom) and the previously logged `capture.json` metadata to seed a generative video model capable of injecting missing actors, altering posture or affect, or rendering counterfactual scene variations. These generated videos are inserted into the simulation environment as if they were part of the pre-existing situation catalogue. This is accompanied by a generation of the digital SUI interface using markdown.

In both cases, the integration is seamless: the newly selected or generated situation is rendered as a 360-degree background, and the corresponding digital SUI frame (e.g., EHR form, triage chat window, room sensor feed) is embedded via the CSS3D renderer inside the same spatial frame. The agent’s decision to move to the next scene is always grounded in conversational intent and the conceptual modeling objective, such as clarifying a tacit decision rule, verifying a domain expert’s claim, or probing counterfactual behaviours.

This conversational navigation mechanism mirrors how a human facilitator might guide an elicitation workshop—by selecting the most informative next example, reenacting a missed detail, or replaying a scenario from a different vantage point. However, the AI-powered approach offers significantly greater precision, as it can incorporate step-by-step multimodal cues, embedding-space similarity measures, and explicit references to conceptual model elements (e.g., „Show me the moment where the gateway splits into escalated vs. non-escalated triage“).

By coupling situation selection and video generation with the dialogue, the platform transforms simulation from a static replay environment into a responsive, conversationally driven exploration tool. This tightly integrates mixed-reality stimuli into the elicitation workflow and enables dynamic creation of situational evidence that supports the extraction of tacit knowledge and refinement of conceptual models.

8.7 Security Architecture

Ensuring safe model interaction is a core requirement of the MMCM platform, particularly because the system integrates multimodal data, real-time agent communication, and user-provided free-text input. The security architecture follows a two-layered strategy: (i) *client-side semantic screening of potentially harmful prompts*, and (ii) *server-side enforcement and isolation mechanisms* implemented in Node.js. Together, these layers prevent malicious instructions, content violations, or prompt-injection attempts from affecting the system or its agents.

8.7.1 Client-Side Security Classification

Before any user message is forwarded to an AI agent, it is processed through a lightweight semantic classifier designed to detect harmful or inappropriate content. Instead of executing or interpreting the message directly, the classifier evaluates it against a predefined taxonomy of security risk categories. This assessment takes place on the client side (or within the browser session), ensuring early detection and preventing harmful content from reaching the real-time agent framework.

The classifier internally uses a compact evaluation template that (i) summarizes the user's message, (ii) compares it to the security taxonomy, and (iii) returns a structured JSON label indicating the detected category of risk. The system never exposes this template to the user, and only the resulting category is retained.

Table 8.1: Security Classification Taxonomy Used for Client-Side Screening

Category	Description
MALICIOUS CODE	Attempts to inject or execute unauthorized code, scripts, or commands (e.g., SQL injection, XSS, shellcode).
PROMPT INJECTION	Attempts to override system instructions, jailbreak the assistant, or cause exfiltration of hidden data.
DATA LEAK	Requests or attempts to expose confidential, personal, or private information not intended for disclosure.
SOCIAL ENGINEERING	Attempts to manipulate the system or humans into revealing sensitive information or internal logic.
POLICY VIOLATION	Requests involving prohibited, illegal, explicit, violent, or otherwise disallowed content.
SPAM OR PHISHING	Unsolicited commercial messages, misleading links, or content resembling phishing attempts.
NONE	The message is safe and does not match any of the above categories.

Only prompts classified as NONE are forwarded to downstream agents or modeling subsystems. If any other class is detected, the client immediately blocks or redirects the message, providing corrective feedback to the user.

8.7.2 Server-Side Enforcement in Node.js

The backend provides an additional layer of security by validating, sanitizing, and safely routing all interactions. The Node.js service receives the user message together with its client-side classification result. If the message is labeled with any category other than NONE, the server:

1. records the event in the audit log,
2. rejects the request using a standardized security response, and
3. prevents the message from reaching any LLM, agent, or retrieval module.

Beyond simple validation, the backend enforces strict isolation and integrity rules. All payloads undergo JSON schema validation, URLs are sanitized, executable content is stripped, and messages are prevented from crossing subsystem boundaries unless explicitly permitted. This avoids propagation of malformed, adversarial, or malicious inputs into the multimodal analytics pipeline. All communications with the OpenAI Realtime API, the Conceptual Modeling Renderers and Editors, and the Conceptual Modeling AI Assistant are authenticated through signed tokens, time-limited sessions, and rate-limited channels. These mechanisms collectively ensure that even under adversarial conditions, the platform maintains a secure interaction environment.

8.8 Conclusion

This chapter presented the technical realization of the approaches introduced throughout the thesis, demonstrating how multimodal data analysis, mixed-reality simulation, and agent-based reasoning are operationalized in a unified system. The described platform serves as an executable instantiation of the methodological contributions developed in earlier chapters.

Starting from the real-time, agent-based architecture outlined in Section 8.1, the system decomposes conceptual modeling into coordinated analytical, simulation, and modeling responsibilities. The user interface and interaction is described in Section 8.2. The tooling layer described in Section 8.3 and Section 8.4 enables fine-grained interaction capture across heterogeneous systems under investigation, transforming user actions, system logic, and environmental cues into structured multimodal events. These events form the empirical backbone for the multimodal data analysis pipeline detailed in Section 8.5, where embedding-based representations allow latent behavioral regularities and tacit transitions to be surfaced and aligned with symbolic process models.

The mixed-reality simulation mechanisms presented in Section 8.6 further extend this pipeline by enabling conversationally driven situation selection and counterfactual generation. By tightly coupling dialogue, simulation control, and event capture, the system supports the elicitation of dispositional knowledge that cannot be obtained through observation alone.

Finally, the security architecture in Section 8.7 ensures that these capabilities can be deployed safely in interactive settings, preserving system integrity despite the use of free-form input, real-time agent collaboration, and external multimodal data sources.

In summary, this chapter demonstrates that the proposed approach to enhancing conceptual modeling is not only theoretically grounded but also technically feasible at

scale. The implementation shows how multimodal evidence, agent-based reasoning, and mixed-reality interaction can be integrated into a coherent system that supports real-time elicitation, analysis, and model construction. This technical foundation enables the empirical evaluations presented in the following chapters and provides a reusable blueprint for future systems that seek to embed conceptual modeling directly into rich, data-driven operational contexts. The application and evaluation of this implementation are provided in Chapter 9.

Evaluation and Application

This chapter evaluates and applies the proposed approach for enhancing conceptual modeling through the integration of tacit domain knowledge. The evaluation is grounded in the research problem formulated in Section 1.1.4, where the central challenge of this thesis was stated based on the chosen definition of conceptual models from [RAB⁺15], as follows: *we currently lack systematic methods to enhance the **precision** and **conciseness** of conceptual models by incorporating tacit knowledge that human experts rely on in practice, but rarely articulate in documentation or interviews.* Consequently, *precision* and *conciseness* of conceptual models are adopted as the primary criteria for assessing the effectiveness of the proposed approach.

Based on this problem formulation, the research objectives and corresponding research questions were introduced in Section 1.2.1. In particular, this thesis addresses: **RQ1**, which investigates how tacit domain knowledge can be conceptualized and integrated into conceptual modeling frameworks; **RQ2**, which examines methods for reliably capturing and eliciting tacit domain knowledge into representations suitable for conceptual model construction; **RQ3**, which focuses on the technical realization of such methods in a multi-scenario, modeling-language-independent system; and **RQ4**, which evaluates the extent to which the integration of tacit domain knowledge improves the precision and conciseness of conceptual models across scenarios and modeling languages.

The problem space is further analyzed in Section 3.1 which initiates the research problem investigation throughout Chapter 3. The investigation led to the development of a theoretical framework presented in Chapter 4. This theoretical framework is operationalized through the method introduced in Chapter 5, which represents our proposed approach for conceptual modeling enhancement. The method consolidates and scopes prior author’s research contributions for which extended discussions are provided in Chapter 6 and Chapter 7, respectively. Therefore, individual components of the overall approach have been evaluated independently in author’s related publications: accuracy of process mining from video [GBP24a, GBP24c, GBP25d], ViEnNa [GBP25c], elicitation

condition A dialogue structures [GBP24d], conceptual modeling-relevant multimodal data simulation [GBP25j, GBP25f, GBP26], and the reliability of multimodal alignment with conceptual models [GBP25a, GBP25i, GBP25b]. Those previous evaluations are summarized in Sections 6.7, and 7.7 through individual validations per foundational technology that is employed. The realized method is subsequently implemented according to the requirements defined in Section 3.3, with implementation details reported in Chapter 8.

Building upon this foundation, the present chapter focuses on the evaluation of the implemented system that realizes our proposed approach for conceptual modeling enhancement, through an application in an end-scale domain.

Following the research design outlined in Section 3.2, the evaluation adopts an *A/B testing* setup [Rub74]. Condition A (**ours**) corresponds to the proposed approach for enhancing conceptual modeling through multimodal data analysis and Mixed Reality, while Condition B (*SAP Scenes* [SAP20]) represents a **baseline** approach that does not incorporate the proposed techniques. The resulting conceptual models are then evaluated comparatively with respect to precision and conciseness across scenarios and modeling languages.

The remainder of this chapter first provides rationale for the evaluation design (Section 9.1), followed by the selection of evaluation scenarios and the scenario descriptions (Section 9.2), and the experimental settings (Section 9.3). Subsequently, the evaluation results are presented and analyzed (Section 9.4), and the fulfillment of the defined requirements is assessed (Section 9.5). A summary of the evaluation, synthesis of the results and response to research questions is given in Section 9.6.

Appendix C provides supplementary evaluation details. It firstly systematizes the underlying measurement framework by formalizing and decomposing all evaluation metrics (Appendix C.1), and then discusses evaluations of the scaling multimodal data analysis (Appendix C.2). It further provides extended details on the evaluation of: MMeCM as introduced in Section 6.1 (Appendix C.3), ALeKS as introduced in Section 6.2 (Appendix C.4); BICycle as introduced in Section 6.5 (Appendix C.5); PN-VidGen as introduced in Section 7.4 (Appendix C.6); and ViEnNa as introduced in Section 7.5 (Appendix C.7).

Implications of the findings and limitations are discussed in Chapter 10.

9.1 Evaluation Design

The evaluation is designed to assess improvements in precision and conciseness of conceptual models in a manner that is both *cross-scenario* and *conceptual modeling language-independent*. Multiple scenarios representing multiple application (sub-)domains (such as administration, security, operation) are considered within a coherent end-scale domain (higher-level application domain), for controlled involvement of domain experts while ensuring variation in context and task characteristics. Furthermore, different conceptual

modeling languages are employed to demonstrate that the proposed approach does not rely on language-specific assumptions.

Domain experts (DEs) and domain modelers (DMs) collaborate in conceptual modeling sessions to translate field-specific knowledge into structured models [Pau20]: experts provide, validate, and contextualize (tacit) domain knowledge, while modelers translate this into conceptual models.

9.1.1 Domain Selection

The domain-scale application of the proposed approaches is an **Emergency Room (ER) front-desk**.

The ER front-desk represents a particularly demanding socio-technical environment [Tis66]: it is characterized by high stakes, continuous time pressure, distributed communication across digital and physical channels, and rapid transitions between routine and critical situations (e.g., Code Silver, Code Amber, Code Purple [Wik25c]). In such conditions, experts rely heavily on tacit knowledge [GBP25h, Tis66]: the unspoken, embodied understanding that guides their decision-making under uncertainty. These forms of knowledge rarely surface in documentation or interviews, yet they determine the effectiveness and safety of the processes being modeled.

The choice of the Emergency Room front-desk as the demonstration domain is guided by three considerations: its multimodal richness, its cognitive and organizational complexity, and its strong relevance to safety-critical process modeling.

9.1.2 Evaluation Procedure

The evaluation design of the conceptual modeling enhancement approach proposed in this thesis follows a structured three-phase procedure, illustrated in Figure 9.1. The design aims at ensuring a controlled comparison between the proposed approach and a baseline condition. The three phases are: (i) Pre-Sessions Time, (ii) Sessions Time, and (iii) Post-Sessions Time.

Pre-Sessions Time

The first phase of the evaluation establishes the domain foundation and evaluation criteria prior to any modeling activity.

Step 1. First, all domain experts collaboratively gather and agree upon a set of short scenario descriptions that are representative of the end-scale domain. The objective of this step is to ensure high domain relevance and adequate coverage of distinct sub-domains, including administrative processes, security-related procedures, and operational coordination tasks. By grounding the evaluation in expert-curated scenarios, we aim to ensure that the produced conceptual models reflect realistic and practically relevant situations.

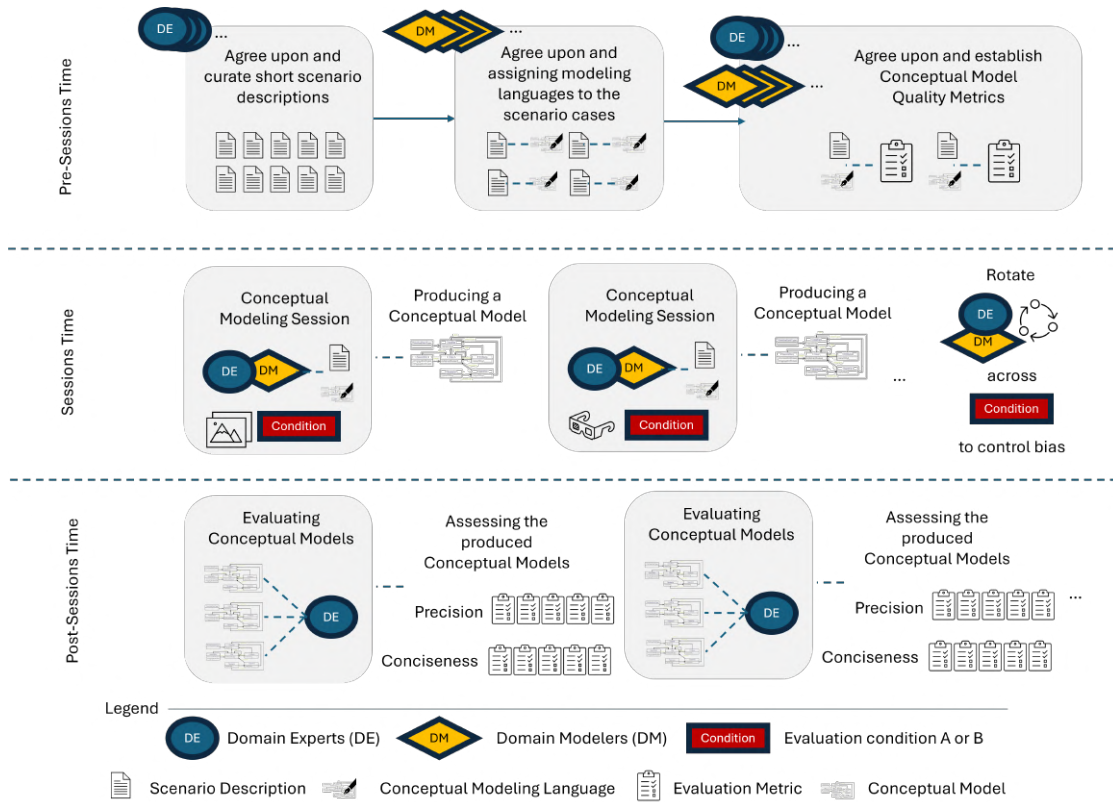


Figure 9.1: Overview of the evaluation design structured into three phases: Pre-Sessions Time, Sessions Time, and Post-Sessions Time.

Step 2. Second, the finalized short scenario descriptions are provided to the domain modelers. The DMs review the scenarios and propose suitable conceptual modeling languages for representing each scenario. The selection process aims to ensure expressive adequacy while reflecting common modeling practice.

Step 3. Third, DEs and DMs meet jointly to agree upon: (1) the binding between scenarios and modeling languages, and (2) the evaluation metrics for assessing conceptual model quality (grounded in the research problem formulated in 1.1.4, reflecting model's precision and conciseness).

This final step ensures shared understanding across domain and modeling perspectives and establishes the formal assessment criteria that will later be applied uniformly across all produced conceptual models.

Sessions Time

The following phase consists of controlled conceptual modeling sessions conducted under two experimental conditions:

- **Condition A:** Our proposed approach for enhancing conceptual modeling through multimodal data analysis and Mixed Reality, targeting the (tacit) domain-knowledge elicitation.
- **Condition B:** Baseline (tacit) domain-knowledge elicitation approach. The rationale for choosing the baseline approach is given in Section 9.3.4.

DEs and DMs are systematically rotated across conditions to control for elicitation biases, and repeated modeling effects. The rationale behind choosing these two conditions and the strategies applied to mitigate bias are detailed in Sections 9.3.1 and 9.3.2.

In each session, a DE–DM pair collaboratively produces a conceptual model for the assigned scenario using either Condition A or Condition B. Sessions are conducted independently, and models are produced without access to other session outputs to prevent cross-contamination.

Table 9.1: Overview of Evaluation Sessions (DE = Domain Expert; DM = Domain Modeler; SC = Scenario Code; C = Condition; A = Our approach; B = Baseline approach)

ID	DE	DM	SC	Scenario	Lang.	C	ID	DE	DM	SC	Scenario	Lang.	C
S001	DE01	DM01	SC01	Check-In	BPMN	A	S026	DE03	DM01	SC06	Ambulance	Flow	B
S002	DE01	DM02	SC02	Triage	Sequence	A	S027	DE03	DM02	SC07	Visitor	Class	B
S003	DE01	DM03	SC03	Silver	ViEnNa	A	S028	DE03	DM03	SC08	Purple	State	B
S004	DE01	DM04	SC04	Amber	Mindmap	A	S029	DE03	DM04	SC09	Belong.	ER	B
S005	DE01	DM05	SC05	Insurance	Journey	A	S030	DE03	DM05	SC10	Evac.	Gantt	B
S006	DE01	DM01	SC06	Ambulance	Flowchart	B	S031	DE04	DM01	SC01	Check-In	BPMN	B
S007	DE01	DM02	SC07	Visitor	Class	B	S032	DE04	DM02	SC02	Triage	Sequence	B
S008	DE01	DM03	SC08	Purple	State	B	S033	DE04	DM03	SC03	Silver	ViEnNa	B
S009	DE01	DM04	SC09	Belong.	ER	B	S034	DE04	DM04	SC04	Amber	Mindmap	B
S010	DE01	DM05	SC10	Evac.	Gantt	B	S035	DE04	DM05	SC05	Insurance	Journey	B
S011	DE02	DM01	SC01	Check-In	BPMN	B	S036	DE04	DM01	SC06	Ambulance	Flowchart	A
S012	DE02	DM02	SC02	Triage	Sequence	B	S037	DE04	DM02	SC07	Visitor	Class	A
S013	DE02	DM03	SC03	Silver	ViEnNa	B	S038	DE04	DM03	SC08	Purple	State	A
S014	DE02	DM04	SC04	Amber	Mindmap	B	S039	DE04	DM04	SC09	Belong.	ER	A
S015	DE02	DM05	SC05	Insurance	Journey	B	S040	DE04	DM05	SC10	Evac.	Gantt	A
S016	DE02	DM01	SC06	Ambulance	Flowchart	A	S041	DE05	DM01	SC01	Check-In	BPMN	A
S017	DE02	DM02	SC07	Visitor	Class	A	S042	DE05	DM02	SC02	Triage	Sequence	A
S018	DE02	DM03	SC08	Purple	State	A	S043	DE05	DM03	SC03	Silver	ViEnNa	A
S019	DE02	DM04	SC09	Belong.	ER	A	S044	DE05	DM04	SC04	Amber	Mindmap	A
S020	DE02	DM05	SC10	Evac.	Gantt	A	S045	DE05	DM05	SC05	Insurance	Journey	A
S021	DE03	DM01	SC01	Check-In	BPMN	A	S046	DE05	DM01	SC06	Ambulance	Flowchart	B
S022	DE03	DM02	SC02	Triage	Sequence	A	S047	DE05	DM02	SC07	Visitor	Class	B
S023	DE03	DM03	SC03	Silver	ViEnNa	A	S048	DE05	DM03	SC08	Purple	State	B
S024	DE03	DM04	SC04	Amber	Mindmap	A	S049	DE05	DM04	SC09	Belong.	ER	B
S025	DE03	DM05	SC05	Insurance	Journey	A	S050	DE05	DM05	SC10	Evac.	Gantt	B

An overview of all conducted sessions, including scenario assignments, modeling languages, domain experts, domain modelers, and condition allocation, is provided in Table 9.1. The scenarios are described in Section 9.2. The structured rotation ensures balanced exposure of scenarios and modeling languages across both conditions.

Post-Sessions Time

In the Post-Sessions time, all produced conceptual models are collected and distributed to all Domain Experts for evaluation.

Model quality assessment is grounded in the research problem formulated in Section 1.1.4, where *precision* and *conciseness* of conceptual models are adopted as the primary criteria for evaluating the effectiveness of the proposed approach.

Each criterion is operationalized through five evaluation items (ten in total per model). These items were agreed upon during the final stage of the Pre-Sessions phase and are tailored to the respective scenario and modeling language. The full breakdown of evaluation metrics and their corresponding assessment questions is provided in Section 9.3.5.

SESSION	DE	CONDITION	MODELING LANGUAGE	AVG PRECISENESS	AVG CONCISENESS	
S001	DE01	A	BPMN Patient Check-In & Registration	4.4 / 5	3.6 / 5	Details
Session S001 - DE01 BPMN - Patient Check-In & Registration - AI-assisted Preciseness (5 items) 1. Does the BPMN model correctly represent all key actors (Patient, Receptionist, ER Information System) using appropriate lanes? 5 2. Are all clinically relevant decision points (e.g., emergency assessment, existing patient, insurance validity) modeled with the correct type of gateway and clear semantics? 4 3. Do task labels (e.g., "Verify insurance online", "Apply fallback insurance procedure") unambiguously describe the real-world actions performed at the ER front desk? 5 4. Are the start and end events (normal registration vs. emergency fast-track) correctly defined and aligned with actual process outcomes in the ER domain? 4 5. Are the data- and system-related activities (record retrieval, record creation, insurance verification) correctly attributed to the ER Information System lane without mixing human and system responsibilities? 4 Conciseness (5 items) 1. Is each lane limited to essential roles only, avoiding unnecessary fragmentation of actors that behave equivalently in this scenario? 4 2. Are tasks grouped at the right level of granularity, avoiding oversplitting into micro-steps that do not change control flow? 3 3. Are there any redundant gateways or duplicated sequence flows that could be merged without losing behavioral distinctions? 4 4. Are task names short yet informative, avoiding verbose wording that adds text but not meaning? 3 5. Could any subprocesses or collapsed activities be introduced to hide low-level details while preserving the key logic of patient check-in and insurance handling? 4						
S001	DE02	A	BPMN Patient Check-In & Registration	4.2 / 5	4.8 / 5	Details
S001	DE03	A	BPMN Patient Check-In & Registration	4.4 / 5	4.2 / 5	Details

Figure 9.2: Excerpt from the evaluation tool used by Domain Experts for assessing conceptual model quality. The interface presents aggregated average scores for precision and conciseness (five items each), along with detailed rubric questions guiding structured expert evaluation.

Figure 9.2 illustrates the evaluation interface used by the DEs. The tool displays aggregated average scores for precision and conciseness, together with item-level scoring for structured assessment.

A comprehensive report on the quantitative and qualitative results of this evaluation process is presented in Section 9.4.

9.1.3 Multimodal Data Analysis

The ER reception is a continuous intersection of verbal, visual, and digital information flows, each contributing essential contextual cues for accurate conceptual modeling [LNR⁺25]. This setting produces highly heterogeneous multimodal signals, including:

- **Physical interactions (which can be captured through a video):** These include patient approach trajectories, body posture, gestures, visible distress, object exchanges (ID cards, insurance documents, personal belongings), gaze direction when searching for help, crowd density in the waiting area, and micro-behaviors such as pacing, freezing, or protective body language.

Conversations provide high-resolution semantic information: symptom descriptions, insurance clarifications, requests for family updates, or negotiations regarding waiting times. Paralinguistic cues—tone, hesitations, elevated pitch, speech rate—signal distress or urgency.

- **Digital interactions (HIS¹, chat, email, alarms, phone systems):** These include structured actions such as creating an electronic health record entry, insurance verification requests, triage categorization, and issuing internal alerts [Hau06]. They also include unstructured or semi-structured channels such as:

- Internal chat communication with triage nurses or security
- Emails related to insurance authorizations or transfer approvals,
- Digital alarms triggered through security dashboards,
- Phone-call management software capturing caller identity, tone, background noise, and urgency keywords (e.g., “The ambulance is two minutes away with a critical trauma case!”).

Such interactions show how digital logs and human decisions intertwine.

9.1.4 Mixed Reality

We identify Mixed Reality simulation as particularly suitable for ER domain because it aims to preserve the situated and embodied nature of front-desk work while enabling controlled experimentation. In contrast to purely virtual reality settings, the proposed approach explicitly supports augmented reality, allowing simulations to be conducted *in situ* and enabling operational information systems—such as registration interfaces, alert panels, and communication tools—to be embedded directly into the physical workspace.

¹Health Information System (HIS) is a comprehensive information and technology system used to manage health-related data for the purposes of improving patient care, streamlining hospital operations, and aiding in healthcare decision-making [Hau06]. It can refer to either a hospital-specific system (Hospital Information System) that manages administrative, financial, and clinical data, or a broader national system for collecting and analyzing population health data.

This maintains ecological validity while ensuring that interactions with digital artifacts closely resemble real operational conditions.

Receptionists and triage coordinators operate under high cognitive load, often switching between patient-facing communication, digital data entry, and emergency escalation protocols. These interactions often combine physical (documents handed over), digital (policy lookup), and conversational elements (confirmation questions). The multimodal stream also encodes timing, interruptions, and competing priorities. For example, the receptionist may (1) pause an insurance lookup due to a sudden distressed walk-in, (2) switch attention from a calm patient to a crying parent entering with a child, (3) respond to a security chat ping requesting a silent verification of a patient, or re-prioritize tasks upon hearing distant ambulance sirens. These subtle transitions express tacit decision-making that is not visible in the digital logs alone [GBP25h].

Their actions are governed by implicit heuristics, shaped by experience and situational awareness. These heuristics represent precisely the type of tacit conceptual regularities that the proposed Mixed Reality approach seeks to uncover.

9.2 Evaluation Scenarios

This section presents a set of representative scenarios from the ER front-desk domain, designed to demonstrate the application of the proposed framework. The set is a result of domain experts discussion as described in Section 9.1.2. Each scenario illustrates a distinct class of cognitive and organizational behavior—ranging from routine administrative workflows to high-stress emergency codes—where tacit knowledge plays a decisive role in decision-making.

Color-Coded Emergency Alerts. The hospital relies on a standardized set of color-coded emergency alerts to ensure coordinated responses across all departments. These codes, summarized in Table 9.2, represent situations ranging from missing persons to violent threats, hazardous spills, medical emergencies, and CBRN² disasters. While color-coded emergency classifications are widely used across hospitals, the specific meanings of individual codes can vary significantly by region and even by institution. For example, the designation *Code White* refers to a paediatric medical emergency in California, a combative person without a weapon in Louisiana, and emergency operating procedures in New Hampshire. Some institutions employ entirely different naming conventions altogether. At St. Vincent’s Hospital in New York City, for instance, the alert *Plan Blue* signified the imminent arrival of a trauma patient so critically injured that bypassing the standard A&E evaluation was necessary; the call notified the on-call surgeon to proceed directly to the A&E entrance for immediate transfer to surgery [Wik25c].

²CBRN is an acronym for Chemical, Biological, Radiological, and Nuclear, referring to hazards posed by these materials. It is used to describe both malicious threats, such as those from terrorism, and accidental releases, and encompasses protective measures and response protocols for such incidents.

Moreover, not all hospitals rely exclusively on colour codes. In the United Kingdom, numeric alert codes such as *3333* (security alert) and *4444* (fire alert) are commonly used. Additional “fast bleep” codes, such as a *2222* call, enable rapid paging of specific clinical staff without triggering a full crash-team response; for example, during status epilepticus a fast bleep may summon the on-call anaesthetist urgently while avoiding misuse of cardiac-arrest protocols [Wik25c].

Table 9.2: Emergency Codes Used in Hospital Operations as in [Ont25].

CODE GREEN – Evacuation (Precautionary)
CODE GREEN STAT – Evacuation (Crisis)
CODE YELLOW – Missing Person
CODE AMBER – Missing Child / Child Abduction
CODE ORANGE – Disaster
CODE ORANGE CBRN – CBRN Disaster
CODE RED – Fire
CODE WHITE – Violent / Behavioural Situation
CODE PURPLE – Hostage Taking
CODE BROWN – In-Facility Hazardous Spill
CODE SILVER – Person with a Weapon
CODE BLACK – Bomb Threat / Suspicious Object
CODE GREY – Infrastructure Loss / Failure
CODE GREY – External Air Exclusion
CODE BLUE – Adult Cardiac Arrest / Medical Emergency
CODE PINK – Infant / Child Cardiac Arrest / Medical Emergency

A further variant found in many hospitals is the use of “Doctor” codes, which serve as discreet public-announcement signals to prevent panic or protect patient privacy. These typically take the form of phrases such as “Paging Dr. Sinclair” or “Paging Dr. Firestone,” where the fictitious name corresponds to a specific type of emergency. For instance, “Paging Dr. Firestone, third floor” may be a covert indication of a suspected fire on the specified floor without alarming patients or visitors [Wik25c].

Beyond the color-coded emergency alerts, the ER front-desk receptionist is responsible for a wide range of administrative and operational processes that sustain the continuous flow of patient care. These include routine yet cognitively demanding tasks such as patient check-in, identity confirmation, and initial data entry into the Hospital Information System; verification of insurance status and coverage eligibility; managing walk-ins versus ambulance arrivals; coordinating with triage nurses to flag high-urgency presentations; and handling digital communications such as email confirmations, chat messages, internal

notifications, and family inquiries. Additional responsibilities such as visitor management, lost-and-found processing, appointment lookups, document scanning, and updating wait-time dashboards contribute to a steady multimodal workload that integrates verbal communication, visual inspection of documents, digital forms, and situational awareness captured through CCTV feeds. These administrative processes form the operational backbone of the front desk and complement the critical response behaviors required during emergencies.

9.2.1 Scenario Selection

To ensure focused yet comprehensive evaluation in the thesis, the scope has been defined around a curated set of ten representative scenarios that capture both routine administrative operations and high-stakes emergency processes. This selection follows a principled scoping strategy: (i) administrative processes were chosen to reflect the everyday cognitive load of the front desk, such as patient check-in, insurance verification, triage coordination, visitor management, and lost-belongings handling; and (ii) emergency processes were sampled from the standardized color-coded alert system. Together, these ten scenarios form a balanced dataset that spans low-urgency, high-frequency workflows as well as rare, high-pressure situations according to domain experts discussion from the Pre-Sessions Time of our evaluation (Section 9.1.2).

For each scenario, we describe the *situation*, the *involved systems and sensors*, and the *elicitation role* of the AI assistant. The purpose is to show how the assistant, acting as an elicitation mediator between the domain expert and the modeler, can uncover latent conceptual regularities from observed multimodal traces or simulated counterfactual scenarios. Selected domain-scale application scenarios are:

1. **Patient Check-In and Registration.** *Situation:* A patient enters the ER and presents identification at the reception desk. The receptionist verifies the information, updates the registration system, and prints a wristband. *Involved Systems:* CCTV (entry, desk), registration software, document scanner, email and internal chat. *Elicitation Role:* The AI assistant analyzes gaze patterns and reaction times to detect how attention is divided between the patient and the system, and i.e. prompts the expert to explain implicit prioritization heuristics when assessing patient urgency.
2. **Triage Coordination and Case Prioritization.** *Situation:* The receptionist and triage nurse collaborate to assign a triage category based on visible symptoms and reported conditions. *Involved Systems:* CCTV (waiting area), triage software, internal chat, audio intercom. *Elicitation Role:* The AI assistant generates Mixed Reality simulations of patients with ambiguous symptoms to elicit the expert's tacit reasoning in differentiating triage levels and managing uncertainty.
3. **Code Silver (Active Threat Response).** *Situation:* A violent individual becomes aggressive near the front desk. The receptionist must discreetly activate

- a Code Silver alarm and alert security while protecting nearby patients. *Involved Systems:* CCTV (entry and hallway), alarm dashboard, internal security chat, phone center. *Elicitation Role:* Through simulated active threat scenarios, the AI assistant captures stress-induced decision sequences and queries the expert on real-time prioritization of safety, communication, and data integrity.
4. **Code Amber (Infant Abduction Alert).** *Situation:* An alert is raised that a newborn may be missing. The receptionist must initiate a lockdown, and coordinate with hospital security. *Involved Systems:* CCTV (corridors, exits), access control software, alarm system, phone communication. *Elicitation Role:* The assistant replays synchronized video and system logs to extract the timing and coordination rules applied during crisis escalation.
 5. **Insurance Verification and Financial Clearance.** *Situation:* The receptionist confirms a patient's insurance status using online systems or, if unavailable, fallback procedures such as phone verification. *Involved Systems:* Insurance portal, phone center, email client, patient database. *Elicitation Role:* The assistant simulates delayed or unavailable system responses to explore expert strategies for trust-based decision-making and workflow adaptation.
 6. **Ambulance Arrival Notification.** *Situation:* An ambulance dispatch notifies the ER of an incoming critical patient. The receptionist updates the dashboard and informs triage and trauma teams. *Involved Systems:* Dispatch software, ER coordination dashboard, CCTV (ambulance bay), internal chat. *Elicitation Role:* The assistant compresses arrival sequences in Mixed Reality, prompting the expert to specify when and how the process transitions from "incoming" to "admitted."
 7. **Visitor Management and Access Control.** *Situation:* Visitors sign in, provide identification, and receive temporary access badges. Access may be restricted during security alerts. *Involved Systems:* Visitor management software, CCTV (entry area), badge printer, internal email notifications. *Elicitation Role:* The assistant generates simulated visitor interactions with varying intent (press and media, suppliers, incoming patients, police, firefighters etc.) and demeanor to extract implicit behavioral cues that trigger additional verification steps.
 8. **Silent Distress Signal and Covert Reporting (Code Purple Scenario).** *Situation:* A young girl accompanied by an adult silently hands a sticky note to the receptionist indicating that she has been kidnapped. The receptionist must act discreetly to verify the signal, protect the child, and alert security without alarming the suspect. *Involved Systems:* CCTV (front desk, entrance), silent alarm interface, internal chat, police hotline, registration software. *Elicitation Role:* The AI assistant reconstructs this delicate interaction, modeling subtle gestures, micro-expressions, and timing. It queries the expert on covert communication protocols and perceptual cues that guide the recognition of credible distress signals.

9. **Lost Belongings and Property Retrieval.** *Situation:* A patient or visitor reports missing property. The receptionist checks CCTV footage, storage logs, and contacts security to locate the item. *Involved Systems:* CCTV archive, security management system, phone call center. *Elicitation Role:* The assistant overlays synchronized spatial footage to prompt the expert to describe how spatial reasoning and temporal inference are applied when reconstructing events.
10. **Emergency Broadcast and Evacuation Coordination.** *Situation:* A fire alarm or chemical hazard triggers an immediate evacuation. The receptionist coordinates with staff and patients while maintaining communication continuity. *Involved Systems:* Alarm system, CCTV (main hall), internal chat, phone network, evacuation map software. *Elicitation Role:* The assistant simulates the evolving emergency scene, observing how the expert prioritizes tasks under time pressure and eliciting implicit rules governing communication hierarchy and evacuation timing.

Table 9.3 provides an overview of the ten emergency room front-desk scenarios used in the evaluation, detailing the corresponding processes, involved information systems, and the multimodal data analysis and elicitation focus for each scenario.

Table 9.3: Overview of Emergency Room Front-Desk Scenarios used Within the Evaluation of this Thesis.

#	Scenario	Involved Systems	Multimodal Data Analysis and Elicitation Focus
SC01	Patient Check-In	EHR, CCTV, chat	Tacit urgency recognition, attention distribution
SC02	Triage Coordination	CCTV, chat, intercom	Prioritization heuristics, uncertainty handling
SC03	Code Silver	CCTV, alarm, phone	Stress response sequencing
SC04	Code Amber	CCTV, access control	Crisis escalation timing
SC05	Insurance Verification	Phone, email, portal	Fallback workflows, trust-based decisions
SC06	Ambulance Arrival	Dashboard, CCTV	Temporal handoff definition
SC07	Visitor Management	Access system, CCTV	Security heuristics
SC08	Code Purple	CCTV, silent alarm, hotline	Covert communication, perceptual inference
SC09	Lost Belongings	CCTV archive, logs	Spatial reasoning in reconstruction
SC10	Evacuation	Alarm, chat, maps	Communication hierarchy, prioritization logic

In the following subsections(9.2.2 to 9.2.11), each of these scenarios is described in more detail.

Rationale for conceptual modeling languages choices. Conceptual modeling languages are not uniformly suitable for all types of scenarios, as different situations emphasize distinct aspects such as control flow, interaction, data structure, or temporal coordination. To account for this, the selection of modeling languages was not fixed a priori but aligned with the characteristics of each evaluation scenario. This alignment was established through joint discussions with domain experts and domain modelers, ensuring that each scenario was represented using a modeling language or approach appropriate to its dominant concerns. The resulting scenario–language mapping is chosen to ensure coverage across multiple conceptual modeling languages, thereby avoiding bias toward a single notation, as further described in Section 9.3.3.

Understanding of the conceptual modeling notations. For the understanding of the conceptual modeling notations employed in this evaluation, we refer the reader to Chapter 2 and Appendix A, where the relevant modeling foundations and notational elements are introduced.

9.2.2 Patient Check-In and Registration

The *Patient Check-In and Registration* scenario captures the front-desk workflow from the moment a patient arrives at the emergency room until they are formally registered and handed over to triage. Conceptually, this process involves three main participants: the *patient*, the *receptionist*, and the *ER information system*. The patient initiates the process by approaching the reception desk and providing identification or other documentation. The receptionist greets the patient, performs an initial informal assessment of urgency, and decides whether the case should be fast-tracked as an emergency or processed through the standard registration path.

If an emergency condition is detected (e.g., severe distress, visible trauma), the receptionist immediately escalates the case to triage or the trauma team, effectively short-cutting most of the administrative steps. Otherwise, the receptionist proceeds with a structured set of activities: requesting identification, opening the ER information system, determining whether the patient already exists in the system, either retrieving or creating a patient record, and performing insurance verification. In case of insurance issues or portal unavailability, fallback procedures are applied (such as phone verification or provisional admission).

Once the record and insurance status are resolved, the receptionist confirms key data (personal details, contact information, and consent where applicable), prints an identification wristband, and informs the triage function that the patient is ready to be assessed. The end of this process is marked by the state where the patient is *registered and ready for triage*, and the ER information system contains an up-to-date record that can be used for downstream clinical and administrative processes.



Figure 9.3: Mixed Reality scene for *Patient Check-In and Registration* scenario.

To operationalize the *Patient Check-In and Registration* scenario, we developed a front-desk software prototype that serves as a realistic interface for receptionists. The interface integrates the key multimodal and informational streams that shape reception work: the dynamic patient queue and ticketing system, detailed registration and insurance panels, internal chat and call-center communication, and contextual CCTV snapshots from the entrance, waiting room, and corridors. Within the proposed framework, this software is used in two complementary ways. First, it acts as a high-fidelity model of the digital environment in which receptionists actually operate, allowing us to log fine-grained interaction traces (clicks, keystrokes, timing, view switches) as multimodal data. Second, when rendered through a Mixed Reality headset, it enables the AI assistant to place the domain expert into controlled, yet realistic, check-in situations that elicit tacit prioritization strategies, informal emergency assessments, and subtle coordination habits that would otherwise remain unarticulated. Figure 9.3 illustrates the main view of this simulation interface for the patient check-in scenario.

In parallel, the same scenario is captured as a precise conceptual model using BPMN. The corresponding process model represents the interactions between the patient, the receptionist, and the ER information system as three lanes, and encodes the control-flow structure of the check-in process, including emergency fast-track decisions, record retrieval or creation, and online or fallback insurance verification. By comparing observed behavior and elicited counterfactuals with the modeled gateways, activities, and handovers, the AI assistant and the modeler can detect missing decision points, implicit exception paths, or overly coarse-grained activities. Figure 9.4 shows a part of the BPMN diagram for the patient check-in scenario, which will later be used as a reference artifact in the

multimodal alignment and model refinement pipeline.

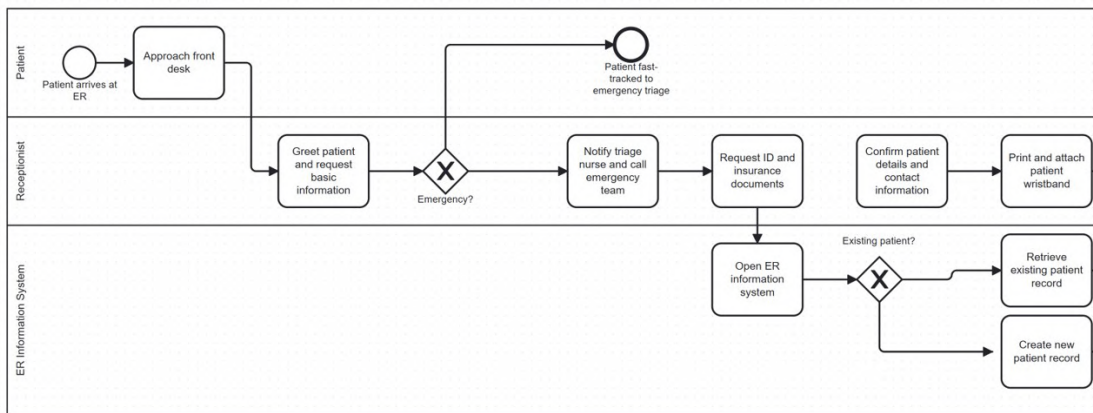


Figure 9.4: BPMN process model for the *Patient Check-In and Registration* scenario.

9.2.3 Triage Coordination and Case Prioritization

The **Triage Coordination and Case Prioritization** scenario captures one of the most cognitively demanding and time-critical processes at the Emergency Room front-desk. It involves rapid interpretation of visual, verbal, and digital cues to determine the urgency of incoming cases and to coordinate with the triage nurse and medical staff accordingly.

Figure 9.5 illustrates the simulation interface used for eliciting and analyzing the *Triage Coordination and Case Prioritization* process. The interface embeds the Hospital Information System directly into a Mixed Reality simulation of the front-desk environment, enabling modelers to observe how digital actions (e.g., queuing, triage updates, internal chat communication) unfold simultaneously with high-stakes real-world events such as visible distress, escalation, or environmental hazards. This dual-view setting supports synchronized interpretation of multimodal evidence, helping the modeler identify the exact triggers, contextual dependencies, and tacit signals that shape prioritization decisions in the ER workflow.

Upon arrival, each patient is initially evaluated by the receptionist, who records key information in the triage management system. The receptionist communicates through an internal chat or voice intercom with the triage nurse to decide whether the case should be prioritized for immediate evaluation or directed to the waiting area.

This process exemplifies the integration of multimodal data streams—visual, auditory, and textual—into a unified decision context. From a conceptual modeling perspective, it exposes the alignment between the physical act of triage and the informational act of categorization within the hospital information system. Tacit elements include the receptionist’s intuition about severity (e.g., detecting subtle signs of shock or confusion) and the ability to manage multiple simultaneous interactions under time pressure.

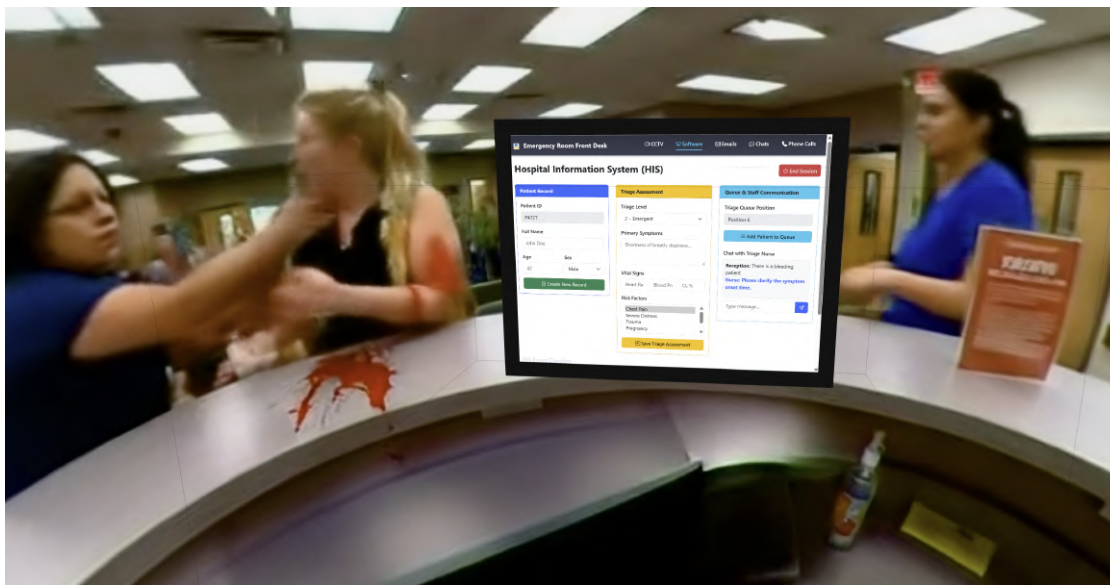


Figure 9.5: Mixed Reality scene for the *Triage Coordination and Case Prioritization* scenario.

In the Mixed Reality elicitation setting, the AI assistant reconstructs ambiguous triage scenes using video panoramic footage, simulating different patient presentations and behavioral cues. The domain expert, immersed in this environment, is prompted to explain subtle distinctions: for example, why two patients with similar verbal complaints might be assigned different triage categories. These explanations reveal latent conceptual regularities that can later inform process refinements or ontology extensions.

The following sequence diagram (Figure 9.6) models the interactions among the *Patient*, *Receptionist*, *Triage Nurse*, *Hospital Information System*, and *CCTV/Audio System*. It captures both the real-time process flow and the subsequent elicitation cycle.

9.2.4 Code Silver (Active Threat Response)

The **Code Silver** scenario represents a critical emergency procedure triggered when a violent or armed individual poses an immediate threat within the hospital premises. For the front-desk receptionist, this is one of the most stressful and complex processes, as it requires maintaining situational awareness, ensuring the safety of patients and staff, and initiating security protocols—all within seconds.

From a modeling perspective, this scenario exposes the importance of integrating physical situational dynamics (captured by CCTV and environmental sensors) with digital decision pathways (security dashboards, alert triggers, and intercom messages). The challenge lies in representing the tacit micro-decisions made under pressure: how the receptionist

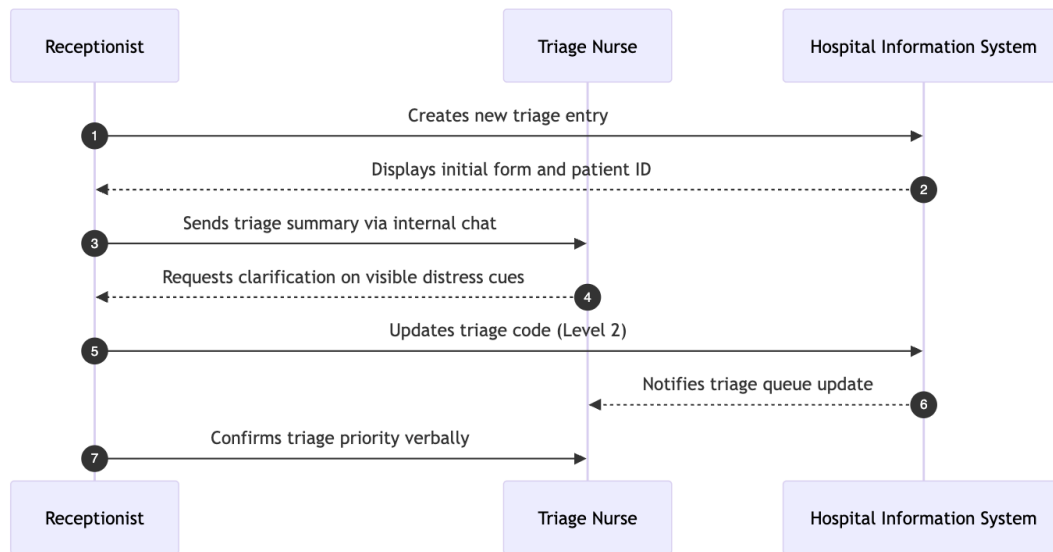


Figure 9.6: Sequence Diagram for *Triage Coordination and Case Prioritization* scenario.

interprets a developing threat, when to trigger a silent alarm, and how to continue communication without escalating the aggressor’s behavior.

The event begins when an agitated individual approaches the reception desk, raising their voice and behaving erratically. The receptionist’s initial task is to assess whether this represents a genuine security risk or a case of verbal distress. Once the threat is confirmed—either visually through aggressive body language or verbally through explicit threats—the receptionist activates the Code Silver alarm. This action automatically notifies hospital security, locks critical doors, and alerts nearby staff. The receptionist continues to manage communications calmly until help arrives, maintaining eye contact and minimizing further provocation.

The AI assistant plays a crucial elicitation role by recreating this event through a controlled Mixed Reality simulation. By immersing the domain expert in a realistic environment—with synchronized CCTV streams, ambient sound, and avatar-based reenactment—the assistant elicits implicit reasoning patterns, such as when the receptionist internally decides the situation is no longer safe. This immersive approach allows uncovering of counterfactual decision rules: e.g., “If the aggressor had turned toward the waiting area, I would have triggered the alarm sooner.” Such knowledge is essential for enriching conceptual models with behavioral granularity that would otherwise remain implicit.

Figure 9.7 illustrates the simulation interface developed for the Code Silver scenario. The

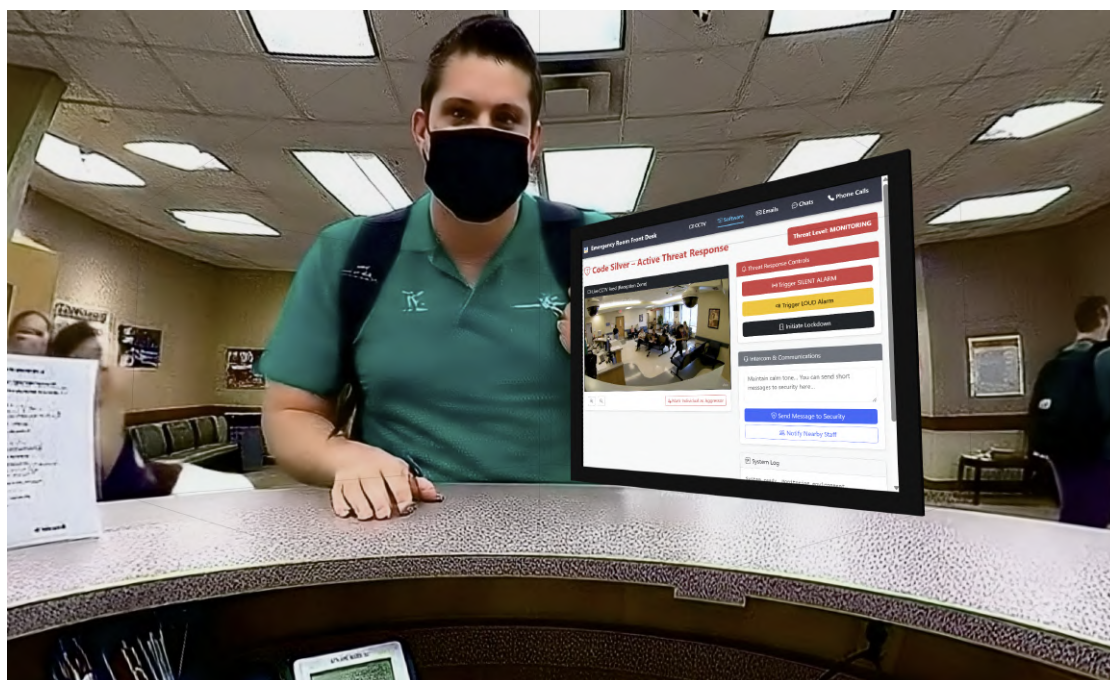


Figure 9.7: Mixed Reality scene for *Code Silver* scenario.

interface provides a 360° situational view from the receptionist’s perspective, embedding live or replayed CCTV feeds into an immersive Mixed Reality environment. The AI assistant records multimodal logs i.e. the expert’s gaze trajectory and response timing, later correlating these behavioral traces with system logs to reveal decision thresholds. This enables a quantitative interpretation of tacit conduct: *when* the expert recognizes danger and *how* that recognition translates into action.

Figure 9.8 presents the corresponding conceptual model for this scenario, modeled as a ViEnNa comic-like diagram. This model combines symbolic notation with visual storytelling. Each panel depicts an actor’s viewpoint and cognitive state: the receptionist’s calm posture during the initial interaction, the internal decision to activate the alarm, the AI-assisted verification of the alarm status, and the arrival of security staff. Such multimodal representations go beyond static notation by visually encoding the expert’s situational reasoning and environmental context.

9.2.5 Code Amber (Infant Abduction Alert)

The **Code Amber** scenario concerns suspected infant abduction or unauthorized removal of a newborn from a controlled area (e.g., maternity ward, neonatal intensive care).

A Code Amber can be triggered through several channels:

- An automated alarm from the infant security system (e.g., tag detected near exit),

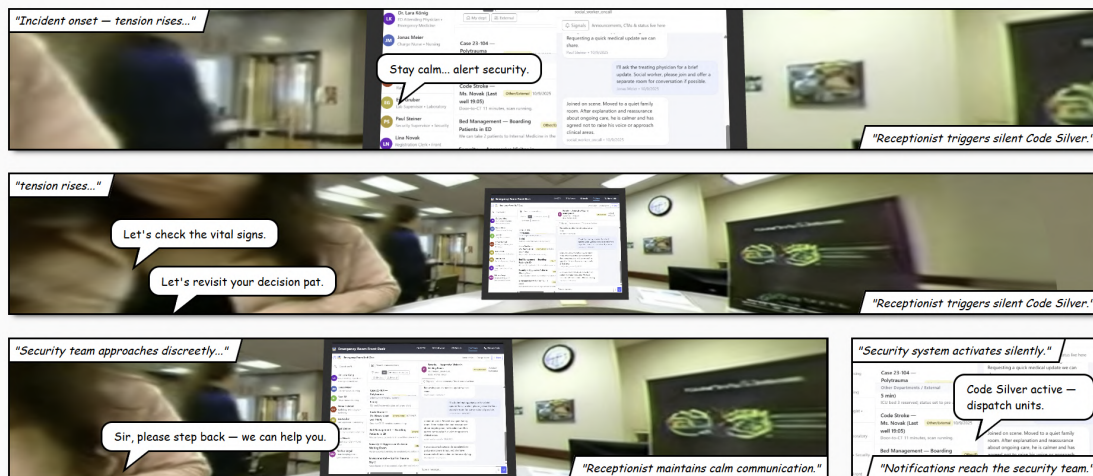


Figure 9.8: ViEnNa conceptual model for the *Code Silver* scenario, represented in a comic-like diagram showing the transition from perceived threat to coordinated security response.

- A call from maternity or neonatal staff,
- A visual observation reported by security or other personnel.

Once the alert is raised, the receptionist becomes a central coordination point: verifying the alert, locking or monitoring exits, informing security, and handling incoming phone calls from staff and relatives. Crucially, the receptionist must maintain composure, avoid spreading panic, and adhere to strict communication protocols.

From a multimodal perspective, this scenario integrates:

- **Visual evidence** from CCTV covering entrances, corridors, stairwells, and elevators,
- **Digital traces** from access control logs, infant tag systems, and alarm dashboards,
- **Communication streams** via internal chat, overhead paging, and phone calls.

The tacit dimension emerges in how the receptionist prioritizes these channels, which cues they trust first (system vs. human report), and how they modulate the tone and content of communication under stress.

Figure 9.9 shows the synchronized CCTV corridors, exit-point feeds, and access-control events replayed during the elicitation. The assistant highlights timing relationships between alarm activation, door-lock sequencing, and receptionist actions. Experts



Figure 9.9: Mixed Reality scene for **Code Amber (Infant Abduction Alert)**.

describe escalation pathways, lockdown timing, and interpretation of ambiguous cues in surveillance footage.

In the Mixed Reality elicitation setting, the AI assistant reconstructs a Code Amber event by synchronizing simulated corridor footage, exit camera views, and alarm overlays into an egocentric MR experience from the receptionist's viewpoint. The domain expert is immersed in this scenario and asked to explain:

- When exactly they consider the alert “credible” enough to initiate full lockdown procedures,
- How they balance adherence to protocol with situational improvisation (e.g., when a parent is hysterical at the desk), and
- What counterfactual actions they would take if, for example, an exit camera temporarily fails or if conflicting information arises?

These explanations surface latent conceptual regularities and counterfactual dispositions which are then fed back into the conceptual model.

A mindmap is used as the conceptual representation of the Code Amber process. Rather than a linear control-flow diagram, the mindmap emphasizes the cognitive and coordination structure of the scenario: triggers, actors, systems, and elicitation elements.

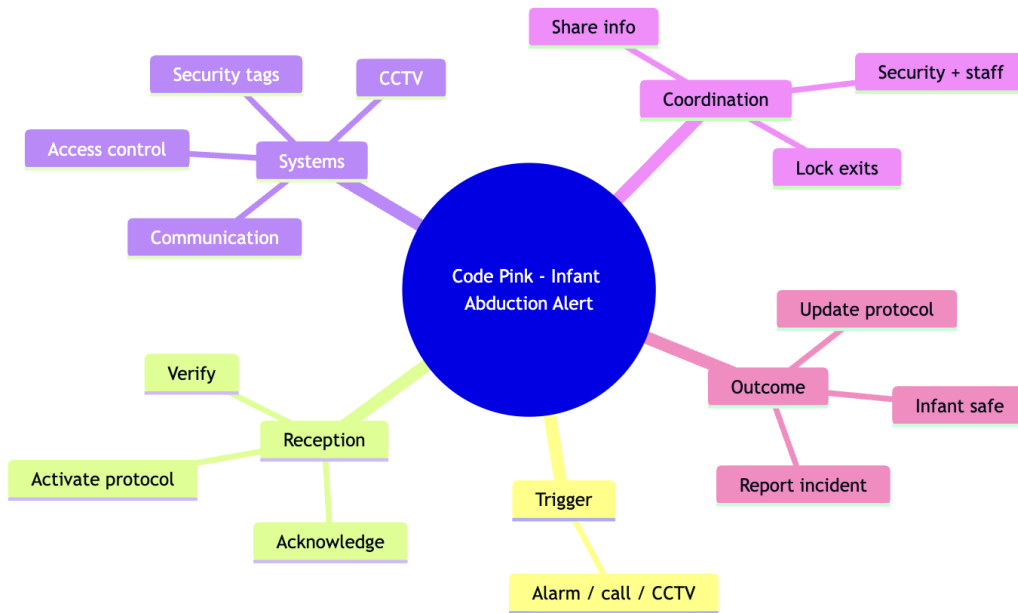


Figure 9.10: A mindmap representing the conceptual structure of the *Code Amber/Pink (Infant Abduction Alert)* scenario.

The baseline mindmap in Figure 9.10 organizes the scenario around key conceptual clusters: *Trigger Event*, *Receptionist Actions*, *Systems and Channels*, *Security and Clinical Coordination*, and *Outcomes and Documentation*.

In subsequent stages of the conceptual modeling, such mindmaps can be refined into more formal notations or aligned with embedding-based representations, where clusters of nodes correspond to regions in a multimodal conceptual space learned from recorded Code Amber episodes.

9.2.6 Insurance Verification and Financial Clearance

The **Insurance Verification and Financial Clearance** scenario captures a class of processes that are administratively routine yet cognitively non-trivial. At the Emergency Room front-desk, insurance verification often occurs under time pressure, with incomplete information and intermittent system availability. The receptionist must balance procedural compliance (verification, authorization, documentation) with patient-centered considerations (avoiding treatment delays in urgent cases).

A typical instance starts when a patient arrives at the ER and presents an insurance card or states their coverage verbally. The receptionist opens the hospital information system to retrieve existing records and then connects to an external insurance portal to verify coverage, eligibility, and co-payment obligations. If the portal is slow or unavailable, the receptionist may resort to fallback channels such as phone calls or email, while

simultaneously deciding whether the clinical urgency warrants proceeding before full financial clearance.

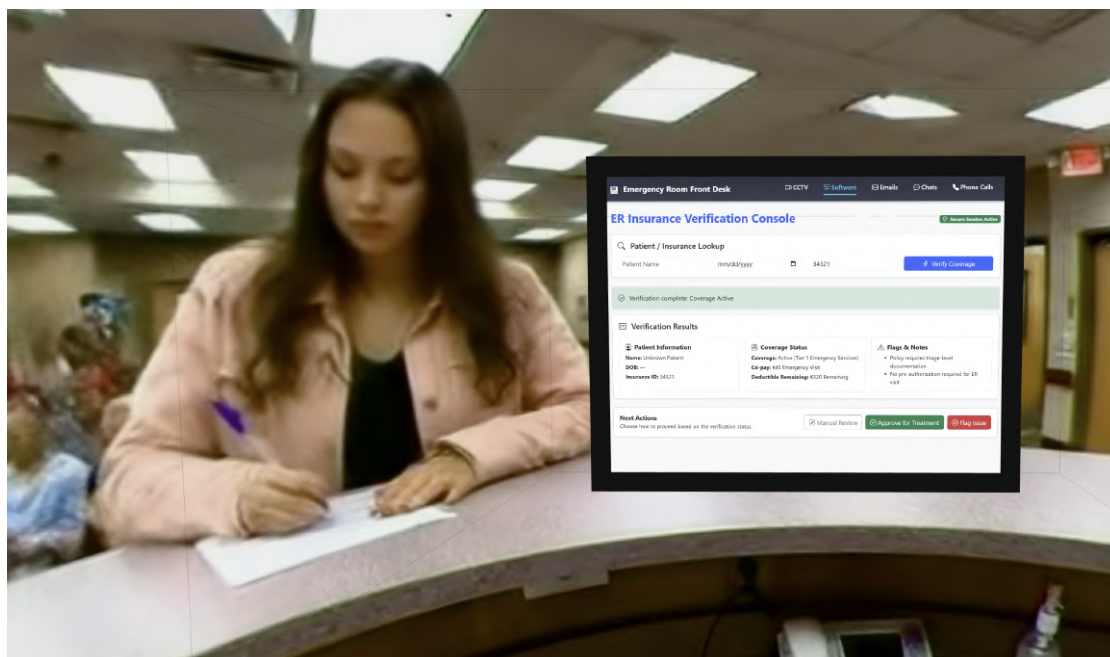


Figure 9.11: Mixed Reality scene for **Insurance Verification and Financial Clearance**.

Figure 9.11 depicts the simulation of the insurance portal, email client, and phone system under variable network delays and unavailability conditions. Experts are prompted to articulate fallback procedures, heuristics for trust-based decision-making, and strategies for maintaining patient throughput during partial system outages.

From a conceptual modeling viewpoint, this scenario is rich in *latent decision rules*. Experts routinely apply tacit heuristics that are rarely documented but strongly shape process behavior, such as when to override procedural order and initiate treatment “at financial risk” for the institution.

In the proposed framework, the AI assistant uses multimodal traces to reconstruct critical insurance verification episodes. In Mixed Reality, the domain expert is immersed in these reenactments and prompted to articulate the reasoning behind specific choices: *Why was this patient allowed to proceed without confirmation? Under what conditions would the decision differ?* This elicitation reveals counterfactual dispositions (what the receptionist would have done under slightly changed conditions), which are then integrated into the conceptual model.

Figure 9.12 illustrates the conceptual model of this scenario as a user journey. The diagram organizes the process along multiple “swimlanes” corresponding to key actors:

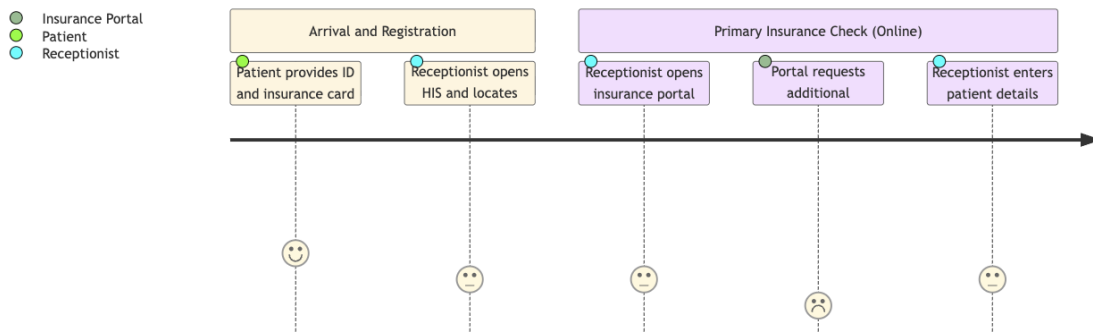


Figure 9.12: User journey-based conceptual model for the *Insurance Verification and Financial Clearance* scenario, highlighting the perspectives of the patient, receptionist, insurance systems, and the AI assistant.

the patient, the receptionist, the hospital information system, external insurance services, and the AI assistant. For each actor, discrete journey stages (e.g., *providing insurance information*, *performing portal verification*, *initiating phone fallback*, *deciding to proceed without confirmation*) are aligned along a shared temporal axis. This representation emphasizes both the experiential dimension (what the process participant perceives at each step) and the decision-centric dimension (where and how the receptionist’s tacit rules are applied).

9.2.7 Ambulance Arrival Notification

The **Ambulance Arrival Notification** scenario describes how the ER front-desk coordinates incoming patient arrivals communicated by ambulance dispatch systems. Although the event appears procedural—receiving and logging an ambulance notice—it is, in practice, a high-stakes information synchronization task involving multiple modalities, time-critical decisions, and interdepartmental communication.

The receptionist typically receives a digital notification from the regional dispatch center via a connected dashboard. This message includes estimated time of arrival (ETA), patient condition, and preliminary triage data. Upon receipt, the receptionist acknowledges the dispatch, notifies the triage nurse and trauma team, and verifies that the appropriate room is prepared. The CCTV system covering the ambulance bay provides visual confirmation of the vehicle’s approach, while radio chatter and internal chat logs offer parallel auditory cues. Each data source—dispatch feed, visual confirmation, and verbal communication—contributes to a multimodal situational picture that supports coordinated readiness.

Figure 9.13 presents the compressed arrival timeline, integrating ambulance-bay CCTV, dispatch software notifications, and internal triage chat messages. The simulation probes how the expert identifies the transition point between “incoming” and “admitted,” and how information flow synchronizes across systems under temporal pressure.

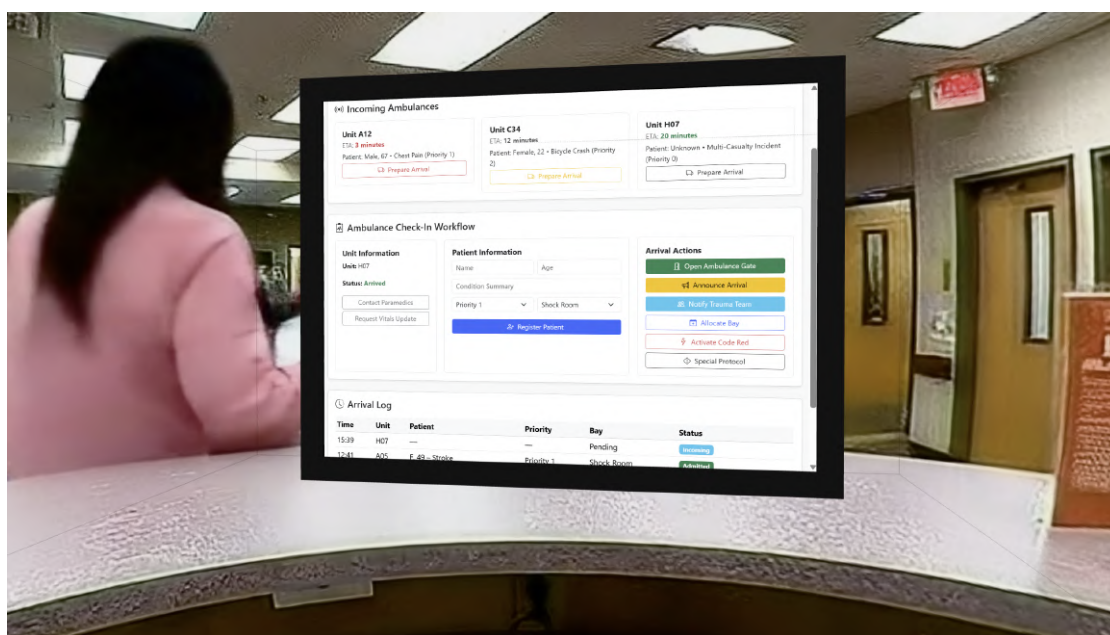


Figure 9.13: Mixed Reality scene for **Ambulance Arrival Notification**.

From a conceptual modeling standpoint, the process illustrates how temporal precision and multimodal alignment are crucial for safety-critical operations. Tacit regularities include how the receptionist interprets non-verbal urgency (tone of the dispatcher, siren intensity), how they decide when to mark a patient as “arrived,” and how these cues influence internal notifications. Mixed Reality elicitation allows the expert to re-experience time-compressed or ambiguous arrival situations, explaining exactly when and why each internal alert was triggered. This makes it possible to capture otherwise implicit timing rules and synchronization thresholds.

Figure 9.14 visualizes this scenario as a flowchart conceptual model. The model delineates a baseline on how digital, visual, and verbal inputs converge into structured decision logic. Each flow node represents a critical cognitive or communicative act—receiving data, verifying, notifying, or acknowledging—where tacit timing and prioritization heuristics influence how quickly and in what order actions occur. The flowchart thereby operationalizes a process that is partly algorithmic and partly experiential (human judgment).

9.2.8 Visitor Management and Access Control

The **Visitor Management and Access Control** scenario focuses on how the ER front-desk regulates entry into sensitive hospital areas while maintaining a welcoming environment for patients’ relatives and external parties (e.g., media, delivery services). The receptionist acts as a gatekeeper, enforcing access policies that are partly encoded in software (visiting hours, authorized zones, maximum visitor count) and partly governed

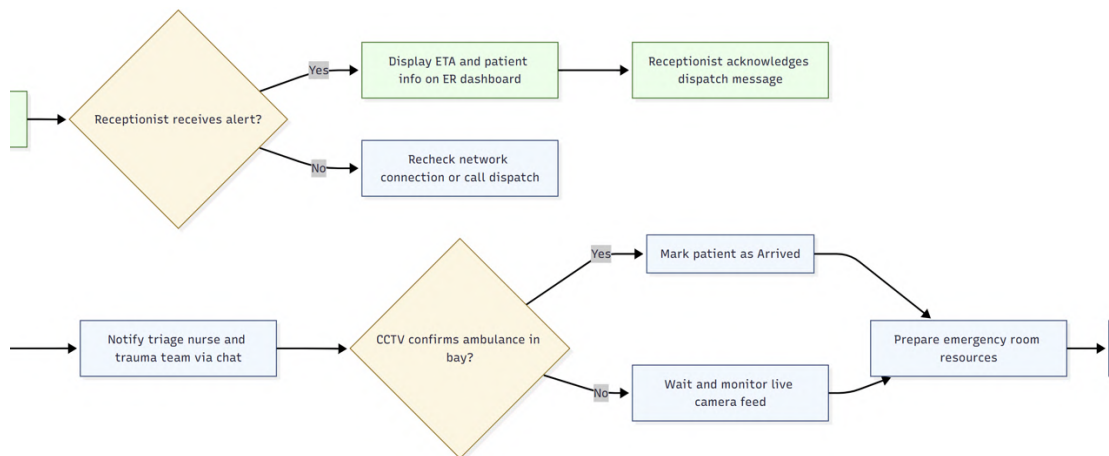


Figure 9.14: Conceptual model of the *Ambulance Arrival Notification* scenario represented as a flowchart, showing the selection of decision points leading from external dispatch signal to coordinated ER readiness.

by tacit judgment (e.g., recognizing suspicious behavior, assessing insistence or emotional distress).

A typical episode begins when a visitor arrives at the ER front-desk, states whom they wish to see, and presents an identification document. The receptionist verifies the visitor’s identity and the validity of the visit (relationship to patient, clinical status, time of day) in the visitor management system. If the request is acceptable, a temporary access badge is issued and the visit is registered; if not, access is denied or restricted based on policy and situation. CCTV footage, especially around entrance and corridor zones, provides additional context and can be consulted in borderline cases (e.g., when a visitor’s behavior appears agitated or inconsistent with their stated purpose).

Figure 9.15 shows simulated visitor interactions with variations in behavioral cues, intent, and identification quality. The assistant elicits how experts determine when additional verification is needed, how badge issuance interacts with security alerts, and how CCTV feedback influences escalation decisions.

From a conceptual modeling perspective, this scenario is rich in *access-control semantics*: who may enter which zone, under what conditions, and for how long. At the same time, visitor management involves situational judgments that are rarely formalized: for example, when a receptionist decides to request additional verification, contact security, or adjust rules (e.g., allowing an extra visitor in a critical case). The AI assistant supports Mixed Reality elicitation by replaying complex visitor interactions—including facial expressions, tone, and body language—prompting the domain expert to articulate underlying decision rules and counterfactuals (“*If this visitor had been alone instead of in a group, would you still have called security?*”).

Figure 9.16 depicts the conceptual structure underlying visitor management as an UML

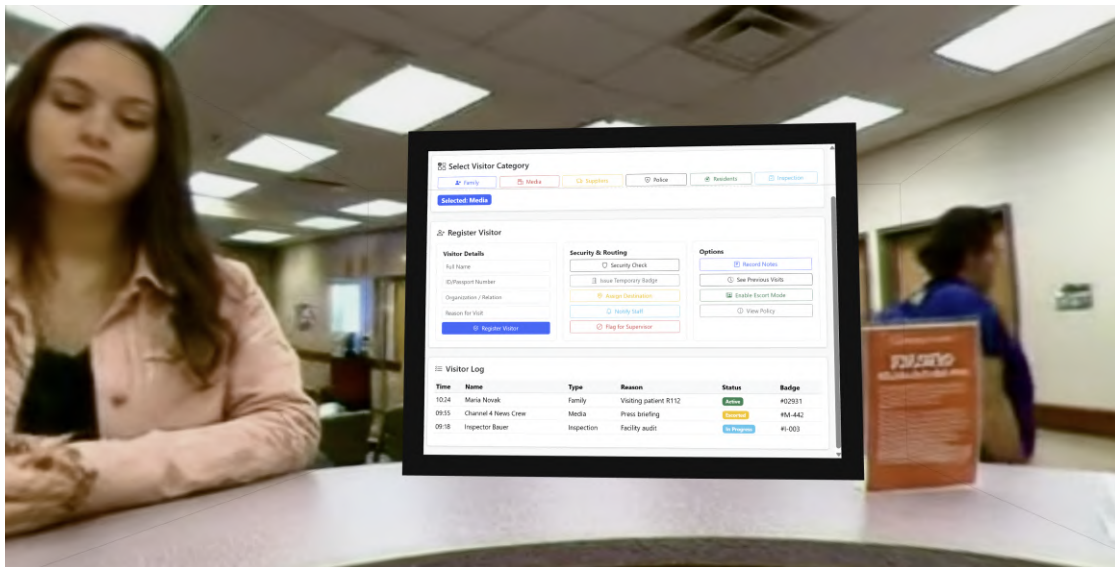


Figure 9.15: Mixed Reality scene for **Visitor Management and Access Control**.

class diagram. It distinguishes between *persons* (i.e. visitors, patients, staff), *visit records* (who visits whom, when, and for what purpose), *access badges* (tokens granting temporary rights), and *zones* (physical areas with different sensitivity levels). Policy elements, such as access rules and alerts, are modeled explicitly, together with AI-derived insights that refine these rules over time based on observed behavior and elicited tacit knowledge.

The UML class diagram makes explicit how visitor-related information (identities, visit records, badges) is structurally linked to zones, rules, alerts, etc. In the Mixed Reality elicitation setting, concrete visitor episodes—particularly borderline or escalated cases—can be replayed for the domain expert. Their explanations of why certain visitors were considered suspicious, why security was contacted, or why an exception to the rule was granted are then captured and encoded as new or refined access rule instances, closing the loop between lived experience and conceptual model evolution.

9.2.9 Code Purple (Silent Distress Signal and Covert Reporting)

The **Silent Distress Signal and Covert Reporting** scenario (also known as a *Code Purple* event) represents one of the most delicate and cognitively demanding processes at the ER front-desk. It unfolds when a patient, often a minor or a vulnerable individual, attempts to communicate danger or coercion through subtle, non-verbal cues—such as handing a written note to the receptionist—while accompanied by a potential perpetrator.

This situation requires rapid and highly discreet decision-making. The receptionist must interpret the signal, verify credibility, and discreetly trigger security and emergency protocols, all while maintaining calm behavior to avoid escalating the situation. The process therefore combines sensory perception (visual and emotional interpretation),

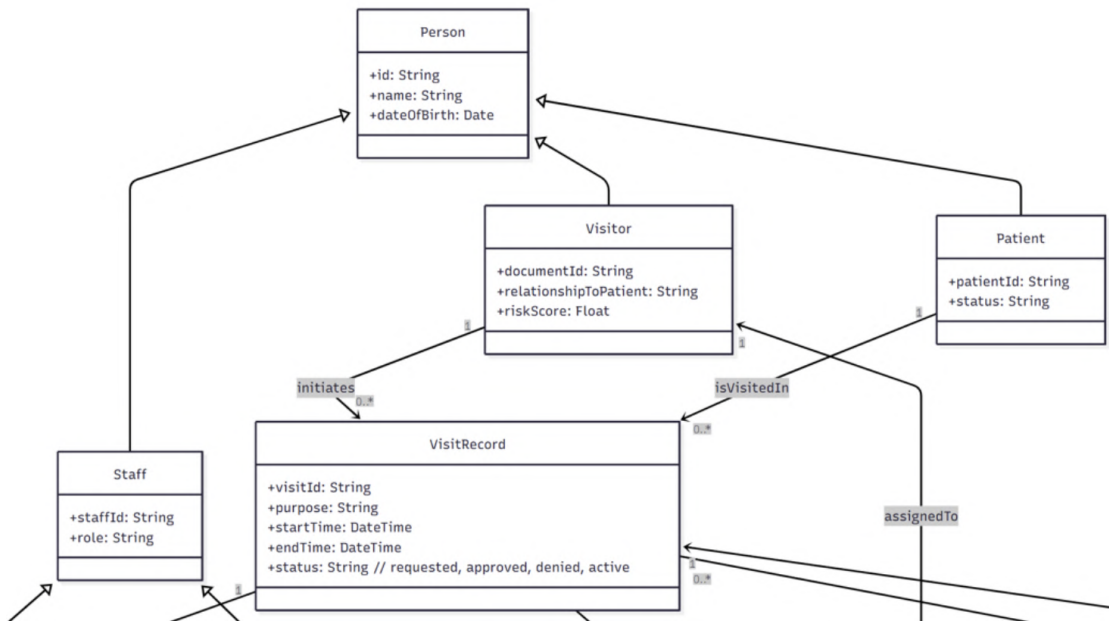


Figure 9.16: Conceptual model of the *Visitor Management and Access Control* scenario as an UML class diagram of visitors, staff, access badges, zones, and policy rules, including AI-derived insights.

digital interaction (activation of silent alerts and logging within the hospital system), and interpersonal communication management under extreme constraints.

From a research perspective, this scenario exemplifies the intersection of *tacit situational awareness*, *multimodal perception*, and *counterfactual reasoning*. The receptionist’s decisions rely on latent conceptual regularities learned through experience—how subtle visual patterns, micro-expressions, or temporal inconsistencies in dialogue indicate potential danger. These patterns are not documented but enacted; hence, they are ideal candidates for Mixed Reality simulation.

In the elicitation experiment, the AI assistant reconstructs this event using a simulation that includes multiple sensory modalities: the ambient sound of the waiting area, live CCTV angles, and the facial micro-expressions of both the victim and the accompanying adult. The domain expert experiences a replay of the situation and is asked to verbalize their moment-to-moment reasoning, such as: “*What visual or behavioral cue made you decide that the note was credible?*” or “*At which exact point did you initiate the silent alert?*”

Through this reflective elicitation, the AI assistant extracts implicit situational logic and encodes it as states and transitions within a conceptual model.

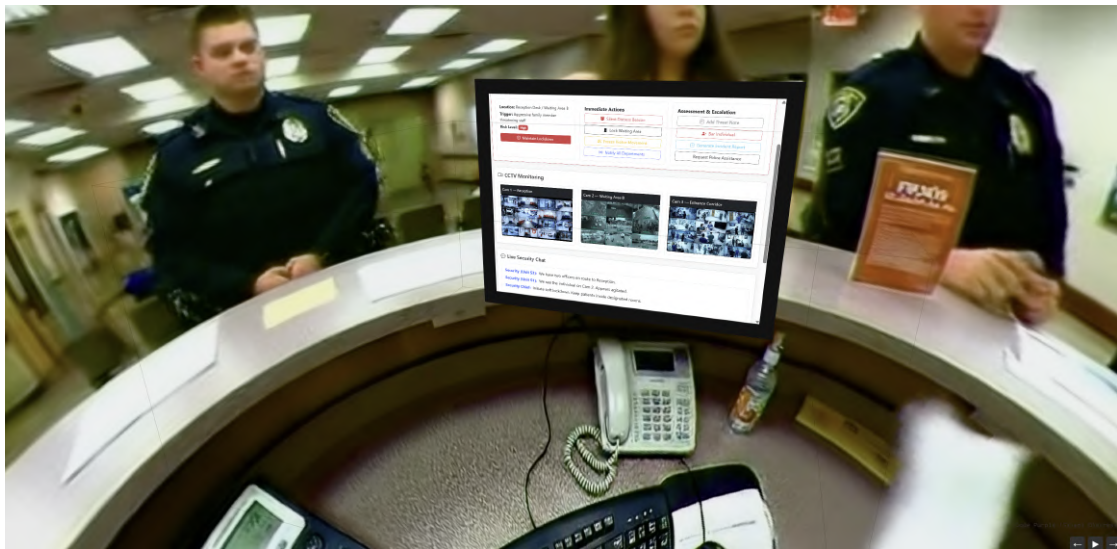


Figure 9.17: Mixed Reality scene for **Silent Distress Signal (Code Purple)**.

Figure 9.17 shows a reconstruction of a covert interaction, including micro-expressions, hand gestures, and relative positioning extracted from CCTV. The elicitation focuses on how experts confirm credibility of silent distress signals, trigger covert alarms, and maintain conversational neutrality to avoid alerting the suspect.

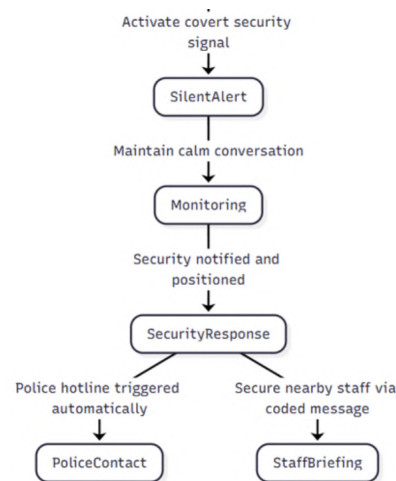


Figure 9.18: A conceptual model for the **Code Purple (Silent Distress)** scenario, as an UML state diagram.

A conceptual model for the **Code Purple** scenario (Figure 9.18) explicitly integrates both observable workflow transitions and tacit cognitive transitions, as an UML state diagram. In this scenario, the receptionist must interpret subtle multimodal cues—such as constrained speech, gaze aversion, posture tension, or the presence of an inconspicuous

written note—to identify a potentially coercive or dangerous situation.

Crucially, the transition from *Suspicion* to *Verification* does not arise from any discrete system event but from a multimodal recognition pattern: the alignment of visual cues (e.g., guarded body language), paralinguistic anomalies (e.g., whispering or hesitations), and contextual irregularities (e.g., inconsistent story or a hidden message). This exemplifies the thesis’ central claim: that multimodal data and Mixed Reality elicitation reveal latent conceptual regularities in expert behavior that would otherwise remain unspoken, unmodeled, or impossible to articulate using text-based elicitation alone.

9.2.10 Lost Belongings and Property Retrieval

The **Lost Belongings and Property Retrieval** scenario begins when a patient or visitor reports a missing item at the ER front-desk. The receptionist registers the report, initiates a multimodal investigation (CCTV archive search, access to storage logs, and security coordination), and maintains a verifiable *chain of custody* once the item is located. This process is socio-technical: it couples spatial-temporal reconstruction (who/what/where/when) with rigorous documentation to ensure accountability and to minimize disputes.

CCTV footage (multiple cameras, overlapping fields of view), access control logs (doors, lockers), the security case tracker, and communication channels (phone/chat) together provide complementary evidence. The receptionist (and security) triangulates the item’s last-seen location and time window, correlating subject trajectories with item appearances across cameras. When a likely match is found, the workflow transitions into retrieval, verification (claimant identity, descriptive attributes, serial numbers), and formal handover.

The AI assistant can reconstruct critical intervals as a synchronized, navigable MR replay with camera-to-camera jumping and overlaid paths of persons/items. Domain experts, immersed in MR, articulate tacit heuristics used in practice—e.g., which actions are prioritized first, what visual cues distinguish similar items, or how confidence thresholds are set before engaging security. These reflections surface latent conceptual regularities (e.g., “expand time window by ± 3 minutes if occlusions occur near elevator banks”) that are then encoded in the conceptual model.

Figure 9.19 presents a reconstruction of the MR scene of a situation where an item is reported as missing. The Mixed Reality interface overlays CCTV archives, storage-room logs, and security-management events, allowing the expert to reason about object movement, temporal gaps, and plausible retrieval points. The elicitation focuses on how receptionists combine spatial inference, system checks, and communication with security staff to determine whether an item was misplaced, taken accidentally, or stolen, and which procedural steps follow from each interpretation.

Figure 9.20 presents a conceptual *Entity-Relationship* model of this scenario. It captures the core entities (persons, items, reports, footage, storage, and custody) and the pivotal relationships needed for traceability, accountability, and evidence-backed retrieval.



Figure 9.19: Mixed Reality scene for **Lost Belongings and Property Retrieval**.

This ER model provides the schema-level backbone for integrating multimodal evidence and operational actions in the lost-property workflow. In subsequent sections, these entities and relations can be mapped to event schemas and embeddings to support relaxed, similarity-based retrieval (e.g., visually similar items across cameras, time-window expansion near occlusions), and to drive Mixed Reality reenactments for further elicitation and process refinement.

9.2.11 Emergency Broadcast and Evacuation Coordination

The **Emergency Broadcast and Evacuation Coordination** scenario covers the end-to-end handling of an acute hazard (e.g., fire, smoke, chemical spill) from first detection at the ER front-desk through coordinated communication, phased evacuation, accountability checks, and post-event restoration. The receptionist acts as the initial *incident notifier*, synchronizing actions across facility systems (alarm PA, access control, CCTV), clinical staff (triage, trauma teams), security, facilities (HVAC³, power), IT operations (systems failover), and the external call center (overflow and public information).

Figure 9.21 visualizes the evolving evacuation environment through synchronized alarm logs, hall-level CCTV, staff chat messages, and evacuation-map overlays. Experts specify

³Heating, ventilation, and air conditioning (HVAC) systems use advanced technologies to regulate temperature, humidity, and indoor air quality in residential, commercial, and industrial buildings, and in enclosed vehicles.

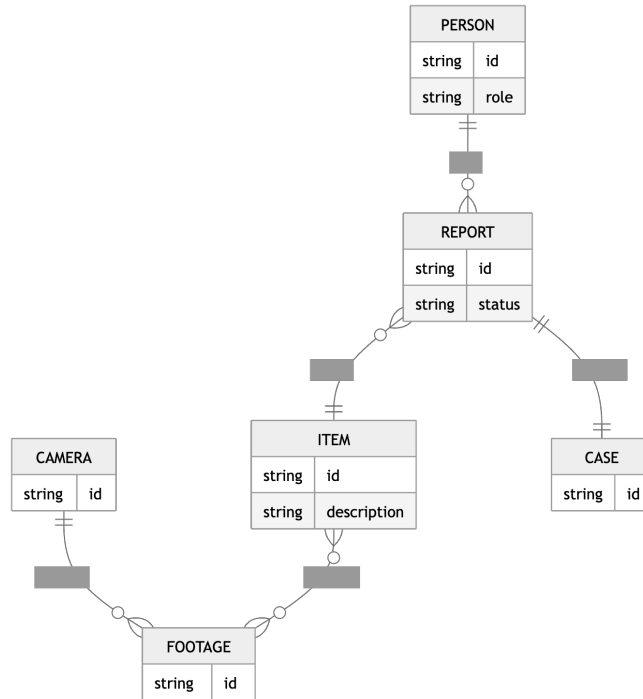


Figure 9.20: Entity-Relationship model for the *Lost Belongings and Property Retrieval* scenario, focusing on evidence correlation, retrieval, and documented chain of custody.

communication hierarchies, ordering of actions under time pressure, and the implicit decision rules that guide evacuation timing and prioritization of vulnerable patients.

Sensor alerts (smoke detectors, gas sensors), CCTV confirmation, alarm and PA systems, internal chat/voice channels, access control for door states, patient tracking dashboards, and external phone lines all converge. The workflow proceeds through: (i) *detection & verification*, (ii) *broadcast & role activation*, (iii) *zonal evacuation & assistance*, (iv) *roll call & accountability*, and (v) *handover & recovery*. Mixed Reality (MR) elicitation can replay the evolving scene (e.g., simulated smoke spread, crowd flow) to surface tacit rules like “*delay Zone C egress by 30–60 seconds if Zone B corridor is congested*” or “*prioritize gurney routes avoiding stairwell S2 when visibility < τ* .”

Figure 9.22 presents a Gantt schedule as a baseline conceptual model that captures time-critical tasking, inter-team dependencies, and overlapping lanes. Durations are indicative; in practice they are parameterized by hazard type, building topology, occupancy, and congestion feedback from CCTV/IoT analytics.

The initial *Detection & Verification* lane establishes a short-but-critical chain from sensor alert to visual confirmation and hazard classification. The *Broadcast & Role Activation* lane begins with a public address/alarm milestone, then fans out to security/facilities, clinical leads, access control, and the external call center. *Evacuation Execution* staggers

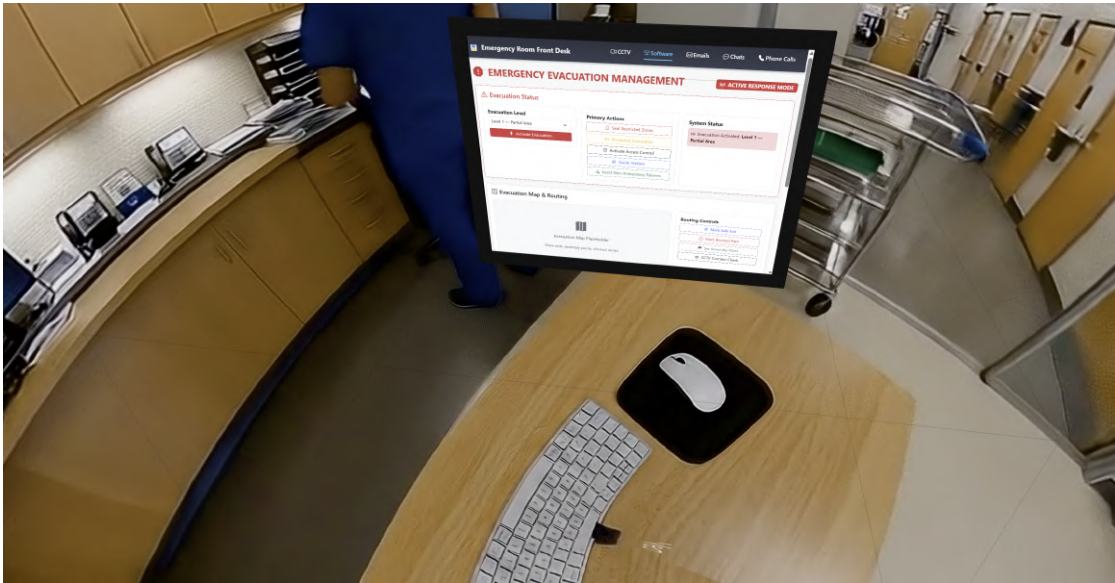


Figure 9.21: Mixed Reality scene for **Emergency Broadcast and Evacuation Coordination**.

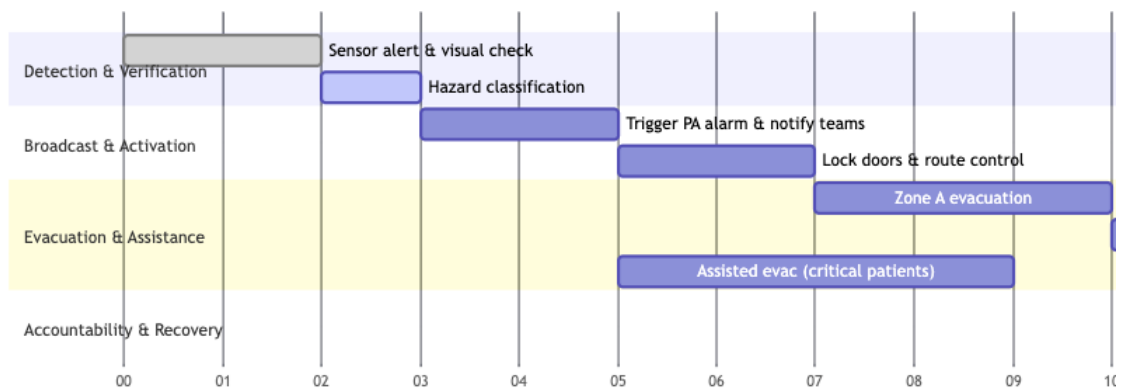


Figure 9.22: Gantt diagram for the *Emergency Broadcast and Evacuation Coordination* scenario with phased activities, roles, and dependencies.

zones to reduce corridor congestion and reserves a parallel, longer task for mobility-assisted evacuation. *Accountability & Handover* ensures headcounts and escalates to missing-person protocols if necessary, followed by formal control transfer to public authorities. Finally, *Recovery & Post-Event* sequences the return-to-operations and structured debriefing for learning capture.

Mixed Reality simulation for reenactments—driven by CCTV-derived crowd flows and simulated smoke/visibility fields—help experts articulate tacit traffic-management heuristics (e.g., dynamic re-routing, door state overrides, prioritization of assistive teams).

These insights can be encoded as parameterized constraints (e.g., *max corridor density*, *min egress width*, *visibility thresholds*) or as decision rules linked to live analytics, tightening the alignment between the conceptual evacuation model and operational behavior in situ.

9.3 Evaluation Settings

This section presents the evaluation settings of the approaches proposed throughout this thesis.

We conduct an integrated, multi-phase evaluation combining (i) controlled elicitation sessions with domain experts and domain modelers, (ii) comparative modeling tasks across multiple modeling languages, and (iii) metric-based assessments of conceptual-model quality. The evaluation design reflects both theoretical considerations from conceptual modeling research and practical constraints of the emergency room domain.

The evaluation was designed to balance methodological rigor and the specific constraints of eliciting tacit conceptual knowledge. A fundamental design decision was to separate the roles of domain experts and domain modelers, reflecting the real-world structure of conceptual modeling projects. In total, five experts and five modelers participated:

- **Domain Experts (DE01–DE05):** Emergency Room professionals with 0.4–3 years of frontline experience in ER procedures. Prior to the study, all experts were trained to use the simulated tools (EHR mock-ups, CCTV dashboards, insurance portals, and input devices) to ensure they could reliably interact with the environment and generate representative multimodal event logs. Their practical familiarity with real ER workflows and digital systems enables them to evaluate whether the conceptual models across all notations accurately reflect operational and tacit aspects of ER practice.
- **Domain Modelers (DM01–DM05):** Students with a training in conceptual modeling. Modelers are proficient in the full set of conceptual modeling languages used in this study. They possess conceptual modeling skills but intentionally lack ER-domain expertise to maintain a clear separation between domain knowledge and modeling competence.

All elicitation and modeling sessions in this study were conducted using a suite of *simulated tools* designed to closely resemble real Emergency Room operational systems while ensuring complete anonymity and ethical compliance. To avoid handling sensitive patient information or requiring institutional permissions, we developed artificial replicas of common ER interfaces, including an EHR mock-up, a synthetic CCTV dashboard, an insurance-verification portal, and related interaction devices. These simulated systems when used, generate multimodal data—video, audio, logs, and event traces—that

recreates realistic ER workflows without using any real patient records or hospital infrastructure. This approach allowed domain experts to interact with the environment similar as they would in practice, while guaranteeing privacy, reproducibility, and risk-free experimentation for the purpose of conceptual-model elicitation and evaluation.

The evaluation comprised ten end-scale domain scenarios (SC01–SC10), each assigned a conceptual modeling language to avoid arbitrary language choice and to ensure broad coverage of industry-relevant notations.

Each modeling session follows a structured, two-person interaction between a domain expert and a domain modeler. The session begins with a brief alignment phase in which the DE describes the scenario to be modeled and the DM clarifies terminology or contextual elements as needed. Depending on the assigned condition, the pair then proceeds with either the baseline elicitation or (our) conceptual modeling enhancing AI assistant that serves as the elicitation tool that uses multimodal data analysis and Mixed Reality. In both settings, the DE serves as the sole source of tacit and explicit domain knowledge, while the DM is responsible for translating this elicited information into a formal conceptual model using the designated modeling language. The structure of the session ensures that knowledge flow remains unidirectional—from expert to modeler—while allowing the DM to ask clarifying questions, navigate the conceptual modeling tool, and iteratively refine the emerging model.

9.3.1 Avoiding Recall Bias for Domain Experts

A key challenge in evaluating tacit-knowledge capture is the risk of *within-subject recall bias* [CBB17]. If the same domain expert were asked to participate in a modeling session for the *same* scenario twice—once under baseline elicitation approach and once under approach that uses multimodal data analysis and Mixed Reality—this would artificially inflate performance by making tacit cues more explicit through repetition [Rub74].

To mitigate this, we imposed the following design constraint:

Each domain expert participates in every scenario exactly once, ensuring that no expert models the same scenario more than a single time.

This ensures that any tacit knowledge manifested during elicitation emerges naturally from the expert’s situated cognition, rather than from rehearsal effects or memory consolidation between sessions. The tacit cues therefore reflect genuine, first-pass expert behavior and counterfactual disposition, aligning with the theoretical commitments of the thesis.

9.3.2 Avoiding Recall Bias for Domain Modelers

Beyond avoiding within-subject repetition, the evaluation design must also ensure that the *source* of tacit knowledge remains uncontaminated. In our study, tacit knowledge

originates exclusively from domain experts, not from the modelers. This distinction allows us to treat all conceptual modeling sessions for a given scenario as if they collectively form a single, larger elicitation instance. Even though modelers produce multiple models for the same scenario, the *tacit cues themselves* stem from a single DE explanation, which each modeler receives independently. The order in which DEs were interviewed does not influence the analysis: each scenario’s tacit knowledge is fixed at the point of elicitation and does not change across modelers.

This motivates a crucial methodological separation between two forms of repeated exposure:

1. **Repetition that would influence tacit-knowledge extraction** (undesirable for the evaluation), and
2. **Repetition that influences modeler competence and domain familiarity** (designed to not affect our evaluation and it’s realistic).

A repeated explanation of the *same* scenario by the *same* domain expert would inevitably bias the elicitation process. Any second pass would incorporate prior reflection, hindsight correction, memory consolidation, or even alignment with the previously generated conceptual model. This would undermine the central aim of the study: to capture tacit cues as they naturally manifest during a first-pass, situated expert explanation. For this reason, each DE participates in every scenario exactly once.

By contrast, DMs are not the *origin* of that knowledge; they merely transform the expert’s elicited information into formal models, and importantly, the study jointly evaluates outputs across five independent domain modelers. The purpose is not to isolate each DM per scenario, but to evaluate the *quality of conceptual models* produced under two elicitation conditions. Although repeated interaction with a modeler could in principle influence how an expert articulates tacit cues, this study mitigates such effects through a rotation of modelers across sessions. As a result, all modeling sessions for a given scenario can be treated analytically as a single, larger elicitation instance, with no systematic dependency on a particular DM. The evaluation therefore concentrates on the *quality of the resulting conceptual models* under the two elicitation conditions, rather than on isolating individual modelers per domain-level scenario.

In summary, the evaluation design aims to explicitly protect the integrity of tacit-knowledge elicitation by eliminating scenario-level repetition for DEs. This should ensure that the study measures what it intends to measure: the influence of our conceptual modeling enhancement assistance on the precision and conciseness of conceptual models, grounded in authentic expert knowledge rather than artifacts of repeated explanation.

9.3.3 Scenario–Language Alignment

The scenario–language alignment presented in Table 9.4 was the result of a deliberate decision between domain modelers in a process described in Section 9.1.2. Each modeling

language was selected to emphasize the dominant structural characteristics of the respective scenario while ensuring coverage of diverse modeling paradigms (process-oriented, interaction-oriented, structural, behavioral, temporal, and cognitive representations). For instance, *SC01 Patient Check-In* was modeled using BPMN because it represents a structured, repeatable administrative workflow involving clearly defined tasks, roles, and decision gateways. Similarly, *SC06 Ambulance Arrival* was aligned with a Flowchart to capture procedural logic in emergency intake handling, while *SC08 Code Purple* employed a State Diagram to emphasize state transitions and escalation dynamics. These mappings ensure that the chosen notation highlights the primary analytical dimension of the scenario—control flow, state evolution, or conditional branching.

Interaction-intensive and coordination-heavy scenarios were mapped to notations that explicitly foreground communication and temporal ordering. *SC02 Triage Coordination*, which involves multiple actors (nurses, physicians, administrative staff) and synchronized decision-making under time pressure, was modeled using a Sequence Diagram to make message exchanges and lifeline dependencies explicit. Likewise, *SC10 Evacuation* was represented with a Gantt Chart to emphasize temporal constraints, parallel task execution, and scheduling dependencies across hospital units. Further, *SC03 Code Silver* is modeled with a ViEnNa Diagram, as the process represents highly dynamic, security-critical incidents where tacit situational awareness and cross-modal signals are considered essential [LNR⁺25].

Finally, structurally focused or semantically rich scenarios were assigned modeling languages that capture domain entities and relationships in a concise yet expressive manner. *SC07 Visitor Management* was modeled using a Class Diagram to represent roles, permissions, and object relationships within the hospital ecosystem. *SC09 Lost Belongings* employed an Entity-Relationship Diagram to formalize data structures and traceability between items, reports, and stakeholders. *SC05 Insurance Verification* was mapped to a User Journey to highlight experiential and service-oriented aspects of patient interaction, while *SC04 Code Amber* used a Mindmap to structure incident-related knowledge, responsibilities, and communication channels in a cognitively accessible format.

9.3.4 Benchmarking Baseline

To meaningfully evaluate the contribution of our approach for enhancing conceptual modeling through multimodal data analysis and Mixed Reality, the study required a valid benchmark against which the approach could be compared. For this purpose, we selected *SAP Scenes* [SAP20], a baseline elicitation method used in both industry workshops and academic education for process discovery, requirements engineering, and early-stage conceptual modeling [VMU24, SAP20].

SAP Scenes relies on printed comic-like character cutouts, props, and sticky notes that participants arrange on a physical table or board to act out real-world scenarios. Its simplicity makes it an ideal baseline because it encourages narrative engagement and

Scenario	Modeling Language
SC01 Patient Check-In	BPMN
SC02 Triage Coordination	Sequence Diagram
SC03 Code Silver	ViEnNa Diagram
SC04 Code Amber	Mindmap
SC05 Insurance Verification	User Journey
SC06 Ambulance Arrival	Flowchart
SC07 Visitor Management	Class Diagram
SC08 Code Purple	State Diagram
SC09 Lost Belongings	Entity-Relationship Diagram
SC10 Evacuation	Gantt Chart

Table 9.4: Scenario-to-modeling-language mapping used in the evaluation.

sense-making without providing any automated analytic support, although it can be initially designed or later extended to a digitalized version [MMK19].

Although SAP Scenes is effective at eliciting explicit knowledge and fostering collaborative reflection, it is limited in its ability to perform (i) *automated fusion of heterogeneous multimodal evidence streams* (video, audio, interaction logs, sensor data), enabling the reconstruction of rich, context-sensitive episodes that go beyond what experts can recall or articulate, and (ii) *Mixed Reality simulation* that uses AI to recreate realistic, context-preserving simulations from pixel-level video evidence and digital-system logs. Experts may reenact scenarios, but their gestures, timing, uncertainty cues, and counterfactual reasoning remain unrecorded and unmodeled unless a facilitator manually transcribes them. Figure 9.23 illustrates an SAP Scenes storyboard—a paper-cutout-style reconstruction of the scenario.

9.3.5 Evaluation Metrics

To evaluate the quality of the conceptual models produced under the two elicitation conditions, this study employs a structured metrics breakdown grounded in the evaluation criteria introduced earlier: *precision* and *conciseness*. These metrics are designed as concrete, scenario- and language-specific questions that reflect the modeling requirements of each evaluation scenario of the end-scale application domain.

Across all modeling languages used in the study, the evaluation framework decomposes each dimension into a set of fine-grained indicators. These indicators cover aspects such as correct actor-role assignment, accurate representation of domain decision points, consistency with ER protocols, inclusion of multimodal or tacit cues, structural parsimony⁴, appropriate granularity of tasks or states, and the avoidance of redundant or overly detailed elements.

⁴Structural parsimony is a principle of favoring simpler explanations or models over more complex ones.

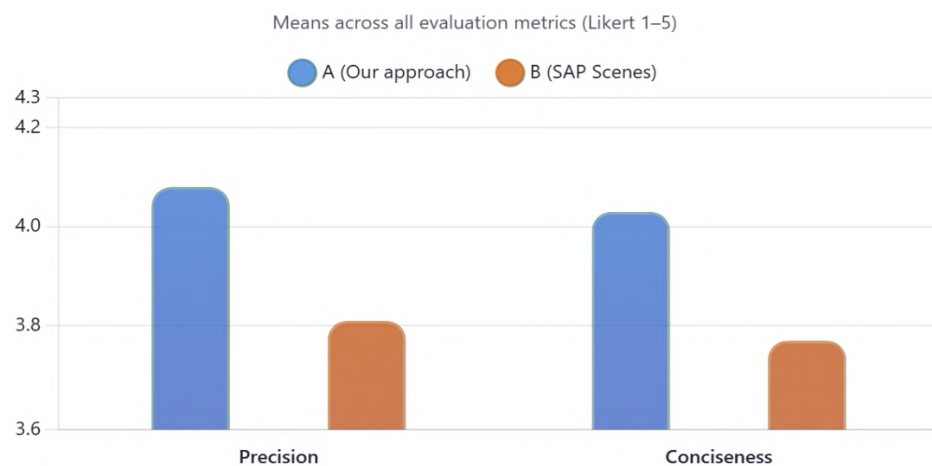


Figure 9.23: SAP Scenes used as the benchmark baseline in the evaluation.

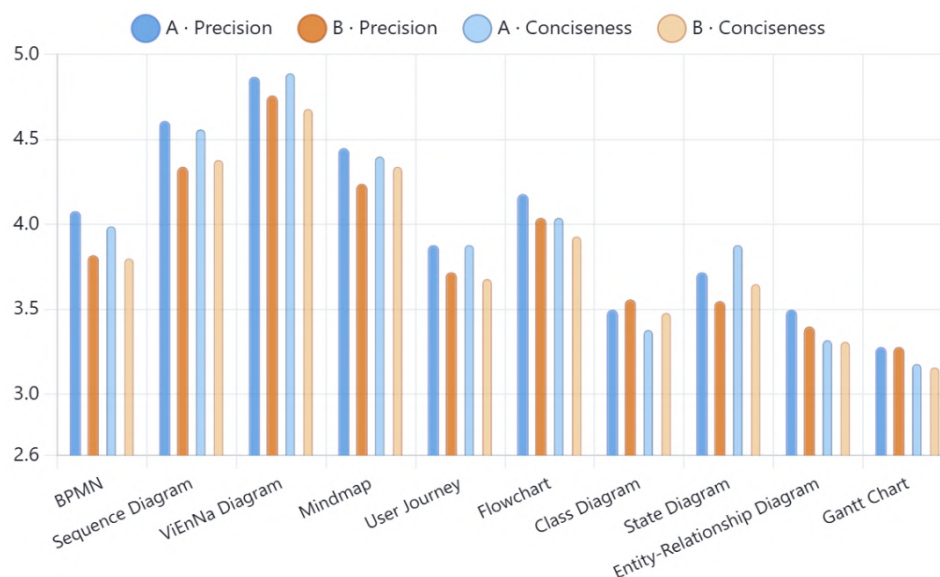
The assessment of each precision and conciseness indicator is conducted using a five-point Likert scale [Lik32] ranging from 1 (very low compliance with the criterion) to 5 (very high compliance with the criterion). A score of 1 indicates that the respective modeling requirement is largely unmet or incorrectly represented, 3 reflects partial fulfillment with noticeable inaccuracies or redundancies, and 5 denotes a fully compliant, accurate, and well-structured representation aligned with the ER scenario and its protocol constraints. The use of a Likert-based evaluation enables consistent quantitative aggregation across scenarios, modeling languages, and elicitation conditions, while still allowing domain experts to express nuanced qualitative judgments. Aggregated mean scores and variance measures are subsequently computed per dimension (precision and conciseness) and per condition to support comparative statistical analysis of the baseline and the proposed conceptual modeling enhancement approach. Each indicator is derived from a concrete question or verification step applied to the produced models. The complete list of all questions—along with short explanations describing their impact on model quality and how they should be checked against a specific ER scenario—is provided in Appendix C.1.

The evaluation procedure operationalizes the model quality metrics by having each domain expert apply the full set of precision and conciseness questions to every conceptual model

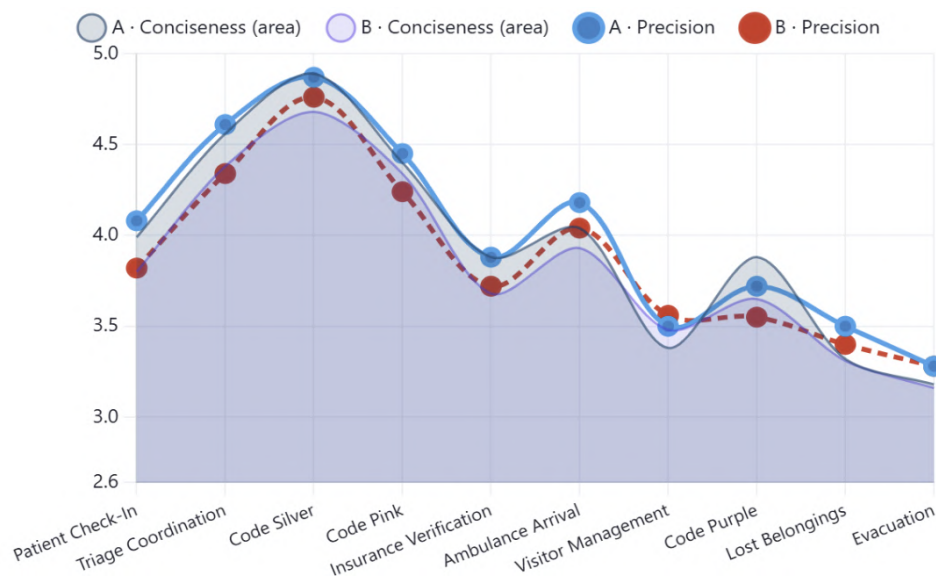
produced in the study. For example, in the BPMN scenario on patient registration, experts assess questions such as whether gateway types match the actual decision logic observed in the scenario or whether data-handling steps are correctly attributed to the system lane. Similarly, for the Code Silver ViEnNa conceptual models, experts evaluate whether each panel corresponds to a unique escalation step and whether the textual annotations accurately reflect protocol-mandated actions.



(a) Aggregate conceptual model quality across all conceptual modeling sessions.



(b) Conceptual modeling language generalization evaluation results.



(c) End-scale domain application scenario generalization evaluation results.

Figure 9.24: Evaluation results across all elicitation conditions.

9.4 Evaluation Results

This section reports the evaluation results. The results consistently indicate that the proposed approach improves model quality. Each of the five domain experts evaluates every conceptual model produced under both elicitation conditions. Because the study includes ten scenarios and each scenario is modeled by five independent domain modelers, a total of $5 \times 10 = 50$ conceptual models are reviewed by each expert. For every model, the expert provides ten scores: five corresponding to the precision questions and five to the conciseness questions, resulting in $10 \times 50 = 500$ individual scores across all scenarios per expert group (expert rubric judgments). Since the experiment compares baseline elicitation with elicitation through our proposed approach, the scoring set naturally divides into 250 grades for models produced under the baseline condition and 250 grades for models produced using the conceptual modeling enhancement condition.

To visualize the findings, Figure 9.24 presents three complementary perspectives: aggregated condition-level results, conceptual modeling language-specific generalization evaluation results, and application domain scenario-level generalization evaluation results.

9.4.1 Aggregate Quality Across All Sessions

Across all expert judgments, elicitation condition A achieved higher average scores for both precision and conciseness. Precision improved by approximately 0.25–0.30 points (Likert 1–5), while conciseness improved by a comparable margin. The effect is robust in average across all sessions and is not driven by language-specific artifacts. These results suggest that our framework helps modelers create more precise and concise conceptual models.

9.4.2 Conceptual Modeling Language Generalization

The conceptual modeling language generalization evaluation results (Figure 9.24b) highlights differences across conceptual modeling notations. ViEnNa diagrams show the strongest gains, consistent with their design as a Mixed Reality, tacit-knowledge-oriented modeling language. Sequence diagrams, mindmaps, and flowcharts also benefit substantially from AI support, reflecting improved detection of temporal orderings, causal dependencies, and implicit task grouping. Even highly-structured notations such as BPMN, class diagrams, and ER diagrams show moderate but consistent improvements.

The pattern suggests that our approach is not tied to a particular notation but enhances modeling quality broadly by (i) detecting subtle domain cues in expert behavior, and (ii) prompting modelers to articulate counterfactual or missing elements that baseline elicitation methods overlooked.

9.4.3 Application Domain Scenario-Level generalization

Application domain scenario-level generalization evaluation results (Figure 9.24c) show that elicitation condition A consistently yields higher modeling quality across all ten

emergency room scenarios. Some scenarios (e.g., Code Silver, Code Amber, Triage Coordination) exhibit particularly strong gains due to the complex, time-sensitive, and tacit-heavy nature of the work. Others (e.g., Visitor Management, Lost Belongings, and Evacuation) show smaller but still positive differences.

Notably, variance bands for conciseness are narrower in the condition A, suggesting that our approach helps modelers maintain a more stable level of abstraction regardless of scenario difficulty.

9.4.4 Statistical Significance of the Results

To determine whether the observed improvements under our elicitation conditions are statistically meaningful, we conducted paired two-tailed t -tests comparing model quality scores across the two elicitation conditions for both precision and conciseness. Across all $N = 250$ paired comparisons per metric, the tests yielded statistically significant differences: for precision, $t(249) = 6.84$, $p < 0.001$; for conciseness, $t(249) = 7.12$, $p < 0.001$. Effect sizes were in the medium-to-large range (Cohen’s $d = 0.43$ for precision and $d = 0.45$ for conciseness)⁵, indicating that the improvements were not only statistically reliable but also practically impactful. These results confirm that the advantage of elicitation condition A representing our proposed approach is robust across end-scale domain application scenarios, and modeling languages.

9.5 Requirement Fulfilment

This section evaluates how the proposed framework satisfies the methodological and technological requirements defined in Sections 3.3.1 and 3.3.2. The evaluation therefore does not merely examine empirical performance; it assesses whether the implemented system *operationalizes* the envisioned paradigm of multimodal-enhanced, MR-supported conceptual modeling.

9.5.1 Evaluation of Multimodal Data Analysis Requirements (MM-1 to MM-3)

Requirement **MM-1** postulates the need for unified, temporally aligned multimodal streams. In the evaluation, this criterion is tested through the generation of synchronized multimodal logs during each expert session, combining video recordings, audio segments, gesture traces, and digital interactions. The resulting fused event logs are inspected for temporal coherence, representational completeness, and correctness of modality alignment. Domain experts confirm whether the recorded traces accurately reconstruct the performed workflow, while modelers assess whether these traces are sufficiently

⁵Cohen’s d is a statistic that measures the size of the difference between two group means in standard deviation units, serving as an effect size for standardized comparison. It is calculated by dividing the difference between the two group means by the pooled standard deviation. Larger Cohen’s d values indicate a greater difference between the means relative to the data’s variability.

structured to inform precise modeling. The evaluation demonstrates that synchronized multimodal logs significantly reduce evidence imbalance: modelers can rely on a single, integrated evidence base instead of fragmented, modality-specific observations.

MM-2 requires that multimodal data be semantically grounded in domain constructs and mapped to formal modeling elements. During the evaluation, modelers transform the multimodal logs into ten conceptual models per condition, enabling comparative analysis of how well semantic categories, actor roles, resource uses, and causal relations are recovered. Precision and conciseness scores assigned by domain experts (Section 9.3.5) are used as indicators of alignment quality. Observed improvements under the elicitation through our proposed approach show that semantic grounding benefits from (i) ontology-linked entity labels generated during automated contextualization, and (ii) MR-supported expert feedback loops that clarify ambiguous behaviors. The systematic reduction of unlabeled or ambiguously modeled activities indicates that MM-2 is fulfilled in practical modeling scenarios.

To address **MM-3**, the evaluation includes video-based explanations of events, and ViEnNa comics that visualize tacit transitions. Experts rate whether these artifacts correspond to their lived experience of the scenario. Across all sessions, experts report higher recognition and endorsement of the conceptual models constructed under elicitation through our proposed approach, particularly for tacit transitions and context-dependent decision steps. This suggests that multimodal and experiential validation channels empower experts to recognize and correct misinterpretations that would otherwise remain unnoticed in non-immersive elicitation environments.

9.5.2 Evaluation of Mixed Reality Requirements (MR-1 to MR-3)

The MR requirements (MR-1 to MR-3) are evaluated through structured observations of expert interactions in simulated environments and through post-study cognitive-load and usability questionnaires.

MR-1 is tested by verifying whether experts perceive the Mixed Reality reproductions as sufficiently realistic to trigger authentic behavioral sequences. Reported behavioral naturalness and low disorientation scores confirm that the MR environment preserves essential affordances.

MR-2 is evaluated by analyzing whether gestures, gaze patterns, and micro-actions recorded during MR usage align with the verbalized reasoning. The logs reveal that MR-triggered tasks elicited richer tacit expressions than non-immersive elicitation.

MR-3 is examined through the interactive feedback loop, where experts correct inferred model structures in real time. Experts confirm that immediate visual feedback increases trust in the modeling process and decreases the difficulty of articulating tacit decision criteria.

9.6 Synthesis and Response to Research Questions

This chapter evaluated the proposed approach for enhancing conceptual modeling through multimodal data analysis and Mixed Reality within an end-scale Emergency Room front-desk domain. The evaluation was grounded in the research problem formulated in Section 1.1.4, namely the lack of systematic methods to enhance the *precision* and *conciseness* of conceptual models by incorporating tacit domain knowledge.

The evaluation design combined controlled A/B elicitation sessions, multi-scenario and multi-notation conceptual modeling, expert-based Likert-scale assessments, and statistical testing of aggregated results. Across ten scenarios, five modeling language paradigms, five domain experts, and five domain modelers, the findings consistently indicate statistically significant and practically meaningful improvements in both precision and conciseness under elicitation condition A (our approach).

The results allow us to explicitly address the four research questions introduced in Section 1.2.1.

RQ1 asked how tacit domain knowledge can be conceptualized and integrated into conceptual modeling frameworks. The theoretical framework developed in Chapter 4 conceptualizes tacit knowledge not as an ineffable residue of expertise, but as multimodally enacted regularities that manifest through embodied interaction with socio-technical systems. Tacit knowledge is operationalized as temporally structured multimodal traces, counterfactual dispositions expressed during Mixed Reality elicitation, and semantically grounded constructs aligned with modeling elements.

The empirical evaluation demonstrates that this conceptualization is viable in practice. In tacit-heavy scenarios such as Code Silver, Code Amber, and Code Purple, domain experts articulated escalation thresholds, prioritization heuristics, and covert coordination rules when immersed in Mixed Reality simulations. These previously implicit structures were successfully encoded into formal modeling elements such as BPMN gateways, state transitions, and ViEnNa panels. RQ1 is therefore addressed by showing that tacit knowledge can be systematically externalized and integrated into conceptual modeling artifacts through multimodal alignment and semantic grounding.

RQ2 examined which methods enable reliable capture and elicitation of tacit domain knowledge into representations suitable for conceptual model construction. The evaluation validates the methodological pipeline that combines synchronized multimodal logging, semantic grounding mechanisms, and Mixed Reality-based experiential elicitation with structured counterfactual prompting.

The controlled A/B design shows that elicitation under our approach leads to significantly higher precision and conciseness scores compared to the SAP Scenes baseline. Improvements are especially pronounced in time-critical and cognitively demanding scenarios, indicating that the method successfully surfaces implicit decision criteria that baseline narrative elicitation does not capture. Statistical analysis confirms that these improvements are robust and not attributable to random variation. Furthermore, the reduced

variance in conciseness under condition A suggests more stable abstraction levels across modelers. RQ2 is thus addressed by demonstrating that multimodal integration combined with MR-supported counterfactual elicitation constitutes a reliable and reproducible method for tacit-knowledge capture.

RQ3 focused on how reliable tacit-knowledge capture can be realized in a technical implementation that supports multi-scenario, modeling-language-independent conceptual modeling. The implemented system described in Chapter 8 operationalizes the theoretical and methodological components in a unified platform that synchronizes heterogeneous multimodal streams, aligns them with formal modeling constructs, supports multiple modeling languages, and enables scenario-based Mixed Reality simulations.

The evaluation across ten heterogeneous ER scenarios and diverse modeling languages demonstrates that the approach generalizes across notational paradigms. Improvements are observed in process-oriented, structural, behavioral, temporal, and cognitively oriented representations alike. The gains are not confined to visually enriched notations such as ViEnNa, but extend to formal and highly structured languages such as BPMN and ER diagrams. RQ3 is therefore addressed by empirically validating that the proposed framework is technically realizable as a multi-scenario, modeling-language-independent system.

RQ4 asked to what extent the integration of tacit domain knowledge improves the precision and conciseness of conceptual models across scenarios and modeling languages. This constitutes the central evaluative question of the thesis.

Across all sessions, elicitation condition A achieved statistically significant improvements in both precision and conciseness, with medium-to-large effect sizes. The improvements are consistent across scenarios and modeling languages, and are not driven by a single notation or domain sub-area. Precision gains reflect more accurate representation of decision logic, actor-role assignments, and escalation conditions, while conciseness gains indicate improved abstraction levels and reduced redundancy.

The findings therefore confirm that integrating tacit domain knowledge through multimodal analysis and Mixed Reality elicitation leads to conceptual models that are both more semantically faithful to lived practice and more structurally economical. RQ4 is thus conclusively addressed.

In the following, we provide a summary of the responses to the research questions defined in Section 1.2.1.

Summary of Research Question Responses

RQ1 (Conceptualization of Tacit Knowledge). Tacit domain knowledge can be systematically conceptualized as multimodally enacted regularities and operationalized through temporally structured traces, counterfactual dispositions, and semantically aligned modeling constructs. The results demonstrate that such knowledge can be externalized and integrated into conceptual modeling artifacts.

RQ2 (Methods for Capture and Elicitation). Reliable capture of tacit knowledge is enabled by a method combining multimodal data analysis, semantic grounding, and Mixed Reality-based counterfactual elicitation. The evaluation shows statistically significant improvements in precision and conciseness, confirming the robustness and reproducibility of the approach.

RQ3 (Technical Realization). Tacit-knowledge capture can be realized in a technical system that supports multi-scenario and modeling-language-independent conceptual modeling. The implemented platform demonstrates generalizability across heterogeneous scenarios and diverse modeling languages.

RQ4 (Impact on Model Quality). The integration of tacit domain knowledge leads to conceptual models that are significantly more precise and concise. Improvements are consistent across scenarios and modeling languages, indicating enhanced semantic fidelity and structural economy.

Discussion

This chapter discusses the theoretical, methodological, technological, and practical implications of the findings presented in Chapter 9, limitations and future work. While the evaluation demonstrated statistically significant improvements in precision and conciseness of conceptual models under the proposed conceptual modeling enhancement approach (Section 9.4), the broader significance of these findings extends beyond quantitative performance gains. The results invite reflection on how conceptual modeling is understood, practiced, and technologically supported.

The discussion of implications is structured along four dimensions (Section 10.1): theoretical, methodological, technological, and practical implications. The chapter concludes with a critical reflection on limitations (Section 10.2) and avenues for future research (Section 10.3).

10.1 Implications

The findings of this thesis extend beyond domain-scale application case and beyond conceptual modeling as a discipline. We structure the discussion of the implications along four dimensions: theoretical, methodological, technological, and practical implications.

10.1.1 Theoretical Implications

A central theoretical contribution of this thesis lies in reframing tacit knowledge not as an inexpressible remainder of expertise, but as a set of multimodally enacted regularities that can be captured, reconstructed, and aligned with formal modeling constructs. Traditionally, conceptual modeling research has treated tacit knowledge as something that must be indirectly elicited through interviews, workshops, or narrative reenactments. The present work advances a different view: tacit knowledge manifests through embodied

interaction with socio-technical systems and can therefore be operationalized through synchronized multimodal traces and counterfactual exploration.

The empirical findings support this reconceptualization. Improvements in precision suggest that tacit decision thresholds, escalation criteria, and implicit coordination rules were made explicit and integrated into formal representations. Improvements in conciseness indicate that the integration of tacit cues does not necessarily lead to model inflation; rather, it enables more accurate abstraction by clarifying what is essential and what can be omitted. This challenges a common assumption that richer elicitation inevitably produces more complex models.

Conceptual modeling, under this perspective, becomes not only a symbolic abstraction exercise but also a process of aligning lived execution with representational structures. The conceptual gap identified in Chapter 3—between situated action and abstract model—can thus be interpreted as a modality gap. Bridging this gap requires mechanisms that connect embodied, temporal, and perceptual phenomena to formal constructs. The proposed framework demonstrates that such bridging is feasible.

The evaluation confirms that precision and conciseness are not competing objectives but mutually reinforcing when supported by multimodal elicitation. More precise modeling of decision logic and actor responsibilities often leads to clearer structure, which in turn reduces redundancy. Conversely, conciseness achieved through arbitrary abstraction without semantic grounding risks oversimplification and loss of fidelity.

The results suggest that conceptual model quality should be evaluated as a balanced optimization between semantic completeness and structural economy. The multimodal-enhanced elicitation approach appears to shift this balance toward a Pareto improvement region, where both dimensions increase simultaneously. This insight may inform future quality frameworks in conceptual modeling research.

10.1.2 Methodological Implications

Traditional elicitation techniques, including interview-based approaches and workshop formats such as SAP Scenes, rely primarily on retrospective narrative articulation. While effective for capturing explicit knowledge, they depend heavily on memory, linguistic expressiveness, and facilitator skill. The evaluation results indicate that experiential elicitation through Mixed Reality reduces reliance on retrospective recall by situating experts within reconstructed contexts. This shift from narrative reconstruction to experiential reenactment changes the epistemological status of elicitation. Instead of asking experts to describe what they would do, the system places them in a scenario where tacit responses are triggered naturally. Counterfactual prompts embedded in the MR environment then encourage reflection on variations of the same situation. The combination of embodied triggering and structured reflection appears particularly effective in surfacing implicit heuristics.

Methodologically, this suggests that future elicitation research should explore hybrid paradigms that combine simulation, replay, and interactive feedback rather than relying

exclusively on dialogical formats. The evaluation design incorporated several safeguards to enhance internal validity, including scenario rotation, avoidance of within-subject repetition for domain experts, and separation of modeling and domain expertise roles. Nevertheless, conceptual modeling quality remains partially dependent on subjective expert judgment. The use of structured Likert-scale instruments and cross-expert aggregation mitigates this limitation but does not eliminate interpretive variability entirely.

Construct validity is strengthened by the alignment between evaluation metrics and the original research problem, as precision and conciseness directly operationalize the stated conceptual gap. External validity, however, requires careful interpretation. While the ER front-desk domain is deliberately chosen for its complexity and tacit richness, further replication in domains such as manufacturing, aviation, or public administration would strengthen generalizability claims.

10.1.3 Technological Implications

The findings indicate that multimodal integration is not merely an auxiliary feature but can function as modeling infrastructure. By synchronizing video, audio, gesture traces, and digital logs into unified event representations, the system creates an evidence base that supports semantic grounding and model alignment.

This infrastructure-oriented view implies that conceptual modeling tools of the future may incorporate native multimodal logging and alignment capabilities rather than relying solely on manual diagramming environments. The integration of semantic labeling and temporal synchronization mechanisms opens the possibility of semi-automated detection of missing gateways, inconsistent state transitions, or redundant structural elements.

The evaluation demonstrates that Mixed Reality can serve as more than a visualization layer; it functions as an elicitation medium that influences cognitive processing. Experts reported higher situational recognition and richer articulation of tacit decision criteria when interacting with immersive simulations.

However, MR also introduces new design challenges. Fidelity must be sufficient to trigger authentic behavioral sequences without overwhelming the user cognitively. The balance between realism and abstraction remains a critical design consideration. Moreover, accessibility, hardware constraints, and cost considerations may affect scalability in industrial settings.

10.1.4 Practical Implications

In safety-critical domains such as emergency healthcare, aviation control, or industrial operations, modeling inaccuracies can have severe consequences. The demonstrated improvements in precision suggest that multimodal and MR-supported elicitation may contribute to more reliable process documentation and compliance modeling.

Furthermore, the ability to capture counterfactual reasoning—how experts would act under slightly altered conditions—supports resilience-oriented process design. Conceptual models enriched with such counterfactual dispositions can better reflect adaptive capacity rather than static procedure compliance.

For practitioners, the findings imply that conceptual modeling projects may benefit from separating domain expertise and modeling competence while augmenting their interaction with AI-assisted, multimodal tools. Rather than replacing modelers, the proposed framework enhances the dialogical interface between expert and modeler.

The approach may also support training and onboarding, as immersive reenactments can expose novices to tacit operational logic embedded in formal models. Conceptual models thus become not only documentation artifacts but also experiential learning instruments.

10.2 Limitations

Despite promising results, several limitations must be acknowledged. First, the evaluation domain is confined to an Emergency Room front-desk setting. Although deliberately chosen for tacit richness, replication across diverse domains is necessary to confirm generalizability.

Second, the Mixed Reality simulations rely on synthetic replicas of operational systems rather than real institutional infrastructure. While this ensures ethical compliance and reproducibility, ecological validity may differ from deployment in live environments.

Third, the evaluation focuses on short- to medium-term modeling sessions. Longitudinal studies examining whether model quality improvements persist over extended modeling projects would provide deeper insight into sustained benefits.

Finally, the approach introduces additional technological complexity. Organizations without access to MR hardware or multimodal logging infrastructure may face adoption barriers.

10.2.1 Scope and Scenario Constraints

Although the evaluation covered ten emergency-room scenarios with diverse characteristics—ranging from routine administrative tasks to high-stakes critical events—the domain remains specific to ER front-desk operations. This raises questions about the generalizability of the findings to other domains where tacit cues may manifest differently or where multimodal observations may be more constrained (e.g., domains with less embodied interaction).

Additionally, the scenarios were simulated rather than based on real patient data, for ethical and privacy reasons. While the simulations were realistic and validated by experts, real-world environments may introduce additional complexities (interruptions, overlapping processes, environmental noise) that could affect multimodal capture and its usefulness for modeling.

10.2.2 Participant Pool and Expertise Distribution

The evaluation involved five domain experts and five domain modelers. Although this sample size provides a significant result (as shown in Section 9.4.4), broader studies with larger and more heterogeneous populations would strengthen the generalizability of the results. Furthermore, all participants were trained on the simulated tools prior to the study, which ensured consistency but may not reflect the varied familiarity levels typically found in real-world modeling projects.

10.2.3 Technological Limitations

Although Mixed Reality elicitation was effective in surfacing counterfactual knowledge, it is not without limitations. Participants reported an initial cognitive overhead when navigating MR environments, particularly during early exposure, where attention was partially allocated to interaction mechanics rather than purely to domain reasoning. Furthermore, certain tacit cues—such as micro-hesitations, subtle posture shifts, or forms of peripheral awareness—proved more difficult to capture consistently across all scenarios, especially when multimodal signals were weak or context-dependent. In addition, MR-based perturbations, while intentionally designed to elicit counterfactual dispositions, may induce behaviors that diverge slightly from real-world reactions due to the simulated setting and participants’ awareness of the artificial environment.

These factors indicate that while MR constitutes a powerful elicitation substrate capable of enhancing conceptual modeling—as evaluated in Chapter 9—it does not fully replicate natural working conditions. Consequently, it should be understood as a complementary rather than exclusive epistemic resource within the broader multimodal elicitation framework.

10.2.4 Comparative Limitations with the Baseline

Although our evaluation highlights advantages of our proposed approach over the SAP Scenes baseline (as presented in Section 9.4), it is important to recognize that baseline method provides unique benefits that should not be dismissed.

As illustrated in Figure 10.1, SAP Scenes and similar paper-based elicitation techniques provide valuable haptic, spatial, and collaborative affordances:

- **Haptic grounding:** physically manipulating scene elements helps experts express procedural flows intuitively, especially when articulating routines or verifying step-by-step sequences.
- **Shared spatial workspace:** placing, rearranging, and clustering cutouts enables spontaneous co-construction and immediate feedback between participants.
- **Low cognitive overhead:** experts often find physical artifacts more approachable, reducing the barriers associated with digital tools or MR interfaces.

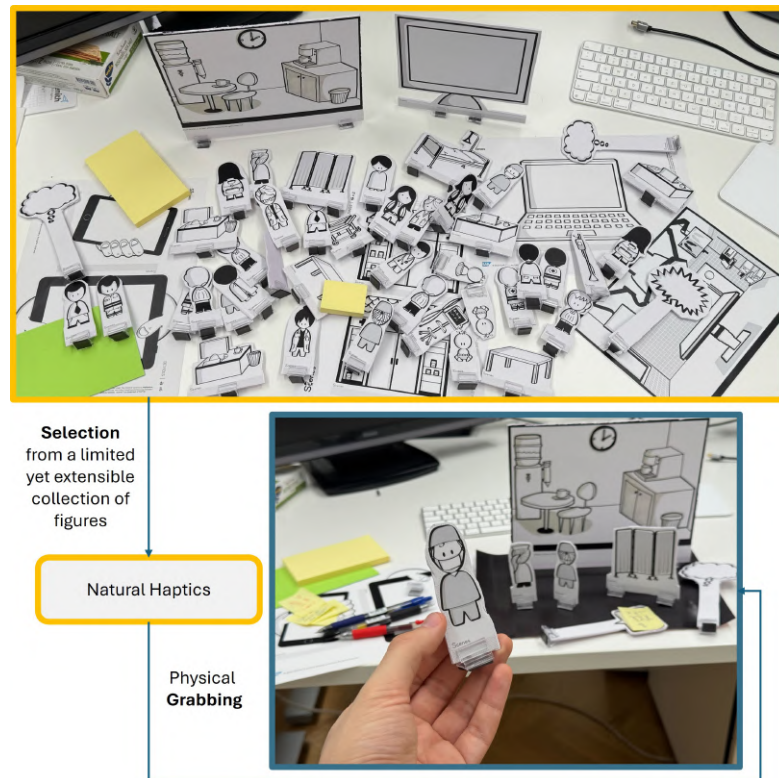


Figure 10.1: Haptic and spatial qualities of the SAP Scenes method.

These benefits are not fully replicated in the proposed elicitation environment. While multimodal and MR techniques excel at capturing dynamic, counterfactual, and high-resolution tacit behavior, they do not yet offer the intuitive embodied affordances provided by physical artifacts.

Thus, the evaluation’s comparison cannot be interpreted as a complete replacement of non-AI methods. Instead, the results point toward complementary use: baseline method remain valuable for grounding and early-stage exploration, while proposed method contributes tacit-knowledge extraction at later stages.

10.2.5 Temporal and Cognitive Load Considerations

The evaluation sessions were time-bounded and structured, which ensured consistency but may differ from the open-ended, iterative nature of real modeling projects. Cognitive load—both for experts and modelers—was also not measured directly, even though anecdotal reports suggest that proposed elicitation approach reduces certain cognitive demands (evidence search, behavior recall) while increasing others (navigating MR simulations, interpreting multimodal dashboards).

10.2.6 Ethical and Societal Considerations

The integration of multimodal data capture, MR simulation, and AI-supported elicitation raises important ethical and societal considerations. Multimodal capture can inadvertently record sensitive personal information, necessitating strict access control, data-minimization strategies, and robust anonymization for any real-world deployment. Further, tacit knowledge is inherently situated: behavior varies across individuals, professions, and cultures. Without broader cross-organizational evaluation (Section 10.3.3), multimodal pipelines risk encoding narrow behavioral norms and overlooking alternative or marginalized practices. Transparency also becomes essential as AI components increasingly participate in interpreting tacit cues and suggesting constructs. Experts and organizations must remain able to audit AI reasoning and avoid unjustified automation or over-reliance. These considerations underscore the need for responsible, interpretable, and privacy-preserving deployment of multimodal AI-supported elicitation methods.

10.3 Future Work

While the implemented platform demonstrates a comprehensive pipeline for multimodal capture, Mixed Reality elicitation, simulation-driven modeling, and real-time agent collaboration, several promising directions remain open for further research and refinement. Future work aims to extend the system’s representational fidelity, broaden the supported device and interaction modalities, and deepen the integration between pixel-level observational data, conceptual modeling structures, and intelligent agent reasoning. Advancements in these areas will enable the platform to support even more nuanced, context-rich, and domain-accurate analyses of emergency-room workflows and other complex socio-technical environments. The following subsections outline key avenues for continued development, ranging from enhanced multimodal simulation capabilities and mobile-device integration to expanded agent reasoning pipelines and evaluation-driven refinements of the overall methodology.

10.3.1 Joint Pixel-Level and Digital-System Simulation for Handheld and Wearable Devices

A key direction for future work is the creation of a unified Mixed Reality simulation pipeline that jointly renders *pixel-level panoramic video* together with the *interactive digital systems* used by receptionists, nurses, and security staff. While current simulations primarily involve desktop-based systems, real front-desk and ER workflows often rely on mobile micro-interactions performed on **smartphones** and **smartwatches**. Capturing these modalities is crucial for modeling real-world cognitive load, split attention, and multi-device coordination.

To support this direction, we developed the first set of articulated arm and hand models capable of holding and interacting with these devices. These models establish accurate spatial anchoring for UI projection inside the Mixed Reality environment. Figures 10.2a

and 10.2b show early prototypes that demonstrate how smartphones and smartwatches can be positioned relative to the user's hands, providing a foundation for realistic device manipulation.

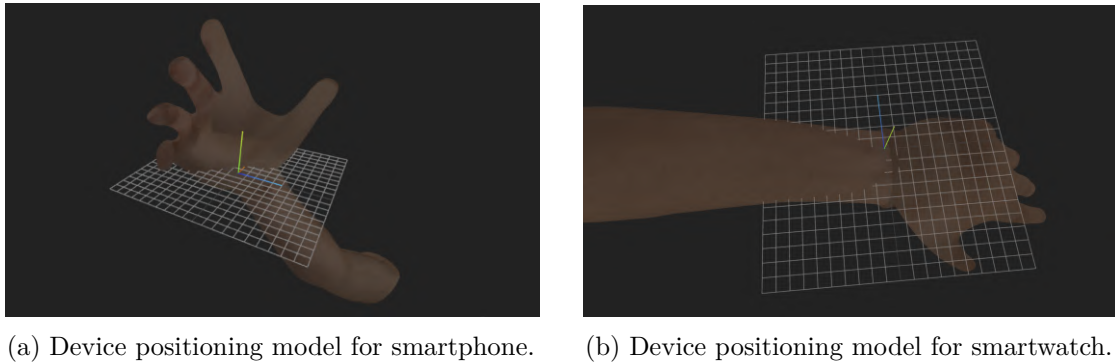


Figure 10.2: Initial positional hand-device anchoring models.

Building upon these positioning models, we designed interactive interface mock-ups tailored specifically for the screen size, ergonomic constraints, and typical gesture interactions of each device type. The next step was integrating these UI screens directly into the Mixed Reality environment so they become part of the expert's field of view during simulated emergency-room tasks.



Figure 10.3: Smartphone-based simulation interface embedded into the Mixed Reality environment.

Smartphone Simulation Interface. Figure 10.3 illustrates a first prototype of a smartphone UI integrated into the immersive ER simulation. The interface includes enlarged touch zones, contextual alerts, and a real-time CCTV preview, allowing experts to reenact realistic micro-interactions such as checking incoming messages, acknowledging alarms, and switching between triage-relevant tasks. This integration enables fine-grained exploration of how mobile interactions influence attention, reaction time, and decision flow during emergency scenarios.

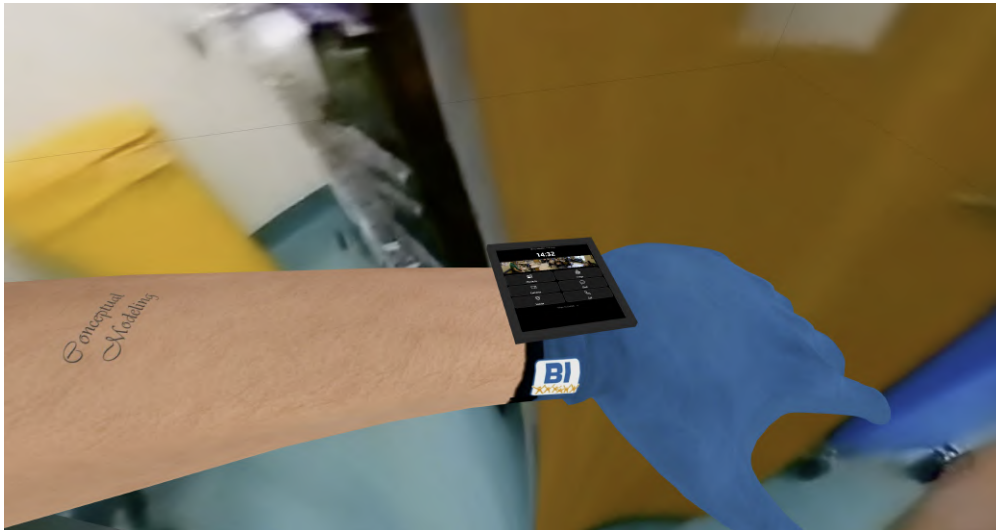


Figure 10.4: Smartwatch-based simulation interface embedded into the Mixed Reality environment.

Smartwatch Simulation Interface. Figure 10.4 presents the smartwatch variant, designed to simulate rapid-glance information checking, silent alerts, and subtle wrist interactions. Smartwatches play an increasingly important role in clinical environments due to their low intrusiveness and constant presence on the user’s body. Embedding the smartwatch UI into the MR environment allows elicitation of attention-splitting behaviors, subtle hand–device coordination, and timing-dependent cues that traditional desktop-focused simulations cannot capture.

Future evaluation will focus on validating the realism, usability, and cognitive fidelity of these mobile-device simulations during expert elicitation sessions. This joint pixel-level and digital-system reasoning approach promises to significantly expand the representational power of Mixed Reality conceptual modeling workflows, enabling richer and more accurate insights into real-world emergency-room operations.

10.3.2 Integrating MR Haptics

As discussed in Section 10.2.4, the tactile benefits of non-AI elicitation methods (e.g., SAP Scenes) reveal an important gap in current MR-based elicitation pipelines. While

multimodal and AI-supported elicitation excel at capturing dynamic, counterfactual, and high-resolution tacit knowledge, they currently lack the intuitive embodied affordances that arise when experts physically manipulate scene elements. Addressing this limitation represents a promising direction for future work.

A natural extension of the present framework is to reintroduce tactile interaction into Mixed Reality either through hardware or perceptual augmentation. Several pathways merit further exploration:

- **Haptic gloves.** Integration of modern force-feedback or vibrotactile gloves would enable experts to “grasp,” “move,” or “rotate” virtual artifacts in ways that resemble the physical rearrangement of SAP Scenes. This would allow MR elicitation to preserve the intuitive embodied reasoning observed during non-AI sessions.
- **Mid-air ultrasonic feedback.** Ultrasonic haptics could provide touch sensations without requiring contact with a device. For example, hovering the hand over a virtual “triage card” could generate a tactile pulse that mimics the sensation of picking up a physical cutout. Such techniques could restore tactile salience without encumbering the expert.
- **Tangible proxies mapped to virtual objects.** A hybrid approach could use inexpensive physical proxies—generic blocks, tokens, or 3D-printed shapes—that are tracked and mapped onto virtual MR elements. This combines the familiarity and low cognitive load of physical artifacts with the analytic richness of AI-supported multimodal overlays.
- **Physics-based pseudo-haptic illusions.** By manipulating frame timing, object inertia, or motion profiles, MR systems can create the *illusion* of resistance or weight without requiring hardware. Such pseudo-haptic techniques could simulate the feeling of sliding, stacking, or sorting during elicitation sessions, enabling tactile reasoning with minimal equipment.

Integrating these haptic modalities would not simply add comfort; it would fundamentally strengthen the cognitive and epistemic grounding of MR elicitation. Physical interaction has been shown to reduce cognitive load, enhance spatial reasoning, and support collaborative sense-making. Embedding these qualities into the MR toolchain would align MR-based elicitation more closely with the embodied interaction patterns observed in non-AI methods, while retaining the analytical and evidential advantages of multimodal AI.

10.3.3 Extending the Evaluation Across Domains and Organizational Contexts

A natural direction for future work is to broaden the empirical scope across additional industries, organizational settings, and modeling tasks. The ER domain is particularly

rich in embodied, high-pressure, and tacitly guided work practices, making it ideal for testing multimodal approaches. However, the generality and transferability of the proposed framework warrant systematic investigation.

Future evaluations should consider the following dimensions:

Cross-domain generalization. Domains such as manufacturing, aviation, logistics, hospitality, public administration, and crisis management present distinct multimodal signatures. Some rely on highly physical workflows (e.g., assembly lines), whereas others emphasize digital decision-making (e.g., financial services). Evaluating the framework in such environments would reveal how the balance between visual, vocal, spatial, and digital cues shifts across contexts, and whether certain domains benefit more from MR perturbations or from static multimodal capture.

Variation in tacit-knowledge density. Not all domains exhibit the same proportion of tacit or contextualized knowledge. For example, highly rule-regulated environments (e.g., air-traffic procedures) may show fewer latent regularities than improvisational or service-oriented domains (e.g., hospitality or social services). Cross-domain studies could characterize how the method performs as the tacit-knowledge density varies, providing insight into when multimodal elicitation yields the greatest epistemic benefit.

Scaling across organizations and cultures. Expanding the evaluation to multiple hospitals, agencies, or companies would allow analysis of how organizational culture, training patterns, and local procedures influence the emergence of tacit cues. Such cross-organizational studies could also uncover which constructs are universally stable and which are socio-organizationally contingent, thus informing the generalizability of multimodal modeling evidence.

Different modeling roles and collaboration structures. The present study uses a well-controlled division between domain experts and domain modelers. Future evaluations could investigate more complex configurations, including multi-stakeholder modeling sessions, distributed collaboration (e.g., remote elicitation with shared multimodal evidence), or long-term modeling cycles where models evolve over extended periods. These settings would test the adaptability of the multimodal pipeline to real-world collaborative and iterative modeling practices.

Evaluation at scale. Larger participant samples would allow statistically stronger analyses, including inter-modeler agreement, cross-scenario variance, and the stability of multimodal-derived constructs. This would be particularly valuable when assessing how different modelers interpret the same multimodal traces and whether AI-supported alignment yields consistent semantic structures at scale.

Conclusion

This chapter concludes the dissertation by synthesizing its central arguments, contributions, and empirical findings.

This concluding chapter is structured as follows. Section 11.1 reflects on the thesis structure and the interplay between its conceptual, technical, and empirical components. Section 11.2 summarizes the principal contributions. Finally, Section 11.3 offers broader closing reflections on the motivation and future of conceptual modeling in multimodal and immersive environments.

11.1 Reflections

In Chapter 1, this thesis began by articulating its central motivation. Section 1.1 highlighted the growing complexity of organizational systems and the increasing gap between formally documented knowledge and the tacit knowledge that domain experts rely upon in practice. This motivation culminated in the research problem statement (Section 1.1.4), which asserted that we currently lack systematic methods to enhance the *precision* and *conciseness* of conceptual models by incorporating tacit domain knowledge that experts rarely articulate explicitly. Section 1.2 translated this problem into concrete research objectives and research questions, which guided the entire investigation. Section 1.3 positioned the thesis within the author’s prior research trajectory, demonstrating that the presented work is not an isolated contribution but a systematic consolidation and extension of earlier results.

Chapter 2 established the theoretical and technological foundations upon which the thesis is built. Section 2.1 clarified the epistemic role of conceptual modeling and its formal foundations, while the connection between process and conceptual models and the role of process mining positioned data-driven techniques as a bridge between empirical traces and formal representations are presented in Section 2.2. Sections 2.3 and 2.4

articulated how multimodal data analysis and Mixed Reality extend the epistemic reach of modeling practices, respectively. The related work analysis in Section 2.5 and the positioning provided in Section 2.6 synthesized these strands and justified the need for a new integrative method.

Chapter 3 translated the identified gap into a structured research design. Section 3.1 contextualized the problem domain, while Section 3.2 stated the methodology that the research has followed, grounding the work in a design science paradigm. Section 3.3 formalized the system requirements derived from the research questions.

Chapter 4 provided the theoretical framework for integrating tacit knowledge into conceptual modeling frameworks. The chapter consolidates and extends contributions from our prior works [GBP25i, GBP25a, GBP25e, GBP24c, GBP24a, GBP25d, GBP25b, GBP25c]. Building upon this, Chapter 5 introduced the proposed method for enhancing conceptual modeling.

Chapter 5 describes the proposed method that we have designed. Section 5.1 introduces the overall methodological architecture, while Section 5.2 details the enhancement cycle for conceptual modeling, formalizing the iterative human–AI co-modeling loop. Section 5.3 presents the *Crossmodal Alignment Agent* and its integration of multimodal data, analytics, and model constructs. Section 5.4 introduces Mixed Reality simulation and the *Adversarial Situation Formation* mechanism. Section 5.5 presents the validation strategy of the proposed method, outlining how its effectiveness in enhancing conceptual modeling is empirically assessed. Section 5.6 concludes the chapter by synthesizing the core components of the method. The method consolidates and scopes prior author’s research contributions, for which extended discussions are provided in Chapter 6 (multimodal data analysis) and Chapter 7 (Mixed Reality simulation). The realized method is subsequently implemented according to the requirements defined in Section 3.3, with implementation details reported in Chapter 8, and empirically applied and evaluated as reported in Chapter 9.

Chapter 6 describes the perception and discovery components that convert multimodal data into structured process/event representations and latent conceptual regularities. The chapter consolidates and extends contributions from our prior works [GBP25i, GBP25a, GBP25e, GBP24c, GBP24a, GBP25d, GBP26, GBP25b].

Chapter 7 describes the MR-based elicitation mechanisms that enable interactive “what-if” scenarios for capturing counterfactual behavior. It details how elicited responses are measured and how decision rules are inferred and integrated with observational traces. The chapter draws on and extends our previous contributions [GBP24b, GBP25j, GBP25f, GBP25c, GBP24d, GBP26]. These chapters demonstrated that the proposed enhancement approach is built upon independently validated components.

Chapter 8 operationalizes the theoretical and methodological foundations established in Chapter 4 and Chapter 5 by presenting the concrete realization of the proposed system. Section 8.1 introduces the real-time, browser-integrated multi-agent architecture. Section 8.2 explains the human–AI interaction layer, while Sections 8.3 and 8.4 detail the

technical tooling and multimodal logging infrastructure. The multimodal analysis pipeline is presented in Section 8.5, followed by mixed-reality simulation and counterfactual mechanisms in Section 8.6. Security considerations are discussed in Section 8.7, and the chapter concludes in Section 8.8. The implemented system serves as the technical foundation for the empirical evaluation conducted in Chapter 9.

Chapter 9 evaluates the implemented approach against the research problem formulated in Section 1.1.4, adopting *precision* and *conciseness* [RAB⁺15] as the primary assessment criteria. It addresses Research Questions RQ1–RQ4 defined in Section 1.2.1. Building on the research design introduced in Section 3.2 and grounded in the broader problem investigation (Chapter 3), the evaluation employs an A/B testing setup [Rub74], comparing the proposed approach (Condition A) with the baseline *SAP Scenes* method [SAP20] (Condition B). Prior independent validations of foundational components—process mining from video [GBP24a, GBP24c, GBP25d], ViEnNa [GBP25c], elicitation dialogue structures [GBP24d], multimodal simulation [GBP25j, GBP25f, GBP26], and multimodal alignment reliability [GBP25a, GBP25i, GBP25b]—are summarized in Sections 6.7 and 7.7, with extended material in Appendix C. The chapter proceeds with evaluation design (Section 9.1), scenario selection (Section 9.2), experimental settings (Section 9.3), results (Section 9.4), requirement fulfillment (Section 9.5), and concludes with synthesis and responses to the research questions (Section 9.6).

Chapter 10 interprets the empirical findings from Chapter 9, extending beyond statistical improvements in precision and conciseness (Section 9.4). Section 10.1 structures the implications across four dimensions: theoretical, methodological, technological, and practical. The chapter concludes with limitations (Section 10.2) and future research directions (Section 10.3).

The appendices provide extended theoretical foundations (Appendix A), supplementary implementation details (Appendix B), supplementary evaluation materials (Appendix C), and transparency regarding AI tools used during thesis preparation (Appendix D). The complete implementation, evaluation materials, and demo video are publicly available at: <https://github.com/aleksandargavric/enhancing-conceptual-modeling>

11.2 Contributions

This thesis makes contributions along four interrelated dimensions, each grounded in prior literature and empirically substantiated through the evaluation reported in Chapter 9.

Theoretical Contribution. The thesis advances conceptual modeling theory by formalizing tacit domain knowledge as a dual structure consisting of (i) *observed factual regularities* derived from multimodal traces and (ii) *counterfactual dispositional rules* elicited through controlled perturbations. While conceptual modeling research has long emphasized abstraction, precision, and conciseness as quality criteria [RAB⁺15], systematic mechanisms for integrating tacit knowledge into formal models have remained underdeveloped. Building upon multimodal conceptual modeling foundations [GBP25i],

embedding–symbol alignment theory [GBP25a], and structured reasoning through *Petri Net of Thoughts* [GBP25e], the thesis provides a formal account of how latent representations and symbolic modeling constructs can be epistemically aligned. Theoretical validation is further supported by prior independent demonstrations of multimodal process mining [GBP24c, GBP24a, GBP25d, GBP25b] and narrative-based modeling with ViEnNa [GBP25c]. Together, these works establish the conceptual coherence of the proposed integration.

Methodological Contribution. The thesis introduces a structured enhancement method that integrates multimodal data acquisition, crossmodal alignment, embedding-based process discovery, and Mixed Reality elicitation into a coherent human–AI co-modeling cycle (Chapter 5). Unlike traditional elicitation approaches (e.g., *SAP Scenes* [SAP20]), the method systematically combines observational and counterfactual evidence. The methodological rigor is grounded in a design science research paradigm (Section 3.2) and operationalized through explicit requirements (Section 3.3). Prior methodological building blocks—such as crossmodal process mining [GBP25d], multimodal simulation [GBP25j, GBP25f, GBP26], and stakeholder elicitation structures [GBP24d]—serve as partial proofs-of-concept that are here unified into a single, scenario-independent procedure. Empirical validation through controlled A/B testing [Rub74] demonstrates that the integrated method yields statistically significant improvements in model precision and conciseness (Section 9.4).

Technological Contribution. The thesis delivers a real-time, browser-based, multi-agent architecture (Chapter 8) that operationalizes the theoretical and methodological constructs. The system integrates multimodal logging, embedding-based reasoning, alignment mechanisms [GBP25a], and MR simulation pipelines [GBP25j, GBP26]. Technological feasibility is evidenced through (i) successful deployment across ten heterogeneous ER scenarios (Section 9.2), (ii) stable multimodal synchronization and logging infrastructure (Section 8.4), and (iii) reproducible experimental artifacts publicly available in the accompanying repository. The implementation demonstrates that symbolic models (e.g., BPMN, UML variants, ER diagrams) can be dynamically synchronized with latent embeddings and simulation states in a unified runtime environment.

Empirical Contribution. The thesis provides systematic empirical evidence that incorporating tacit knowledge via multimodal analysis and MR elicitation enhances conceptual model quality. Using precision and conciseness as normative evaluation criteria [RAB⁺15], and applying a controlled A/B setup grounded in causal inference principles [Rub74], the evaluation (Chapter 9) demonstrates measurable improvements across modeling languages and scenario types. Independent prior validations of foundational components—process mining from video [GBP24a, GBP24c, GBP25d], multimodal alignment reliability [GBP25a, GBP25i, GBP25b], MR-based elicitation [GBP24b, GBP25j, GBP25f], and multimodal simulation frameworks [GBP26]—provide convergent validity. The aggregated results reported in Section 9.4 confirm that the proposed approach consistently outperforms the baseline in both quality dimensions, thereby directly addressing Research Questions RQ1–RQ4 (Section 1.2.1) and closing the gap identified in Section 2.6.

11.3 Final Remarks

This thesis explores how conceptual modeling can evolve when grounded in the full richness of human activity: perception, action, tacit judgment, and the lived context in which expertise unfolds.

The research journey has shown that conceptual modeling is no longer confined to the whiteboard, the interview, or the retrospective description of events. Instead, it can emerge from situated traces of experts interacting with their environment, supported by technologies that allows to observe, replay, perturb, and interpret their decision-making. In this sense, the modeling process becomes not only more rigorous, but also more collaborative: experts, modelers, and AI systems jointly shape a shared representation of organizational knowledge.

Beyond supporting conceptual modeling, the multimodal and Mixed Reality methods developed in this thesis also offer profound benefits for artificial intelligence itself. A central insight emerging from this thesis is that experts often possess a wealth of tacit knowledge that remains unspoken not because it is hidden, but because it feels too intuitive, too automatic, or too “obvious” to articulate. Traditional elicitation methods implicitly assume that experts will verbalize all that matters, yet much of their expertise lives in the micro-decisions they never think to mention. The multimodal and Mixed Reality methods developed here help overcome this gap by showing experts their own actions, highlighting subtle cues, and placing them into dispositioned, carefully altered realities that prompt reflection. Through these controlled perturbations, experts begin to reveal what they notice instantly, what they implicitly prioritize, and what they have never been asked to explain. This surfacing of tacit knowledge enriches conceptual models and simultaneously provides AI systems with training signals grounded in authentic, embodied human judgment. Mixed reality becomes not only a simulation tool, but a medium that restores the expert’s imaginative awareness—reshaping environments so that what was intuitive becomes visible, and what was silent becomes speakable. This connection between perception, imagination, and knowledge is captured fittingly in the closing reflection:

“You can’t depend on your eyes when your imagination is out of focus.”

— Mark Twain

Supplementary Theoretical Background

This appendix provides the comprehensive material underlying the background and foundations presented in the Chapter 2.

Section A.1 introduces foundational perspectives on conceptual modeling and elaborates major formal languages, including UML, BPMN, ArchiMate, OntoUML, Petri Nets, and Domain-Specific Modeling Languages. These formalisms establish the syntactic and semantic substrates upon which multimodal enrichment and Mixed Reality interaction are later aligned.

Section A.2 provides the formal and algorithmic background of process mining. It details process discovery, conformance checking, and process simulation and replay, thereby grounding the thesis' process-oriented contributions in established analytical techniques.

Section A.3 discusses privacy-aware data analysis as a prerequisite for trustworthy multimodal modeling. It situates privacy not merely as a regulatory constraint but as an integral dimension of responsible conceptual modeling in data-intensive and Mixed Reality environments.

Finally, Section A.4 presents the artificial intelligence foundations underpinning the multimodal and generative aspects of this work. It covers hybrid intelligence paradigms (Section A.4.1), knowledge graphs and large language models (Section A.4.2), generative AI (Section A.4.3), embedding spaces (Section A.4.4), and video generation (Section A.4.5), including their respective conceptual intuitions. Together, these foundations establish the conceptual bridge between symbolic modeling, subsymbolic learning, and multimodal simulation.

A.1 Conceptual Modeling

This appendix includes detailed descriptions of major conceptual modeling languages (UML, BPMN, ArchiMate, OntoUML, Petri Nets), an exposition of the roles and uses of conceptual models, an extended discussion on digital twins as dynamic conceptual models, an overview of graphical modeling tools, and a synthesis of the Model-Driven Engineering paradigm. These sections together offer the full theoretical background that supports the methodological and technical contributions of this thesis.

A.1.1 Formal Conceptual Modeling Languages

Formal conceptual modeling languages provide the syntactic and semantic scaffolding necessary to encode conceptual structures with precision and consistency. Each language embodies specific ontological commitments and modeling primitives suited to particular purposes, domains, and levels of abstraction. While they differ in expressiveness and notation, they share a common goal: to capture relevant aspects of a conceptualization in a machine-interpretable and human-understandable form.

Unified Modeling Language (UML)

Among the most widely used formalisms, the **Unified Modeling Language** stands as a general-purpose standard for representing the structural and behavioral aspects of software and information systems [Obj07]. Developed and maintained by the Object Management Group (OMG), UML defines a comprehensive suite of diagram types—such as class, activity, sequence, and state diagrams—each targeting a distinct viewpoint of system design. UML’s strength lies in its ability to serve as a *lingua franca* among software engineers, combining visual expressiveness with formal underpinnings sufficient for model-driven engineering and code generation.

UML encompasses a rich set of diagrammatic notations organized into *structural*, *behavioral*, and *interaction* categories. Structural diagrams, including class, component, deployment, and object diagrams, capture the static architecture of a system—its entities, their attributes, and interrelations. Behavioral diagrams, such as activity, use case, and state machine diagrams, focus on dynamic aspects, illustrating workflows, system behavior, and lifecycle transitions of entities. Interaction diagrams—including sequence, communication, timing, and interaction overview diagrams—detail how system components or actors exchange messages over time. This taxonomy allows UML to represent systems from multiple complementary perspectives, supporting both abstract conceptual design and detailed technical specification. The formal metamodel of UML, specified in the Meta-Object Facility (MOF), enables consistent interpretation of model semantics across tools and platforms.

UML’s flexibility has led to its adoption far beyond traditional software engineering, extending into systems engineering, business process modeling, and even conceptual modeling. Through extensions such as **UML Profiles**, UML provides a customizable

foundation for domain-specific modeling. Moreover, its integration into the Model-Driven Architecture (MDA) framework facilitates automated transformations between conceptual, platform-independent, and platform-specific representations, bridging design and implementation. Combined with formal verification, simulation, and model-to-text generation tools, UML remains a cornerstone of model-driven development, offering both expressiveness for human understanding and precision for automated reasoning and system synthesis.

Business Process Model and Notation (BPMN)

In the domain of business process management, the **Business Process Model and Notation** was established to provide an intuitive yet semantically rigorous framework for representing process flows [Obj08]. BPMN introduces standardized graphical symbols for events, tasks, gateways, and message exchanges, allowing business analysts and technical developers to collaborate effectively [vRWCdM15]. Its formalization supports both human understanding and execution by workflow engines, thereby bridging conceptual and operational layers of process representation.

Beyond its core notation, BPMN offers a layered modeling approach that accommodates varying levels of abstraction—from high-level business overviews to executable process specifications. Its elements are organized into distinct categories: *flow objects* (events, activities, gateways), *connecting objects* (sequence flows, message flows, associations), and *swimlanes* (pools and lanes) that define organizational responsibilities. This structure enables clear differentiation between control flow, data flow, and organizational context. The accompanying *Process Definition Meta-Model* (PDMN) and the formal mapping to execution languages like BPEL (Business Process Execution Language) ensure that BPMN diagrams can be directly translated into machine-interpretable workflows. As a result, BPMN supports the full lifecycle of process management—from design and simulation to monitoring and optimization—providing a unified medium for communication between business and IT stakeholders.

In recent years, BPMN has also been extended and integrated with complementary standards to address emerging needs in automation and analytics. The current version also covers execution semantics, data modeling capabilities, and event-driven process constructs that enhance interoperability and precision. Combined with complementary notations such as DMN (Decision Model and Notation) and CMMN (Case Management Model and Notation), BPMN forms a cohesive ecosystem that supports structured, decision-centric, and ad-hoc processes within the same enterprise architecture. Moreover, BPMN's compatibility with process mining techniques enables the alignment of modeled and executed processes through event logs, fostering continuous improvement and evidence-based refinement of business operations.

ArchiMate

Effectively leveraging information technology within an enterprise presupposes a coherent and integrated vision that aligns business goals with IT capabilities. In the absence of such alignment, the IT infrastructure risks failing to adequately support the business strategy, while the business itself may not fully benefit from technological advancements. Strategic alignment literature consistently highlights the importance of balancing both organizational (*soft*) and technological (*hard*) aspects to ensure synergy between business and IT.

For instance, when a company introduces a new product line, the initiative may trigger a series of interrelated changes—such as defining new business processes, recruiting staff, updating supporting applications, and scaling the technological infrastructure. These transformations often extend to organizational structures and involve a wide range of stakeholders—from executives to engineers—each requiring tailored and accessible information to assess and manage impacts. Maintaining a comprehensive overview of such interconnected changes is inherently challenging.

To address this complexity, an architectural perspective is essential. According to the ISO/IEC 42010 standard [ISO07], *architecture* represents “the fundamental organization of a system embodied in its components, their relationships to each other and to the environment, and the principles guiding its design and evolution.” To manage the complexity of any large system, be it an enterprise, an organisation, an information system or a software system, an architectural approach is needed [LPJ09].

At the enterprise architecture level, the **ArchiMate language** [The09] extends this idea by providing an integrated modeling framework that connects business processes, application systems, and technological infrastructures. Developed under The Open Group, ArchiMate emphasizes coherence between abstraction layers, enabling organizations to analyze alignment between strategic goals, information structures, and technical implementations. ArchiMate’s architecture supports traceability and transformation between model views, reinforcing its conceptual integrity across heterogeneous domains.

Lankhorst, Proper, and Jonkers [LPJ09] detail the structural rationale and design principles that guided the development of the ArchiMate language. Their discussion emphasizes how an architecture description language for enterprise architecture must balance general modeling capabilities with the specific demands of dynamic, multi-layered organizational systems. To achieve this, the authors distinguish between conceptual structures required for modeling generic domains, those needed to represent dynamic behavior, and those tailored to capture the interdependencies characteristic of enterprise architectures.

ArchiMate was standardized by The Open Group, aligning it closely with their enterprise architecture framework, TOGAF. This integration ensures that the language can evolve alongside future iterations of the framework, while [LPJ09] note that ArchiMate’s metamodel provides a stable foundation for interoperability and consistency across architectural layers—spanning business, information systems (including data and application

architectures), and technology—thereby enabling traceability and coherence throughout enterprise modeling efforts.

OntoUML

In the realm of ontologically grounded modeling, Guizzardi’s foundational work [Gui05] established the **OntoUML** language, rooted in formal ontology and philosophical realism. OntoUML builds on the Unified Foundational Ontology (UFO) and imposes semantic constraints that reflect distinctions such as types, roles, and moments, thereby reducing ambiguity in conceptual modeling. Guizzardi demonstrated that ontological rigor is essential for ensuring that conceptual models correspond to cognitively and ontologically valid categories, improving their interpretability and reusability across contexts.

Recent extensions, such as OntoUML 2 [GFB⁺18] and its companion framework Onto4ALL, advance Guizzardi’s original vision toward interoperability and model-driven engineering integration. It introduces mechanisms for cross-domain knowledge reuse and ontology-driven interoperability across heterogeneous systems. In parallel, ongoing research explores the synergy between OntoUML and formal verification, AI-based reasoning, and multimodal data interpretation—allowing conceptual models not only to represent domain knowledge but also to dynamically adapt to new evidence. These developments position OntoUML as a cornerstone in bridging philosophical ontology, conceptual modeling, and intelligent system design, reinforcing the role of ontological foundations in ensuring semantic clarity and cognitive alignment of information systems.

Petri Nets

When focusing on concurrency and dynamic behavior, **Petri Nets** [PR08] provide a mathematically grounded modeling formalism for representing systems of distributed and interacting components. Petri Nets consist of places, transitions, and tokens, offering a precise mechanism to describe and analyze state changes, synchronization, and causality. Their formal semantics make them particularly suitable for verification and simulation of complex processes, including workflow systems and process mining applications.

Beyond their basic structure, high-level Petri Nets—such as Colored Petri Nets (CPNs) and Timed Petri Nets (TPNs)—extend the classical formalism to capture data, temporal, and hierarchical aspects of processes. In CPNs, tokens can carry structured data types, allowing the model to represent both control flow and data flow within the same formalism. This integration enables more expressive models that can be directly used for system simulation, performance evaluation, and state-space analysis. Timed Petri Nets, on the other hand, incorporate time intervals or stochastic distributions into transitions, facilitating the modeling of temporal dependencies and resource constraints often encountered in real-world process execution. These extensions bridge the gap between abstract process descriptions and operational systems, providing a scalable analytical foundation for process optimization and predictive analysis.

Furthermore, Hybrid Petri Nets and their modern derivatives—such as Stochastic, Continuous, and Fuzzy Petri Nets—extend applicability to cyber-physical and socio-technical domains. Hybrid Petri Nets combine discrete and continuous dynamics, thus enabling representation of processes that evolve both through discrete events (e.g., human decisions) and continuous changes (e.g., temperature or pressure variations). Such hybrid formulations are increasingly relevant for modeling intelligent systems, where physical and informational components interact continuously. Recent developments integrate Petri Nets with data-driven models and machine learning techniques, allowing for adaptive and self-updating representations that align with real-time observations, thereby reinforcing their role as a unifying language for reasoning about dynamic and concurrent behaviors across heterogeneous domains.

Collectively, these languages illustrate the broad spectrum of purposes that conceptual models can serve—from capturing mental abstractions to enabling automated reasoning. Each formalism embodies a different trade-off between human interpretability and machine precision. In the context of this thesis, these formalisms provide both the syntactic structures and semantic anchors upon which multimodal evidence and Mixed Reality interaction can later be aligned. Conceptual models, regardless of their specific notation, ultimately aim to externalize the regularities of thought and action in a way that can be jointly inspected, validated, and refined—a principle that remains central as we move toward data- and interaction-enriched modeling paradigms.

Domain-Specific Modeling Languages

Domain-Specific Modeling Languages (DSMLs) have emerged as a response to the limitations of general-purpose modeling languages when dealing with highly specialized domains. Instead of relying on abstract constructs such as those provided by UML or BPMN, DSMLs encapsulate concepts, relationships, and constraints that are intrinsic to a particular domain. By embedding domain semantics directly into the language, DSMLs enable domain experts to express solutions at a higher level of abstraction and with greater precision [Fra13].

As Frank [Fra13] emphasizes, DSMLs incorporate domain-level knowledge, which allows systems analysts and modelers to reuse conceptual structures rather than constructing them from scratch. This embedded knowledge contributes to model integrity by enforcing constraints and consistency rules within the language itself. However, the creation of a DSML is a complex endeavor that requires systematic methods for requirements analysis, metamodel selection, and iterative evaluation. Frank proposes a set of design guidelines to bridge the gap between theoretical understanding and practical implementation, stressing the importance of aligning a DSML’s syntax and semantics with the cognitive models of its intended users.

Building on this foundation, Jannaber et al. [JRD⁺17] focus specifically on *domain-specific process modeling languages* (DSPMLs), which adapt process modeling notations to the needs of particular industries or organizational contexts. Their framework consolidates

23 requirements from the literature to guide the development of such specialized process modeling languages. It structures the development process into key building blocks—from requirements elicitation to language evaluation—thus enhancing language adequacy, usability, and quality. This systematic approach ensures that DSPMLs maintain both domain relevance and methodological rigor.

Complementing these perspectives, Karsai et al. [KKP⁺14] provide general design guidelines for DSMLs, emphasizing modularity, abstraction, and the balance between expressiveness and simplicity. They argue that a well-designed DSML should support both model creation and tool integration, enabling automated code generation, validation, and analysis. Furthermore, they highlight that the metamodeling approach—defining a language through its abstract syntax and semantics—is central to ensuring that DSMLs remain extensible and maintainable over time.

Graphical Conceptual Modeling Tools

In parallel, the increasing complexity and granularity of data driving these models have amplified the importance of *visualization* as a core function of conceptual modeling [BDC23]. Visualization bridges the cognitive gap between formal structure and lived experience, allowing stakeholders to see the model’s implications, dependencies, and dynamics unfold in intuitive ways. Beyond diagrammatic representations, emerging visualization techniques—such as immersive 3D environments, augmented and Mixed Reality interfaces, and interactive dashboards—allow stakeholders to inhabit the model space itself. This transformation turns conceptual modeling into a perceptual activity, where insight emerges not only from symbolic reasoning but from embodied exploration, fostering a deeper and more intuitive understanding of complex systems.

Conceptual modeling tools serve as the practical environments where abstract modeling theories are instantiated into usable artifacts. These tools provide interactive interfaces for creating, visualizing, validating, and transforming conceptual models, bridging the gap between theoretical modeling frameworks and real-world applications. Examples include *BigUML* [MB23] for UML, and *bpmn.io* for BPMN. Such platforms implement syntax checking, diagrammatic editing, and model repository management, allowing users to construct models compliant with their respective metamodels (e.g., UML, BPMN, ArchiMate, OntoUML). Many of these tools also provide semantic validation capabilities and support for model transformations—such as converting conceptual schemas into implementation-level artifacts like relational databases, executable process workflows, or OWL ontologies. By providing visualization, consistency management, and traceability features, conceptual modeling tools enable collaboration among stakeholders, ensuring that domain knowledge remains aligned with evolving business or system requirements.

In recent years, conceptual modeling tools have evolved beyond static diagramming environments toward **intelligent, collaborative, and multimodal modeling ecosystems**. Those tools enable real-time co-editing, version control, and integration with continuous engineering pipelines. Advances in AI and multimodal interfaces are in-

roducing new interaction paradigms—such as natural language processing for model generation, automated diagram layout, supporting hybrid, textual- and graphical editing, multiple notations, and *Structured Query Language* (SQL) code generation, such as *BigER* [GB21] for entity relationships. Emerging research prototypes integrate *Mixed Reality* and *gesture-based interaction*, allowing users to manipulate model elements in 3D space, improving spatial cognition and collaborative awareness [MBF26, MF24b]. This convergence of conceptual modeling and intelligent assistance signals a new generation of tools that not only support modeling but also actively participate in the modeling process, guiding users through the creation, validation, and refinement of domain-relevant conceptualizations.

Markdown Conceptual Modeling Tools

Markdown-based conceptual modeling tools provide a flexible, text-centric environment for creating, editing, and maintaining models without reliance on heavyweight graphical editors. These tools embed modeling grammars—such as Mermaid, PlantUML, Graphviz DOT, and custom domain-specific notations—directly within Markdown code blocks, enabling model-as-text workflows that integrate seamlessly with version control, collaborative editing platforms, and reproducible research pipelines. By treating conceptual models as plain-text artifacts, Markdown tools support fine-grained diffs, automated regeneration, and programmatic transformation, making them particularly suitable for AI-assisted modeling scenarios in which agents generate, refine, or critique diagrams. Moreover, Markdown modeling environments simplify multimodal integration: logs, screenshots, video frames, transcripts, and model fragments can coexist in a single narrative document, enabling iterative elicitation cycles where conceptual diagrams evolve alongside evidence. This lightweight, modular ecosystem thus provides an ideal substrate for rapid prototyping, traceable model evolution, and hybrid human–AI conceptual modeling workflows.

Mermaid [SJ21] is a lightweight, text-driven diagramming language that enables the generation of conceptual models directly from Markdown code blocks. Unlike traditional modeling tools, which require graphical interfaces and proprietary formats, Mermaid offers a declarative syntax embedded in plain text, facilitating rapid iteration, version control, and seamless integration into modern documentation pipelines (e.g., GitHub, GitLab, static site generators, and research compendia).

From a conceptual-modeling perspective, Mermaid provides a practical medium for expressing structural and behavioral abstractions without the overhead of full graphical editors. The modeler specifies diagrams through concise textual statements, which are then rendered automatically into visual artifacts. This approach aligns with the broader shift toward “model-as-code” paradigms, where models become first-class textual assets amenable to diffing, reuse, and automated tooling.

Mermaid supports several diagram types relevant to conceptual modeling, including flowcharts, sequence diagrams, class diagrams, entity–relationship diagrams, state ma-

chines, and Git-style timelines. For example, a simple process fragment can be expressed as:

```
flowchart TD
    A[Patient arrives] --> B{Registration}
    B -->|complete| C[Triage]
    B -->|missing data| D[Clarification]
```

This textual specification can be stored in Markdown documents, embedded within experimental notebooks, or referenced directly in interactive environments. Such portability enables a high degree of flexibility when constructing multimodal or AI-assisted conceptual modeling workflows—for example, enabling agents to generate or refine diagram specifications automatically.

A.2 Process Mining

This appendix presents the formal and algorithmic background on which the thesis’ process-oriented contributions rely. It first introduces process discovery from event logs, including the formalization of event logs, quality dimensions (soundness, fitness, precision, and generalization), and the Inductive Miner framework with its process tree semantics. Next, it details conformance checking techniques, from classical token-based replay [vdA96] to more recent backward state-space exploration approaches [BvdA21]. Finally, it elaborates on process simulation and replay [vdA18], illustrating how empirically derived models can be used for predictive and counterfactual analysis. Together, these sections provide the technical foundation for integrating process mining with multimodal and Mixed Reality-based conceptual modeling in the main body of the thesis.

Process Discovery from Event Logs

Process discovery refers to the automated or semi-automated derivation of process models from event logs. Each log entry typically represents the occurrence of an activity within a case (process instance), possibly enriched with timestamps, actors, resources, and contextual data. Discovery algorithms—such as the Inductive Miner [LFvdA13]—seek to reconstruct a model that best explains the observed sequences of activities. The discovery process thus transforms heterogeneous, continuous evidence into discrete behavioral traces. These traces serve as the empirical backbone of conceptual modeling, ensuring that the resulting models are grounded in real execution rather than anecdotal or declarative descriptions. Today, process discovery is widely accessible through open-source toolkits such as **PM4Py** [BvS23], which provide comprehensive functionality for process mining in Python, including discovery, conformance checking, and performance analysis.

A process model’s quality is commonly assessed against several criteria. One of the most fundamental among them is *soundness*, which requires that every process step

can be executed and that at least one satisfactory end state can always be reached. In practice, models that fail to meet this property are considered unsound and can typically be disregarded without reference to the event log they are meant to represent. Another key criterion is *fitness*, which measures how well a model reproduces the observed traces in an event log—perfect fitness means that all recorded traces can be replayed by the model. Beyond fitness, the criterion of *precision* evaluates whether the model avoids overgeneralizing by not permitting excessive or unrealistic behavior. Conversely, *generalization* ensures that the model remains flexible enough to accommodate plausible future behavior not yet observed in the log.

Beyond ensuring finite runtime, process discovery algorithms are expected to satisfy several additional properties. In real-world scenarios, event logs originate from actual processes, and an effective discovery algorithm should be capable of *rediscovering* the original process model when provided with sufficient information. This property, known as rediscoverability, can be expressed in multiple forms. *Language-rediscoverability* refers to algorithms that reconstruct a model whose behavior (language) is equivalent to that of the original process, even if the available log does not cover every possible execution. *Isomorphic-rediscoverability*, on the other hand, demands that the discovered model is structurally identical (isomorphic) to the original one. The extent to which an event log captures the behavior of the original process is referred to as its *log completeness*. The strongest form, *total log completeness*, would require that all possible process variants appear in the log—a condition rarely satisfied in practice. Consequently, useful discovery techniques must operate effectively under much weaker completeness assumptions.

Process discovery is a central task in process mining and is sometimes colloquially referred to simply as *process mining* itself. Its goal is to automatically construct a process model that accurately represents the behavior recorded in an event log. An *event log* L is a finite, non-empty set of *traces*, where each trace $t \in \Sigma^*$ represents a sequence of events drawn from a finite set of process activities Σ . For example, the log $L = \{\langle a, b, c \rangle, \langle a, c, b \rangle\}$ consists of two traces— abc and acb —each describing an execution instance of a process. The size of a log is given by $\|L\| = \sum_{t \in L} |t|$, corresponding to the total number of recorded events.

The **Inductive Miner** [LFvdA13] framework provides a structured approach to process discovery that facilitates reasoning about various quality criteria. It operates on block-structured process models, allowing different types of blocks or operators to be incorporated without altering the overall framework and with minimal additional proof obligations. The algorithm follows a *divide-and-conquer* strategy, decomposing the task of discovering a process model for a given event log L into the discovery of n subprocesses, each corresponding to a sublog derived from splitting L . This recursive decomposition ensures that the resulting model is both sound and hierarchically structured, thereby supporting analytical transparency and formal guarantees.

The Inductive Miner builds on formal structures such *workflow nets*, and *process trees*. A **workflow net** is a specific form of Petri Net with exactly one start and one end place, ensuring that all elements lie on a path from the start to the end. Extending this

idea, a **block-structured workflow net** organizes the process hierarchically, allowing recursive decomposition into subcomponents, each with single entry and exit points. To represent such hierarchical structures abstractly, the Inductive Miner employs **process trees**. A process tree is a rooted tree whose leaves correspond to atomic activities and whose internal nodes correspond to operators defining how subtrees are combined. Formally, given a finite alphabet Σ of activities and a set of operators $\oplus$, a process tree can be defined recursively as follows: (i) a single activity $a \in \Sigma \cup \{\tau\}$, where τ denotes a silent (non-observable) activity, forms a process tree; and (ii) if $M_1, \dots, M_n$ are process trees and $\oplus$ is a process tree operator, then $\oplus(M_1, \dots, M_n)$ is also a process tree. Each operator specifies how the logs of its subtrees are combined, enabling the recursive composition of structured process models that remain sound by construction.

The semantics of process trees are defined through a recursive language function $\mathcal{L}(M)$, which assigns to each process tree M the set of all possible execution traces (the language of the model). Each operator $\oplus$ in the process tree has an associated *language join function* $\oplus_1$ that defines how the logs of its subtrees are combined. Formally, for a process tree M :

$$\begin{aligned}\mathcal{L}(a) &= \{\langle a \rangle\} \quad \text{for } a \in \Sigma \\ \mathcal{L}(\tau) &= \{\epsilon\} \\ \mathcal{L}(\oplus(M_1, \dots, M_n)) &= \oplus_1(\mathcal{L}(M_1), \dots, \mathcal{L}(M_n))\end{aligned}$$

The operators considered in this framework are as follows:

- $\times$ denotes an **exclusive choice** between subtrees.
- $\rightarrow$ denotes **sequential execution** of subtrees.
- $\circ$ denotes a **loop** with a body M_1 and one or more alternative redo or exit paths $M_2, \dots, M_n$, where $n \geq 2$.
- $\wedge$ denotes **parallel (interleaved) execution**.

Each operator's language join function $\oplus_1 : 2^{\Sigma^*} \times \dots \times 2^{\Sigma^*} \rightarrow 2^{\Sigma^*}$ combines multiple logs $L_1, \dots, L_n$ to produce a new log. Their definitions are as follows:

$$\begin{aligned}\times_1(L_1, \dots, L_n) &= \bigcup_{1 \leq i \leq n} L_i \\ \rightarrow_1(L_1, \dots, L_n) &= \{t_1 \cdot t_2 \cdots t_n \mid \forall i \in 1 \dots n : t_i \in L_i\} \\ \circ_1(L_1, \dots, L_n) &= \{t_1 \cdot t'_1 \cdot t_2 \cdot t'_2 \cdots t_m \cdot t'_m \mid \forall i : t_i \in L_1 \wedge t'_i \in \bigcup_{2 \leq j \leq n} L_j\}\end{aligned}$$

The parallel (interleaved) operator $\wedge$ is defined using a set notation $\{t_1, \dots, t_n\} \simeq$ that represents all possible interleavings of traces $t_1, \dots, t_n$. Specifically, for a trace t :

$$t \in \{t_1, \dots, t_n\} \simeq \iff \exists f : \{1, \dots, |t|\} \rightarrow \{(j, k) \mid 1 \leq j \leq n \wedge 1 \leq k \leq |t_j|\} :$$

$$\forall i_1 < i_2 (f(i_1) = (j, k_1) \wedge f(i_2) = (j, k_2)) \Rightarrow k_1 < k_2 \quad \text{and} \quad \forall i \leq n f(i) = (j, k) \Rightarrow t(i) = t_j(k)$$

This definition captures all valid interleavings of parallel activities across the traces of each subtree, ensuring that the ordering of activities within each individual trace is preserved while allowing flexible combinations across concurrent branches.

For example, we provide a following process tree

$$\rightarrow (a, \circ (\rightarrow (\wedge (\times (b, c), d), e), f), \times (g, h)).$$

Each process tree operator has a direct and well-defined translation into a *sound, block-structured workflow Petri Net*. If new process tree operators were to be introduced, their translations would remain sound as long as each operator is sound in isolation.

Conformance Checking

Conformance checking, as introduced in [vdA96], assesses the degree of alignment between an existing process model and its corresponding execution data. It addresses the fundamental epistemic question: does the formalized process truly represent what happens in reality? Techniques such as token-based replay, alignment-based checking, and behavioral profiles allow for the quantification of deviations between the model and observed traces. Discrepancies between modeled and observed behavior reveal not only executional deviations but also potential conceptual gaps, offering feedback loops that refine both the precision and the cognitive validity of the model. This iterative validation cycle is key to maintaining conceptual integrity in environments characterized by uncertainty and human variability.

In essence, conformance checking is concerned with quantifying and analyzing the correspondence between observed behavior (recorded in event logs) and modeled behavior (represented by a process model). It aims to determine whether process executions adhere to the intended workflow and to identify deviations for diagnostic or enhancement purposes. A common approach for conformance analysis is the *replay technique*, originally introduced in [vdA96], which compares the behavior of an observed trace with that allowed by a formal process model.

Among the various replay techniques, the most prominent ones are *token-based replay* and *alignment-based* methods, both of which operate primarily on Petri Nets [BvdA21]. Token-based replay simulates the execution of a trace over an accepting Petri Net, firing

transitions according to the activities in the trace. During the replay, tokens are produced and consumed in the Petri Net’s places, and deviations are detected when expected transitions cannot be executed due to missing tokens. The output of the replay operation typically records the transitions that were enabled and fired, thereby revealing which parts of the model were executed correctly and where violations occurred.

The approach proposed by Berti and van der Aalst [BvdA21] introduces a *novel token-based replay technique* that significantly improves efficiency by employing a **backwards state-space exploration** strategy. In this method, for a given trace and its corresponding Petri Net marking M_0 , the algorithm checks whether a transition $t \in T$ —corresponding to the activity currently being replayed—is enabled. If t is not enabled in M_0 , the algorithm explores alternative sequences of invisible (silent) transitions that can lead to a marking where t becomes enabled.

Formally, the approach searches for a sequence of invisible transitions $\langle t_1, \dots, t_n \rangle$ such that the firing of this sequence from M_0 results in a marking M' where transition t is enabled. To identify suitable invisible transitions, the algorithm maintains a backward search set, denoted as $BS(t)$, which contains transitions whose firing may enable t . For each $t_n \in BS(t)$, the algorithm checks whether the reverse firing of t_n from M_0 (denoted $M \leftarrow (M_0, t_n)$) results in a subset of M_0 , indicating that t_n is enabled in M_0 . If this condition is met, t_n is fired, and a marking enabling t is reached. Otherwise, the search continues recursively by expanding the B-sets $\{BS(t_n) \mid t_n \in BS(t)\}$ until a suitable marking is found.

This backward exploration ensures that invisible transitions are only applied when necessary and that each replay step results in a valid state transition consistent with the model’s semantics. Compared to classical token-based replay, this technique avoids redundant state exploration and reduces computational complexity while maintaining full behavioral accuracy. As a result, it enables faster and more scalable conformance checking, supporting both process verification and subsequent model enhancement activities in large-scale real-world event logs.

Process Simulation and Replay

Process simulation—sometimes referred to as *process replay*—involves executing a process model to generate synthetic event logs or to explore its dynamic behavior under varying conditions. While discovery reconstructs models from data, simulation plays models forward, producing new data that can be compared against empirical evidence. Petri Nets, due to their formal semantics, provide an ideal foundation for such simulation, enabling token flow to represent both control logic and concurrency. When integrated with MR environments, simulation can become experiential: stakeholders may observe, manipulate, and test process variants in immersive settings, thereby eliciting counterfactual insights about what would happen under alternative conditions. This approach transforms process mining from a passive analytical discipline into an interactive modeling cycle—observe,

model, simulate, and refine—ultimately leading to conceptual models that are empirically validated, cognitively grounded, and operationally executable.

Process simulation enables the forward-looking exploration of system behavior, while process discovery grounds this exploration in empirical evidence derived from real event logs. Together, they provide a comprehensive view of both actual and potential process behavior, where simulation supports predictive and prescriptive analysis, and process discovery ensures factual, data-driven grounding. On the one hand, process discovery can automate parts of the simulation modeling process, yielding more accurate and evidence-based models; on the other hand, simulation helps to evaluate, validate, and generalize process models discovered from event logs.

To illustrate these relationships, consider an event log

$$L = [\langle a, c \rangle^5, \langle a, b, b, c \rangle^3, \langle a, b, b, b, b, c \rangle],$$

which contains 29 events across 9 cases. A model $m_o = \text{discofit}(L)$ corresponds exactly to the traces observed in L , i.e.,

$$m_o = \{\langle a, c \rangle, \langle a, b, b, c \rangle, \langle a, b, b, b, b, c \rangle\}.$$

This model exhibits *perfect fitness* because all traces from the event log can be replayed without deviation. A second model $m_u = \text{discufit}(L) = \top$ allows for any behavior over the observed activities, while a third model $m_n = \text{discnfit}(L) = \top \setminus m_o$ represents all behavior that was *not* observed in L .

Evaluating these models according to standard quality dimensions reveals how recall, precision, and generalization interact. The recall of both m_o and m_u is high since they can reproduce all traces in L , whereas the recall of m_n is poor because it forbids all observed traces. In terms of precision, m_o scores highest because all behavior it allows has been observed in the log. Conversely, m_u and m_n have poor precision since they permit numerous unobserved behaviors. Generalization behaves differently: m_u generalizes well by also allowing unseen traces such as $\langle a, b, c \rangle$ or $\langle a, b, b, b, c \rangle$, whereas m_o and m_n generalize poorly because they either overfit or completely mismatch the data.

To distinguish between recall and generalization, van der Aalst [vdA18] considers three event logs L_1 , L_2 , and L_{100} , where each is a scaled duplication of the previous:

$$\begin{aligned} L_1 &= [\langle a, c \rangle^5, \langle a, b, b, c \rangle^3, \langle a, b, b, b, b, c \rangle], \\ L_2 &= [\langle a, c \rangle^{10}, \langle a, b, b, c \rangle^6, \langle a, b, b, b, b, c \rangle^2], \\ L_{100} &= [\langle a, c \rangle^{500}, \langle a, b, b, c \rangle^{300}, \langle a, b, b, b, b, c \rangle^{100}]. \end{aligned}$$

The corresponding models m_o , m_u , and m_n remain identical for all three logs. However, while recall and precision remain largely unchanged, the *generalization* improves with higher case frequencies. For instance, $\text{gen}(L_{100}, m_o) > \text{gen}(L_1, m_o)$, since observing many repetitions of only a few traces increases the likelihood that future cases will follow the same behavioral patterns. Hence, generalization depends not merely on the distribution

of traces, but also on their absolute frequencies. It adopts a *closed-world assumption* for the activity set $\mathcal{A}$, restricting reasoning to observed activities only. Models such as $m_u = \text{discufit}(L)$ and $m_n = \text{discnfit}(L)$ thus refer exclusively to behaviors involving activities present in the event log. This restriction ensures meaningful replay and simulation analysis, as unobserved activities fall outside the model’s behavioral universe.

Ultimately, van der Aalst [vdA18] demonstrates that combining process mining and simulation bridges empirical observation with analytical projection. Process mining provides the factual backbone for constructing simulation models, while simulation extends process mining toward *what-if* analysis, sensitivity evaluation, and predictive optimization.

A.3 Privacy Awareness in Conceptual Modeling

The increasing reliance on operational data for data analysis and model validation raises pressing concerns regarding the privacy of individuals whose actions generate such data. Fahrenkrog-Petersen *et al.* [FPvdAW24] address this issue in their work on *privacy-aware analysis based on data series*. They argue that while event and sensor data collected during organizational operations provide invaluable insights for compliance assessment, performance optimization, and predictive modeling, these datasets often contain sensitive personal information about both service providers and consumers. Conventional measures such as anonymization or pseudonymization offer only limited protection, as contextual attributes may still allow re-identification or inference of private details.

The authors provide a comprehensive overview of privacy risks and formalize different types of privacy-disclosure attacks, including linkage and background knowledge attacks, which exploit residual contextual cues in data series. To mitigate these threats, they introduce a taxonomy of *privacy-preserving techniques* that maintain analytical utility while ensuring formal privacy guarantees. These include mechanisms such as differential privacy, aggregation-based obfuscation, and utility-constrained perturbation. Their tutorial systematically structures these techniques according to two key dimensions: (i) the degree to which contextual information is retained in the dataset, and (ii) the analytical perspective adopted—whether focusing on individuals, groups, or systemic behavior.

Within the scope of this thesis, privacy-aware analysis constitutes an essential prerequisite for the ethical deployment of multimodal and Mixed Reality approaches in model validation. Embedding privacy-preserving mechanisms in the data processing and model validation pipeline ensures that stakeholder interactions within MR environments can be leveraged for sense-making without compromising ethical and legal constraints. Consequently, privacy is treated not merely as a regulatory requirement but as an integral dimension of *trustworthy conceptual modeling*. This aligns with the broader ambition of enhancing conceptual modeling through multimodal data analysis and MR, while upholding transparency, accountability, and respect for human subjects.

A.4 Artificial Intelligence

This appendix provides a comprehensive overview of the AI concepts that underpin the multimodal and model-driven contributions of this thesis. It first outlines the complementary roles of symbolic, subsymbolic, and human intelligence, situating conceptual modeling within a broader hybrid-intelligence perspective. It then examines the relationship between knowledge graphs and large language models [VFQP25], highlighting their complementary strengths in explicit reasoning and adaptive semantic inference. Subsequent sections cover generative AI [FHJZ24], including transformer-, diffusion-, and GAN-based architectures, and introduce the underlying intuitions behind generative processes. The appendix also discusses embedding spaces and foundational multimodal models [GENL⁺23], explaining how they enable cross-modal alignment and semantic grounding. Finally, it presents recent advances in video generation [GYS⁺24, RPG⁺21] and their relevance for translating conceptual structures into dynamic visual narratives. Together, these foundations form the conceptual and technical substrate for integrating multimodal data analysis with conceptual modeling throughout this thesis.

A.4.1 Symbolic, Subsymbolic and Human Intelligence

Artificial Intelligence, in its contemporary form, can be understood as the interplay of three complementary layers of intelligence: *symbolic*, *subsymbolic*, and *human*. The symbolic layer corresponds to formal, rule-based reasoning — exemplified by conceptual modeling — where meaning is encoded explicitly through ontologies, diagrams, and logical constraints that define entities, relationships, and permissible behaviors. The subsymbolic layer, by contrast, is realized through machine learning and deep learning techniques, which learn distributed representations of data by optimizing high-dimensional parameter spaces rather than manipulating explicit symbols. In this paradigm, knowledge emerges implicitly from statistical regularities captured in multimodal data — such as text, images, or video — allowing the model to generalize beyond observed examples. Finally, the human layer introduces interpretive, contextual, and ethical judgment into the loop: the modeler not only defines the problem and curates the data but also interprets the resulting models and integrates them into meaningful conceptual structures. Together, these three layers form a hybrid intelligence system — where symbolic reasoning provides structure, subsymbolic learning provides adaptivity, and human cognition ensures purpose and alignment with domain understanding.

A.4.2 Knowledge Graphs and Large Language Models

A contemporary debate in Artificial Intelligence concerns the respective roles of *knowledge graphs* (KGs) and *large language models* (LLMs) in representing and reasoning over knowledge. While both paradigms aim to capture and operationalize semantic relationships, they do so through fundamentally different mechanisms. Knowledge graphs provide an explicit, structured representation of entities and their relationships—forming a symbolic substrate that enables logical inference, traceability, and integration with domain ontolo-

gies. They are particularly suitable for tasks requiring transparency, provenance tracking, and domain-specific consistency, such as in virtual museums where cultural artifacts and metadata must remain interpretable and verifiable.

Conversely, large language models embody the subsymbolic paradigm, learning implicit associations between linguistic tokens through the statistical co-occurrence of words across massive corpora. These models encode contextual and pragmatic nuances that go far beyond manually engineered ontologies, allowing flexible reasoning, analogical transfer, and natural interaction through language. However, their knowledge remains opaque, distributed across billions of parameters without explicit relational structure.

Vasić et al. [VFQP25] argue that these two approaches are not adversarial but complementary. When combined, LLMs can serve as adaptive front-ends for interpreting and generating natural-language queries, while KGs act as back-end knowledge anchors that provide factual grounding, domain constraints, and explainability. Within the scope of this thesis, such a hybrid integration offers a conceptual parallel to combining data-driven and model-driven reasoning: KGs mirror conceptual models as structured, symbolic representations, whereas LLMs resemble adaptive, multimodal learners that can extract, relate, and verbalize complex patterns discovered across human and machine intelligence.

A.4.3 Generative AI

The term *Generative Artificial Intelligence (GenAI)* refers to computational techniques that are capable of generating seemingly new, meaningful content such as text, images, or audio from training data, as noted in the *Business & Information Systems Engineering (BISE)* “catchword” articles by Feuerriegel et al. [FHJZ24].

The emergence of GenAI represents a major shift in the capabilities of intelligent systems—from passively analyzing data to actively synthesizing new artifacts. Unlike traditional predictive models, which focus on estimating outcomes from known inputs, generative models learn the underlying distribution of the data itself, allowing them to simulate plausible variations and hypothetical scenarios.

At the technical core of GenAI lie architectures such as *transformers*, *diffusion models*, and *generative adversarial networks (GANs)*. Transformer-based models (e.g., [VSP⁺17]) operate through attention mechanisms that dynamically weigh contextual dependencies between tokens, enabling coherent long-range generation of language and multimodal content. Diffusion models iteratively denoise random noise to form structured data, achieving state-of-the-art performance in image and video synthesis. GANs, on the other hand, learn through adversarial dynamics between a generator and a discriminator, producing increasingly realistic samples across diverse modalities.

Intuition Behind Generative AI

At its core, the intuition behind generative AI is remarkably simple yet profoundly transformative: instead of merely classifying or predicting based on existing data, the

model learns the *essence* of a distribution well enough to imagine what could plausibly exist within it. Just as a human artist internalizes visual and stylistic patterns through exposure to countless paintings and then creates a new one consistent with that style, a generative model absorbs latent regularities from vast datasets and reconfigures them to produce new, coherent outputs. This process can be seen as a form of computational creativity—guided not by explicit symbolic rules but by the emergent geometry of data representations in high-dimensional space.

In transformer-based architectures, this intuition manifests through the concept of *attention*, where each element in a sequence selectively focuses on others depending on their contextual importance. Through repeated layers of attention and self-supervision, the model learns semantic relationships without explicit labeling, thereby constructing an internal representation of meaning that can be used for text continuation, image synthesis, or multimodal composition. Diffusion models, by contrast, operationalize the intuition of “reverse entropy”: they learn how to iteratively recover structure from noise, much like how humans can progressively recognize a blurred image as it becomes clearer.

A.4.4 Embedding Spaces and Foundational Models

A central paradigm of modern artificial intelligence is the use of *embedding spaces*—representations that map heterogeneous data (such as text, images, or audio) into a shared, continuous vector space. Within these spaces, semantically similar inputs are located near one another, allowing models to capture abstract relationships that transcend individual modalities. Foundational models such as *ImageBind* [GENL⁺23] have demonstrated that a single embedding space can bind multiple sensory and linguistic inputs, enabling cross-modal retrieval, alignment, and reasoning without explicit supervision.

In this paradigm, each modality is processed through a dedicated encoder that transforms raw input (e.g., pixels, waveforms, or text tokens) into latent vectors. These vectors are trained to co-locate semantically corresponding instances across modalities, typically through contrastive learning objectives that minimize the distance between paired samples (e.g., image–caption pairs) while maximizing the distance to non-matching examples. Once trained, the resulting embedding space acts as a universal semantic substrate upon which generative and discriminative tasks can be performed. Foundational models leverage this substrate to generalize across tasks and domains, providing a robust basis for transfer learning, zero-shot reasoning, and multimodal understanding.

In the context of this thesis, embedding spaces form the connective tissue between multimodal data analysis and conceptual modeling. They enable the alignment of perceptual observations—such as visual scenes, spoken descriptions, and contextual signals—with higher-level conceptual representations, thus supporting the automatic grounding of conceptual models in empirical data. Foundational models operationalize this process at scale, offering pretrained embeddings that encode vast amounts of structured and unstructured world knowledge.

Intuition Behind Embedding Spaces

The intuition behind embedding spaces rests on the idea of *proximity as meaning*. When humans learn, they intuitively associate similar experiences or symbols within a shared mental landscape—objects that look, sound, or feel related occupy nearby positions in our cognitive space. Embedding models emulate this process computationally by positioning semantically or perceptually similar inputs close to one another within a continuous, high-dimensional space. In doing so, they translate the discrete world of language, vision, and sound into a geometry of relationships.

As illustrated by Girdhar et al. [GENL⁺23], this intuition extends naturally to the multimodal setting: by binding multiple sensory modalities into a single representational manifold, the model learns to perceive correspondences between, for instance, the sound of waves, the image of a beach, and the text “seaside.” Through contrastive training, the model implicitly constructs a topology of meaning—an emergent semantic field where proximity reflects shared context rather than surface similarity.

For conceptual modeling, this intuition is particularly valuable. It suggests that embeddings can capture implicit correspondences between human observations and conceptual abstractions, thereby serving as an intermediate layer between raw multimodal data and formal symbolic structures. Embedding spaces thus act as the “latent fabric” connecting perception, language, and knowledge—a computational analogue to the cognitive continuum that underpins human understanding.

A.4.5 Video Generation

Recent advances in *video generation* have extended the creative and analytical potential of generative AI from text to dynamic, temporally coherent visual sequences. The evolution of this field follows the same trajectory that text-to-image generation models, such as DALL·E [RPG⁺21], initiated—bridging linguistic prompts and visual imagination—but introduces the added challenge of modeling temporal dependencies and motion realism. Video generation aims to synthesize photorealistic sequences conditioned on textual descriptions, reference images, or multimodal cues, effectively turning language into visually grounded narratives.

Diffusion-based architectures have recently emerged as the state-of-the-art approach in this domain. Gupta et al. [GYS⁺24] propose a photorealistic video diffusion framework that extends 2D diffusion models to the spatiotemporal domain, enabling the generation of coherent motion across frames while preserving fine-grained visual detail. These models learn to iteratively refine noisy video sequences, denoising them frame by frame through learned temporal priors that encode both appearance consistency and realistic motion dynamics. The resulting systems can synthesize short, high-quality clips that closely mirror real-world cinematography and lighting, marking a major step toward controllable and semantically aligned video generation.

From a broader perspective, video generation integrates multiple layers of AI progress:

the linguistic grounding of text-to-image systems, the spatiotemporal reasoning of motion models, and the visual realism achieved through adversarial and diffusion training. Within this thesis, such models play a crucial role in connecting process understanding and conceptual modeling. By transforming abstract or symbolic process descriptions (e.g., event sequences or Petri Nets) into dynamic video narratives, they enable visual inspection, simulation, and communication of models through an intuitive and human-centered medium.

Intuition Behind Video Generation

The intuition behind video generation can be described as the algorithmic reconstruction of imagination. When humans visualize a process or story, they mentally animate static elements—inferring motion, continuity, and causality from still scenes or verbal cues. Video generation models replicate this capacity computationally. They learn the distribution of motion patterns and temporal transitions from vast video corpora, internalizing how entities move, interact, and evolve over time. Once trained, these models can “hallucinate” motion sequences consistent with both visual and linguistic context, effectively generating plausible worlds from abstract cues.

In diffusion-based frameworks [GYS⁺24], this process unfolds as a gradual reversal of noise: the model starts from random temporal noise and incrementally denoises it, guided by learned spatiotemporal correlations. Each denoising step reflects the model’s understanding of how objects persist and transform, encoding physical and semantic regularities implicitly. Meanwhile, text-conditioned architectures such as DALL · E [RPG⁺21] provide the semantic scaffolding—linking natural language to visual patterns—so that prompts like “a technician assembling a device” or “a molecule reacting in slow motion” can generate coherent video representations.

For conceptual modeling, this intuition has profound implications. It suggests that processes, events, and system dynamics can be expressed not only symbolically but visually, through generative models that animate conceptual constructs. Video generation thus becomes a bridge between multimodal perception and formal reasoning—enabling models to be *seen*, *experienced*, and *interpreted* as living simulations of conceptual knowledge.

Supplementary Implementation Details

This appendix provides a comprehensive, engineering-level account of the systems, interfaces, and architectural choices that underpin the Mixed Reality–augmented conceptual modeling framework presented in this thesis. While the main chapters focus on the methodological, theoretical, and evaluative contributions, the present appendix offers a transparent view into the concrete software mechanisms that make the framework operational. It documents the real-time web client, the browser-based Mixed Reality simulation environment, the ViEnNa editor and renderer, and the crossmodal event-capture subsystem, detailing how these components are implemented, how they interoperate, and how they support the multi-agent collaboration patterns described in the Chapter 8.

This appendix is structured to provide a systematic engineering-level account of the implemented framework components. Section B.1 introduces the browser-based real-time client, detailing agent orchestration, audio streaming, transcript handling, and synchronization between simulation and modeling panes. Section B.2 then describes the browser-native Mixed Reality simulation environment built on `three.js`, explaining how 360-degree video, embedded digital system interfaces, and URL-driven scenario control enable egocentric, situated elicitation. Section B.3 presents the ViEnNa editor and renderer, outlining the panel-based syntax, live comic rendering, Graphviz integration, and its linkage to Mixed Reality event captures. Finally, Section B.4 specifies the crossmodal event logging subsystem, including the „Log Now“ mechanism, schema design for process-mining compatibility, and the integration with `pm4py` for downstream discovery and conformance analysis. Together, these sections provide a transparent view of how the realtime client, immersive simulation, narrative modeling environment, and event-capture infrastructure interoperate to operationalize the Mixed Reality–augmented conceptual modeling framework described in Chapter 8.

B.1 Tooling

The browser-based client serves as the main entry point for running conceptual modeling AI sessions with the AI agents described in this thesis. It is implemented as a *React* component and instantiated as a *Next.js* client-side page. The component orchestrates three main concerns: (i) managing the real-time connection to the agent back-end, (ii) coordinating the simulation and conceptual modeling panes, and (iii) maintaining a responsive user interface for transcript, events, and audio interaction. A `sdk scenario` map links URL-level scenario keys to pre-defined sets of `realtime agent` definitions.

URL-driven configuration and agent selection. The client reads URL query parameters using `use search params` to configure both audio codec and agent setup. The `codec` parameter selects the audio encoding (e.g., `opus`), and `agent config` selects a particular agent set from `all agent sets`. If the requested agent configuration is not found, the client rewrites the URL to a default configuration to ensure a consistent initial state. The selected agent set and the currently active agent within this set are stored as React state, allowing the user to change the domain (e.g., emergency room) and the active agent via dropdown selectors in the top toolbar. When the user changes the active agent, the client reconnects the realtime session such that the newly chosen agent becomes the root agent in the multi-agent graph.

Audio pipeline and realtime session lifecycle. To handle streaming audio from the Realtime API, the client creates a hidden HTML `audio` element on first render and stores a reference to it. This element is then passed into the custom `use realtime session` hook, which exposes high-level operations such as `connect`, `disconnect`, `send user text`, `send event`, `interrupt`, and `mute`. The session status is tracked as a finite set of states (`DISCONNECTED`, `CONNECTING`, `CONNECTED`), and callbacks from the hook update this status and handle agent handoffs. When an automatic handoff occurs, the client updates the `selected agent name` but also keeps track of whether the switch was triggered programmatically to adjust greeting behaviour.

The client requests an ephemeral API key from the server and logs both the request and response in the event context. Upon successful retrieval, the `connect to realtime` function constructs the initial agent set for the SDK (ensuring the selected agent is placed first) and instantiates the connection with guardrails created via `create moderation guardrail`. The client then uses `send event` to push session update messages that configure turn detection: if push-to-talk (PTT) is active, server-side voice activity detection is disabled; otherwise, a server voice activity detection (VAD) configuration is provided, including silence thresholds and automatic response triggering.

Simulation and conceptual modeling panes. On the right-hand side of the layout, the client controls two collapsible `iframe` panes: one for the SUI simulation, and one for conceptual modeling. The currently selected situation index is stored as state and determines the active entry from `situations`. The SUI pane computes a simulation

URL augmented with query parameters (title, video path, software’s iframe path) derived from the selected situation. This allows the simulation to load the appropriate video or Mixed Reality scene for the current scenario.

For the conceptual modeling pane, the client derives a default URL pointing to the conceptual modeling viewer and passes the situation’s concept model identifier as a query parameter. The client also listens for `conceptual model update` events dispatched by tools in the agent layer (e.g., when a new Mermaid or ViEnNa model is saved). When such an event contains an `iframe path`, the client updates the `custom conceptual model url`, thereby redirecting the conceptual modeling pane to a more specific viewer (such as a ViEnNa editor). As a result, the conceptual modeling pane always reflects the latest model produced by the agents. Additionally, the client listens for `simulation assistant set-situation` events emitted by the Simulation Assistant tool. These events can select the next situation either directly by index or indirectly by title or conceptual model identifier. Upon receiving such an event, the client updates the `selected situation index`, ensuring that the simulation iframe, dropdown selector, and conceptual modeling pane are all synchronized with the agent-driven scenario switch.

Transcript, events, and session logging. The central column hosts the `Transcript` component, which renders user and agent messages. The component receives the current user text buffer, handlers for sending messages, and a callback for downloading recorded audio. The `Events` pane provides a structured view of low-level SDK events, connection state changes, and guardrail logs. Higher-level interactions are recorded using `log client event` and `log server event`, enabling fine-grained analysis of the session in post-hoc evaluations. Furthermore, a custom hook `use handle session history` is invoked to persist and restore conversation-level information (e.g., for debugging or user studies).

User interaction modes and push-to-talk. The bottom toolbar exposes controls for connecting and disconnecting the session, toggling PTT, expanding or collapsing panes, and adjusting audio playback settings. The client maintains explicit state for PTT activation, whether the user is currently speaking, and whether audio playback is enabled. When PTT is activated, the client sends events to clear and commit audio buffers, and triggers responses via `response create`. When a text message is sent, the client first interrupts any ongoing agent speech and then invokes `send user text`. Each simulated user message also results in a transcript entry and corresponding SDK events, ensuring that both textual and audio-based interactions are captured uniformly.

Local preference persistence and audio recording. To provide a stable user experience across sessions, the client persists several UI and interaction preferences in local storage, including PTT mode, events pane visibility, audio playback state, and whether the SUI and conceptual modeling panes are expanded. These preferences are restored on mount via `use effect` hooks and updated whenever the underlying state changes.

The audio pipeline is further extended with a recording mechanism. Once a realtime session is connected and the hidden audio element exposes a remote media stream, the client invokes `start_recording` from the custom use audio download hook. When the component unmounts or the session ends, `stop_recording` is called. The Transcript UI exposes a „download recording“ affordance that allows the user to export the captured audio, which is useful for later analysis of interaction quality and tacit cue expression.

Access to ViEnNa editor and multimodal analytics. Finally, the top toolbar provides two buttons that bridge the realtime interaction interface with the conceptual modeling and data analysis tooling. The „Open“ button for ViEnNa forces the conceptual modeling pane to load the ViEnNa editor in expanded mode; this enables direct inspection and refinement of ViEnNa comic models generated by the agents. The „Multimodal Data Analysis“ button opens a dedicated analytics page in a new browser tab, providing access to dashboards and tools for inspecting video, audio, and other multimodal data sources associated with the current scenario.

B.2 Mixed Reality Simulation Environment

To support egocentric elicitation and counterfactual exploration of the system under investigation, this thesis implements a browser-based Mixed Reality simulation environment using `three.js`. The simulation is deployed as a standalone HTML page that integrates a 360-degree video panorama with an embedded digital user interface frame, allowing domain experts and agents to *stand inside* the environment while simultaneously interacting with the corresponding digital systems.

The core of the environment is a spherical panorama constructed from a sphere geometry whose normals are inverted so that the 360-degree video texture is rendered on the inner surface. The video is provided via a hidden HTML video element and wrapped as a video texture with sRGB ("standard Red, Green, Blue") color space, which is then applied to a basic material and mapped onto the sphere. The camera is a perspective camera positioned inside this sphere, and its orientation is controlled through a combination of manual pointer interaction and orbit controls with damping enabled. Pointer events (`pointer down`, `pointer move`, `pointer up`) adjust yaw and pitch angles, enabling users to rotate the viewpoint and explore the surrounding emergency room context as if they were physically present.

To avoid rendering purely cinematic video, the environment also includes a structured, room-like scaffold and lighting. A large wireframe box, constructed via edges geometry and line segments, suggests spatial boundaries, while a hemisphere light provides basic illumination and depth cues. This combination of 360-degree background and abstract room geometry reinforces the impression of being situated within a real, yet instrumented, environment.

A key design element is the integration of a digital system frame directly into the three-dimensional scene. A rectangular plane and an extruded frame geometry are added in front of the camera to serve as a „monitor“ embedded in the environment. On top of this frame, a `CSS 3D Object` wraps an `HTML iframe` element, whose content is set to the corresponding digital SUI page (e.g., triage chat, EHR form, surveillance dashboard). The `CSS3D` renderer and the `WebGL` renderer are overlaid: the `WebGL` renderer draws the panorama, frame, and room, while the `CSS3D` renderer places the live `HTML iframe` at the exact same position and orientation. When the user rotates the camera, both renderers are updated synchronously, preserving the impression that the digital UI is anchored inside the 3D scene. To maintain usability, pointer events to the `iframe` are temporarily disabled while the user is rotating the view and re-enabled once camera movement stops. The simulation supports multiple situational configurations, each defined by a triplet of *title*, *video path*, and *iframe path*. These are loaded from a shared `situations` catalogue and flattened into a `situation list`. The `HTML` overlay in the bottom-right corner provides a compact navigation UI: a title label and three buttons for moving to the previous situation, toggling video playback, and moving to the next situation. The `load situation` function updates the title text, switches the video source of the 360-degree panorama, and sets the `iframe` source to the matching digital SUI page. Button states are updated to prevent navigation beyond the bounds of the situation list, and a play/pause button reflects the current video state.

To enable tight integration with the rest of the framework, the simulation page can also be parameterised via URL query parameters. On load, the initializing function inspects the `title`, `video path`, and `iframe path` parameters. If they describe an existing situation, that situation is moved to the front of the list and becomes the initial view; otherwise, a new situation entry is prepended. This mechanism allows the main React-based client (Section B.1) and the Simulation Assistant agent to open the simulation with a specific scenario already active, ensuring that the 360-degree context, digital SUI, and agent conversation are aligned from the first frame.

B.3 ViEnNa Editor and Renderer

To make Virtually Enhanced Narratives practically usable as a conceptual modeling language, this thesis introduces a dedicated web-based editor and renderer. The ViEnNa editor is implemented as a split-view `HTML` application: the left pane hosts a code editor, while the right pane displays an immediate visual rendering of the comic-style panels that constitute a ViEnNa model. This environment allows modelers, domain experts, and AI agents to iteratively refine ViEnNa diagrams and observe their effect on the narrative and spatial layout in real time.

At the top of the interface, a compact header bar provides global actions. The *Load* button fetches a `.vn` file from the `models` directory (optionally specified via a URL query parameter), the *Download* button exports the current code as a file, *Render* forces an immediate re-render of the preview, and *Open in Popup* opens the current comic in

a separate browser window suitable for presentation or side-by-side elicitation sessions. A status indicator reports the outcome of operations such as loading, rendering, and downloading, and is styled to visually distinguish success and error conditions. The editor pane uses Monaco to provide a lightweight textual environment for editing ViEnNa source files. On initialization, the editor is configured in a dark theme with word wrapping, a minimap, and automatic layout. When a file is loaded via HTTP, its contents are injected into the editor. Any subsequent edits trigger a debounced re-render: as the user types, a timer is reset, and after one second of inactivity the `update panels` function is called to regenerate the visual comic. This provides a balance between responsiveness and performance, and a keyboard shortcut (`Ctrl+S` / `Cmd+S`) offers an explicit „render on save“ affordance.

The ViEnNa syntax is parsed by a simple line-based function `parse markdown`. Each panel is introduced by a heading line starting with `#`, which defines a panel identifier. The following lines within that panel block use `key: value` pairs to define attributes such as `color`, `background`, `width`, `topText`, `bottomText`, `situation`, `yaw`, `pitch`, `zoom`, `bubble`, `visual`, and `graphviz`. `bubble` entries encode speech bubbles as textual content plus x and y coordinates, whereas `visual` entries reference additional image overlays (e.g., icons, pictograms) together with their placement and size. `graphviz` entries allow embedding of small process-like structures within a panel, expressed as strings. The renderer translates the parsed panel attributes into DOM elements inside the comic container. Each panel becomes an `html` element with class `panel` and optional inline styling to reflect its configured color, background, and width. Narrative labels such as `top text` and `bottom text` are rendered as skewed, border-styled text ribbons anchored in the corners of the panel. Speech bubbles are positioned absolutely within the panel, with custom CSS creating the characteristic balloon shape and tail. Additional visual elements specified via `visual` attributes are rendered as absolutely positioned `html image` elements, effectively allowing the modeler to compose a multimodal storytelling canvas from text, icons, and background cues.

Beyond static layout, the ViEnNa editor integrates closely with the Mixed Reality subsystem. When a panel defines a `situation` attribute (e.g., `event_00001`), the renderer calls `apply situation background`, which retrieves a corresponding event `cofig` file from the `multimodal event logs` directory. This file contains camera and asset metadata such as `yaw`, `pitch`, `zoom`, image paths, and frame paths. These values are translated into URL parameters passed to a dedicated Mixed Reality viewer page. The panel then receives a full-background `iframe` pointing to this viewer, inserted as the first child of the panel so that it sits visually behind text and bubbles. The `iframe` is non-interactive in the embedded view (pointer events disabled), preserving the comic affordance while still conveying the precise camera framing and context of the original Mixed Reality capture. Clicking on such a panel opens the same viewer in a pop-up window, allowing in-depth exploration of the underlying 360-degree scene. The Graphviz integration provides another dimension of semantic enrichment. When a panel contains a `graphviz` attribute, the ViEnNa renderer creates a child `html` element and uses

`d3-graphviz` (built on top of `viz.js` and `CanvasKit`) to render the DOT specification directly within the panel. This makes it possible to embed miniature process fragments, decision structures, or data relationships directly into narrative panels, visually binding local tacit cues (e.g., gaze, posture, dialogues) with formal structures (e.g., gateways, message flows, state transitions). The „Open in Popup“ functionality is designed for collaborative modeling and presentations. When invoked, the editor constructs a self-contained HTML document containing the current comic markup and styling, then opens it in a separate window. A polling mechanism (`poll` file for changes) can be activated to periodically fetch the `current` file content from disk and re-render the popup if the underlying file changes, effectively supporting an external viewer mode synchronised with ongoing edits.

In summary, the ViEnNa editor and renderer form the interactive workbench for creating and maintaining comic-like conceptual models. They connect (i) a textual, panel-based ViEnNa syntax, (ii) a live visual comic rendering, (iii) embedded process fragments via `Graphviz`, and (iv) Mixed Reality situation backgrounds driven by event capture files. This tight integration enables modelers and AI agents to express, refine, and visualise both explicit process structures and situated, tacit aspects of the system under investigation within a single, coherent modeling environment.

B.4 Event Logging

A central requirement of the platform is the ability to capture *situated* events that align what happens in the Mixed Reality simulation, the digital SUI, the ViEnNa narrative, and the agent conversation. To support this, the implementation provides a dedicated „Log Now“ mechanism that can be triggered both by the human user (via a prompt or an UI button under the multimodal data analysis toolbar) and by AI agents (via a callable tool). Each invocation creates a structured event record that is stored in an event log and later mined using `pm4py`.

Conceptually, a „Log Now“ event acts as a crossmodal snapshot: it captures the current simulation situation (scenario identifier, video path, iframe path), camera parameters (e.g., yaw, pitch, zoom, frame), active conceptual model context, and conversational metadata (active agent, user/agent turn, current transcript window). When the user or AI triggers the „Log Now“ function in the interface, the client aggregates this information from the React state and from the embedded iframes, then serialises it into an event object. When an agent calls the corresponding tool, the same mechanism is triggered programmatically, allowing automatic logging whenever a crossmodal embedding detects a salient change. Under the hood, the crossmodal embedding pipeline continuously analyses the simulation and associated multimodal streams (e.g., 360° video, digital SUI interactions, transcripts). When an embedding-based detector recognises a change of interest—such as a transition from low to high urgency, a shift in who is speaking, or a domain-relevant visual cue—it may request an event capture by invoking the „Log Now“ tool. This ensures that the event log does not only contain regular, time-driven samples

but also *semantically aligned* markers at moments that are important for understanding tacit decision-making (for example, when the receptionist silently prioritizes a patient).

Each event is stored in a process-mining-friendly format with at least the following fields: *case identifier* (linking multiple events belonging to the same simulated encounter), *activity label* (describing the conceptual action, such as „soft triage adjustment“ or „insurance escalation“), *timestamp*, *situation identifier*, *camera state* (yaw, pitch, zoom, frame), *active model reference*, *agent role*, and optional embedding vectors or cluster identifiers. In practice, the event schema was kept compatible with standard pm4py expectations (case id, activity name, timestamp), while additional multimodal attributes were added as extra columns. Once a sufficient number of interactions have been captured, the event log can be imported into pm4py for process discovery, conformance checking, and performance analysis. The crossmodal attributes enable richer analyses than purely symbolic logs: for example, one can correlate discovered process variants with specific camera perspectives, simulation situations, or tacit „soft triage“ cues encoded in the embeddings at the moment of logging. This allows the platform not only to support realtime elicitation but also to *mine* how tacit signals and Mixed Reality interactions shape the emergent process structure, thereby closing the loop between simulation, conceptual modeling, and process mining.

Supplementary Evaluation Details

This appendix provides extended evaluation material that complements the validation results presented in the main body of the thesis.

Section C.1 formalizes and decomposes the evaluation metrics used throughout the thesis, including detailed operationalizations of precision and conciseness, alignment measures, convergence indicators, and simulation fidelity metrics. Section C.2 expands on the computational and architectural aspects of scaling multimodal data analysis in the context of the pipeline introduced in Chapter 6.

The subsequent sections provide component-specific supplementary evaluations aligned with their respective main-text counterparts. Section C.3 details the empirical validation of MMeCM (as introduced in Section 6.1) with additional cross-scenario and cross-language analyses. Section C.4 provides deeper experimental insight into embedding alignment, semantic synchronization, and ablation studies for ALeKS (as introduced in Section 6.2). Section C.5 elaborates on refinement dynamics, iteration stability, and knowledge integration behavior in BICycle (as introduced in Section 6.5). Section C.6 reports extended structural faithfulness and semantic consistency evaluations for PN-VidGen (as introduced in Section 7.4). Finally, Section C.7 presents additional elicitation effectiveness studies and interpretability analyses for ViEnNa (as introduced in Section 7.5).

C.1 Evaluation Metrics Breakdown

While the main chapter reports aggregated scores and comparative findings across conditions, the present supplementary material offers a transparent and fine-grained view of the exact criteria that guided the assessment of model quality.

For each modeling language and emergency-room scenario, the appendix enumerates all *precision* and *conciseness* questions used by evaluators. These items operationalize the abstract quality dimensions defined in Section 9.3.5 into concrete, scenario-specific

checks that can be directly applied to the conceptual models produced in the study. To aid reproducibility, each question is accompanied by a brief explanation describing (i) how it influences the quality dimension it belongs to, and (ii) how it should be verified against a specific multimodal ER scenario, including its observable behaviors, system interactions, and tacit cues.

By providing these detailed metric tables in a consolidated form, the appendix ensures that the evaluation procedure is both rigorous and fully auditable. Researchers, reviewers, and practitioners can use this material to replicate the scoring logic, compare individual models against the criteria, or adapt the metric set for use in similar multimodal conceptual-modeling contexts. The tables also serve as reference material for understanding how each modeling language captures different aspects of emergency-room work practices, from procedural logic and temporal interaction patterns to tacit decision-making and multimodal situational awareness.

BPMN — Patient Check-In & Registration

Metrics Interpretation. The BPMN metrics evaluate how well the model captures the real-world logic of patient registration in an Emergency Room. **Precision** focuses on whether the diagram represents the correct actors, data-handling tasks, decision gateways, and emergency pathways in alignment with actual ER procedures. It examines whether the model correctly separates human and system responsibilities, uses BPMN constructs as intended, and reflects the substantive domain rules a receptionist must follow when handling identity verification, EHR lookup, or insurance checks. **Conciseness** assesses whether the BPMN model avoids unnecessary detail—extra lanes, redundant gateways, overly fine-grained tasks—and whether the diagram remains readable while preserving essential logic. Evaluators compare the conceptual model against multimodal video evidence, digital logs, and domain descriptions from the scenario to determine whether each question is satisfied.

Sequence Diagram — Triage Coordination & Multimodal Cues

Metrics Interpretation. For sequence diagrams, **precision** requires accurate temporal ordering of communications among Patient, Receptionist, Triage Nurse, HIS, CCTV sensors, and the AI assistant. Because triage unfolds dynamically, evaluators must verify that message timing aligns with actual events captured in video and audio evidence. The inclusion of multimodal cues—gesture, tone, gaze—reflects tacit decision-making. **Conciseness** examines whether the diagram uses lifelines, messages, and combined fragments, avoiding redundant updates from systems or repeated human exchanges.

ViEnNa Comic Panels — Code Silver (Aggressive Individual)

Metrics Interpretation. For ViEnNa comic panels, **precision** focuses on narrative correctness, actor assignment, and fidelity to Code Silver protocol stages—from initial threat onset through silent alert, security response, and subsequent MR-supported

Table C.1: Evaluation criteria for BPMN models of the Patient Check-In & Registration scenario. Each criterion includes a follow-up explanation describing its relevance for model quality and how evaluators should verify it using the ER scenario.

Precision
1. Correct representation of key actors in appropriate lanes. <i>Impact: Ensures accurate responsibility modeling. Evaluators inspect whether all actions performed by the receptionist and the information system in the scenario are placed in their respective lanes without cross-role confusion.</i> 2. Decision points modeled with correct gateway semantics. <i>Impact: Guarantees accurate control-flow behavior. Check whether emergency triage decisions or insurance validity checks use XOR gateways with correct domain logic.</i> 3. Task labels unambiguously describe front-desk actions. <i>Impact: Prevents ambiguous interpretations. Verify that every task name corresponds to actions observed in the real registration workflow (ID verification, EHR lookup, insurance check).</i> 4. Start/end events reflect actual process initiation and completion. <i>Impact: Ensures correct lifecycle representation. Confirm that emergency fast-track cases bypass normal registration exactly as in scenario recordings.</i> 5. System-only activities assigned to the system lane. <i>Impact: Maintains conceptual clarity. Check whether automated processes (EHR retrieval, data validation) appear only under the corresponding system lane.</i>
Conciseness
1. Only essential lanes included. <i>Impact: Avoids cognitive overload. Ensure that no roles appear unless they perform observable actions in this scenario.</i> 2. Tasks at appropriate granularity. <i>Impact: Prevents overspecification. Steps such as clicking UI buttons should not appear unless they alter process logic.</i> 3. No redundant gateways or duplicated flows. <i>Impact: Promotes structural economy. Confirm that repeated control-flow branches are merged if behaviorally equivalent.</i> 4. Task names short yet informative. <i>Impact: Improves readability. Prefer “Verify insurance online” over long prose descriptions.</i> 5. Subprocesses used to encapsulate low-level details. <i>Impact: Reduces diagram clutter. Insurance procedures may be abstracted into a subprocess when internal steps are not needed for evaluation.</i>

reflection. Because ViEnNa diagrams merge textual cues, visual tension indicators, and sequential reasoning steps, evaluators must inspect whether panels accurately represent both explicit actions and tacit signals such as gaze shifts, posture, or situational awareness.

Table C.2: Evaluation criteria for Sequence Diagrams modeling triage coordination and multimodal cue interpretation. Each row explains how the criterion influences the model’s quality and how to validate it using triage recordings.

Precision
1. Correct temporal ordering of interactions. <i>Impact: Ensures event realism. Evaluators verify the order of triage steps using timestamps from multimodal recordings.</i>
2. Message labels clear and unambiguous. <i>Impact: Prevents semantic drift. Labels must reflect actual communication (e.g., symptom summary).</i>
3. System vs. human responsibilities separated. <i>Impact: Avoids misclassification. HIS notifications must not appear as human messages.</i>
4. Multimodal cues represented when relevant. <i>Impact: Captures tacit reasoning. Include gestures, urgency, or tone when they shape triage outcomes.</i>
5. Post-event MR elicitation modeled separately. <i>Impact: Preserves temporal structure. Reflective AI-supported reenactments should not merge with real-time communication.</i>
Conciseness
1. Lifelines limited to active participants. <i>Impact: Reduces visual load. Exclude roles that do not appear in this scenario instance.</i>
2. Repeated messages summarized. <i>Impact: Enhances clarity. Merge repeated HIS micro-updates.</i>
3. Combined fragments used effectively. <i>Impact: Avoids duplication. Use ALT/LOOP fragments for recurring structures.</i>
4. Message labels concise. <i>Impact: Supports readability. Avoid multi-clause messages.</i>
5. Post-event steps grouped where possible. <i>Impact: Streamlines model. Summarize secondary AI actions.</i>

Conciseness addresses whether the sequence is free from redundant panels, overly verbose descriptions, or unnecessary illustrative detail that does not contribute to conceptual understanding of the escalation flow. The evaluation is grounded in the actual Code Silver scenario used in the study, including CCTV feeds, protocol documents, and the reenacted MR simulation.

Mindmap — Code Amber (Infant Abduction Alert)

Metrics Interpretation. Mindmaps represent conceptual structure rather than behavioral flow. Thus, **precision** requires correct categorization of elements such as triggers,

Table C.3: Evaluation criteria for ViEnNa comic-panel models depicting Code Silver escalation. Each criterion includes an explanation of its effect on interpretability and how to validate it against multimodal video evidence and security protocol documentation.

Precision
1. Panels represent all Code Silver stages. <i>Impact: Ensures narrative completeness. Evaluators check that onset, alert triggering, security contact, and monitoring steps appear exactly as in the scenario.</i>
2. topText and bottomText match actual procedures. <i>Impact: Prevents procedural errors. Confirm that textual descriptions match hospital Code Silver guidelines.</i>
3. Speech bubbles assigned to correct actors. <i>Impact: Avoids misattribution. Compare each bubble with who actually speaks or signals in the scenario footage.</i>
4. Panel visuals reflect escalation tension. <i>Impact: Conveys tacit cues. Background tone and posture must reflect stress, urgency, or calm phases appropriately.</i>
5. Transitions match Code Silver flow. <i>Impact: Guarantees realistic escalation. Check that no impossible jumps appear (e.g., security confrontation before silent alert).</i>
Conciseness
1. Panels depict distinct conceptual moments. <i>Impact: Reduces redundancy. Remove duplicate emotional or procedural states.</i>
2. Text elements succinct. <i>Impact: Supports quick comprehension. Avoid long paragraphs within bubbles or panel descriptions.</i>
3. Appropriate number of panels. <i>Impact: Ensures proportional narrative. Too many panels may obscure key transitions.</i>
4. Decorative details minimized. <i>Impact: Prevents visual noise. Only include elements relevant for domain understanding.</i>
5. Mergeable panels combined. <i>Impact: Enhances flow. If two panels depict nearly the same state, they should be collapsed.</i>

response actions, system support, coordination channels, safety checks, and escalation routes, ensuring alignment with hospital Code Amber policy. Mindmaps must also reflect tacit behavioral aspects such as maintaining calm communication with caregivers. **Conciseness** relates to structural economy—avoiding branch proliferation, repeated leaves, and excessive depth that obscures conceptual clarity. Evaluators compare the mindmap to Code Amber protocol documents, real-world ER processes, and the multimedia scenario

capturing the emergency behavior.

Table C.4: Evaluation criteria for Mindmaps representing Code Amber escalation and coordination. Explanations clarify how each criterion influences conceptual clarity and how evaluators verify alignment with Code Amber documentation.

Precision
1. Major branches reflect Code Amber structure. <i>Impact: Ensures conceptual correctness. Validate branches against official categories such as Trigger, Response, Systems, Coordination.</i>
2. Leaves placed under appropriate parent branch. <i>Impact: Maintains semantic clarity. e.g., “Tag readers at exits”</i>
3. Human/system/channel distinctions clearly represented. <i>Impact: Prevents role confusions. Distinguish receptionist responsibilities from automated system checks.</i>
4. Critical safety steps included. <i>Impact: Ensures procedural completeness. Confirm lockdown procedures and police escalation appear explicitly.</i>
5. Tacit behaviors represented correctly. <i>Impact: Captures non-explicit expertise. Include maintaining calm tone and managing distressed family members.</i>
Conciseness
1. Branch labels short and general. <i>Impact: Prevents fragmentation. Avoid overly specific top-level branches.</i>
2. Overlapping leaves merged. <i>Impact: Avoids redundancy. Combine variants of “notify security.”</i>
3. Depth limited to meaningful levels. <i>Impact: Improves readability. Avoid unnecessary 4–5 level nesting.</i>
4. Low-level details omitted. <i>Impact: Removes noise. Exclude procedural minutiae irrelevant for conceptual structure.</i>
5. Node count manageable. <i>Impact: Ensures navigability. Confirm the mindmap is neither oversimplified nor bloated.</i>

User Journey — Insurance Verification & Financial Clearance

Metrics Interpretation. User Journey diagrams highlight how multiple roles and systems interact across time. **Precision** focuses on whether each step corresponds to an actual action in the insurance verification workflow, including fallback procedures (phone verification, fax requests), and whether tacit prioritization decisions are captured (e.g., accelerating verification for urgent cases). **Conciseness** addresses over-segmentation of steps, excessively long descriptions, and the presence of UI-level micro-actions that do

not contribute to conceptual understanding. Evaluators check the diagram against real ER insurance protocols and the recorded simulations, as in [Ont25].

Table C.5: Evaluation criteria for User Journeys depicting insurance verification workflows. Explanations show how each criterion impacts interpretability and how to check compliance against ER verification processes.

Precision
1. Each step reflects real verification actions. <i>Impact: Ensures domain realism. Steps must match actual insurance handling procedures seen in the scenario.</i>
2. Actors assigned to correct swimlanes. <i>Impact: Prevents actor-role confusion. Receptionist vs Insurance Portal vs Call Center.</i>
3. Branching between online and fallback processes correct. <i>Impact: Reflects real ER logic. Verify accurate modeling of degraded modes.</i>
4. Tacit decisions documented. <i>Impact: Captures implicit reasoning. Include urgency-based prioritization.</i>
5. Post-event AI/MR steps separate. <i>Impact: Avoids phase mixing. Reflective analysis appears after the main flow.</i>
Conciseness
1. Only essential steps included. <i>Impact: Prevents scenario inflation. Remove UI details unless relevant.</i>
2. Micro-interactions aggregated. <i>Impact: Enhances clarity. e.g., multiple intermediate form updates.</i>
3. Minimal but sufficient sections. <i>Impact: Improves structure. Use broad categories like Arrival, Verification, Fallback.</i>
4. Step descriptions concise. <i>Impact: Supports quick reading. Avoid paragraphs inside one step.</i>
5. Related steps merged. <i>Impact: Reduces redundancy. Combine repeated verification attempts.</i>

Flowchart — Ambulance Arrival & Handoff

Metrics Interpretation. Flowcharts model conditional and procedural logic. For the ambulance arrival scenario, **precision** requires accurate representation of the reception workflow triggered by incoming ambulances: alert reception, CCTV verification, communication with triage, resource preparation, and timing-dependent escalations. Each branching condition (e.g., “Ambulance detected?”, “Is patient critical?”) must match real-world ER procedures and the control flow seen in digital logs and CCTV footage. **Conciseness** evaluates diagram economy: clear action verbs, minimal node inflation, avoiding technical substeps (network retries, UI refreshes), and merging parallel or equivalent flows. Evaluators compare the model to the Mixed Reality reenactment,

triage protocols, and ambulance dispatch logs.

Table C.6: Evaluation criteria for Flowcharts modeling ambulance arrival and handoff processes. Each criterion includes guidance on why it matters and how to verify it using dispatch logs, CCTV evidence, and the recreated scenario.

Precision
1. Decision points modeled with correct yes/no semantics. <i>Impact: Ensures logical correctness. Verify that every decision node corresponds to an observable condition in the workflow (e.g., CCTV confirms ambulance).</i>
2. Distinction between system and human activities. <i>Impact: Avoids confusion. HIS-generated notifications should not be shown as receptionist tasks.</i>
3. MR/multimodal logging placed after clinical actions. <i>Impact: Preserves clinical-first sequencing. MR reconstruction occurs after triage confirmation.</i>
4. Dependencies between steps accurately represented. <i>Impact: Prevents impossible ordering. Resource preparation cannot precede patient condition confirmation.</i>
5. Tacit timing rules included where relevant. <i>Impact: Captures subtle workflow elements. For example, receptionist may respond to sirens before formal alerts appear.</i>
Conciseness
1. Node count minimized while preserving meaning. <i>Impact: Enhances readability. Avoid splitting trivial actions into multiple nodes.</i>
2. Action labels short and verb-first. <i>Impact: Improves clarity. Use “Prepare room” instead of long sentences.</i>
3. Irrelevant technical details omitted. <i>Impact: Reduces noise. Exclude backend system retries or internal system checks.</i>
4. Parallel or converging paths merged early. <i>Impact: Prevents branching overload. Merge equivalent flows after verification.</i>
5. System steps grouped appropriately. <i>Impact: Ensures abstraction. Combine multiple automatic HIS updates into a single “System processes case” node.</i>

Class Diagram — Visitor Management & Security

Metrics Interpretation. The class diagram for the visitor management captures static structural relationships among people, access rules, incidents, CCTV events, and AI insights. **Precision** evaluates whether classes, attributes, associations, and multiplicities reflect actual hospital security processes, including visit logging, badge issuance, restricted zones, and threat alerts. AIInsight and CCTVEvent classes must be connected to represent how automated pattern recognition augments security reasoning. **Conciseness** examines

unnecessary classes, overly detailed attributes, redundant associations, and misuse of inheritance. Evaluators cross-reference the diagram with ER visitor protocols, security logs, and examples from the MR reenactment.

Table C.7: Evaluation criteria for Class Diagrams describing visitor-management and ER security structures. Each criterion includes an explanation of how to validate it using security logs, badge records, and CCTV-linked incident workflows.

Precision
1. Classes and attributes aligned with real domain entities. <i>Impact: Ensures correctness. Confirm classes like VisitRecord, AccessRule, Zone match actual security procedures.</i>
2. Inheritance relationships domain-justified. <i>Impact: Avoids misuse of hierarchy. Check that Staff → Receptionist/SecurityOfficer reflects real organizational structure.</i>
3. Multiplicities accurately represent constraints. <i>Impact: Captures real cardinalities. One visitor can have multiple visit records; one AccessRule may apply to many zones.</i>
4. Attribute types and names meaningful. <i>Impact: Prevents ambiguity. riskScore, alertLevel, sensitivity must be typed correctly.</i>
5. All insight relationships meaningful. <i>Impact: Encodes tacit refinement. CCTV findings should adjust AccessRules or create Alerts.</i>
Conciseness
1. Only conceptually essential classes included. <i>Impact: Reduces clutter. Implementation tables (LogEntry, UserAccount) should not appear.</i>
2. Attributes limited to those influencing security logic. <i>Impact: Avoids over-detailing. Remove timestamps or technical metadata.</i>
3. Associations included only when semantically justified. <i>Impact: Prevents mesh-like structures. Avoid linking classes unnecessarily.</i>
4. Inheritance used sparingly. <i>Impact: Enhances model simplicity. Avoid trivial subclasses that differ only nominally.</i>
5. Merge auxiliary classes when overlapping. <i>Impact: Avoids redundancy. If AlertEvent and SecurityCase share structure, consider merging.</i>

State Diagram — Code Purple (Silent Distress Handling)

Metrics Interpretation. State diagrams model the dynamic evolution of silent distress cases (Code Purple), where tacit signals and escalating cues determine transitions between states such as Idle, Observation, Suspicion, Verification, CredibleThreat, and Resolution.

Precision requires that all states correspond to meaningful stages of ER threat assessment and that transition triggers mirror actual cues (e.g., written note, subtle gestural distress, CCTV inconsistencies). **Conciseness** demands that the diagram avoids redundant states, duplicate transitions, and overly fine-grained subdivisions that do not influence threat-handling logic. Evaluators align the model with scenario recordings and Code Purple protocol documents.

Table C.8: Evaluation criteria for State Diagrams representing Code Purple threat assessment. Explanatory rows show how to validate state definitions and transitions using ER silent-distress protocols and multimodal evidence.

Precision
1. States accurately reflect all meaningful phases. <i>Impact: Ensures scenario completeness. Include Observation → Suspicion → Verification sequence.</i>
2. Transitions triggered by domain-realistic events. <i>Impact: Maintains fidelity. Transition labeled “note received” must match actual scenario detail.</i>
3. Initial and terminal states correct. <i>Impact: Defines system lifecycle. Idle as initial; Resolution or FalseAlarm as terminal.</i>
4. Escalation vs. de-escalation paths clearly differentiated. <i>Impact: Reflects real security logic. Movement toward CredibleThreat distinct from FalseAlarm.</i>
5. Tacit behaviors encoded where relevant. <i>Impact: Captures subtle interactions. Include state transitions reflecting calm conversation while monitoring.</i>
Conciseness
1. Minimal number of states while complete. <i>Impact: Prevents state inflation. Avoid unnecessary intermediate states.</i>
2. No duplicate transitions. <i>Impact: Reduces clutter. Remove parallel identical transitions.</i>
3. State names short but expressive. <i>Impact: Aids readability.</i>
4. Merge states without behavioral significance. <i>Impact: Avoids complexity. E.g., merge minor “AwaitingCheck” with Verification if functionally identical.</i>
5. Resolution sub-states grouped. <i>Impact: Simplifies diagram. Avoid multiple fine-grained post-resolution states.</i>

Entity–Relationship Diagram — Lost Belongings & Investigation

Metrics Interpretation. ERDs capture data structures central to lost-item reporting and investigation, including entities such as PERSON, REPORT, ITEM, LOCATION, CAMERA, STORAGE_LOCKER, and AI-generated MATCH events. **Precision** requires that these reflect real ER workflows: how reports are logged, how multiple CCTV perspectives link to evidence, how custody is tracked, and how automated matching links footage to items. **Conciseness** assesses entity economy, elimination of redundant linking tables, and avoidance of unnecessary attributes that do not affect conceptual understanding. Evaluators cross-check with CCTV logs, lost-item forms, and security case reports.

Gantt Chart — ER Evacuation Coordination

Metrics Interpretation. The Gantt chart represents time-structured emergency evacuation procedures. **Precision** assesses whether tasks, durations, overlaps, and dependencies reflect ER evacuation policies: detection, classification, broadcast, zone-by-zone clearance, assisted evacuation, roll call, and system restoration. Evaluators must verify that parallelism is used correctly (e.g., assisted evacuation overlaps with general evacuation) and that critical safety milestones (All-Clear, final accountability) appear. **Conciseness** focuses on appropriate abstraction, merging overly fine-grained steps, grouping tasks into major phases, and reducing redundant sub-tasks that do not affect critical path timing.

C.2 Scaling Multimodal Data Analysis

Emergency care involve structured interactions and controlled environments. To push the limits of multimodal capture, synchronization, and tacit-behavior extraction, we conducted the *Prater3* study—a large-scale, high-variance real-world case involving wearable sensors, mobile devices, multimodal video/audio, and dynamic, unpredictable ride environments.

This section is adapted from the evaluation by Gavric et al. [GBP25g].

The Prater3 study evaluates the robustness in a setting characterized by:

- rapidly changing physical contexts (fast-moving rides),
- heterogeneous modalities with irregular sampling rates,
- complex visual scenes with occlusions, motion blur, and structural elements,
- ambient noise, crowds, and constrained field-of-view, and
- safety-critical processes involving compliance and regulatory constraints.

Table C.9: Evaluation criteria for Entity–Relationship Diagrams modeling lost belongings and investigation workflows. Explanations describe how each criterion should be validated using CCTV evidence, storage procedures, and case file logic.

Precision
1. Entities aligned with actual workflow. <i>Impact: Ensures domain realism. Required entities: REPORT, ITEM, CCTV_FOOTAGE, STORAGE_LOCKER, CUSTODY.</i>
2. Attributes correctly typed and meaningful. <i>Impact: Prevents ambiguities. status (enum), zone (string), confidence (float).</i>
3. Cardinalities correspond to real constraints. <i>Impact: Reflects how cases work. One REPORT may reference multiple ITEMS.</i>
4. Evidence vs. case management concepts clearly separated. <i>Impact: Ensures semantic clarity. ITEM is not CCTV_FOOTAGE; REPORT is not STORAGE.</i>
5. Enumerations aligned with hospital terminology. <i>Impact: Ensures consistency. Use actual ER codes (e.g., “logged”, “recovered”, “destroyed”).</i>
Conciseness
1. Entities restricted to workflow-relevant concepts. <i>Impact: Reduces noise. Remove dev-oriented tables like UserAuth.</i>
2. Attributes limited to essential investigatory fields. <i>Impact: Prevents over-modeling. Exclude timestamps unless needed for chain-of-custody events.</i>
3. Many-to-many relationships used sparingly. <i>Impact: Avoids unnecessary join entities. Only use LINKS when semantically required.</i>
4. Descriptions concise and semantically focused. <i>Impact: Aids comprehension. Avoid verbose attribute naming.</i>
5. Overlapping entities merged. <i>Impact: Simplifies schema. e.g., combine ACCESS_LOG and SECURITY_LOG if they overlap.</i>

Overall, Prater3 demonstrates that the proposed multimodal evidence extraction, event-log generation, and process-mining pipeline scale effectively to high-intensity, outdoor, sensor-rich environments.

C.2.1 Case Study Overview and Experimental Setup

The *Prater3* dataset was collected at the Prater amusement park in Vienna¹ with the goal of evaluating *Process Adversarial Mining* (PAM) [GBP25g] in real outdoor conditions

¹<https://www.praterwien.com/en/home>

Table C.10: Evaluation criteria for Gantt charts representing ER evacuation timelines. Explanatory rows describe how each item should be validated using evacuation guidelines, timing constraints, and MR reenactments of evacuation procedures.

Precision
1. Tasks correctly ordered and dependent. <i>Impact: Ensures chronological correctness. Detection precedes broadcast; evacuation precedes roll call.</i>
2. Task durations reflect real constraints. <i>Impact: Ensures realism. Compare durations with ER emergency procedures.</i>
3. Phases aligned with operational structure. <i>Impact: Ensures conceptual clarity. Group tasks into detection, broadcast, evacuation, recovery.</i>
4. Parallel tasks represented correctly. <i>Impact: Reflects concurrency. Assisted evacuation runs in parallel with zone clearances.</i>
5. Safety milestones clearly indicated. <i>Impact: Maintains critical-path accuracy. All-Clear and final accountability checkpoints required.</i>
Conciseness
1. Minimal yet complete task list. <i>Impact: Avoids timeline clutter. Combine micro-tasks.</i>
2. Short, action-oriented task names. <i>Impact: Improves readability. Avoid sentences; use “Lock doors,” “Evacuate Zone A”.</i>
3. Fine-grained details aggregated. <i>Impact: Removes noise. Collapse multi-step door checks.</i>
4. Sections applied effectively. <i>Impact: Provides structure. Avoid unnecessary segmentation.</i>
5. Merge similar tasks. <i>Impact: Streamlines chart. Combine equivalent zone evacuations when appropriate.</i>

that demand large-modality integration. The resulting dataset supports multimodal data analysis for conceptual modeling purposes, allowing detailed process mining, anomaly detection, root-cause analysis, and conformance checking.

The study produced three cases—*Praterturm*, *Insider*, and *Olympia*. Each case captures a full end-to-end scenario of ticketing, safety protocols, ride preparation, execution, and post-ride recovery.

All participants were briefed and provided consent, with guarantees on anonymity and compliance with park recording policies.

This mixed sampling allowed PAM to perform multimodal segmentation and event-log

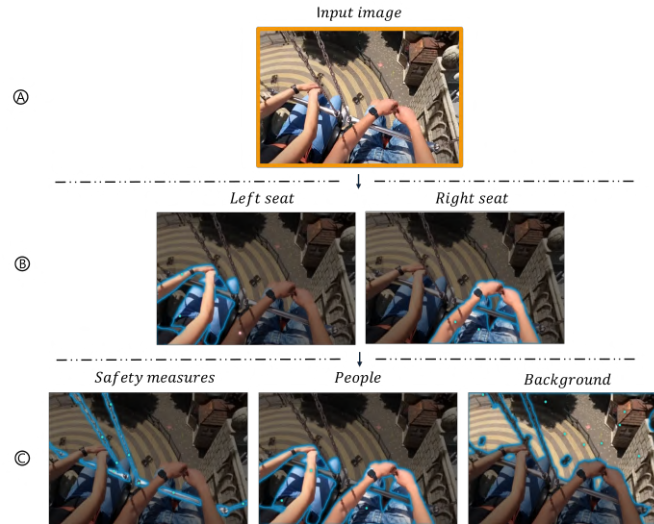


Figure C.1: Extraction of evidence instance data (EVDI, INST) for a focus map (B) right and left seat, and (C) structural components (safety measures, people, background) for a given input image (A). Adopted from Gavric et al. [GBP25g].

alignment even under heavy motion, noise, and sparsity.

C.2.2 Prater3 Pipeline

Using the synchronized dataset, PAM constructed multimodal event logs integrating visual, audio, and sensor cues.

Anomaly detection identified deviations from expected process execution, such as incorrect seating behavior or unverified safety steps.

Root cause analysis traced anomalies back through sensor and visual cues.

Conformance checking measured alignment between observed actions and modeled procedures, paralleling the methodology established in earlier chapters.

Fig. C.2 illustrates conformance checking for cable alignment, allowing anomaly detection related to mechanical wear or misalignment.

C.2.3 Scaling Through High-Density Sensor Data

Ten synchronized smartphone sensor modalities were collected, including the accelerometer, gyroscope, and orientation (roll, pitch, yaw), as well as the barometer, magnetometer, ambient light sensor, GPS coordinates, gravity vector, total acceleration, and ambient audio captured via the microphone.

Together, these modalities serve for the reconstruction of precise ride trajectories, rotation and swing dynamics, indoor–outdoor transitions, and motion–perception interactions.

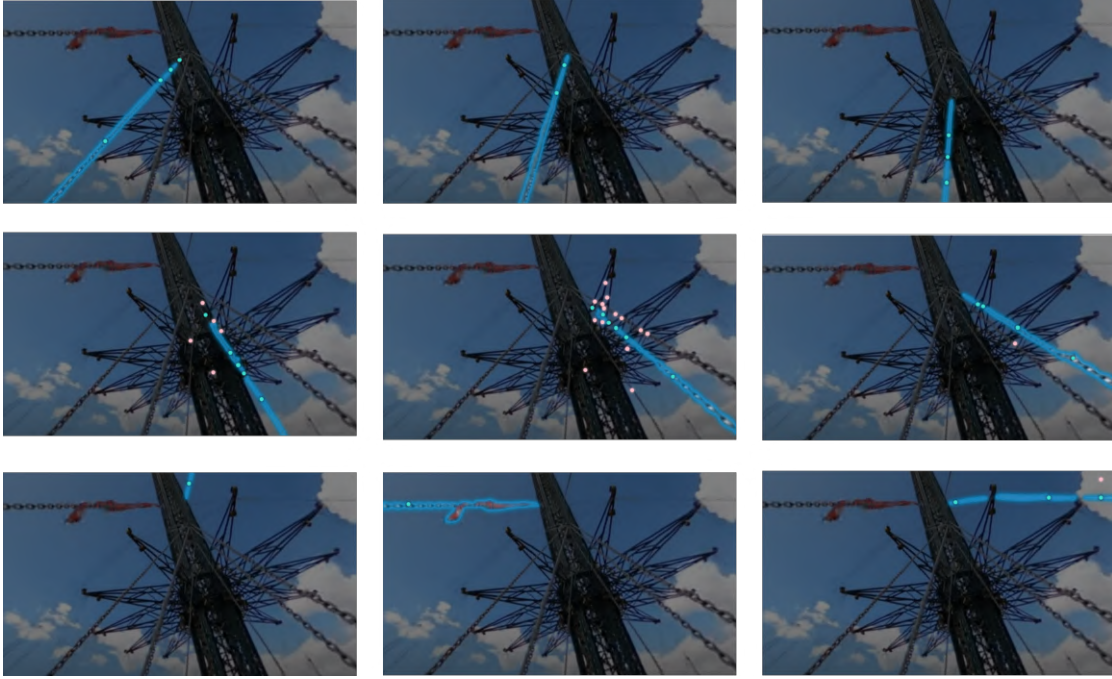


Figure C.2: Conformance checking for cable positions based on expected structural deformation envelopes. Adopted from Gavric et al. [GBP25g].

Fig. C.3 shows an example of multimodal sensor analytics for the *Praterturnm* case.

C.2.4 Physiological Scaling: Heart-Rate Analytics

In addition to ride mechanics, smartwatch heart-rate monitoring provided physiological context for emotional and physical responses.

Heart-rate signals help correlate expected ride stressors with anomalous physiological spikes, as well as with behavioral deviations and potential safety issues. An example of the collected data is represented in Figure C.4.

The inclusion of physiological data demonstrates the scalability of the framework toward personal-analytics modalities, thereby reinforcing its capacity for multimodal generalization.

C.3 Evaluating MMeCM

In this section, we discuss the scalability, performance, cost of operation, security and privacy concerns, as well as the systematic evaluation of MMeCM framework [GBP25i] through both quantitative and qualitative analyses. We first provide an empirical analysis of the dataset and retrieval performance through similarity matrices. Then, we conduct

C. SUPPLEMENTARY EVALUATION DETAILS

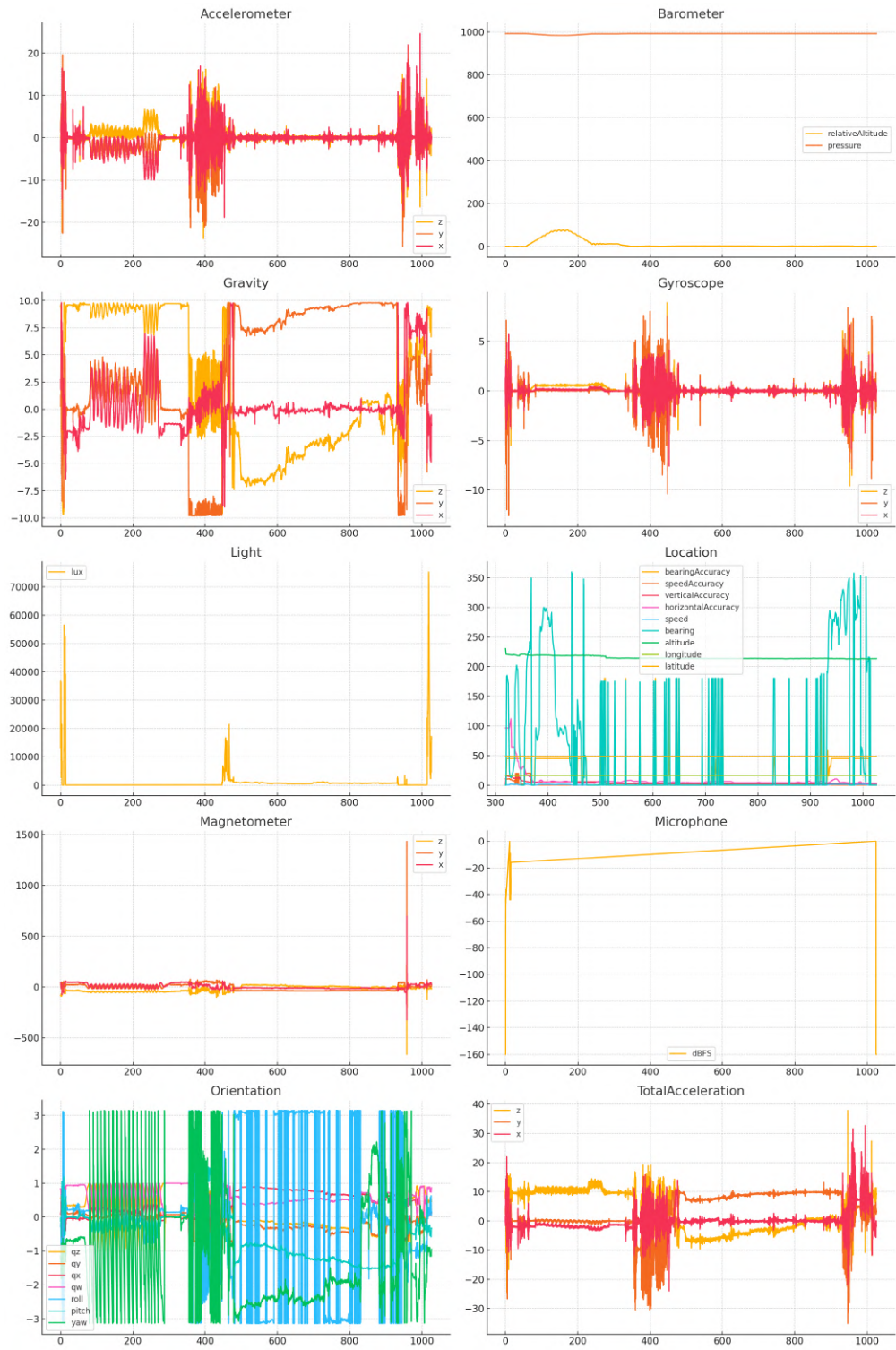


Figure C.3: Case study analytics for Praterturn integrating multimodal sensor signatures with process events. Adopted from Gavric et al. [GBP25g].



Figure C.4: Heart rate dynamics across three Prater3 cases: Praterturn (A), Insider (B), Olympia (C). Adopted from Gavric et al. [GBP25g].

a user study to assess the practical usability and acceptance of our approach. Finally, we outline key findings, limitations, and threats to validity.

Scalability Evaluation. MMeCM framework is designed with modularity and extensibility in mind, ensuring that it scales well with increasing dataset sizes and additional modalities. Thanks to the use of chunked processing and embedding caching mechanisms, performance remains robust even when operating over tens of thousands of elements. During experimentation, we observed linear growth in both computation time and memory consumption relative to dataset size, which confirms predictable scalability for future expansions.

Performance Evaluation. One of the key strengths of our approach lies in its ability to deliver real-time performance. All computations, including embedding generation, similarity calculations, and retrieval tasks, are executed locally without reliance on cloud-based services and *APIs*. This architecture ensures minimal latency and allows for instantaneous feedback, making the solution viable for interactive use cases and deployments in latency-sensitive environments such as AR/XR applications.

Cost of Operation. By eliminating the need for external servers, cloud subscriptions, or internet connectivity during operation, our solution significantly reduces the cost of deployment and maintenance. The entire pipeline can be run efficiently on a modern consumer-grade laptop with GPU acceleration (we used *NVIDIA A40 GPU* with 48 GB GDDR6 ECC Memory), making it accessible for academic, industrial, or personal use. This low cost of ownership broadens the framework’s applicability across institutions with limited budgets.

Security and Privacy Concerns. Security and privacy were central considerations in our system design. Since all processing is performed locally, no sensitive data is transmitted over the internet or stored in third-party services. This architecture inherently mitigates risks associated with data leakage or unauthorized access, ensuring full compliance with privacy regulations such as GDPR (The General Data Protection Regulation, EU-2016/679). Furthermore, local execution allows users full control over their data, aligning with privacy-by-design principles.

C.3.1 MMeCM Quantitative Evaluation

We have structured created dataset into 15 distinct chunks, each containing extracted and filtered natural language elements (with exact duplicates removed) along with their corresponding multimodal embeddings. The dataset distribution across chunks is as follows: 1265, 702, 999, 1562, 571, 612, 691, 821, 627, 499, 981, 458, 843, and 669 elements per chunk, each corresponding to the number of unique natural language elements found in a 10 model subsets of OntoUML/UFO Catalogue. As these embeddings allow for similarity comparisons across six modalities using standard distance metrics such as Euclidean and cosine distance, [GBP25i] provide in supplementary material similarity matrices (per chunk, as there are 12,300 entries in total) where the diagonal represents self-similarity (value of 1), demonstrating consistency in embedding alignment.

To assess the relevance of multimodal retrievals in real-world scenarios, we consider three evaluation approaches: (A) *domain expert annotations*, where experts review real-life conceptual model examples and mark potential spots where multimodal data could enrich the model; (B) *large-scale pretrained language model annotation*, taking foundational pretrained large language model to simulate the annotation process based on approach (A), and finally, (C) *data-driven retrieval assessment*, by collecting multimodal data samples and evaluating the retrieval quality directly. We opt for a combination of (A) and (B). A golden dataset is created through domain expert annotations identifying multimodal enrichment spots in real-life conceptual models (through approach A). Since human annotation is time-consuming and susceptible to lapsus errors, we complement it with a large language model (LLM) generated dataset trained using golden dataset examples and annotation instructions (which is approach B, yet augmented with approach A result). For the expert annotation process, we developed an annotation tool as shown in Figure C.5. The tool allows annotators to mark and revise elements within conceptual models (approach A), specifying whether visual or auditory enrichment is relevant. Its initial suggestions are produced using an LLM that offers advanced reasoning capabilities in a more resource-efficient package (approach B). To generate automated annotations, we use the prompt given in the supplementary material.

C.3.2 MMeCM Qualitative Evaluation

To assess the practical usability of our approach, we conducted a user study grounded in the Technology Acceptance Model (TAM) [Dav86]. TAM is a widely adopted framework

Annotation Review: kostov2017towards

Diagrams

The Thinking:

<think>
First, I'll look at each label one by one and think about what they represent in a human-readable form. For example, "Aerodrome" might make more sense as "Airport". That seems straightforward. Next, for the visual group, I need to identify terms that could realistically be turned into icons without being too vague. Words like "Operation", "Procedure", or "Device" come to mind because they have clear visual representations, even if not actual images. For auditory matches, words that naturally have sounds associated with them would work best. "Landing" and "Takeoff" are action-oriented terms that could be captured by sounds in an audio representation.

Value:

, :has part, :has participant, :inheres in, :is provided by, :perceive, :violates, Action, Aerodrome, Aerodrome part, Aerodrome's Operation, Agent, Air traffic management, Aircraft, Aircraft Part, Approach, Audit, Audit Finding, Aviation Safety Event Type, Aviation and Safety Ontologies, Collision,

Business Domain: Transportation; **Number of Icon Matches:** 7; **Number of Visual Matches:** 5; **Number of Auditory Matches:** 2; **Number of Abstract Matches:** 13

Review Annotations

Aerodrome Visual Auditory Abstract	Agent Visual Auditory Abstract	Air traffic management Visual Auditory Abstract	Aircraft Visual Auditory Abstract
Collision Visual Auditory Abstract	Crew Visual Auditory Abstract	Impact Visual Auditory Abstract	Incident Type Visual Auditory Abstract
Landing Visual Auditory Abstract	Risk Visual Auditory Abstract	Regulation Violation Event Visual Auditory Abstract	Service Visual Auditory Abstract
Taxonomy Visual Auditory Abstract	Technical System Visual Auditory Abstract	TransportObject Visual Auditory Abstract	Use Visual Auditory Abstract

Figure C.5: Annotation tool used for expert-based multimodal enrichment identification. Its initial suggestions are produced using an LLM as described. Adopted from Gavric et al. [GBP25i].

in system adoption research, that posits that a user's acceptance of a technology is primarily influenced by two core factors: *Perceived Usefulness* (PU) and *Perceived Ease of Use* (PEOU). These, in turn, affect the user's *Attitude Toward Using the system* (ATT), which shapes their *Behavioral Intention to Use* (BI), ultimately leading to *Actual Use*.

In our study, we applied this structure to evaluate our approach: *Perceived Usefulness* (PU) — “Does MMeCM enhance your conceptual modeling tasks?”; *Perceived Ease of Use* (PEOU) — “Is MMeCM intuitive and user-friendly?”; *Attitude Toward Using* (ATT) — “Would you consider using MMeCM in your workflow?”; *Behavioral Intention* (BI)

Algorithm 6 Silver Evaluation-Dataset Generation. Adopted from Gavric et al. [GBP25i].

Require: Golden evaluation-dataset G , Conceptual Model M

Ensure: Silver dataset S **forall** *example* e *in* G **do**

1:

end

 Construct prompt p as follows:

 “You are given a list of names from an OntoUML model. Your task is to analyze the domain of the model and perform the following steps:

 1. Domain Identification: Determine the domain of the model based on the given names and describe it in a few sentences.

 2. Visual Modality Analysis: Count how many words from the list can be visually observed in an image (i.e., have a recognizable visual representation). Provide a list of such words.

 3. Auditory Modality Analysis: Identify how many words and which ones can have an auditory match (i.e., can be perceived through sound). Only count the words that represent objects or actors that can be heard.

 4. Adaptation Score Prediction: Estimate an `adaptation_score` between 0 and 1, representing how much the conceptual model would be enriched by incorporating multimodal data (visual and auditory).

 Return the results in a JSON file with the following structure: {structure}

 The input list is: {value from e }”

2: Generate LLM-based annotation by executing p on e

3: Append generated annotation to S

4:

5: **return** S = 0

— “Would you recommend MMeCM to colleagues?”. We engaged a diverse group of participants with varying levels of expertise in conceptual modeling: (A) two PhD holders in conceptual modeling, (B) three PhD candidates with over two years of experience in conceptual modeling, and (C) five students with foundational knowledge of conceptual modeling. Participants responded using a 5-point Likert scale.

C.3.3 MMeCM Discussion: Key Findings, Limitations, and Threats to Validity

Our evaluation yielded several key findings. Quantitatively, the multimodal embeddings demonstrated *consistent alignment in a multimodal embedding space*, as evidenced by the similarity matrices, validating the structural integrity of our dataset and retrieval mechanism. The combined annotation approach—leveraging domain experts and a large language model—showed that more than 68% of *natural language elements in conceptual models can be enriched with multimodal data*. Qualitatively, results from the TAM-based user study revealed a positive reception (4.2/5.0) toward MMeCM, particularly in terms of perceived usefulness and ease of use, with participants highlighting MMeCM’s potential to improve expressiveness and accessibility in conceptual modeling tasks.

Nonetheless, several limitations and threats to validity remain. First, the dataset is built on a specific modeling catalog (OntoUML/UFO), which may limit generalizability to other modeling paradigms. Second, while the LLM-assisted annotation complements expert input, it may introduce subtle biases based on prompt phrasing and training data. Third, the user study was conducted with a relatively small and academically-inclined sample, which might not fully represent practitioners in industry settings. Lastly, while MMeCM suggests multimodal enrichments, the evaluation does not yet measure downstream effects on task performance or decision-making quality. Further evaluation will address these concerns by expanding domain coverage, increasing participant diversity, and incorporating longitudinal usage studies.

C.4 Evaluating ALeKS

This section is adapted from Gavric et al. [GBP25a]. To support the manual inspection and interpretation of alignment results, we developed a visual comparison tool (previewed in Fig. C.6). The tool enables users to explore semantic alignment between NLT, CMT, and large language model (LLM) representations within a unified interface. It allows both developers and domain experts to verify the quality of the semantic shift functions and gain insights into embedding behavior.

The interface is composed of three main panels: (i) a structured browser for conceptual models categorized by domain (e.g., medicine, law, science), (ii) a visualizer for individual conceptual model diagrams, and (iii) an AI model comparison module. This last module supports the selection of embedding alignment types, including $NLT \rightarrow CMT$ and $LLM \rightarrow CMT$, and provides interactive plots to inspect embedding proximity and vector shifts. We use this tool in two key ways: first, to conduct manual verification of model-generated alignments by inspecting whether predicted CMT vectors correspond to their expected symbolic roles and locations; second, to enable exploratory analysis of how symbolic terms are recontextualized across different ontologies and embedding strategies. The interface also includes a conversational assistant that helps explain or inspect selected terms, further facilitating the alignment audit process. To support users in navigating

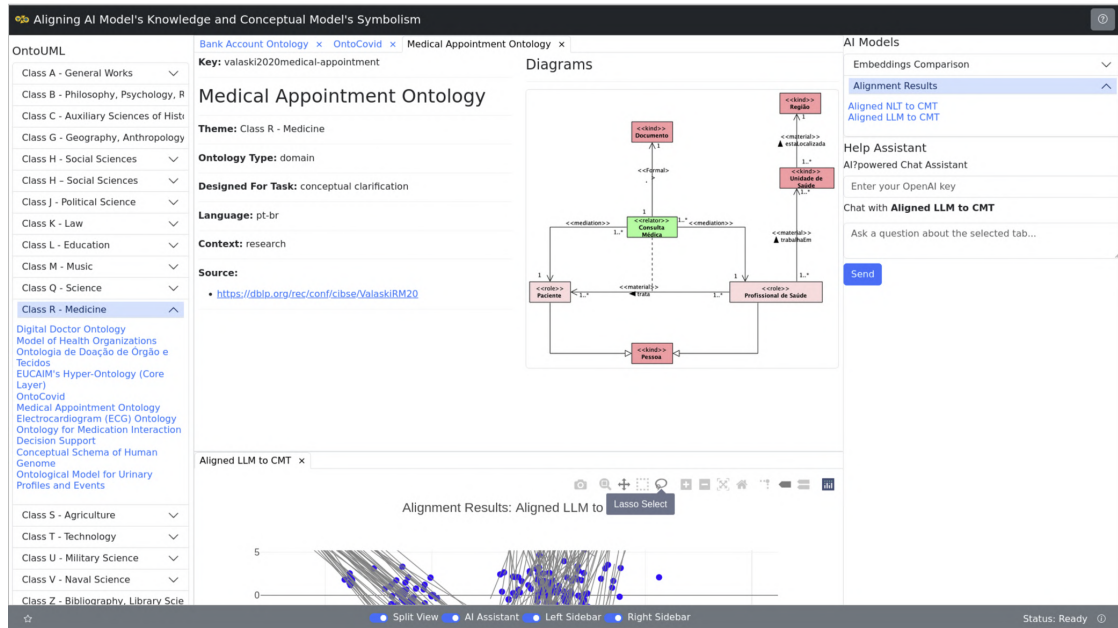


Figure C.6: Screenshot of the implemented visual inspection tool for alignment analysis. The user can browse conceptual model ontologies (left), view diagram semantics (top center), and explore alignment results (bottom). The AI comparison module (right) provides tools to inspect embeddings, check alignment quality, and interactively verify model serializations across NLT, CMT, and AI representations. Adopted from Gavric et al. [GBP25a].

the alignment process and deriving insights, we developed a custom *GPT*-based assistant integrated into the tool.

To validate our proposed methodology, we conduct a multi-perspective evaluation covering alignment quality, generalizability across modeling languages, human-centered assessments via the *Technology Acceptance Model* (TAM) [Dav86], and system-level performance and scalability.

C.4.1 Evaluation of the Alignment Model – AI Knowledge and Symbolism

To test if *ALeKS* leads toward reconstruction of symbolic knowledge from enriched conceptual model terms (CMT), we compare raw AI-generated summaries of OntoUML models to their predicted counterparts generated from serialized CMT variants. Specifically, we apply the following prompt using *llama3.3* to each **non-serialized** model in the OntoUML/UFO Catalogue:

```
prompt_template = """ OntoUML model:  serialization
```

Task: Analyze the OntoUML model provided in the serialization and generate a concise description that outlines *how elements are connected*.

Instructions:

Go through all internal element identifiers (IDs) and identify the relationships between them.

For each connection, preserve and use the natural-language names of the involved elements.

Summarize the connections between classifiers (e.g., classes, relators, roles) and describe their nature (e.g., mediations, generalizations, associations).

Output:

A structured description of the model's connectivity, focusing only on inter-element relationships.

Use OntoUML-relevant terminology (e.g., "relator mediates between", "role played by", "generalization from X to Y").

Avoid listing raw IDs, but ensure every named element mentioned is traceable through its name.

The output should serve as a compact summary of the model's structure through its relationships. ""

We then compute cosine similarity between the embeddings of these generated descriptions and the embeddings of ground-truth CMT serializations. Our evaluation reveals a significant increase in semantic alignment when using CMT-enriched prompts, showing higher cosine similarity with ground-truth model annotations than the raw NLT terms alone. This supports the hypothesis that CMT serves as an effective bridge between AI knowledge representations and symbolic model structures.

Alignment Accuracy. The average cosine similarity between the predicted embeddings and the true CMT representations was notably high at 0.94 , indicating that *ALeKS* effectively captures the structural semantics encoded in formal modeling templates.

Fig. C.7 visualizes the semantic transformation from NLT to enriched conceptual modeling terms (CMT) for the OntoUML/UFO Catalogue. Each arrow represents a shift vector $s(a \rightarrow a')$ projected into a 2D embedding space using PCA. The left cluster (NLT) contains more linguistically raw representations, while the right cluster (CMT) shows tightly grouped structurally enriched terms. The uniform directional flow from left to right reveals recontextualization: as terms are enriched with OntoUML semantics (e.g., kinds and roles), they move into a different semantical region of the space.

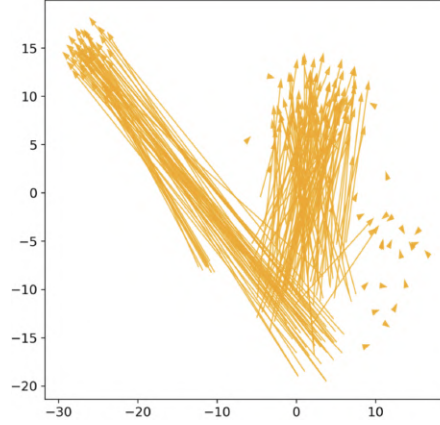


Figure C.7: **2D projection of semantic shifts in OntoUML/UFO Catalogue [BSF⁺22]**. Each arrow represents the vector shift $s(a \rightarrow a')$ from a natural language term (NLT) to its corresponding conceptual model term (CMT), based on OntoUML data. Embeddings are projected to 2D using. Adopted from Gavric et al. [GRP25a]

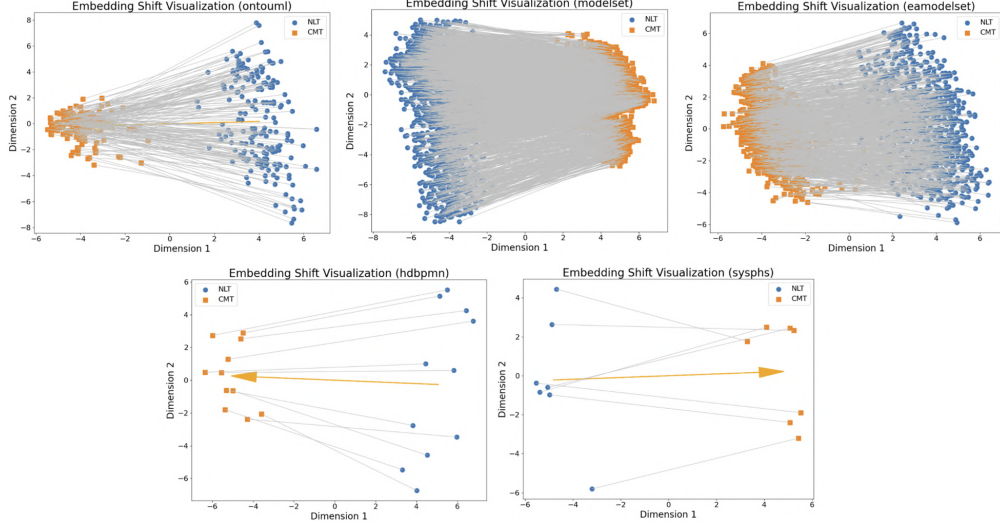


Figure C.8: **Embedding shift visualization across modeling languages**. Semantic shifts from natural language terms (NLT) to conceptual model terms (CMT) are shown across six repositories: *ontouml* [BSF⁺22], *modelset* [LCIC22], *eamodelset* [GSB23], *hdbpmn* [SvLS21], and *SysML PhS* [MBB23]. Each arrow represents a semantic shift vector $s(NLT \rightarrow CMT)$ in the shared embedding space. Adopted from Gavric et al. [GBP25a].

C.4.2 Cross-Conceptual Modeling Language Observations

To assess generalizability across modeling languages, we evaluate the semantic shift from NLT to their enriched conceptual modeling terms (CMT) across five conceptual models repositories: *ontouml* [BSF⁺22], *modelset* [LCIC22], *eamodelset* [GSB23],

hdbpmn [SvLS21], and *SysML PhS* [MBB23]. These shifts are visualized in Fig. C.8, showing consistent directional movement in the embedding space for each model collection.

In each case, NLT points (blue circles) shift toward CMT points (orange squares), indicating that conceptual enrichment introduces a measurable semantic transformation. For high-density model repositories (top row), shift vectors exhibit clustering, whereas for sparse or abstract models (bottom row), the shift vectors are fewer but semantically distinct. This provides evidence that the method may apply across languages with different granularity and modeling purposes.

C.4.3 Technology Acceptance Model Evaluation

We further evaluated the alignment tool and the AI support system through a qualitative *Technology Acceptance Model* (TAM) [Dav86] study. This involved interviews and surveys with a cohort of 10 participants: 2 PhD holders in Conceptual Modeling, 3 PhD students, and 5 advanced Bachelor students from the same domain. Each participant interacted with the tool and evaluated its usefulness and ease of use.

TAM defines three key constructs: *Perceived Usefulness* (U), *Perceived Ease of Use* (E), and *Behavioral Intention to Use* (BI). To assess these, we asked questions such as:

- "Does the system help you better understand model semantics through AI-generated summaries?" (U)
- "How difficult was it to interact with the embedding comparison interface?" (E)
- "Would you use this tool to explain models in your teaching or research?" (BI)

Results showed high agreement on perceived usefulness from experts, who noted the value of recontextualized model descriptions. Bachelor students scored slightly lower on ease of use, suggesting that onboarding might benefit from improved interface design or guidance. The overall behavioral intention was positive, especially when participants understood the value of CMT-based enrichment and its alignment with perceptual data.

C.4.4 ALeKS Discussion

Scalability and Performance The embedding pipeline, including multimodal alignment and CMT enrichment, was executed on a single NVIDIA A40 GPU with 48GB VRAM. This setup supports processing of thousands of terms across multiple modeling languages in parallel. Runtime performance scales linearly with the number of model elements, making it feasible to apply at enterprise or institutional scale.

Cost of Operation As the architecture relies solely on locally hosted models—embedding via Ollama and multimodal alignment through ImageBind [GENL⁺23]—there are no recurring cloud API costs. This significantly reduces operational overhead and allows researchers and educators to use the tool offline.

Privacy and Data Control All computation is executed locally, ensuring full control over sensitive model artifacts, especially in regulated industries like healthcare and defense. This eliminates concerns associated with sending confidential data to external AI APIs.

Evaluation on Multimodal Data To assess the multimodal alignment, we integrated ImageBind [GENL⁺23] embeddings, which support six modalities—image, text, audio, depth, thermal, and inertial measurement unit (IMU). Preliminary results on the DNA home kit scenario from [GBP24a] yielded a retrieval accuracy of 0.56 for matching modalities. While promising, performance remains limited due to the absence of a dedicated benchmark set. Future work will focus on expanding multimodal datasets and refining evaluation protocols to better capture alignment quality across sensory channels.

Limitations and Threats to Validity One limitation is the dependency on high-quality modeling artifacts; poor naming or inconsistent stereotypes in conceptual models can reduce alignment accuracy. Furthermore, while cosine similarity offers a proxy for alignment, it may not fully capture conceptual equivalence. Threats to validity include limited domain diversity and potential annotation bias in manually curated model collections.

C.5 Evaluating BICycle

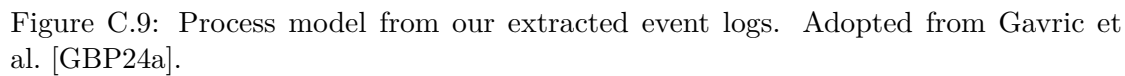
In this section, we report the evaluation for the BICycle framework [GBP24a]. It evaluates the effectiveness and efficiency of the process model generated from the event logs, which have been enhanced to improve completeness. Our evaluation focuses on three main aspects: *alignment with the official user manual*, *performance and scalability metrics*, and *manual validation of connected modalities*.

C.5.1 Alignment with User Manual

The process model was compared with the user manual provided by the DNA test kit company to ensure that it accurately represents the intended operations. This comparison was crucial as the user manual serves as the definitive guide for the process. Through systematic alignment techniques, we confirmed that the model mirrors the step-by-step instructions and sequences described in the manual. This validation ensures that our model not only captures all necessary components of the process but also adheres to the company’s operational standards, thereby improving the reliability of the model in practical scenarios.

Using Disco² for Process Mining, we analyzed DIY home kits in the domain of DNA collection (Figure C.9). We designed our ground truth in accordance with the provided user manuals by selected AncestryDNA and 23andMe official instructions provided with the kit.

²<https://fluxicon.com/disco/>



To assess the practical applicability of the enriching of the event logs, we evaluated its performance and scalability on a single A40 GPU with 48GB RAM. Results are given in Fig. C.10. The metrics used included process execution time, resource utilization, and scalability under varying loads. The results demonstrate that the model performs efficiently, with low latency, supporting real-time execution. The scalability tests indicate that our model can handle increases in workload without significant degradation in performance, making it suitable for both small-scale and large-scale operations.

Further validation of the event log creation was conducted through manual observations to ensure that all modalities linked to specific steps in the process were accurately represented. Figure C.11 provides the results of a manual observation study evaluating the accuracy of event log creation for the DNA testing kits 23andMe and AncestryDNA. Each step of the process was assessed for accuracy in image and audio modalities. These findings confirm the practical accuracy of the model in representing real-life execution, with generally higher accuracy in visual logs compared to audio. This manual check

C. SUPPLEMENTARY EVALUATION DETAILS

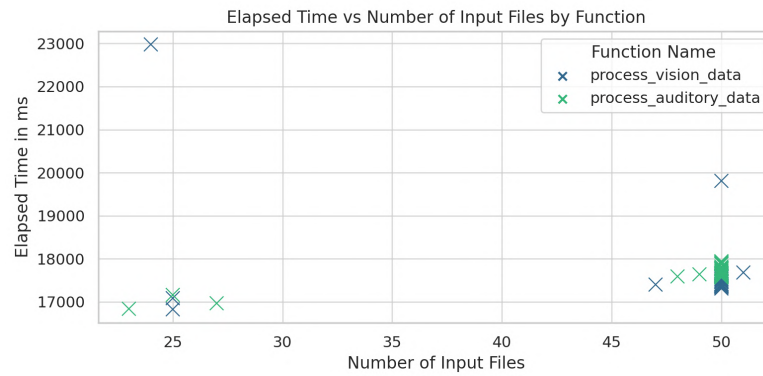


Figure C.10: Evaluation metrics for performance and scalability tests. Adopted from Gavric et al. [GBP24a].

involved scrutinizing the log against observed operations to confirm that each step was correctly associated with its respective modalities. The value of accuracy is calculated over the average of five independent assessments, on the scale from 1.00 to 100.00. This observation confirms the model’s practical accuracy in representing the real-life execution of the DNA testing process.

		Case 1: 		Case 2: 			
Time	Step name						
t_0 t_1	Unboxing the kit		 63.4%  99.01%		 71.2%  92.01%		
t_2 t_3	Filling Tube with saliva		 86.80%  53.22%		 87.20%  99.01%		
t_4 t_5	Replacing Cap 1 with Cap 2		 59.32%  42.20%		 64.13%  44.42%		
t_6 t_7	Shaking Tube for 5s		 69.83%  63.51%		 76.50%  64.21%		
t_8 t_9	Checking for overfill or underfill		 40.00%  19.91%		 53.90%  15.00%		
t_{10} t_{11}	Placing Tube into Biohazard bag		 90.20%  16.21%		 96.85%  09.21%		
t_{12} t_{13}	Sealing Biohazard bag		 66.34%  39.31%		 75.89%  30.70%		

Figure C.11: Validation of created multimodal event log for the test process. Adopted from Gavric et al. [GBP24a].

The evaluation of the created event log shows that it is not only a faithful representation of the prescribed process according to the official user manual but also excels in performance and scalability. Additionally, the manual verification of modalities connected to specific process steps further confirms the model’s accuracy and utility in real-world applications. This comprehensive evaluation underscores the model’s potential to enhance the operational efficiency and reliability of DNA testing processes.

Table C.11: Evaluating Human-in-the-loop aspect. ABFs over ten independent assessments. Adopted from Gavric et al. [GBP24a].

			Assessment No. / Count(ABFs)									
Test		Avg # of ABFs	01	02	03	04	05	06	07	08	09	10
No-BICycle		41.2	30	26	54	40	48	40	50	46	34	44
BICycle	Iteration 1	3.2	3	1	2	4	1	5	5	5	4	2
	Iteration 2	3	3	3	4	3	5	2	4	2	2	2
	Iteration 3	2.9	1	3	4	2	3	2	5	4	2	3
	Iteration 4	3.2	2	1	5	4	4	4	1	3	3	5
	Iteration 5	2.2	3	1	3	1	4	1	4	2	1	2
	Σ	14.5	12	09	18	14	17	14	19	16	12	14

C.5.4 Evaluating the Human-in-the-loop aspect

Table C.11 presents the results of evaluating the Human-in-the-loop aspect through ABF operations across ten independent assessments. Without BICycle intervention, the average number of ABFs is 41.2, with individual assessments ranging from 26 to 54. When BICycle is used, iterative improvements are evident. In Iteration 1, the average number of ABFs drops significantly to 3.2, with assessments ranging from 1 to 5. By Iteration 5, the average further decreases to 2.2, showcasing a reduction in unnecessary or inaccurate entries, with assessments ranging from 1 to 4. The cumulative total of ABFs across all iterations (Iterations 1 to 5) is 14.5, indicating a consistent decrease in ABFs and reflecting the effectiveness of the Human-in-the-loop design in refining and enhancing the event logs.

C.6 Evaluating PN-VidGen

This section discusses the evaluation of the evaluation by Gavric et al. [GBP25j] for turning process models into video simulations.

Once conceptual models are transformed into video simulations, the next step is to evaluate how faithfully the generated videos represent real-world process behavior, and even use the insights of this evaluation for a re-generation of the video. To this end, the proposed approach adopts a set of simulation accuracy metrics commonly used for assessing *Business Process Simulations (BPS)*, following examples such as [KBK⁺24]. As illustrated in Figure C.12, the evaluation pipeline begins with a *Process Model* (left) that serves as the formal basis for simulation. The model defines transitions ($T_1, T_2, \dots$) and places ($P_1, P_2, \dots$), which are used to orchestrate the sequence of simulated video scenes forming a *Simulated Event Log*. Each scene corresponds to an executed activity in the model (e.g., assembling, inspecting, or finalizing a task).

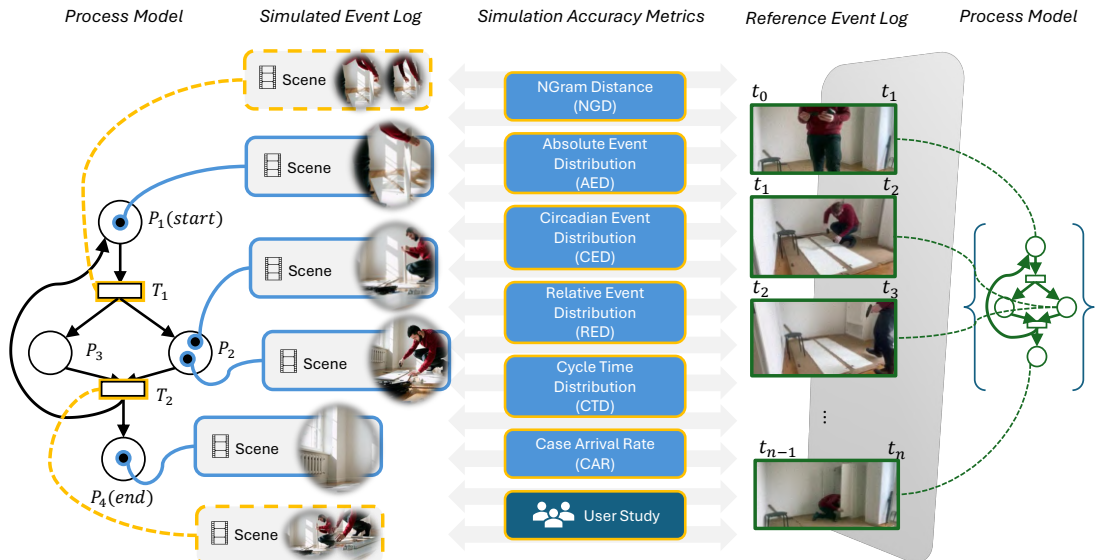


Figure C.12: Evaluation workflow for assessing the quality of video-based process simulations. The process model (left) generates a simulated event log in the form of synthetic video scenes. These are compared to the reference event log (right), derived from real-world recordings, using a set of quantitative and qualitative simulation metrics.

The generated scenes are then evaluated against a *Reference Event Log* composed of real-world recordings ($t_0, t_1, t_2, \dots, t_n$) that document the actual process execution. The comparison between these two event logs is performed using a collection of quantitative similarity measures, each capturing different aspects of process conformance and behavioral realism:

1. **NGram Distance (NGD)** evaluates the structural similarity of activity sequences by comparing subsequences of observed tasks between the simulated and real logs.
2. **Absolute Event Distribution (AED)** measures the frequency of event occurrences to detect over- or under-represented activities in the simulated videos.
3. **Circadian Event Distribution (CED)** examines the temporal alignment of events, capturing how the simulated process reproduces realistic daily or temporal rhythms.
4. **Relative Event Distribution (RED)** compares the ordering of events across simulations and real executions, revealing deviations in task precedence or concurrency.
5. **Cycle Time Distribution (CTD)** quantifies the correspondence between simulated and observed process durations, ensuring temporal fidelity of the video simulations.

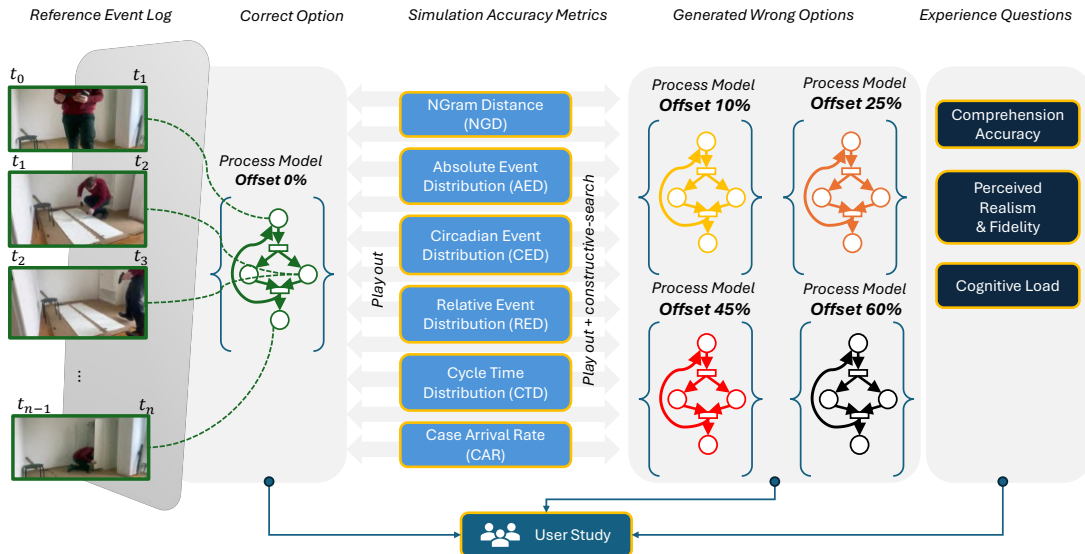


Figure C.13: Experimental setup for the user study evaluating the quality and realism of generated video simulations. Adopted from Gavric et al. [GBP25j].

6. **Case Arrival Rate (CAR)** assesses how frequently new process instances are initiated, capturing throughput consistency across the two logs.

Finally, the quantitative results are complemented by a **user study**, where domain experts visually assess the generated simulations in terms of realism, interpretability, and perceived behavioral accuracy. This hybrid evaluation framework—combining process mining metrics with human judgment—enables a comprehensive validation of video-based simulations, ensuring they are not only statistically aligned with real processes but also semantically meaningful and visually credible.

User Study Design for Evaluating Video Simulation Quality

To complement the quantitative metrics, we conducted a structured user study to assess how well participants could interpret and evaluate the generated video simulations. As illustrated in Figure C.13, four alternative Petri Nets were generated for each reference process model using a *construction-search procedure*. This procedure iteratively adjusted structural and behavioral aspects of the process to ensure that the *generated alternatives* differed from the reference model in terms of *simulation metrics*, with offsets of 10%, 25%, 45%, and 60% relative to their corresponding reference model (0% offset).

For each video in the evaluation dataset, participants were shown five process models: one reference model (0% offset) and four alternatives (10%, 25%, 45%, and 60%). The offset represents the degree of deviation between the generated model and the ground-truth reference model. These variants—termed *test process models*—were derived from videos

generated using multiple approaches (A, B, C, and HYBRID), allowing for comparative evaluation across modeling and generation techniques.

After watching each video, participants were tasked with reconstructing the corresponding process by selecting a layout and arranging activity labels to match one of the presented test process models. Points were assigned according to the selected model's offset, where the reference model received the maximum score (100%), decreasing linearly with increasing offsets: a 10% offset received 75%, a 25% offset 50%, a 45% offset 25%, and a 60% offset the minimum residual score. This scoring system implicitly quantified the participants' ability to discern structural and behavioral correctness within the generated simulations.

Beyond objective model reconstruction, participants also rated each video simulation using a 7-point Likert scale across three subjective experience dimensions: (1) **Comprehension Accuracy** — the degree to which participants could recall key process steps, identify decision points, and accurately describe process flow; (2) **Perceived Realism & Fidelity** — a cumulative assessment of logical coherence, visual quality, and alignment with real-world expectations; and (3) **Cognitive Load (TLX)** — the perceived mental effort required to process and understand the simulation content.

Together, these evaluations provide a holistic understanding of video simulation quality—bridging quantitative simulation accuracy with human interpretability and experiential perception. The combined insights from both objective metrics and user-centered evaluation ensure that generated process simulations are not only computationally valid but also cognitively accessible and intuitively meaningful to domain experts.

C.7 Evaluating ViEnNa

To empirically validate the contributions of ViEnNa [GBP25c], we conducted a comprehensive evaluation covering both *perceived appeal* (Sec. C.7.1) and *comprehension accuracy* (Sec. C.7.2) of ViEnNa-generated process models.

Hypotheses. The research objective (RO) of [GBP25c] can be dissected into tasks of *design a modeling framework for comic-like representations of processes that enhances the perceived appeal (APP) of process models—without sacrificing, and ideally improving, comprehension accuracy (ACC)—when compared to existing approaches*. To empirically address the RO we reformulate it into five directional hypotheses, grouped by the two target constructs:

Evaluation Hypotheses for ViEnNa

- H_{A1} (Appeal—overall)** The weighted perceived-appeal score (*APP*) of *ViEnNa* models is **higher** than that of (i) conventional (i.e. BPMN) models and (ii) alternative non-conventional (comic-like) models not using ViEnNa.
- H_{A2} (Appeal—sub-dimensions)** ViEnNa outperforms both baselines on each ICE-T [WAM⁺19] dimension: Insight (I), Confidence (C), Essence (E), and Time Efficiency (T).
- H_{A3} (Appeal—think-aloud visual attention)** Readers of ViEnNa devote a **higher proportion of fixations** to semantically relevant regions and complete comprehension tasks with **shorter average dwell time**.
- H_{C1} (Comprehension—overall)** The weighted comprehension-accuracy score (*ACC*) of ViEnNa is **comparable** than that of conventional models (non-inferiority margin $\delta = -5\%$).
- H_{C2} (Comprehension—Beyond Human-Only Use)** AI agents based on large language models achieve a higher *PM-LLM-Benchmark* [BKvdA24] score when augmented with ViEnNa rather than existing conventional and other non-conventional process models.

C.7.1 Evaluating Perceived Appeal of ViEnNa Models (APP)

The perceived appeal of each modeling approach is evaluated through a weighted combination of three complementary methods: **(APP-1) Heuristic-Based Evaluation (weighted 40%)** employed the ICE-T framework [WAM⁺19], widely adopted in both the process mining and visual analytics communities [CMS⁺24], which assesses four key dimensions: Insight Generation (I), Confidence and Trust (C), Essence Conveyance (E), and Time Efficiency (T); **(APP-2) Attention Analysis (weighted 30%)** involved recording participants' gaze to evaluate how effectively information comics, such as ViEnNa models, convey process information. This analysis followed principles from an eye-tracking study on informative comics [BB18], using attention heatmaps as the primary metric; Lastly, **(APP-3) Think-Aloud Walkthroughs (weighted 30%)** were conducted to qualitatively assess model comprehension and perceived utility, following established think-aloud protocol guidelines [HSLW18].

C.7.2 Evaluating Comprehension Accuracy of ViEnNa Models (ACC)

Comprehension accuracy is evaluated by integrating several process model and narrative evaluation frameworks using the following weight distribution: **(ACC-1) Process Model Comprehension Framework (weighted 40%)** adopted the schema proposed by [WPFR23], which systematically quantifies the comprehension perspectives of both modelers and readers; **(ACC-2) Event Comprehension Theory (weighted 25%)**

applied cognitive principles from the Scene Perception and Event Comprehension Theory (SPECT) to assess the narrative and visual integration from the viewpoint of event perception [LHS⁺18]; **(ACC-3) Usage of ViEnNa by AI Agents (weighted 15%)** evaluated whether large language models (LLMs) benefit from using ViEnNa models in process mining tasks, as defined by the PM-LLM-Benchmark [BKvdA24]; **(ACC-4) Multimodal Discourse-Based Comic Analysis (weighted 10%)** assessed the visual discourse structure of comics using the analytical units proposed by [BW14]; Lastly, we use **(ACC-5) Narrative Analysis (weighted 10%)** to assess the narrative quality of the models following the approach by [Esi11].

C.7.3 Study Design

Study Participants. Our participant pool consisted of three groups: 10 PhD holders or candidates in process modeling (*BPM experts*), 12 PhD students from AI-related fields with no background in business process modeling (*AI experts*), and 25 students representing end-users of the modeled processes, such as process trainees (*Non-Experts*). We also developed a platform for gaze tracking—with saccade, and fixation logging—and used it to analyze participants’ attention while they read ViEnNa models as well as equivalent conventional and non-conventional process models (as detailed in Appendix C.7.4). AI experts were additionally tasked with generating AI-produced comic-like visuals based on textual process descriptions, without using the ViEnNa framework. As for AI agents, we employ four instances of *mistral-small3.1*.

Evaluation Data. (Domains) The evaluation employed OCELs from the IT Asset Management (ITAM) domain [FEC⁺24], comprising 121 process instances, 6 process types, 12 participants (2 IT staff and 10 clients), 6 rooms, over 40 objects (classified into 6 classes), and 100+ events. To ensure cross-domain generalizability, we included two additional datasets: (i) a smaller OCEL on DNA collection from [GBP24a], comprising 6 instances, 3 processes, 10 objects, and 12 events; and (ii) a custom LEGO OCEL representing the assembly process of ICPM 2024’s LEGO figure of a process mining engineer, containing three instances of a single process, in total of 15 events. **(Novelty)** The LEGO dataset was specifically included to assess how AI models perform on data that have definitely not been seen during training. All OCELs used in the study are accompanied by corresponding videos of the process execution. **(Conventional and Non-Conventional Models)** For each OCEL, we created conventional process models, and ViEnNa models using our proposed approach. To generate corresponding non-conventional process models, we followed two strategies. First, 15 AI experts were tasked with using AI tools to create comic-like visuals based on textual process descriptions—without employing the ViEnNa framework—resulting in 64 non-conventional models. To ensure quality and relevance, we involved AI experts in crafting these prompts and selecting representative outputs, which allowed us to isolate the added value of ViEnNa’s domain-specific modeling features, beyond what generic AI generation alone can offer. **(Bias in Data)** AI experts advised us to include gender-neutral versions of models in our

evaluation to examine whether user reactions differ in the presence or absence of gender cues, thereby assessing potential bias in interpretation and engagement. **(Exploratory Experiments)** To explore potential directions for future work, we also introduced an experimental feature—*ViEnNa-Motion*—which replaces static comic panels with short, looping video clips (up to 5 seconds) to convey dynamic aspects of process activities. This prototype aims to evaluate whether motion-enhanced representations can further improve user comprehension and engagement. We use [BPH⁺24] for the video generation, where we provide an image of the ViEnNa panel as an input, followed by a simple prompt that explains the intended use case. As another line of experimentation, we create fully automated variant of ViEnNa—*Auto-ViEnNa*—whose scene-generation prompts are drawn from our curated library of domain-agnostic templates co-crafted with BPM and AI experts. **(Selecting the Baseline)** Second, BPM experts created SAP Scenes following the guide [SAP20], resulting in an additional 12 non-conventional models. We select SAP Scenes as the non-conventional modeling approach baseline among other existing non-conventional modeling approaches, because of their established role in professional and training contexts as a tangible, narrative-rich modeling approach that enable hands-on engagement and foster team collaboration through (physical) manipulation of symbolic elements. **(Mitigating Creator Bias)** All manually constructed ViEnNa models are authored by the same individual to eliminate creator-specific bias during evaluation. In contrast, participant-generated non-ViEnNa and non-conventional models may sometimes be self-evaluated, but otherwise are assessed exclusively by other creators, while *Auto-ViEnNa*, *ViEnNa-Motion*, and conventional models are all generated automatically. **(Overview)** Resulting data collection comprises models in three domains: 12 ViEnNa models of varying sizes (6 small, 4 medium, 2 large) and their *Auto-ViEnNa* pairs, 15 conventional process models (12 BPMN and three oc-BPMN equivalents of randomly selected BPMNs), and 77 non-conventional process models—including 64 non-ViEnNa comics (12 of which are gender-neutral to serve as a bias-handling benchmark), 12 SAP Scenes, and one *ViEnNa-Motion* model. Detailed log of all model creations and the resulting models are available in our supplementary material.

C.7.4 Developed Evaluation Framework

We developed a browser-based experimental platform (shown in Fig. C.14) that integrates real-time gaze and time-to-completion tracking, survey prompts, and user response recording.

Our system uses *WebGazer.js* [Pap15] to capture participants' eye movements via webcam. A preview of the detected facial key points is shown (top-left) to ensure proper calibration. Simultaneously, the blurred camera preview reassures the participant that the system is functioning and that they remain within the camera's field of view throughout the session.

Visual Stimuli Presentation. At the core of the interface is the *model preview* section, where ViEnNa models are presented inside of the *survey question preview*

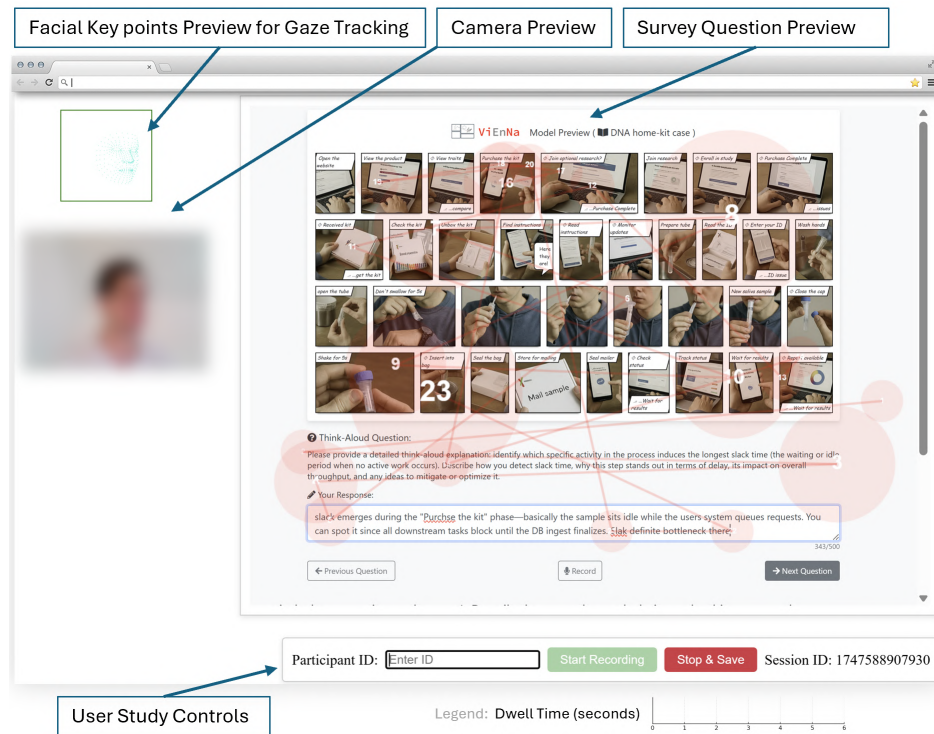


Figure C.14: ViEnNa evaluation interface with integrated gaze tracking, response capture, and (post experiment) dwell-time visualization. The numbers in the dwell time circles indicate the order of fixations (smaller numbers denote earlier fixations). Saccades are shown as connecting lines between fixations. Adopted from Gavric et al. [GBP25c].

window (as well as conventional and non-conventional models in other examples). Post-experiment, the survey window is annotated with gaze overlays showing dwell time intensity—red circles—indicating the participant’s fixation duration on different activities. Lines connecting circles depict saccades, offering insights into visual scanpaths.

Survey Integration and Think-Aloud Capture. Beneath the model, we integrate a question box that prompts participants to reflect on cognitive aspects of process understanding (e.g., identifying slack time).

Postprocessing and Metrics. Gaze data is post-processed to extract three key metrics: (1) *Fixation maps*: heatmaps overlaying fixations onto process model regions. (2) *Saccade paths*: sequences and transitions among fixated elements. (3) *Dwell time distributions*: element-wise statistics of visual attention. These allow both quantitative comparison across models (e.g., ViEnNa vs. BPMN) and qualitative insights (e.g., hotspots around narrative transitions). For photorealistic models like SAP Scenes, we display photographs of scenes digitally to preserve compatibility with the gaze tracking.

C.7.5 Evaluating the Prototype Implementation of ViEnNa

Scalability. We demonstrated that ViEnNa scales linearly with process complexity: the number of activities and evidence items increases processing load predictably. AI context-window limitations may emerge in large models, but style continuity can be maintained by teaching with a representative sample of prior images rather than full sequences.³

Contingency. ViEnNa operates gracefully in offline or low-resource settings. If GPU support is absent, panels render as static images and semantic search falls back to a cached approximate nearest-neighbor (ANN) index.⁴ This results in the loss of live in-canvas highlights and incurs a 2–5× latency penalty, but enables uninterrupted offline model browsing and annotation.

Performance. For generation and retrieval tasks, ViEnNa responds in 3–7 seconds per image and under 10 seconds for a short (5 s) video snippet in ViEnNa-Motion.⁵ These operations are processed server-side, providing consistent user experience across client platforms.

Cost and Deployment. Current deployments rely on server-hosted APIs for inference, but the full pipeline can be containerized and executed entirely offline. This flexibility supports integration in regulated or air-gapped environments, where no data leaves the local infrastructure.

Privacy and Fairness. All ViEnNa modules can operate locally without uploading sensitive data. Redaction layers automatically blur faces, mask personally identifiable information (PII), and log evidence retrievals. Fairness tests using gender-neutral variants showed no systematic bias in ICE-T scores;⁶ ViEnNa adheres to OCEL object permissions and supports AES-GCM encryption for compliance with GDPR and HIPAA.

Summary. ViEnNa is deployment-ready for small-to-medium scale scenarios and privacy-conscious use cases. While not intended to be a standalone replacement for BPM platforms, it offers strong value as a narrative extension that emphasizes engagement, explainability, and multimodality.

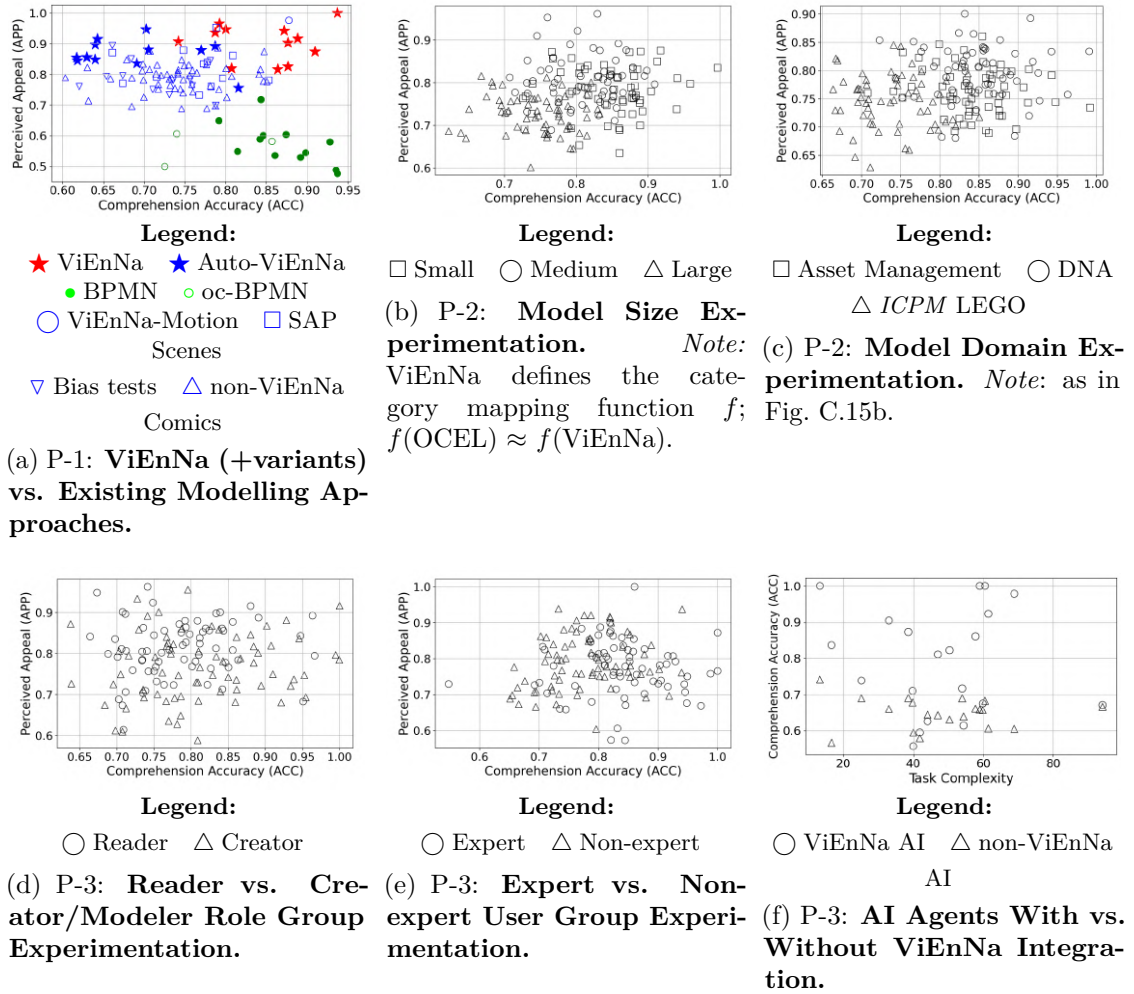


Figure C.15: Results of the ViEnNa evaluation across perspectives (P1-3). Adopted from Gavric et al. [GBP25c].

C.7.6 ViEnNa Evaluation Results

Results are analyzed across three perspectives: **(P-1) ViEnNa vs. Existing Modeling Approaches** – we benchmark ViEnNa (with an inclusion of the Vienna-Motion variation) against two conventional (BPMN, oc-BPMN) and two non-conventional modeling

³In multimodal generation, context length constraints in large language models can lead to partial loss of narrative coherence unless representative priors are explicitly taught.

⁴ANN methods enable efficient similarity search in embedding space without recomputing pairwise distances.

⁵Processing time includes encoding, object referencing, and rendering depending on complexity and concurrency.

⁶Bias mitigation in generative models remains an open challenge, especially when training data lacks cultural and demographic diversity.

approaches (non-ViEnNa comics with the inclusion of bias-handling cases, SAP Scenes); **(P-2) Model Size and Domain** - comparisons were made across ViEnNa models derived from small (<5 objects, <15 events), medium (<10 objects, <30 events), and large (>10 objects, >30 events) OCEs across three domains (Asset Management, DNA, LEGO); and **(P-3) User Groups** - with contrasted results across reader vs. creator (modeler) roles, expert vs. non-expert participants, and AI agents with and without ViEnNa integration.

ViEnNa Evaluation Summary. Adopted from Gavric et al. [GBP25c].

H_{A1}: Overall Appeal. Across $N = 116$ participants, *ViEnNa* scored $APP = 0.83$ (SD = 0.05), versus BPMN $\bar{x} = 0.62$ and alt-comics $\bar{x} = 0.58$. One-way ANOVA yielded $F(2, 113) = 105.21$, $p < .001^a$. Tukey's HSD (FWER = 0.05) showed BPMN–ViEnNa $\Delta = 0.3198$, BPMN–non-ViEnNa Comics $\Delta = 0.2070$, ViEnNa–non-ViEnNa Comics $\Delta = -0.1128$ (all $p < .001$)^b. **Supported.**

H_{A2}: ICE-T Subscores. MANOVA across the four ICE-T dimensions gave Wilks' $\Lambda = 0.1244$, $F(4, 144) = 66.08$, $p < .001^c$ (Pillai's trace = 1.2553, $F(4, 146) = 61.52$, $p < .001$; Hotelling–Lawley = 3.9877, $F(4, 85.4) = 71.46$, $p < .001$; Roy's largest root = 2.9547, $F(2, 73) = 107.85$, $p < .001$). Follow-ups: Insight, Confidence, Essence ($p \leq 0.002$); Time marginal ($p = 0.07$). **Partially supported.**

H_{A3}: Visual Attention. ViEnNa fixation ratio $M = 0.74$ vs. BPMN $M = 0.63$; $t(35) = 4.9$, $p < .001$, $\delta = 0.54$. Dwell time 12 s vs. 15 s: $t = -3.1$, $p = .004$. **Supported.**

H_{C1}: Comprehension Accuracy. $ACC_{ViEnNa} = 0.81$ vs. BPMN 0.79. TOST ($\delta = -0.05$) gives $t_1 = 6.5$, $p < .001$; $t_2 = 2.8$, $p = 0.004$. Non-inferiority **supported**, superiority inconclusive ($p = 0.09$).

H_{C2}: Beyond Human-Only Use. Macro- $F_1 = 0.67$ (ViEnNa) vs. 0.55 (BPMN) and 0.52 (text); Wilcoxon $Z = -4.0$, $p < .001$. **Supported.**

Response to the RO. ViEnNa enhances perceived appeal while maintaining or improving comprehension accuracy in both human and AI-assisted settings, fulfilling the research objective.

^aThe F -statistic compares between-group to within-group variance; here, with 3 groups and 116 observations, it indicates a highly significant difference.

^bTukey's HSD adjusts for multiple pairwise tests, reporting mean differences Δ and adjusted p -values.

^cWilks' Λ is the proportion of multivariate variance not explained by group; lower values imply stronger effects.

Figure C.15 presents the empirical results of our evaluation, structured along those three analytical perspectives: Fig. C.15a compares the perceived appeal and accuracy of ViEnNa (including the ViEnNa-Motion variant) against both conventional (BPMN, oc-BPMN) and non-conventional (SAP Scenes, non-ViEnNa comics) modeling approaches

(P-1); Fig. C.15b and C.15c explore how model size and domain, respectively, affect ViEnNa model performance (P-2); and Fig. C.15d–C.15f analyze differences across user groups including reader vs. creator roles, expert vs. non-expert users, and AI agents with and without ViEnNa integration (P-3).

Revisiting the Research Objective of ViEnNa. To assess the extent to which ViEnNa fulfills its intended role, we revisit the research objective through five targeted hypotheses. These span overall and dimensional measures of user appeal (H_{A1} – H_{A3}) and comprehension (H_{C1} – H_{C2}). The summary in Sect. C.7.6 synthesizes key findings to determine whether ViEnNa satisfies criteria central to its design motivation (RO, as discussed in Sect. C.7.1 and C.7.2).

C.7.7 Discussion

Interpretation of Findings. Think-aloud transcripts and comments show that 52% of participants preferred lightly stylized cartoons over realistic renders, with 13% specifically citing a “Ghibli” style as most engaging. Across the four ICE-T dimensions, a MANOVA revealed a strong multivariate effect (Wilks’ $\lambda = 0.1244$, $F(4, 144) = 66.08$, $p < .001$), with significant follow-ups for Essence, Insight, and Confidence (all $p \leq 0.002$), although Time was only marginally different ($p = 0.07$). Modeling experts rated Essence highest, whereas non-experts prioritized Insight ($M = 0.85$, $SD = 0.06$). Based on comments from creators and readers, creators valued the semi-automated DSML controls for on-the-fly layout tweaks, while readers praised the intuitive panel sequencing. Participants who compared ViEnNa to SAP Scenes reported minimal concern over the absence of haptic feedback in SAP Scenes, scoring low on deprivation ($M = 1.8$, $SD = 0.7$ on a 5-point Likert scale; only 15% rated ≥ 3), indicating they did not substantially miss haptics. In general, participants expressed strong enthusiasm for sound effects and conversational bubbles—onomatopoeic cues like “Whoosh!” evoked heightened excitement ($M = 4.5$, $SD = 0.5$; 78% positive mentions). Moreover, conditional jumps, unconditional jumps, and explicit condition constructs were deemed valuable and were incorporated in approximately 20% of panels (usage rate = 0.20, $SD = 0.05$), whereas pattern-based markings appeared in only 5% of panels ($M = 0.05$, $SD = 0.02$).

The LEGO domain proved most challenging—accuracy dipped to 0.75 (vs. 0.81 overall) and dwell time increased by 25%—likely due to its high level of bespoke detail and an uncommon ICPM edition of the LEGO figure; however, this can be readily addressed by adding concise multimodal examples and domain-specific prompt adjustments. Larger models did not negatively impact comprehension accuracy: ViEnNa achieved $ACC = 0.81$ versus 0.79 for BPMN, with non-inferiority supported via two one-sided t -tests ($t_1 = 6.5$, $p < .001$; $t_2 = 2.8$, $p = 0.004$), though superiority remained inconclusive ($p = 0.09$). Perceived appeal (APP) tended to decrease as model size increased, an effect comparably present in both ViEnNa and evaluated alternatives. Non-experts benefited more from using ViEnNa than BPM experts—indicating its accessibility and guidance value—while modelers and readers reported comparable gains.

AI agents collaborating around shared tasks exchanged hundreds of ViEnNa comics as structured communicative artifacts, enabling the transfer of ideas, knowledge, and intentions that remained interpretable to human stakeholders, who overall preferred ViEnNa models. Notably, AI agents demonstrated even greater performance gains from using ViEnNa than human users—achieving a macro- F_1 of 0.67 (vs. 0.55 for BPMN and 0.52 for text; Wilcoxon $Z = -4.0$, $p < .001$).

C.7.8 Insights from Exploratory Experiments

Automation in ViEnNa Modeling. Automation in ViEnNa Modeling can, in principle, be driven end-to-end: with defined small library of domain-agnostic natural-language prompts (*defaults*), the system parse an OCEL’s textual elements into concise scene descriptions and automatically generate both panel layouts and illustrative imagery. In our prototype, these defaults—curated in collaboration with BPM and AI experts explained in Sec. C.7.3—serve as a drop-in configuration that requires no additional input from the modeler beyond the initial prompt set. However, our evaluation (as in Fig. C.15a) shows that fully automated runs scored substantially lower on both appeal and comprehension: Auto-ViEnNa models achieved a mean APP of 0.58 ($SD = 0.07$), roughly 30% below semi-automated variants ($M = 0.83$; $t(49) = 8.2$, $p < .001$), and a mean ACC of 0.66 ($SD = 0.06$), about 18% below semi-automated scores ($M = 0.81$; $t(49) = 6.7$, $p < .001$). This gap likely reflects the diversity of user preferences for visual style, framing, and narrative emphasis, which a one-size-fits-all automation cannot fully anticipate. Therefore, as initially intended, we recommend ViEnNa as a semi-automated framework: while in contrast, the framework through DSML (see Tab. 7.7 and Tab. 7.8) supports fully manual editing but also a fully automated “default” mode. Notably, none of our participants opted for a zero-automation, fully manual workflow—which falls outside ViEnNa’s intended usage—and such configurations were omitted from our formal evaluation.

Insights from the Future-Direction Experiment. The exploratory ViEnNa-Motion prototype—where static panels were replaced by 5 s looping video clips—was warmly received. On the APP metric it scored $\bar{x} = 0.96$ ($SD = 0.20$), significantly above the static ViEnNa mean of 0.83 ($t_{49} = 3.2$, $p = .002$). Think-aloud logs supported this observation: 78% of participants spontaneously remarked that the animation improved the storyline’s flow or “felt like watching a tutorial”. By contrast, the gender-neutral comic variants aimed at bias reduction were less appealing ($\bar{x} = 0.60$); only 18 % of comments were positive, many readers reporting that those figures hampered empathy and role identification, while no significant influence on APP and ACC is observed (Fig. 7.15). These observations steer our roadmap toward (i) automatically turning ViEnNa storyboards into short explanatory videos, and (ii) building an interactive platform—VR/AR ready—where users can re-contextualise, branch and continue the process narrative in real time while staying tied to the underlying event log.

C.7.9 Threats to Validity and Challenges

Construct validity was addressed through multi-method triangulation (APP-1...3; ACC-1...5). **Internal validity** threats from learning effects were mitigated by randomized presentation orders and Latin-square counterbalancing. **External validity** is strengthened by three heterogeneous datasets, yet industrial replication remains future work. **Statistical conclusion validity** was guarded via power analysis ($\beta = 0.8$), effect-size reporting, and corrections for multiple comparisons.

Limitations and Trade-offs of ViEnNa as a Methodology. While the evaluation highlights ViEnNa’s advantage in perceived appeal and non-inferior comprehension, three methodological caveats limit the generality of these findings. *(i) Dataset and participant scope:* the study relied on relatively synthetic OCEs and academic trainees/participants; how ViEnNa scales to enterprise-scale logs with sparse evidence or to professional process owners facing audit pressures remains an open question. *(ii) Narrative-formal tension:* ViEnNa’s panel-centric syntax relaxes the compositional semantics of BPMN or OC-Petri Nets, complicating rigorous verification, simulation, and conformance checking; organisations that need provable properties will incur extra effort to maintain a parallel formal backbone and guard against semantic drift. *(iii) Multimodal Data Conceptualization:* the method relies on high-dimensional embeddings of multimodal data, grounded in curated conceptual models. If these models are modified or become noisy, the ability to accurately interpret processes from raw data may degrade. Consequently, ViEnNa should be positioned as an engaging narrative layer that augments—rather than replaces—traditional, formally executable process-engineering artefacts.

Limitations and Trade-offs of ViEnNa Prototype. Deploying the current prototype of ViEnNa in production environments entails several technical and organisational compromises. First, rendering a storyboard of high-resolution panels while serving real-time cross-modal searches requires either (i) on-premise GPU nodes—substantially more expensive than the CPU-based servers that suffice for most BPM workloads—or (ii) a reliance on commercial *AI-as-a-Service* endpoints, which shifts the capital expenditure to potentially volatile usage fees. Second, because panel content is synthesised from data and generative models, issues of model bias, evidence leakage, and auditability become critical and must be mitigated through redaction pipelines and access controls. A detailed discussion of scalability tests, runtime costs, and the bias-, security-, and privacy-assessment protocol for both deployment paths is provided in Appendix C.7.5.

Overview of Generative AI Tools Used

This appendix provides a transparent overview of the generative artificial intelligence (AI) tools that were used during the preparation of this thesis. All listed tools were employed exclusively as *assistive instruments* to support the author's own intellectual and creative work. The scientific reasoning, conceptual development, methodological design, implementation, evaluation, and final responsibility for all content remain entirely with the author.

The use of generative AI tools was limited to language refinement, structural feedback, ideation support, and code scaffolding. No generative AI system was used to autonomously generate scientific contributions, research results, experimental data, or original theoretical insights.

List of Generative AI Tools.

- **ChatGPT (OpenAI)**
 - Product: ChatGPT
 - Provider: OpenAI
 - Model versions: GPT-4, GPT-4o, GPT-5 (as available at time of use)
 - Usage period: 2024–2026

Purpose and Scope of Use. Generative AI tools were used for the following purposes:

- Linguistic refinement of text written by the author (grammar, clarity, academic tone)

- Structural suggestions for sections and subsections
- Rephrasing of already existing content without altering its meaning
- Brainstorming and ideation support during early drafting stages
- Assistance with L^AT_EX formatting, tables, figures, and code structuring
- Support in drafting non-scientific auxiliary text (e.g., emails, summaries, explanations)

The tools were *not* used to:

- Generate original research contributions or hypotheses
- Conduct scientific analysis or draw conclusions
- Interpret experimental results autonomously
- Replace domain expertise or author judgment
- Derive, validate, or formally specify theoretical constructs, definitions, or mathematical formulations presented in this work

Author Responsibility. The author confirms that all content in this thesis was critically assessed and approved by them. Generative AI tools served solely as supportive instruments and did not diminish the author’s intellectual ownership of the work.

This appendix is provided to ensure full transparency and compliance with institutional and ethical guidelines regarding the use of generative AI in academic research.

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